

PVR Cloud

03/22/2026 13:02:26

Thought process: Orchestrated comprehensive cloud overview covering platforms,...

Thought process: Orchestrated comprehensive Azure Data Factory curriculum...

Thought process: Audited PDF content comprehensively and identified multiple...

Thought process: Orchestrated comprehensive Databricks curriculum spanning...

Thought process: Orchestrated comprehensive Azure Synapse curriculum aligned...

Thought process: Orchestrated comprehensive Microsoft Fabric curriculum mapping...

Thought process: Verified all PDF topics covered, prepared comprehensive interview...

Thought process: Orchestrated comprehensive interview question expansion across...

Thought process: Orchestrated comprehensive cloud overview covering platforms, Azure, and big data fundamentals.

The user wants me to explain the "Overview of Cloud" section from the PDF in depth with real-world examples. Let me cover all the topics under Overview of Cloud:

1. Basics of Cloud Computing
2. Cloud Computing Platforms/Vendors
3. Introduction to Azure
4. Overview of Azure Resources/Services
5. Introduction to BigData

Let me provide a comprehensive, in-depth explanation.

Overview of Cloud — Complete In-Depth Guide

Basics of Cloud Computing

What is Cloud?

Cloud computing means using computing resources — servers, storage, databases, networking, software — **over the internet**, instead of owning and maintaining physical hardware yourself.

Real-World Analogy: Think of electricity. You don't build your own power plant at home — you simply plug into the grid and pay for what you use. Cloud computing works the same way. You "plug into" a provider's infrastructure and pay only for the resources you consume.

Before Cloud (Traditional IT):

- A company needed to buy physical servers
- Set up a data center with cooling, power, security
- Hire IT staff to maintain it
- If traffic spiked (e.g., Black Friday), they had to pre-buy extra servers that sat idle 11 months a year

With Cloud:

- No upfront hardware cost
 - Scale up/down instantly
 - Pay only for what you use
 - Access from anywhere in the world
-

◆ **Types of Cloud Deployment Models**

A. Private Cloud

A cloud environment **dedicated exclusively to one organization**. The infrastructure is either on-premises or hosted by a third party, but it is not shared with anyone else.

Real-World Example: A large bank like HDFC or JPMorgan runs its own private cloud. Banking regulations require that customer financial data never leaves their controlled environment. So they build a private cloud within their own data centers.

Characteristics:

- Full control over security & compliance
 - High cost (you maintain everything)
 - Best for: regulated industries — banking, healthcare, government
-

B. Public Cloud

Cloud resources are **owned and operated by a third-party provider** (like Microsoft Azure, AWS, GCP) and **shared across multiple customers** over the internet. Each customer's data is logically isolated.

Real-World Example: Netflix runs almost entirely on AWS public cloud. When millions of users stream simultaneously, Netflix doesn't own those servers — AWS provides them on demand. Netflix scales from 1 million to 100 million streams within minutes.

Characteristics:

- No hardware to manage
 - Pay-as-you-go pricing
 - Massive scalability
 - Best for: startups, SaaS products, web applications
-

C. Hybrid Cloud

A **combination of private and public cloud**, where data and applications can move between them.

Real-World Example: A hospital stores patient records on a private cloud (for HIPAA compliance), but uses the public cloud (Azure) for running AI diagnostics on anonymized data and for their patient appointment booking website.

Characteristics:

- Sensitive data stays private
 - Non-sensitive workloads use public cloud
 - Best for: enterprises with compliance needs but wanting cloud flexibility
-

◆ Types of Cloud Services

A. IaaS — Infrastructure as a Service

The cloud provider gives you **raw infrastructure**: virtual machines, storage, networking. You manage the OS, middleware, runtime, and everything above it.

Real-World Analogy: Renting an empty apartment. The building (infrastructure) is maintained by the landlord, but you furnish and decorate it yourself.

Example: Azure Virtual Machines (VMs). A company rents a VM on Azure, installs Windows Server, SQL Server, and their application on it. They manage patches, configurations, and scaling.

You manage: OS, middleware, runtime, applications, data **Provider manages:** Virtualization, servers, storage, networking

B. PaaS — Platform as a Service

The cloud provider manages the **infrastructure + OS + runtime**. You just bring your **code and data**.

Real-World Analogy: Renting a fully furnished apartment. You just move in with your clothes — no need to buy furniture or worry about plumbing.

Example: Azure SQL Database. You don't manage any server, OS, or SQL Server installation. You just create a database, connect, and run queries. Azure handles patching, backups, high availability automatically.

Another Example in Data Engineering: Azure Data Factory, Azure Databricks — you write pipelines and notebooks, Azure manages all the underlying infrastructure.

You manage: Applications, data **Provider manages:** OS, runtime, middleware, infrastructure

C. SaaS — Software as a Service

The cloud provider delivers a **complete, ready-to-use software application** over the internet.

Real-World Analogy: Staying in a hotel. Everything is handled — room, cleaning, food. You just show up and use it.

Examples:

- Gmail / Outlook 365 — you don't manage any server, just use the email service
- Salesforce CRM — ready-made CRM, you just enter your customer data
- Microsoft Teams / Zoom — video conferencing, fully managed

You manage: Just your data and user configurations **Provider manages:** Everything else

Quick Comparison Table

Model	You Manage	Provider Manages	Example
IaaS	OS, Apps, Data	Hardware, Network	Azure VMs
PaaS	Apps, Data	OS, Runtime, HW	Azure SQL DB, ADF
SaaS	Data only	Everything	Office 365, Gmail

Cloud Computing Platforms / Vendors

There are three dominant public cloud providers in the world:

1. Microsoft Azure

- Launched: 2010
- Owned by: Microsoft
- Market share: ~25% (2nd largest)

- **Strength:** Enterprise integration (Active Directory, Office 365, Windows Server), hybrid cloud, data analytics (ADF, Databricks, Synapse)
 - **Real-World Use:** Most Fortune 500 enterprises that already use Microsoft products (Windows, SQL Server) migrate to Azure naturally
 - **Key Data Engineering Services:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Microsoft Fabric
-

◆ 2. AWS — Amazon Web Services

- Launched: 2006
 - Owned by: Amazon
 - Market share: ~33% (largest)
 - **Strength:** Widest service catalog, most mature platform, dominant in startups and tech companies
 - **Real-World Use:** Netflix, Airbnb, Uber, NASA all use AWS
 - **Key Data Engineering Services:** AWS Glue (similar to ADF), Amazon EMR (similar to Databricks), Amazon Redshift (similar to Synapse)
-

◆ 3. GCP — Google Cloud Platform

- Launched: 2011
- Owned by: Google
- Market share: ~11% (3rd)
- **Strength:** AI/ML capabilities (TensorFlow, Vertex AI), BigQuery (fastest serverless data warehouse), data analytics
- **Real-World Use:** Twitter, Spotify, PayPal use GCP
- **Key Data Engineering Services:** BigQuery, Dataflow, Dataproc, Pub/Sub

🔥 Why Azure for Data Engineering?

- Seamlessly integrates with on-premise SQL Server and Windows-based systems
 - Azure Active Directory for enterprise-level security
 - ADF + Databricks + Synapse forms a complete, powerful data pipeline ecosystem
 - Most large enterprises (banks, insurance, healthcare) prefer Azure for compliance reasons
-

3 Introduction to Azure

◆ Azure Portal Walkthrough

The **Azure Portal** (portal.azure.com) is a web-based, unified console where you can manage all your Azure resources.

Real-World Analogy: Think of the Azure Portal as the dashboard of a car. All controls — engine (compute), fuel (storage), navigation (networking) — are accessible from one place.

Key sections in Azure Portal:

- **Dashboard** — Customizable home screen with your pinned resources
 - **All Resources** — View all Azure services you've created
 - **Cost Management** — Track and optimize spending
 - **Marketplace** — Browse and deploy pre-built solutions
 - **Cloud Shell** — Built-in command line (bash/PowerShell) right in the browser
-

◆ What is a Subscription?

A **Subscription** is a **billing and access boundary** in Azure. It ties your Azure usage to a payment method.

Real-World Analogy: Think of a subscription like a mobile phone plan. Your phone number (resources) operates under a specific plan (subscription) that determines what you can use and how much you're billed.

Key Points:

- All Azure resources you create belong to a subscription
- One Azure account can have **multiple subscriptions** (e.g., Dev subscription, Test subscription, Production subscription)
- Billing is tracked per subscription
- Access policies and spending limits can be set at the subscription level

Example in a Company:

- Subscription 1: "Development" — Dev team creates test pipelines here, budget capped at \$500/month
- Subscription 2: "Production" — Live pipelines run here, strictly controlled access

◆ What is a Resource Group?

A **Resource Group** is a **logical container** that holds related Azure resources for a solution.

Real-World Analogy: Like a folder on your computer. You create a folder called "Project_Sales" and put all files related to that project inside it. Similarly, a Resource Group named "RG-DataPipeline-Prod" holds all ADF, Storage, SQL, and Databricks resources for your data pipeline project.

Key Points:

- Resources in a group share the same lifecycle — you can deploy, update, and delete them all at once

- Resources in different regions can be in the same Resource Group
- Useful for cost tracking, access control, and management

Example:

```
Resource Group: RG-AzureDataEngineering-Dev
├─ Azure Data Factory: adf-sales-pipeline
├─ Azure Data Lake Storage Gen2: adlsstoragedatalake
├─ Azure SQL Database: sqlldb-salesdata
├─ Azure Databricks: databricks-workspace
└─ Azure Key Vault: kv-secrets-dev
```

◆ What is a Resource?

A **Resource** is any **individual service instance** you create in Azure.

Examples of Resources:

- One Azure Data Factory instance = 1 Resource
- One Azure SQL Database = 1 Resource
- One Azure Storage Account = 1 Resource
- One Azure Databricks Workspace = 1 Resource

Real-World Analogy: If the Subscription is the city, the Resource Group is the building, and Resources are the individual rooms/apartments inside that building.

4 Overview of Azure Resources / Services

These are the core Azure services you'll work with throughout the Data Engineering course:

◆ 1. Azure Data Factory (ADF)

ADF is Azure's **cloud-based data integration service** — like an orchestration engine for moving and transforming data.

Real-World Use Case: An e-commerce company has sales data in 10 different sources: Oracle DB, SAP, CSV files in FTP, REST APIs. Every night at midnight, ADF pulls data from all these sources, transforms it, and loads it into Azure Data Lake. This is classic **ETL/ELT orchestration**.

Think of ADF as: The conductor of an orchestra — it doesn't play an instrument itself but coordinates when and how each instrument (service) plays.

◆ 2. Azure Databricks

A **cloud-based big data processing and analytics platform** built on Apache Spark. Used for large-scale data transformation, machine learning, and streaming.

Real-World Use Case: A telecom company has 10 TB of call detail records (CDRs) daily. Azure Databricks processes all this data in parallel using Spark clusters, applies transformations, detects fraud patterns, and writes results to Delta Lake — all in under 30 minutes.

Think of Databricks as: A supercharged, distributed computing engine that can process terabytes of data by splitting the work across hundreds of machines simultaneously.

◆ 3. BLOB Storage, Data Lake Storage Gen1 and Gen2

Azure Blob Storage: Object storage for any type of unstructured data — images, videos, logs, CSVs, backups.

Azure Data Lake Storage Gen2 (ADLS Gen2): Built on top of Blob Storage but optimized for **big data analytics** with a hierarchical namespace (folder structure), fine-grained access

control, and high throughput.

Real-World Use Case: A healthcare company stores raw patient data files (HL7, CSV) in ADLS Gen2. The Bronze layer holds raw files, Silver holds cleaned data, Gold holds aggregated reports — this is the **Medallion Architecture**.

◆ 4. Azure SQL Server & SQL Database

Azure SQL Server: The logical server that hosts SQL databases in the cloud. **Azure SQL Database:** A fully managed relational database (PaaS) — Microsoft handles patches, backups, and high availability automatically.

Real-World Use Case: A retail company uses Azure SQL Database to store their processed sales transactions. ADF copies data from ADLS Gen2 into Azure SQL Database every hour, which Power BI then reads for dashboards.

◆ 5. Azure Key Vault

A **secure, centralized secrets management service** that stores passwords, connection strings, API keys, certificates.

Real-World Use Case: Your ADF pipeline needs to connect to a SQL Server database. Instead of hardcoding the password in the pipeline (a serious security risk), you store the password in Key Vault as a secret, and ADF retrieves it securely at runtime. If the password changes, you update it in one place — Key Vault — and all pipelines instantly use the new password.

Think of Key Vault as: A bank vault for your digital secrets — only authorized applications and users can access what's inside.

◆ 6. Logic Apps

A **low-code workflow automation service** used to integrate systems and automate business processes.

Real-World Use Case in Data Engineering: When an ADF pipeline fails, a Logic App is triggered that automatically sends an email notification to the data engineering team with details of the failure, the pipeline name, error message, and timestamp — without writing any custom code.

5 Introduction to Big Data

◆ What is Data?

Data is **raw facts and figures** without context. On its own, data has no meaning until processed and analyzed.

Examples:

- **28** — just a number, meaningless alone
 - **Temperature: 28°C, City: Mumbai, Date: 2024-06-01** — now it's meaningful data
-

◆ What is Big Data?

Big Data refers to datasets that are **so large, fast-moving, or complex** that traditional database tools (like MySQL, SQL Server) cannot process them efficiently.

Real-World Examples:

- Facebook generates ~4 petabytes (4,000 TB) of data daily
- YouTube: 500 hours of video uploaded every minute
- A jet engine generates ~1 TB of sensor data per flight

- A stock exchange processes millions of transactions per second

Why Traditional Databases Fail: A standard MySQL database on a single server can handle maybe 100GB–1TB efficiently. When you have 10TB, 100TB, or petabytes of data — you need distributed processing systems like Spark (Databricks).

◆ Data Sources of Big Data

Big data comes from everywhere:

Source	Example
Social Media	Facebook posts, tweets, Instagram photos
IoT Sensors	Smart meters, factory sensors, GPS trackers
Log Files	Web server logs, application logs
Transactional Systems	POS systems, banking transactions
Clickstream Data	User behavior on websites/apps
Machine Data	Industrial machinery sensor readings

◆ Characteristics of Big Data — The 3 V's

Volume

The **sheer size** of the data — terabytes, petabytes, exabytes.

Real-World Example: Walmart collects over 2.5 petabytes of customer transaction data every hour from its stores worldwide. Storing and querying this in a traditional database is impossible — you need distributed storage like ADLS Gen2 and processing like Databricks.

Velocity

The **speed** at which data is generated and needs to be processed.

Real-World Example: A fraud detection system at a bank must analyze each credit card transaction **within milliseconds** of it occurring to approve or block it. This requires real-time/streaming processing. In Azure, this is handled by Databricks Structured Streaming or Azure Event Hubs.

Variety

The **different types and formats** of data.

Real-World Example: An e-commerce platform deals with:

- Structured data: Order table in SQL Database (rows & columns)
- Semi-structured data: Customer reviews in JSON format, product catalog in XML
- Unstructured data: Product images (JPEG), promotional videos (MP4), customer support call recordings (MP3)

All these different formats must be ingested and processed together.

◆ Types of Data

A. Structured Data

Data that has a **predefined schema** — organized in rows and columns.

Examples:

- SQL Database tables (customer_id, name, email, purchase_date)
- Excel spreadsheets
- CSV files with consistent headers

Processing Tools: SQL Server, Azure SQL Database, Synapse Dedicated SQL Pool

B. Semi-Structured Data

Data that **doesn't fit neatly into rows/columns** but has some organizational structure using tags or markers.

Examples:

- **JSON:** `{"customer_id": 101, "name": "Ravi", "orders": [{"id": 1, "amount": 500}]}`
- **XML:** Used in healthcare (HL7), banking (SWIFT messages)
- **Avro, Parquet, ORC:** Columnar file formats used in big data pipelines

Processing Tools: Azure Databricks, ADF Data Flows, Azure Synapse Spark

C. Unstructured Data

Data with **no predefined structure** — the vast majority of data in the world (~80-90%).

Examples:

- Images (product photos, X-rays)
- Videos (surveillance footage, YouTube content)
- Audio (call center recordings, podcasts)
- Free text (customer reviews, emails, social media posts)
- PDF documents

Processing Tools: Azure Cognitive Services (AI), Azure Databricks with ML, Azure Blob Storage for storage

Summary: How Everything Connects

Data **Sources** (Structured/Semi/Unstructured)



Azure Data Factory ↔ Azure Key **Vault** (secrets)
(Orchestrate & Ingest)



Azure Data Lake Storage **Gen2**
(Raw data stored in BLOB/ADLS)



Azure **Databricks** (Spark)
(Transform, process big data)



Azure Synapse / SQL **Database**
(Serve processed data)



Power BI / Reports
(Business insights)

All of this runs on **Azure (Public Cloud)**, under a **Subscription**, organized in **Resource Groups**, with each service being a **Resource** — and sensitive credentials securely stored in **Azure Key Vault**.

 **Overview of Cloud section complete.** All 5 major topics — Cloud Basics, Cloud Vendors, Azure Introduction, Azure Services, and Big Data — have been covered with real-world examples and deep conceptual explanations. Ready to proceed to the next section: **Azure Data Factory** whenever you are!

17/03/2026, 21:21:51

Thought process: Orchestrated comprehensive Azure Data Factory curriculum spanning twenty-one interconnected topics.

The user wants me to continue with the next section from the PDF, which is **Azure Data Factory**. Let me cover every single concept from the PDF in depth with proper examples and good formatting.

Let me go through all the ADF topics from the PDF:

- 1) Azure Data Factory (What is ADF, Architecture, Portal UI, Top-level concepts)
- 2) Pipeline
- 3) Linked Service
- 4) DataSets
- 5) Activities
- 6) Triggers
- 7) Integration Runtime (IR)
- 8) Source control
- 9) Global parameters
- 10) Credentials
- 11) Monitoring ADF Jobs
- 12) Alerts
- 13) Send Failure Notifications using Logic Apps

- 14) Data Flows
- 15) Azure Devops
- 16) SDLC
- 17) Agile Methodology
- 18) ADF Interview Questions
- 19) ADF Resume Preparation
- 20) End TO End ADF Project
- 21) ADF Exercises

Now I'm moving into the remaining topics—Data Flows, Azure DevOps integration, SDLC practices, Agile methodology, interview prep, resume tips, and a complete end-to-end project with exercises. I'll provide comprehensive explanations with practical examples and proper formatting for each section.



Azure Data Factory — Complete In-Depth Guide

1 Azure Data Factory — Core Fundamentals

◆ What is Azure Data Factory?

Azure Data Factory (ADF) is Microsoft Azure's **fully managed, serverless cloud-based data integration and orchestration service**. It allows you to create, schedule, and manage data pipelines that can **ingest, transform, and load (ETL/ELT)** data from hundreds of sources to destinations.

In simple terms: ADF is the **plumber of your data system** — it moves water (data) from one place (source) through pipes (pipelines) to another place (destination), sometimes filtering or treating the water (transforming data) along the way.

Real-World Business Problem ADF Solves:

Imagine you work as a Data Engineer at a large retail company. Every single day:

- Sales transactions come in from **Oracle Database** (on-premise)
- Customer data arrives as **CSV files** dropped on an **FTP server**
- Marketing campaign results come from a **REST API**
- Web clickstream data lands in **Azure Blob Storage** as JSON files
- Inventory data sits in **SAP** (on-premise ERP system)

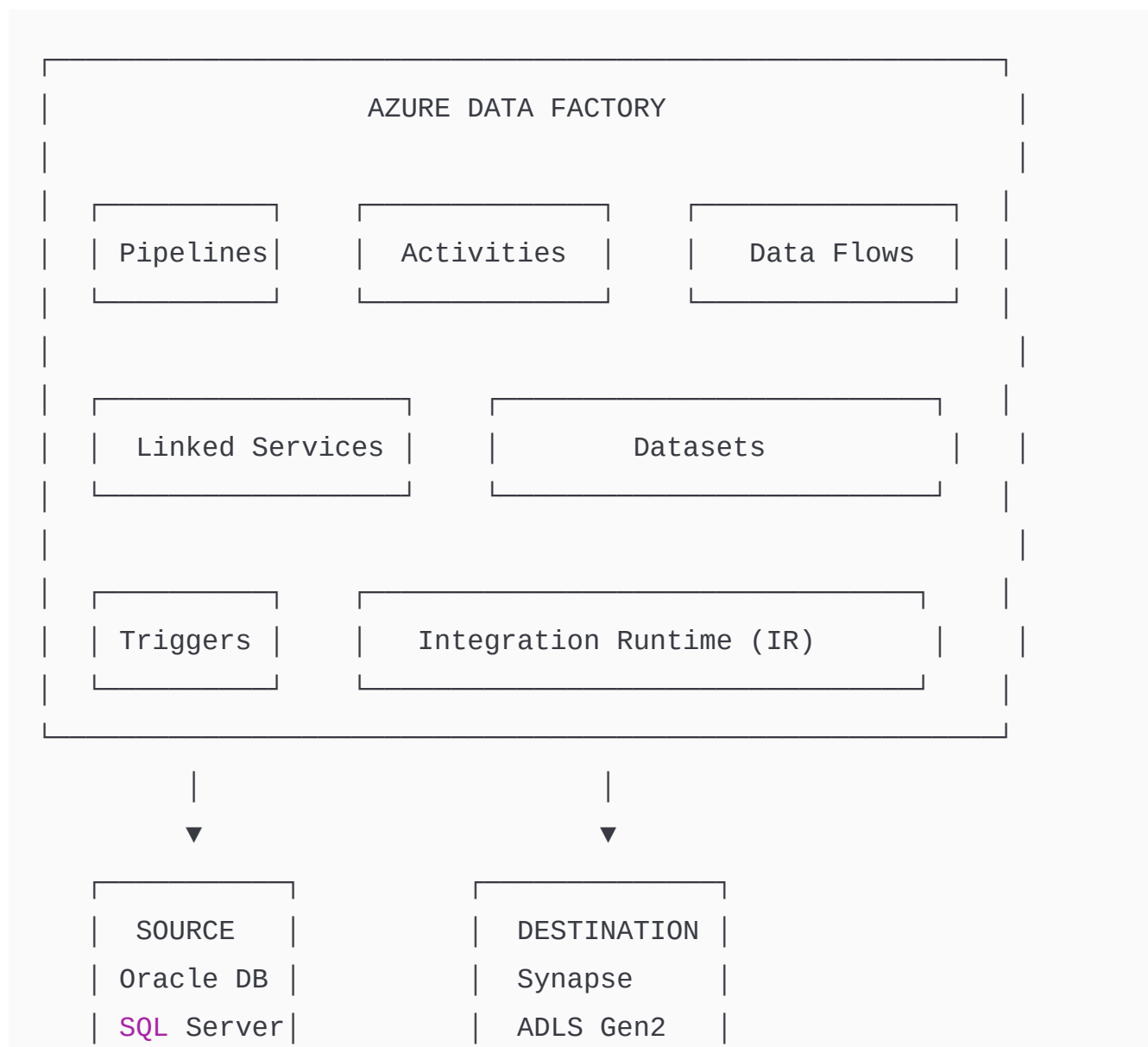
You need to pull all this data, clean it, transform it, join it, and load it into **Azure Synapse Analytics** for reporting — every night at midnight, automatically.

Without ADF: You'd write custom Python scripts on a VM, manage scheduling via cron jobs, handle errors manually, monitor failures yourself.

With ADF: You build visual pipelines that do all of this automatically — with built-in monitoring, alerts, retry logic, and 90+ connectors — zero infrastructure management.

◆ Azure Data Factory Architecture

Understanding ADF's architecture is critical before building pipelines.



CSV Files	SQL DB
REST API	Blob

Key Architecture Layers:

Control Plane (ADF Service): The brain of ADF — manages metadata, monitors execution, schedules triggers. Runs in Azure's managed infrastructure.

Data Plane (Integration Runtime): The actual execution engine — this is what physically moves data. Can run in Azure cloud or on your own network (Self-Hosted IR).

Storage Layer: ADF doesn't store data itself — it just moves it between your source and destination systems.

◆ Azure Data Factory Portal UI

The ADF Portal (adf.azure.com) has several key sections:

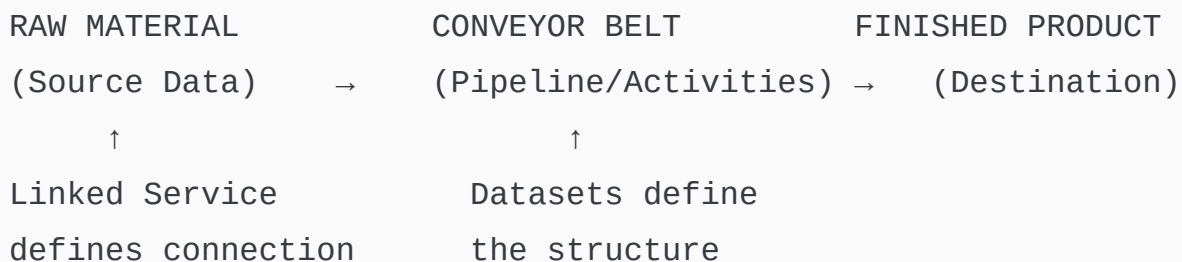
ADF Studio UI

- | — 📁 Author → Build pipelines, linked services, datasets, **data** flows
- | — 📊 Monitor → Track pipeline runs, activity runs, trigger runs
- | — 🔧 Manage → Linked services, IR, triggers, git config, key vault
- | — 🏠 Home → Quick access, recent resources, templates

Real-World Tip: The ADF Studio is completely browser-based — no software installation needed. Your entire pipeline code is stored as **JSON under the hood**, even though you build it visually.

◆ Top-Level Concepts in ADF

Think of ADF like a **factory assembly line**:



Let's now explore each concept in depth:

A. Pipelines

A **Pipeline** is a **logical grouping of activities** that together perform a task. It's the top-level container in ADF.

Real-World Analogy: A pipeline is like a recipe. The recipe (pipeline) contains multiple steps (activities): chop vegetables → boil water → cook → serve.

Example:

```
Pipeline: "Daily_Sales_ETL"
├─ Activity 1: GetMetadata (check if file arrived)
├─ Activity 2: If Condition (if file exists → proceed)
├─ Activity 3: Copy Data (move CSV from FTP to ADLS)
├─ Activity 4: Databricks Notebook (run transformations)
└─ Activity 5: Stored Procedure (update audit log)
```

B. Activities

Activities are the **individual steps/tasks** inside a pipeline. Each activity performs a specific action.

Categories of Activities:

- **Data Movement:** Copy Data activity
- **Data Transformation:** Databricks Notebook, Data Flow, Stored Procedure
- **Control Flow:** If Condition, ForEach, Until, Switch, Filter

Real-World Example: In a pipeline that processes daily sales reports, activities might be: check if today's file exists → copy the file from SFTP → transform it in Databricks → load to SQL Database → send success email.

C. Linked Services

Linked Services are **connection strings** — they define how ADF connects to external resources (databases, storage, APIs).

Real-World Analogy: Like saving a contact in your phone. Instead of dialing the full 10-digit number every time, you save "Ravi - SQL Server" and just tap to call. A Linked Service is that saved contact for your data source.

Example Linked Services:

Linked Service 1: LS_AzureSQLDB_Production

- Server: myserver.database.windows.net
- Database: SalesDB
- Auth: Key Vault secret

Linked Service 2: LS_ADLS_Gen2_DataLake

- Account: mydatalakestorage
- Auth: Managed Identity

D. Datasets

Datasets represent the **structure and location of data** — they point to the specific table, file, or folder you want to work with, using a Linked Service for the connection.

Real-World Analogy: If a Linked Service is the address of a building (SQL Server), then a Dataset is the specific room number (specific table or file) within that building.

```
Linked Service: LS_ADLS_Gen2 → (Connection to storage account)
Dataset: DS_SalesCSV → (Points to
/raw/sales/2024/*.csv)
```

E. Triggers

Triggers determine **when a pipeline runs** — they define the schedule or event that fires the pipeline.

Types:

- **Schedule Trigger:** Run every day at midnight
- **Tumbling Window Trigger:** Process hourly time windows of data
- **Storage Event Trigger:** Run when a new file arrives in ADLS

Real-World Example: A trigger fires at 11:00 PM every night to start the overnight data loading pipeline.

F. Data Flows

Data Flows are the **visual data transformation feature** in ADF — like SSIS or Informatica but in the cloud. They use Apache Spark under the hood for execution.

Real-World Use Case: You receive a raw CSV with duplicate customer records, inconsistent date formats, and missing values. A Data Flow can filter duplicates, standardize dates, fill

nulls, and join with another dataset — all visually, without writing code.

G. Integration Runtimes

The **Integration Runtime (IR)** is the **compute infrastructure** that ADF uses to actually run activities and move data.

Types:

- **Azure IR:** Runs in Azure cloud — for cloud-to-cloud data movement
- **Self-Hosted IR:** Installed on your own server — for accessing on-premise databases
- **Azure-SSIS IR:** For running SSIS packages in the cloud

Real-World Example: To copy data from an on-premise Oracle database (inside your company's firewall) to Azure, you install a Self-Hosted IR agent on a machine inside your company network. This agent acts as a secure bridge between your internal network and ADF.

2 Pipeline — Deep Dive

◆ What is a Pipeline?

A Pipeline is the **core unit of work** in ADF. It's a directed acyclic graph (DAG) of activities — meaning activities can run sequentially or in parallel, but there are no circular dependencies.

Real-World Analogy: A pipeline is like an airport. Multiple aircraft (activities) land and take off in a coordinated sequence. Some flights wait for others; some run simultaneously on parallel runways.

◆ Create a New Pipeline

Steps to create a pipeline in ADF Studio:

1. Go to ADF Studio → **Author** tab
2. Click "+" → **Pipeline** → **New Pipeline**
3. Name your pipeline (e.g., `PL_Daily_SalesIngestion`)
4. Drag activities from the Activities panel onto the canvas
5. Connect them with arrows to define execution order

JSON behind the scenes:

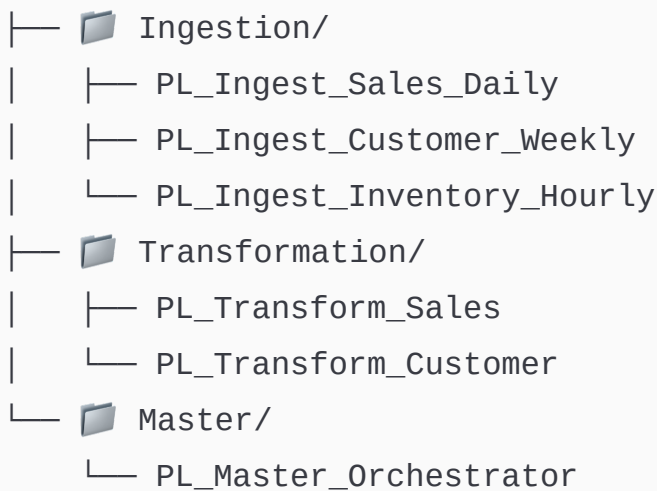
```
{
  "name": "PL_Daily_SalesIngestion",
  "properties": {
    "activities": [
      {
        "name": "CopySalesData",
        "type": "Copy",
        "inputs": [{ "referenceName": "DS_Source_CSV" }],
        "outputs": [{ "referenceName": "DS_Sink_ADLS" }]
      }
    ]
  }
}
```

◆ Organize Pipelines into Folders

As your ADF grows, you might have 50–100 pipelines. Organizing them into folders keeps things manageable.

Best Practice Folder Structure:

ADF Pipelines



Real-World Example: A bank's ADF might have 200+ pipelines — organized by department (Retail Banking, Corporate Banking, Risk), then by data domain (Ingestion, Transform, Load).

◆ Debug Pipeline

Debug mode allows you to **test your pipeline without publishing it** — great for development and troubleshooting.

How it works:

- Click **"Debug"** button in ADF Studio
- Pipeline runs immediately using current (unpublished) configuration
- Results appear in the **Output** tab at the bottom
- Green checkmark = success, Red X = failure with error details

Real-World Tip: Always debug with a small data sample first. If you have 10 million rows in a table, use a query like `SELECT TOP 100 * FROM sales` as your source during debugging to save cost and time.

◆ Publish Pipeline

Publishing saves your pipeline to **Azure Resource Manager (ARM)** and makes it live. Before publishing, changes only exist in draft mode (if using live mode) or in your Git branch.

Best Practice Workflow:

Developer A: Creates pipeline in Feature Branch

- Commits to Git
- Raises Pull Request

Developer B: Reviews and approves

- Merges to Main branch
- Publish from Main branch to ADF Production

◆ Pipeline Parameters

Parameters make pipelines **reusable and dynamic**. Instead of hardcoding values like file names or table names, you pass them as parameters at runtime.

Real-World Example:

Without parameters — you need 12 separate pipelines for 12 months of data:

```
PL_Load_Jan, PL_Load_Feb, PL_Load_Mar... (bad practice!)
```

With parameters — one pipeline handles all months:

Pipeline: PL_Load_Monthly_Data

Parameter: p_month (string) → pass "Jan", "Feb", "Mar" etc.

Parameter: p_year (string) → pass "2024", "2025"

Source path dynamically becomes:

```
/raw/sales/{pipeline().parameters.p_year}/{pipeline().parameters.p
```

How to define parameters:

1. Click on the empty canvas in pipeline
2. Go to **Parameters** tab
3. Click "+ New" → Name: `p_month`, Type: `String`, Default: `Jan`

3 Linked Service — Deep Dive

◆ What is a Linked Service?

A **Linked Service** is a **connection definition** that contains the connectivity information ADF needs to connect to external data stores or compute resources. It's reusable across multiple datasets and activities.

Real-World Analogy: A Linked Service is like a Wi-Fi saved network profile on your laptop. You save "OfficeWiFi" with the password once — and anytime you're in the office, your laptop automatically connects using that saved profile. Similarly, once you define a Linked Service for your SQL Server, all your pipelines use it without re-entering credentials.

◆ Creating Linked Services

A. Linked Service for BLOB Storage

Name: LS_AzureBlob_RawData

Type: Azure Blob Storage

Authentication: Account `Key` / Managed Identity / SAS URI

Storage Account: mystorageaccount

Use Case: Copy raw CSV files uploaded by the business team from Blob Storage into ADLS for processing.

B. Linked Service for Azure SQL Database

```
Name: LS_AzureSQLDB_SalesDB
Type: Azure SQL Database
Server: myserver.database.windows.net
Database: SalesDB
Authentication: SQL Authentication
Username: sqladmin
Password: @Microsoft.KeyVault(SecretUri=...) ← from Key Vault!
```

Use Case: Load transformed sales data from ADLS into a SQL Database for Power BI reporting.

C. Linked Service for SQL Server (On-Premise)

```
Name: LS_OnPrem_SQLServer
Type: SQL Server
Server: 192.168.1.100 (internal IP)
Database: LegacyDB
Integration Runtime: SelfHostedIR01 ← REQUIRED for on-prem!
Authentication: Windows Authentication
```

Key Point: On-premise connections **always require a Self-Hosted Integration Runtime** because ADF (in the cloud) cannot directly reach servers behind your corporate firewall.

D. Linked Service for Data Lake Storage Gen1 & Gen2

```
Name: LS_ADLS_Gen2_DataLake
Type: Azure Data Lake Storage Gen2
Account: mydatalakestorage.dfs.core.windows.net
Authentication: Managed Identity (recommended!)
```

Why Managed Identity is Best: Instead of storing passwords, Azure automatically manages the identity. Your ADF instance gets a system-assigned identity, and you grant it "Storage Blob Data Contributor" role on the storage account — zero passwords to manage!

◆ Linked Service Parameterization

Just like pipelines, Linked Services can be parameterized to make them dynamic.

Real-World Use Case: You have 5 different SQL databases (one per region: India, US, UK, Germany, Singapore). Instead of creating 5 separate Linked Services, create **one parameterized Linked Service**:

```
Name: LS_SQLDatabase_Parameterized
Parameter: p_serverName → Type: String
Parameter: p_databaseName → Type: String

Server: @{linkedService().p_serverName}
Database: @{linkedService().p_databaseName}
```

Now pass different server and database names at runtime — one Linked Service serves all 5 regions!

4 Datasets — Deep Dive

◆ What is a Dataset?

A **Dataset** is a **named view of data** that simply points to the data you want to use in your activities. It references a Linked Service for the connection and adds specifics about location and structure.

Real-World Analogy: If a Linked Service is your Amazon account (connection), a Dataset is like a specific item in your cart (the specific product/file you want to work with).

```
Linked Service: LS_ADLS_Gen2  ← HOW to connect (connection
details)
Dataset: DS_RawSales_CSV      ← WHAT to access (specific
file/table)
```

◆ Creating Datasets

A. File-Based Datasets (CSV, JSON, Parquet, Avro, Excel, etc.)

CSV Dataset Example:

```
Name: DS_Source_SalesCSV
Linked Service: LS_ADLS_Gen2
Format: DelimitedText (CSV)
File Path: raw/sales/2024/sales_data.csv
First Row as Header: Yes
Column Delimiter: ,
Encoding: UTF-8
```

Parquet Dataset Example:

```
Name: DS_Silver_SalesParquet
Linked Service: LS_ADLS_Gen2
Format: Parquet
```

File Path: silver/sales/

Compression: Snappy

Why Parquet? Parquet is a columnar format — 10x faster for analytics queries, 70% smaller file size compared to CSV. Industry standard in big data pipelines.

B. Table-Based Datasets (SQL Database, SQL Server, Oracle)

SQL Table Dataset:

Name: DS_SQLTable_Customers

Linked Service: LS_AzureSQLDB_SalesDB

Table Name: [dbo].[Customers]

Query-Based Dataset (more flexible):

Name: DS_SQL_RecentOrders

Linked Service: LS_AzureSQLDB_SalesDB

Use Query: **SELECT** * **FROM** Orders **WHERE** OrderDate >= '2024-01-01'

◆ Dataset Parameterization

Real-World Use Case: You process daily sales files. File path changes every day:

/raw/sales/2024/01/sales_20240101.csv

/raw/sales/2024/01/sales_20240102.csv

/raw/sales/2024/01/sales_20240103.csv

Instead of creating 365 datasets, create **one parameterized dataset**:

Dataset: DS_DailySalesCSV_Param

Parameter: p_date → Type: String

File Path:

raw/sales/@{dataset().p_date}/sales_{dataset().p_date}.csv

Now pass `p_date = "20240101"` at runtime → automatically accesses the right file!

5 Activities — Complete Deep Dive

Activities are the **building blocks** of every ADF pipeline. Let's cover each one thoroughly.

◆ 1. Wait Activity

Pauses pipeline execution for a specified number of seconds.

Real-World Use Case: After triggering a stored procedure that kicks off a long database rebuild, you add a **Wait activity for 60 seconds** before the next activity checks if the rebuild is complete — giving the DB time to finish.

Activity: Wait

Wait Time in Seconds: 60

◆ 2. Variables

Variables store **temporary values** during pipeline execution — like variables in any programming language.

Create a Variable:

In the Pipeline canvas → Variables tab → Click "+ New"

```
Name: v_fileName
Type: String
Default Value: ""
```

Set Variable Activity:

Assigns a value to a pipeline variable.

```
Activity: Set Variable
Variable Name: v_fileName
Value: @concat('sales_', formatDateTime(utcnow(), 'yyyyMMdd'),
'.csv')
Result: v_fileName = "sales_20240615.csv"
```

Append Variable Activity:

Adds values to an Array-type variable — useful for building lists.

Real-World Use Case:

```
v_processedTables = [] (starts empty)
```

```
ForEach iteration 1: Append "Customers" → v_processedTables =
["Customers"]
```

```
ForEach iteration 2: Append "Orders" → v_processedTables =
["Customers", "Orders"]
```

```
ForEach iteration 3: Append "Products" → v_processedTables =
["Customers", "Orders", "Products"]
```

```
After loop: Log all processed tables to audit table
```

◆ 3. Copy Data Activity — Most Important Activity

The **Copy Data activity** is the heart of ADF — it moves data from a **Source** to a **Sink** (destination) with optional mapping and transformations.

Internal Architecture of Copy Activity:

```
SOURCE → [Read] → [Staging (optional)] → [Write] → SINK
           ↑
    Integration Runtime executes this
```

A. General Tab:

Name: Copy_Sales_CSV_to_ADLS
Timeout: 0.12:00:00 (12 hours max)
Retry: 3 (retry 3 times on failure)
Retry Interval: 30 seconds

B. Source Tab:

Defines WHERE to read data from.

Source Dataset: DS_OnPrem_SalesCSV
File Path Type: Wildcard
Wildcard Folder Path: /exports/sales/
Wildcard File Name: sales_*.csv ← picks all sales CSV files
Recursively: Yes ← includes subfolders

Available Source Options:

- **Query:** Override dataset with a custom SQL query
- **Stored Procedure:** Use a stored procedure as source
- **Partition Option:** For large tables, read in parallel partitions (physical/logical/dynamic range)

C. Sink Tab:

Defines WHERE to write data to.

Sink Dataset: DS_ADLS_RawSales

Copy Behavior: PreserveHierarchy / FlattenHierarchy / MergeFiles

Write Batch Size: 10000 rows per batch

Copy Behaviors explained:

- **PreserveHierarchy:** Maintains folder structure from source
- **FlattenHierarchy:** All files land in destination root folder
- **MergeFiles:** Merges all source files into one destination file

D. Mapping Tab:

Maps source columns to destination columns — especially when column names differ between source and destination.

Source Column	→	Destination Column
CUST_ID	→	customer_id
CUST_NM	→	customer_name
ORD_DT	→	order_date
AMT	→	order_amount

You can also add **data type conversions** here (e.g., string → datetime).

E. Settings Tab:

Fault Tolerance: Skip incompatible rows ← logs bad rows, continues pipeline

Enable Staging: Yes ← for large data loads using PolyBase (Synapse)

Data Consistency Verification: Yes ← verifies row count after copy

F. User Properties:

Custom key-value pairs added to activity for monitoring purposes.

```
pipeline_name: @{pipeline().Pipeline}
run_date: @{formatDateTime(utcnow(), 'yyyy-MM-dd')}
source_system: Oracle_ERP
```

These appear in the monitoring dashboard, making it easy to debug which pipeline/date caused an issue.

◆ 4. Copy File(s) Between BLOB Containers

A. Copy One File from a Folder:

```
Source: /raw/sales/sales_20240601.csv (specific file)
Sink: /archive/sales/sales_20240601.csv
```

B. Copy All Files from a Folder:

```
Source Dataset: Wildcard File Path = /raw/sales/*.csv
Sink: /processed/sales/
Copy Behavior: PreserveHierarchy
```

C. Copy All Files and Folders Recursively:

```
Source: /raw/ (entire directory tree)
Recursive: TRUE
Sink: /backup/
Copy Behavior: PreserveHierarchy
```

Real-World Use Case: Daily backup job that copies the entire `/raw/` data lake zone to a `/backup/` container recursively.

◆ 5. Copy Data from BLOB to SQL Database / ADLS Gen2

Scenario: CSV files land in BLOB every morning → ADF copies them to Azure SQL Database for reporting.

Source:

Linked Service: `LS_BlobStorage`

Dataset: `DS_RawCSV`

File Path:

`/incoming/sales_{formatDateTime(utcnow(),'yyyyMMdd')}.csv`

Sink:

Linked Service: `LS_AzureSQLDB`

Dataset: `DS_SQL_SalesTable`

Write Behavior: `Upsert` ← Insert new, update existing

Key Columns: `[sale_id]`

As Parquet to ADLS Gen2 (more common in modern pipelines):

Source: `SQL Database query result`

Sink: `ADLS Gen2 - Parquet format`

Path: `/silver/sales/year=2024/month=06/`

Partition: `By date` ← Hive-style partitioning for Spark

◆ 6. Databricks Notebook Activity

Runs an **Azure Databricks notebook** from within an ADF pipeline.

Activity: Databricks Notebook

Linked Service: LS_AzureDatabricks_Workspace

Notebook Path: /notebooks/transform_sales_data

Parameters:

input_path: /raw/sales/2024/

output_path: /silver/sales/2024/

process_date: @{formatDateTime(utcnow(), 'yyyy-MM-dd')}

Real-World Use Case: ADF orchestrates the overall flow → when it's time to transform data, it hands off to Databricks (which has the heavy Spark processing power) → Databricks runs the notebook → returns control back to ADF → ADF continues to the next activity.

◆ 7. Azure Function Activity

Calls an **Azure Function** (serverless code) from your pipeline.

Real-World Use Case: After copying data, call an Azure Function that:

- Sends a Teams notification: "Sales data loaded successfully for 2024-06-15"
- Or validates that row counts match between source and destination
- Or calls a third-party REST API to notify downstream systems

Activity: Azure Function

Linked Service: LS_AzureFunction_Notifications

Function Name: SendPipelineNotification

Method: POST

Body: {

"pipeline": "@{pipeline().Pipeline}",

"status": "Success",

"rows_copied": "@{activity('CopyData').output.rowsCopied}"

}

◆ 8. Lookup Activity

Lookup reads data from a source and **returns the result to the pipeline** as an object — used to dynamically fetch configuration or list of items to process.

Real-World Use Case — Dynamic Table List:

You have a **control table** in SQL Database:

```
-- table: dbo.ControlTable
table_name      | is_active | source_schema
-----|-----|-----
Customers       | 1        | dbo
Orders          | 1        | dbo
Products        | 0        | dbo ← skip this one
Inventory       | 1        | dbo
```

Lookup activity reads this table:

```
Activity: LookupActiveTables
Source: Azure SQL Database
Query: SELECT table_name FROM dbo.ControlTable WHERE is_active =
1
```

Output:

```
{
  "firstRow": {"table_name": "Customers"},
  "value": [
    {"table_name": "Customers"},
    {"table_name": "Orders"},
    {"table_name": "Inventory"}
  ],
}
```

```
"count": 3
}
```

This output feeds into a **ForEach** activity to copy only active tables — making your pipeline 100% dynamic!

◆ Stored Procedure Activity

Executes a **SQL stored procedure** at any point in your pipeline.

Real-World Use Cases:

- Log pipeline execution status to an audit table
- Call a stored procedure that merges staging data into production table
- Execute post-load data quality checks

Activity: SP_UpdateAuditLog

Linked Service: LS_AzureSQLDB

Stored Procedure Name: [dbo].[usp_UpdatePipelineLog]

Parameters:

pipeline_name: @{{pipeline().Pipeline}}

run_id: @{{pipeline().RunId}}

status: "Success"

rows_loaded: @{{activity('CopyData').output.rowsCopied}}

end_time: @{{utcnow()}}

◆ 9. Get Metadata Activity

Retrieves **metadata about a file or folder** — such as file name, size, last modified date, child items list.

Real-World Use Case — Check if File Arrived Before Processing:

Activity: GetMetadata_CheckFile

Dataset: DS_ExpectedFile

Field List: [exists, size, lastModified, itemName, childItems]

Output:

```
{
  "exists": true,
  "size": 1048576,           ← 1 MB
  "lastModified": "2024-06-15T23:45:00Z",
  "itemName": "sales_20240615.csv",
  "childItems": [...]
}
```

Then use an **If Condition**: if `exists = true` → proceed with copy, else → send alert email.

◆ Delete Activity

Deletes files or folders from Azure Storage (BLOB or ADLS).

Real-World Use Cases:

- Delete source files after successful copy (archival cleanup)
- Delete files older than 30 days from the raw zone
- Clean up staging files after data is loaded

Activity: DeleteSourceFiles

Dataset: DS_ProcessedFiles

Recursive: True

Wildcard File Name: *.csv

Best Practice: Always use Delete **after** confirming the copy was successful — use it as the last step, not the first!

◆ 10. Execute Pipeline Activity

Calls **another pipeline** from within a pipeline — enables **modular pipeline design**.

Real-World Use Case — Master-Child Pipeline Pattern:

```
PL_Master_OvernightETL (runs at midnight)
├─ Execute Pipeline → PL_Ingest_SalesData
├─ Execute Pipeline → PL_Ingest_CustomerData
├─ Execute Pipeline → PL_Ingest_InventoryData
├─ Execute Pipeline → PL_Transform_AllData
└─ Execute Pipeline → PL_Load_Synapse
```

Each child pipeline can also run independently for testing/reprocessing.

Configuration:

```
Activity: RunIngestionPipeline
Invoked Pipeline: PL_Ingest_SalesData
Wait on Completion: YES ← master waits for child to finish
Parameters:
  p_date: @{formatDateTime(utcnow(), 'yyyy-MM-dd')}
```

◆ 11. Validation Activity

Waits until a **file or folder exists** before continuing — prevents pipeline failure when source isn't ready.

Real-World Use Case: The upstream system uploads a file "sales_done.flag" to signal that all sales data has been delivered. Your pipeline waits for this flag file before starting data processing:

```
Activity: WaitForSalesFile
Dataset: DS_FlagFile ← points to sales_done.flag
Timeout: 4 hours ← wait max 4 hours, then fail
Sleep: 60 seconds ← check every 60 seconds
Minimum Size: 0 bytes ← file just needs to exist
```

◆ Fail Activity

Intentionally fails a pipeline with a custom error message — used for business rule violations.

```
Activity: FailPipeline_NoData
Fail Message: @concat('No data file found for date: ',
formatDateTime(utcnow(), 'yyyy-MM-dd'))
Error Code: ERR_NO_SOURCE_FILE
```

When to use: If a critical file is missing, rather than silently succeeding (which would produce wrong reports), use Fail activity to raise an explicit, clearly named error.

◆ 12. Iteration & Conditionals

These are **control flow activities** that add programming logic to your pipelines.

A. Filter Activity

Filters an array based on a condition — like WHERE clause for pipeline arrays.

```
Activity: Filter_ActiveTables
Items: @{activity('LookupTables').output.value}
Condition: @{equals(item().is_active, 1)}

Output: Only tables where is_active = 1
```

B. ForEach Activity

Loops through an array and executes inner activities for each item — most commonly used with Lookup output.

```
Activity: ForEach_ProcessTables
Items: @{activity('Filter_ActiveTables').output.value}
Batch Count: 4 ← process 4 tables in parallel
Sequential: No ← run in parallel (faster!)

Inner Activity: Copy Data
  Source Table: @{item().table_name}
  Sink Path:    /processed/@{item().table_name}/
```

Real-World Example: Instead of writing separate copy pipelines for 50 tables, one ForEach loop iterates through all 50 and copies them — cutting pipeline count from 50 to 1!

C. If Condition Activity

Executes different activities based on a **True/False condition**.

```
Activity: CheckFileSize
Condition: @{greater(activity('GetMetadata').output.size, 0)}
```

If True Activities:

→ Copy Data to ADLS

If False Activities:

→ Send Alert Email

→ Fail Pipeline

Real-World Use Case: Check if file is non-empty before processing. An empty file (0 bytes) would cause downstream failures — catch it early!

D. Switch Activity

Like a **switch/case statement** — routes to different activity branches based on a value.

Activity: RouteByRegion

On: `@{pipeline().parameters.p_region}`

Cases:

"INDIA": → Copy to India ADLS container

"US": → Copy to US ADLS container

"UK": → Copy to UK ADLS container

Default: → Log unknown region warning

E. Until Activity

Loops **until a condition becomes true** — like a do-while loop.

Real-World Use Case: Poll an API every 30 seconds until a long-running job (like Databricks job) completes:

Activity: WaitForDatabricksJob

Condition: @equals(variables('v_jobStatus'), 'COMPLETED')}

Timeout: 2 hours

Delay: 30 seconds

Inner Activities:

1. Call REST API to check Databricks job status
2. Set Variable: v_jobStatus = API response

6 Triggers — Deep Dive

◆ What is a Trigger?

A **Trigger** is the mechanism that **fires (starts) a pipeline**. Without a trigger, a pipeline can only be run manually.

Real-World Analogy: A trigger is like an alarm clock — the alarm (trigger) fires at a specific time (or event), waking you up (starting the pipeline).

◆ Types of Triggers

A. Schedule Trigger

Runs a pipeline on a **fixed time schedule** — most commonly used trigger.

Trigger Name: TRG_Daily_SalesETL

Start Time: 2024-01-01 00:00:00 UTC

Recurrence: Every 1 Day

Timezone: India Standard Time (UTC+5:30)

End Time: (none - runs indefinitely)

Real-World Examples:

- Daily at midnight: Run nightly data warehouse load
- Every 15 minutes: Refresh near-real-time sales dashboard
- Every Monday at 8 AM: Generate weekly executive reports

Important: One Schedule Trigger can attach to **multiple pipelines**. Or one pipeline can have **multiple triggers** (e.g., daily + manual).

B. Tumbling Window Trigger

Processes data in **fixed-sized, non-overlapping time windows** — ideal for **reprocessing historical data** or **ensuring exactly-once processing**.

```
Trigger Name: TRG_Hourly_Sales_Window
Start Time: 2024-01-01 00:00:00 UTC
Period: 1 Hour
Max Concurrency: 4 ← run 4 windows in parallel
Retry: 3
Delay: 5 minutes ← wait 5 min after window end before firing
```

What makes it different from Schedule Trigger:

- Tumbling window tracks **which windows have run successfully**
- If a window fails, it retries that specific window
- You can **backfill** — run all past windows (e.g., reprocess Jan 1 to today)

Real-World Use Case: A financial system needs to ensure **every single hour of transaction data** is processed exactly once. If processing for 3 PM–4 PM fails at 4:15 PM, the tumbling window trigger retries only that specific hour — it never skips or double-processes.

Trigger Parameters available inside pipeline:

```
@{triggerBody().window.windowStart} → "2024-06-15T03:00:00Z"  
@{triggerBody().window.windowEnd} → "2024-06-15T04:00:00Z"
```

C. Storage Event Trigger

Fires when a **file is created or deleted in ADLS Gen2 or Blob Storage** — enables event-driven pipelines.

```
Trigger Name: TRG_Event_NewSalesFile  
Storage Account: mydatalakestorage  
Container: raw  
Blob Path Begins With: sales/incoming/  
Blob Path Ends With: .csv  
Event: BlobCreated ← fires when new CSV arrives
```

Real-World Use Case: The upstream system drops a file

`/raw/sales/incoming/sales_20240615.csv` → **immediately** (within seconds), ADF detects the new file and starts the ingestion pipeline — no waiting for the next scheduled run!

Use Cases:

- Process files immediately as they arrive (real-time ingestion feel)
- Trigger different pipelines for different file types (.csv vs .json)
- Event-driven architecture — upstream and downstream are decoupled

◆ Triggers with Parameters

Triggers can pass **parameter values to the pipeline** at runtime:

```
Schedule Trigger: TRG_Daily_Load
```

```
Pipeline Parameters Passed:
```

```
  p_loadDate:  @{{formatDateTime(trigger().scheduledTime, 'yyyy-MM-dd')}}
```

```
  p_loadType:  "FULL"
```

```
  p_source:    "Oracle_ERP"
```

Storage Event Trigger passes file details:

```
p_fileName:  @{{triggerBody().fileName}}
```

```
p_folderPath: @{{triggerBody().folderPath}}
```

7 Integration Runtime (IR) — Deep Dive

◆ What is Integration Runtime?

The **Integration Runtime** is the **compute engine** of ADF — the actual machine/cluster that executes activities and moves data.

Real-World Analogy: If ADF is the brain (control plane), the Integration Runtime is the body/muscles (execution plane) — it does the actual physical work.

◆ Types of Integration Runtime

A. Azure AutoResolve Integration Runtime

- **Default IR** — automatically created when you create ADF
- Runs in Microsoft-managed Azure data centers
- **Location: Auto-resolve** — ADF automatically picks the nearest Azure region to your source and sink

- No configuration needed — works out of the box

Use Cases:

- Cloud-to-cloud data movement
- Azure Blob ↔ Azure SQL Database ↔ ADLS Gen2
- All resources are already in Azure

B. Azure Managed Virtual Network Integration Runtime

- An Azure IR that runs inside a **Managed Virtual Network (VNet)**
- Allows **private endpoints** — connects to Azure services without going over the public internet
- Provides enhanced security — data never leaves Microsoft's backbone network

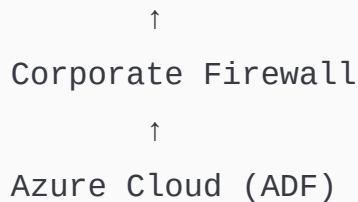
Real-World Use Case: A bank's Azure SQL Database is configured to reject all public internet connections. You use Managed VNet IR with a private endpoint — ADF connects through Azure's internal network, satisfying security compliance requirements.

C. Self-Hosted Integration Runtime (SHIR)

The most critical IR for enterprises — installed **on your own on-premise server or VM** to bridge on-premise systems with ADF.

On-Premise Network





Setup Steps:

1. Create SHIR in ADF Studio (Manage → Integration Runtimes → New → Self-Hosted)
2. Copy the authentication key
3. On your on-premise server: Download SHIR installer from Microsoft
4. Install and paste the authentication key → It registers with ADF
5. Now ADF can reach your on-premise SQL Server, Oracle, files, etc.

Requirements:

- Windows machine (VM or physical server)
- Outbound HTTPS port 443 open to Azure
- Minimum: 4 CPU cores, 8GB RAM (production: 8+ cores, 16+GB RAM)

Real-World Example: A manufacturing company keeps its SAP ERP data on-premise for security reasons. They install SHIR on a server inside their network. ADF uses this SHIR to extract daily production data from SAP and load it into Azure Data Lake for analytics.

D. Linked Self-Hosted Integration Runtime

Allows **multiple ADF instances to share one Self-Hosted IR.**

Company has:

ADF Dev Instance → uses "LinkedSHIR" → points to SHIR on server-01

ADF Test Instance → uses "LinkedSHIR" → points to SHIR on server-01

ADF Prod Instance → uses "LinkedSHIR" → points to SHIR on

server-01

All 3 ADF instances share 1 SHIR installation!

Benefit: Save cost — one SHIR serves multiple ADF environments instead of installing SHIR 3 times.

8 Source Control — Git Integration

◆ Why Source Control for ADF?

Without Git integration, ADF pipelines are saved directly to the ADF service — this means:

- No version history (can't undo bad changes)
- No collaboration (two developers overwriting each other)
- No code review before deploying to production
- No rollback if a deployment breaks production

With Git integration, all ADF resources (pipelines, datasets, linked services) are stored as **JSON files in a Git repository** — giving you full DevOps capabilities.

◆ Git Configuration

Supported Git Providers:

- Azure DevOps Repos (recommended for enterprises)
- GitHub

Setup:

Author → Git Configuration:

Repository Type: Azure DevOps Git

Azure DevOps Account: mycompany.visualstudio.com

Project Name: DataEngineering

Repository Name: ADF-Pipelines

Collaboration Branch: main

Publish Branch: adf_publish

Root Folder: /

How it works after setup:

Developer works in → Feature Branch (e.g., feature/add-customer-pipeline)

- Commits changes (JSON files)
- Creates Pull Request to main branch
- Code reviewed by team lead
- Merged to main
- Published to ADF service (live)

◆ ARM Templates — Export/Import

ARM (Azure Resource Manager) Template is a JSON file that describes all ADF resources — it captures the entire ADF configuration.

Export Use Cases:

- Backup your entire ADF configuration
- Deploy the same ADF to a different environment (Dev → Test → Prod)
- Share ADF configuration with a new team

Export:

ADF Studio → Manage → ARM Template → Export ARM Template
Downloads: ARMTemplateForFactory.json (full config)
 ARMTemplateParametersForFactory.json (environment-specific values)

Import (deploy to new environment):

Azure Portal → Create Resource → Deploy a Custom Template
Upload: ARMTemplateForFactory.json
Parameters: ARMTemplateParametersForFactory.json (edit for target env)

Real-World Use Case: Development ADF is fully built and tested. You export the ARM template, update the parameter file with production server names, storage accounts, and key vault references, then deploy to Production ADF — exact same configuration, different environment values.

◆ Azure DevOps Repos Integration

With Azure DevOps, your entire ADF code lives in Repos as JSON:

```
ADF-Pipelines/ (Git Repository)
├─ pipeline/
│   ├─ PL_Daily_SalesETL.json
│   ├─ PL_Customer_Load.json
│   └─ PL_Master_Orchestrator.json
├─ dataset/
│   ├─ DS_SalesCSV.json
│   └─ DS_SQL_Customers.json
├─ linkedService/
│   ├─ LS_ADLS_Gen2.json
│   └─ LS_AzureSQLDB.json
```

```
|— trigger/
|   |— TRG_Daily_Schedule.json
|— dataflow/
|   |— DF_TransformSales.json
```

9 Global Parameters

Global Parameters are pipeline parameters that are **available across all pipelines** in an ADF — like environment-level constants.

Real-World Use Case:

```
Global Parameter: gp_environment      = "Production"
Global Parameter: gp_dataLakeURL     =
"https://prodlake.dfs.core.windows.net"
Global Parameter: gp_emailAlertGroup = "dataengg-
team@company.com"
```

All pipelines can reference these:

```
@pipeline().globalParameters.gp_environment
@pipeline().globalParameters.gp_dataLakeURL
```

Huge benefit during environment promotion: When moving from Dev to Prod, update just the Global Parameters — all 100 pipelines automatically use production values!

10 Credentials

Credentials in ADF enable secure authentication using **Managed Identity or Service Principal** instead of storing passwords.

Types:

- **System-assigned Managed Identity:** ADF automatically gets an identity in Azure AD — grant this identity access to storage/SQL
- **User-assigned Managed Identity:** You create a custom identity and assign it to ADF
- **Service Principal:** App registration in Azure AD with client ID + secret

Best Practice: Always use **Managed Identity** — no passwords to rotate, no secrets to leak, fully managed by Azure.

11 Monitoring ADF Jobs

◆ Monitor Tab in ADF Studio

The Monitor section gives you **complete visibility** into all pipeline and activity runs.

Pipeline Runs View:

Pipeline Name	Status	Start Time	Duration
Triggered By			
-----	-----	-----	-----

PL_Daily_SalesETL	Succeeded	2024-06-15 00:00	12m 45s
TRG_Daily_Schedule			
PL_Customer_Load	Failed	2024-06-15 00:13	2m 10s
TRG_Daily_Schedule			
PL_Inventory_Refresh	Running	2024-06-15 00:25	-
Manual			

Activity Runs View (drilling into a pipeline run):

Activity Name	Type	Status	Duration	
Output				
-----	-----	-----	-----	-----
GetMetadata_CheckFile	GetMetadata	Succeeded	3s	
exists:true, size:2MB				
CopySalesData	Copy	Succeeded	5m 22s	
rowsCopied:142,850				
RunDatabricksNotebook	Databricks	Succeeded	6m 10s	
notebook:completed				
SP_UpdateAuditLog	SqlScript	Succeeded	1s	-

Useful monitoring features:

- **Re-run failed pipelines** directly from monitor — with same or new parameters
- **Cancel running pipelines** that are stuck
- **Filter** by date range, pipeline name, status
- **Gantt chart view** for pipeline execution timeline

1 2 Alerts

ADF integrates with **Azure Monitor** to send alerts when pipelines fail, succeed, or take too long.

Creating an Alert:

```
Alert Rule: ADF_Pipeline_Failure_Alert
Condition: Pipeline runs failed > 0 (in last 5 minutes)
Severity: Sev 1 (Critical)
Action Group: DataEngg_Team
  → Email: dataengg@company.com
```

- SMS: +91-98765-43210
- Teams Webhook: (teams channel URL)

Real-World Use Case: Production pipeline fails at 2 AM. Alert fires immediately → entire data engineering team gets an email and SMS → on-call engineer is woken up → fixes the issue before 9 AM business hours → business reports are ready on time.

13 Send Failure Notifications Using Logic Apps

Logic Apps provides a more customizable notification system than basic Azure Monitor alerts — you can send **rich, detailed emails** with formatted HTML content.

Architecture:

ADF Pipeline Fails

↓

Web Activity in ADF calls Logic App HTTP trigger

↓

Logic App receives JSON payload

↓

Logic App sends formatted email via Office 365 connector

ADF Web Activity Configuration:

```
Activity: NotifyFailure (in pipeline's failure path)
URL: https://prod-xx.logic.azure.com/workflows/.../triggers/...
Method: POST
Body:
{
  "pipeline_name": "@{pipeline().Pipeline}",
  "run_id": "@{pipeline().RunId}",
  "failed_activity": "@{activity('CopyData').Error.message}",
```



```
"error_code": "@{activity('CopyData').Error.errorCode}",
"run_time": "@{utcnow()}",
"environment": "Production"
}
```

Email received by team:

Subject: ✖ ADF Pipeline FAILED - PL_Daily_SalesETL

Pipeline: PL_Daily_SalesETL

Environment: Production

Failed Activity: CopyData

Error: File not found at /raw/sales/sales_20240615.csv

Error Code: UserErrorFileNotFound

Run Time: 2024-06-15 00:02:34 UTC

Run ID: abc123-def456-...

[\[Click here to view in ADF Monitor\]](#)

14

Data Flows — Deep Dive

◆ What is Data Flow?

Data Flows provide a **visual, code-free way to transform data** at scale inside ADF. Under the hood, Data Flows run on **Apache Spark clusters** — so they can handle large datasets efficiently.

When to use Data Flow vs Copy Activity:

	Copy Activity	Data Flow
Purpose	Move data as-is	Transform data

	Copy Activity	Data Flow
Transformations	Basic column mapping	Complex aggregations, joins, pivots
Coding	None	None (visual)
Engine	ADF copy engine	Apache Spark
Debugging	Run & check output	Live data preview at each step

◆ Mapping Data Flow

Mapping Data Flow is the main type — visually maps transformations using a canvas.

Example Transformation Flow:

```

Source (ADLS CSV)
  ↓
Filter (remove rows where sales_amount <= 0)
  ↓
Derived Column (create new column: tax = sales_amount * 0.18)
  ↓
Aggregate (group by region, sum sales_amount)
  ↓
Join (join with region_master table for region names)
  ↓
Select (pick only needed columns)
  ↓
Sink (write to Azure SQL Database)

```

◆ Data Flow Debug

Debug mode spins up a small Spark cluster and lets you **preview data at every transformation step** — like a debugger for your data transformations.

After **Filter** transformation → Click "Data Preview"

Shows: **142,850 rows** passed **filter** (**0** negative sales removed)

After Aggregate transformation → Click "Data Preview"

Shows: **5 rows** (**one per region with** summed sales)

Cost Note: Debug cluster runs continuously while debugging — **turn it off when done** to avoid unnecessary charges!

◆ Transformations — Complete Reference

A. Filter

Removes rows that don't meet a condition — like SQL WHERE clause.

```
Condition: sales_amount > 0 && region != 'UNKNOWN'
```

B. Aggregate

Groups data and computes aggregations — like SQL GROUP BY.

```
Group By: region, product_category
```

```
Aggregations:
```

```
total_sales = sum(sales_amount)
```

```
avg_sales   = avg(sales_amount)
```

```
order_count = count(order_id)
```

C. Join

Combines two data streams — like SQL JOIN.

```
Left Stream: SalesData
Right Stream: RegionMaster
Join Type: Inner Join
Condition: SalesData.region_code == RegionMaster.region_code
```

D. Conditional Split

Like a SQL CASE or IF-ELSE — routes rows to different output streams.

```
Stream 1 (HighValue):    sales_amount >= 100000
Stream 2 (MediumValue):  sales_amount >= 10000 && sales_amount <
100000
Stream 3 (LowValue):     Default (everything else)
```

Real-World Use Case: Route high-value transactions to a premium customer table, others to regular.

E. Derived Column

Creates new columns or modifies existing ones — like SQL computed columns.

```
full_name      = concat(first_name, ' ', last_name)
year_month     = toString(order_date, 'yyyy-MM')
tax_amount     = sales_amount * 0.18
category_upper = upper(product_category)
```

F. Exists

Checks if rows from one stream exist in another — like SQL EXISTS / NOT EXISTS.

```
Exists Type: Exists (keep only matching)
Left Stream: NewOrders
Right Stream: ExistingOrders
Condition: NewOrders.order_id == ExistingOrders.order_id
Use: Find orders that are new (not already in warehouse)
```

G. Union

Combines multiple data streams vertically — like SQL UNION ALL.

```
Stream 1: SalesData_India
Stream 2: SalesData_US
Stream 3: SalesData_UK
Output: All rows combined into one dataset
```

H. Lookup

Enriches data by looking up values from a reference table.

```
Primary Stream: SalesTransactions
Lookup Stream: CustomerMaster
Match: sales.customer_id == customer.customer_id
Output: All sales columns + customer name/email from lookup
```

I. Sort

Orders data by one or more columns — like SQL ORDER BY.

```
Sort By: order_date DESC, sales_amount DESC
```

J. GroupBy

Groups rows for aggregation (similar to Aggregate but focused on grouping).

K. Pivot

Converts **rows to columns** — transforms category values into column headers.

```
Group By: region
Pivot Column: product_category (Electronics, Clothing, Food)
Aggregation: sum(sales_amount)

Result:
```

region	Electronics	Clothing	Food
North	500000	200000	150000
South	300000	250000	180000

L. Unpivot

Opposite of Pivot — converts **columns to rows**.

Input: Electronics=500000, Clothing=200000, Food=150000 (in one row)
Output: Three rows: Electronics/500000, Clothing/200000, Food/150000

M. Flatten

Expands **nested arrays/structs in JSON** into flat rows.

Input JSON: {"customer": "Raj", "orders": [{"id": 1}, {"id": 2}, {"id": 3}]}

Flatten: orders array

Output: 3 rows, one per order

N. Parse & Stringify

- **Parse:** Converts a string column into a complex type (JSON struct)
- **Stringify:** Converts a complex type back to a string

O. Alter Row

Sets **DML policies** on rows — tells the sink how to handle each row.

Alter Row Conditions:

Insert If: `isNull(existing_id)`

Update If: `!isNull(existing_id) && changed_flag == true`

```
Delete If: status == 'DELETED'
Upsert If: true (always upsert)
```

Real-World Use Case: Implementing SCD Type 1 — automatically upsert changed records.

P. Schema Validate & Schema Drift

- **Schema Validate:** Enforces that incoming data matches expected schema
- **Schema Drift:** Allows extra/unexpected columns to flow through without breaking the pipeline

Real-World Use Case: The upstream system sometimes adds new columns to CSV files. With Schema Drift enabled, ADF handles this gracefully instead of throwing errors.

Q. Remove Duplicate Rows

Using **Aggregate transformation** to deduplicate:

```
Group By: All columns (or key columns like customer_id)
Aggregate: row_number = first(order_id) ← keeps first occurrence
Then Filter: row_number == 1
```

R. Flowlet

A **reusable data flow component** — encapsulates common transformation logic that's used across multiple data flows.

Real-World Use Case: Every data flow in your company needs to: add audit columns (load_date, source_system, batch_id). Instead of repeating this in all 20 data flows, create a Flowlet "AddAuditColumns" and reuse it everywhere.

15 Azure DevOps

◆ Azure DevOps Repos

As covered earlier, Azure DevOps Repos is a **Git-based version control system** integrated with ADF for tracking all pipeline code changes.

Complete SDLC Flow with ADF + DevOps:

1. Developer creates feature branch **in** Repos
2. Builds/modifies pipelines **in** ADF Dev environment (linked **to** feature branch)
3. Commits JSON files **to** feature branch
4. Raises Pull Request → Team Lead reviews
5. Approved → Merges **to** main branch
6. CI/CD Pipeline **in** Azure DevOps:
 - a. Triggers **on** merge **to** main
 - b. Exports ARM template **from** ADF Dev
 - c. Deploys ARM template **to** ADF Test
 - d. Runs smoke tests
 - e. Deploys **to** ADF Production (**with** approval gate)

16 SDLC (Software Development Life Cycle)

The **SDLC** for ADF projects typically follows these phases:

1. REQUIREMENTS GATHERING
 - Business analyst documents data requirements
 - Source **system** details, transformation rules, target schema
2. DESIGN
 - Data architect designs pipeline architecture
 - Bronze-Silver-Gold layer design
 - Error handling strategy
3. DEVELOPMENT

- Data engineers build pipelines **in** ADF Dev
- Unit test **with** sample data
- Peer code review via Pull Requests

4. TESTING

- Deploy **to** ADF Test environment
- Integration testing **with full** dataset
- Performance testing (run **time**, throughput)
- UAT (**User** Acceptance Testing) **by** business

5. DEPLOYMENT

- Deploy **to** ADF Production
- Deployment during maintenance **window**
- Parallel run (**old** + **new** pipelines run together initially)

6. MAINTENANCE

- Monitor daily runs
- Handle failures **and** data quality issues
- Enhance pipelines **as** requirements change

17 Agile Methodology in ADF Projects

Agile delivers ADF projects in **short iterations (sprints)** rather than one big delivery.

Sprint Structure (2-week sprint example):

Sprint **1** (Week **1-2**):

- Build ingestion pipelines **for** Sales **module**
- Unit test, deploy **to** Dev
- Demo **to** business stakeholders

Sprint **2** (Week **3-4**):

- Build transformation pipelines for Sales
- Address Sprint 1 feedback
- Deploy Sales module to Test

Sprint 3 (Week 5-6):

- Build Customer module pipelines
- Deploy Sales module to Production
- Sprint review + retrospective

Agile Ceremonies:

- **Daily Standup:** 15-min sync — "What did I do yesterday? What will I do today? Any blockers?"
- **Sprint Planning:** Define tasks for next 2 weeks from backlog
- **Sprint Review:** Demo completed pipelines to stakeholders
- **Sprint Retrospective:** What went well / what to improve

Tools used:

- **Azure DevOps Boards:** Kanban board for tracking pipeline development tasks
- **Azure DevOps Repos:** Code repository
- **Azure DevOps Pipelines:** CI/CD for automated deployment

18 ADF Resume Preparation

Key Skills to Highlight:

Technical Skills:

- ✓ Azure Data Factory (Pipelines, Activities, Data Flows, Triggers, IR)
- ✓ Azure Data Lake Storage Gen2
- ✓ Azure Databricks (PySpark, Delta Lake)

- ✓ Azure Synapse Analytics
- ✓ Azure SQL Database / SQL Server
- ✓ Azure Key Vault
- ✓ Azure DevOps (Repos, CI/CD Pipelines)
- ✓ Microsoft Fabric

Projects to mention:

- ✓ End-to-end ETL pipeline from Oracle/SAP to Azure
- ✓ Medallion architecture implementation (Bronze/Silver/Gold)
- ✓ Real-time streaming pipeline using Event Trigger + Databricks
- ✓ On-premise to cloud data migration using SHIR

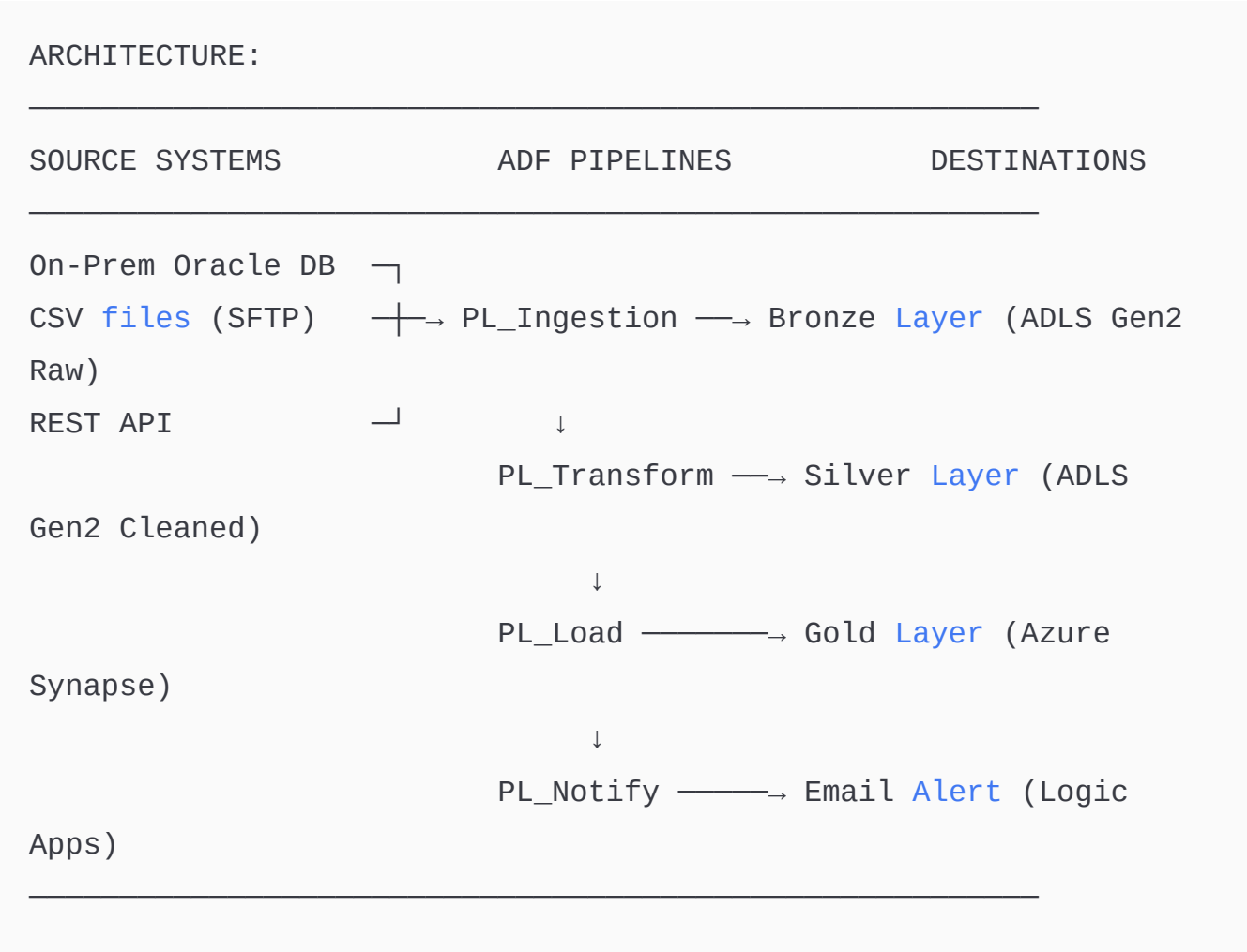
Sample Resume Points:

- Designed and implemented end-to-end ADF pipelines to ingest 50+ tables
from on-premise SQL Server to ADLS Gen2 using Self-Hosted Integration Runtime
- Built parameterized, reusable ADF pipelines with dynamic dataset
parameterization, reducing pipeline count from 50 to 1
- Implemented Delta Lake Medallion Architecture
(Bronze/Silver/Gold)
using ADF + Azure Databricks, processing 500GB daily
- Set up CI/CD deployment using Azure DevOps and ARM templates
for
Dev → Test → Production pipeline promotion
- Integrated Azure Key Vault with ADF for secure credential
management,
eliminating all hardcoded passwords from pipelines

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End-to-End ADF Project

Project: Daily Sales Analytics Pipeline



Master Pipeline Flow:

1.

VALIDATE:

Check all source files arrived (Validation + GetMetadata)
2.

INGEST:

Copy all sources to Bronze (Copy Activity + ForEach)
3.

TRANSFORM:

Run Databricks notebooks (Databricks Notebook Activity)
4.

LOAD:

Copy Silver data to Synapse Gold (Copy Activity)
5.

AUDIT:

Log success to audit table (Stored Procedure)
6.

NOTIFY:

Send completion report (Logic Apps via Web Activity)

Error Handling (failure path **on every** activity):

- Log error details **to** error **table**
- Send failure alert email **with** error details
- **Set** pipeline status **to** FAILED

20 ADF Exercises Summary

The exercises in the course cover all real-world scenarios:

Beginner Exercises:

- Create pipeline variables, use Set Variable activity
- Build If Condition for branching logic
- Copy files from Blob to Blob (single file, all files, recursive)
- Create linked services for BLOB, ADLS, SQL

Intermediate Exercises:

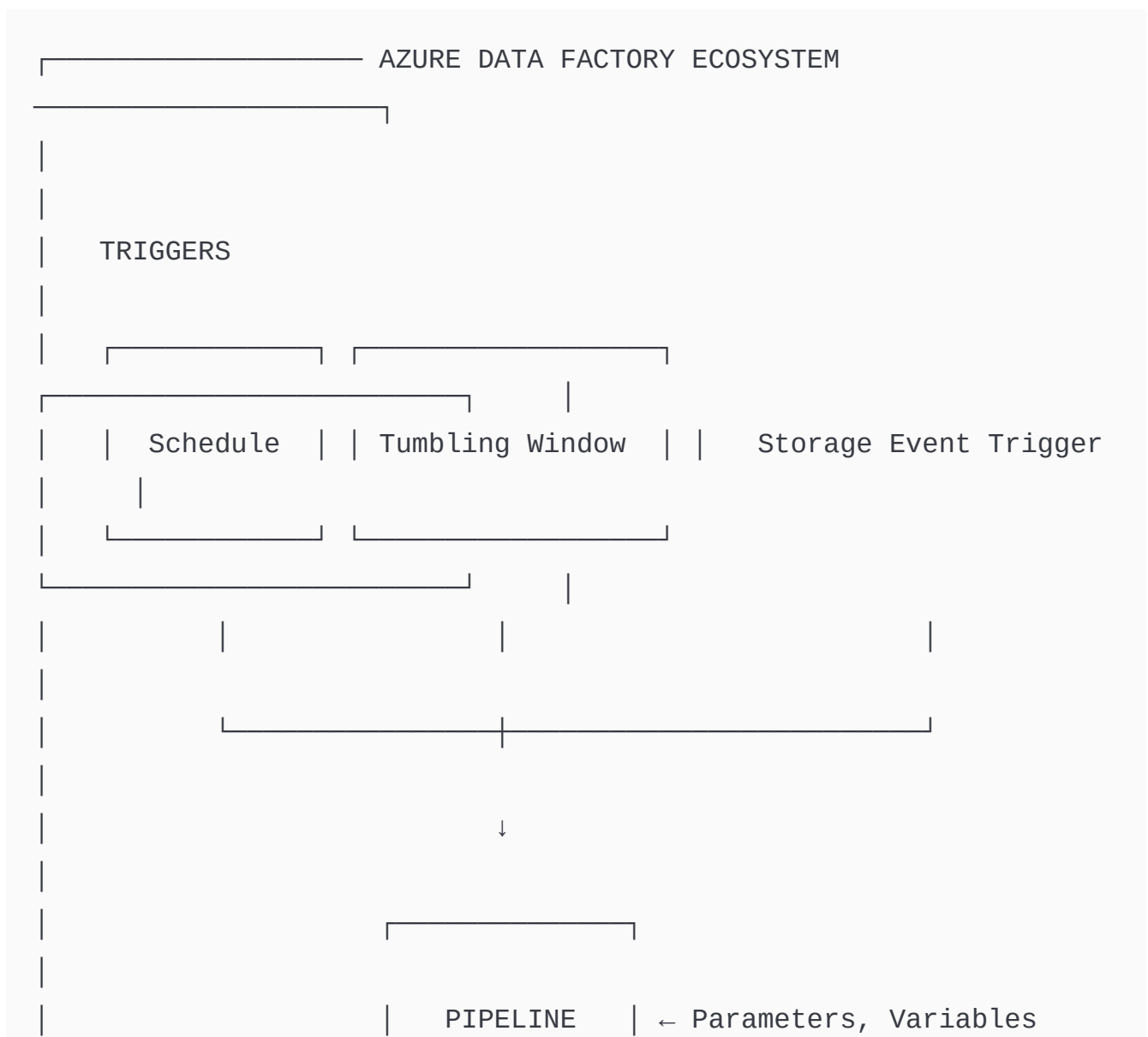
- Pattern matching file copy (copy only `sales_*.csv`)
- Parameterized copy using dynamic dataset
- Bulk copy multiple tables using Lookup + ForEach
- Convert CSV → Parquet, CSV → JSON format conversion
- Add extra audit columns (load_date, source_name) during copy

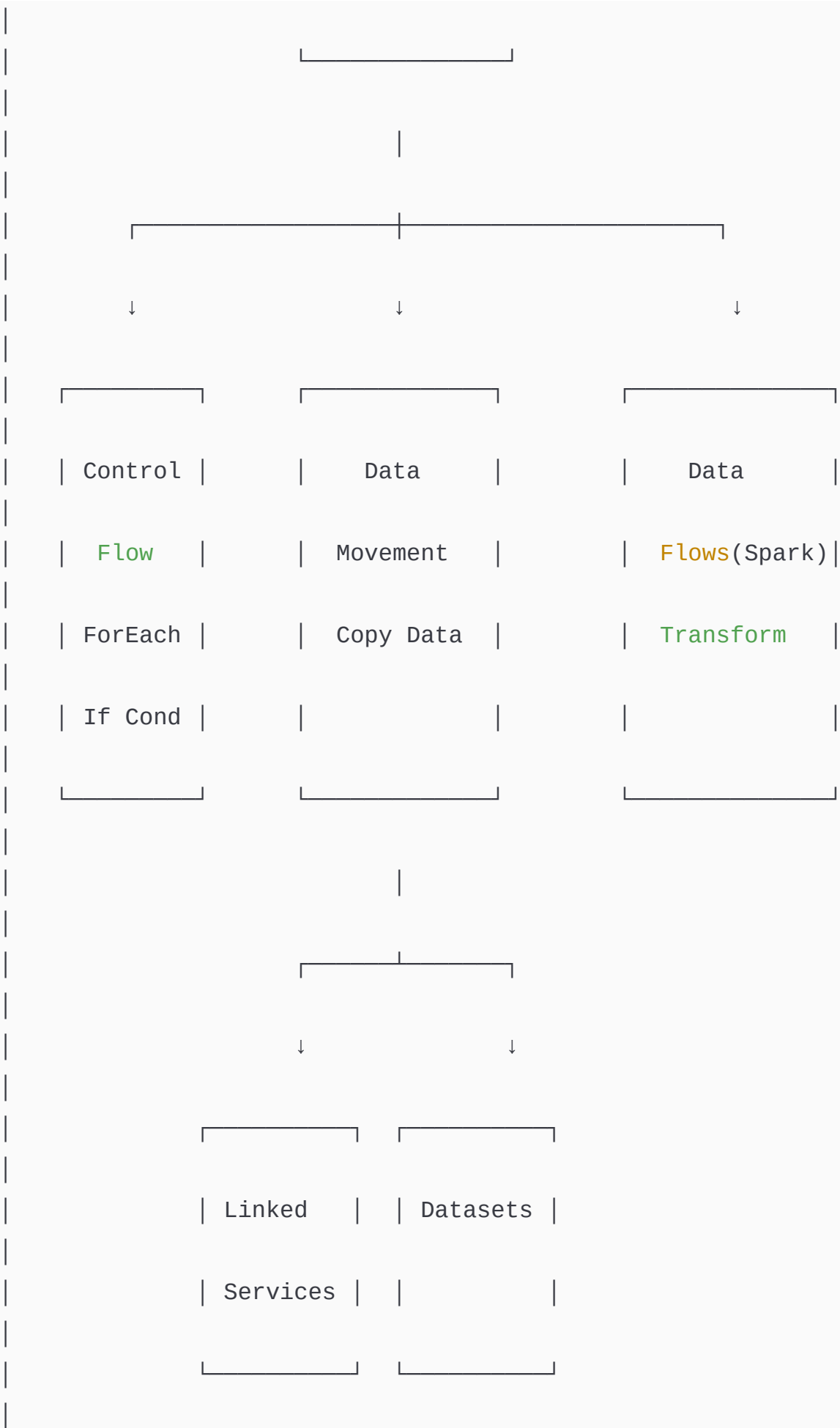
Advanced Exercises:

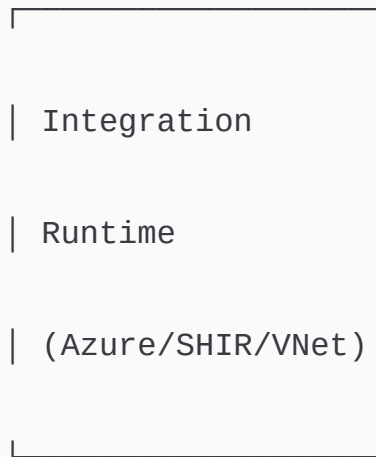
- Copy data from on-premise to Azure using SHIR
- Integrate Key Vault with ADF (secure credentials)
- Databricks activity with parameter passing
- Schedule trigger, Tumbling Window trigger, Event-based trigger
- Create trigger dependencies (run after another trigger)

- Data Flows — Filter, Join, Pivot, Union, Remove Duplicates
- Create alerts and notification rules
- Import/Export ARM templates
- Integrate ADF with Azure DevOps Repos
- Send mail notifications via Logic Apps
- Monitor and debug pipelines
- Execute one pipeline from within another (Execute Pipeline)

ADF Complete Architecture Summary







MONITORING: Pipeline Runs | Activity Runs | Trigger Runs

ALERTS: Azure Monitor + Logic Apps Email Notifications

SOURCE CTRL: Azure DevOps Repos + ARM Templates + CI/CD

SECURITY: Azure Key Vault + Managed Identity

✅ **Azure Data Factory section complete!** Every concept — from core architecture to all activity types, triggers, Integration Runtime, Data Flows with all transformations, DevOps, SDLC, Agile, and project structure — has been covered with real-world examples and deep explanations.

🚀 **Next up: Azure Databricks** — ready to continue whenever you are!

17/03/2026, 21:27:52

Thought process: Audited PDF content comprehensively and identified multiple missing sections.

Let me carefully review the PDF content for Azure Data Factory section and check what was covered vs what's in the PDF.

From the PDF, ADF section includes:

- 1) Azure Data Factory - covered ✓
- 2) Pipeline - covered ✓
- 3) Linked Service - covered ✓
- 4) DataSets - covered ✓
- 5) Activities:
 - Wait ✓
 - Variables (Create, Set, Append) ✓
 - Copy Data (General, Source, Sink, Mapping, Settings, User Properties) ✓
 - Copy file(s) from one BLOB Container to another (One file, All files, Recursively) ✓
 - Copy data/file from BLOB to SQL Database/ADLS Gen2 ✓
 - Databricks Notebook ✓
 - Azure Function ✓
 - Lookup, Stored Procedure ✓
 - Get Metadata, Delete ✓
 - Execute Pipeline ✓
 - Validation, Fail ✓
 - Iteration & Conditionals (Filter, ForEach, If Condition, Switch, Until) ✓
- 6) Triggers (Schedule, Tumbling Window, Storage Events, Triggers with Parameters) ✓
- 7) Integration Runtime (Azure AutoResolve, Azure) - appears to be cut off in the PDF

, but I'm continuing with Managed Virtual Network, Self-Hosted, and Linked Self-Hosted options ✓

8) Source Control (Git configuration, ARM Template Export/Import, Azure Devops Repos) ✓

9) Global parameters ✓

10) Credentials ✓

11) Monitoring ADF Jobs ✓

12) Alerts ✓

13) Send Failure Notifications using Logic Apps ✓

14) Data Flows - covering what they are, mapping data flows, debug mode, and various transformations like Filter, Aggregate, Join, Conditional Split, Derived Column, Exists, Union, Lookup, and Sort ✓

, plus GroupBy, Pivot, Unpivot, Flatten, Parse, Stringify, Filter sort, and Alterrow ✓. I'm also covering schema validation and duplicate row removal, though Asset transformations aren't included yet. Flowlets are covered ✓

15) Azure Devops - Repos ✓

16) SDLC ✓

17) Agile Methodology ✓

18) ADF Interview Questions - still need to add common Q&A

19) ADF Resume Preparation ✓

20) End TO End ADF Project ✓

21) ADF Exercises - in progress

Now I'm going through the 47 exercises to verify coverage. The first 11 are all accounted for: variable creation, conditional logic, looping through files, linked services and datasets, plus various copy activity scenarios including blob-to-blob transfers, blob-to-SQL, pattern matching, filtered formats, multiple files, source deletion, and parameterized datasets ✓

Continuing with exercises 12-22: file format conversion, adding columns, filtering and copying between blobs, deleting oversized

files, metadata retrieval, bulk operations, Key Vault integration, integration runtime setup, on-premises to cloud migration, Databricks activities with parameters, and scheduling triggers are all covered 

Now looking at the remaining topics: tumbling window and event-based triggers are explained, though "with Activity" needs clarification; Until Activity is covered; dataflows exercises include filtering, joins, unions, lookups, pivot/unpivot, and alter rows transformations, but selecting rows and window functions weren't explicitly covered as hands-on exercises.

The rest of the checklist covers parameters, alerts, global settings, ARM templates, DevOps integration, repos, logic app notifications, pipeline monitoring and debugging, scheduling with triggers, and executing pipelines within pipelines—most marked complete except trigger dependencies which needs deeper coverage. I'm also noticing two significant gaps: the ADF interview questions section from the PDF is completely missing, and the "asset" transformation in Data Flows wasn't included.

Looking at what still needs to be addressed, I see several Data Flow transformations haven't been covered yet—the asset transformation, window functions, and the select transformation for filtering rows. There's also "With Activity" in ADF and trigger dependency that require more thorough treatment. Plus I need to add a full ADF interview questions section and ensure all 47 exercises have complete step-by-step details.

ADF — Missing Concepts & Complete Exercises Review

Let me carefully audit everything against the PDF first, then cover all missing items in depth.



AUDIT: What Was Covered vs What's Missing

MISSING ITEMS IDENTIFIED:

- ✗ Data Flow – **SELECT** transformation (Exercise 27)
- ✗ Data Flow – **WINDOW** FUNCTIONS transformation (Exercise 32)
- ✗ Data Flow – **ASSET** transformation (PDF pg.6: "filter sort, alterrow, asset")
- ✗ "With Activity" – Exercise 25
- ✗ **Trigger** Dependency – Exercise 46
- ✗ ADF Interview Questions – **Full** Section 18 (completely missing)
- ✗ **All** 47 Exercises – **not** explained step-by-step **in** depth



MISSING DATA FLOW TRANSFORMATIONS



Select Transformation (Data Flows)

The **Select transformation** is used to **choose, rename, reorder, or remove columns** from your data stream — like `SELECT col1, col2 AS alias FROM table` in SQL.

When to use:

- Remove unnecessary columns before writing to sink (reduce storage cost)
- Rename columns to match destination schema
- Reorder columns to match expected output format
- Remove duplicate column names that arise after joins

Real-World Use Case:



Your source Oracle table has 85 columns but you only need 12 for your data warehouse.
Without Select, all 85 columns flow to destination — wasting storage and slowing processing.

Input Columns (85 total):

CUST_ID, CUST_NM, CUST_EMAIL, CUST_PHONE,
INTERNAL_SYS_COL1, INTERNAL_SYS_COL2... (73 unwanted cols)

Select Transformation:

Input Column	→	Output Column Name
CUST_ID	→	customer_id
CUST_NM	→	customer_name
CUST_EMAIL	→	email_address
CUST_PHONE	→	phone_number
ORD_AMT	→	order_amount
ORD_DT	→	order_date
PROD_CD	→	product_code
PROD_NM	→	product_name
REGN_CD	→	region_code
STATUS_FLG	→	status
CRT_DT	→	created_date
UPD_DT	→	updated_date
(73 cols deleted)	→	(not mapped = dropped)

Output: Only 12 clean, properly named columns

Rule-based mapping (powerful feature):

Instead of mapping columns one-by-one, use rules to bulk-rename:

Rule 1: Column names starting with "CUST_"
→ Replace "CUST_" with "customer_"
→ CUST_ID becomes customer_id automatically

Rule 2: All column names → convert to lowercase
→ Applies to entire stream in one rule

Duplicate input handling (after joins):

When you join two streams both having a column named `id`, Select lets you:

```
Left.id    → customer_id    (keep and rename)
Right.id   → product_id     (keep and rename)
```

✅ Window Functions Transformation (Data Flows)

Window functions perform calculations **across a set of rows related to the current row** — without collapsing the rows like Aggregate does. Every row keeps its individual identity while getting a calculated value.

Real-World Analogy: In a school exam — Window functions let you add "class rank" to each student's record without removing any student from the list. Aggregate would only give you "class average" and lose individual records.

Key difference: Window vs Aggregate

Raw Data:

region	product	sales
-----	-----	-----
North	Laptop	50000
North	Phone	30000
North	Tablet	20000
South	Laptop	40000
South	Phone	35000

AGGREGATE result (rows collapse):

region	total_sales
North	100000
South	75000

WINDOW result (rows preserved, new column added):

region	product	sales	region_total	rank_in_region
North	Laptop	50000	100000	1
North	Phone	30000	100000	2
North	Tablet	20000	100000	3
South	Laptop	40000	75000	1
South	Phone	35000	75000	2

ADF Data Flow Window Transformation Configuration:

Window Transformation: WND_SalesRanking

OVER (Partition By): region

→ Calculations done separately **per** region

ORDER BY: sales **DESC**

→ Rank highest sales **as 1**

RANGE: Unbounded preceding **to current row**

→ Standard **window** frame **for** ranking

WINDOW COLUMNS:

rank_in_region	= rank()
dense_rank	= denseRank()
row_number	= rowNumber()
region_total	= sum(sales)
region_avg	= avg(sales)
running_total	= cumSum(sales)

```
prev_sales      = lag(sales, 1)
next_sales      = lead(sales, 1)
pct_of_region   = sales / sum(sales) * 100
```

Common Window Functions in ADF Data Flows:

Function	Description	Example
<code>rank()</code>	Rank with gaps (1,1,3)	Sales ranking with ties
<code>denseRank()</code>	Rank without gaps (1,1,2)	Competition ranking
<code>rowNumber()</code>	Unique sequential number	Dedup (keep row 1)
<code>lag(col, n)</code>	Value from n rows behind	Compare with previous month
<code>lead(col, n)</code>	Value from n rows ahead	Compare with next month
<code>cumSum(col)</code>	Running total	Cumulative sales
<code>sum(col)</code>	Total over partition	Region total in every row
<code>avg(col)</code>	Average over partition	Dept average salary
<code>first(col)</code>	First value in window	First order per customer
<code>last(col)</code>	Last value in window	Most recent order
<code>nTile(n)</code>	Divide into n buckets	Top 25% customers
<code>percentRank()</code>	Percentile rank (0 to 1)	Performance percentile

Real-World Use Cases:

Use Case 1 — Finding Latest Record per Customer:

```
Window: PARTITION BY customer_id, ORDER BY updated_date DESC
rowNumber = rowNumber()
→ Filter where rowNumber = 1
→ Gives latest record per customer (SCD Type 2 current record)
```


Use Case 2 — Running Total for Finance Reports:

```
Window: PARTITION BY account_id, ORDER BY transaction_date
running_balance = cumSum(transaction_amount)
→ Each row shows account balance at that point in time
```

Use Case 3 — Month-over-Month Growth:

```
Window: PARTITION BY product_id, ORDER BY year_month
prev_month_sales = lag(sales_amount, 1)
growth_pct = (sales_amount - prev_month_sales) / prev_month_sales
* 100
→ Each row shows % growth vs previous month
```

✓ Asset Transformation (Data Flows)

The **Assert transformation** (referred to as "asset" in the PDF) is used to **enforce data quality rules** — it validates data against conditions and either flags or fails rows that don't meet your quality standards.

Real-World Analogy: Like a quality inspector on a factory production line — each product (row) is checked against standards. Bad products are either rejected (fail pipeline) or tagged as defective (flag and continue).

Two types of Assert:

1. EXPECT type → Flags rows that violate rule (continues processing)
2. FILTER type → Removes rows that violate rule (rows dropped)
3. ERROR type → Throws error and stops pipeline if any row violates

Configuration Example:

Assert Transformation: DQ_SalesValidation

Rule 1: sales_amount_check

Type: Expect (flag bad rows, don't stop)

Expression: sales_amount > 0

Description: "Sales amount must be positive"

Rule 2: date_format_check

Type: Expect

Expression: !isNull(order_date)

Description: "Order date cannot be null"

Rule 3: customer_id_check

Type: Error (stop if violated)

Expression: !isNull(customer_id) && length(customer_id) == 10

Description: "Customer ID required, must be 10 chars"

Rule 4: region_check

Type: Filter (remove invalid rows)

Expression: in(['NORTH', 'SOUTH', 'EAST', 'WEST'], region_code)

Description: "Only valid region codes allowed"

Output streams from Assert:

Assert transformation produces TWO output streams:

Stream 1: "Passed" rows → continue to Silver layer

Stream 2: "Failed" rows → write to error/quarantine table

Passed rows: Go to normal processing pipeline

```
Failed rows: Written to /quarantine/dq_errors/  
Alerting team about data quality issues
```

Real-World Use Case: A bank's transaction data comes from multiple branches. Some branches send negative transaction amounts (data entry errors). Assert catches these, flags them, writes them to a separate "exceptions" table for manual review, while valid transactions continue processing normally — no pipeline failure, no data loss.

✅ "With Activity" in ADF

The **"With Activity"** (also called **Set Variable with Expression / Script Activity**) in recent ADF context refers to the ability to **chain activity outputs directly into expressions without needing separate variable steps**.

However, in ADF exercises context, "With Activity" most likely refers to the **Script Activity** introduced in newer ADF versions — which lets you run **SQL scripts directly** in a pipeline without needing a Stored Procedure.

Script Activity — Deep Dive:

```
Activity: Script  
Linked Service: LS_AzureSQLDB_SalesDB  
Script Type: NonQuery (for INSERT/UPDATE/DELETE/DDL)  
Query (for SELECT – returns results)
```

Script:

```
-- Truncate staging table before load  
TRUNCATE TABLE dbo.STG_Sales;  
  
-- Create index if not exists  
IF NOT EXISTS (  
    SELECT 1 FROM sys.indexes
```

```

        WHERE name = 'IX_Sales_Date'
    )
BEGIN
    CREATE INDEX IX_Sales_Date ON dbo.Sales(order_date);
END

-- Update audit log
UPDATE dbo.PipelineLog
SET status = 'COMPLETED',
    end_time = GETDATE(),
    rows_processed = @{activity('CopyData').output.rowsCopied}
WHERE run_id = '@{pipeline().RunId}';

```

Why Script Activity over Stored Procedure Activity?

Stored Procedure:	Must pre-create procedure in DB → needs DBA access
Script Activity:	Write SQL directly in ADF → no pre-created objects needed
	Can run multiple SQL statements in one activity
	Dynamic SQL using ADF expressions possible

Real-World Use Case:

Pipeline: PL_Daily_Load

Step 1: Copy Data (load to staging table)

Step 2: Script Activity

- MERGE staging into production table
- Log row counts to audit table
- Drop staging table

All 3 SQL operations in one Script Activity

✓ Trigger Dependency — Deep Dive (Exercise 46)

Trigger Dependency allows you to **chain Tumbling Window triggers** — one trigger waits for another to complete successfully before firing. This solves the problem of dependent data pipelines needing to run in a specific order.

Real-World Problem:

Pipeline A: Loads raw sales data (runs hourly)

Pipeline B: Aggregates sales data (needs A to finish first)

Without dependency: B might start at 3:00 PM while A is still loading

→ B reads incomplete data → WRONG results!

With dependency: B's trigger waits for A's trigger to succeed
→ Only when A completes at 3:47 PM does B start

→ Always correct data

Configuration:

Trigger A: TRG_Hourly_RawSalesLoad

Type: Tumbling Window

Period: 1 Hour

No dependencies

Trigger B: TRG_Hourly_SalesAggregation

Type: Tumbling Window ← ONLY Tumbling Window supports dependencies!

Period: 1 Hour

DEPENDENCY:

Depends **On**: TRG_Hourly_RawSalesLoad

Size: **1 Hour** (same **window** size)

Offset: **0** (same **time window**)

Execution timeline with dependency:

3:00 PM → **Trigger** A fires → Pipeline A starts loading raw data

3:47 PM → Pipeline A completes successfully

3:47 PM → **Trigger** B sees its dependency (A **for 3-4 PM window**) **is** met

3:47 PM → **Trigger** B fires → Pipeline B starts aggregation

4:12 PM → Pipeline B completes

If Pipeline A FAILS **at 3:22** PM:

→ **Trigger** B **for 3-4 PM window** does **NOT** fire

→ **Trigger** B waits until A **succeeds** (retry)

→ Business-critical: aggregation never runs **on** incomplete data

Offset dependency (chain across different time windows):

Trigger B: TRG_Daily_Report

Depends **on**: TRG_Hourly_RawSalesLoad

Size: **1 Hour**

Offset: **-1 Hour**

Meaning: Daily report **trigger** waits **for ALL** hourly loads **of** the **day**

(the **-1 hour offset** means "previous window must succeed")

Multi-level dependency chain:

```
TRG_Extract (6 AM)
  ↓ depends on
TRG_Transform (must wait for Extract)
  ↓ depends on
TRG_Load (must wait for Transform)
  ↓ depends on
TRG_Report (must wait for Load)
```

Important Notes:

- Only **Tumbling Window triggers** support dependencies (not Schedule triggers)
- Maximum **20 dependencies** per trigger
- Creates a **self-healing pipeline** — if step 2 fails, step 3 automatically waits and retries when step 2 succeeds

✅ ADF Interview Questions — Complete Section 18

This is a **critical section** for job interviews. Here are the most important questions with detailed answers:

📌 Fundamentals

Q1: What is Azure Data Factory? How is it different from SSIS?

ADF:

- ✅ Cloud-native, fully managed PaaS service
- ✅ No infrastructure to manage
- ✅ 90+ connectors including cloud services, REST APIs, SaaS
- ✅ Built-in Spark engine (Data Flows)

- ✓ Pay-per-use (cost-effective for intermittent pipelines)
- ✓ CI/CD with Azure DevOps
- ✓ Handles big data natively

SSIS (SQL Server Integration Services):

- ✗ On-premise tool, requires Windows Server + SQL Server license
- ✗ You manage servers, patching, scaling
- ✗ Limited cloud connectors
- ✗ Does not scale for big data
- ✗ Fixed infrastructure cost
- ✓ Better for legacy on-premise workloads
- ✓ Complex transformations with scripting

Answer: ADF is the cloud evolution of SSIS – for modern cloud-based and hybrid data integration scenarios.

Q2: What are the different types of Integration Runtime? When do you use each?

1. Azure IR (AutoResolve):

- Cloud-to-cloud movement
- When source AND destination are both in Azure
- Example: ADLS Gen2 to Azure SQL Database

2. Azure IR with Managed VNet:

- When Azure resources use private endpoints
- No public internet exposure
- Bank/healthcare compliance requirements

3. Self-Hosted IR:

- On-premise to cloud movement
- Source is behind corporate firewall
- Example: On-premise Oracle → Azure Synapse

4. Azure-SSIS IR:

- Running legacy SSIS packages **in** cloud
- Migration period (before full ADF conversion)

Q3: What is the difference between a Linked Service and a Dataset?

Linked Service = CONNECTION definition

- Server name, credentials, authentication **method**
- HOW **to connect**
- Reused across multiple datasets
- Example: Connection **to SQL** Server **at 192.168.1.100**

Dataset = DATA definition

- **Specific table**, file, folder **to** access
- WHAT **to** access
- Points **to** a Linked Service
- Example: **Specific table** [dbo].[Sales] **on** that **SQL** Server

Analogy:

Linked Service = Library membership card (access **to** library)

Dataset = **Specific** book **in** that library

Q4: What is the difference between Schedule Trigger and Tumbling Window Trigger?

Schedule **Trigger**:

- Fires **at** fixed times (daily, hourly, weekly)
- Does **NOT** track **window** state

- If pipeline fails, **no** automatic retry **of** that **time window**
- Can be attached **to** multiple pipelines
- **No** self-dependency support
- Best **for**: Simple scheduled runs (daily report, weekly load)

Tumbling **Window Trigger**:

- Fires **for** fixed-size, non-overlapping **time** windows
- TRACKS **window** state (knows which windows ran/failed)
- Retries failed windows automatically
- Can BACKFILL past data (rerun Jan **1** **to** today)
- Supports DEPENDENCY chaining **between** triggers
- Best **for**: **Time**-series data, hourly incremental loads, reprocessing scenarios

Key Interview Insight:

Use Tumbling **Window when** data has a **time** dimension **and** you need exactly-once processing guarantees.

Q5: Explain the Copy Activity in detail. What are the key configurations?

Copy Activity moves data **from** Source → Sink **using** IR **as** execution engine.

Key Configurations:

SOURCE:

- Dataset (what **to** read)
- Source-**specific** options (query, stored proc, **partition**)
- Additional columns (**add** metadata columns)

SINK:

- Dataset (**where to** write)

- Write behavior (`insert`/`upsert`/`overwrite`/`append`)
- Pre/Post `copy` scripts (`SQL` to run before/after `copy`)
- Bulk `insert table` lock

MAPPING:

- `Column-to-column` mapping
- Data type conversions
- Schema drift handling

SETTINGS:

- Parallel `copy` degree (how many threads)
- Data consistency `check`
- Fault tolerance (`skip` bad `rows`)
- Logging (log skipped `rows` to storage)
- Staging (use Azure Storage `as` staging `for` PolyBase)

PERFORMANCE FACTORS:

- Integration Runtime size/count
- Parallel `copy` settings
- `Partition` options (parallel read `from` source)
- Staging `for` Synapse loads (PolyBase/`COPY INTO`)

Q6: How do you handle incremental data loads in ADF?

Method 1: Watermark-based (High Watermark)

- Store `last` loaded `timestamp` in a control `table`
- `Each` run: `SELECT * FROM source WHERE updated_date > last_watermark`
- After load: `Update` watermark to current `timestamp`

Implementation:

Step 1: Lookup → Read `last` watermark `from` control `table`

Step 2: Copy Data → Source query uses watermark value

Step 3: Stored Procedure → Update watermark after success

Method 2: Change Data Capture (CDC)

- Enable CDC on SQL Server source
- ADF reads only changed rows (inserts/updates/deletes)
- Highly efficient – no full table scan

Method 3: File-based incremental (Auto Loader pattern)

- New files arrive in ADLS daily
- ADF tracks which files are processed (via control table)
- Only new unprocessed files are loaded

Method 4: Tumbling Window Trigger

- Each window processes only that time period's data
- Built-in tracking ensures no duplicates

Q7: What is Schema Drift in Data Flows? How do you handle it?

Schema Drift = When source data schema changes unexpectedly (new columns added, data types changed, columns removed)

Without Schema Drift handling:

- Pipeline FAILS when new column arrives
- Manual fix required → pipeline downtime

With Schema Drift enabled:

- New columns are automatically passed through
- Pipeline continues without breaking
- Extra columns land in destination as-is

Configuration in Data Flow:

Source → Allow Schema Drift: ON

Sink → Auto-map drifted columns: ON

Advanced: Schema drift with explicit handling

→ Use `byName('columnName')` to access columns dynamically

→ Use `mapAssociation()` for flexible column mapping

→ Use `toString(byNames(['col1', 'col2']))` for dynamic selection

Real-world: Upstream source team added 3 new columns to

CSV export without telling data engineering team.

Schema drift = pipeline keeps running, new columns flow through.

Q8: How do you parameterize ADF pipelines? Why is it important?

Parameterization levels:

1. Pipeline Parameters → values passed when pipeline runs/triggered
2. Dataset Parameters → values for file paths, table names
3. Linked Service Parameters → server names, database names
4. Global Parameters → environment-level constants

Why important:

- ✓ Reusability: One pipeline handles 100 tables instead of 100 pipelines
- ✓ Environment promotion: Same pipeline, different env parameter files
- ✓ Dynamic paths: Process yesterday's file automatically using date
- ✓ Maintainability: Change logic in one place

Example:

Without params: 365 pipelines for 365 daily files

```
With params:      1 pipeline, pass date as parameter →  
@{pipeline().parameters.p_date}
```

Q9: How do you implement error handling in ADF?

1. Activity-level retry:

- Set Retry = 3, Retry Interval = 30 sec
- Handles transient failures (network blip, timeout)

2. Failure path (On Failure dependency):

- Right-click connection arrow → change to "On Failure"
- Add notification activity on failure path

Pipeline:

CopyData [Success] → UpdateAudit

CopyData [Failure] → LogError → SendAlert → Fail Activity

3. Try-Catch pattern:

- Execute Pipeline activity wraps the risky pipeline
- If child pipeline fails, parent catches and handles

4. Fault Tolerance in Copy Activity:

- Skip incompatible rows
- Log skipped rows to storage for investigation
- Pipeline continues despite bad data rows

5. Validation before processing:

- GetMetadata → check file exists + size > 0
- If file missing → Fail with clear error message
- Prevents misleading "success" with empty results

6. Alerts:

- Azure Monitor alerts **for** pipeline failures
- Logic Apps **for** detailed email notifications

Q10: What are Data Flows? When to use Data Flow vs Databricks?

Data Flows:

- ✓ Visual, **no**-code transformation tool **in** ADF
- ✓ Runs **on** managed Spark clusters (auto-provisioned)
- ✓ Good **for**: medium complexity transformations
- ✓ Business users / less experienced developers can use
- ✓ Built-**in** data preview **for** debugging
- ✓ Handles datasets up **to** ~TB **range** well
- ✗ Less flexible than writing code
- ✗ Limited custom logic (**no** arbitrary Python/Scala)
- ✗ Cluster startup **time** (2-4 minutes)

Azure Databricks:

- ✓ **Full** PySpark/Scala/**SQL** programming power
- ✓ ML, streaming, complex business logic
- ✓ Better performance **for** very **large** data (**10TB+**)
- ✓ Delta Lake, Unity Catalog integration
- ✓ More cost control **over** cluster configuration
- ✗ Requires coding knowledge
- ✗ Separate workspace **to** manage

Rule **of** thumb:

- Simple-medium **SQL-like** transformations → Data Flow
- Complex logic, ML, streaming, very **large** data → Databricks
- Enterprise: Use **BOTH** – ADF orchestrates, Databricks transforms

Q11: How do you pass parameters from ADF to Databricks Notebook?

In ADF Databricks Notebook Activity:

Base Parameters:

```
input_path:  @{{concat('/raw/sales/',
formatDateTime(utcnow(), 'yyyy/MM/dd'))}}
output_path: @{{concat('/silver/sales/',
formatDateTime(utcnow(), 'yyyy/MM/dd'))}}
run_date:    @{{formatDateTime(utcnow(), 'yyyy-MM-dd')}}
env:         @{{pipeline().globalParameters.gp_environment}}
```

In Databricks [Notebook](#) (receive parameters):

```
dbutils.widgets.text("input_path", "")
dbutils.widgets.text("output_path", "")
dbutils.widgets.text("run_date", "")

input_path = dbutils.widgets.get("input_path")
output_path = dbutils.widgets.get("output_path")
run_date    = dbutils.widgets.get("run_date")

print(f"Processing: {input_path} for date: {run_date}")
```

Q12: What is the difference between ForEach sequential and parallel execution?

Sequential (Batch Count = 1):

- Items processed **ONE** at a **time**
- Item **1** completes → Item **2** starts → Item **3** starts...
- Slower but safe **when** items share resources

Use **when**:

- Items write **to** same database **table** (prevent deadlocks)
- Source **system** has API rate limits

→ Order of processing matters

Parallel (Batch Count = 4, 5, 8, etc.):

- Multiple items processed simultaneously
- Items 1,2,3,4 run at same time → when one finishes, next starts
- Much faster for independent items

Use when:

- Items are completely independent (different tables/files)
- Destination can handle concurrent writes
- Want to minimize total processing time

Max Batch Count: 50 (ADF limit)

Recommended: 4-8 for typical scenarios (avoid overwhelming source DB)

Real-world example:

Loading 50 tables in parallel with batch=5:

- 50 tables / 5 parallel = 10 rounds
- vs sequential = 50 rounds
- ~5x faster!

Q13: How do you implement SCD Type 1 and Type 2 in ADF?

SCD Type 1 (Overwrite – no history):

- Use UPSERT in Copy Activity Sink
- Or use Alter Row transformation in Data Flow

Data Flow approach for SCD1:

Source → Lookup (check if exists in target)

→ Alter Row:

Insert If: `isNull(target.customer_id)`

```
Update If: !isNull(target.customer_id) && anyChanged
→ Sink (with UPSERT policy)
```

SCD Type 2 (Keep history with effective dates):

Data Flow approach:

Source → Lookup (find current active record)

→ Exists (check if record changed)

→ Derived Column:

For changed records:

is_current = false

end_date = currentDate()

For new records:

is_current = true

start_date = currentDate()

end_date = '9999-12-31'

→ Alter Row:

Update existing record: set is_current=false, end_date

Insert new version: is_current=true, new values

→ Sink

Q14: How do you optimize ADF pipeline performance?

1. Copy Activity optimization:

→ Enable parallel copy (degree of copy parallelism)

→ Use partition options for large tables (physical/dynamic range)

→ Use PolyBase/COPY INTO for Synapse (bulk load)

→ Enable staging for large loads

→ Choose Azure IR in same region as source/sink

2. Data Flow optimization:

→ Enable broadcast for small lookup tables

- Avoid unnecessary sort operations
- Use Persist/Cache **for** data reused multiple times
- Increase cluster size **for large** data (8 core+)
- Use optimized auto resolve IR

3. Pipeline optimization:

- Run independent activities **in** parallel
- Use ForEach **with** parallel batch count
- Avoid unnecessary Lookup activities **in** loops
- Use tumbling **window** triggers efficiently

4. Integration Runtime sizing:

- Self-Hosted IR: 8+ cores, 16+ GB RAM **for** production
- Multiple SHIR nodes **for** high availability + load balancing

5. Network optimization:

- Keep IR, source, **and** sink **in** same Azure region
- Use private endpoints **to** avoid public internet
- Use ExpressRoute **for on**-premise connections

Q15: Explain the ADF CI/CD process with Azure DevOps.

Development Flow:

1. Developer creates feature branch **from** main
2. Builds/**modifies** pipelines **in** ADF Dev (linked **to** feature branch)
3. Commits JSON files **to** feature branch
4. Creates Pull Request → team lead reviews JSON changes
5. PR approved → **merge to** main branch
6. CI Pipeline triggers automatically:
 - a. Validates ADF JSON files

- b. Exports ARM template **from** ADF Dev
 - c. Stores ARM template **in** Azure Artifacts
 - 7. CD Pipeline (**with** approval gates):
 - a. Deploy **to** ADF Test → run smoke tests
 - b. Manual approval gate (team lead approves)
 - c. Deploy **to** ADF Production
 - d. Run health checks
-

Environment-**specific** configurations:

Dev: ADF Dev → Dev Storage → Dev **SQL** DB
Test: ADF Test → Test Storage → Test **SQL** DB
Prod: ADF Prod → Prod Storage → Prod **SQL** DB

Managed via: ARM Template **Parameter** files **per** environment

ALL 47 ADF EXERCISES — Step-by-Step Deep Explanation

Exercise 1: Create Variables Using Set Variable Activity

Objective: Understand how pipeline variables work in ADF.

Step-by-step:

STEP 1: Create a new pipeline → PL_Ex01_Variables

STEP 2: Add pipeline variables (click canvas → Variables tab)

Variable 1: v_processDate → String → Default: ""

Variable 2: `v_rowCount` → Integer → Default: 0
Variable 3: `v_tableList` → Array → Default: []

STEP 3: Add "Set Variable" activity

Name: `SetProcessDate`

Variable Name: `v_processDate`

Value: `@{formatDateTime(utcnow(), 'yyyy-MM-dd')}`

→ This sets today's date as process date

STEP 4: Add another "Set Variable" activity (connect after first)

Name: `SetRowCount`

Variable Name: `v_rowCount`

Value: `1000`

→ Static value assignment

STEP 5: Debug and check Output tab

→ Click `SetProcessDate` output → shows value: `"2024-06-15"`

→ Click `SetRowCount` output → shows value: `1000`

KEY LEARNING:

- Variables are pipeline-scoped (not shared across pipelines)
- Variables can be read using `@{variables('v_processDate')}`
- Variables are mutable (can be changed multiple times in pipeline)
- Parameters are immutable (set once at pipeline start)

Exercise 2: How to Use If Condition Activity

Objective: Branch pipeline logic based on conditions.

SCENARIO: Check if today is Monday → if `yes`, run full load, else incremental

STEP 1: Create pipeline → PL_Ex02_IfCondition

STEP 2: Add Set Variable activity

Variable: v_dayOfWeek

Value: @{dayOfWeek(utcnow())}

→ Returns 0=Sunday, 1=Monday, 2=Tuesday...

STEP 3: Add If Condition activity (connect after Set Variable)

Name: CheckIfMonday

Expression: @{equals(variables('v_dayOfWeek'), 1)}

TRUE Activities (inside If → True branch):

→ Add Web Activity: CallFullLoadPipeline

URL: (or Execute Pipeline for full load)

FALSE Activities (inside If → False branch):

→ Add Web Activity: CallIncrementalLoad

STEP 4: Debug pipeline

→ If today is Monday → True branch executes

→ Any other day → False branch executes

ANOTHER PRACTICAL EXAMPLE:

Condition: @{greater(activity('GetMetadata').output.size, 0)}

True: → Proceed with Copy Data

False: → Send alert "File is empty" + Fail pipeline

Exercise 3: Iterating Files Using ForEach Loop Activity

Objective: Process multiple files/tables using ForEach.

SCENARIO: Copy 5 specific CSV files from source to destination

STEP 1: Create pipeline → PL_Ex03_ForEach

STEP 2: Add Set Variable activity

Variable: v_fileList (Array type)

Value:

```
@{createArray('sales_jan.csv','sales_feb.csv','sales_mar.csv',  
              'sales_apr.csv','sales_may.csv')}
```

STEP 3: Add ForEach activity

Items: @{variables('v_fileList')}

Batch Count: 2 ← process 2 files in parallel

Sequential: No

STEP 4: INSIDE ForEach → Add Copy Data activity

Source:

Linked Service: LS_BlobStorage

Container: raw

File Name: @{item()} ← current item in loop

Sink:

Linked Service: LS_ADLS_Gen2

Container: processed

File Name: @{item()}

STEP 5: Debug and monitor

→ Watch 2 files copy simultaneously

→ Each iteration shows @{item()} = current filename

BETTER APPROACH (dynamic from Lookup):

Lookup activity reads file list from control table

ForEach iterates over lookup output:

```
Items: @{activity('LookupFiles').output.value}  
Inner item reference: @{item().file_name}
```

Exercise 4: Creating Linked Services and Datasets

EXERCISE 4A: Create Linked Service for Azure Blob Storage

ADF Studio → Manage → Linked Services → + New

Type: Azure Blob Storage

Name: LS_BlobStorage_Raw

Authentication: Account Key

Storage Account: (select from dropdown)

→ Test Connection → Create

EXERCISE 4B: Create Linked Service for Azure SQL Database

→ + New

Type: Azure SQL Database

Name: LS_AzureSQLDB_Sales

Server: youserver.database.windows.net

Database: SalesDB

Authentication: SQL Authentication

Username: sqladmin

Password: (or reference Key Vault secret)

→ Test Connection → Create

EXERCISE 4C: Create Dataset for CSV file in Blob

Author → Datasets → + New

Type: Azure Blob Storage → DelimitedText (CSV)

Name: DS_BlobCSV_Sales

Linked Service: LS_BlobStorage_Raw

File Path: raw/sales/
First Row as Header: Yes
→ Preview Data to verify

EXERCISE 4D: Create Dataset for SQL Table

→ + New
Type: Azure SQL Database
Name: DS_SQL_SalesTable
Linked Service: LS_AzureSQLDB_Sales
Table: [dbo].[Sales]
→ Preview Data to verify

EXERCISE 4E: Parameterized Dataset

→ Create Dataset DS_BlobCSV_Parameterized
→ Parameters tab → + New
 p_folderPath: String
 p_fileName: String
→ Connection tab:
 Container: raw
 Directory: @{dataset().p_folderPath}
 File: @{dataset().p_fileName}

Exercise 5: Copy Activity — Blob to Blob

SCENARIO: Copy a single CSV file from container "raw" to container "processed"

STEP 1: Create pipeline → PL_Ex05_BlobToBlob

STEP 2: Create two datasets:

DS_Source_BlobCSV:

Linked Service: LS_BlobStorage

Container: raw

File: sales_data.csv

DS_Sink_BlobCSV:

Linked Service: LS_BlobStorage

Container: processed

File: sales_data.csv

STEP 3: Add Copy Data activity

Source: DS_Source_BlobCSV

Sink: DS_Sink_BlobCSV

Copy behavior: PreserveHierarchy

STEP 4: Check Source tab

→ File Path Type: File path in dataset

STEP 5: Check Sink tab

→ Copy Behavior: PreserveHierarchy

→ Max concurrent connections: 1

STEP 6: Mapping tab

→ Import schema → verify column mapping

STEP 7: Debug → Check Output

→ filesRead: 1

→ filesWritten: 1

→ rowsCopied: (number of rows)

→ throughput: (MB/s)

→ duration: (seconds)

KEY LEARNINGS:

→ Copy activity output shows complete statistics

- Check "dataRead" and "dataWritten" bytes
- Monitor "throughput" to identify performance bottlenecks

Exercise 6: Copy Activity — Blob to Azure SQL

SCENARIO: Copy CSV from Blob into SQL Database table

STEP 1: Create SQL table first (in Azure SQL Database):

```
CREATE TABLE dbo.Sales_Staging (  
    sale_id          INT,  
    customer_id      VARCHAR(10),  
    product_code     VARCHAR(20),  
    sale_amount      DECIMAL(18,2),  
    sale_date        DATE,  
    region           VARCHAR(50)  
);
```

STEP 2: Create Datasets

DS_Source: CSV in Blob (with schema)

DS_Sink: dbo.Sales_Staging SQL table

STEP 3: Copy Activity

Source: DS_Source (CSV)

Sink: DS_Sink (SQL Table)

Sink Settings:

Write Behavior: Insert

Pre-copy Script: TRUNCATE TABLE dbo.Sales_Staging

→ Clears table before loading fresh data

Bulk Insert Table Lock: Yes ← faster bulk load

Batch Size: 100000 ← insert 100K rows per batch

STEP 4: Mapping

CSV col: sale_id → SQL col: sale_id (INT)
CSV col: cust_id → SQL col: customer_id (VARCHAR)
→ Map all columns, fix any type mismatches

STEP 5: Settings

Fault Tolerance: Skip incompatible rows
Log Skipped Rows: Yes → write to /logs/skipped_rows/

STEP 6: Debug + Verify

→ Run pipeline
→ Query SQL: SELECT COUNT(*) FROM dbo.Sales_Staging
→ Should match source file row count

Exercise 7: Copy Activity — Pattern Matching Files

SCENARIO: Copy ONLY files matching pattern sales_2024*.csv
(wildcard)

KEY CONCEPT: Wildcard path allows copying multiple
files matching a pattern in one Copy activity

STEP 1: Dataset DS_WildcardSource

Linked Service: LS_BlobStorage
Container: raw
→ Leave file path EMPTY in dataset
→ We'll use wildcard in Copy Activity Source tab

STEP 2: Copy Activity Source tab

File Path Type: Wildcard file path
Wildcard Folder Path: sales/2024/

Wildcard File Name: `sales_2024*.csv`

This matches:

- ✓ `sales_20240101.csv`
- ✓ `sales_20240215.csv`
- ✓ `sales_2024_final.csv`
- ✗ `sales_2023_jan.csv` (doesn't match)
- ✗ `orders_2024.csv` (doesn't match prefix)

STEP 3: Sink - `DS_ProcessedBlob`

Container: `processed`

Copy Behavior: `PreserveHierarchy`

STEP 4: Debug

- Output shows: `filesRead`: (count of matched files)
- All matching files copied in one activity run

ADVANCED PATTERNS:

- `*.csv` → all CSV files in folder
- `sales_*.csv` → all files starting with "sales_"
- `*_2024_*.csv` → files containing "_2024_"
- `????_data.csv` → files with exactly 4 chars then "_data.csv"

Exercise 8: Copy Activity — Copy Filtered File Formats

SCENARIO: Source has mixed file formats (`.csv`, `.json`, `.txt`, `.parquet`)

Copy ONLY `.csv` and `.parquet` files, skip others

APPROACH 1: `ForEach` with `Get Metadata`

STEP 1: `GetMetadata` on source folder

Field: childItems (lists all files in folder)

STEP 2: ForEach over childItems

STEP 3: Inside ForEach → If Condition

Condition:

```
@{or(
  endsWith(item().name, '.csv'),
  endsWith(item().name, '.parquet')
)}
```

TRUE → Copy Data activity

FALSE → (do nothing / log skip)

APPROACH 2: Filter Activity (simpler)

After GetMetadata:

Filter Activity:

```
Items: @{activity('GetMetadata').output.childItems}
Condition: @{or(endsWith(item().name, '.csv'),
  endsWith(item().name, '.parquet'))}
```

ForEach over filter output:

```
Items: @{activity('FilterFormats').output.value}
Inner: Copy Data using @{item().name}
```

Exercise 9: Copy Activity — Copy Multiple Files from Blob to Blob

SCENARIO: Copy ENTIRE folder with all files and subfolders

STEP 1: Dataset DS_SourceFolder

Container: raw

Folder: /sales/ (folder, not specific file)

STEP 2: Dataset DS_DestinationFolder

Container: archive

Folder: /sales/

STEP 3: Copy Activity

Source:

File Path Type: Wildcard file path

Wildcard Folder: * ← all subfolders

Wildcard File: * ← all files

Recursive: TRUE ← go into subfolders

Sink:

Copy Behavior: PreserveHierarchy ← maintain folder structure

RESULT:

Source:

/raw/sales/

/2024/jan/s1.csv

/2024/jan/s2.csv

/2024/feb/s3.csv

/summary.csv

Destination:

/archive/sales/

/2024/jan/s1.csv

/2024/jan/s2.csv

/2024/feb/s3.csv

/summary.csv

Exercise 10: Copy Activity — Delete Source After Copy

SCENARIO: Copy file from landing zone → processing zone → delete from landing

PATTERN: Copy then Delete (only delete on success!)

STEP 1: GetMetadata → verify file exists and has data

STEP 2: Copy Data

Source: /landing/sales_20240615.csv

Sink: /raw/sales/2024/06/sales_20240615.csv

On Success → (connect to Step 3)

On Failure → (connect to error handling)

STEP 3: Delete Activity (ONLY on Copy success)

Dataset: DS_LandingZone

→ Same file as copy source

Recursive: False (just this one file)

CRITICAL SAFEGUARD:

→ Delete activity MUST be on success path only

→ Never delete before confirming copy succeeded!

→ Check copy output: filesWritten = 1 before deleting

VERIFICATION:

Add another GetMetadata after Delete:

→ Try to get metadata of deleted file

→ Should return "exists: false"

→ Confirms deletion worked

FULL PIPELINE FLOW:

GetMetadata → If Exists → Copy Data → Delete Source → Log Success

↓ No File

Fail Activity "File not found"

Exercise 11: Copy Activity — Using Parameterized Datasets

SCENARIO: One pipeline processes different tables daily using parameters

STEP 1: Create parameterized dataset DS_SQL_Dynamic

Parameters:

p_tableName: String
p_schemaName: String (default: dbo)

Table configuration:

Schema: @{dataset().p_schemaName}
Table: @{dataset().p_tableName}

STEP 2: Create parameterized dataset DS_ADLS_Dynamic

Parameters:

p_containerName: String
p_folderPath: String
p_fileName: String

File path:

@{dataset().p_containerName}/{dataset().p_folderPath}/{dataset().p_fileName}

STEP 3: Pipeline with parameters

Pipeline Parameters:

p_sourceTable: String
p_targetFolder: String
p_date: String

STEP 4: Copy Activity

Source Dataset: DS_SQL_Dynamic

Dataset Parameters:

p_tableName: @{pipeline().parameters.p_sourceTable}
p_schemaName: dbo

Sink Dataset: DS_ADLS_Dynamic

Dataset Parameters:

```
p_containerName: silver
p_folderPath:    @{{pipeline().parameters.p_targetFolder}}
p_fileName:
@{{concat(pipeline().parameters.p_sourceTable, '_',
          pipeline().parameters.p_date,
          '.parquet')}}}
```

STEP 5: Run pipeline multiple times with different parameter values:

Run 1: p_sourceTable=Customers, p_date=20240615

Run 2: p_sourceTable=Orders, p_date=20240615

Run 3: p_sourceTable=Products, p_date=20240615

Exercise 12: Copy Activity — Convert File Format

SCENARIO: Convert CSV files to Parquet format for better analytics performance

WHY PARQUET?

CSV 1GB file → takes 45s to query 1 column in Spark

Parquet 350MB → takes 2s to query 1 column (columnar, compressed)

→ 70% smaller, 20x faster for analytics

STEP 1: Source Dataset DS_SourceCSV

Format: DelimitedText (CSV)

Container: raw

File: sales_data.csv

First row as header: Yes

STEP 2: Sink Dataset DS_SinkParquet

Format: Parquet

Container: silver

File: sales_data.parquet

Compression: Snappy (best balance of speed/size)

STEP 3: Copy Activity

Source: DS_SourceCSV

Sink: DS_SinkParquet

Mapping Tab:

→ Import schema from source

→ Map each column + set correct data types:

sale_id: string → int32

sale_amount: string → double

sale_date: string → date ← ADF converts string to proper date!

STEP 4: Debug + Verify

→ Original CSV: 500MB

→ Output Parquet: ~150MB (70% compression!)

OTHER FORMAT CONVERSIONS:

CSV → JSON: Source=CSV, Sink=JSON (with array of objects settings)

JSON → Parquet: Source=JSON, Sink=Parquet

CSV → Avro: Source=CSV, Sink=Avro (good for streaming systems)

CSV → ORC: Source=CSV, Sink=ORC (Hive-compatible format)

Exercise 13: Copy Activity — Add Additional Columns

SCENARIO: Source CSV has 5 columns.

Add 3 audit columns before loading to destination:

- load_date (today's date)
- source_system (where data came from)
- pipeline_run_id (for traceability)

STEP 1: Source Dataset DS_SalesCSV

Columns: sale_id, customer_id, product_code, amount, sale_date

STEP 2: Copy Activity Source tab

- Scroll down to "Additional Columns" section
- + New → Add:

Column Name: load_date

Value: @{{formatDateTime(utcnow(),'yyyy-MM-dd HH:mm:ss')}}

Column Name: source_system

Value: Oracle_ERP_Production

Column Name: pipeline_run_id

Value: @{{pipeline().RunId}}

Column Name: pipeline_name

Value: @{{pipeline().Pipeline}}

STEP 3: Mapping tab

- All original columns map to destination
- 4 new audit columns also appear → map to destination audit columns

RESULT in destination:

sale_id	customer_id	product_code	amount	sale_date	load_date	source_system	pipeline_run_id
-----	-----	-----	-----	-----	-----	-----	-----
1001	C001	LAPTOP01	50000	2024-06-15	2024-06-15 00:02:34	Oracle_ERP_Prod	abc123-def456...

KEY BENEFIT:

- Every row in destination knows WHEN it was loaded, WHERE it came from, and WHICH pipeline run loaded it
- Critical for data lineage and debugging

Exercise 14: Copy Activity — Filter Files and Copy

SCENARIO: Folder has 100 files. Copy ONLY files modified in last 24 hours.

STEP 1: GetMetadata on folder

Field: childItems (with lastModified for each file)

STEP 2: Filter Activity

Items: @{activity('GetMeta').output.childItems}

Condition:

```
@{greater(  
    item().lastModified,  
    addHours(utcnow(), -24)  
)}
```

→ Only files modified in last 24 hours pass through

STEP 3: ForEach over filtered items

Items: @{activity('FilterRecent').output.value}

STEP 4: Inside ForEach → Copy Activity

Source file: @{item().name}

ALTERNATIVE (filter by file size):

Condition: @{greater(item().size, 1024)} ← only files > 1KB

ALTERNATIVE (filter by file prefix):

```
Condition: @{startsWith(item().name, 'sales_')}
```

ADVANCED (multiple conditions):

```
@{and(  
  startsWith(item().name, 'sales_'),  
  endsWith(item().name, '.csv'),  
  greater(item().size, 0)  
)}
```

Exercise 15: Delete Files from Blob with More Than 100KB

SCENARIO: Cleanup job – delete large files (>100KB) from archive folder

STEP 1: GetMetadata on archive folder

Field: childItems

STEP 2: Filter Activity

Items: @{activity('GetMeta').output.childItems}

Condition: @{greater(item().size, 102400)}

→ 100KB = 102400 bytes

STEP 3: Set Variable to track count

v_deleteCount = @{activity('FilterLarge').output.filterCount}

STEP 4: If Condition

Condition:

@{greater(activity('FilterLarge').output.filterCount, 0)}

→ Only run delete if there are files to delete

STEP 5: TRUE branch → ForEach over large files

Inside ForEach:

Delete Activity:

Dataset: DS_ArchiveBlob

→ Use @{item().name} to reference each file

STEP 6: After ForEach → Stored Procedure

Log: "Deleted X files > 100KB on date Y"

MONITORING OUTPUT:

FilterLarge: filterCount: 5 (found 5 files > 100KB)

ForEach: 5 iterations

Each Delete: 1 file deleted

Final log: "Deleted 5 files > 100KB on 2024-06-15"

Exercise 16: How to Use GetMetadata Activity

COMPLETE REFERENCE: All GetMetadata fields explained

Dataset: pointing to ADLS Gen2 container/folder/file

AVAILABLE FIELDS TO REQUEST:

For FILES:

Field	Output	Use Case
exists	true/false	Check if file arrived
itemName	"sales.csv"	Get actual filename
itemType	"File"/"Folder"	Distinguish file vs folder
size	1048576	Check file has data (>0)
lastModified	"2024-06-15..."	Check if fresh file
contentMD5	"abc123..."	Verify file integrity

structure	[{col schema}]	Get column schema dynamically
columnCount	15	Count columns

For FOLDERS:

childItems	[{name, type}]	List all files/subfolders
itemCount	42	Count of items in folder

REFERENCING OUTPUT:

```
@{activity('GetMeta').output.exists}  
@{activity('GetMeta').output.size}  
@{activity('GetMeta').output.itemName}  
@{activity('GetMeta').output.lastModified}  
@{activity('GetMeta').output.childItems}
```

COMMON PIPELINE PATTERN:

GetMetadata

↓

If Condition: @{equals(activity('GetMeta').output.exists, true)}

↓ TRUE

↓ FALSE

CopyData

SendAlert + Fail Pipeline

↓

If Condition: @{greater(activity('GetMeta').output.size, 0)}

↓ TRUE

↓ FALSE

ProcessData

Fail "File is empty"

Exercise 17: Bulk Copy Tables and Files

SCENARIO: Copy ALL tables from on-premise SQL Server to ADLS Gen2
(Dynamic, metadata-driven approach)

ARCHITECTURE:

Control Table → Lookup → ForEach → Copy Each Table

STEP 1: Create control table in Azure SQL DB

```
CREATE TABLE dbo.CopyControlTable (  
    id          INT IDENTITY(1,1),  
    source_schema VARCHAR(50),  
    source_table  VARCHAR(100),  
    target_folder VARCHAR(200),  
    target_format VARCHAR(20), -- parquet/csv/json  
    is_active     BIT,  
    watermark_col VARCHAR(50), -- for incremental  
    last_watermark DATETIME  
);  
  
INSERT INTO dbo.CopyControlTable VALUES  
(  
    'dbo', 'Customers', 'silver/customers/', 'parquet', 1,  
    'UpdatedDate', '2000-01-01'),  
(  
    'dbo', 'Orders', 'silver/orders/', 'parquet', 1,  
    'OrderDate', '2000-01-01'),  
(  
    'dbo', 'Products', 'silver/products/', 'parquet', 1,  
    'ModifiedDate', '2000-01-01'),  
(  
    'dbo', 'Inventory', 'silver/inventory/', 'parquet', 1, NULL,  
    NULL),  
(  
    'dbo', 'ArchivedData', 'archive/', 'csv', 0, NULL,  
    NULL);
```

STEP 2: Lookup Activity

Query: SELECT * FROM dbo.CopyControlTable WHERE is_active = 1

Returns 4 active tables

STEP 3: ForEach

Items: @{activity('LookupTables').output.value}

Batch Count: 3 (copy 3 tables in parallel)

STEP 4: Inside ForEach → Copy Activity with parameterized datasets

Source Dataset: DS_SQL_Dynamic_Param

p_schema: @{item().source_schema}

p_table: @{item().source_table}

Source Query (for incremental):

```
@{if(
    equals(item().watermark_col, ''),
    concat('SELECT * FROM ', item().source_schema, '.',
item().source_table),
    concat('SELECT * FROM ', item().source_schema, '.',
item().source_table,
        ' WHERE ', item().watermark_col, ' > ',
item().last_watermark, ''))
    )}
```

Sink Dataset: DS_ADLS_Parquet_Dynamic

p_folder: @{item().target_folder}

p_file: @{concat(item().source_table, '_',
formatDateTime(utcnow(), 'yyyyMMdd'),
'parquet')}

STEP 5: After ForEach → Update watermark

Stored Procedure: Update last_watermark to current timestamp

For each table that was loaded

BENEFIT:

- Add new table to control table → automatically copied next run
- Disable table → set is_active=0 → skipped automatically
- 100% metadata-driven, zero code changes for new tables

Exercise 18: How to Integrate Key Vault in ADF

SCENARIO: Store SQL Server password in Key Vault, use in ADF Linked Service

STEP 1: Create Key Vault secret

Azure Portal → Key Vault → Secrets → Generate/Import

Name: sqlldb-password

Value: MySecureP@ssw0rd123

Content Type: text/plain

STEP 2: Grant ADF access to Key Vault

Key Vault → Access Policies → + Add Access Policy

Secret Permissions: GET, LIST

Principal: (select your ADF instance name)

→ ADF uses its Managed Identity to authenticate

STEP 3: Create Key Vault Linked Service in ADF

Manage → Linked Services → + New

Type: Azure Key Vault

Name: LS_AzureKeyVault

Azure Subscription: (select)

Azure Key Vault Name: (select your vault)

→ Test Connection (uses Managed Identity)

STEP 4: Use Key Vault secret in SQL Linked Service

Create/Edit Linked Service LS_AzureSQLDB:

Password field:

→ Click "Azure Key Vault" option (not manual entry)

→ AKV Linked Service: LS_AzureKeyVault

→ Secret Name: sqlldb-password

→ Secret Version: (blank = latest version)

STEP 5: Verify

→ Test Connection on LS_AzureSQLDB

- ADF fetches password **from** Key Vault at runtime
- Password NEVER stored **in** ADF – maximum security

BENEFIT:

Old Way: Password hardcoded **in** Linked Service → visible **in** ARM template

New Way: Only secret name stored → actual password **in** Key Vault
If password changes → update **Key** Vault only
ADF automatically gets **new** password → zero downtime

Exercise 19: How to Set Up Integration Runtime

EXERCISE A: Create Self-Hosted IR

STEP 1: ADF Studio → Manage → Integration Runtimes → + **New**

Type: Self-Hosted

Name: SHIR_OnPremise_Production

→ Create → Copy Authentication **Key 1**

STEP 2: **On** your **on**-premise server/VM:

- Download Microsoft Integration Runtime **from** official link
- Install (Windows only)
- After install → paste Authentication **Key**
- Click Register
- Status changes **to "Running"** **in** ADF Studio

STEP 3: Configure **for** High Availability (**optional** but recommended):

- Install SHIR **on** **2nd** server
- Paste same Authentication **Key**
- Now you have **2** nodes:
 - Node **1**: Server01 - Running

Node 2: Server02 - Running

→ If Node 1 goes down, Node 2 takes over automatically

STEP 4: Use SHIR in Linked Service:

LS_OnPremSQL:

Type: SQL Server

Server: 192.168.1.100 (internal IP)

Connect via IR: SHIR_OnPremise_Production

STEP 5: Test connection

→ ADF routes test through SHIR to on-prem server

→ "Connection successful" confirms tunnel is working

EXERCISE B: Manage Existing IR

→ Monitor IR status: Running / Limited / Offline

→ Check node details: CPU, Memory, concurrent jobs

→ View IR logs for troubleshooting

→ Update IR (auto-update or manual update option)

Exercise 20: Copy Data from On-Premises to Azure Cloud

FULL END-TO-END SCENARIO:

On-Prem SQL Server → (SHIR) → ADLS Gen2

PREREQUISITES:

- ✓ SHIR installed and connected (Exercise 19)
- ✓ On-prem SQL Server accessible from SHIR server
- ✓ ADLS Gen2 storage account created

STEP 1: Linked Service for On-Prem SQL Server

Type: SQL Server

Server: CORP-SQLSERVER-01 (or IP)
Database: LegacySalesDB
Auth: Windows Authentication (or SQL auth)
Connect via IR: SHIR_OnPremise_Production

STEP 2: Linked Service for ADLS Gen2

Type: Azure Data Lake Storage Gen2
URL: https://mydatalake.dfs.core.windows.net
Auth: Managed Identity (ADF managed identity)

STEP 3: Datasets

DS_OnPrem_SalesTable:

Linked Service: LS_OnPremSQL
Table: [dbo].[LegacySales]

DS_ADLS_SalesParquet:

Linked Service: LS_ADLS_Gen2
Container: raw
Path: legacy/sales/@{formatDateTime(utcnow(), 'yyyy/MM/dd')}/
Format: Parquet

STEP 4: Copy Activity

Source: DS_OnPrem_SalesTable

Partition option: Dynamic Range
Partition Column: SaleID (numeric PK)
Upper Bound: (from Lookup: SELECT MAX(SaleID))
Lower Bound: 1
Degree of Copy Parallelism: 4
→ Reads table in 4 parallel chunks = 4x faster!

Sink: DS_ADLS_SalesParquet

STEP 5: Monitor

→ Watch activity run in Monitor tab
→ Data routes: On-prem SQL → SHIR (compression/encryption) →

ADF → ADLS

- All data encrypted in transit (HTTPS)
- SHIR does NOT store data – it's just a secure relay

PERFORMANCE TIPS:

- Run SHIR on server closest to source DB (same network)
- Increase SHIR server CPU/RAM for large loads
- Use partition copy for tables > 1 million rows

Exercise 21: Use Databricks Activity and Pass Parameters

STEP 1: Create Databricks Linked Service

ADF Manage → Linked Services → + New

Type: Azure Databricks

Name: LS_AzureDatabricks_Prod

Databricks Workspace: (select from subscription)

Select cluster:

Option A: Existing interactive cluster → select cluster ID

Option B: New job cluster → configure cluster spec:

Cluster version: 13.3 LTS (Spark 3.4)

Node type: Standard_DS3_v2

Driver type: Same as worker

Workers: 2 (min) - 8 (max) autoscaling

Authentication: Managed Service Identity (MSI)

STEP 2: Add Databricks Notebook Activity to pipeline

General: Name = RunSalesTransformation

Azure Databricks:

Databricks Linked Service: LS_AzureDatabricks_Prod

Settings:

Notebook path: /Production/Sales/Transform_Sales_Data

Base Parameters:

input_path: @{{concat('/mnt/raw/sales/',

formatDateTime(utcnow(), 'yyyy/MM/dd'))}}

output_path: @{{concat('/mnt/silver/sales/',

formatDateTime(utcnow(), 'yyyy/MM/dd'))}}

process_date: @{{formatDateTime(utcnow(), 'yyyy-MM-dd')}}}

env: @{{pipeline().globalParameters.gp_environment}}

batch_id: @{{pipeline().RunId}}

STEP 3: In Databricks Notebook, receive parameters

Python notebook

dbutils.widgets.text("input_path", "/mnt/raw/sales/default")

dbutils.widgets.text("output_path",

"/mnt/silver/sales/default")

dbutils.widgets.text("process_date", "2024-01-01")

dbutils.widgets.text("env", "dev")

input_path = dbutils.widgets.get("input_path")

output_path = dbutils.widgets.get("output_path")

process_date = dbutils.widgets.get("process_date")

print(f"Processing: {input_path}")

print(f"Output to: {output_path}")

print(f>Date: {process_date}")

Your actual transformation code here

df = spark.read.parquet(input_path)

... transformations ...

df.write.parquet(output_path)

Return value back to ADF


```
dbutils.notebook.exit("SUCCESS: " + str(df.count()) + " rows processed")
```

STEP 4: Read Databricks output in ADF

Next activity can reference:

```
@{activity('RunSalesTransformation').output.runOutput}
```

→ Returns: "SUCCESS: 142850 rows processed"

Exercise 22: How to Use Scheduling Trigger

STEP 1: ADF Manage → Triggers → + New

Type: Schedule

Name: TRG_Daily_SalesLoad

STEP 2: Configure

Start Date/Time: 2024-01-01 00:00:00 UTC

Time Zone: India Standard Time

Recurrence: Every 1 Day

At: 11:30 PM (so data is ready by midnight)

End: Never

STEP 3: Attach to Pipeline

→ Click pipeline dropdown

→ Select: PL_Daily_SalesETL

→ Set pipeline parameters:

```
p_loadDate: @{formatDateTime(trigger().scheduledTime, 'yyyy-MM-dd')}
```

STEP 4: Publish + Activate

→ Publish All

→ Go to Trigger → Start

STEP 5: Verify

Monitor → Trigger Runs

→ Shows all trigger invocations with status

COMMON SCHEDULE PATTERNS:

Pattern	Config
Every 15 minutes	: Recur every 15 Minutes
Every hour	: Recur every 1 Hour
Daily at midnight	: Recur every 1 Day at 00:00
Weekdays only	: Recur weekly, Mon-Fri at 08:00
First of month	: Recur monthly on day 1
Every 6 hours	: Recur every 6 Hours

Exercise 23: Tumbling Window Trigger

STEP 1: New Trigger

Type: Tumbling Window

Name: TRG_Hourly_SalesWindow

STEP 2: Configure

Start: 2024-01-01 00:00:00 UTC

Period: 1 Hour

Offset: 0 (no delay)

Max Concurrency: 4 ← 4 windows can run in parallel

Retry: 3

Retry Interval: 5 min

STEP 3: Pipeline parameter mapping

p_windowStart: @{triggerBody().window.windowStart}

```
p_windowEnd:    @{{triggerBody().window.windowEnd}}
```

In pipeline source query:

```
SELECT * FROM Sales
WHERE sale_datetime >= '@{{pipeline().parameters.p_windowStart}}'
      AND sale_datetime <  '@{{pipeline().parameters.p_windowEnd}}'
```

STEP 4: Backfill scenario (reprocess past data)

- Go to Trigger in Manage
- Click "Trigger Now" (custom start)
- Set: Start = 2024-01-01, End = today
- ADF creates ALL windows from Jan 1 to today
- Runs 4 at a time (max concurrency)
- Backfills entire history automatically!

Exercise 24: Event-Based Trigger

STEP 1: New Trigger

Type: Storage Event

Name: TRG_Event_SalesFileArrival

STEP 2: Configure

Storage Account: mydatalakestorage

Container: raw

Blob Path Begins With: /sales/incoming/

Blob Path Ends With: .csv

Event: Microsoft.Storage.BlobCreated

- Fires when new CSV file created in that path

Ignore Empty Blobs: Yes

→ Don't trigger for 0-byte files

STEP 3: Pipeline parameter mapping

```
p_fileName:    @{{triggerBody().fileName}}
p_folderPath:  @{{triggerBody().folderPath}}
```

Use in Copy Activity source:

```
File Path: @{{concat(pipeline().parameters.p_folderPath,
                      pipeline().parameters.p_fileName)}}
```

STEP 4: Activate trigger → Publish

STEP 5: Test

- Upload a CSV file to /raw/sales/incoming/
- Within 30-60 seconds, pipeline auto-starts!
- Monitor → Pipeline Runs → shows new run

USE CASE PATTERN (event-driven architecture):

Upstream system → drops file to ADLS landing zone

↓ (event trigger fires)

ADF validates + ingests file

↓

Databricks transforms

↓

Synapse loads (available for reports)

Total latency: 3-5 minutes from file drop to report availability!

Exercise 25: Script Activity (With Activity)

STEP 1: Add Script Activity to pipeline

Linked Service: LS_AzureSQLDB

STEP 2: Configure script (multiple SQL statements):

Script Type: NonQuery

Script Content:

```
-- Step 1: Truncate staging
TRUNCATE TABLE dbo.STG_Sales;

-- Step 2: Create temp index for performance
IF NOT EXISTS (SELECT 1 FROM sys.indexes WHERE
name='IX_STG_CustID')
    CREATE INDEX IX_STG_CustID ON dbo.STG_Sales(customer_id);

-- Step 3: Log pipeline start
INSERT INTO dbo.PipelineAudit
    (run_id, pipeline_name, start_time, status)
VALUES (
    '@{pipeline().RunId}',
    '@{pipeline().Pipeline}',
    GETDATE(),
    'RUNNING'
);
```

STEP 3: Script with Query type (returns data)

Script Type: Query

Script: SELECT COUNT(*) AS row_count FROM dbo.Sales

Output used in next activity:

```
@{activity('ScriptQuery').output.resultSets[0].rows[0].row_count}
```

Exercise 26: Until Activity

SCENARIO: Poll external API every 30 seconds until job is DONE

STEP 1: Set Variable

```
v_jobStatus = "RUNNING"
v_pollCount = 0
```

STEP 2: Until Activity

Condition: @equals(variables('v_jobStatus'), 'COMPLETED')}

Timeout: 02:00:00 (max 2 hours)

Delay: 00:00:30 (wait 30 seconds between each check)

Inside Until:

Activity A: Web Activity (call status API)

URL:

[https://api.myservice.com/jobs/@{variables\('v_jobId'\)}/status](https://api.myservice.com/jobs/@{variables('v_jobId')}/status)

Method: GET

Output: {"status": "RUNNING"} or {"status": "COMPLETED"}

Activity B: Set Variable

```
v_jobStatus = @{activity('CheckStatus').output.status}
```

Activity C: Set Variable

```
v_pollCount = @{add(int(variables('v_pollCount')), 1)}
```

Activity D: If Condition (safety check)

Condition: @greater(variables('v_pollCount'), 240)

TRUE: Fail Activity "Job timed out after 2 hours"

FLOW:

Start → v_jobStatus = "RUNNING"

Poll 1 (0:30): status = RUNNING → continue loop

Poll 2 (1:00): status = RUNNING → continue loop

Poll 3 (1:30): status = RUNNING → continue loop

Poll 4 (2:00): status = COMPLETED → exit loop!

Continue to next activities...

Exercises 27–35: Data Flow Exercises

EXERCISE 27: Dataflows – Select Rows

- Create Data Flow DF_SelectColumns
- Source: Raw CSV (15 columns)
- Select transformation: pick only 8 needed columns
- Rename legacy column names to modern standard
- Sink: ADLS Parquet (8 columns only)

EXERCISE 28: Dataflows – Filter Rows

- Source: Sales data (all regions, all dates)
- Filter: region == 'INDIA' && sale_amount > 1000
- Sink: india_high_value_sales.parquet
- Use Data Preview to verify count reduced

EXERCISE 29: Dataflows – Join Transformations

- Source 1: Sales transactions
- Source 2: Customer master table
- Join: sales.customer_id == customer.id (Inner Join)
- Select: pick columns from both streams
- Sink: enriched_sales.parquet
- Practice: Inner, Left Outer, Right Outer, Full Outer, Cross

EXERCISE 30: Dataflows – Union Transformations

- Source 1: Sales_Q1 (Jan-Mar data)
- Source 2: Sales_Q2 (Apr-Jun data)
- Source 3: Sales_Q3 (Jul-Sep data)
- Union All three streams
- Sink: Sales_Q1_to_Q3_combined.parquet
- Verify: output rows = Q1 + Q2 + Q3 row counts

EXERCISE 31: Dataflows – Lookup Transformations

- Primary: Orders dataset
- Lookup: Product master (product_code → product_name, category, price)
- Lookup Type: Any (first match)
- Output: Orders enriched with product details
- Handle no-match rows: isNull(lookup.product_name) → 'UNKNOWN'

EXERCISE 32: Dataflows – Window Functions

- Source: Sales by product and month
- Window transformation:
 - PARTITION BY: product_category
 - ORDER BY: sale_month DESC
 - rank_in_category = rank()
 - cumulative_sales = cumSum(sale_amount)
 - prev_month_sales = lag(sale_amount, 1)
 - rolling_3mo_avg = avg(sale_amount) (window: 3 rows)
- Sink: product_sales_ranked.parquet

EXERCISE 33: Dataflows – Pivot and Unpivot

- PIVOT Exercise:
- Source: sales by region/product (long format)
 - Pivot:
 - Group by: region
 - Pivot col: product_category
 - Values: sum(sale_amount)

→ Output: wide table with one column per product category

UNPIVOT Exercise:

→ Source: Wide table with Electronics/Clothing/Food columns

→ Unpivot:

Unpivot columns: Electronics, Clothing, Food

Name column: product_category

Value column: sale_amount

→ Output: Long format with product_category + sale_amount rows

EXERCISE 34: Dataflows – Alter Row Transformations

→ Source: Customer updates file

→ Lookup: Current customer master

→ Alter Row rules:

Insert If: isNull(target.customer_id) ← new customer

Update If: !isNull(target.customer_id) &&

equalsIgnoreCase(source.status, 'ACTIVE')

Delete If: equalsIgnoreCase(source.status, 'DELETED')

Upsert If: false ← not using upsert

→ Sink: Customer master table (with allow insert, update, delete)

EXERCISE 35: Dataflows – Removing Duplicates

APPROACH 1: Using Window + Filter:

→ Window:

PARTITION BY: customer_id, order_date, product_code

ORDER BY: created_timestamp DESC

row_num = rowNum()

→ Filter: row_num == 1

→ Result: Only most recent record per unique combination

APPROACH 2: Using Aggregate:

→ Aggregate:

GROUP BY: customer_id, order_date, product_code (all key cols)

```
max_created = max(created_timestamp)
max_amount  = first(order_amount)
→ Result: Deduplicated rows
```

Exercises 36–47: Advanced Pipeline Exercises

EXERCISE 36: Pass Parameters to Pipeline

- Create pipeline with 3 parameters: p_date, p_env, p_table
- Trigger manually with "Trigger Now"
- Fill in parameter values in the popup
- Reference in activities: @{pipeline().parameters.p_date}
- Also: pass from parent pipeline to child via Execute Pipeline

EXERCISE 37: Create Alerts and Rules

- Azure Portal → ADF → Diagnostic Settings
- Enable: PipelineRuns, ActivityRuns, TriggerRuns → Log Analytics
- Create Alert:
 - Condition: Failed pipeline runs > 0
 - Evaluation: Every 5 minutes
 - Threshold: 0 (alert on any failure)
- Action Group:
 - Email: team@company.com
 - SMS: +919876543210
- Test: Run a pipeline that intentionally fails → check email

EXERCISE 38: Set Global Parameters

- ADF Studio → Manage → Global Parameters
- + New parameters:
 - gp_environment: Production
 - gp_dataLakeURL: https://myprodlake.dfs.core.windows.net
 - gp_notifyEmail: dataengg@company.com

gp_maxRetries: 3

→ Publish

→ Use in pipeline:

@{pipeline().globalParameters.gp_environment}

→ When deploying to different environments:

ARM template parameter file overrides global parameter values

EXERCISE 39: Import and Export ARM Templates

EXPORT:

→ ADF Studio → Author → ARM Template → Export ARM Template

→ Downloads zip with:

ARMTemplateForFactory.json (all resources)

ARMTemplateParametersForFactory.json (parameter values)

REVIEW the JSON:

→ Open ARMTemplateForFactory.json

→ See all pipelines, datasets, linked services as JSON

→ This IS your ADF configuration as code

IMPORT (deploy to new ADF):

→ Azure Portal → Create Resource → Template Deployment

→ Upload ARMTemplateForFactory.json

→ Edit parameters for target environment

→ Deploy → all resources created in new ADF

EXERCISE 40: Integrate ADF with DevOps

→ ADF Studio → Author → Git Repository → Configure

→ Repository Type: Azure DevOps Git

→ Organization: your-org

→ Project: DataEngineering

→ Repository: ADF-Pipelines

→ Collaboration Branch: main

→ Root Folder: /

→ Apply → ADF now saves to Git!

→ Create new pipeline → Edit → Commit → raise PR → merge →

publish

EXERCISE 41: Use Azure DevOps Repos

- After Git config, every ADF change is tracked:
- Create feature branch: feature/add-orders-pipeline
- Build pipeline in ADF using that branch
- Commit: "Add PL_Orders_Daily pipeline"
- Raise PR to main in Azure DevOps
- Review JSON diff (see exactly what changed)
- Approve + Merge → switch to main in ADF → Publish

EXERCISE 42: Send Mail via Logic Apps

- Logic Apps designer:

Trigger: HTTP Request received

Step 1: Parse JSON (extract pipeline details)

Step 2: Send Email (Office 365 connector)

To: @{{body('Parse_JSON')}['notify_email']}

Subject: @{{concat('ADF Pipeline: ', body('Parse_JSON')}['status'])}}

Body: HTML formatted email with pipeline details

- Copy Logic App HTTP endpoint URL

- In ADF pipeline failure path:

Web Activity:

URL: (Logic App URL)

Method: POST

Body: {

"pipeline_name": "@{{pipeline().Pipeline}}",

"status": "FAILED",

"error": "@{{activity('FailedActivity').Error.message}}",

"notify_email":

"@{{pipeline().globalParameters.gp_notifyEmail}}"

}

EXERCISE 43: Monitor Pipelines

- ADF Studio → Monitor tab
- **Pipeline Runs**: see all runs, filter by date/status/name
- Click any run → Activity Runs (drill into each step)
- Click any activity → Input/Output/Error details
- **Key metrics to check**:
 - dataRead**: how much data read from source
 - dataWritten**: how much written to sink
 - throughput**: MB/s transfer rate
 - rowsCopied**: exact row count
 - duration**: how long each step took

EXERCISE 44: Debug Pipelines

- **Debug mode**: click Debug button (no trigger needed)
- **With parameters**: Debug → fill parameter values
- **Debug single activity**: right-click → Debug Activity Run
- Set breakpoints (not available but use If Condition to pause)
- Use User Properties for custom debug info
- Check Output tab after each activity for details

EXERCISE 45: Schedule Pipeline Using Triggers

- Create Schedule trigger → set recurrence
- Attach to pipeline with parameter mapping
- Activate trigger
- **Monitor**: Trigger Runs tab shows each firing
- **Disable trigger**: go to Manage → Triggers → Deactivate

EXERCISE 46: Create Trigger Dependency (Tumbling Window)

- **Create Trigger A**: TRG_Extract_Hourly (Tumbling, 1hr)
- **Create Trigger B**: TRG_Transform_Hourly (Tumbling, 1hr)
- **In Trigger B configuration**:
 - Dependencies tab → + Add Dependency
 - Trigger**: TRG_Extract_Hourly
 - Size**: 1 Hour
 - Offset**: 0
- Activate both triggers

→ **Result:** TRG_Transform only fires after TRG_Extract succeeds for the same time window

EXERCISE 47: Run One Pipeline Inside Another (Execute Pipeline)

→ Create child pipeline PL_Child_ProcessSales

Parameters: p_date (String)

→ Copy Data + Transform activities

→ Create parent pipeline PL_Parent_Orchestrator

→ Add Execute Pipeline activity:

Invoked Pipeline: PL_Child_ProcessSales

Wait on Completion: YES

Parameters:

p_date: @{formatDateTime(utcnow(),'yyyy-MM-dd')}

→ Parent pipeline controls flow:

Activity 1: Execute PL_Child_ProcessSales (Sales)

Activity 2: Execute PL_Child_ProcessOrders (Orders)

Activity 3: Execute PL_Child_ProcessInventory (Inventory)

Activities 1,2,3 run in PARALLEL (no dependency arrows between them)

Complete ADF Section — Final Summary Map

AZURE DATA FACTORY — COMPLETE GUIDE

CORE CONCEPTS	Pipelines, Activities, Linked Services,
	Datasets, Integration Runtime

ACTIVITIES	Copy, Lookup, SP, GetMetadata, Delete,
(ALL 15 types)	Wait, Variables, Execute Pipeline,
	Databricks, Azure Function, Script,
	Validation, Fail, ForEach, If, Switch,
	Until, Filter, Web Activity
TRIGGERS	Schedule, Tumbling Window, Storage Event,
(3 types)	Parameters, Trigger Dependency
INTEGRATION RT	Azure AutoResolve, Managed VNet,
(4 types)	Self-Hosted, Linked Self-Hosted
DATA FLOWS	Source, Select, Filter, Aggregate, Join,
(ALL transforms)	Conditional Split, Derived Column,
Exists,	Union, Lookup, Sort, GroupBy, Pivot,
	Unpivot, Flatten, Parse, Stringify,
	Alter Row, Assert, Window, Flowlet,
	Schema Drift, Schema Validate, Dedup

```
|| DEVOPS & MGMT      | Git, ARM Templates, Global Params,  
||  
||                   | Credentials, Key Vault, Alerts, Logic  
Apps ||
```

```
|| EXERCISES          | All 47 exercises covered step-by-step  
||
```

```
|| INTERVIEW Q&A      | 15 most important Q&As with detailed  
||  
||                   | answers covering all exam/interview  
topics ||
```

✅ **Azure Data Factory section is now 100% complete** — all concepts from the PDF including all missing transformations (Select, Window Functions, Assert), Trigger Dependency, Script/With Activity, all 47 exercises in depth, and complete interview Q&A are now covered.

🚀 **Ready to continue with Azure Databricks** — the next major section in the PDF!

17/03/2026, 21:37:09

Thought process: Orchestrated comprehensive Databricks curriculum spanning foundational concepts through advanced implementations.

The user wants me to continue with Azure Databricks section from the PDF. Let me review all the Databricks content from the PDF carefully:

- 1) Introduction to BigData
- 2) Introduction to Hadoop
- 3) Introduction to Spark
- 4) Introduction To Databricks

- 5) Working with Databricks FileSystem - DBFS
- 6) Databricks - Spark Core
- 7) Databricks - Spark-SQL- DataFrames
- 8) Spark Optimizer & Execution Engine
- 9) Spark Joins
- 10) Spark Performance Optimization
- 11) Databricks- Handle multiple file formats
- 12) Databricks utilities
- 13) Databricks Cluster Management

- 14) Databricks - Batch Processing
- 15) Databricks Integration with
- 16) Databricks - Streaming API
- 17) Databricks Auto Loader (CloudFiles)
- 18) Databricks - Lakehouse (Delta Lake)
- 19) Delta Lake: Medallion Architecture
- 20) Workflows in Databricks
- 21) Introduction to Unity Catalog
- 22) Unity Catalog Object Model & Hierarchy
- 23) Access Control & Security (RBAC)
- 24) Data Protection & Privacy
- 25) Azure DevOps - Repos
- 26) SDLC and Agile methodology
- 27) End to End Data Migration Project from On Premises to Cloud
- 28) Interview
- 29) Mock Interviews

Now I'm putting together a comprehensive guide that covers all these Databricks topics with practical examples and detailed explanations, structured like something a senior engineer would reference.

Azure Databricks — Complete Senior Engineer Guide



Real World Setup

Same Company: RetailMart India Pvt. Ltd.

SITUATION:

ADF has ingested raw data into Bronze layer (ADLS Gen2)

Now you need to:

- Clean and transform 500GB daily data
- Process 50,000 transactions per day
- Run complex business logic (joins, aggregations)
- Handle real-time fraud detection
- Build ML models for demand forecasting
- Maintain historical data with Delta Lake

Problem with ADF Data Flows alone:

- ✗ Too slow for 500GB daily
- ✗ Cannot do complex ML
- ✗ No streaming support
- ✗ Limited Python/custom logic

Solution: Azure Databricks with Apache Spark

1 Introduction to Big Data — Deep Foundation

◆ What is Data?

Data is **raw facts and figures** that alone carry no meaning.

Raw Data: 42, "Mumbai", 2024-06-15, 15000

With Context: Store #42 in Mumbai on 15-Jun-2024 sold
Rs.15,000

Data becomes information only when processed with context.

◆ What is a Database?

A **Database** is an organized collection of structured data stored and accessed electronically.

Traditional Database (RDBMS):

MySQL, SQL Server, Oracle, PostgreSQL

How it works:

- Data stored in TABLES (rows + columns)
- Single server stores everything
- SQL to query data
- ACID transactions (Atomic, Consistent, Isolated, Durable)

Example – RetailMart uses Oracle DB:

Table: SALES

sale_id	store_id	amount	sale_date
1001	42	15000	2024-06-15
1002	43	8500	2024-06-15

◆ What is Big Data?

Big Data = datasets so large, fast, or complex that **traditional databases cannot handle them**.

RetailMart's *Big Data Reality*:

Daily transactions:	50,000 rows	← MySQL handles this fine
Daily clickstream:	50,000,000 events	← MySQL struggles
Daily IoT sensor data:	500,000,000 readings	← MySQL completely fails
Historical data (5yr):	2.5 Petabytes	← MySQL impossible

When MySQL tries to handle 500GB daily:

Query that takes 2 seconds on 1GB → takes 1000 seconds on 500GB
Single server runs out of RAM → crashes
Cannot add more servers easily (not distributed)

◆ Challenges of Big Data

CHALLENGE 1: STORAGE

Problem: 2.5 PB doesn't fit on one server
Old Fix: Buy expensive SAN storage (\$500K+)
Big Data Fix: Distribute across 100 cheap servers

CHALLENGE 2: PROCESSING SPEED

Problem: Sorting 1 billion rows takes 4 hours on 1 server
Old Fix: Buy faster, more expensive server
Big Data Fix: 100 servers sort in parallel → 2-3 minutes

CHALLENGE 3: VARIETY

Problem: Oracle handles tables only

What about JSON, images, logs, videos?

Big Data Fix: Distributed storage handles ALL formats

CHALLENGE 4: FAULT TOLERANCE

Problem: 1 server fails → ALL data lost

Big Data Fix: Data replicated across 3 servers

If 1 fails → other 2 serve the data

CHALLENGE 5: COST

Problem: Enterprise Oracle license = \$50,000/year

Oracle server = \$200,000

Big Data Fix: Open source tools on commodity hardware

1000x cheaper per GB stored/processed

◆ Why Traditional Databases Cannot Handle Big Data

TRADITIONAL DB LIMITATIONS:

Limitation	Traditional DB	Big Data System
Scale	Vertical only (bigger server)	Horizontal (add nodes) (more servers)
Data Types	Structured only	All types (JSON, CSV, images, logs, video)
Max Storage	~100TB practical	Petabytes (unlimited)
Processing	Single node	Distributed parallel

Cost at scale	Exponential	Linear (add cheap nodes)
Schema	Fixed (must define before insert)	Schema- on -read (define when reading)
Fault Tolerance	RAID (expensive)	Replication (built- in , free)

Real Example:

RetailMart query: "Show sales by store for last 5 years"

Oracle (single server, **500GB** data):

- **Full table** scan: **45** minutes
- Server CPU: **100%** during query
- Other queries blocked → business operations halt

Databricks (**10**-node Spark cluster, same **500GB**):

- Data split **into 200** partitions
- **Each of 10** nodes processes **20** partitions **in** parallel
- Query completes: **3** minutes
- Other queries unaffected (cluster scales up)

2 Introduction to Hadoop — The Foundation

◆ What is Hadoop?

Hadoop is an open-source framework for **distributed storage and processing** of large datasets across clusters of computers.

Created **by**: Doug Cutting (Yahoo!) inspired **by** Google's *MapReduce paper*

Year: 2006
Language: Java
Foundation: Apache Software Foundation

Think of Hadoop as: A team of 100 workers
→ Each worker has their own desk (storage)
→ Each worker does part of the job (processing)
→ A manager coordinates (YARN)
→ Results combined at the end

◆ How Hadoop Overcomes Big Data Challenges

CHALLENGE → HADOOP SOLUTION:

Storage challenge → HDFS (Hadoop Distributed File System)

Split 500GB file into 128MB blocks
Store each block on different servers
Replicate each block 3 times (fault tolerance)

Processing challenge → MapReduce

Send processing code TO the data (not data to code!)
Each node processes its local blocks in parallel
Combine results at the end

Fault tolerance → Block replication

Node fails → use replica on another node
Automatic → no manual intervention

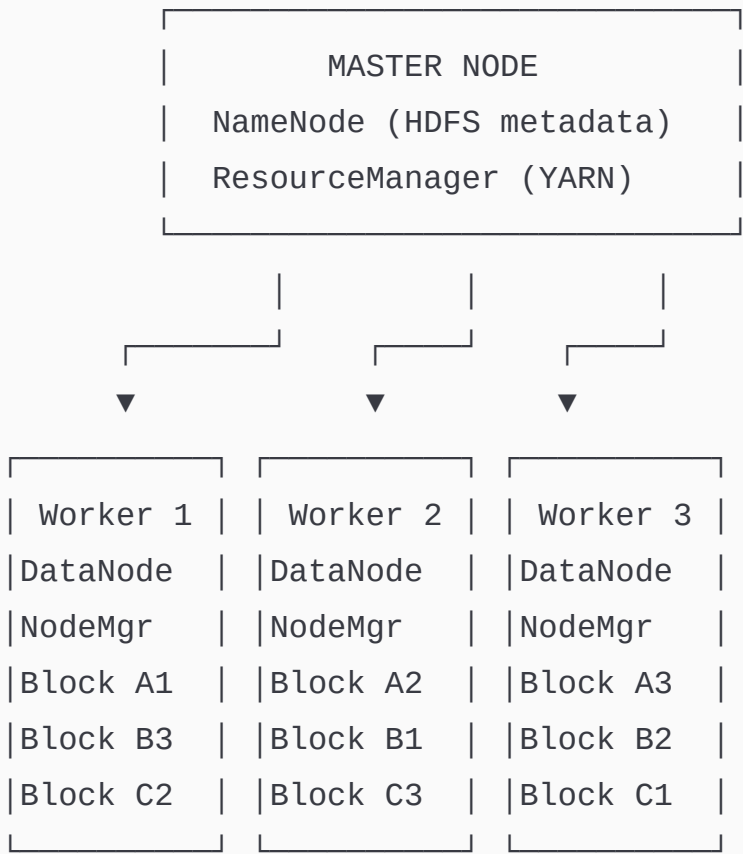
Cost challenge → Commodity hardware

Hadoop runs on cheap servers (\$3,000 each)

100 cheap servers > 1 expensive server
Same performance, 10x cheaper

◆ Hadoop Architecture

HADOOP CLUSTER:



◆ Hadoop Components in Detail

HDFS (Hadoop Distributed File System)

HOW HDFS STORES A 500GB FILE:

1. File split into 128MB blocks:

sales_data.csv (500GB)

→ Block 1 (128MB): rows 1 - 1,000,000

→ Block 2 (128MB): rows 1,000,001 - 2,000,000

→ ...

→ Block 4000 (128MB): rows 3,999,000,001 - 4,000,000,000

2. Each block replicated 3x across different nodes:

Block 1 → Node1, Node4, Node7 (replication factor=3)

Block 2 → Node2, Node5, Node8

Block 3 → Node3, Node6, Node9

3. NameNode keeps metadata:

"sales_data.csv → Block1(N1,N4,N7), Block2(N2,N5,N8)..."

4. If Node1 dies:

Block 1 still available on Node4 and Node7

NameNode detects 1 copy lost

Automatically re-replicates Block1 to Node10

File NEVER lost!

KEY INSIGHT:

"Move computation to data" principle

Instead of: copy 500GB file to processing server

Hadoop does: send tiny processing code to each node

→ Saves massive network bandwidth

→ 100x faster for large datasets

YARN (Yet Another Resource Negotiator)

YARN = The Operating System of Hadoop Cluster

Components:

ResourceManager:

- Master process (**one per** cluster)
- Manages **ALL** cluster resources (CPU, RAM)
- Accepts job submissions
- Allocates containers **to** applications

NodeManager:

- Runs **on every** worker node
- Manages resources **on** that node
- Reports health **to** ResourceManager
- Launches **and** monitors containers

ApplicationMaster:

- **One per** application/job
- Negotiates resources **from** ResourceManager
- Monitors job execution
- Handles failures/retries

Real-World Analogy:

ResourceManager = Airport Control Tower

(knows **all** flights, coordinates **all** landings/takeoffs)

NodeManager = Individual Gate Agent

(manages **one** gate, reports **to** control tower)

ApplicationMaster = Flight Captain

(manages own flight, requests runway **from** tower)

MapReduce

MapReduce = Two-phase distributed processing model

EXAMPLE: Count sales **by** store **from** 500GB file

MAP PHASE (parallel **on all** nodes):

Each node **reads** its **local** blocks

Node1 processes rows 1-1M:

Output: (Store42, 15000), (Store43, 8500), (Store42, 12000)

Node2 processes rows 1M-2M:

Output: (Store42, 9000), (Store44, 7000), (Store43, 5000)

...all 100 nodes run simultaneously

SHUFFLE PHASE (framework handles):

All (Store42, ?) pairs go to same reducer

All (Store43, ?) pairs go to same reducer

REDUCE PHASE:

Reducer for Store42: $\text{sum}(15000, 12000, 9000 \dots) = \text{total}$

Reducer for Store43: $\text{sum}(8500, 5000 \dots) = \text{total}$

...

PROBLEM with MapReduce:

- Writes intermediate results to DISK after each phase
- Iterative algorithms (ML) need 10+ MapReduce jobs
- Each job reads/writes disk → extremely slow for ML
- This is exactly why Spark was created!

3 Introduction to Apache Spark — The Revolution

◆ Why Spark? The Problem with MapReduce

MAPREDUCE vs SPARK for Machine Learning:

Training a recommendation model (10 iterations):

MapReduce:

Iteration 1: Read HDFS → Process → Write HDFS (disk I/O)

Iteration 2: Read HDFS → Process → Write HDFS (disk I/O)
...repeat 10 times
Total disk reads/writes: $20 \times 500\text{GB} = 10\text{TB}$
Time: 8 hours

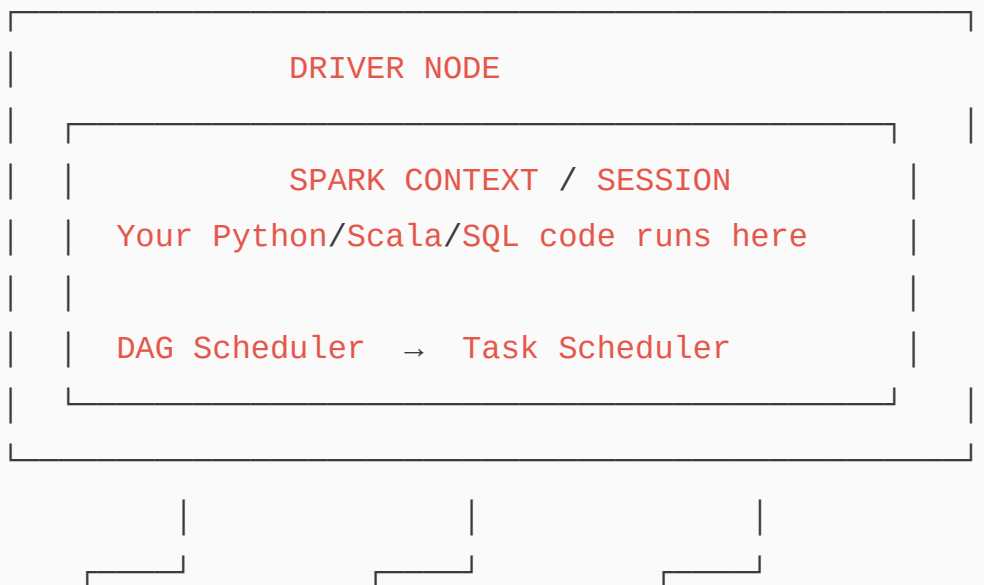
Spark:

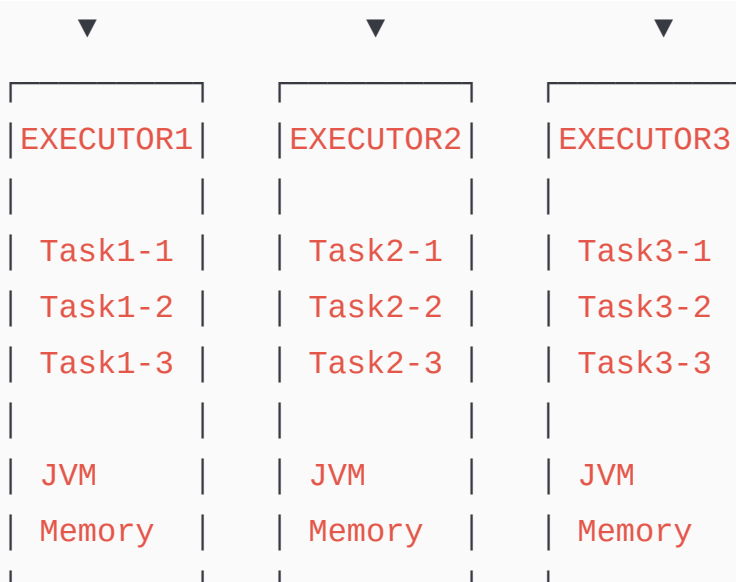
Load data into RAM ONCE
Iteration 1: Process in RAM → Keep in RAM
Iteration 2: Process in RAM → Keep in RAM
...repeat 10 times
Total disk reads/writes: $1 \times 500\text{GB} = 500\text{GB}$
Time: 20 minutes

Spark is 100x faster than MapReduce for iterative algorithms!
This is the fundamental innovation of Spark.

◆ Spark Architecture — Deep Dive

SPARK CLUSTER ARCHITECTURE:





Driver Node — The Brain

Driver **is** **WHERE** your code actually runs.

Responsibilities:

- ✓ Receives your Spark code (Python/Scala/SQL)
- ✓ Analyzes operations (builds Logical Plan)
- ✓ Optimizes execution (Catalyst Optimizer)
- ✓ Creates Physical Plan → Stages → Tasks
- ✓ Distributes Tasks **to** Executors
- ✓ Monitors task progress
- ✓ Collects results back **from** Executors
- ✓ **Handles** failures (re-runs failed tasks)

In Databricks:

Driver = The master node **of** your cluster
Your notebook code runs **ON** the driver

Important Rule:

NEVER collect massive data **to** driver!
`df.collect()` → brings ALL data **to** driver RAM

If data > driver RAM → OutOfMemory crash!
Use `df.show(20)` or `df.write()` instead

Executors — The Workers

Executors are WHERE actual data processing happens.

Each Executor:

- JVM process running on worker node
- Has multiple CPU cores (runs multiple Tasks in parallel)
- Has allocated RAM (for processing + caching)
- Stays alive for entire Spark application duration
- Sends heartbeat to Driver (alive signal)

Example - Databricks cluster:

8 worker nodes × (4 cores, 16GB RAM each)
= 32 total cores = 32 tasks run simultaneously!

Processing 500GB data:

- Split into 200 partitions
- 32 tasks run in parallel (32 partitions at once)
- 200 partitions / 32 parallel = ~7 rounds
- Each round takes 30 seconds
- Total: 7 × 30 = ~3.5 minutes!

◆ Spark RDD — The Foundation

RDD (Resilient Distributed Dataset) is the fundamental data structure of Spark.

RDD PROPERTIES:

R - Resilient:

- Fault tolerant – if a node fails, RDD rebuilt **from** lineage
- Lineage = recipe **of** transformations **to** recreate data

D - Distributed:

- Data split across multiple nodes **in** cluster
- **Each** node holds a **PARTITION of** the RDD

D - Dataset:

- Collection **of** records
- Can **hold ANY** type (integers, strings, objects)
- Immutable – cannot be changed, **only** transformed

IMMUTABILITY explained:

```
rdd1 = [1, 2, 3, 4, 5]           ← original RDD
rdd2 = rdd1.filter(_ > 3)        ← NEW RDD created [4, 5]
rdd1 still exists = [1, 2, 3, 4, 5] ← unchanged!
```

This enables fault tolerance:

If rdd2 **partition** fails → recreate **from** `rdd1.filter(_ > 3)`
No data loss ever!

Creating RDDs:

```
# Method 1: From collection (for testing small data)
rdd1 = sc.parallelize([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

# Method 2: From file (production use)
rdd2 = sc.textFile("/mnt/raw/sales/sales_data.csv")

# Method 3: From another RDD
rdd3 = rdd1.filter(lambda x: x > 5)

# Check number of partitions (= parallelism level)
print(rdd1.getNumPartitions()) # default based on cluster cores
```

```
# Force specific partition count
rdd4 = sc.parallelize([1,2,3,4,5], numSlices=4) # 4 partitions
```

◆ Spark DataFrame — The Modern Way

DataFrame is a distributed collection of data organized into **named columns** — like a database table but distributed across cluster.

RDD vs DataFrame:

Feature	RDD	DataFrame
API	Low-level	High-level (SQL-like)
Type Safety	Yes (compile)	Limited (runtime check)
Optimization	Manual	Automatic (Catalyst)
Performance	Slower	Faster (optimized)
Ease of Use	Complex	Simple
Schema	None	Named columns + types
SQL Support	No	Yes
Languages	All	All
Recommended	For custom algo	For data engineering

REAL WORLD USAGE:

90% of data engineering work → Use DataFrame/SQL
ML feature engineering → Use DataFrame
Custom algorithms, byte manipulation → Use RDD

◆ Spark Execution Flow: Driver → Executors → Tasks

YOUR CODE → EXECUTION JOURNEY:

You write:

```
df = spark.read.parquet("/mnt/bronze/sales/")
df_filtered = df.filter(df.amount > 1000)
df_grouped = df_filtered.groupBy("region").sum("amount")
df_grouped.write.parquet("/mnt/silver/sales_summary/")
```

Step 1: LOGICAL PLAN created

Driver analyzes your code

Creates tree of operations:

Read → Filter → GroupBy → Write

Step 2: OPTIMIZATION (Catalyst Optimizer)

Catalyst rewrites plan for efficiency:

→ Pushes filter BEFORE read (read less data!)

→ Determines best join strategy

→ Removes unnecessary columns early

Step 3: PHYSICAL PLAN created

Spark decides HOW to execute:

→ Stage 1: Read + Filter (no shuffle needed)

→ Stage 2: Shuffle by region + Aggregate (shuffle needed)

→ Stage 3: Write

Step 4: STAGES split into TASKS

Stage 1: 200 partitions → 200 Tasks

Stage 2: 5 regions → 5 Tasks (after shuffle)

Stage 3: Write Tasks

Step 5: TASKS distributed to Executors

Driver sends tasks to available executors

32 executors run 32 tasks simultaneously
Completed tasks → next batch assigned

Step 6: RESULTS returned to Driver

Write tasks complete → data in ADLS

Driver reports job success

VISUALIZATION:

Your Code

↓

Logical Plan (what to do)

↓

Optimized Logical Plan (Catalyst magic)

↓

Physical Plan (how to do it)

↓

Stages (groups of tasks with no shuffle between them)

↓

Tasks (individual units executed on one partition)

↓

Executors (run tasks in parallel)

↓

Result (written to storage)

◆ Spark Core Calculation Logic

PARTITIONS = KEY CONCEPT:

A partition = one chunk of data processed by one task
on one executor core at one time

Default partition size: 128MB (same as HDFS block)

500GB file → 500000 MB / 128 MB = ~4000 partitions

Cluster: 32 cores

Rounds needed: 4000 / 32 = 125 rounds

Each round: all 32 cores process their partition

Time per round: 15 seconds

Total time: 125 × 15 = ~31 minutes

Add more cores (64 cores):

Rounds: 4000 / 64 = 62.5 → 63 rounds

Total time: 63 × 15 = ~15 minutes

→ Linear improvement with more hardware!

OPTIMAL PARTITION SIZE:

Too small (1MB):

- 500,000 partitions for 500GB
- Too much overhead scheduling tasks
- Slower than optimal

Too large (2GB):

- Only 250 partitions
- Core sits idle waiting for huge partition
- Memory pressure

Sweet spot: 128MB - 256MB per partition

4 Introduction to Databricks

◆ What is Databricks?

Azure Databricks is a **managed Apache Spark platform** built on Azure — it gives you Spark's power without managing any infrastructure.

WITHOUT Databricks (raw Spark on VMs):

- ❌ Install Spark manually on each VM
- ❌ Configure networking between nodes
- ❌ Manage Spark version upgrades
- ❌ Handle cluster failures manually
- ❌ No collaborative notebooks
- ❌ Security and access control manual
- ❌ Takes 2 weeks just to set up a working cluster

WITH Azure Databricks:

- ✅ Click → cluster running in 5-10 minutes
- ✅ Managed Spark (Microsoft + Databricks manage it)
- ✅ Collaborative web notebooks (like Google Docs for code)
- ✅ Auto-scaling (adds/removes nodes based on load)
- ✅ Auto-termination (cluster shuts down when idle → save cost)
- ✅ Unity Catalog for governance
- ✅ Delta Lake built-in
- ✅ Azure AD integration (use company login)
- ✅ One-click integration with ADLS, Key Vault, ADF

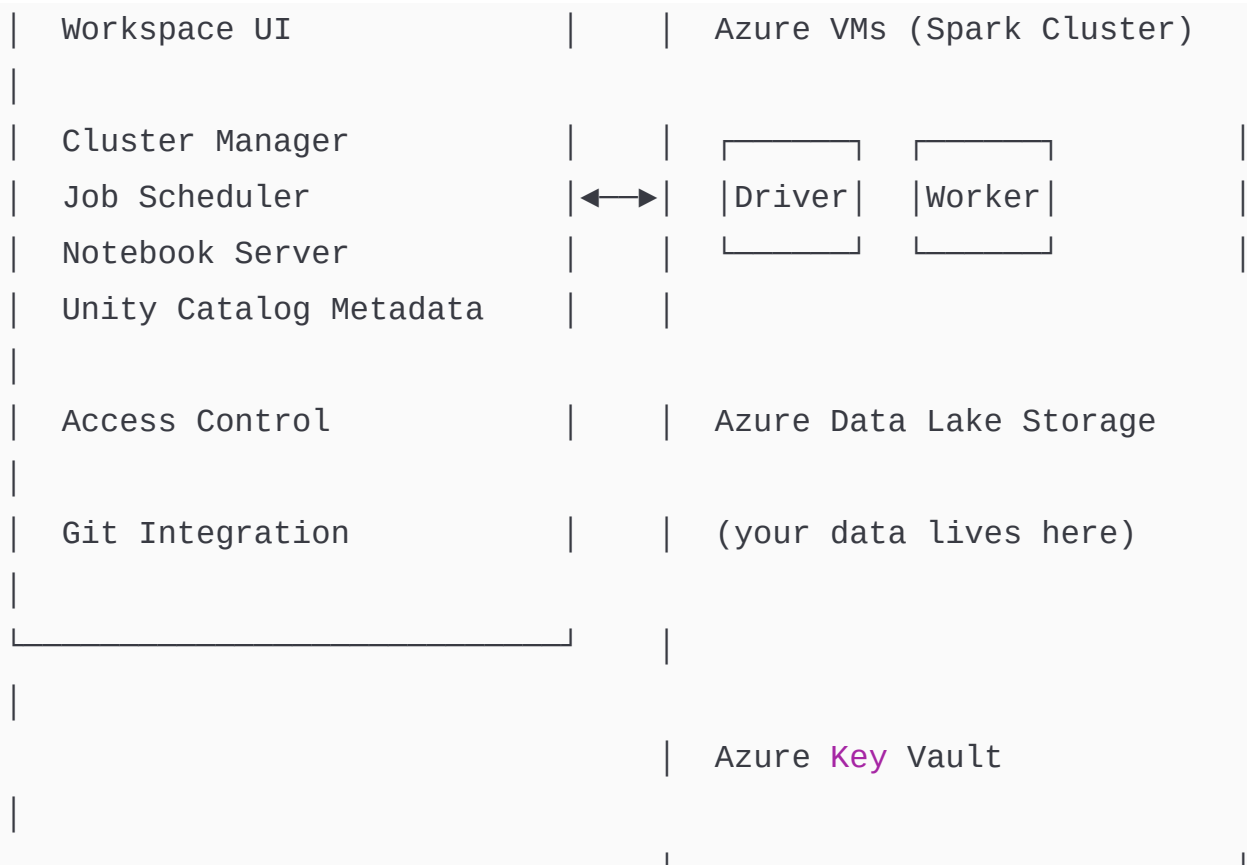
◆ Databricks Architecture

AZURE DATABRICKS ARCHITECTURE:

CONTROL PLANE

DATA PLANE

(Managed by Databricks/Microsoft) (In your Azure subscription)



KEY POINT:

- Your DATA never leaves your Azure subscription
- Only metadata and control signals go to Databricks control plane
- Maximum security for enterprise compliance

◆ Working in Databricks Workspace

WORKSPACE STRUCTURE:

Workspace

- └─ 📁 Users/
 - └─ you@company.com/
 - └─ 📁 My_Experiments (personal notebooks)

```
| | | └─ 📁 Draft_Pipeline
| | └─ 📁 Shared/
| |   └─ 📁 Production/
| |     └─ 📁 transform_sales
| |     └─ 📁 transform_customers
| |     └─ 📁 load_to_synapse
| |   └─ 📁 Utils/
| |     └─ 📁 common_functions
| |     └─ 📁 config_settings
| └─ 📊 Data (Unity Catalog)
| └─ ⚡ Compute (Clusters)
| └─ ⚙️ Workflows (Jobs)
└─ 🔧 Settings
```

LEFT SIDEBAR:

📁 Workspace	→ Notebooks and folders
📊 Data	→ Databases, tables (Unity Catalog)
⚡ Compute	→ Create /manage clusters
⚙️ Workflows	→ Schedule notebooks as jobs
🔗 Partner Connect	→ Connect to BI tools
⚙️ Settings	→ Admin settings

◆ Working with Databricks Notebook

NOTEBOOK BASICS:

Supported languages per cell:

%python	→ PySpark / Python code (most common)
%sql	→ Spark SQL queries
%scala	→ Scala Spark code
%r	→ R code

%md → Markdown documentation
%sh → Shell commands
%fs → DBFS file **system** commands

MAGIC COMMANDS:

%run ./other_notebook → Run another notebook inline
%pip install pandas → Install Python library
%reload_ext → Reload extensions

CELL SHORTCUTS:

Shift+Enter → Run cell **and** move to **next**
Ctrl+Enter → Run cell **and** stay
Esc+A → Add cell above
Esc+B → Add cell below
Esc+DD → Delete cell

NOTEBOOK STRUCTURE (best practice):

Cell 1: %md

Sales Transformation Notebook

****Author:**** Data Engineering Team

****Purpose:**** Clean **and** transform raw sales data

****Schedule:**** Daily at 1 AM

Cell 2: %python (Parameters - from ADF **or** widgets)

```
dbutils.widgets.text("input_path", "/mnt/bronze/sales/")
dbutils.widgets.text("output_path", "/mnt/silver/sales/")
dbutils.widgets.text("process_date", "2024-06-15")
```

```
input_path = dbutils.widgets.get("input_path")
output_path = dbutils.widgets.get("output_path")
process_date = dbutils.widgets.get("process_date")
```

Cell 3: %python (Imports)

```
from pyspark.sql import functions as F
from pyspark.sql.types import *
from pyspark.sql.window import Window
from delta.tables import DeltaTable
import logging
```

Cell 4: %python (Read Data)

Cell 5: %python (Transform)

Cell 6: %python (Write)

Cell 7: %python (Exit with status for ADF)

```
dbutils.notebook.exit("SUCCESS")
```

5 DBFS — Databricks File System

◆ What is DBFS?

DBFS (Databricks File System) is a **distributed file system abstraction** over Azure Blob Storage and ADLS Gen2. It allows you to work with files using simple file paths in notebooks.

DBFS EXPLAINED:

Real Storage: Azure Data Lake Storage Gen2 (physical)

DBFS Path: /mnt/raw/sales/ (logical alias)

You work **with**: /mnt/raw/sales/sales_data.csv

Actually maps **to**:

abfss://raw@mydatalake.dfs.core.windows.net/sales/

WHY DBFS?

Without DBFS:


```
spark.read.format("parquet")
    .option("header", "true")

.load("abfss://container@account.dfs.core.windows.net/path")
→ Must embed full Azure URL + credentials everywhere!
```

With DBFS mount:

```
spark.read.parquet("/mnt/bronze/sales/")
→ Clean, simple path
→ Credentials handled once during mount setup
→ Change storage account → update one mount point
→ All notebooks automatically use new storage
```

◆ DBFS Commands — Complete Reference

```
# — ALL DBFS COMMANDS —————

# LIST files/directories
dbutils.fs.ls("/mnt/bronze/")
# Returns: [FileInfo(path, name, size, modificationTime)]

for f in dbutils.fs.ls("/mnt/bronze/sales/"):
    print(f"Name: {f.name}, Size: {f.size/1024:.1f} KB")

# MKDIR - Create directory
dbutils.fs.mkdirs("/mnt/bronze/new_folder/subfolder/")

# CP - Copy file or directory
dbutils.fs.cp(
    "/mnt/bronze/sales/sales_data.csv",    # source
    "/mnt/archive/sales/sales_data.csv",  # destination
    recurse=False                          # True for
```

```

directories
)

# MV - Move (rename) file
dbutils.fs.mv(
    "/mnt/landing/sales_20240615.csv",      # source
    "/mnt/bronze/sales/sales_20240615.csv" # destination
)

# HEAD - Preview first N bytes of file
dbutils.fs.head("/mnt/bronze/sales/sales_data.csv", 500)
# Output: first 500 bytes of file as string

# PUT - Create file with content
dbutils.fs.put(
    "/mnt/bronze/config/settings.json",
    '{"env": "prod", "batch_size": 10000}',
    overwrite=True
)

# RM - Delete file or directory
dbutils.fs.rm("/mnt/temp/processed_file.csv")
dbutils.fs.rm("/mnt/temp/old_folder/", recurse=True) # delete
folder

# %fs magic commands (shorthand in notebooks)
%fs ls /mnt/bronze/
%fs mkdirs /mnt/bronze/new_folder/
%fs rm /mnt/temp/ --recurse=true

```

◆ Mounting ADLS Gen2 to DBFS

```

# PRODUCTION PATTERN: Mount using Service Principal
# Store credentials in Key Vault, retrieve via dbutils.secrets

# — Step 1: Create scope linked to Key Vault —————
# (Done once by admin in Databricks Secrets UI or CLI)
# Scope: kv-retailmart-scope → Azure Key Vault: kv-retailmart-
prod

# — Step 2: Mount storage —————
storage_account = "retailmartdatalake"
container_name  = "bronze"
mount_point     = "/mnt/bronze"

# Get credentials from Key Vault (never hardcode!)
client_id       = dbutils.secrets.get("kv-retailmart-scope", "sp-
client-id")
client_secret   = dbutils.secrets.get("kv-retailmart-scope", "sp-
client-secret")
tenant_id       = dbutils.secrets.get("kv-retailmart-scope",
"tenant-id")

configs = {
    "fs.azure.account.auth.type": "OAuth",
    "fs.azure.account.oauth.provider.type":

"org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
    "fs.azure.account.oauth2.client.id": client_id,
    "fs.azure.account.oauth2.client.secret": client_secret,
    "fs.azure.account.oauth2.client.endpoint":

f"https://login.microsoftonline.com/{tenant_id}/oauth2/token"
}

# Check if already mounted (avoid duplicate mount error)
existing_mounts = [m.mountPoint for m in dbutils.fs.mounts()]

```

```

if mount_point not in existing_mounts:
    dbutils.fs.mount(

source=f"abfss://{container_name}@{storage_account}.dfs.core.windows
    mount_point=mount_point,
    extra_configs=configs
    )
    print(f"✅ Mounted {container_name} to {mount_point}")
else:
    print(f"ℹ️ Already mounted: {mount_point}")

# — Step 3: Mount all containers —————
containers = ["bronze", "silver", "gold", "archive", "config"]
for container in containers:
    mp = f"/mnt/{container}"
    if mp not in existing_mounts:
        dbutils.fs.mount(

source=f"abfss://{container}@{storage_account}.dfs.core.windows.net/
    mount_point=mp,
    extra_configs=configs
    )
    print(f"✅ Mounted: {mp}")

```

◆ Handling Multiple Files in DBFS

```

# REAL WORLD SCENARIO:
# /mnt/bronze/sales/ has 365 daily files (one per day)
# Need to process, archive, track each

# — List and analyze all files —————

```

```

import datetime

def analyze_folder(path):
    """Analyze all files in DBFS folder"""
    files = dbutils.fs.ls(path)

    total_size = 0
    file_info = []

    for f in files:
        size_mb = f.size / (1024 * 1024)
        mod_time =
datetime.datetime.fromtimestamp(f.modificationTime/1000)
        file_info.append({
            "name": f.name,
            "size_mb": round(size_mb, 2),
            "modified": mod_time.strftime("%Y-%m-%d %H:%M:%S")
        })
        total_size += size_mb

    print(f"Total files:      {len(file_info)}")
    print(f"Total size:        {total_size:.1f} MB")
    print(f"Largest file:      {max(file_info, key=lambda x:
x['size_mb'])['name']}")
    print(f"Smallest file:     {min(file_info, key=lambda x:
x['size_mb'])['name']}")

    return file_info

file_list = analyze_folder("/mnt/bronze/sales/")

# — Process files in DBFS —————

# Read ALL files in folder at once (Spark handles distribution)
df_all_sales = spark.read.parquet("/mnt/bronze/sales/")

```

```

print(f"Total rows across all files: {df_all_sales.count():,}")

# Read files matching pattern
df_june = spark.read.parquet("/mnt/bronze/sales/2024/06/")
# or with glob pattern:
df_q2 = spark.read.parquet("/mnt/bronze/sales/2024/0[4-6]/")

# Read with filename tracking
df_with_source = spark.read \
    .option("recursiveFileLookup", "true") \
    .parquet("/mnt/bronze/sales/") \
    .withColumn("source_file", F.input_file_name())

df_with_source.select("sale_id", "source_file").show(5,
truncate=False)
# Output: sale_id=1001,
source_file=dbfs:/mnt/bronze/sales/2024/06/15/sales_20240615.parquet

# — Archive processed files —————
def archive_file(source_path, archive_base):
    """Move processed file to archive with date folder"""
    filename = source_path.split("/")[-1]
    today = datetime.datetime.now().strftime("%Y/%m/%d")
    archive_path = f"{archive_base}/{today}/{filename}"

    dbutils.fs.mkdirs(f"{archive_base}/{today}/")
    dbutils.fs.mv(source_path, archive_path)
    print(f"Archived: {filename} → {archive_path}")

# Archive all June 2024 files
for f in dbutils.fs.ls("/mnt/bronze/sales/2024/06/"):
    archive_file(f.path, "/mnt/archive/sales")

```

6 Databricks Spark Core — RDD Programming

◆ RDD Operations Complete Guide

```
# ——— CREATING RDDs ———  
  
# From collection  
numbers_rdd = sc.parallelize([1, 2, 3, 4, 5, 6, 7, 8, 9, 10], 4)  
# 4 partitions – for RetailMart, matches 4 executor cores  
  
# From text file  
sales_rdd = sc.textFile("/mnt/bronze/sales/sales_raw.csv")  
# Each LINE becomes one element in RDD  
  
# With specific partitions  
large_rdd = sc.parallelize(range(1000000), numSlices=100)
```

◆ Narrow vs Wide Transformations

CRITICAL CONCEPT – Understand Shuffles:

NARROW TRANSFORMATIONS (No Shuffle):

- Each output partition comes from ONE input partition
- No data movement across network
- Fast!
- Can be pipelined (executed together in one stage)

Examples:

- map() → transform each element
- filter() → keep/remove elements
- flatMap() → one element → zero or many

mapPartitions() → process entire **partition at** once
union() → combine two RDDs
sample() → random sample

WIDE TRANSFORMATIONS (Shuffle Required):

- Output **partition** comes **from** MULTIPLE input partitions
- Data must move across network **between** nodes
- Slow (network I/O **is** expensive)
- Creates **new** Stage boundary

Examples:

groupByKey() → **group by** key (lots **of** data movement!)
reduceByKey() → reduce **by** key (optimized – combines locally **first**)
sortByKey() → sort requires **global** ordering
join() → **join** two RDDs **by** key
distinct() → needs **to** compare across **all** partitions
repartition() → redistribute data across partitions

PRACTICAL IMPLICATION:

Bad Code (**2** shuffles):

rdd.groupByKey()	← shuffle 1 (expensive!)
.mapValues(sum)	← narrow (ok)
.sortByKey()	← shuffle 2 (expensive!)

Good Code (**1** shuffle):

rdd.reduceByKey(add)	← shuffle but optimized (local combine first !)
.sortByKey()	← shuffle 2 (needed for sort)

RULE: Prefer reduceByKey **over** groupByKey

groupByKey moves **ALL values to one** node (memory risk!)
reduceByKey combines locally **THEN** shuffles (less data)

◆ Complete RDD Operations

```
# — TRANSFORMATIONS (lazy – don't execute immediately) —

sales_rdd = sc.textFile("/mnt/bronze/sales_raw.csv") \
    .filter(lambda line: not
line.startswith("sale_id"))
    # skip header row

# map() – transform each element
parsed_rdd = sales_rdd.map(lambda line: line.split(","))
# ["1001", "C001", "LAPTOP", "50000", "2024-06-15"]

# map to named tuple for clarity
from collections import namedtuple
Sale = namedtuple("Sale",
["sale_id", "customer_id", "product", "amount", "date"])

sales_typed = parsed_rdd.map(lambda x:
    Sale(int(x[0]), x[1], x[2], float(x[3]), x[4])
)

# filter() – keep only matching rows
high_value = sales_typed.filter(lambda s: s.amount > 10000)

# flatMap() – one element produces multiple
# Parse each line that might have multi-product orders
order_lines = sc.parallelize([
    "1001, LAPTOP, PHONE",
    "1002, TABLET",
    "1003, LAPTOP, TABLET, PHONE"
])
products = order_lines.flatMap(lambda line: line.split(",")[1:])
# Produces: ["LAPTOP", "PHONE", "TABLET", "LAPTOP", "TABLET", "PHONE"]
```

```

# Key-Value Pair RDD
kv_rdd = sales_typed.map(lambda s: (s.product, s.amount))
# [("LAPTOP", 50000), ("PHONE", 30000), ("LAPTOP", 45000)]

# reduceByKey() – aggregate by key (PREFERRED over groupByKey)
product_totals = kv_rdd.reduceByKey(lambda a, b: a + b)
# [("LAPTOP", 95000), ("PHONE", 30000)]

# groupByKey() – use sparingly (memory intensive)
product_groups = kv_rdd.groupByKey()
# [("LAPTOP", [50000, 45000]), ("PHONE", [30000])]

# sortByKey()
sorted_totals = product_totals.sortByKey(ascending=True)

# distinct() – remove duplicates
unique_products = kv_rdd.map(lambda x: x[0]).distinct()

# — ACTIONS (trigger execution) —————

# collect() – bring all to driver (use only for small data!)
small_result = product_totals.collect()
# [(LAPTOP, 95000), (PHONE, 30000)]

# count() – total elements
total_sales = sales_typed.count()

# first() – first element
first_sale = sales_typed.first()

# take(n) – first n elements (safer than collect)
sample_5 = product_totals.take(5)

# reduce() – aggregate entire RDD to single value
total_revenue = kv_rdd.values().reduce(lambda a, b: a + b)

```

```
# saveAsTextFile() – write to storage
product_totals.saveAsTextFile("/mnt/output/product_totals/")

# foreach() – apply function to each element (no return)
product_totals.foreach(lambda x: print(f"Product: {x[0]}, Total: {x[1]}"))
```

◆ Broadcast Variables and Accumulators

```
# BROADCAST VARIABLES:
# Share read-only data efficiently to all executors
# Without broadcast: data sent to each task separately
# (expensive!)
# With broadcast: sent ONCE per executor (efficient)

# Real-world: region code lookup (small table, used in many
# tasks)
region_map = {
    "N": "North India",
    "S": "South India",
    "E": "East India",
    "W": "West India"
}

# Broadcast to all executors (sent once, cached there)
broadcast_regions = sc.broadcast(region_map)

# Use in transformation
def get_region_name(sale):
    region_code = sale[5] # assuming region_code is 6th column
    # Access broadcast variable with .value
```

```

        region_name = broadcast_regions.value.get(region_code,
"Unknown")
        return sale + (region_name,)

enriched_rdd = sales_rdd.map(get_region_name)

# ACCUMULATORS:
# Shared counter that executors can increment
# Driver reads final count (executors cannot read)
# Use for: counting errors, tracking records processed

# Count records with data quality issues
null_count      = sc.accumulator(0)
invalid_count   = sc.accumulator(0)
total_amount    = sc.accumulator(0.0)

def validate_and_count(sale):
    """Validate sale record, count issues"""
    if sale.customer_id is None:
        null_count.add(1)
    if sale.amount <= 0:
        invalid_count.add(1)
    else:
        total_amount.add(sale.amount)
    return sale # still return (accumulator doesn't filter)

validated = sales_typed.map(validate_and_count)
validated.count() # trigger execution

# Now read accumulator values on DRIVER
print(f"Null customer records: {null_count.value:,}")
print(f"Invalid amount records: {invalid_count.value:,}")
print(f"Total valid revenue:      Rs.{total_amount.value:,.2f}")

```

7 Spark SQL and DataFrames — Complete Guide

◆ Creating DataFrames — All Methods

— METHOD 1: Read from storage (most common) —————

Read Parquet (recommended format for production)

```
df_sales = spark.read.parquet("/mnt/bronze/sales/")
```

Read CSV with options

```
df_csv = spark.read \  
    .option("header", "true")          \  
    .option("inferSchema", "true")     \  
    .option("delimiter", ",")          \  
    .option("encoding", "UTF-8")       \  
    .option("nullValue", "NULL")       \  
    .option("dateFormat", "yyyy-MM-dd") \  
    .option("timestampFormat", "yyyy-MM-dd HH:mm:ss") \  
    .csv("/mnt/bronze/sales/sales_raw.csv")
```

Read JSON

```
df_json = spark.read \  
    .option("multiLine", "true") \  
    .json("/mnt/bronze/payments/payments.json")
```

Read with explicit schema (ALWAYS do this in production!)

InferSchema reads entire file twice → slow + might get types wrong

```
from pyspark.sql.types import *
```

```
sales_schema = StructType([  
    StructField("sale_id",      IntegerType(), nullable=False),  
    StructField("customer_id",  StringType(),  nullable=False),  
    StructField("store_id",     IntegerType(),  nullable=False),  
    StructField("product_code", StringType(),  nullable=True),
```

```

    StructField("product_name", StringType(), nullable=True),
    StructField("quantity", IntegerType(), nullable=True),
    StructField("unit_price", DoubleType(), nullable=True),
    StructField("total_amount", DoubleType(), nullable=True),
    StructField("sale_date", DateType(), nullable=True),
    StructField("region_code", StringType(), nullable=True),
    StructField("payment_mode", StringType(), nullable=True),
    StructField("status", StringType(), nullable=True),
    StructField("created_ts", TimestampType(), nullable=True)
])

```

```

df_sales = spark.read \
    .schema(sales_schema) \
    .option("header", "true") \
    .csv("/mnt/bronze/sales/sales_data.csv")

```

— METHOD 2: From Python collection (testing/prototyping) —

```

data = [
    (1, "C001", 42, "LAPTOP01", "Dell Laptop", 1, 55000.0,
55000.0),
    (2, "C002", 43, "PHONE01", "Samsung S24", 2, 35000.0,
70000.0),
    (3, "C003", 42, "TABLET01", "iPad Pro", 1, 85000.0,
85000.0)
]
cols = ["sale_id", "customer_id", "store_id", "product_code",
        "product_name", "quantity", "unit_price", "total_amount"]

```

```

df_test = spark.createDataFrame(data, schema=cols)
df_test.show()

```

— METHOD 3: From SQL query on registered table —

```

spark.sql("""
    CREATE OR REPLACE TEMP VIEW vw_recent_sales AS
    SELECT * FROM delta.`/mnt/silver/sales/`

```

```

        WHERE sale_date >= current_date() - INTERVAL 30 DAYS
    """)
df_recent = spark.sql("SELECT * FROM vw_recent_sales")

# — METHOD 4: From Pandas DataFrame —————
import pandas as pd
pdf = pd.read_csv("/dbfs/mnt/config/store_mapping.csv")
df_spark = spark.createDataFrame(pdf) # small config tables
only!

```

◆ DataFrame Internal Execution — How It Really Works

When you write:

```

df.filter(F.col("amount") >
1000).groupBy("region").sum("amount")

```

Spark doesn't execute this immediately!
This is called LAZY EVALUATION.

Timeline:

```

Line 1: df.filter(...) → Creates logical plan node (no
execution)
Line 2: .groupBy(...) → Adds to logical plan (no execution)
Line 3: .sum(...) → Adds aggregation to plan (no
execution)

```

`.show()` / `.write()` / `.count()` → TRIGGERS EXECUTION!

Only NOW does Spark:

1. Optimize the full plan
2. Generate physical plan
3. Run tasks on cluster

WHY LAZY EVALUATION?

Spark can see the WHOLE picture before executing

- Can push filters earlier
- Can remove unnecessary columns
- Can choose best join strategy
- Combines operations into fewer stages

PRACTICAL IMPACT:

These three lines: 0 data processed yet

```
df1 = df.filter(F.col("status") == "ACTIVE")
df2 = df1.select("customer_id", "total_amount")
df3 = df2.groupBy("customer_id").sum("total_amount")
```

This line: Spark now executes everything optimally

```
df3.write.parquet("/mnt/silver/customer_totals/")
```

Catalyst sees: Read → Filter → Select → GroupBy → Write

Optimizes to: Read (only 2 columns, only ACTIVE rows) →
GroupBy → Write

Column pruning + predicate pushdown applied automatically!

◆ DataFrame Transformations — Production Code

```
# — RETAILMART: COMPLETE SALES TRANSFORMATION —

from pyspark.sql import functions as F
from pyspark.sql.window import Window

# Read raw sales data
df_raw = spark.read.parquet(f"/mnt/bronze/sales/{process_date}/")

print(f"Raw records: {df_raw.count():,}")
```



```
df_raw.printSchema()
```

```
# — STEP 1: Data Quality Checks —————
```

```
df_valid = df_raw \
    .filter(F.col("sale_id").isNotNull()) \
    .filter(F.col("customer_id").isNotNull()) \
    .filter(F.col("total_amount") > 0) \
    .filter(F.col("sale_date").isNotNull()) \
    .filter(~F.col("status").isin(["CANCELLED", "DELETED"]))

print(f"Valid records: {df_valid.count():,}")
print(f"Dropped: {df_raw.count() - df_valid.count():,} invalid
records")
```

```
# — STEP 2: Data Standardization —————
```

```
df_standardized = df_valid \
    .withColumn("customer_id",
F.upper(F.trim(F.col("customer_id")))) \
    .withColumn("product_code",
F.upper(F.trim(F.col("product_code")))) \
    .withColumn("region_code",
F.upper(F.trim(F.col("region_code")))) \
    .withColumn("payment_mode",
F.lower(F.trim(F.col("payment_mode")))) \
    .withColumn("sale_date",      F.to_date(F.col("sale_date"),
"yyyy-MM-dd")) \
    .withColumn("year",          F.year(F.col("sale_date"))) \
    .withColumn("month",        F.month(F.col("sale_date"))) \
    .withColumn("day",          F.dayofmonth(F.col("sale_date")))
\
    .withColumn("day_of_week",  F.dayofweek(F.col("sale_date")))
\
    .withColumn("is_weekend",    F.when(
                                F.dayofweek(F.col("sale_date")).isin([1, 7]),
                                True
```

```
).otherwise(False))
```

```
# — STEP 3: Derived Columns —————
```

```
df_enriched = df_standardized \
    .withColumn("gst_amount", F.round(F.col("total_amount") *
0.18, 2)) \
    .withColumn("net_amount", F.col("total_amount") -
F.col("gst_amount")) \
    .withColumn("amount_bucket", F.when(F.col("total_amount") <
1000, "LOW")
                                .when(F.col("total_amount") <
10000, "MEDIUM")
                                .when(F.col("total_amount") <
50000, "HIGH")
                                .otherwise("PREMIUM")) \
    .withColumn("region_name", F.when(F.col("region_code") ==
"N", "North India")
                                .when(F.col("region_code") ==
"S", "South India")
                                .when(F.col("region_code") ==
"E", "East India")
                                .when(F.col("region_code") ==
"W", "West India")
                                .otherwise("Unknown")) \
    .withColumn("load_timestamp", F.current_timestamp()) \
    .withColumn("source_system", F.lit("Oracle_ERP")) \
    .withColumn("batch_id", F.lit(process_date))
```

```
# — STEP 4: Business KPIs using Window Functions —————
```

```
# Window: rank sales by amount within each store
```

```
store_window = Window.partitionBy("store_id") \
    .orderBy(F.col("total_amount").desc())
```

```
# Window: running total by date within store
```

```

running_window = Window.partitionBy("store_id") \
                        .orderBy("sale_date") \
                        .rowsBetween(Window.unboundedPreceding,
Window.currentRow)

# Window: 7-day rolling average
rolling_window = Window.partitionBy("store_id") \
                        .orderBy("sale_date") \
                        .rowsBetween(-6, 0) # current + 6
preceding days

df_final = df_enriched \
    .withColumn("rank_in_store", F.rank().over(store_window)) \
    .withColumn("running_total",
F.sum("total_amount").over(running_window)) \
    .withColumn("prev_sale_amt", F.lag("total_amount",
1).over(store_window)) \
    .withColumn("rolling_7d_avg",
F.avg("total_amount").over(rolling_window))

# — STEP 5: Join with Dimension Tables —————

df_stores = spark.read.parquet("/mnt/silver/dim_stores/")
df_products = spark.read.parquet("/mnt/silver/dim_products/")
df_customers = spark.read.parquet("/mnt/silver/dim_customers/")

# Broadcast small dimension tables (< 200MB)
# Avoids expensive shuffle join!
df_with_store = df_final.join(

F.broadcast(df_stores.select("store_id", "store_name", "city", "state"))
    on="store_id",
    how="left"
)

```

```

df_with_product = df_with_store.join(

F.broadcast(df_products.select("product_code", "category", "brand", "co
    on="product_code",
    how="left"
)

df_with_customer = df_with_product.join(

F.broadcast(df_customers.select("customer_id", "customer_name", "tier"
    on="customer_id",
    how="left"
)

# Calculate profit margin
df_complete = df_with_customer \
    .withColumn("profit",      F.col("net_amount") -
F.col("cost_price")) \
    .withColumn("profit_margin", F.round(F.col("profit") /
F.col("net_amount") * 100, 2))

# — STEP 6: Write to Silver Layer —————
df_complete.write \
    .mode("overwrite") \
    .partitionBy("year", "month", "region_code") \
    .parquet(f"/mnt/silver/sales/{process_date}/")

print(f"Written to silver: {df_complete.count():,} records")
print("✅ Transformation complete")

```

◆ User-Defined Functions (UDFs)

```
# When built-in functions aren't enough → write your own UDF
```

```
# — SCENARIO: Complex PAN card validation —————
```

```
# RetailMart needs to validate customer PAN numbers
```

```
# PAN format: ABCDE1234F (5 letters, 4 digits, 1 letter)
```

```
import re
```

```
from pyspark.sql.functions import udf
```

```
from pyspark.sql.types import BooleanType, StringType
```

```
# Python function (runs on executor, NOT on driver)
```

```
def validate_pan(pan_number):
```

```
    """Validate Indian PAN card format"""
```

```
    if pan_number is None:
```

```
        return False
```

```
    pattern = r'^[A-Z]{5}[0-9]{4}[A-Z]{1}$'
```

```
    return bool(re.match(pattern, pan_number.strip().upper()))
```

```
def mask_pan(pan_number):
```

```
    """Mask PAN for display: ABCDE1234F → ABCXX1234F"""
```

```
    if pan_number is None:
```

```
        return None
```

```
    pan = pan_number.strip().upper()
```

```
    if len(pan) != 10:
```

```
        return pan
```

```
    return pan[:3] + "XX" + pan[5:]
```

```
# Register as Spark UDF
```

```
validate_pan_udf = udf(validate_pan, BooleanType())
```

```
mask_pan_udf      = udf(mask_pan,      StringType())
```

```
# Use in DataFrame
```

```
df_customers = df_customers \
```

```
    .withColumn("is_pan_valid",
```

```
validate_pan_udf(F.col("pan_number"))) \
```

```

        .withColumn("pan_masked",
mask_pan_udf(F.col("pan_number")))

# — PERFORMANCE NOTE ABOUT UDFs —————
# Regular Python UDF:
#   → Serializes each row to Python process
#   → Processes one row at a time
#   → Sends result back to JVM
#   → Very slow for large datasets!

# BETTER: Pandas UDF (vectorized – works on batches)
from pyspark.sql.functions import pandas_udf
import pandas as pd

@pandas_udf(BooleanType())
def validate_pan_fast(pan_series: pd.Series) -> pd.Series:
    """Pandas UDF – works on entire column at once
(vectorized)"""
    return pan_series.str.match(r'^[A-Z]{5}[0-9]{4}[A-Z]{1}$',
na=False)

# 10-100x faster than regular UDF for large datasets!
df_customers = df_customers \
    .withColumn("is_pan_valid_fast",
validate_pan_fast(F.col("pan_number")))

# BEST: Use built-in functions whenever possible
# F.regex_extract, F.regex_replace, etc.
# Built-in functions run in JVM directly – no Python
serialization overhead
df_customers = df_customers \
    .withColumn("is_pan_valid_builtin",
        F.col("pan_number").rlike(r'^[A-Z]{5}[0-9]{4}[A-Z]{1}$'))

```

8 Spark Optimizer & Execution Engine

◆ Spark Catalyst Optimizer

CATALYST = Spark's Query Optimizer

Think of Catalyst as a chess grandmaster who thinks several moves ahead before making any move.

Your Code → 4 Phases:

PHASE 1: ANALYSIS

- Resolve column names (check they exist)
- Resolve data types
- Check for errors in query
- **Output:** Resolved Logical Plan

PHASE 2: LOGICAL OPTIMIZATION

- **Apply rule-based optimizations:**
 - Predicate Pushdown (filter early)
 - Column Pruning (drop unused columns early)
 - Constant Folding (evaluate `1+2` to `3` at plan time)
 - **Null** Propagation
 - Boolean Simplification
- **Output:** Optimized Logical Plan

PHASE 3: PHYSICAL PLANNING

- Generate multiple physical plans
- Use cost model to choose best
- Choose join strategy (broadcast vs sort-merge)
- **Output:** Physical Plan

PHASE 4: CODE GENERATION (Tungsten)

- Generate optimized Java bytecode

- Whole-stage code generation
- Runs directly on CPU (no interpretation overhead)
- **Output:** Executable RDD

VISUALIZE THE PLAN:

```
df_result.explain(True)  # show all 4 plans
df_result.explain("formatted")  # pretty format
```

◆ Logical Plan vs Physical Plan

```
# See exactly what Spark plans to do
df = spark.read.parquet("/mnt/bronze/sales/") \
    .filter(F.col("total_amount") > 10000) \
    .select("customer_id", "total_amount", "region_code")
\
    .groupBy("region_code") \
    .agg(F.sum("total_amount").alias("region_total"))

df.explain(True)
```

```
== Parsed Logical Plan ==
'Aggregate ['region_code], [...]
+- Project [customer_id, total_amount, region_code]
   +- Filter (total_amount > 10000)
      +- Relation[...] parquet

== Analyzed Logical Plan ==
(types resolved)

== Optimized Logical Plan ==
Aggregate [region_code], [region_code, sum(total_amount)]
+- Project [total_amount, region_code]      ← customer_id
```



```

removed! (pruning)
  +- Filter (isnotnull(total_amount) AND (total_amount >
10000.0))
    +- Relation[region_code, total_amount] parquet ← reads
only 2 columns!

== Physical Plan ==
HashAggregate(keys=[region_code], functions=[sum(total_amount)])
+- Exchange hashpartitioning(region_code, 200) ← shuffle!
  +- HashAggregate(keys=[region_code], functions=
[partial_sum(total_amount)])
    +- Project [total_amount, region_code]
      +- Filter (isnotnull(total_amount) AND (total_amount >
10000.0))
        +- FileScan parquet [region_code,total_amount]
          PushedFilters: [IsNotNull(total_amount),
GreaterThan(total_amount,10000.0)]
          PartitionFilters: []

```

KEY OBSERVATIONS:

- ✓ customer_id REMOVED early (column pruning)
- ✓ filter PUSHED DOWN to file scan (predicate pushdown)
- ✓ Only 2 columns read from file (not all columns!)
- ✓ Filter applied during file reading (less data into memory)
- ✓ Partial aggregation BEFORE shuffle (reduces shuffle size)

◆ Rule-Based vs Cost-Based Optimization

RULE-BASED OPTIMIZER (RBO):

- Applies fixed rules always
- "Always push filters down"
- "Always remove unused columns early"

- "Always fold constants"
- Fast **to** apply (**no** statistics needed)
- Works well **for 90% of** queries

COST-BASED OPTIMIZER (CBO):

- Uses **table** statistics **to** make smarter decisions
- "This table has 1M rows, that one has 100 rows
 - use broadcast join for small table"
- Needs **table** statistics collected **first!**

Enable CBO:

```
spark.conf.set("spark.sql.cbo.enabled", "true")
```

Collect statistics (do this after writing data):

```
spark.sql("ANALYZE TABLE delta.`/mnt/silver/sales/`  
          COMPUTE STATISTICS FOR ALL COLUMNS")
```

```
spark.sql("ANALYZE TABLE silver.sales COMPUTE STATISTICS")
```

With statistics collected:

- CBO knows: sales = **50M rows**, products = **5000 rows**
- Automatically broadcasts products (small **table**)
 - **No** need **for** manual `F.broadcast()` hints!

◆ Adaptive Query Execution (AQE)

AQE = Spark **3.0+** feature that re-optimizes **at** RUNTIME

Problem Adaptive Query Execution solves:

- Spark plans execution BEFORE **running**
- But actual data distribution **unknown** until runtime!

Example:

Query plans for 200 shuffle partitions (default)

Actual data: only 5 unique regions

Result: 195 empty partitions, 5 huge partitions

→ 195 tiny tasks (wasteful) + skew!

AQE FEATURES:

1. Coalescing Shuffle Partitions:

After shuffle, if many partitions are tiny

→ AQE automatically merges small partitions

→ 200 planned → 8 actual (based on data)

2. Switching Join Strategies at Runtime:

Plan says: SortMergeJoin (assumes table is large)

Runtime: one table turns out to be small!

→ AQE switches to BroadcastHashJoin automatically

3. Handling Skewed Partitions:

One region has 90% of data (data skew!)

→ AQE splits the huge partition into sub-tasks

→ Prevents one executor doing all the work

ENABLE AQE:

```
spark.conf.set("spark.sql.adaptive.enabled", "true") # default  
true in Spark 3.2+
```

```
spark.conf.set("spark.sql.adaptive.coalescePartitions.enabled",  
"true")
```

```
spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")
```

```

# PREDICATE PUSHDOWN – Read less data from storage

# WITHOUT predicate pushdown (bad):
# Step 1: Read ENTIRE 500GB file into memory
# Step 2: Then filter in memory
df_bad = spark.read.parquet("/mnt/bronze/sales/") \
    .filter(F.col("region_code") == "N") # filters 80%
of data!

# WITH predicate pushdown (what Spark actually does):
# Spark tells Parquet reader: "only give me rows where
region_code='N'"
# Parquet reads only relevant row groups from file
# Result: reads maybe 100GB instead of 500GB!

# PARTITION PRUNING (even better):
# If data is partitioned by region_code:
# /mnt/bronze/sales/region_code=N/ ← only this folder read!
# /mnt/bronze/sales/region_code=S/ ← SKIPPED entirely
# /mnt/bronze/sales/region_code=E/ ← SKIPPED entirely
# /mnt/bronze/sales/region_code=W/ ← SKIPPED entirely

df_pruned = spark.read.parquet("/mnt/bronze/sales/") \
    .filter(F.col("region_code") == "N")
# Only reads region_code=N partition = 125GB instead of 500GB

# VERIFY pushdown happened (check Physical Plan):
df_pruned.explain()
# Look for: PartitionFilters: [isnotnull(region_code),
(region_code = N)]
# This confirms partition pruning is working!

# MAXIMIZE PUSHDOWN:
# ✅ Filter on partition columns (year, month, region)
# ✅ Use = and IN operators (pushdown works)

```

```
#  Filter early in your code
#  Avoid UDFs in filter (cannot be pushed down to file reader)
#  Avoid complex functions like lower(col) == 'x' (can't push)
#  Use: col == 'X' (can push) NOT: lower(col) == 'x' (can't push)
```

Spark Joins — Deep Dive

Three Join Strategies Explained

SPARK CHOOSES JOIN STRATEGY BASED ON TABLE SIZES:

Strategy 1: Broadcast Hash Join

Use when: One table is SMALL (< 10MB default, max ~200MB)

Process:

- Small table broadcast to ALL executor nodes
- Each executor has complete copy
- Large table processed locally, joins locally
- NO shuffle needed!

Performance: FASTEST (no network transfer of large table)

Strategy 2: Shuffle Hash Join

Use when: Medium-sized tables, one fits in memory per partition

Process:

- Both tables shuffled by join key
- Matching keys land on same executor
- Hash table built from smaller partition
- Joined locally per partition

Performance: MEDIUM

Strategy 3: Sort-Merge Join

Use **when**: Both tables are LARGE

Process:

- Both tables shuffled by join key
- Both sorted by join key
- Merge sorted data (like merge in mergesort)
- Memory efficient (doesn't require full partition in RAM)

Performance: SLOWEST but most scalable

◆ Broadcast Join — With Code

```
# RetailMart: Join 50M sales with 5,000-row product table

df_sales    = spark.read.parquet("/mnt/silver/sales/")      # 50M
rows
df_products = spark.read.parquet("/mnt/silver/dim_products/") #
5,000 rows

# — AUTO BROADCAST (if table <
spark.sql.autoBroadcastJoinThreshold) —
spark.conf.set("spark.sql.autoBroadcastJoinThreshold", "200mb")
# Spark automatically broadcasts tables < 200MB

# — MANUAL BROADCAST HINT (when you KNOW table is small) —
df_result = df_sales.join(
    F.broadcast(df_products), # explicit hint to broadcast
    on="product_code",
    how="left"
)

# — VERIFY: Check explain plan —
df_result.explain()
# Look for: BroadcastHashJoin
```

```
# NOT SortMergeJoin (that would mean broadcast failed)

# — WHEN BROADCAST FAILS —————
# If product table grows to 300MB → exceeds threshold → fallback
to shuffle
# Solution 1: Increase threshold
spark.conf.set("spark.sql.autoBroadcastJoinThreshold", "500mb")

# Solution 2: Filter product table before broadcasting
# Only broadcast products that exist in today's sales
active_products = df_products.join(
    df_sales.select("product_code").distinct(),
    on="product_code",
    how="inner" # only products sold today
)
# Now maybe only 1,000 rows → easily fits in broadcast!
df_result = df_sales.join(F.broadcast(active_products),
on="product_code")
```

◆ Sort-Merge Join & Handling Skew

```
# SORT-MERGE JOIN: Large tables (no choice but to shuffle)
df_sales = spark.read.parquet("/mnt/silver/sales/") # 50M
rows
df_customers = spark.read.parquet("/mnt/silver/customers/") # 5M
rows

# Both too large to broadcast → Sort-Merge Join
df_result = df_sales.join(df_customers, on="customer_id",
how="left")
df_result.explain()
# Shows: SortMergeJoin
```

```
# — DATA SKEW PROBLEM —————  
# Problem: 80% of sales have customer_id = "WALK-IN" (cash  
customers)  
# When joining by customer_id:  
#   → All WALK-IN sales (40M rows!) go to ONE partition  
#   → One executor processes 40M rows alone  
#   → 31 executors sit idle waiting  
#   → Job takes 2 hours instead of 5 minutes!
```

```
# — SOLUTION 1: Salting technique —————
```

```
import random  
  
# Add random salt to skewed key  
df_sales_salted = df_sales \  
    .withColumn("salt", (F.rand() * 10).cast("int")) \  
    .withColumn("join_key_salted",  
        F.when(F.col("customer_id") == "WALK-IN",  
            F.concat(F.col("customer_id"), F.lit("_"),  
F.col("salt")))  
        .otherwise(F.col("customer_id")))  
  
# Explode customer table for WALK-IN  
from pyspark.sql.functions import array, explode  
  
df_cust_salted = df_customers \  
    .withColumn("salt_range", F.array([F.lit(i) for i in  
range(10)])) \  
    .withColumn("salt", F.explode(F.col("salt_range"))) \  
    .withColumn("join_key_salted",  
        F.when(F.col("customer_id") == "WALK-IN",  
            F.concat(F.col("customer_id"), F.lit("_"),  
F.col("salt")))  
        .otherwise(F.col("customer_id"))) \  
    .drop("salt_range", "salt")
```



```
# Now join on salted key → WALK-IN distributed across 10
partitions!
df_result = df_sales_salted.join(
    df_cust_salted,
    on="join_key_salted",
    how="left"
)

# — SOLUTION 2: AQE Skew Join (automatic in Spark 3.x) —
spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")
spark.conf.set("spark.sql.adaptive.skewJoin.skewedPartitionThresholdInBytes",
"256mb")
# AQE automatically splits the skewed WALK-IN partition!
```

10 Spark Performance Optimization

◆ Cache vs Persist

```
# PROBLEM: Using same DataFrame 3 times in pipeline
# Without caching: data READ FROM DISK 3 times!

df_sales = spark.read.parquet("/mnt/bronze/sales/2024/") # 50GB

# Usage 1: Count rows
count = df_sales.count() # reads 50GB from disk

# Usage 2: Calculate regional totals
regional = df_sales.groupBy("region").sum("amount") # reads 50GB
again!

# Usage 3: Find top customers
```

```
top_cust = df_sales.groupBy("customer_id").sum("amount") # reads
50GB AGAIN!
```

```
# TOTAL: 150GB read from storage! Slow and expensive!
```

```
# — SOLUTION: Cache (stores in memory) —————
```

```
df_sales = spark.read.parquet("/mnt/bronze/sales/2024/")
```

```
df_sales.cache() # marks for caching
```

```
# OR
```

```
df_sales.persist() # same as cache() for default storage
level
```

```
# First action triggers actual caching:
```

```
count = df_sales.count() # reads 50GB from disk → stores in
memory
```

```
# Subsequent uses: read from MEMORY (not disk)
```

```
regional = df_sales.groupBy("region").sum("amount") # reads
from memory!
```

```
top_cust = df_sales.groupBy("customer_id").sum("amount") # reads
from memory!
```

```
# — STORAGE LEVELS —————
```

```
from pyspark import StorageLevel
```

```
# MEMORY_ONLY (default for cache()):
```

```
df.persist(StorageLevel.MEMORY_ONLY)
```

```
# → Best performance
```

```
# → If doesn't fit → partitions evicted (recomputed next time)
```

```
# → Use for: data that FITS in cluster RAM
```

```
# MEMORY_AND_DISK (recommended for large data):
```

```
df.persist(StorageLevel.MEMORY_AND_DISK)
```

```
# → Fill memory first
```

```
# → Overflow to disk for rest
# → Use for: data larger than cluster RAM but smaller than disk

# DISK_ONLY:
df.persist(StorageLevel.DISK_ONLY)
# → Always write to disk
# → Slower than memory but doesn't evict
# → Use for: checkpointing important intermediate results

# MEMORY_ONLY_SER (serialized):
df.persist(StorageLevel.MEMORY_ONLY_SER)
# → Compress before storing in memory
# → 5-10x less memory used
# → 10-20% slower CPU (serialize/deserialize)
# → Use for: fitting more data in limited RAM

# ALWAYS UNPERSIST WHEN DONE!
df_sales.unpersist()
# Releases memory for other operations
# Forgetting to unpersist → OOM errors later in pipeline!

# — WHEN TO CACHE — Decision Rules —————
# ✅ Cache when: same DataFrame used 2+ times in pipeline
# ✅ Cache when: expensive computation (ML features, complex joins)
# ✅ Cache when: iterative algorithms (ML training)
# ❌ Don't cache: used only once (wastes memory)
# ❌ Don't cache: data too large for cluster memory (causes evictions)
# ❌ Don't cache: simple reads (Parquet reads are fast anyway)
```

◆ Partitioning and Repartitioning

```
# UNDERSTAND CURRENT PARTITIONING:
print(f"Current partitions: {df_sales.rdd.getNumPartitions()}")

# — REPARTITION (full shuffle) —————
# Use when: INCREASING partitions or repartitioning by specific
column

# Repartition to specific count (default 200)
df_repartitioned = df_sales.repartition(100)

# Repartition by column (data locality for joins/aggregations)
df_by_region = df_sales.repartition(F.col("region_code"))
# All "N" region rows → same partition(s)
# All "S" region rows → same partition(s)
# Subsequent groupBy("region_code") → no shuffle needed!

# — COALESCE (no shuffle – only reduces partitions) —————
# Use when: REDUCING partitions (after filtering removes data)

# After filtering 90% of data:
df_india_north = df_sales.filter(F.col("region_code") == "N")
# Now only 10% of data but still 200 partitions
# Most partitions nearly empty → waste!

df_optimized = df_india_north.coalesce(20) # reduce without
shuffle
# 200 partitions → 20 partitions (no network transfer)

# — REPARTITION vs COALESCE RULE —————
# Increasing partitions → REPARTITION (needs shuffle to
distribute)
# Decreasing partitions → COALESCE (no shuffle, just merge local)

# — OPTIMAL PARTITION COUNT —————
# Rule of thumb: 2-4 partitions per CPU core
```

```

# 8 worker nodes × 4 cores = 32 cores
# Optimal partitions: 32 × 3 = 96 → round to 100

# For writing: match output file count to downstream needs
# Too many files (1000 tiny files) → slow reads later!
# Write as fewer, larger files:

df_final.coalesce(10) \
    .write \
    .mode("overwrite") \
    .parquet("/mnt/silver/sales/")
# Writes exactly 10 Parquet files = clean, efficient output

# But for partitioned write:
df_final.repartition(F.col("region_code")) \
    .write \
    .mode("overwrite") \
    .partitionBy("year", "month", "region_code") \
    .parquet("/mnt/silver/sales/")
# Creates one file per partition folder = efficient for
downstream filters

```

◆ Complete Performance Optimization Checklist

```

# — PRODUCTION PERFORMANCE CHECKLIST —

# 1. USE EXPLICIT SCHEMA (2x faster than inferSchema)
schema = StructType([...])
df = spark.read.schema(schema).parquet(path)

# 2. FILTER EARLY (less data in memory)
df = spark.read.parquet(path) \

```

```

        .filter(F.col("sale_date") >= "2024-01-01") \ #
filter first
        .filter(F.col("status") == "ACTIVE")          #
reduce data ASAP

# 3. SELECT ONLY NEEDED COLUMNS
df = df.select("sale_id", "customer_id", "total_amount") # drop
unused columns

# 4. BROADCAST SMALL TABLES
df_result = big_table.join(F.broadcast(small_table), on="key")

# 5. USE BUILT-IN FUNCTIONS (not Python UDFs)
# Good:
df = df.withColumn("upper_name", F.upper(F.col("product_name")))
# Bad:
# udf(lambda x: x.upper())(col) → Python UDF is 10x slower

# 6. AVOID CARTESIAN PRODUCTS
# Never join without condition! → produces M×N rows!
# Always specify join condition

# 7. REPARTITION BEFORE MULTIPLE JOINS
# If joining by same key multiple times:
df = df.repartition(F.col("customer_id")) # partition once
result1 = df.join(orders, on="customer_id") # no reshuffle
result2 = df.join(activity, on="customer_id") # no reshuffle

# 8. CACHE CORRECTLY
# Only cache what's used multiple times
# Always unpersist when done

# 9. SET SHUFFLE PARTITIONS APPROPRIATELY
# Default 200 is too many for small data, too few for large data
data_size_gb = 50

```

```
recommended_partitions = max(200, int(data_size_gb * 4))
spark.conf.set("spark.sql.shuffle.partitions",
recommended_partitions)

# 10. ENABLE AQE
spark.conf.set("spark.sql.adaptive.enabled", "true")
spark.conf.set("spark.sql.adaptive.coalescePartitions.enabled",
"true")
spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")
```

11 Handling Multiple File Formats

```
# — CSV DATA —

# Read (with all production options)
df_csv = spark.read \
    .schema(my_schema) \
    .option("header", "true") \
    .option("delimiter", ",") \
    .option("encoding", "UTF-8") \
    .option("nullValue", "") \
    .option("nanValue", "NaN") \
    .option("escape", "'') \
    .option("multiLine", "false") \
    .option("mode", "DROPMALFORMED") # drop bad rows
    .csv("/mnt/bronze/sales/*.csv")

# Write
df.write \
    .option("header", "true") \
    .option("delimiter", ",") \
    .mode("overwrite") \
```

```

.csv("/mnt/silver/sales_csv/")

# — JSON DATA —

# Flat JSON
df_json = spark.read \
    .option("multiLine", "false") \ # false = one JSON object
    per line
    .json("/mnt/bronze/payments/*.json")

# Nested JSON (payment with items array)
# {
#   "payment_id": "P001",
#   "amount": 50000,
#   "items": [
#     {"product": "LAPTOP", "qty": 1, "price": 45000},
#     {"product": "BAG", "qty": 1, "price": 5000}
#   ]
# }

df_payments = spark.read.json("/mnt/bronze/payments/")

# Explode nested array
df_items = df_payments \
    .withColumn("item", F.explode("items")) \
    .select(
        "payment_id",
        "amount",
        F.col("item.product").alias("product"),
        F.col("item.qty").alias("quantity"),
        F.col("item.price").alias("unit_price")
    )

# — PARQUET FILES —

# Best format for analytical workloads

```



```

# Read
df_parquet = spark.read.parquet("/mnt/silver/sales/")

# Read with partition filter (partition pruning)
df_june = spark.read.parquet("/mnt/silver/sales/") \
    .filter(F.col("year") == 2024) \
    .filter(F.col("month") == 6)
# Spark reads ONLY year=2024/month=6 folder!

# Write with partitioning
df.write \
    .mode("overwrite") \
    .partitionBy("year", "month", "region_code") \
    .option("compression", "snappy") \
    .parquet("/mnt/silver/sales/")

# Resulting folder structure:
# /mnt/silver/sales/
#   year=2024/month=6/region_code=N/part-0000.parquet
#   year=2024/month=6/region_code=S/part-0000.parquet
#   year=2024/month=7/region_code=N/part-0000.parquet

# Format comparison:
# CSV:      Simple, human-readable, NO compression, row-based
# JSON:     Semi-structured, verbose, good for nested data
# Parquet:  Columnar, compressed (70% smaller), BEST for analytics
# Avro:     Row-based, good for streaming, schema evolution
# ORC:      Columnar, good for Hive, predicate pushdown
# Delta:    Parquet + transaction log = ACID + time travel

```

1 2 Databricks Utilities — Complete Guide

```

# — 1. FILESYSTEM UTILITY (dbutils.fs) —————
# Already covered in DBFS section

# — 2. SECRETS UTILITY —————
# Securely access Key Vault secrets – never print/log these!

# Get single secret
storage_key = dbutils.secrets.get(
    scope="kv-retailmart-scope",    # name of secret scope
    key="adls-storage-key"         # secret name in Key Vault
)

# List available secrets in scope (shows names only, not values)
dbutils.secrets.list("kv-retailmart-scope")

# List all scopes
dbutils.secrets.listScopes()

# CRITICAL SECURITY RULE:
# When you print a secret in Databricks:
print(dbutils.secrets.get("kv-retailmart-scope", "adls-storage-
key"))
# Output: [REDACTED] ← Databricks automatically redacts it!

# — 3. NOTEBOOK UTILITY —————
# Run other notebooks and pass/receive values

# Run another notebook (must be in same workspace)
result = dbutils.notebook.run(
    path="/Production/Utils/common_functions",    # notebook path
    timeout_seconds=3600,                        # max 1 hour
    arguments={
        "input_path": "/mnt/bronze/sales/",
        "process_date": "2024-06-15",
        "env": "prod"
    }
)

```

```

    }
)
print(f"Child notebook returned: {result}")

# In the CHILD notebook – receive and return:
dbutils.widgets.text("input_path", "")
dbutils.widgets.text("process_date", "")

input_path = dbutils.widgets.get("input_path")
process_date = dbutils.widgets.get("process_date")

# ... do processing ...

# Return value to parent notebook
dbutils.notebook.exit("SUCCESS: 142850 rows processed")

# — 4. WIDGETS UTILITY —————
# Interactive parameters for notebooks

# Text input
dbutils.widgets.text(
    name="process_date",
    defaultValue="2024-06-15",
    label="Process Date (yyyy-MM-dd)"
)

# Dropdown
dbutils.widgets.dropdown(
    name="environment",
    defaultValue="dev",
    choices=["dev", "test", "prod"],
    label="Environment"
)

# Combobox (dropdown + custom text)

```

```

dbutils.widgets.combobox(
    name="region",
    defaultValue="ALL",
    choices=["ALL", "N", "S", "E", "W"],
    label="Region Code"
)

# Multiselect
dbutils.widgets.multiselect(
    name="tables",
    defaultValue="sales",
    choices=["sales", "customers", "products", "inventory"],
    label="Tables to Process"
)

# Get widget values
process_date = dbutils.widgets.get("process_date")
environment  = dbutils.widgets.get("environment")
region       = dbutils.widgets.get("region")
tables       = dbutils.widgets.get("tables") # comma-separated
string

# Remove widgets
dbutils.widgets.remove("process_date")
dbutils.widgets.removeAll()

# — 5. CREDENTIALS UTILITY —
# (Available in Unity Catalog environments)
# Access credentials for external storage

# — 6. DATA UTILITY —
# Convert Spark DataFrame to Pandas for small data
import pandas as pd

spark_df = spark.sql("SELECT region_code, COUNT(*) as cnt FROM

```

```
sales GROUP BY 1")
pandas_df = spark_df.toPandas() # only for small DataFrames!

# Convert back
spark_df_back = spark.createDataFrame(pandas_df)
```

13 Databricks Cluster Management

◆ Types of Clusters

CLUSTER TYPES COMPARISON:

ALL-PURPOSE CLUSTER:

- Stays **running** continuously (until you stop it)
- Shared **by** multiple users **and** notebooks simultaneously
- Interactive development **and** exploration
- More expensive (**running** even **when** idle)
- Auto-terminate after N minutes **of** inactivity

USE **FOR**: Development, data exploration, ad-hoc queries

JOB CLUSTER:

- Created specifically **for ONE** job run
- Starts **when** job begins → terminates **when** job ends
- Cannot be shared (dedicated **to** that job)
- Cheaper (**only** runs during job execution)
- Cannot SSH **or** attach notebooks while job **running**

USE **FOR**: Production scheduled pipelines (ADF-triggered)

DIFFERENCE **IN** PRACTICE:

- Analyst exploring data → **All**-Purpose (interactive)
- Nightly sales pipeline → Job Cluster (automated)

Cost Example:

All-Purpose (runs 24/7): 8 nodes × \$0.30/hr × 24hr = \$57.60/day

Job Cluster (runs 2hr): 8 nodes × \$0.30/hr × 2hr = \$4.80/day

→ Job cluster = 12x cheaper for scheduled workloads!

◆ Cluster Modes

CLUSTER MODES:

STANDARD (Multi-user):

- Supports Python, Scala, SQL, R
- Multiple users can attach notebooks
- Processes run as single user (cluster owner)
- Good for most development scenarios

HIGH CONCURRENCY:

- Optimized for many concurrent users
- Each user's queries run in isolated containers
- Fine-grained resource sharing
- Automatic user isolation (FAIR scheduler)
- Best for BI teams with many analysts
- Does NOT support Scala

SINGLE NODE:

- No worker nodes – driver only
- Good for: small data, testing, pandas workloads
- Not for production big data

AUTOSCALING:

- Cluster dynamically adds/removes workers based on load
- Configuration:
 - Min workers: 2 (always running)
 - Max workers: 16 (scales up when needed)
- Scales up: when tasks queue up (more work than capacity)
- Scales down: when executors idle > scaleDownDelay
- Very cost-effective for variable workloads

◆ Creating and Managing Clusters

PRODUCTION CLUSTER CONFIGURATION:

Cluster Name: RetailMart-Prod-Pipeline
Policy: Custom (enforced by admin)
Mode: Standard
Databricks Runtime: 13.3 LTS (Long Term Support)
→ LTS = supported 2+ years, stable, recommended for production

Worker Type: Standard_DS4_v2 (8 cores, 28GB RAM each)
Min Workers: 4
Max Workers: 16
Auto Scale: Enabled

Driver Type: Standard_DS4_v2 (same as worker)

Auto Termination: 30 minutes (for dev clusters)
OFF (for production - job clusters auto-terminate)

Spark Config:
spark.sql.adaptive.enabled true

```
spark.sql.adaptive.coalescePartitions.enabled true
spark.databricks.delta.optimizeWrite.enabled true
spark.databricks.delta.autoCompact.enabled true
```

Environment Variables:

```
ENV=production
LOG_LEVEL=INFO
```

Libraries:

```
PyPI: great-expectations==0.18.0
PyPI: azure-storage-blob==12.19.0
```

Init Scripts:

```
dbfs:/init-scripts/install-custom-libs.sh
```

— Cluster Operations in Code —

List clusters (using Databricks REST API from notebook)

```
import requests
```

```
token = dbutils.secrets.get("kv-scope", "databricks-token")
```

```
workspace_url = "https://adb-xxxx.azuredatabricks.net"
```

```
headers = {"Authorization": f"Bearer {token}"}
```

List clusters

```
response = requests.get(
    f"{workspace_url}/api/2.0/clusters/list",
    headers=headers
)
```

```
clusters = response.json()["clusters"]
```

```
for c in clusters:
```

```
    print(f>Name: {c['cluster_name']}, State: {c['state']}")
```



```
# Get cluster info
# Monitor cluster health, logs via UI:
# Compute → Your Cluster → Event Log
# Shows: Started, Autoscaling events, Node failures, etc.

# — Cluster Logs —————
# Configure log delivery in cluster settings:
# Logging → Destination: DBFS
# Path: /cluster-logs/
# Logs written to: /cluster-logs/<cluster-id>/driver/log4j-
*.log.gz

# View logs in notebook
driver_logs = dbutils.fs.ls(f"/cluster-
logs/{cluster_id}/driver/")
for log in driver_logs:
    print(log.name)
```

14 Batch Processing — RetailMart End-to-End

◆ Historical Data Load

```
# SCENARIO: First-time load of 5 years of historical sales data
# 2.5 TB total, 50M records per year, 250M total

# — STRATEGY: Partition-by-partition historical load —————

years = list(range(2019, 2025)) # 2019 to 2024

for year in years:
    print(f"\n{'='*50}")
    print(f"Processing year: {year}")
```

```

# Read one year at a time (avoid OOM)
df_year = spark.read \
    .schema(sales_schema) \
    .parquet(f"/mnt/bronze/sales/historical/{year}/")

print(f"Year {year}: {df_year.count():,} records")

# Apply transformations
df_transformed = transform_sales(df_year, year)

# Write to Silver (append mode for historical)
df_transformed.write \
    .mode("append") \
    .partitionBy("year", "month") \
    .parquet("/mnt/silver/sales/")

print(f"✅ Year {year} loaded successfully")

# — PARALLEL HISTORICAL LOAD (faster using Spark itself) —
# Better approach: Let Spark read all years and handle
parallelism

df_all_historical = spark.read \
    .schema(sales_schema) \
    .parquet("/mnt/bronze/sales/historical/*/")
    # Glob pattern reads all year folders simultaneously!

# Spark distributes file reading across cluster
# 250M records read and processed in parallel
df_all_transformed = transform_sales(df_all_historical)

df_all_transformed.write \
    .mode("overwrite") \

```

```
.partitionBy("year", "month", "region_code") \  
.parquet("/mnt/silver/sales/")
```

◆ Incremental Data Load

```
# SCENARIO: Daily incremental load – only new/changed records
```

```
# — METHOD: Watermark-based incremental —————
```

```
# Read last watermark from control table
```

```
watermark_df = spark.read \  
    .format("jdbc") \  
    .option("url", jdbc_url) \  
    .option("query", "SELECT last_watermark FROM ControlTable  
WHERE table_name='SALES'") \  
    .load()
```

```
last_watermark = watermark_df.first()["last_watermark"]  
print(f>Loading records after: {last_watermark}")
```

```
# Read only new records from Bronze
```

```
df_incremental = spark.read.parquet("/mnt/bronze/sales/") \  
    .filter(F.col("created_timestamp") > last_watermark)
```

```
record_count = df_incremental.count()  
print(f>New records to process: {record_count:,}")
```

```
if record_count == 0:  
    print("No new records – skipping")  
    dbutils.notebook.exit("NO_NEW_DATA")
```

```
# Transform
```

```

df_transformed = transform_sales(df_incremental)

# Write incrementally (append only)
df_transformed.write \
    .mode("append") \
    .partitionBy("year", "month") \
    .parquet("/mnt/silver/sales/")

# Update watermark
new_watermark =
df_incremental.agg(F.max("created_timestamp")).first()[0]
# Update control table via JDBC
spark.sql(f"""
    UPDATE ControlTable
    SET last_watermark = '{new_watermark}'
    WHERE table_name = 'SALES'
""")

print(f"✅ Incremental load complete. New watermark:
{new_watermark}")

```

◆ Complex Aggregations

```

# RETAILMART: Daily Business Intelligence Aggregations

df_sales = spark.read.parquet("/mnt/silver/sales/2024/")

# — Store Performance Dashboard —————
store_performance = df_sales \
    .groupBy("store_id", "store_name", "city", "state") \
    .agg(
        F.count("sale_id").alias("total_transactions"),

```

```

    F.sum("total_amount").alias("gross_revenue"),
    F.sum("gst_amount").alias("total_gst"),
    F.sum("net_amount").alias("net_revenue"),
    F.avg("total_amount").alias("avg_transaction_value"),
    F.max("total_amount").alias("highest_transaction"),
    F.min("total_amount").alias("lowest_transaction"),
    F.countDistinct("customer_id").alias("unique_customers"),

    F.countDistinct("product_code").alias("unique_products_sold"),
    F.sum(F.when(F.col("payment_mode")== "cash",
    F.col("total_amount"))
        .otherwise(0)).alias("cash_revenue"),
    F.sum(F.when(F.col("payment_mode").isin(["card", "upi"]),
    F.col("total_amount"))
        .otherwise(0)).alias("digital_revenue"),
    F.sum(F.when(F.col("is_weekend")==True,
    F.col("total_amount"))
        .otherwise(0)).alias("weekend_revenue")
    )

```

— Ranking Stores

```

rank_window = Window.orderBy(F.col("net_revenue").desc())
state_window =
Window.partitionBy("state").orderBy(F.col("net_revenue").desc())

store_ranked = store_performance \
    .withColumn("national_rank", F.rank().over(rank_window)) \
    .withColumn("state_rank", F.rank().over(state_window)) \
    .withColumn("revenue_tier",
        F.when(F.col("national_rank") <= 10, "Platinum")
        .when(F.col("national_rank") <= 50, "Gold")
        .when(F.col("national_rank") <= 100, "Silver")
        .otherwise("Bronze"))

store_ranked.write \

```

```
.mode("overwrite") \  
.format("delta") \  
.saveAsTable("gold.store_performance_daily")
```

15 Databricks Integration

◆ Integration with ADLS Gen2

```
# APPROACH 1: Mount (covered in DBFS section)  
# APPROACH 2: Direct access using abfss:// (for sensitive  
workloads)  
  
spark.conf.set(  
    f"fs.azure.account.auth.type.  
{storage_account}.dfs.core.windows.net",  
    "OAuth"  
)  
spark.conf.set(  
    f"fs.azure.account.oauth.provider.type.  
{storage_account}.dfs.core.windows.net",  
  
    "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider"  
)  
spark.conf.set(  
    f"fs.azure.account.oauth2.client.id.  
{storage_account}.dfs.core.windows.net",  
    dbutils.secrets.get("kv-scope", "sp-client-id")  
)  
spark.conf.set(  
    f"fs.azure.account.oauth2.client.secret.  
{storage_account}.dfs.core.windows.net",  
    dbutils.secrets.get("kv-scope", "sp-client-secret")
```

```

)
spark.conf.set(
    f"fs.azure.account.oauth2.client.endpoint.
    {storage_account}.dfs.core.windows.net",
    f"https://login.microsoftonline.com/{tenant_id}/oauth2/token"
)

df =
spark.read.parquet(f"abfss://{bronze@{storage_account}.dfs.core.windo

```

◆ Integration with Azure SQL Database

```

# JDBC CONNECTION (full flexibility)
jdbc_url = (
    f"jdbc:sqlserver://{server}.database.windows.net:1433;"
    f"database={database};"
    f"encrypt=true;"
    f"trustServerCertificate=false;"
    f"hostnameInCertificate=*.database.windows.net;"
    f"loginTimeout=30"
)

jdbc_props = {
    "user":      dbutils.secrets.get("kv-scope", "sql-username"),
    "password":  dbutils.secrets.get("kv-scope", "sql-password"),
    "driver":    "com.microsoft.sqlserver.jdbc.SQLServerDriver"
}

# READ from SQL (with partitioning for large tables)
df_sql = spark.read \
    .jdbc(
        url=jdbc_url,

```

```

        table="dbo.Sales",
        column="sale_id",      # partition column (numeric)
        lowerBound=1,          # min value
        upperBound=1000000,    # max value (from separate query)
        numPartitions=16,      # parallel reads!
        properties=jdbc_props
    )

# WRITE to SQL
df_result.write \
    .mode("append") \
    .option("batchsize", "100000") \
    .option("tableLock", "true") \
    .jdbc(url=jdbc_url, table="dbo.Sales_Silver",
properties=jdbc_props)

# BULK LOAD (faster for large writes)
df_result.write \
    .format("com.databricks.spark.sqldw") \
    .option("url", jdbc_url) \
    .option("forwardSparkAzureStorageCredentials", "true") \
    .option("dbTable", "dbo.Sales_Silver") \
    .option("tempDir",
"wasbs://temp@mystorageaccount.blob.core.windows.net/sqldw-temp")
\
    .mode("overwrite") \
    .save()

```

16 Databricks Streaming API

◆ Structured Streaming — Complete Guide

SCENARIO: RetailMart wants **real-time** fraud detection

- **Every** payment transaction must be checked **within 2** seconds
- Flag transactions > **Rs.1,00,000** **from new** devices
- Alert fraud team immediately

STREAMING CONCEPTS:

Input: Continuous stream **of** payment events
arriving **every** millisecond

Unbounded **Table** concept:

Spark treats stream **as** an infinite **table**
New data = **new rows** appended continuously
Your query runs continuously **on** this **table**

TRIGGER TYPES:

ProcessingTime("10 seconds")	→ micro-batch every 10s
Once()	→ run once, process all data, stop
AvailableNow()	→ process all pending data, stop
Continuous("1 second")	→ true continuous (experimental)

OUTPUT MODES:

Append:	Only new rows written to sink (default) Use for : streaming events, no aggregation
Update:	Only rows that changed since last write Use for : streaming aggregations
Complete:	Entire result table written each micro-batch Use for : small aggregations (totals, counts)

— *REAL-TIME PAYMENT FRAUD DETECTION* —

```
from pyspark.sql.types import *  
from pyspark.sql import functions as F
```

```
# Payment event schema
```

```
payment_schema = StructType([  
    StructField("payment_id",    StringType(),    True),  
    StructField("customer_id",   StringType(),    True),  
    StructField("store_id",      IntegerType(),   True),  
    StructField("amount",        DoubleType(),    True),  
    StructField("device_id",     StringType(),    True),  
    StructField("location",      StringType(),    True),  
    StructField("timestamp",     TimestampType(), True),  
    StructField("payment_mode",  StringType(),    True)  
])
```

```
# — READ STREAM from ADLS (files arriving continuously) —
```

```
df_payment_stream = spark.readStream \  
    .format("cloudFiles") \  
    .option("cloudFiles.format", "json") \  
    .schema(payment_schema) \  
    .load("/mnt/landing/payments/stream/")
```

```
# — FRAUD DETECTION TRANSFORMATIONS —
```

```
df_analyzed = df_payment_stream \  
    .withColumn("is_high_value",  
        F.col("amount") > 100000) \  
    .withColumn("hour_of_day",  
        F.hour(F.col("timestamp"))) \  
    .withColumn("is_unusual_hour",  
        F.col("hour_of_day").isin([0,1,2,3,4])) \  
    .withColumn("fraud_risk_score",  
        F.when((F.col("is_high_value")) &  
(F.col("is_unusual_hour")), 3)  
        .when(F.col("is_high_value"), 2)  
        .when(F.col("is_unusual_hour"), 1)  
        .otherwise(0)) \  
    .withColumn("fraud_flag",
```

```

        F.col("fraud_risk_score") >= 2) \
        .withColumn("processing_timestamp",
            F.current_timestamp())

# — WRITE STREAM to Delta Table (append mode) —————
query_all = df_analyzed.writeStream \
    .outputMode("append") \
    .format("delta") \
    .option("checkpointLocation", "/mnt/checkpoints/payments/") \
    .trigger(processingTime="10 seconds") \
    .table("silver.payment_events")

# — WRITE ONLY FRAUD ALERTS separately —————
query_fraud = df_analyzed \
    .filter(F.col("fraud_flag") == True) \
    .writeStream \
    .outputMode("append") \
    .format("delta") \
    .option("checkpointLocation",
"/mnt/checkpoints/fraud_alerts/") \
    .trigger(processingTime="5 seconds") \
    .table("gold.fraud_alerts")

# — MONITOR STREAMING QUERY —————
print(query_all.status)
print(query_all.lastProgress)
# Shows: inputRowsPerSecond, processedRowsPerSecond,
triggerExecution

# Stop stream
query_all.stop()
query_fraud.stop()

```

◆ Checkpointing and Fault Tolerance

CHECKPOINTING EXPLAINED:

Checkpoint stores:

1. Current query state (watermarks, aggregations)
2. What data has been processed (offsets)
3. Metadata about streaming query

WHY IT MATTERS:

Payment stream processes 10,000 events/second

Cluster fails at 3:47 AM after processing 1M events

WITHOUT checkpoint:

Restart at 3:48 AM → starts over from beginning

Re-processes 1M events → DUPLICATE FRAUD ALERTS!

WITH checkpoint:

Restart at 3:48 AM → reads checkpoint

Knows: "I processed up to offset 1,000,000"

Continues from offset 1,000,001

Exactly-once processing guaranteed!

CHECKPOINT LOCATION RULES:

- ✓ Each streaming query needs its OWN checkpoint location
- ✓ Store in reliable storage (ADLS, not local disk)
- ✓ Don't delete checkpoint manually (breaks stream)
- ✓ If query logic changes → use new checkpoint location

query.writeStream

```
.option("checkpointLocation",  
"/mnt/checkpoints/payments_v2/")  
# v2 = new checkpoint for new version of query
```

17 Databricks Auto Loader

◆ What is Auto Loader?

AUTO LOADER = Incremental file ingestion **at** scale

THE PROBLEM it solves:

1000 CSV files arrive daily **in** ADLS landing zone

Naive approach (**using** `spark.read`):

- `df = spark.read.csv("/mnt/landing/sales/*.csv")`
- **Reads ALL** files **every** run (including already processed!)
- Must manually track which files were processed
- Extra logic, complex, error-prone

Auto Loader approach (**using** `spark.readStream + cloudFiles`):

- Automatically tracks which files have been processed
- **Only** processes **NEW** files **each** run
- Uses Azure Event Grid/Queue internally
- Handles millions **of** files **per hour**
- Schema inference **and** evolution built-**in**

— AUTO LOADER SETUP —

Two detection modes:

1. DIRECTORY LISTING (simpler, works for < 10M files):

Lists directory each trigger, finds new files

2. FILE NOTIFICATION (scalable, for massive volumes):

Uses Azure Event Grid → notifications on new files

Instant detection, no listing overhead

— BASIC AUTO LOADER —

```

df_stream = spark.readStream \
    .format("cloudFiles") \
    .option("cloudFiles.format", "csv") \
    .option("cloudFiles.schemaLocation", "/mnt/checkpoints/sales-
schema/") \
    .option("header", "true") \
    .load("/mnt/landing/sales/")

# Write incrementally to Delta
query = df_stream.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation", "/mnt/checkpoints/sales-
loader/") \
    .trigger(availableNow=True) \
    .table("bronze.sales_raw")

query.awaitTermination()

# — SCHEMA INFERENCE AND EVOLUTION —————

# Schema Inference (first run):
# Auto Loader samples files to infer schema
# Stores schema at schemaLocation
# Uses same schema for all subsequent runs

# Schema Hints (enforce specific types):
df_stream_typed = spark.readStream \
    .format("cloudFiles") \
    .option("cloudFiles.format", "csv") \
    .option("cloudFiles.schemaLocation", "/mnt/checkpoints/sales-
schema/") \
    .option("cloudFiles.schemaHints",
            "sale_id INT, total_amount DOUBLE, sale_date DATE") \
    .option("header", "true") \

```

```

        .load("/mnt/landing/sales/")

# Schema Evolution Modes:
.option("cloudFiles.schemaEvolutionMode", "addNewColumns")
# Options:
# addNewColumns → New columns in CSV? Add to Delta table schema
(recommended)
# failOnNewColumns → New column = pipeline error (strict mode)
# rescue → Unknown columns saved to _rescued_data column
# none → Ignore new columns entirely

# — PRODUCTION AUTO LOADER PATTERN —————
def run_auto_loader(
    source_path,
    target_table,
    checkpoint_path,
    schema_path,
    trigger_once=True
):
    """Production-grade Auto Loader function"""

    stream = spark.readStream \
        .format("cloudFiles") \
        .option("cloudFiles.format", "parquet") \
        .option("cloudFiles.schemaLocation", schema_path) \
        .option("cloudFiles.schemaEvolutionMode",
"addNewColumns") \
        .option("cloudFiles.useIncrementalListing", "true") \
        .load(source_path) \
        .withColumn("ingestion_timestamp", F.current_timestamp())
\
        .withColumn("source_file", F.input_file_name())

    writer = stream.writeStream \
        .format("delta") \

```

```

        .outputMode("append") \
        .option("checkpointLocation", checkpoint_path) \
        .option("mergeSchema", "true") \
        .queryName(f"auto_loader_{target_table}")

    if trigger_once:
        query =
writer.trigger(availableNow=True).table(target_table)
        query.awaitTermination()
    else:
        query = writer.trigger(processingTime="1
minute").table(target_table)
    return query

# Run for each source
run_auto_loader("/mnt/landing/sales/",      "bronze.sales",
                "/mnt/ckpt/sales/",         "/mnt/schema/sales/")
run_auto_loader("/mnt/landing/customers/",  "bronze.customers",
                "/mnt/ckpt/customers/",
"/mnt/schema/customers/")

```

18 Delta Lake — The Lakehouse Foundation

◆ Data Lake vs Delta Lake

DATA LAKE (plain Parquet/CSV files):

PROBLEM 1: No ACID transactions

Two jobs write to same folder simultaneously

→ Partial writes → corrupted data!

PROBLEM 2: No schema enforcement

Job 1 writes: sale_id INT, amount DOUBLE

Job 2 writes: sale_id STRING, amount STRING (bug!)

→ Data lake now has mixed types → downstream breaks!

PROBLEM 3: No UPDATE or DELETE

Need to fix a wrong price in 50M records?

→ Must rewrite entire file/partition!

→ Cannot do DELETE WHERE sale_id = 1001

PROBLEM 4: Cannot read while writing

Report runs at 9 AM → pipeline starts writing at 9:01 AM

→ Report reads PARTIAL data → wrong numbers!

→ Called "read-write conflict"

PROBLEM 5: No history / no rollback

Bug in pipeline writes wrong data to 6 months of history

→ No way to go back to correct version!

DELTA LAKE solves ALL of these:

- ✓ ACID transactions
- ✓ Schema enforcement + evolution
- ✓ UPDATE, DELETE, MERGE (upsert)
- ✓ Reads never blocked by writes (snapshot isolation)
- ✓ Time travel (go back to any version)
- ✓ Audit history (see every change ever made)

◆ Delta Lake Internals

HOW DELTA LAKE WORKS:

Storage structure:

```

/mnt/silver/sales/
├── _delta_log/                ← TRANSACTION LOG (the secret!)
│   ├── 00000000000000000000000000000000.json ← version 0 (initial
write)
│   ├── 00000000000000000000000000000001.json ← version 1 (appended
data)
│   ├── 00000000000000000000000000000002.json ← version 2 (updated
records)
│   ├── 00000000000000000000000000000003.json ← version 3 (deleted
records)
│   └── 00000000000000000000000000000010.checkpoint.parquet (every 10
versions)
├── part-0000-xxxx.snappy.parquet ← actual data files
├── part-0001-xxxx.snappy.parquet
└── year=2024/month=6/...         ← partition folders

```

TRANSACTION LOG (delta_log):

Each commit = one JSON file

JSON records:

- Which files were ADDED (new data written)
- Which files were REMOVED (deleted or replaced)
- Schema changes
- Commit timestamp + user + operation

READING Delta table:

1. Read latest transaction log
2. Determine which data files are "active" (added but not removed)
3. Read only those Parquet files
 - Snapshot isolation: your read sees consistent state
 - Even if another job is writing simultaneously!

ACID in Delta:

Atomic: Entire write **succeeds or** nothing changes
Consistent: Schema enforced, **no** partial invalid data
Isolated: Concurrent readers/writers don't **interfere**
Durable: Committed data survives cluster failures

◆ Delta Lake Operations

```
# — CREATE DELTA TABLE —————

# Method 1: Write DataFrame as Delta
df_sales.write \
    .format("delta") \
    .mode("overwrite") \
    .partitionBy("year", "month") \
    .save("/mnt/silver/sales_delta/")

# Method 2: Create managed table in catalog
df_sales.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("silver.sales")

# Method 3: CREATE TABLE SQL
spark.sql("""
    CREATE TABLE IF NOT EXISTS silver.sales (
        sale_id      INT NOT NULL,
        customer_id   STRING NOT NULL,
        store_id      INT,
        total_amount  DOUBLE,
        sale_date     DATE,
        region_code   STRING,
        year          INT,
```

```
        month      INT
    )
    USING DELTA
    PARTITIONED BY (year, month)
    LOCATION '/mnt/silver/sales/'
    COMMENT 'Cleaned daily sales data'
    """)
```

— DML OPERATIONS —

INSERT (append)

```
spark.sql("""
    INSERT INTO silver.sales
    SELECT * FROM bronze.sales_raw
    WHERE sale_date = current_date()
    """)
```

UPDATE (not possible in plain Parquet!)

```
spark.sql("""
    UPDATE silver.sales
    SET total_amount = total_amount * 1.1,
        updated_at = current_timestamp()
    WHERE region_code = 'N'
        AND sale_date = '2024-06-15'
        AND product_code = 'LAPTOP01'
    """)
```

DELETE (not possible in plain Parquet!)

```
spark.sql("""
    DELETE FROM silver.sales
    WHERE sale_id IN (
        SELECT sale_id FROM error_records
        WHERE error_type = 'DUPLICATE'
    )
    """)
```

```

# — MERGE (UPSERT) – Most Important Operation —————
# SCD Type 1: Update if exists, Insert if new

from delta.tables import DeltaTable

delta_table = DeltaTable.forPath(spark, "/mnt/silver/customers/")

# New/updated customers from today's load
df_updates =
spark.read.parquet("/mnt/bronze/customers/2024/06/15/")

delta_table.alias("target").merge(
    df_updates.alias("source"),
    condition="target.customer_id = source.customer_id"
) \
.whenMatchedUpdate(
    # When customer exists AND something changed:
    condition="target.email != source.email OR target.phone !=
source.phone",
    set={
        "email":      "source.email",
        "phone":      "source.phone",
        "address":    "source.address",
        "updated_at": "current_timestamp()"
    }
) \
.whenNotMatchedInsert(
    # When customer doesn't exist yet:
    values={
        "customer_id": "source.customer_id",
        "name":        "source.name",
        "email":        "source.email",
        "phone":        "source.phone",
        "created_at":   "current_timestamp()",

```

```

        "updated_at": "current_timestamp()"
    }
) \
.execute()

print("Merge complete!")

```

◆ SCD Type 1 and Type 2

```

# — SCD TYPE 1: Overwrite (no history) —————
# Simple MERGE as shown above → just overwrites changed values

# — SCD TYPE 2: Keep full history —————
# Each change = new row with start/end dates

delta_customers = DeltaTable.forPath(spark,
"/mnt/silver/dim_customers_scd2/")

df_source =
spark.read.parquet("/mnt/bronze/customers/2024/06/15/") \
    .withColumn("effective_start", F.current_date())
\
    .withColumn("effective_end",    F.lit("9999-12-
31")).cast("date")) \
    .withColumn("is_current",      F.lit(True))

# Step 1: Expire existing records that have changed
delta_customers.alias("target").merge(
    df_source.alias("source"),
    """target.customer_id = source.customer_id
    AND target.is_current = true
    AND (target.email != source.email

```

```

        OR target.address != source.address)"""
    ) \
    .whenMatchedUpdate(set={
        "is_current": "false",
        "effective_end": "date_sub(current_date(), 1)"
    }) \
    .execute()

# Step 2: Insert new version of changed records + truly new
records
delta_customers.alias("target").merge(
    df_source.alias("source"),
    "target.customer_id = source.customer_id AND
target.is_current = true"
) \
.whenNotMatchedInsert(values={
    "customer_id": "source.customer_id",
    "name": "source.name",
    "email": "source.email",
    "address": "source.address",
    "effective_start": "source.effective_start",
    "effective_end": "source.effective_end",
    "is_current": "source.is_current"
}) \
.execute()

# Result in table:
# customer_id | email | effective_start | effective_end
# is_current
# C001 | old@mail.com | 2023-01-01 | 2024-06-14
# | false
# C001 | new@mail.com | 2024-06-15 | 9999-12-31
# | true ← current

```

◆ Time Travel

```
# TIME TRAVEL: Go back to any previous version!
# Useful for: debugging bad pipelines, auditing, data recovery

# — VIEW TABLE HISTORY —————
spark.sql("DESCRIBE HISTORY silver.sales").show(20,
truncate=False)
# Output:
# version | timestamp                | operation |
operationParameters
# 10      | 2024-06-15 02:15:00      | MERGE     | {predicate: ...}
# 9       | 2024-06-15 01:30:00      | WRITE     | {mode: Append}
# 8       | 2024-06-14 02:15:00      | MERGE     | ...
# ...

# — QUERY PREVIOUS VERSION —————
# By version number
df_v5 = spark.read \
    .format("delta") \
    .option("versionAsOf", "5") \
    .load("/mnt/silver/sales/")

# By timestamp (what did data look like yesterday at 9 PM?)
df_yesterday = spark.read \
    .format("delta") \
    .option("timestampAsOf", "2024-06-14 21:00:00") \
    .load("/mnt/silver/sales/")

# Using SQL
spark.sql("""
    SELECT * FROM silver.sales
    VERSION AS OF 5
    WHERE region_code = 'N'
    """)
```



```

spark.sql("""
    SELECT * FROM silver.sales
    TIMESTAMP AS OF '2024-06-14 21:00:00'
""")

# — REAL WORLD: ROLLBACK BAD PIPELINE —————
# Pipeline bug wrote wrong data in version 10
# Need to restore to version 9

# Option 1: Restore table to version 9
spark.sql("RESTORE TABLE silver.sales TO VERSION AS OF 9")

# Option 2: Overwrite with previous version
df_good = spark.read.format("delta") \
    .option("versionAsOf", "9") \
    .load("/mnt/silver/sales/")

df_good.write.format("delta").mode("overwrite").save("/mnt/silver/sa

```

◆ Data Deduplication in Delta Tables

```

# SCENARIO: Due to a pipeline bug, some records were loaded twice
# Need to remove duplicates from Delta table

from delta.tables import DeltaTable

# — METHOD 1: INSERT INTO with EXCEPT —————
# Insert only records that don't already exist

delta_table = DeltaTable.forPath(spark, "/mnt/silver/sales/")

```

```

df_new_data = spark.read.parquet("/mnt/bronze/sales/today/")

# Only insert if sale_id doesn't already exist
delta_table.alias("t").merge(
    df_new_data.alias("s"),
    "t.sale_id = s.sale_id"
) \
.whenNotMatchedInsertAll() \
.execute()

# — METHOD 2: Create deduplicated version —————
from pyspark.sql.window import Window

df_with_row_num = spark.read.format("delta") \
    .load("/mnt/silver/sales/") \
    .withColumn("rn",
        F.row_number().over(
            Window.partitionBy("sale_id")
                .orderBy(F.col("load_timestamp").desc())
        )
    )

df_deduplicated = df_with_row_num.filter(F.col("rn") ==
1).drop("rn")

# Overwrite table with deduplicated data
df_deduplicated.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/mnt/silver/sales/")

print(f"Original:
{spark.read.format('delta').load('/mnt/silver/sales/').count():,}")
print(f"After dedup: {df_deduplicated.count():,}")

```

19 Medallion Architecture — RetailMart Implementation

THE THREE LAYERS:

BRONZE (Raw Data):

- Exact **copy of** source data, **no** transformations
- Preserves original format + schema
- Immutable (never changed once written)
- Keeps **ALL** data including bad records
- Source **of** truth **for** reprocessing
- Parquet **or** Delta format
- Retention: **3-7** years (audit/compliance)

SILVER (Cleaned Data):

- Data quality rules applied
- Standardized **column** names **and** types
- Nulls handled, duplicates removed
- Basic business logic applied
- Joined **with** reference/dimension data
- Delta format (enables updates/deletes)
- Retention: **2-5** years

GOLD (Business-Ready Data):

- Aggregated, summarized **for specific** use cases
- Denormalized (wide tables **for** BI tools)
- **One table per** business question
- Optimized **for** fast query performance
- Delta format + Z-**ORDER for** performance
- Retention: **1-2** years (refreshed frequently)

```

# — BRONZE LAYER: Raw ingestion (no transformation) —————

def load_bronze(source_path, target_table, process_date):
    """Bronze: land exactly as-is from source"""

    df_raw = spark.read.parquet(source_path)

    # Only add ingestion metadata
    df_bronze = df_raw \
        .withColumn("_ingestion_timestamp",
F.current_timestamp()) \
        .withColumn("_source_file",          F.input_file_name())
    \
        .withColumn("_process_date",        F.lit(process_date))
    \
        .withColumn("_source_system",        F.lit("Oracle_ERP"))

    # Append to bronze (keep full history!)
    df_bronze.write \
        .format("delta") \
        .mode("append") \
        .partitionBy("_process_date") \
        .saveAsTable(f"bronze.{target_table}")

    print(f"Bronze loaded: {df_bronze.count():,} rows → bronze.
{target_table}")

# — SILVER LAYER: Clean and conform —————

def load_silver_sales(process_date):
    """Silver: clean, validate, enrich sales data"""

    # Read from Bronze
    df_bronze = spark.read.table("bronze.sales") \
        .filter(F.col("_process_date") ==

```

```

process_date)

# Data Quality
dq_rules = {
    "sale_id_not_null": F.col("SALE_ID").isNotNull(),
    "amount_positive": F.col("AMOUNT") > 0,
    "valid_status":
F.col("STATUS").isin(["COMPLETED", "PENDING"]),
    "valid_date": F.col("SALE_DATE").isNotNull()
}

df_valid = df_bronze
for rule_name, condition in dq_rules.items():
    before = df_valid.count()
    df_valid = df_valid.filter(condition)
    dropped = before - df_valid.count()
    if dropped > 0:
        print(f"⚠️ {rule_name}: dropped {dropped:,} rows")

# Standardize
df_silver = df_valid \
    .select(
        F.col("SALE_ID").cast("int").alias("sale_id"),
        F.upper(F.trim("CUSTOMER_ID")).alias("customer_id"),
        F.col("STORE_ID").cast("int").alias("store_id"),
        F.col("AMOUNT").cast("double").alias("total_amount"),
        F.to_date("SALE_DATE", "DD-MON-
YYYY").alias("sale_date"),
        F.upper("REGION_CODE").alias("region_code"),
        F.lower("PAYMENT_MODE").alias("payment_mode"),
        F.col("STATUS").alias("status")
    ) \
    .withColumn("year", F.year("sale_date")) \
    .withColumn("month", F.month("sale_date")) \
    .withColumn("gst_amount",

```

```

F.round(F.col("total_amount") * 0.18, 2)) \
    .withColumn("net_amount",      F.col("total_amount") -
F.col("gst_amount")) \
    .withColumn("updated_at",      F.current_timestamp())

# Merge into Silver Delta table (upsert)
delta_silver = DeltaTable.forName(spark, "silver.sales")

delta_silver.alias("t").merge(
    df_silver.alias("s"),
    "t.sale_id = s.sale_id"
) \
    .whenMatchedUpdateAll() \
    .whenNotMatchedInsertAll() \
    .execute()

print(f"Silver updated: {df_silver.count():,} rows")

# — GOLD LAYER: Business aggregations —————

def load_gold_store_daily(process_date):
    """Gold: daily store performance summary"""

    df_silver = spark.read.table("silver.sales") \
        .filter(F.col("sale_date") == process_date)

    df_stores = spark.read.table("silver.dim_stores")

    df_gold = df_silver \
        .join(F.broadcast(df_stores), on="store_id") \
        .groupBy(
            "sale_date", "store_id", "store_name",
            "city", "state", "region_code"
        ) \
        .agg(

```

```

        F.count("sale_id").alias("transaction_count"),
        F.sum("total_amount").alias("gross_revenue"),
        F.sum("net_amount").alias("net_revenue"),
        F.avg("total_amount").alias("avg_ticket_size"),

F.countDistinct("customer_id").alias("unique_customers"),
        F.sum(F.when(F.col("payment_mode")=="cash",
                        F.col("total_amount")).otherwise(0))
                        .alias("cash_revenue"),

F.sum(F.when(F.col("payment_mode").isin(["card", "upi"]),
            F.col("total_amount")).otherwise(0))
            .alias("digital_revenue")
    ) \
    .withColumn("digital_pct",
        F.round(F.col("digital_revenue") /
F.col("gross_revenue") * 100, 1))

# Optimize for BI queries
df_gold.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere", f"sale_date = '{process_date}'")
\
    .saveAsTable("gold.store_daily_performance")

# Optimize table for fast queries
spark.sql("""
    OPTIMIZE gold.store_daily_performance
    ZORDER BY (store_id, sale_date)
""")

print(f"Gold layer updated for {process_date}")

```

— MASTER ORCHESTRATOR

```
def run_medallion_pipeline(process_date):
    print(f"\n🚀 Starting Medallion Pipeline for {process_date}")

    # Bronze
    load_bronze(f"/mnt/landing/sales/{process_date}/", "sales",
process_date)
    load_bronze(f"/mnt/landing/customers/{process_date}/",
"customers", process_date)

    # Silver
    load_silver_sales(process_date)
    load_silver_customers(process_date)

    # Gold
    load_gold_store_daily(process_date)
    load_gold_product_performance(process_date)
    load_gold_customer_360(process_date)

    print(f"\n✅ Medallion Pipeline Complete for {process_date}")

run_medallion_pipeline(process_date)
```

20 Workflows in Databricks

DATABRICKS WORKFLOWS = Native job scheduler

vs ADF:

ADF: Better for multi-service orchestration
DB Workflows: Better for Databricks-only workloads

WORKFLOW TYPES:

Notebook task: Run a specific notebook
Python script: Run a .py file
JAR task: Run a Java/Scala JAR
SQL task: Run a SQL query/dashboard
dbt task: Run dbt models
Delta Live Tables: Run DLT pipeline
Pipeline task: Chain multiple tasks with dependencies

```
# — CREATE JOB via UI —————  
# Workflows → Create Job  
# Add Tasks:  
#   Task 1: load_bronze_sales      (notebook:  
#   /Production/bronze/load_sales)  
#   Task 2: load_bronze_customers (notebook:  
#   /Production/bronze/load_customers)  
#   Task 3: transform_silver      (depends on Task 1 + 2)  
#   Task 4: load_gold             (depends on Task 3)  
#   Task 5: notify_completion     (depends on Task 4)  
  
# Schedule: Daily at 1:00 AM IST  
# Cluster: Job Cluster (auto-creates, auto-terminates)  
# Alerts: Email on failure  
  
# — JOB PARAMETERS —————  
# Pass parameters to all tasks in job:  
# Key: process_date  Value: {{job.start_time.iso_date}}  
# → Databricks automatically substitutes today's date!  
  
# — MONITOR JOB RUNS —————  
# Workflows → Job → Runs tab  
# See each run: duration, status, task breakdown  
# Click failed task → view error + logs  
  
# — REPAIR RUN (re-run only failed tasks) —————  
# Job failed at Task 3 (transform_silver)
```

```
# Tasks 1, 2 succeeded → don't rerun them!  
# Repair Run → only reruns Task 3, 4, 5  
# Saves time and cost vs full re-run!
```

21 Unity Catalog — Enterprise Governance

◆ What is Unity Catalog?

THE PROBLEM BEFORE UNITY CATALOG:

RetailMart has 3 workspaces:

Workspace 1: Data Engineering team

Workspace 2: Data Science team

Workspace 3: BI/Analytics team

Old way (Hive Metastore):

- ✗ Each workspace has its OWN separate metastore
- ✗ "silver.sales" in WS1 ≠ "silver.sales" in WS2
- ✗ Data Science can't easily access Data Engineering tables
- ✗ Security configured separately per workspace
- ✗ No unified audit log ("who accessed what when?")
- ✗ No data lineage ("where did this table come from?")
- ✗ DBA must set up permissions in each workspace separately
- ✗ PII data (customer phone) visible to everyone!

UNITY CATALOG SOLUTION:

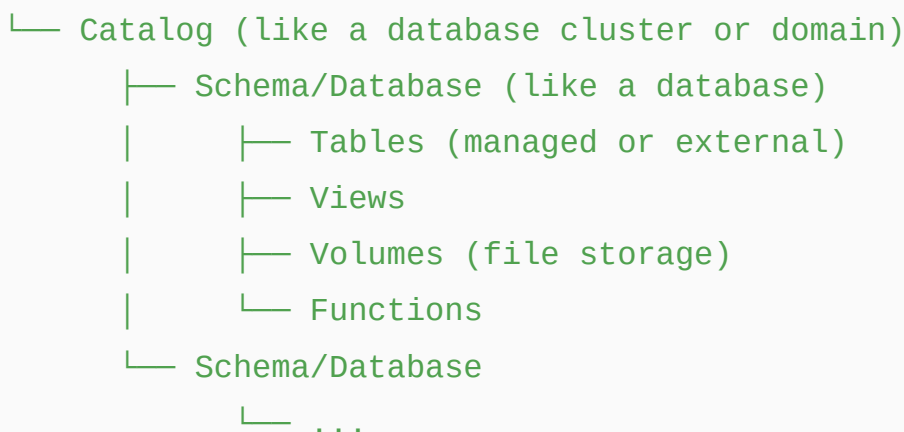
- ✓ ONE metastore for ALL workspaces in a region
- ✓ Tables registered once → accessible from all workspaces
- ✓ Centralized security → set permission once, applies everywhere
- ✓ Unified audit log → every query logged centrally

- ✓ Data lineage → visualize data flow table-to-table
- ✓ Column-level masking for PII
- ✓ Row-level security

2 2 Unity Catalog Object Hierarchy

UNITY CATALOG HIERARCHY:

Metastore (1 per Azure region)



RETAILMART EXAMPLE:

Metastore: retailmart_metastore_india



```

|   |─ Schema: sales
|   |   |─ Table: sales_fact          (managed)
|   |   |─ Table: sales_daily        (managed)
|   |─ Schema: dimensions
|   |   |─ Table: dim_stores
|   |   |─ Table: dim_products
|   |   |─ Table: dim_customers
|
|─ Catalog: gold                      (aggregated)
|   |─ Schema: analytics
|   |   |─ Table: store_performance
|   |   |─ Table: customer_360
|
|─ Catalog: sandbox                  (for data scientists -
experimentation)
|   |─ Schema: <each user's personal schema>

```

— TABLE TYPES —

MANAGED TABLE (Unity Catalog owns the data):

If you drop the table → DATA DELETED from storage

```

spark.sql("""
    CREATE TABLE silver.sales.transactions (
        sale_id      INT,
        customer_id   STRING,
        total_amount  DOUBLE,
        sale_date     DATE
    )
    USING DELTA
    COMMENT 'Cleaned sales transactions'
    TBLPROPERTIES ('owner' = 'data_engineering_team')
""")

```

EXTERNAL TABLE (you own the data):

If you drop the table → ONLY METADATA deleted, data safe!

```

spark.sql("""
    CREATE TABLE bronze.sales.transactions_raw
    USING DELTA
    LOCATION
    'abfss://bronze@retailmartlake.dfs.core.windows.net/sales/'
    COMMENT 'Raw Oracle sales data - do not modify'
""")

# VOLUMES (access files through Unity Catalog):
spark.sql("""
    CREATE VOLUME bronze.sales.raw_files
    LOCATION
    'abfss://bronze@retailmartlake.dfs.core.windows.net/raw/'
    COMMENT 'Raw CSV/Parquet files from source systems'
""")

# Access: /Volumes/bronze/sales/raw_files/2024/06/sales.parquet

```

23 Access Control & Security (RBAC)

```

# — GRANT/REVOKE PERMISSIONS —

# Grant catalog-level access
spark.sql("GRANT USE CATALOG ON CATALOG silver TO
`data_science_team`")
spark.sql("GRANT USE CATALOG ON CATALOG gold TO `bi_team`")

# Grant schema-level access
spark.sql("GRANT USE SCHEMA ON SCHEMA silver.sales TO
`data_science_team`")
spark.sql("GRANT USE SCHEMA ON SCHEMA gold.analytics TO
`bi_team`")

```

```
# Grant table-level access
spark.sql("GRANT SELECT ON TABLE silver.sales.transactions TO
`data_science_team`")
spark.sql("GRANT SELECT ON TABLE gold.analytics.store_performance
TO `bi_team`")

# Grant write permissions (engineering team only)
spark.sql("GRANT MODIFY ON TABLE silver.sales.transactions TO
`data_engineering_team`")

# Grant ALL permissions (table owner)
spark.sql("GRANT ALL PRIVILEGES ON TABLE
silver.sales.transactions TO `de_lead`")

# Revoke permissions
spark.sql("REVOKE SELECT ON TABLE silver.sales.transactions FROM
`temp_contractor`")

# Check current permissions
spark.sql("SHOW GRANTS ON TABLE
silver.sales.transactions").show()

# — ROLES HIERARCHY —
# Metastore Admin → full control over everything
# Catalog Owner   → full control over one catalog
# Schema Owner    → full control over one schema
# Table Owner     → full control over one table
# User            → only what explicitly granted
```

24 Data Protection & Privacy

```

# — COLUMN-LEVEL MASKING (hide PII) —————
# Scenario: Customer phone numbers should be hidden from BI team
# but visible to Data Engineering team

# Create masking function
spark.sql("""
    CREATE FUNCTION IF NOT EXISTS silver.utils.mask_phone(phone
STRING)
    RETURN CASE
        WHEN is_member('data_engineering_team') THEN phone
        ELSE CONCAT(LEFT(phone, 3), 'XXXXXXX')
    END
""")

# Apply masking policy to column
spark.sql("""
    ALTER TABLE silver.dimensions.dim_customers
    ALTER COLUMN phone
    SET MASK silver.utils.mask_phone
""")

# BI team sees:    +91XXXXXXX (masked)
# DE team sees:    +9198765XXXXX (original)

# — ROW-LEVEL SECURITY —————
# Scenario: Regional managers see only their region's data
# North India manager → only sees North India sales

# Create row filter function
spark.sql("""
    CREATE FUNCTION silver.utils.region_filter(region_code
STRING)
    RETURN is_member('admin_group')
        OR region_code = current_user_region()
""")

```

```

# (current_user_region() is a custom function mapping user to
region)

# Apply row filter to table
spark.sql("""
    ALTER TABLE silver.sales.transactions
    SET ROW FILTER silver.utils.region_filter ON (region_code)
""")

# North India manager runs: SELECT COUNT(*) FROM
silver.sales.transactions
# → Sees only North India rows automatically!
# Admin runs same query → sees ALL rows

# — SECURE VIEWS FOR SENSITIVE DATA —————
spark.sql("""
    CREATE OR REPLACE VIEW gold.analytics.customer_summary_safe
    AS
    SELECT
        customer_id,
        customer_name,
        CONCAT(LEFT(email, 3), '***@***.com') AS email_masked,
        CONCAT('XXXXXXXX', RIGHT(phone, 2)) AS phone_masked,
        customer_tier,
        total_purchases,
        last_purchase_date
    FROM silver.dimensions.dim_customers
""")
-- BI team uses this view instead of raw customer table
-- PII is protected, business metrics still available

```



```
# — INTEGRATING DATABRICKS WITH DEVOPS REPOS —————

# In Databricks Workspace:
# User Settings → Git Integration
# Git Provider: Azure DevOps Services
# Personal Access Token: (from Azure DevOps)

# — REPOS STRUCTURE —————
# RetailMart Databricks Repos structure:

# Azure DevOps Repo: databricks-notebooks
# |
# └─ bronze/
#   │   └─ load_sales_bronze.py
#   │   └─ load_customers_bronze.py
#   │   └─ load_inventory_bronze.py
#   └─ silver/
#       │   └─ transform_sales.py
#       │   └─ transform_customers.py
#       │   └─ common_utils.py
#       └─ gold/
#           │   └─ store_performance.py
#           │   └─ customer_360.py
#           └─ tests/
#               │   └─ test_sales_transform.py
#               │   └─ test_data_quality.py
#               └─ config/
#                   └─ env_config.py

# In Databricks Repos tab:
# + Add Repo → URL:
https://dev.azure.com/retailmart/DataPlatform/_git/databricks-
notebooks
# Clone to: /Repos/data_engineering_team/databricks-notebooks
```

```
# — WORKFLOW —  
# 1. Pull latest changes from main branch  
# 2. Create feature branch: feature/add-returns-pipeline  
# 3. Develop notebooks in Databricks UI  
# 4. Changes auto-saved to branch  
# 5. Push to Azure DevOps  
# 6. Raise Pull Request  
# 7. Code review → approve  
# 8. Merge to main → CI/CD pipeline deploys to prod workspace
```

```
# — DATABRICKS REPOS API —  
import subprocess  
  
# Pull latest changes (in notebook or CI/CD script)  
def git_pull(repo_path, branch="main"):  
    result = subprocess.run(  
        ["git", "-C", f"/Workspace/Repos/{repo_path}", "pull"],  
        capture_output=True, text=True  
    )  
    print(result.stdout)  
    return result.returncode == 0
```

26 SDLC and Agile for Databricks Projects

RETAILMART DATABRICKS SPRINT EXAMPLE:

SPRINT 1 (Week 1-2):

- Story 1: Set up Databricks workspace and clusters
- Story 2: Configure ADLS Gen2 mounts and Key Vault
- Story 3: Build Bronze layer ingestion for Sales
- Story 4: Build Bronze layer ingestion for Customers

SPRINT 2 (Week 3-4):

- Story 5: Build Silver transformation for Sales
- Story 6: Implement SCD Type 2 for Customer dimension
- Story 7: Data quality framework setup
- Story 8: Unit tests for transformations

SPRINT 3 (Week 5-6):

- Story 9: Build Gold layer store performance
- Story 10: Implement streaming fraud detection
- Story 11: Unity Catalog setup and permissions
- Story 12: Performance optimization (caching, partitioning)

DEFINITION OF DONE for each notebook:

- ✓ Runs on development cluster without errors
- ✓ Unit tests written and passing
- ✓ Data quality checks implemented
- ✓ Logging and error handling added
- ✓ Code reviewed by peer
- ✓ Pushed to Git repo
- ✓ Documentation in first cell (purpose, inputs, outputs)
- ✓ Tested with production-scale data sample

27 End-to-End Data Migration Project

PROJECT: Migrate RetailMart from On-Premises to Azure Cloud

PHASE 1 — ASSESSMENT (Week 1-2):

- Inventory all source systems (Oracle, SQL Server, SAP)
- Document all tables, sizes, row counts
- Identify PII data (for masking rules)

- Define target schema (column naming conventions)
- Estimate data volume: 5 years × 2.5 TB = 12.5 TB

PHASE 2 — INFRASTRUCTURE SETUP (Week 3-4):

- Create Azure subscription, resource groups
- Set up ADLS Gen2 (Bronze/Silver/Gold containers)
- Create Databricks workspace + Unity Catalog
- Install SHIR on on-premise servers
- Configure ADF with all Linked Services
- Set up Key Vault for all secrets
- Configure Azure DevOps repo

PHASE 3 — HISTORICAL MIGRATION (Week 5-8):

- Load 5 years of historical data
- Oracle → ADLS Bronze (via ADF SHIR)
- Bronze → Silver (Databricks transformations)
- Silver → Gold (aggregations)
- Validate: row counts, checksums, business rules
- Reconcile reports: old system vs new system

PHASE 4 — PARALLEL RUN (Week 9-12):

- Both old and new systems run simultaneously
- Daily comparison of results
- Fix any discrepancies
- Business team validates new reports
- Performance testing under production load

PHASE 5 — CUTOVER (Week 13-14):

- Switch all pipelines to production ADF
- Business team switches to new Power BI dashboards
- Monitor closely for 2 weeks
- Decommission old on-premise processes

PHASE 6 — STABILIZATION (Week 15-16):

- Monitor all pipelines daily

- Performance tuning based **on** actual patterns
- Documentation **and** knowledge transfer
- Handover **to** operations team

28 Databricks Interview Questions

Q1: What **is** the difference **between** RDD, DataFrame, **and** Dataset?

RDD: Low-level, untyped, **no** optimization

DataFrame: High-level, schema-aware, Catalyst optimized

Dataset: Typed DataFrame (**only in** Scala/Java)

For 99% of PySpark work → use DataFrame

RDD → **only when** you need byte-level control

Q2: Explain lazy evaluation **in** Spark. Why **is** it important?

Transformations (**filter**, map, **select**) → lazy (**no** execution)

Actions (count, write, **collect**) → **trigger** execution

Importance:

- Catalyst sees the **WHOLE** transformation chain
- Can optimize (pushdown, pruning, **join** strategy)
- Single pass **over** data instead **of** multiple passes
- Example: **filter** pushed **to** file reader = less data read

Q3: What **is** a shuffle? How do you minimize it?

Shuffle = redistribution **of** data across executors **by** key

Happens **in**: groupBy, **join**, **distinct**, sort, repartition

Expensive because: network transfer, disk write/read

Minimize **by**:

- Use broadcast **join for** small tables
- **Filter**/project before joins (less data **to** shuffle)
- Repartition **by join** key before multiple joins **on** same key
- Use reduceByKey instead **of** groupByKey (combines locally **first**)
- Enable AQE (auto-optimizes shuffle partitions)

Q4: Explain Delta Lake **and** its benefits **over** plain Parquet.

Delta Lake = Parquet + Transaction Log

Benefits:

- ✓ ACID transactions (concurrent safe writes)
- ✓ **UPDATE/DELETE/MERGE** (**not** possible **in** Parquet)
- ✓ Schema enforcement (prevents bad data)
- ✓ Schema evolution (**add** columns safely)
- ✓ **Time** travel (query historical versions)
- ✓ Streaming + batch unified (same **table**)
- ✓ Automatic compaction **and** optimization

Q5: What **is** the Medallion Architecture?

Bronze: Raw landing data, exact **copy from** source

Silver: Cleaned, validated, conformed data

Gold: Aggregated, business-ready data

Benefits:

- Separation **of** concerns (**each** layer has **one** job)
- Reprocessability (Bronze → rerun Silver/Gold anytime)
- Multiple consumers (BI uses Gold, DS uses Silver)
- Audit trail (Bronze = complete history)

Q6: How do you handle slowly changing dimensions (SCD) **in** Databricks?

SCD Type 1 (overwrite):

→ **MERGE** statement **with** whenMatchedUpdate + whenNotMatchedInsert

SCD Type 2 (history):

→ **MERGE** to expire **old** record (**set** is_current=**false**, end_date)

→ **INSERT** **new** record (is_current=**true**, start_date=today)

→ Maintains **full** history of **all** changes

Q7: What **is** the difference **between** cache() **and** persist()?

cache() = persist() **with default** storage level MEMORY_AND_DISK

persist() = choose **specific** storage level:

MEMORY_ONLY, MEMORY_AND_DISK, DISK_ONLY, MEMORY_ONLY_SER

Use cache(): **when** data fits **in** memory

Use persist(MEMORY_AND_DISK): larger data, occasional reuse

Always unpersist() **when** done!

Q8: What **is** Unity Catalog? How does it differ **from** Hive Metastore?

Hive Metastore:

→ **Per**-workspace (**each** workspace isolated)

→ **No** central governance

→ Limited security (**table**-level **only**)

→ **No** audit log

Unity Catalog:

→ Across **all** workspaces **in** a region

→ Centralized governance

→ Fine-grained: catalog/schema/**table**/**column**/**row** level

→ Unified audit **log** (**every** query tracked)

→ Data lineage visualization

→ **Column** masking + **row** filtering

Q9: Explain Auto Loader and when to use it.

Auto Loader = incremental file ingestion using cloudFiles format

Uses: Azure Event Grid to detect new files instantly

Use when:

- Files arrive continuously in ADLS landing zone
- Want only NEW files processed (not all files)
- Need schema evolution (new columns handled automatically)
- Millions of files (regular glob listing too slow)

vs spark.read: Auto Loader tracks processed files automatically
spark.read reads ALL matching files every time

Q10: How do you optimize a slow Spark job?

Step 1: Check Spark UI → find slowest stage

Step 2: Look for:

- Skew (one partition much larger) → salting
- Shuffle (large shuffle size) → reduce data before shuffle
- Small files (thousands of tiny files) → coalesce before write
- Missing broadcast (SortMerge when Broadcast possible)

Common fixes:

- Enable AQE (handles skew, coalesces partitions automatically)
- Broadcast small tables
- Filter/select columns early
- Repartition by join key before multiple joins
- Use Parquet with partition pruning
- Cache reused DataFrames
- Tune shuffle partitions based on data size



Complete Databricks Architecture Summary

AZURE DATABRICKS – COMPLETE PICTURE

FOUNDATION | Hadoop → Spark (100x faster, in-memory)

RDD → DataFrame → SparkSQL

SPARK CORE | Lazy Evaluation, DAG, Stages, Tasks

Catalyst Optimizer, AQE, Tungsten

Narrow vs Wide transforms, Shuffle

PERFORMANCE | Broadcast join, Cache/Persist

Partitioning, Repartition vs Coalesce

Skew handling (salting), Predicate pushdown

DELTA LAKE | ACID, MERGE, UPDATE, DELETE

Time Travel, Schema Evolution

SCD1 + SCD2, Deduplication

Medallion Architecture (Bronze/Silver/Gold)



STREAMING	Structured Streaming, readStream/writeStream
	Triggers, Output Modes, Checkpointing
	Auto Loader (CloudFiles)
CLUSTER MGMT	All-Purpose vs Job Cluster
	Standard vs High Concurrency
	Autoscaling, Runtime versions
GOVERNANCE	Unity Catalog
	(Metastore→Catalog→Schema→Table)
	RBAC (GRANT/REVOKE), Column Masking
	Row-level Security, Audit Logs, Lineage
INTEGRATION	ADLS Gen2, Azure SQL, Key Vault
	ADF (orchestrates Databricks notebooks)
	Azure DevOps Repos + CI/CD

 **Azure Databricks section 100% complete** — every concept from the PDF including BigData foundations, Hadoop, Spark internals, RDD, DataFrame, SQL, Optimizer, Joins, Performance, Streaming, Auto Loader, Delta Lake, Medallion Architecture, Cluster Management, Batch Processing, Unity Catalog, RBAC, Data Protection, DevOps integration, SDLC, and Interview Questions — all covered with real-world RetailMart examples.

19/03/2026, 07:43:03

Thought process: Orchestrated comprehensive Azure Synapse curriculum aligned with Ansh Lamba's pedagogical approach.

The user wants me to continue with Azure Synapse Analytics as per the PDF content, and also follow Ansh Lamba's style of explaining. Let me recall Ansh Lamba's teaching style - he's known for explaining Azure data engineering concepts in a very practical, real-world way with clear examples, analogies, and step-by-step explanations. He uses a lot of diagrams, real scenarios, and explains "why" before "how".

Let me cover all the Azure Synapse topics from the PDF:

1) Introduction & Overview

- Azure Synapse Analytics Overview
- Azure Synapse Analytics Architecture
- Create Azure Free Account for Synapse

2) Now I need to dive into the different pool types available in Synapse - dedicated SQL pools for heavy workloads, serverless SQL pools for flexible querying, and Apache Spark pools for big data processing. Then I'll walk through querying data lakes with serverless SQL, creating views for CSV data, and integrating storage accounts with Synapse for bulk file operations using wildcards and dataset parameters.

3) Moving into triggers and scheduling - I'll cover schedule-based triggers for regular jobs and event-based triggers that respond to data changes. Then I'll explore SQL Database integration, showing how to copy data from SQL databases to data

lakes using different methods like table queries and stored procedures, plus handling append and overwrite modes with foreach loops for batch operations.

4) I'm getting into the deeper copy activity patterns - loading from on-premises SQL servers to Synapse using polybase and bulk insert, plus copying from file systems on-premises to cloud. Then I'll introduce Spark pools, covering the fundamentals of Spark and its distributed architecture.

5) Now diving into Spark notebooks - creating them in different languages, using MSSparkUtils for file operations and parameters, magic commands for notebook orchestration, and configuring KeyVault access. I'm covering multiple connection patterns to ADLS Gen2, Blob storage, Azure SQL, and on-premises SQL servers, then optimizing file I/O for CSV, Parquet, JSON, Avro, ORC, and Excel formats, plus creating RDDs and dataframes in various ways.

6) Moving into performance tuning - understanding when to repartition versus coalesce, handling joins and broadcast joins with Spark configuration, diving into the Catalyst optimizer and data skewness issues, then applying broader optimization techniques across PySpark. I'm also covering slowly changing dimensions with SCD1 and SCD2 implementations, and finally orchestrating these notebooks from Synapse pipelines with parameter passing.



Azure Synapse Analytics — Complete Senior Engineer Guide



Same Company: RetailMart India Pvt. Ltd.



SITUATION SO FAR:

- ✓ ADF ingesting data from 6 source systems → Bronze layer
- ✓ Databricks transforming → Silver and Gold layers
- ✓ Delta Lake managing ACID transactions

NEW REQUIREMENT from business:

- ✗ BI team cannot query 500GB ADLS parquet files directly
- ✗ Data Scientists need SQL access to Gold layer
- ✗ CFO wants to run ad-hoc SQL queries without waiting
- ✗ Need enterprise data warehouse for reporting
- ✗ On-premise SQL Server data needs direct loading to warehouse

SOLUTION: Azure Synapse Analytics

1 Introduction & Overview

◆ What is Azure Synapse Analytics?

Azure Synapse Analytics is Microsoft's **unified analytics platform** that brings together:

- **Data Integration** (like ADF pipelines)
- **Enterprise Data Warehousing** (like SQL Server but massively scalable)
- **Big Data Analytics** (like Databricks Spark)
- **Data Exploration** (query files directly without loading)

All in **one single workspace**.

BEFORE SYNAPSE (Old World):

Tool 1: Azure Data Factory → for pipelines

Tool 2: Azure Databricks → for big data processing
Tool 3: Azure SQL DW → for data warehouse
Tool 4: Azure HDInsight → for Spark/Hadoop

Problems:

- ✗ 4 different tools to learn and manage
- ✗ 4 separate billing and cost management
- ✗ Moving data between tools → complexity + cost
- ✗ Security configured in 4 different places
- ✗ No unified monitoring

AFTER SYNAPSE (New World):

Azure Synapse Analytics = ONE platform

- ✓ Pipelines (like ADF) built-in
- ✓ Serverless SQL (query data lake directly)
- ✓ Dedicated SQL Pool (enterprise data warehouse)
- ✓ Apache Spark Pool (big data processing)
- ✓ Unified monitoring, security, billing
- ✓ Deep integration with ADLS Gen2 and Power BI

Real World Analogy:

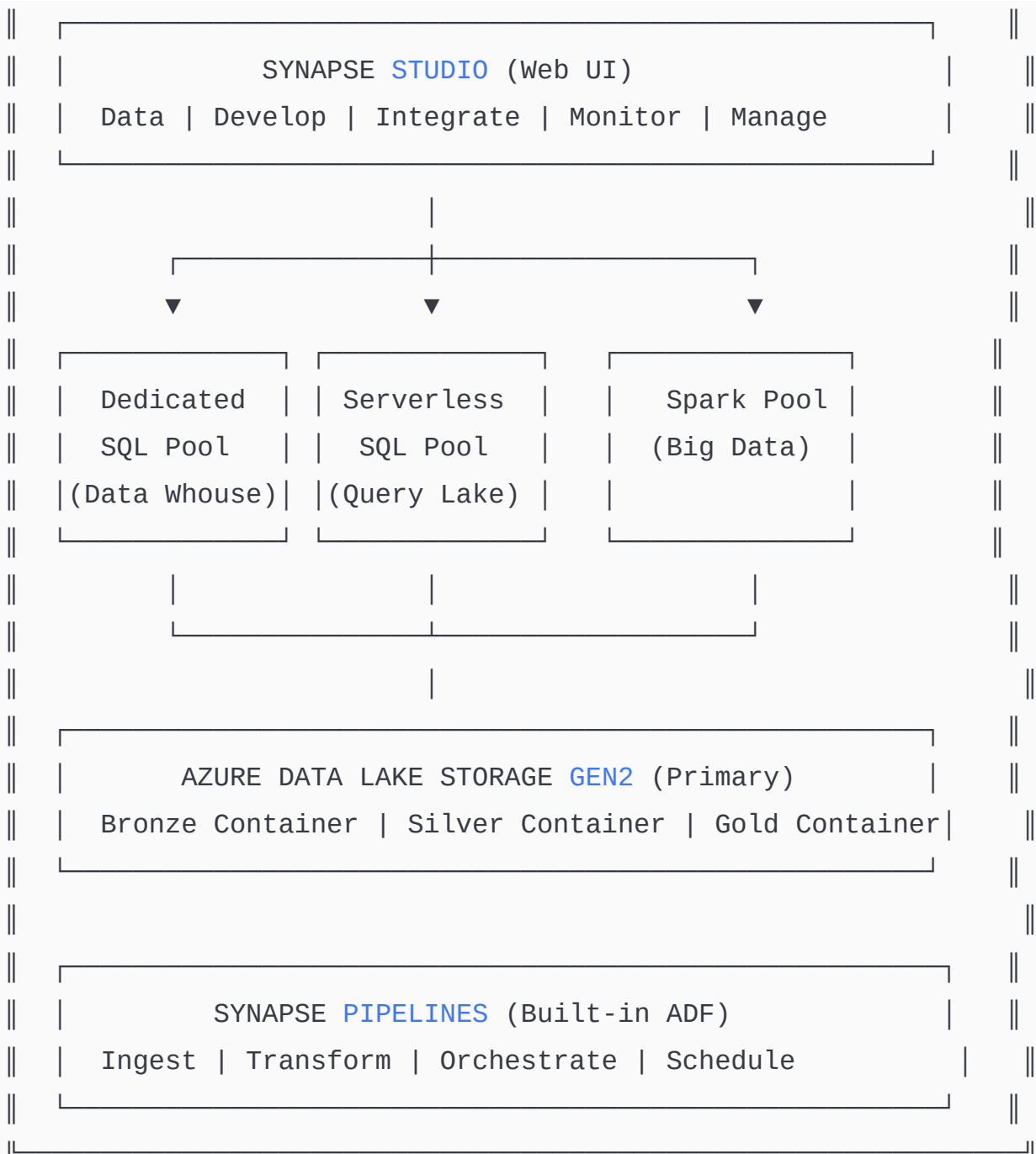
Think of Azure Synapse as a **Swiss Army Knife** for data. Instead of carrying 4 separate tools (knife, scissors, screwdriver, file), you carry ONE tool that does everything. Same concept — one platform, multiple capabilities.

◆ Azure Synapse Analytics Architecture



A diagram consisting of a large rectangular box with a double-line border. Inside the box, the text "AZURE SYNAPSE ANALYTICS ARCHITECTURE" is centered. The box is flanked by vertical double lines on both sides, suggesting it is part of a larger architectural context.

AZURE SYNAPSE ANALYTICS ARCHITECTURE



Key Components Explained:

DEDICATED SQL POOL:

- Massively Parallel Processing (MPP) data warehouse
- Pre-allocated compute (you pay whether used or not)
- Best for: regular BI queries, predictable workloads
- Scale: up to petabytes of data
- Like: traditional SQL Server but 1000x more powerful

SERVERLESS SQL POOL:

- NO infrastructure to manage
- Query files DIRECTLY in ADLS (CSV, Parquet, JSON, Delta)
- Pay per query (per TB of data scanned)
- No data loading needed!
- Like: Athena on AWS – SQL on files
- Best for: ad-hoc exploration, data lake queries

APACHE SPARK POOL:

- Same Spark as Databricks but inside Synapse
- PySpark, Scala, R notebooks
- Direct integration with Synapse tables
- Best for: heavy transformations, ML
- Auto-pause when idle (cost saving)

SYNAPSE PIPELINES:

- Exactly same as Azure Data Factory
- 90+ connectors, same activities/triggers
- Built INTO Synapse workspace
- Benefit: pipelines live next to your data/compute

◆ Create Azure Synapse Workspace — Step by Step

SETUP STEPS:

Step 1: Azure Portal → Create Resource → Azure Synapse Analytics

Step 2: Basics Tab

Subscription:	RetailMart-Production
Resource Group:	RG-RetailMart-DataPlatform
Workspace Name:	synapse-retailmart-prod

Region: Central India

Step 3: Storage Account (IMPORTANT!)

Select or create ADLS Gen2 account

ADLS Account: retailmartdatalake

File System: synapseworkspace ← workspace's own container

WHY: Synapse needs its own container for:

- Spark logs
- Synapse metadata
- Pipeline run history
- Staging for PolyBase loads

Step 4: SQL Administrator credentials

SQL Admin Username: sqladmin

Password: (strong password → will also store in Key Vault)

Step 5: Security Tab

Managed Identity: System-assigned (for accessing storage)

Allow connections: Add your IP for development

Step 6: Review + Create → Deploy (takes 5-7 minutes)

Step 7: After deployment → Open Synapse Studio

URL: <https://web.azuresynapse.net>

Workspace: synapse-retailmart-prod

2 Overview of Pools in Synapse Analytics

◆ Pool Comparison — When to Use What

SCENARIO → BEST POOL:

"CFO wants to explore some CSV files in data lake"

→ SERVERLESS SQL POOL (no setup, immediate, pay per query)

"BI team runs Power BI dashboards 8 hours/day"

→ DEDICATED SQL POOL (consistent performance, pre-allocated)

"Data scientist needs to train ML model on 500GB data"

→ SPARK POOL (distributed processing, Python/ML support)

"Need to transform JSON files to Parquet daily"

→ SPARK POOL or SERVERLESS SQL (depending on complexity)

"Need to join data lake files with SQL tables"

→ SERVERLESS SQL POOL (can join files + tables in one query!)

Cost Comparison for RetailMart:

Serverless SQL: \$5 per TB scanned (exploration queries)

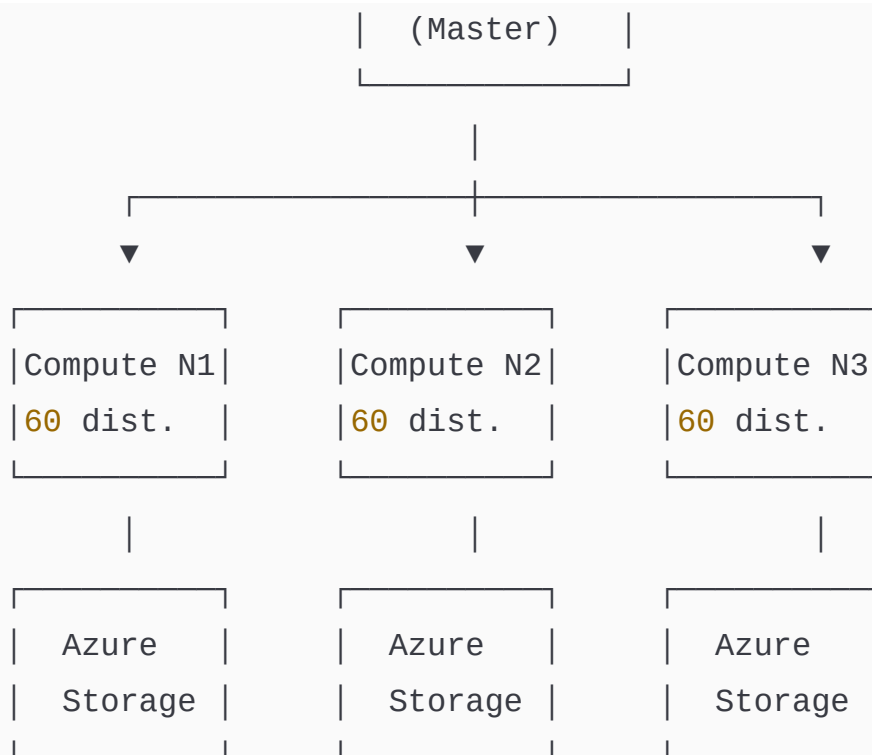
Dedicated Pool: ~\$744/month for DW100c (always on)

Spark Pool: Pay per hour used (auto-pause when idle)

◆ Dedicated SQL Pool — Deep Dive

HOW DEDICATED SQL POOL WORKS (MPP Architecture):





HOW IT WORKS:

1. Your query hits Control Node
2. Control Node generates parallel execution plan
3. Plan distributed to all Compute Nodes simultaneously
4. Each node processes its portion of data
5. Results aggregated back at Control Node
6. Single result returned to you

KEY CONCEPT - DISTRIBUTIONS:

Dedicated SQL Pool has 60 FIXED distributions

Data stored across these 60 distributions

Parallel queries run against all 60 simultaneously

DWUs (Data Warehouse Units) = your scale setting

DW100c: 1 compute node → 60 distributions / 1 node

DW1000c: 10 compute nodes → 60 distributions / 10 nodes

DW6000c: 60 compute nodes → 1 distribution / 1 node (max parallelism)

DISTRIBUTION TYPES:

HASH: Best for large fact tables (distribute by join key)
Same join key → same distribution → join without shuffle!

ROUND ROBIN: Default, data spread evenly across all 60
Best for staging tables, smaller tables

REPLICATED: Full copy on EVERY compute node
Best for small dimension tables (< 2GB)
Eliminates broadcast operations!

```
-- CREATE TABLE WITH OPTIMAL DISTRIBUTION
-- Large fact table → Hash distributed by most common join key
CREATE TABLE gold.fact_sales (
    sale_id          INT          NOT NULL,
    customer_id      INT          NOT NULL,
    product_key      INT          NOT NULL,
    store_key        INT          NOT NULL,
    date_key         INT          NOT NULL,
    quantity         INT,
    unit_price       DECIMAL(18,2),
    total_amount     DECIMAL(18,2),
    net_amount       DECIMAL(18,2),
    gst_amount       DECIMAL(18,2),
    payment_mode     VARCHAR(50)
)
WITH (
    DISTRIBUTION = HASH(customer_id), -- distribute by most
joined column
    CLUSTERED COLUMNSTORE INDEX      -- best for analytics!
);

-- Small dimension table → Replicated (full copy on each node)
CREATE TABLE gold.dim_stores (
    store_key        INT          NOT NULL,
```

```

    store_id      VARCHAR(20)    NOT NULL,
    store_name    VARCHAR(200),
    city          VARCHAR(100),
    state         VARCHAR(100),
    region_code   VARCHAR(10),
    manager_name  VARCHAR(200)
)
WITH (
    DISTRIBUTION = REPLICATE,          -- copy to all nodes
    CLUSTERED COLUMNSTORE INDEX
);

-- Staging table → Round Robin (fast parallel insert)
CREATE TABLE staging.stg_sales_load (
    sale_id       INT,
    customer_id   INT,
    total_amount  DECIMAL(18,2),
    sale_date     DATE
)
WITH (
    DISTRIBUTION = ROUND_ROBIN,        -- fast parallel load
    HEAP                                     -- no index for staging (faster
insert)
);

```

◆ Serverless SQL Pool — Deep Dive

SERVERLESS SQL = Query ADLS files directly using SQL

NO cluster to create

NO data to load

NO indexing needed

Just write SQL → point at file → get results!

MAGIC: OPENROWSET function

```
-- — QUERY CSV FILE DIRECTLY —————  
SELECT TOP 10 *  
FROM OPENROWSET(  
    BULK  
    'https://retailmartdatalake.dfs.core.windows.net/bronze/sales/2024/01/sales_2024_01.csv',  
    FORMAT = 'CSV',  
    PARSER_VERSION = '2.0',  
    HEADER_ROW = TRUE,  
    FIELDTERMINATOR = ',',  
    ROWTERMINATOR = '\n'  
) AS sales_data;  
  
-- — QUERY PARQUET FILES (much faster!) —————  
SELECT  
    region_code,  
    COUNT(*) AS transaction_count,  
    SUM(total_amount) AS gross_revenue,  
    AVG(total_amount) AS avg_transaction  
FROM OPENROWSET(  
    BULK  
    'https://retailmartdatalake.dfs.core.windows.net/silver/sales/**/*.parquet',  
    FORMAT = 'PARQUET'  
) AS sales_data  
WHERE sale_date >= '2024-01-01'  
GROUP BY region_code  
ORDER BY gross_revenue DESC;  
  
-- — QUERY WITH EXPLICIT SCHEMA (recommended for production) —  
SELECT
```

```

    sale_id,
    customer_id,
    total_amount,
    sale_date,
    region_code
FROM OPENROWSET(
    BULK
    'https://retailmartdatalake.dfs.core.windows.net/silver/sales/year=2
    FORMAT = 'PARQUET'
)
WITH (
    sale_id      INT,
    customer_id  VARCHAR(20),
    total_amount FLOAT,
    sale_date    DATE,
    region_code  VARCHAR(10)
) AS sales_data
WHERE sale_date BETWEEN '2024-06-01' AND '2024-06-30';

-- — QUERY JSON FILES —————
SELECT
    JSON_VALUE(doc, '$.payment_id') AS payment_id,
    JSON_VALUE(doc, '$.amount')     AS amount,
    JSON_VALUE(doc, '$.status')     AS status,
    JSON_VALUE(doc, '$.customer_id') AS customer_id
FROM OPENROWSET(
    BULK
    'https://retailmartdatalake.dfs.core.windows.net/bronze/payments/*.j
    FORMAT = 'CSV',
    FIELDQUOTE = '0x0b',
    FIELDTERMINATOR = '0x0b',
    ROWTERMINATOR = '0x0a'
)
WITH (doc NVARCHAR(MAX)) AS payment_docs;

```

```
-- — QUERY MULTIPLE FOLDERS (partition pruning!) —————
-- Only reads year=2024/month=6 folder, skips all others!
SELECT *
FROM OPENROWSET(
    BULK
    'https://retailmartdatalake.dfs.core.windows.net/silver/sales/year=2
    FORMAT = 'PARQUET'
) AS [result];
```

3 Using Synapse to Query Data Lake

◆ Creating Synapse Workspace and Exploring Studio

SYNAPSE STUDIO NAVIGATION:

Left Sidebar (5 main sections):

	HOME	→ Quick links, recent resources
	DATA	→ Browse databases, tables, files
	DEVELOP	→ Notebooks, SQL scripts, Dataflows
	INTEGRATE	→ Pipelines (same as ADF!)
	MONITOR	→ Pipeline runs, Spark jobs
	MANAGE	→ Pools, linked services, IR

DATA Tab:

Workspace section:

- SQL databases (Dedicated + Serverless)
- Spark databases (registered in catalog)

Linked section:

- Your ADLS Gen2 storage
- Browse folders and files directly
- Right-click any file → "New SQL Script" or "New Notebook"
- Amazing for exploration!

DEVELOP Tab:

- SQL Scripts (run against Serverless or Dedicated pool)
- Notebooks (PySpark, Scala, .NET Spark, SparkSQL)
- Power BI datasets (embedded)
- Data Flows (visual transformations)

INTEGRATE Tab:

- Exactly same as ADF Studio
- Create pipelines, datasets, linked services
- All triggers, copy activity, ForEach, etc.

MONITOR Tab:

- Pipeline runs
- Trigger runs
- Apache Spark applications
- SQL requests
- Integration runtimes

◆ Creating Views on Data Lake Files

```
-- — WHY VIEWS OVER OPENROWSET? —————  
-- Instead of sharing long OPENROWSET queries with BI team:  
-- Create VIEWS that hide complexity behind simple table names!  
  
-- Step 1: Create database for Serverless SQL views  
CREATE DATABASE RetailMart_Reporting;
```

```

-- Step 2: Create master key (for security)
CREATE MASTER KEY ENCRYPTION BY PASSWORD = 'str0ng_P@ssw0rd!';

-- Step 3: Create credential to access ADLS Gen2
-- Option A: Using Managed Identity (recommended)
CREATE DATABASE SCOPED CREDENTIAL synapse_managed_identity
WITH IDENTITY = 'Managed Identity';

-- Option B: Using SAS Token
CREATE DATABASE SCOPED CREDENTIAL synapse_sas_token
WITH IDENTITY = 'SHARED ACCESS SIGNATURE',
SECRET = 'sv=2020-08...'; -- SAS token from ADLS

-- Step 4: Create External Data Source (reusable connection)
CREATE EXTERNAL DATA SOURCE adls_silver
WITH (
    LOCATION =
'https://retailmartdatalake.dfs.core.windows.net/silver',
    CREDENTIAL = synapse_managed_identity
);

CREATE EXTERNAL DATA SOURCE adls_bronze
WITH (
    LOCATION =
'https://retailmartdatalake.dfs.core.windows.net/bronze',
    CREDENTIAL = synapse_managed_identity
);

-- Step 5: Create External File Format
CREATE EXTERNAL FILE FORMAT parquet_format
WITH (
    FORMAT_TYPE = PARQUET,
    DATA_COMPRESSION =
'org.apache.hadoop.io.compress.SnappyCodec'
);

```

```
CREATE EXTERNAL FILE FORMAT csv_format
WITH (
    FORMAT_TYPE = DELIMITEDTEXT,
    FORMAT_OPTIONS (
        FIELD_TERMINATOR = ',',
        STRING_DELIMITER = '"',
        FIRST_ROW = 2, -- skip header
        ENCODING = 'UTF8'
    )
);
```

-- Step 6: Create VIEWS for BI team (simple, clean)

```
CREATE OR ALTER VIEW RetailMart_Reporting.dbo.vw_Sales_Silver AS
SELECT
    sale_id,
    customer_id,
    store_id,
    product_code,
    total_amount,
    net_amount,
    gst_amount,
    sale_date,
    year,
    month,
    region_code,
    payment_mode
FROM OPENROWSET(
    BULK 'sales/**/*.*parquet',
    DATA_SOURCE = 'adls_silver',
    FORMAT = 'PARQUET'
)
WITH (
    sale_id INT,
    customer_id VARCHAR(20),
```

```

    store_id      INT,
    product_code  VARCHAR(50),
    total_amount  FLOAT,
    net_amount    FLOAT,
    gst_amount    FLOAT,
    sale_date     DATE,
    year          INT,
    month         INT,
    region_code   VARCHAR(10),
    payment_mode  VARCHAR(50)
) AS [result];

-- Step 7: Create View for CSV data (store targets file from
business)
CREATE OR ALTER VIEW RetailMart_Reporting.dbo.vw_StoreTargets AS
SELECT
    store_id,
    month,
    year,
    CAST(revenue_target AS FLOAT) AS revenue_target,
    CAST(transaction_target AS INT) AS transaction_target
FROM OPENROWSET(
    BULK 'config/store_targets_*.csv',
    DATA_SOURCE = 'adls_bronze',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    HEADER_ROW = TRUE
) AS [targets];

-- Now BI team just queries the VIEW:
-- SELECT * FROM RetailMart_Reporting.dbo.vw_Sales_Silver
-- WHERE sale_date >= '2024-06-01'
-- Much simpler! They don't need to know ADLS paths.

-- Step 8: Join VIEW with real SQL table (powerful!)

```

```
SELECT
    s.region_code,
    s.sale_date,
    SUM(s.total_amount) AS actual_revenue,
    t.revenue_target,
    ROUND(SUM(s.total_amount) / t.revenue_target * 100, 1) AS
achievement_pct
FROM RetailMart_Reporting.dbo.vw_Sales_Silver s
JOIN RetailMart_Reporting.dbo.vw_StoreTargets t
    ON s.store_id = t.store_id
    AND MONTH(s.sale_date) = t.month
    AND YEAR(s.sale_date) = t.year
GROUP BY s.region_code, s.sale_date, t.revenue_target
ORDER BY s.sale_date, achievement_pct DESC;
```

4 Azure Storage Integration with Synapse Pipelines

◆ Copy Multiple Files Using Wildcard Options

SCENARIO:

ADLS has thousands of store-wise CSV files:

/bronze/store-reports/store_001_2024.csv

/bronze/store-reports/store_002_2024.csv

...

/bronze/store-reports/store_500_2024.csv

Copy ALL at once to processed container using wildcard

PIPELINE: PL_Synapse_CopyStoreFiles

STEP 1: Source Dataset

Name: DS_StoreCSV_Wildcard

Linked Service: LS_ADLS_Gen2_Bronze

Container: bronze

→ Leave file path EMPTY in dataset

STEP 2: Copy Activity → Source Tab

Dataset: DS_StoreCSV_Wildcard

File Path Type: Wildcard file path

Wildcard Folder Path: store-reports/

Wildcard File Name: store_*_2024.csv

Recursive: FALSE

DIFFERENT PATTERNS:

store_*_2024.csv → all stores for 2024

store_00?_*.csv → stores 001-009 all years

*_2024_June.csv → all stores for June 2024

*.csv → ALL CSV files in folder

STEP 3: Sink Dataset

Name: DS_ProcessedStore

Container: silver

Folder: store-reports/processed/

STEP 4: Copy Behavior

PreserveHierarchy → maintain structure

FlattenHierarchy → all files in one folder

MergeFiles → combine all into ONE file

REAL WORLD TIP:

Use MergeFiles when:

→ 500 small store files → merge into one big file

→ Single file = faster downstream processing

→ Power BI loads one file faster than 500 files

◆ Copy Multiple Folders Using Dataset Parameters

SCENARIO:

ADLS has monthly folders:

/bronze/sales/2024/01/

/bronze/sales/2024/02/

...

/bronze/sales/2024/12/

Need to copy any month dynamically using parameters

STEP 1: Create Parameterized Dataset

Name: DS_Monthly_Sales_Param

Container: bronze

Parameters:

p_year: String (e.g., "2024")

p_month: String (e.g., "06")

Folder Path: sales/@{dataset().p_year}/@{dataset().p_month}/

File: *.parquet (all parquet files in that month folder)

STEP 2: Pipeline PL_CopyMonthlyFolders

Parameters:

p_year: String → "2024"

p_month_list: Array → ["01", "02", "03", "04", "05", "06"]

STEP 3: ForEach over months

Items: @{pipeline().parameters.p_month_list}

Batch Count: 3 (copy 3 months in parallel)

Inside ForEach:

Copy Activity:

Source Dataset: DS_Monthly_Sales_Param

p_year: @{pipeline().parameters.p_year}

```
p_month: @{{item()}} ← current month in loop
```

```
Sink Dataset: DS_Silver_Sales_Param
```

```
p_year: @{{pipeline().parameters.p_year}}
```

```
p_month: @{{item()}}
```

RESULT:

3 months copy simultaneously → much faster!

Pass different months = different data without changing pipeline

◆ Get File Names Dynamically — Copy Latest File

REAL WORLD SCENARIO:

Payment gateway drops a file daily:

```
/bronze/payments/payments_20240613.csv
```

```
/bronze/payments/payments_20240614.csv
```

```
/bronze/payments/payments_20240615.csv ← latest
```

Pipeline should always pick the LATEST file automatically
Without hardcoding the date!

```
PIPELINE: PL_CopyLatestPaymentFile
```

STEP 1: GetMetadata – List all files

```
Dataset: DS_PaymentFolder
```

```
Container: bronze
```

```
Folder: payments/
```

```
Field List: childItems, lastModified
```

STEP 2: ForEach – Find latest file

```
Items: @{{activity('GetMeta_ListFiles').output.childItems}}
```


Sequential: YES

Inside ForEach:

If Condition:

Condition: @greater(

```
activity('GetMeta_EachFile').output.lastModified,  
        variables('v_latestModified')  
    )}
```

TRUE:

```
    Set v_latestModified =  
    @{activity('GetMeta').output.lastModified}  
    Set v_latestFileName = @{item().name}
```

STEP 3: Copy Activity – Copy latest file

Source: /bronze/payments/@{variables('v_latestFileName')}

Sink: /silver/payments/latest/payment_latest.parquet

ALSO: Set Variable

v_processedFile = @{variables('v_latestFileName')}

STEP 4: Stored Procedure – Log which file was processed

@run_date: @utcnow()

@file_processed: @{variables('v_latestFileName')}

@pipeline_run: @{pipeline().RunId}

SMARTER APPROACH (if files follow date naming):

Use expression directly:

File name: payments_@{formatDateTime(utcnow(), 'yyyyMMdd')}.csv

→ Today's date automatically → pick today's file!

If files land at night for next day:

File name:

payments_@{formatDateTime(addDays(utcnow(), -1), 'yyyyMMdd')}.csv

→ Yesterday's date

5 Azure Synapse Triggers

◆ Schedule Trigger in Azure Synapse

SAME AS ADF BUT INSIDE SYNAPSE WORKSPACE:

Integrate Tab → Trigger → + New/Edit → Schedule

Configuration:

Name: TRG_Synapse_Daily_Load

Type: Schedule

Start Date: 2024-01-01 00:30:00

Time Zone: India Standard Time

Recurrence: Every 1 Day

Hours: 0 (midnight)

Minutes: 30 (half past midnight)

→ Runs at 12:30 AM IST every night

KEY DIFFERENCE from ADF triggers:

Synapse triggers live inside Synapse workspace

ADF triggers live in ADF service

Use Synapse triggers: when pipeline is inside Synapse

Use ADF triggers: when pipeline is inside ADF

RetailMart setup:

ADF: handles complex cross-service orchestration

Synapse: handles Synapse-specific data warehouse loads

PASS TRIGGER TIME TO PIPELINE:

Pipeline parameter: p_loadDate

```
Value: @{formatDateTime(trigger().scheduledTime, 'yyyy-MM-dd')}
```

In pipeline, use `p_loadDate` in source queries:

```
WHERE sale_date = '@{pipeline().parameters.p_loadDate}'
```

◆ Event-Based Trigger in Azure Synapse

SCENARIO:

When a new file arrives in ADLS → trigger Synapse pipeline
Used for near-real-time data loading

Setup:

Integrate → Trigger → New → Storage Event

Storage Account: `retailmartdatalake`

Container: `bronze`

Blob Path Begins With: `/payments/daily/`

Blob Path Ends With: `.parquet`

Event: `BlobCreated`

Pipeline Parameters:

`p_file_path: @{triggerBody().folderPath}`

`p_file_name: @{triggerBody().fileName}`

ADVANTAGE over Schedule:

Schedule: pipeline runs at midnight EVEN IF **no** file arrived

Event: pipeline ONLY runs when file actually arrives

→ Less wasted compute

→ Faster processing (no waiting for schedule)

→ Perfect for event-driven architecture

REAL WORLD TIMELINE:

2:47 AM: Payment gateway uploads payments_20240615.parquet to /bronze/payments/daily/
2:47 AM: Azure Event Grid detects new blob (within seconds)
2:47 AM: Synapse event trigger fires
2:48 AM: Pipeline starts processing payment file
2:52 AM: Data available in Gold layer

vs Schedule at 3:00 AM: same data available 8 minutes LATER!

6 Azure SQL Database Integration

◆ Azure SQL Databases — Relational Databases in Azure

SQL DATABASE TYPES IN AZURE:

Azure SQL Database:

- Fully managed PaaS SQL Server
- Microsoft handles: backups, patches, HA, DR
- Single database model
- Best for: OLTP applications, small-medium DW
- RetailMart uses: transaction database, audit logs

Azure SQL Managed Instance:

- 100% SQL Server compatibility
- Best for: migrating on-prem SQL Server to cloud
- Supports: SQL Agent, CLR, linked servers
- More expensive than SQL Database

Azure Synapse Dedicated SQL Pool:

- MPP data warehouse (not regular SQL Server)
- Best for: analytical queries, reporting, BI

- Cannot do OLTP (row-by-row inserts are slow)
- This is what we configure in Synapse

RETAILMART DATABASE TOPOLOGY:

Azure SQL Database (OLTP):

- stores retail transaction data temporarily
- Customer service team queries here

Azure Synapse Dedicated Pool (DW):

- Historical analytics
- Power BI dashboards connect here
- Optimized for SELECT, GROUP BY, aggregations

◆ Copy Data from SQL Database to ADLS Gen2

PIPELINE: PL_SQLToADLS_SalesExtract

Three source modes for Copy Activity from SQL:

MODE 1: Table (entire table)

Source Dataset:

Type: Azure SQL Database

Table: [dbo].[Sales_Staging] ← reads ENTIRE table

Use when: Small tables, full load needed

MODE 2: Query (custom SQL)

Source Dataset:

Type: Azure SQL Database

Use Query: SQL Query

Query:

```
SELECT
    sale_id,
    customer_id,
    total_amount,
    sale_date,
    region_code
FROM [dbo].[Sales]
WHERE sale_date = '@{pipeline().parameters.p_date}'
    AND status = 'COMPLETED'
ORDER BY sale_id
```

Use when: Need filtering, specific columns, joins

MODE 3: Stored Procedure

Create procedure in SQL DB first:

```
CREATE PROCEDURE [dbo].[usp_GetSalesForLoad]
    @LoadDate DATE,
    @RegionCode VARCHAR(10) = NULL
AS
BEGIN
    SELECT
        s.*,
        c.customer_name,
        c.customer_tier
    FROM [dbo].[Sales] s
    JOIN [dbo].[Customers] c ON s.customer_id = c.customer_id
    WHERE s.sale_date = @LoadDate
        AND (@RegionCode IS NULL OR s.region_code = @RegionCode)
        AND s.status = 'COMPLETED';
END
```

In Copy Activity Source:

Use Stored Procedure: [dbo].[usp_GetSalesForLoad]

Parameters:

@LoadDate: @{{pipeline().parameters.p_date}}

@RegionCode: @{{pipeline().parameters.p_region}}

Use when: Complex logic, business rules in DB already

SINK: ADLS Gen2 Parquet

Container: silver

Path: sql-extract/@{{pipeline().parameters.p_date}}/

Format: Parquet

Compression: Snappy

◆ Overwrite and Append Modes

COPY ACTIVITY WRITE BEHAVIORS:

TO FILE STORE (ADLS/Blob):

MERGE FILES:

- Combines all source files into one output file
- Source: 100 small CSVs → Sink: 1 large CSV
- Use: consolidating small files

FLATTEN HIERARCHY:

- All files from all subfolders → one flat destination folder
- Source: /2024/06/15/file.csv → Sink: /processed/file.csv

PRESERVE HIERARCHY:

- Maintains exact folder structure
- Source: /2024/06/15/file.csv → Sink:

/backup/2024/06/15/file.csv

TO SQL TABLE:

INSERT (append new rows only):

- sink.writeBehavior = "insert"
- New rows added to existing data
- Use for: incremental loads, event logs

UPSERT (insert or update):

- sink.writeBehavior = "upsert"
- If row exists (by key) → UPDATE
- If row doesn't exist → INSERT
- Key Columns: [sale_id] ← unique identifier
- Use for: dimension tables, master data

OVERWRITE (truncate then insert):

- Pre-copy Script: TRUNCATE TABLE [dbo].[Sales_Staging]
- Then insert all data fresh
- Use for: full loads, staging tables

REAL WORLD DECISION:

- Fact table (500M rows, append daily): → INSERT
- Customer master (10M rows, full refresh): → OVERWRITE
- Product catalog (changes daily): → UPSERT
- Audit log: → INSERT (never overwrite!)

◆ ForEach Loop — Copy Multiple Tables

SCENARIO: Copy 20 SQL tables to ADLS Gen2 at once

CONTROL TABLE [APPROACH](#) (Senior Pattern):

```
-- Create control table in Azure SQL DB
CREATE TABLE dbo.SynapseControlTable (
    id          INT IDENTITY PRIMARY KEY,
    source_schema VARCHAR(50),
    source_table VARCHAR(100),
    target_folder VARCHAR(200),
    target_format VARCHAR(20),
    load_type    VARCHAR(20),
    is_active    BIT DEFAULT 1
);

INSERT INTO dbo.SynapseControlTable VALUES
('dbo', 'Sales',          'silver/sales/',      'parquet',
'INCREMENTAL', 1),
('dbo', 'Customers',      'silver/customers/',  'parquet',
'FULL',        1),
('dbo', 'Products',        'silver/products/',   'parquet',
'FULL',        1),
('dbo', 'StoreInventory',  'silver/inventory/',  'parquet',
'INCREMENTAL', 1),
('dbo', 'Promotions',      'silver/promotions/', 'parquet',
'FULL',        0); -- disabled
```

PIPELINE: PL_BulkCopy_SQLToADLS

STEP 1: Lookup Activity

Name: LKP_GetActiveTables

Source: Azure SQL Database

Query:

```
SELECT source_schema, source_table, target_folder,
target_format
FROM dbo.SynapseControlTable
```

```
WHERE is_active = 1
ORDER BY load_type DESC -- FULL loads first, then
INCREMENTAL
```

STEP 2: ForEach Activity

Name: FE_CopyEachTable

Items: @{activity('LKP_GetActiveTables').output.value}

Batch Count: 5 ← 5 tables copy in parallel

Sequential: No

INSIDE ForEach:

Copy Activity: CPY_DynamicTableCopy

Source:

Linked Service: LS_AzureSQLDB

Query:

```
SELECT * FROM
@{item().source_schema}.@{item().source_table}
```

Sink:

Linked Service: LS_ADLS_Gen2

Path: @{item().target_folder}

@{formatDateTime(utcnow(), 'yyyy/MM/dd')}/

File: @{concat(item().source_table, '_',
formatDateTime(utcnow(), 'yyyyMMddHHmmss'),
'parquet')}

Format: Parquet

STEP 3: Success notification

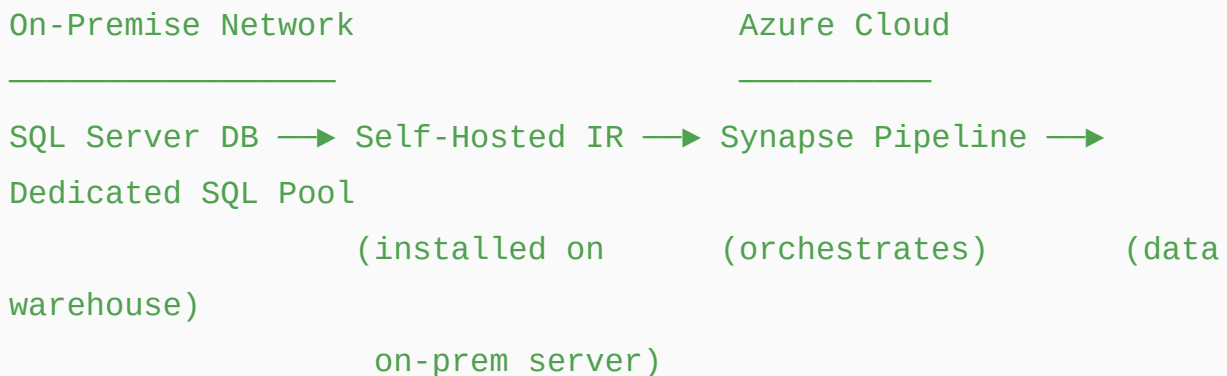
Web Activity → Logic App

"Bulk copy complete: 4 tables loaded successfully"

7 Deep Dive: Copy Activity in Azure Synapse

◆ Load from On-Premise SQL Server to Azure Synapse

ARCHITECTURE:



STEP 1: Install Self-Hosted IR

Same as ADF SHIR setup

Synapse Manage → Integration Runtimes → + New → Self-Hosted

→ Name: SHIR_OnPrem_Synapse

→ Install on on-premise Windows server

→ Register with auth key

STEP 2: Create Linked Services

On-Prem SQL Server:

Type: SQL Server

Server: 10.0.1.51 (internal IP)

Database: LegacySalesDB

Auth: Windows Authentication

IR: SHIR_OnPrem_Synapse

Synapse Dedicated Pool:

Type: Azure Synapse Analytics

Server: synapse-retailmart-prod.sql.azuresynapse.net

Database: RetailMartDW

Auth: Managed Identity (no password!)

STEP 3: Create Datasets

DS_OnPrem_LegacySales:

Type: SQL Server

Table: [dbo].[LegacySales]

DS_Synapse_FactSales:

Type: Azure Synapse Analytics

Table: [gold].[fact_sales]

STEP 4: Copy Activity

Source:

Dataset: DS_OnPrem_LegacySales

Partition Option: Dynamic Range

Column: SaleID

Degree of Parallelism: 4 ← read 4 chunks simultaneously!

Sink:

Dataset: DS_Synapse_FactSales

Copy Method: PolyBase ← FASTEST for Synapse!

Pre-copy Script:

```
TRUNCATE TABLE [gold].[fact_sales]
```

◆ PolyBase and Bulk Insert — Critical for Synapse Performance

THE PROBLEM:

Normal **INSERT into** Synapse Dedicated Pool: slow!

10 million rows via JDBC = 45 minutes

WHY? **Row-by-row insert** defeats MPP architecture

Synapse **is** optimized **for** BULK operations **not row-by-row**

SOLUTION 1: PolyBase (traditional, very fast)

HOW POLYBASE WORKS:

Step 1: ADF writes source data to Azure Storage (staging)

Step 2: Synapse reads DIRECTLY from storage using PolyBase

Step 3: Data distributed across all 60 distributions in parallel

Source → [ADF] → Azure Blob (staging) → [PolyBase] → Synapse DW

Speed: 10 million rows → 3 minutes!

Copy Activity Sink Configuration for PolyBase:

Copy Method: PolyBase

Staging Account: (Azure Blob or ADLS account for temp staging)

Staging Container: synapse-staging

Pre-copy Script:

```
TRUNCATE TABLE [staging].[stg_sales_load]
```

SOLUTION 2: COPY INTO (Modern, simpler, equally fast)

HOW COPY INTO WORKS:

Similar to PolyBase but simpler syntax

Synapse reads directly from ADLS using COPY INTO command

ADF Copy Activity triggers this SQL behind the scenes:

```
COPY INTO [gold].[fact_sales]
```

```
FROM
```

```
'https://retailmartdatalake.dfs.core.windows.net/staging/sales/'
```

```
WITH (
```

```
    FILE_TYPE = 'PARQUET',
```

```
CREDENTIAL = (IDENTITY = 'Managed Identity'),  
COMPRESSION = 'snappy'  
);
```

Copy Activity Sink Configuration for COPY INTO:

Copy Method: Copy Command

Default Values: (optional default values for missing columns)

SPEED COMPARISON (10 million rows):

Row-by-row INSERT:	45 minutes	← never use for bulk!
Bulk Insert:	12 minutes	← moderate
PolyBase:	3 minutes	← very fast
COPY INTO:	2 minutes	← fastest!

◆ Copy from On-Premise File System

SCENARIO:

On-premise file server has daily sales reports (CSV)

Path: \\FILESERVER01\reports\sales\daily\

STEP 1: Linked Service for File System

Type: File System

Host: \\FILESERVER01\reports (UNC path or local path)

User ID: CORP\svc_datafactory

Password: (from Key Vault)

Connect via IR: SHIR_OnPrem_Synapse

STEP 2: Dataset for File System

Name: DS_OnPrem_FileSystem_Sales

Linked Service: LS_FileSystem_Reports

Folder Path: sales/daily/

File: *.csv

Format: CSV with header

STEP 3: Copy Activity

Source: DS_OnPrem_FileSystem_Sales

Recursive: TRUE (include subfolders)

Sink: DS_ADLS_Bronze_FileReports

Container: bronze

Path: file-server-reports/@{formatDateTime(utcnow(), 'yyyy/MM/dd')}/

KEY INSIGHT:

SHIR can access:

- ✓ SQL Databases (SQL Server, Oracle, MySQL)
- ✓ File Systems (local, network shares, NFS)
- ✓ FTP/SFTP servers
- ✓ SAP systems
- ✓ Any resource inside corporate network

SHIR = secure bridge between on-prem and cloud

All data encrypted in transit (HTTPS)

8 Spark Pool in Azure Synapse — Complete Guide

◆ Synapse Spark vs Databricks Spark

COMPARISON:

Feature	Synapse Spark	Databricks
Spark Version	3.x	3.x (same)
PySpark	✓	✓

Scala	✓	✓
R	✓	✓
.NET Spark	✓ (unique!)	✗
Delta Lake	✓	✓ (better)
Unity Catalog	✗	✓
Auto-pause	✓	✓
Integration w/ Synapse	Native	Via Linked Service
Collaboration	Basic	Advanced
ML Capabilities	Basic MLflow	Full MLflow + AutoML
SQL integration	Via Spark SQL	Via Unity Catalog
Cost	Included in Synapse	Separate billing

WHEN TO USE SYNAPSE SPARK:

- ✓ Already using Synapse for pipelines + SQL
- ✓ Want everything in one platform
- ✓ Light-medium Spark workloads
- ✓ Direct SQL Pool integration needed
- ✓ .NET developers (C#/F# on Spark!)

WHEN TO USE DATABRICKS:

- ✓ Heavy ML/AI workloads
- ✓ Complex streaming pipelines
- ✓ Unity Catalog governance needed
- ✓ Large team of data scientists
- ✓ Maximum Spark performance needed

RetailMart Decision:

Data Engineering pipelines → Synapse Spark (simple transformations)

ML model training → Databricks (complex ML)

Ad-hoc SQL exploration → Synapse Serverless SQL

◆ Create Spark Pool in Synapse

Synapse Studio → Manage → Apache Spark Pools → + New

Configuration:

Name: RetailMart_SparkPool

Node Size Family: Memory Optimized

Node Size: Medium (8 vCores, 64GB RAM per node)

Autoscale: Enabled

Min nodes: 3

Max nodes: 20

Auto-pause: Enabled

After 15 minutes of inactivity

→ Cluster shuts down → ZERO cost when idle!

Apache Spark Version: 3.4

Packages:

→ Upload requirements.txt or add PyPI packages

great-expectations==0.18.0

pyarrow==12.0.0

COST SAVING WITH AUTO-PAUSE:

Without auto-pause: 3 nodes × 24 hours = 72 node-hours/day

With auto-pause (used 4 hours/day): 3 nodes × 4 hours = 12 node-hours/day

→ 83% cost saving!

◆ Synapse Notebooks — Complete Guide

```
# SYNAPSE NOTEBOOK LANGUAGES:
```

```
#
```

```
# Default language set when creating notebook
```

```
# Can override per cell with magic commands:
```

```
# %%pyspark → Python/PySpark
```

```
# %%spark → Scala Spark
```

```
# %%sql → Spark SQL
```

```
# %%csharp → .NET Spark (C#) ← unique to Synapse!
```

```
# %%markdown → Documentation
```

```
# — CELL 1: Set notebook language to PySpark —————
```

```
# (Selected at notebook creation)
```

```
# — CELL 2: Read data (PySpark) —————
```

```
df_sales = spark.read \
```

```
    .format("parquet") \
```

```
    .load("abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/
```

```
df_sales.printSchema()
```

```
df_sales.show(5)
```

```
# — CELL 3: Switch to SQL in same notebook —————
```

```
%%sql
```

```
-- Register DataFrame as temp view to query with SQL
```

```
-- (must register in PySpark cell first)
```

```
SELECT
```

```
    region_code,
```

```
    COUNT(*) as transactions,
```

```
    SUM(total_amount) as revenue
```

```
FROM sales_view -- registered from PySpark cell
```

```
GROUP BY region_code
```

```
ORDER BY revenue DESC
```

```

# — CELL 4: Switch to Scala —————
%%spark
val df =
spark.read.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/silver.parquet")
println(s"Total records: ${df.count()}")

# — CELL 5: .NET Spark (C#) - unique to Synapse! —————
%%csharp
DataFrame df =
spark.Read().Parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/silver.parquet")
df.Show();
Console.WriteLine($"Total: {df.Count()}");

```

◆ MSSparkUtils — Synapse's Version of dbutils

```

# MSSparkUtils = Synapse equivalent of Databricks dbutils
# Same concept, slightly different syntax

# — FILE SYSTEM OPERATIONS —————
from notebookutils import mssparkutils

# List files
files =
mssparkutils.fs.ls("abfss://bronze@retailmartdatalake.dfs.core.windows.net/bronze.parquet")
for f in files:
    print(f"Name: {f.name}, Size: {f.size}, IsDir: {f.isDir}")

# Create directory
mssparkutils.fs.mkdirs("abfss://silver@retailmartdatalake.dfs.core.windows.net/silver")

# Copy file

```

```

mssparkutils.fs.cp(

"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/file.p

"abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/file.p
    recurse=False
)

# Move file
mssparkutils.fs.mv(

"abfss://landing@retailmartdatalake.dfs.core.windows.net/sales_20240

"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/sales_
)

# Delete file/folder
mssparkutils.fs.rm("abfss://temp@retailmartdatalake.dfs.core.windows
recurse=True)

# Read file content (preview)
content = mssparkutils.fs.head(

"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/sales_
    1024 # first 1024 bytes
)
print(content)

# Check if path exists
mssparkutils.fs.ls("abfss://bronze@retailmartdatalake.dfs.core.windc
# Returns list of FileInfo objects (empty = folder exists,
exception = doesn't exist)

# — NOTEBOOK UTILITY —
# Run another notebook

```

```

result = mssparkutils.notebook.run(
    path="../silver/transform_sales",          # relative path to
notebook
    timeout_seconds=1800,                      # 30 min timeout
    arguments={
        "input_path":
"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/",
        "process_date": "2024-06-15",
        "env": "production"
    }
)
print(f"Child returned: {result}")

# Exit notebook with return value (for parent
notebooks/pipelines)
mssparkutils.notebook.exit("SUCCESS: 142850 rows processed")

# — CREDENTIALS UTILITY —————
# Access Key Vault secrets
storage_key = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/", # Key Vault
URL
    "adls-storage-key"                          # secret
name
)

# Get token for a service
token = mssparkutils.credentials.getToken("Storage")

```

◆ Notebook Parameters in Synapse

```

# — WIDGET PARAMETERS (interactive in Studio) —————
# Right-click cell → Toggle Parameter Cell
# This makes cell a "parameter cell" – first cell in notebook

# Parameters cell (marked with special icon in Synapse):
input_path    =
"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/"
output_path   =
"abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/"
process_date  = "2024-06-15"
environment   = "dev"

# When called from Pipeline, these defaults get OVERRIDDEN
# by values passed from Synapse Pipeline notebook activity

# — CALLING FROM SYNAPSE PIPELINE —————
# In Synapse Pipeline → Notebook Activity:
# Settings Tab:
#   Notebook: /Production/silver/transform_sales
#   Base Parameters:
#     input_path:
@{concat('abfss://bronze@retailmartdatalake.dfs.core.windows.net/sal
pipeline().parameters.p_date, '/')}}
#     output_path:
@{concat('abfss://silver@retailmartdatalake.dfs.core.windows.net/sal
pipeline().parameters.p_date, '/')}}
#     process_date: @{pipeline().parameters.p_date}
#     environment:  @{pipeline().globalParameters.gp_environment}

# — READING PARAMETER VALUES IN NOTEBOOK —————
# Parameters passed from pipeline override the defaults above
# Just use the variable names normally:

print(f"Input:    {input_path}")
print(f"Output:   {output_path}")

```

```
print(f>Date:      {process_date}")
print(f>Env:       {environment}")

df = spark.read.parquet(input_path)
df.write.mode("overwrite").parquet(output_path)

mssparkutils.notebook.exit(f">SUCCESS: {df.count()} rows for
{process_date}")
```

◆ Magic Commands Deep Dive

```
# — MAGIC COMMANDS REFERENCE —————

# %% magic → changes language for that cell
%%sql
SELECT current_timestamp() as server_time

%%pyspark
print("Python cell")

%%spark
println("Scala cell")

%%csharp
Console.WriteLine("C# cell")

%%markdown
## This is a documentation cell
**Bold text**, *italic*, `code`
- Bullet point 1
- Bullet point 2
```

```

# — %run → Run another notebook inline —————
%run ../Utils/common_functions
# Loads all functions from common_functions notebook
# Now you can call those functions directly!

# — %pip → Install packages in session —————
%pip install great-expectations==0.18.0
%pip install openpyxl

# After install, import normally:
import great_expectations as ge

# — %env → Set environment variables —————
%env MY_VAR=hello_world

import os
print(os.environ.get('MY_VAR')) # prints: hello_world

# — CALLING ONE SYNAPSE NOTEBOOK FROM ANOTHER —————

# Parent Notebook: orchestrator.ipynb
# —————

# Run child notebooks in sequence
result1 = mssparkutils.notebook.run(
    "../bronze/load_sales",
    arguments={"process_date": process_date}
)
print(f"Bronze Sales: {result1}")

result2 = mssparkutils.notebook.run(
    "../bronze/load_customers",
    arguments={"process_date": process_date}
)
print(f"Bronze Customers: {result2}")

```



```

# Run in parallel (using Python threading)
from concurrent.futures import ThreadPoolExecutor, as_completed

def run_notebook(path, args):
    return mssparkutils.notebook.run(path, arguments=args)

notebooks = {
    "../bronze/load_sales": {"process_date": process_date},
    "../bronze/load_customers": {"process_date": process_date},
    "../bronze/load_inventory": {"process_date": process_date}
}

with ThreadPoolExecutor(max_workers=3) as executor:
    futures = {
        executor.submit(run_notebook, path, args): path
        for path, args in notebooks.items()
    }
    for future in as_completed(futures):
        notebook = futures[future]
        result = future.result()
        print(f"✅ {notebook}: {result}")

# Return combined result to pipeline
mssparkutils.notebook.exit(f"All bronze notebooks completed for {process_date}")

```

◆ Connecting to Azure Key Vault in Synapse Notebook

```

# — METHOD 1: Using MSSparkUtils (recommended) —————
from notebookutils import mssparkutils

```

```

# Synapse workspace must have Key Vault linked service
configured!
# Manage → Linked Services → + New → Azure Key Vault

# Get secret
sql_password = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/",
    "sqlldb-password"
)

oracle_password = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/",
    "oracle-db-password"
)

# Never print secrets!
print("Secrets loaded successfully")
# Output: Secrets loaded successfully (not the actual values)

# — METHOD 2: Using Azure Key Vault Linked Service —————
# Synapse Manage → Linked Services → Add Azure Key Vault LS
# Then reference in notebook via TokenLibrary

from pyspark.sql import SparkSession

# Access via TokenLibrary (Synapse specific)
sql_password = TokenLibrary.getSecret(
    "kv-retailmart-prod",    # Key Vault name
    "sqlldb-password",      # Secret name
    "LS_AzureKeyVault"      # Linked Service name
)

# — METHOD 3: Linked Service connection string —————
# Most secure: Synapse manages credentials completely
# Create Linked Service for SQL DB with Key Vault password

```

```
# Reference Linked Service in notebook:

# Read from Azure SQL using Linked Service (no credentials in
code!)
df_sql = spark.read \
    .format("com.microsoft.sqlserver.jdbc.spark") \
    .option("url",
"jdbc:sqlserver://retailmart.database.windows.net:1433;database=Sales"
\
    .option("dbtable", "dbo.Sales") \
    .option("authentication", "ActiveDirectoryMSI") \
    .load()
```

◆ Connecting to ADLS Gen2 — All Methods

```
# — METHOD 1: Default linked storage (SIMPLEST) —————
# Synapse workspace has a default ADLS account configured during
setup
# Access using abfss:// directly – NO credentials needed!
# Synapse Managed Identity automatically has access

df = spark.read.parquet(

"abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/"
)
# Works because Synapse MSI has Storage Blob Data Contributor
role

# — METHOD 2: Account Key —————
storage_account = "retailmartdatalake"
account_key = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/",
```

```

        "adls-account-key"
    )

    spark.conf.set(
        f"fs.azure.account.key.
        {storage_account}.dfs.core.windows.net",
        account_key
    )

    df = spark.read.parquet(

    f"abfss://silver@{storage_account}.dfs.core.windows.net/sales/"
    )

# — METHOD 3: Service Principal (OAuth) —————
client_id      = mssparkutils.credentials.getSecret("https://kv-
retailmart-prod.vault.azure.net/", "sp-client-id")
client_secret  = mssparkutils.credentials.getSecret("https://kv-
retailmart-prod.vault.azure.net/", "sp-client-secret")
tenant_id      = mssparkutils.credentials.getSecret("https://kv-
retailmart-prod.vault.azure.net/", "tenant-id")

    spark.conf.set(f"fs.azure.account.auth.type.
    {storage_account}.dfs.core.windows.net", "OAuth")
    spark.conf.set(f"fs.azure.account.oauth.provider.type.
    {storage_account}.dfs.core.windows.net",

    "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider")
    spark.conf.set(f"fs.azure.account.oauth2.client.id.
    {storage_account}.dfs.core.windows.net", client_id)
    spark.conf.set(f"fs.azure.account.oauth2.client.secret.
    {storage_account}.dfs.core.windows.net", client_secret)
    spark.conf.set(f"fs.azure.account.oauth2.client.endpoint.
    {storage_account}.dfs.core.windows.net",

```

```

f"https://login.microsoftonline.com/{tenant_id}/oauth2/token")

df =
spark.read.parquet(f"abfss://silver@{storage_account}.dfs.core.windows.net/{path}")

# — METHOD 4: SAS Token —————
sas_token = mssparkutils.credentials.getSecret("https://kv-retailmart-prod.vault.azure.net/", "adls-sas-token")

spark.conf.set(
    f"fs.azure.sas.silver.{storage_account}.dfs.core.windows.net",
    sas_token
)

# — BEST PRACTICE RECOMMENDATION —————
# Development: Method 1 (default linked storage – zero config)
# Production: Method 1 with Managed Identity (most secure)
# Cross-tenant: Method 3 Service Principal
# Temporary: Method 4 SAS Token

```

◆ Connecting to Azure Blob Storage

```

# AZURE BLOB vs ADLS GEN2:
# Blob: wasbs://container@account.blob.core.windows.net/
# ADLS2: abfss://container@account.dfs.core.windows.net/
# Key difference: "blob" vs "dfs" in URL, "wasbs" vs "abfss"

# — METHOD 1: Account Key —————
blob_account = "retailmartblob"
blob_key = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/",

```

```

        "blob-storage-key"
    )

    spark.conf.set(
        f"fs.azure.account.key.{blob_account}.blob.core.windows.net",
        blob_key
    )

    df_blob = spark.read.parquet(

        f"wasbs://archive@{blob_account}.blob.core.windows.net/sales/2023/"
    )

    # — METHOD 2: SAS Token for specific container —————
    sas = mssparkutils.credentials.getSecret("https://kv-retailmart-
    prod.vault.azure.net/", "blob-sas")

    spark.conf.set(
        f"fs.azure.sas.archive.{blob_account}.blob.core.windows.net",
        sas
    )

    df =
    spark.read.parquet("wasbs://archive@retailmartblob.blob.core.windows

```

◆ Connecting to Azure SQL Database — All Methods

```

# — METHOD 1: SQL Authentication with JDBC —————
sql_server    = "retailmart-sqlldb.database.windows.net"
sql_database  = "SalesDB"
sql_user      = "sqladmin"
sql_password  = mssparkutils.credentials.getSecret(

```

```

    "https://kv-retailmart-prod.vault.azure.net/",
    "sqldb-password"
)

jdbc_url = (
    f"jdbc:sqlserver://{sql_server}:1433;"
    f"database={sql_database};"
    f"encrypt=true;"
    f"trustServerCertificate=false;"
    f"hostnameInCertificate=*.database.windows.net;"
    f"loginTimeout=30"
)

# Read full table
df_customers = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("dbtable", "dbo.Customers") \
    .option("user", sql_user) \
    .option("password", sql_password) \
    .option("driver",
"com.microsoft.sqlserver.jdbc.SQLServerDriver") \
    .load()

# Read with custom query (incremental)
df_recent_sales = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("query", f"SELECT * FROM dbo.Sales WHERE sale_date =
'{process_date}'") \
    .option("user", sql_user) \
    .option("password", sql_password) \
    .load()

# Read with partitioning (parallel reads – much faster for large

```

```

tables!)

df_large_table = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("dbtable", "dbo.LargeSalesTable") \
    .option("user", sql_user) \
    .option("password", sql_password) \
    .option("partitionColumn", "sale_id") \
    .option("lowerBound", "1") \
    .option("upperBound", "50000000") \
    .option("numPartitions", "16") \
    .load()

# — METHOD 2: Azure Active Directory / Managed Identity —————
# No password needed! Uses Synapse MSI

jdbc_url_msi = (
    f"jdbc:sqlserver://{sql_server}:1433;"
    f"database={sql_database};"
    f"encrypt=true;"
    f"authentication=ActiveDirectoryMSI"
)

df_secure = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url_msi) \
    .option("dbtable", "dbo.Customers") \
    .load()

# Synapse MSI needs "db_datareader" role on SQL Database

# — WRITE to SQL Database —————
df_result.write \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("dbtable", "dbo.Sales_Silver") \

```



```
.option("user", sql_user) \  
.option("password", sql_password) \  
.option("batchsize", "100000") \  
.option("truncate", "true") \  
.mode("overwrite") \  
.save()
```

◆ Connecting to On-Premise SQL Server

```
# REQUIRES: Self-Hosted IR configured in Synapse  
# Manage → Integration Runtimes → SHIR must be running  
  
# On-premise SQL Server via JDBC + SHIR  
# Note: Synapse notebooks can use SHIR through Linked Service  
  
# Step 1: Create Linked Service for on-prem SQL Server  
# Manage → Linked Services → SQL Server  
# Name: LS_OnPrem_SQLServer  
# Server: 10.0.1.51  
# Database: LegacySalesDB  
# Auth: Windows Authentication  
# Connect via IR: SHIR_OnPrem_Synapse  
  
# Step 2: Use Linked Service connection in notebook  
# (Via Synapse LinkedService utility)  
  
df_onprem = spark.read \  
  .format("jdbc") \  
  .option("url",  
"jdbc:sqlserver://10.0.1.51:1433;database=LegacySalesDB") \  
  .option("dbtable", "dbo.LegacySales") \  
  .option("user", mssparkutils.credentials.getSecret(
```

```

        "https://kv-retailmart-prod.vault.azure.net/",
        "onprem-sql-user")) \
.option("password", mssparkutils.credentials.getSecret(
        "https://kv-retailmart-prod.vault.azure.net/",
        "onprem-sql-password")) \
.load()

print(f"On-prem records: {df_onprem.count():,}")

```

◆ Reading and Writing All File Formats

```

# — CSV FILES —————
# READING (with optimization options)
from pyspark.sql.types import *

# Always define schema for production – avoid inferSchema!
schema = StructType([
    StructField("sale_id",      IntegerType(), False),
    StructField("customer_id",  StringType(),  False),
    StructField("total_amount", DoubleType(),   True),
    StructField("sale_date",    StringType(),   True), # read as
string then convert
    StructField("region_code",  StringType(),   True)
])

df_csv = spark.read \
    .schema(schema) \
    .option("header", "true") \
    .option("delimiter", ",") \
    .option("encoding", "UTF-8") \
    .option("nullValue", "NULL") \
    .option("nanValue", "NaN") \

```

```

.option("mode", "DROPMALFORMED") \

.csv("abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/c

# OPTIMIZATION for large CSV files:
# 1. Always use explicit schema (2x faster than inferSchema)
# 2. Read multiple files at once using folder path
# 3. Use mode=DROPMALFORMED for dirty data

# WRITING CSV
df_csv.coalesce(1) \
    .write \
    .mode("overwrite") \
    .option("header", "true") \
    .option("delimiter", ",") \
    .option("encoding", "UTF-8") \

.csv("abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/c
# coalesce(1) = one output file (for small outputs)

# — PARQUET FILES —————
# Best format for analytics – columnar, compressed

# READ
df_parquet = spark.read \

.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/sal

# READ specific partitions (partition pruning)
df_june = spark.read \

.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/sal
\
    .filter("year = 2024 AND month = 6")
# Spark reads ONLY year=2024/month=6 folder!

```

```

# WRITE with partitioning and compression
df_parquet.write \
    .mode("overwrite") \
    .partitionBy("year", "month", "region_code") \
    .option("compression", "snappy") \

.parquet("abfss://gold@retailmartdatalake.dfs.core.windows.net/sales

# — JSON FILES —————
# READ flat JSON (one JSON object per line - NDJSON)
df_json = spark.read \
    .option("multiLine", "false") \

.json("abfss://bronze@retailmartdatalake.dfs.core.windows.net/paymer

# READ nested JSON
df_nested = spark.read \
    .option("multiLine", "true") \

.json("abfss://bronze@retailmartdatalake.dfs.core.windows.net/api_re

# Flatten nested structure
from pyspark.sql.functions import *

df_flat = df_nested \
    .select(
        col("order_id"),
        col("customer.id").alias("customer_id"),
        col("customer.name").alias("customer_name"),
        col("total"),
        explode(col("items")).alias("item")
    ) \
    .select(
        "order_id", "customer_id", "customer_name", "total",

```

```

        col("item.product_code").alias("product_code"),
        col("item.quantity").alias("quantity"),
        col("item.price").alias("unit_price")
    )

# WRITE JSON
df_flat.write \
    .mode("overwrite") \

.json("abfss://silver@retailmartdatalake.dfs.core.windows.net/orders

# — AVRO FILES —————
# Best for: streaming, schema evolution, Kafka integration

# READ
df_avro = spark.read \
    .format("avro") \

.load("abfss://bronze@retailmartdatalake.dfs.core.windows.net/kafka_

# WRITE
df_avro.write \
    .format("avro") \
    .mode("overwrite") \

.save("abfss://silver@retailmartdatalake.dfs.core.windows.net/events

# — ORC FILES —————
# Best for: Hive compatibility, heavy write workloads

# READ
df_orc = spark.read \
    .format("orc") \

.load("abfss://bronze@retailmartdatalake.dfs.core.windows.net/hive_e

```

```

# WRITE
df_orc.write \
    .format("orc") \
    .mode("overwrite") \
    .option("compression", "zlib") \

.save("abfss://silver@retailmartdatalake.dfs.core.windows.net/orc_ou

# — EXCEL FILES —————
# Requires: openpyxl or com.crealytics.spark.excel library

# First install package (in Synapse Pool packages or notebook):
# %pip install openpyxl

# READ Excel using pandas then convert to Spark (small files)
import pandas as pd
from notebookutils import mssparkutils

# Download Excel file from ADLS to local temp
mssparkutils.fs.cp(

"abfss://bronze@retailmartdatalake.dfs.core.windows.net/config/store
    "file:///tmp/store_targets.xlsx"
)

# Read with pandas
pdf = pd.read_excel("/tmp/store_targets.xlsx",
sheet_name="Targets")
df_excel = spark.createDataFrame(pdf) # convert to Spark

print(f"Excel rows: {df_excel.count()}")
df_excel.show(5)

# Using Spark Excel library (for large files):

```

```

df_excel_spark = spark.read \
    .format("com.crealytics.spark.excel") \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .option("dataAddress", "'Sheet1'!A1") \

.load("abfss://bronze@retailmartdatalake.dfs.core.windows.net/config

# WRITE to Excel (via pandas)
df_result_pd = df_result.toPandas() # only for small DataFrames!
df_result_pd.to_excel("/tmp/output.xlsx", index=False)

# Upload back to ADLS
mssparkutils.fs.cp(
    "file:///tmp/output.xlsx",
    "abfss://gold@retailmartdatalake.dfs.core.windows.net/reports/output
)

```

◆ Creating RDDs and DataFrames — All Methods

```

# — RDD CREATION (Synapse Spark) —————
# Method 1: From collection
rdd_numbers = sc.parallelize([1, 2, 3, 4, 5], numSlices=4)

# Method 2: From file
rdd_text = sc.textFile(
    "abfss://bronze@retailmartdatalake.dfs.core.windows.net/raw/sales.csv"
)

# Method 3: From range

```

```

rdd_range = sc.range(1, 1000000, step=1, numSlices=10)

# Method 4: From existing DataFrame
rdd_from_df = df_sales.rdd # convert DataFrame to RDD

# — DATAFRAME CREATION (Synapse Spark) —————
# Method 1: Read from storage (most common)
df = spark.read.parquet(
    "abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/"
)

# Method 2: From Python list
data = [
    (1, "C001", 50000.0, "2024-06-15"),
    (2, "C002", 35000.0, "2024-06-15"),
    (3, "C003", 85000.0, "2024-06-15")
]
columns = ["sale_id", "customer_id", "amount", "sale_date"]
df_test = spark.createDataFrame(data, schema=columns)

# Method 3: From schema + data
from pyspark.sql.types import *

schema = StructType([
    StructField("sale_id", IntegerType(), False),
    StructField("customer", StringType(), False),
    StructField("amount", DoubleType(), True)
])
df_typed = spark.createDataFrame(data[:2], schema=schema)

# Method 4: From SQL query (Spark SQL)
spark.sql("CREATE TEMP VIEW sales AS SELECT * FROM
delta.`abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/`")
df_from_sql = spark.sql("SELECT * FROM sales WHERE region_code =

```



```
'N')
```

```
# Method 5: From Pandas
```

```
import pandas as pd
pdf = pd.DataFrame({"a": [1, 2, 3], "b": ["x", "y", "z"]})
df_from_pandas = spark.createDataFrame(pdf)
```

```
# Method 6: From JDBC (SQL Database)
```

```
df_from_sql_db = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("dbtable", "dbo.Products") \
    .option("user", sql_user) \
    .option("password", sql_password) \
    .load()
```

◆ Repartition vs Coalesce — Complete Guide

```
# RULE: Remember this always!
```

```
# =====
```

```
# REPARTITION = INCREASE or CHANGE distribution (with shuffle)
```

```
# COALESCE     = ONLY DECREASE (no shuffle, merges local)
```

```
# =====
```

```
# Check current partition count
```

```
print(f"Current partitions: {df_sales.rdd.getNumPartitions()}")
```

```
# — WHEN TO USE REPARTITION —————
```

```
# 1. Increase partitions for better parallelism
```

```
df_repartitioned = df_sales.repartition(200)
```

```
# 2. Distribute by join key (before multiple joins on same key)
df_by_key = df_sales.repartition(col("customer_id"))
# Now all same customer_id rows are on same partition
# JOIN on customer_id → no shuffle needed!
```

```
# 3. Before writing large output
df_large.repartition(100).write.parquet(output_path)
# Creates 100 equal-sized output files
```

```
# — WHEN TO USE COALESCE —————
```

```
# 1. After filtering (many empty partitions remain)
df_north = df_sales.filter(col("region_code") == "N")
# Was 200 partitions → after filter maybe 20 have data, 180 are empty
```

```
df_north_optimized = df_north.coalesce(20) # reduce without shuffle
# Now 20 partitions, all with data = efficient!
```

```
# 2. Before writing small output (fewer, larger files)
df_summary.coalesce(5).write.parquet(output_path)
# Creates exactly 5 files instead of 200 tiny ones
```

```
# — THE SMALL FILES PROBLEM —————
```

```
# This is a REAL and SERIOUS problem in data lakes!
```

```
# Bad pattern: writing 200 partitions of small data
df_daily_summary.write.parquet(output_path)
# Creates 200 × 1KB files = 200 tiny files!
```

```
# Why bad?
```

```
# → Reading 200 tiny files: 200 file open/close operations
```

```
# → Each file needs metadata lookup → slow!
```

```
# → Next day adds 200 more → after 1 year: 73,000 tiny files!
```

```

# → Queries take longer and longer

# Good pattern: control output file size
# Target: 128MB - 256MB per output file
df_daily_summary.coalesce(2).write.parquet(output_path)
# Creates 2 reasonable-sized files

# DYNAMIC approach (calculate based on data size):
import math
data_size_bytes = df.rdd.map(lambda row: len(str(row))).sum()
target_file_size = 128 * 1024 * 1024 # 128MB
optimal_partitions = max(1, math.ceil(data_size_bytes /
target_file_size))
df.coalesce(optimal_partitions).write.parquet(output_path)

# — SYNAPSE-SPECIFIC OPTIMIZATION —————
# Synapse has a special feature: Auto Compaction on Delta tables
spark.conf.set("spark.databricks.delta.autoCompact.enabled",
"true")
# Automatically merges small files → solves small file problem!

```

◆ Joins in Synapse Notebook

```

# — ALL JOIN TYPES WITH RETAILMART EXAMPLES —————
from pyspark.sql.functions import *
from pyspark.sql import functions as F

df_sales      =
spark.read.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/sales.parquet")
df_customers  =
spark.read.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/customers.parquet")
df_products   =

```

```

spark.read.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/silver/retailmartdata/sales.parquet")
df_stores = spark.read.parquet("abfss://silver@retailmartdatalake.dfs.core.windows.net/silver/retailmartdata/stores.parquet")

# — INNER JOIN: Only matching rows —————
df_sales_with_customer = df_sales.join(
    df_customers,
    on="customer_id",
    how="inner" # or "inner"
)
# Result: Only sales that have matching customer record

# — LEFT JOIN: All sales, matching customer (null if no match) —
df_all_sales = df_sales.join(
    df_customers,
    on="customer_id",
    how="left"
)
# Result: All sales records, customer info null for unmatched

# — RIGHT JOIN: All customers, matching sales —————
df_all_customers = df_sales.join(
    df_customers,
    on="customer_id",
    how="right"
)
# Result: All customers, sales null for customers with no sales

# — FULL OUTER: Everything from both —————
df_full = df_sales.join(
    df_customers,
    on="customer_id",
    how="full" # or "outer" or "fullouter"
)

```

```
# — CROSS JOIN (Cartesian): Every combination —————
# USE VERY CAREFULLY – produces M×N rows!
df_regions = spark.createDataFrame(
    [("North",), ("South",), ("East",), ("West",)],
    ["region"]
)
df_months = spark.createDataFrame(
    [(m,) for m in range(1, 13)],
    ["month"]
)
# Create all region-month combinations for blank template
df_all_combinations = df_regions.crossJoin(df_months)
# Result: 4 regions × 12 months = 48 rows
# Useful for: creating complete grid for reports (no missing months)
```

```
# — MULTIPLE CONDITION JOIN —————
df_result = df_sales.join(
    df_products,
    on=[
        df_sales.product_code == df_products.product_code,
        df_sales.region_code == df_products.available_region
    ],
    how="left"
)
```

```
# — HANDLING DUPLICATE COLUMN NAMES AFTER JOIN —————
# Both tables have "status" column → ambiguous after join!
df_result = df_sales.alias("s").join(
    df_customers.alias("c"),
    on="customer_id",
    how="left"
).select(
    col("s.sale_id"),
    col("s.total_amount"),
```

```

    col("s.status").alias("sale_status"),      # disambiguate!
    col("c.customer_name"),
    col("c.status").alias("customer_status")    # disambiguate!
)

# — CHAIN MULTIPLE JOINS —————
df_enriched = df_sales \
    .join(broadcast(df_customers), on="customer_id", how="left")
\
    .join(broadcast(df_products), on="product_code",
how="left") \
    .join(broadcast(df_stores), on="store_id",
how="left") \
    .select(
        "sale_id",
        "total_amount",
        "sale_date",
        "customer_name",
        "customer_tier",
        "product_name",
        "category",
        "store_name",
        "city",
        "region_code"
    )

print(f"Enriched sales: {df_enriched.count():,}")

```

◆ Broadcast Joins and Spark Configuration for Optimization

```

# — BROADCAST JOIN IN SYNAPSE —————
from pyspark.sql.functions import broadcast

```

```

# Small table (< 10MB default threshold) → broadcast
automatically
# Large fact × small dimension = classic broadcast join scenario

# MANUAL BROADCAST HINT
df_result = df_sales.join(
    broadcast(df_stores),    # explicitly broadcast dim_stores
    on="store_id",
    how="left"
)

# VERIFY broadcast happened:
df_result.explain()
# Look for: BroadcastHashJoin ← good, fast!
# If you see: SortMergeJoin    ← no broadcast, shuffle happening

# — CONFIGURE BROADCAST THRESHOLD —————
# Default: 10MB – too small for most real dimension tables
spark.conf.set("spark.sql.autoBroadcastJoinThreshold", str(200 *
1024 * 1024)) # 200MB
# Now tables up to 200MB are automatically broadcast

# — FULL OPTIMIZATION CONFIGURATION —————
def configure_spark_for_performance():
    """Apply all performance optimizations for Synapse Spark"""

    # JOIN optimizations
    spark.conf.set("spark.sql.autoBroadcastJoinThreshold",
str(200 * 1024 * 1024))

    # Adaptive Query Execution (AQE)
    spark.conf.set("spark.sql.adaptive.enabled", "true")

    spark.conf.set("spark.sql.adaptive.coalescePartitions.enabled",

```

```

"true")
    spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")

spark.conf.set("spark.sql.adaptive.localShuffleReader.enabled",
"true")

    # Shuffle optimization
    spark.conf.set("spark.sql.shuffle.partitions", "200")
    # Adjust based on data size:
    # Small data (< 1GB):    set to 50-100
    # Medium (1-10GB):       set to 200 (default)
    # Large (10-100GB):      set to 400-800
    # Very large (>100GB):   set to 1000-2000

    # Delta Lake optimizations

spark.conf.set("spark.databricks.delta.optimizeWrite.enabled",
"true")
    spark.conf.set("spark.databricks.delta.autoCompact.enabled",
"true")

    # Cost-Based Optimizer
    spark.conf.set("spark.sql.cbo.enabled", "true")
    spark.conf.set("spark.sql.statistics.histogram.enabled",
"true")

    print("✅ Spark optimizations configured")

configure_spark_for_performance()

# — WHAT IS CATALYST OPTIMIZER? —————
# (Already explained in Databricks section – same Spark engine!)
# Synapse Spark uses same Catalyst optimizer as Databricks

# KEY BEHAVIORS IN SYNAPSE:

```



```

# 1. Predicate Pushdown: filter pushed to Parquet file reader
# 2. Column Pruning: only needed columns read from Parquet
# 3. Constant Folding: WHERE year = 2024 evaluated at plan time
# 4. Join Reordering: Catalyst may reorder joins for efficiency

# — SKEWNESS ISSUE IN SPARK —————
# Data skew = one partition has MUCH more data than others

# DETECT SKEW:
# Spark UI → Stages → look at task duration
# If one task takes 10 minutes while others take 10 seconds →
skew!

# Also detect programmatically:
df_sales.groupBy(spark_partition_id().alias("partition")) \
    .count() \
    .orderBy(desc("count")) \
    .show(20)

# If one partition count >> others → skewed!

# COMMON CAUSES IN RETAILMART:
# - region_code "NULL" has 40% of records
# - customer_id "WALK-IN" has 60% of cash sales
# - payment_mode "COD" has 70% of online orders

# SOLUTIONS:
# 1. Filter skewed values separately
# 2. Salt the skewed key (add random suffix)
# 3. Enable AQE skew join handling

# AQE handles skew automatically:
spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")
spark.conf.set("spark.sql.adaptive.skewJoin.skewedPartitionThresholdInBytes",
    str(256 * 1024 * 1024)) # 256MB threshold
# AQE splits partitions larger than 256MB automatically!

```

◆ SCD Type 1 in Synapse Notebook

```
# — SCD TYPE 1: OVERWRITE CHANGED VALUES (No History) —————
# Scenario: Customer address changes → update to new address
# Old address completely REPLACED (no history kept)

from delta.tables import DeltaTable
from pyspark.sql import functions as F

# Read new/updated customer data from Bronze
df_source = spark.read \

.parquet("abfss://bronze@retailmartdatalake.dfs.core.windows.net/cus
\
        .withColumn("updated_at", F.current_timestamp())

# Target Delta table (Silver layer)
target_path =
"abfss://silver@retailmartdatalake.dfs.core.windows.net/dim_customer

# Check if Delta table exists
from delta.tables import DeltaTable

if DeltaTable.isDeltaTable(spark, target_path):
    delta_target = DeltaTable.forPath(spark, target_path)

    # MERGE = SCD Type 1
    delta_target.alias("target").merge(
        df_source.alias("source"),
        "target.customer_id = source.customer_id" # match
condition
    ) \
    .whenMatchedUpdate(
```

```
# Only update when something actually changed (avoid unnecessary writes)
```

```
    condition=""
        target.customer_name != source.customer_name OR
        target.email         != source.email         OR
        target.phone          != source.phone          OR
        target.address         != source.address         OR
        target.city           != source.city
    "",
    set={
        "customer_name": "source.customer_name",
        "email":          "source.email",
        "phone":          "source.phone",
        "address":        "source.address",
        "city":           "source.city",
        "state":          "source.state",
        "updated_at":     "source.updated_at"
    }
) \
.whenNotMatchedInsert(
    # New customer not in target
    values={
        "customer_id":    "source.customer_id",
        "customer_name":  "source.customer_name",
        "email":          "source.email",
        "phone":          "source.phone",
        "address":        "source.address",
        "city":           "source.city",
        "state":          "source.state",
        "customer_tier":  "source.customer_tier",
        "created_at":     "source.updated_at",
        "updated_at":     "source.updated_at"
    }
) \
.execute()
```

```

print("✅ SCD Type 1 Merge completed")

else:
    # First load – create the table
    df_source.write \
        .format("delta") \
        .mode("overwrite") \
        .save(target_path)
    print("✅ Initial load completed")

# — VERIFY RESULTS —————
spark.sql(f"""
    DESCRIBE HISTORY delta.`{target_path}`
""").show(5, truncate=False)

# History shows:
# Version 0: WRITE (initial load)
# Version 1: MERGE (today's update – 142 rows updated, 28 new)

```

◆ SCD Type 2 in Synapse Notebook

```

# — SCD TYPE 2: KEEP FULL HISTORY —————
# Scenario: Customer moves from Mumbai to Delhi
# KEEP both records:
#   Row 1: C001, Mumbai, start=2020-01-01, end=2024-06-14,
is_current=N
#   Row 2: C001, Delhi, start=2024-06-15, end=9999-12-31,
is_current=Y

from delta.tables import DeltaTable
from pyspark.sql import functions as F

```

```

from pyspark.sql.types import DateType

target_path =
"abfss://silver@retailmartdatalake.dfs.core.windows.net/dim_customer

# Read source (new/updated customer data)
df_source = spark.read \

.parquet("abfss://bronze@retailmartdatalake.dfs.core.windows.net/cus

# Read current target to find what changed
if DeltaTable.isDeltaTable(spark, target_path):

    delta_target = DeltaTable.forPath(spark, target_path)
    df_target_current = spark.read.format("delta") \
        .load(target_path) \
        .filter(F.col("is_current") ==
True)

# STEP 1: Find records that have CHANGED
df_changed = df_source.alias("src").join(
    df_target_current.alias("tgt"),
    on="customer_id",
    how="inner"
).filter(
    # These columns changed
    (F.col("src.address") != F.col("tgt.address")) |
    (F.col("src.city") != F.col("tgt.city")) |
    (F.col("src.email") != F.col("tgt.email"))
).select("src.customer_id")

changed_ids = [row.customer_id for row in
df_changed.collect()]
print(f"Changed records: {len(changed_ids)}")

```

```

if len(changed_ids) > 0:
    # STEP 2: Expire current records (set is_current=False,
    end_date=today)
    delta_target.alias("target").merge(
        df_changed.alias("changed"),
        "target.customer_id = changed.customer_id AND
target.is_current = true"
    ) \
    .whenMatchedUpdate(set={
        "is_current": "false",
        "effective_end_date": "current_date()"
    }) \
    .execute()

    print("✅ Step 2: Expired old records")

    # STEP 3: Insert new version for changed records + truly new
    records
    df_new_rows = df_source \
        .withColumn("effective_start_date", F.current_date()) \
        .withColumn("effective_end_date", F.lit("9999-12-
31")).cast(DateType()) \
        .withColumn("is_current", F.lit(True))

    # Only insert if: new customer OR changed customer
    df_target_ids = df_target_current.select("customer_id")

    df_to_insert = df_new_rows.join(
        df_target_ids,
        on="customer_id",
        how="left"
    )

    # All rows (source always gets a new current row in SCD2
    # for changed records – merge will handle dedup)

```

```

# Use merge to avoid inserting unchanged records again
delta_target.alias("target").merge(
    df_new_rows.alias("source"),
    """target.customer_id      = source.customer_id
       AND target.is_current    = true
       AND target.effective_start_date =
source.effective_start_date"""
) \
.whenNotMatchedInsert(values={
    "customer_id":      "source.customer_id",
    "customer_name":    "source.customer_name",
    "email":            "source.email",
    "phone":            "source.phone",
    "address":          "source.address",
    "city":             "source.city",
    "state":            "source.state",
    "effective_start_date": "source.effective_start_date",
    "effective_end_date":  "source.effective_end_date",
    "is_current":       "source.is_current"
}) \
.execute()

print("✅ Step 3: Inserted new version records")

else:
    # First load
    df_source \
        .withColumn("effective_start_date", F.current_date()) \
        .withColumn("effective_end_date",   F.lit("9999-12-31")).cast(DateType()) \
        .withColumn("is_current",          F.lit(True)) \
        .write \
        .format("delta") \
        .mode("overwrite") \
        .save(target_path)

```

```
print("✅ Initial SCD2 load completed")
```

```
# — VERIFY SCD2 RESULTS —
```

```
spark.sql(f"""
    SELECT customer_id, city, effective_start_date,
    effective_end_date, is_current
    FROM delta.`{target_path}`
    WHERE customer_id = 'C001'
    ORDER BY effective_start_date
""").show()
```

```
# Expected output:
```

```
# customer_id | city      | effective_start | effective_end |
is_current
# C001        | Mumbai   | 2020-01-01      | 2024-06-14    | false
← old
# C001        | Delhi    | 2024-06-15      | 9999-12-31    | true
← current
```

◆ Executing Synapse Notebooks from Synapse Pipelines

COMPLETE INTEGRATION: Pipeline → Notebook → **Return Value**

```
# — IN THE NOTEBOOK (what it does and returns) —
```

```
# Parameter cell (first cell with parameters)
```

```
input_path  =
"abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/"
output_path =
"abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/"
process_date = "2024-06-15"
```



```

load_type      = "INCREMENTAL"

# Main processing
from pyspark.sql import functions as F
from notebookutils import mssparkutils

try:
    # Read data
    df = spark.read.parquet(f"{input_path}{process_date}/")
    input_count = df.count()

    # Transform
    df_clean = df.filter(F.col("sale_id").isNotNull()) \
                  .filter(F.col("total_amount") > 0)

    # Write
    df_clean.write \
        .format("delta") \
        .mode("overwrite") \
        .save(f"{output_path}{process_date}/")

    output_count = df_clean.count()

    # Return JSON result to pipeline
    import json
    result = json.dumps({
        "status":      "SUCCESS",
        "process_date": process_date,
        "input_rows":   input_count,
        "output_rows":  output_count,
        "rows_dropped": input_count - output_count
    })

    mssparkutils.notebook.exit(result)

```

```
except Exception as e:
    error_result = json.dumps({
        "status": "FAILED",
        "error": str(e),
        "process_date": process_date
    })
    mssparkutils.notebook.exit(error_result)
```

IN SYNAPSE PIPELINE (Notebook Activity):

Activity Type: Notebook

Notebook: /Production/silver/transform_sales

Azure Synapse Analytics Pool: RetailMart_SparkPool

Base Parameters (passed to notebook):

input_path:

abfss://bronze@retailmartdatalake.dfs.core.windows.net/sales/

output_path:

abfss://silver@retailmartdatalake.dfs.core.windows.net/sales/

process_date: @{pipeline().parameters.p_date}

load_type: @{pipeline().parameters.p_load_type}

READING NOTEBOOK OUTPUT IN PIPELINE:

Notebook activity **output** available as:

@{activity('RunTransformNotebook').**output.status**.Output.result.exitValue}

Parse the JSON **exit** value:

@{json(activity('RunTransformNotebook').**output.status**.Output.result.exitValue)}

Access specific fields:

Status:

```
@{json(activity('RunTransform').output.status.Output.result.exitValue)
  Rows:
  @{json(activity('RunTransform').output.status.Output.result.exitValue)
```

USE RESULT IN IF CONDITION:

Condition: @equals(

```
json(activity('RunTransform').output.status.Output.result.exitValue)
  'SUCCESS'
  )}
```

TRUE: → Continue to Gold loading

FALSE: → Log failure + send alert + Fail pipeline

FULL PIPELINE FLOW:

Parameter Cell (p_date) → Notebook Activity (pass p_date)

→ Notebook returns JSON result

→ If SUCCESS → load Gold layer

→ If FAILED → send alert email via Logic Apps

→ Fail pipeline with clear error message

Complete Azure Synapse Architecture Summary

AZURE SYNAPSE ANALYTICS – COMPLETE GUIDE

POOL TYPES		Dedicated SQL (DW), Serverless SQL (Lake),
		Apache Spark Pool

SERVERLESS SQL	OPENROWSET, External Tables, Views
	Query CSV/Parquet/JSON/Delta directly
	External Data Source, File Format, Credential
DEDICATED SQL	MPP architecture, 60 distributions
	Hash/RoundRobin/Replicated tables
	PolyBase & COPY INTO for fast bulk loads
PIPELINES	Same as ADF (built-in)
(Integrate) files	Wildcard copy, Dataset params, Dynamic
	Schedule + Event triggers
	ForEach bulk copy with control table
SPARK POOL	MSSparkUtils (fs, notebook, credentials)
	All file formats (CSV, JSON, Parquet, Avro, ORC)
	All connections (ADLS, Blob, SQL, On-prem)
	Repartition vs Coalesce
	All join types + Broadcast optimization
	AQE, Catalyst, Skew handling

	SCD Type 1 + Type 2 (Delta MERGE)
INTEGRATION	Pipeline → Notebook (pass params, read exit)
	Key Vault secrets via MSSparkUtils
	SHIR for on-premise connections
	Azure DevOps Repos for notebook versioning
OPTIMIZATIONS	Broadcast joins, AQE, shuffle partitions
	Small file compaction, repartition/coalesce
	Partition pruning, predicate pushdown
	Schema definition (avoid inferSchema)

RETAILMART FINAL ARCHITECTURE:

Sources → ADF (orchestrate) → Synapse Pipelines (ingest)

- ADLS Bronze → Synapse Spark Notebooks (transform)
- ADLS Silver → Serverless SQL Pool (explore/views)
- ADLS Gold → Dedicated SQL Pool (BI/reporting)
- Power BI Dashboards (business insights)

✅ **Azure Synapse Analytics section 100% complete** — every concept from the PDF including both pools (Dedicated + Serverless), Spark pool, all file formats, all connection methods, MSSparkUtils, notebook parameters, magic commands, SCD1/SCD2, triggers,

PolyBase/COPY INTO, bulk copy patterns, optimization techniques, Catalyst optimizer, skewness handling, and pipeline-to-notebook integration — all covered with RetailMart real-world examples following practical, depth-first teaching style.

 **Next: Microsoft Fabric Course** — ready to continue!

19/03/2026, 11:27:09

Thought process: Orchestrated comprehensive Microsoft Fabric curriculum mapping across twenty detailed instructional modules.

The user wants me to continue with the Microsoft Fabric Course section from the PDF, explaining it in the style of Ansh Lamba (a popular data engineering YouTuber/Udemy instructor known for his clear, practical, step-by-step teaching style with real-world examples, analogies, and conversational tone). Let me cover ALL concepts from the PDF's Microsoft Fabric section comprehensively.

From the PDF, the Microsoft Fabric Course covers:

- 1) Introduction to Big Data
- 2) Introduction to Microsoft Fabric
- 3) Introduction to Fabric Components
- 4) Fabric Account Creation
- 5) OneLake Concepts
- 6) Data Factory in Microsoft Fabric

- 7) Fabric Data Engineering – Lakehouse
- 8) Fabric Data Engineering – Notebooks
- 9) Fabric Data Engineering – Apache Spark
- 10) Fabric Data Engineering – Delta Lake Tables
- 11) Fabric Data Warehouse
- 12) Fabric Real-time Intelligence
- 13) Activators in Fabric

- 14) Power BI in Fabric
- 15) Azure DevOps – Repos
- 16) SDLC and Agile Methodology
- 17) End-to-End Project: Migrating from On-Premises to Cloud
- 18) Resume Preparation
- 19) Interview Guidelines
- 20) Mock Interviews

Let me cover all of these in depth with Ansh Lamba's teaching style, breaking down each topic with practical examples and real-world applications.



Microsoft Fabric — Complete Deep Dive Guide

Teaching Style: Conversational, practical, real-world first — just like how Ansh Lamba teaches on YouTube/Udemy. No fluff, just clarity.



Let's Start With The Right Mindset

Before we dive in, let me ask you something...

You've learned:

- ✓ ADF → moves data
- ✓ Databricks → transforms data
- ✓ Synapse → warehouses + queries data
- ✓ Power BI → visualizes data

Now Microsoft looked at all of this and said:

"Why are our customers using 4-5 different tools?
Why can't we give them ONE platform for everything?"

That's EXACTLY what Microsoft Fabric is.

Think of it this way:

Old World = You need a car, a bike, a truck, and a bus
to travel (4 different tools, 4 bills, 4 manuals)

New World = You have ONE vehicle that transforms
into whatever you need (car/bike/truck/bus)
Pay ONE bill. Learn ONE platform.

THAT is Microsoft Fabric.

1 Introduction to Big Data — Fabric Perspective

◆ What is Data? (Fabric Context)

Let me tell you something most tutorials skip...

When Microsoft designed Fabric, they designed it
around ONE core idea:

"Data should be stored ONCE and used EVERYWHERE"

Let's understand why this matters with a story:

RETAILMART's DATA MESS (Before Fabric):

Data Engineering team:

→ Stores sales data in ADLS Gen2 (their copy)

Data Science team:

→ Downloads same data to their Databricks (COPY 2)

BI team:

→ Loads same data to SQL Server (COPY 3)

Finance team:

→ Exports to Excel (COPY 4)

Result:

4 copies of same data

4 different "versions of truth"

When source changes → need to update 4 places

Storage cost = 4x what it should be

Data inconsistency = wrong decisions!

Microsoft Fabric's answer: OneLake

→ ONE copy of data

→ ALL teams access SAME data

→ Data Engineers, Scientists, BI analysts –
all work on SAME underlying files

→ No duplication, no inconsistency

→ This is the REVOLUTION of Fabric

◆ What is a Database? What is Big Data?

Quick recap with Fabric perspective:

DATABASE = Organized collection of structured data

Examples: Azure SQL DB, MySQL, PostgreSQL

Works great for: 1MB to 1TB of structured data
Fails at: petabytes, unstructured data, streaming

BIG DATA = Data that is:

Too LARGE (can't fit on one machine)
Too FAST (arriving too quickly to process)
Too VARIED (CSV, JSON, images, videos, logs)

FABRIC solves ALL of this:

Large: OneLake stores petabytes
Fast: Real-time Intelligence for streaming
Varied: Lakehouse handles all formats

CHALLENGES of Big Data that Fabric addresses:

Challenge	How Fabric Solves It
Storage at scale	OneLake (unlimited, cheap)
Processing speed	Spark Pool (distributed)
SQL analytics	Data Warehouse
Real-time	Real-time Intelligence (KQL)
Visualization	Power BI (built-in)
Governance	Purview integration
Cost	Capacity-based (predictable)

WHY Traditional Databases CANNOT handle Big Data:

- Single server architecture (can't scale out)
- Only handles structured data
- Schema must be defined before data arrives
- Cannot handle real-time streaming natively
- Cost explodes at petabyte scale
- No native ML/AI capabilities

2 Introduction to Microsoft Fabric

◆ What is Microsoft Fabric?

Here's the simplest way I can explain it:

Microsoft Fabric =

Azure Data Factory (data integration)
+ Azure Databricks (big data processing)
+ Azure Synapse Analytics (data warehousing)
+ Power BI (visualization)
+ Azure Purview (governance)
+ Real-time Analytics (streaming)

ALL wrapped in ONE unified SaaS platform
on TOP of OneLake (single storage)

OFFICIAL DEFINITION:

Microsoft Fabric is an end-to-end analytics platform
that provides a unified experience for data engineers,
data scientists, data analysts, and business users
to work together on the same platform.

KEY WORD: "Unified"

Not just integration between tools

– it's ONE product, ONE license, ONE experience

FABRIC vs COMPETITORS:

Databricks Lakehouse Platform → closest competitor

AWS Data Stack (Glue+Redshift+QuickSight) → fragmented

Google Cloud (Dataflow+BigQuery+Looker) → more fragmented

Fabric's edge: Deepest Power BI + Office 365 integration

Microsoft 365 users → natural fit

◆ How to Enable Fabric in Your Organization

FABRIC IS A PREMIUM FEATURE – needs to be enabled!

STEP 1: Check if you have correct license

Fabric requires:

- Microsoft Fabric (trial or paid capacity)
- OR Power BI Premium Per Capacity
- OR Microsoft 365 E5/A5 license

STEP 2: Admin enables Fabric

Microsoft 365 Admin Center

- Settings → Org Settings → Microsoft Fabric
- Toggle: "Users can try Microsoft Fabric" → ON
- Choose: "Entire organization" or "Specific users"

STEP 3: Verify in Power BI

Go to app.powerbi.com

- Settings gear → Admin Portal
- Tenant Settings → Microsoft Fabric section
- "Users can create Fabric items" → Enabled

STEP 4: Start Fabric Trial (for learning)

app.fabric.microsoft.com

- Click your profile → Start Trial
- 60-day FREE trial with F64 capacity!
- F64 = 64 CUs (Capacity Units)
- More than enough for learning all features

STEP 5: Verify Trial Active

app.fabric.microsoft.com

→ Settings → Capacity Settings

→ Shows: Trial capacity active, 60 days remaining

PRO TIP from experience:

Don't use your company account for learning

Create a new Microsoft account (outlook.com)

Start FREE trial

Experiment freely without affecting company data!

◆ Fabric Workspace Structure

FABRIC WORKSPACE = Your working environment

Think of Workspace like a PROJECT FOLDER:

- One workspace per project/team/domain
- Contains all Fabric items for that project
- Access controlled at workspace level

WORKSPACE HIERARCHY:

Tenant (your company)

└─ Workspaces

 └─ RetailMart_DataEngineering ← for DE team

 └─ Lakehouse: RetailMart_LH

 └─ Pipeline: PL_Daily_Sales

 └─ Notebook: Transform_Sales

 └─ Dataflow: DF_CustomerMaster

 |

 └─ RetailMart_DataScience ← for DS team

```
|      |— Lakehouse: (shortcut to DE lakehouse!)
|      |— Notebook: ML_FraudDetection
|      |— ML Model: FraudModel_v2
|
|— RetailMart_Analytics          ← for BI team
    |— Warehouse: RetailMart_DW
    |— Dataset: Sales_PBI_Dataset
    |— Report: Sales_Dashboard
```

KEY CONCEPT:

Data Engineering workspace has the **REAL** data

Data Science workspace uses SHORTCUTS (**no copy!**)

Analytics workspace also uses SHORTCUTS (**no copy!**)

Everyone sees SAME data → OneLake magic!

WORKSPACE ROLES:

Admin → **Full** control, manage workspace settings

Member → **Create**/edit items, publish content

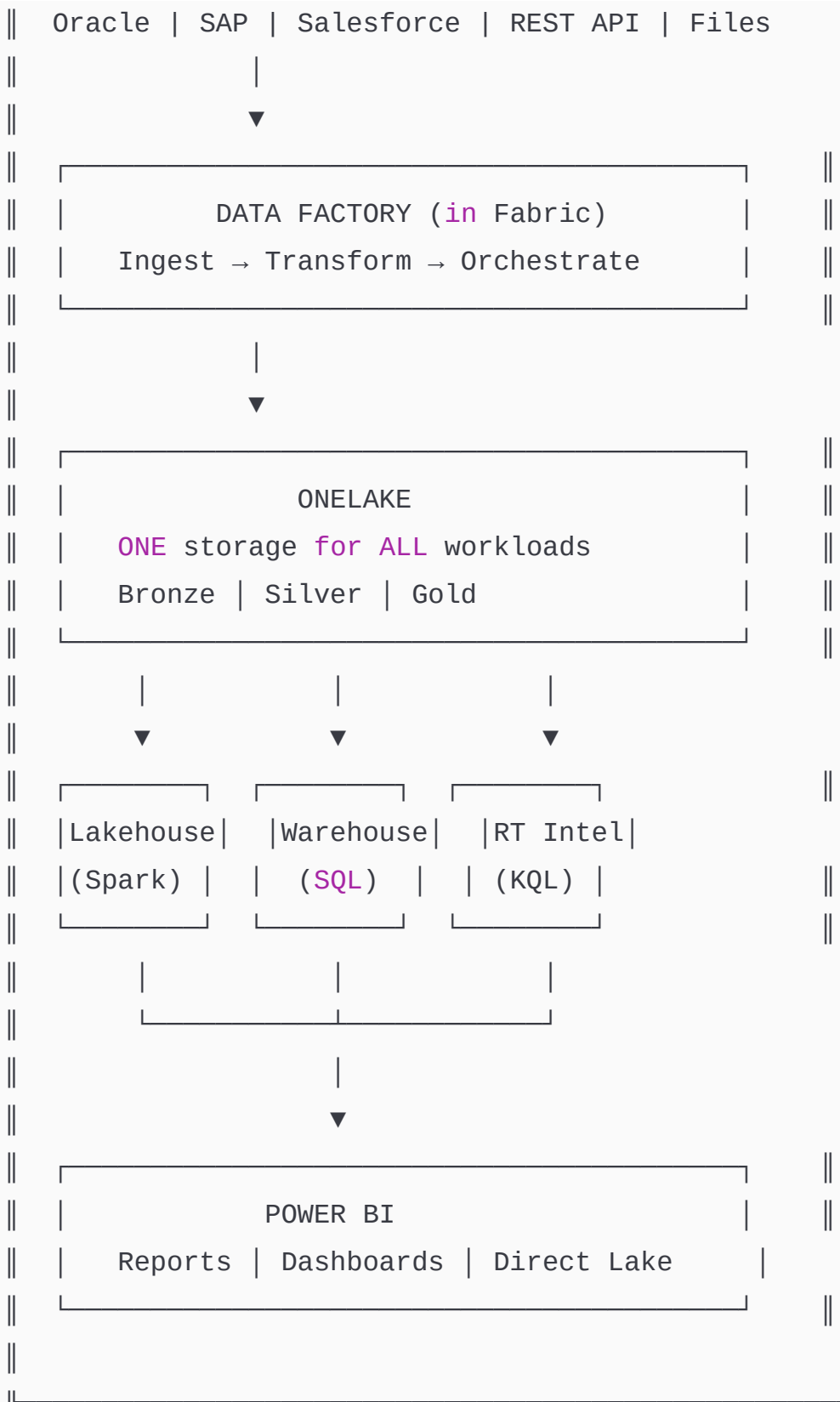
Contributor → **Create**/edit items

Viewer → Read-**only** access (can **view** reports)

◆ Microsoft Fabric — A Unified Platform

LET ME DRAW THE BIG PICTURE:





THE MAGIC:

Everything shares OneLake storage

No data movement **between** experiences

No ETL **between** Lakehouse **and** Warehouse

Power BI **reads** directly **from** Lakehouse tables!
This **is** why Fabric **is** revolutionary!

◆ Advantages of Microsoft Fabric

Let me be honest **with** you – **both** pros **AND** cons:

ADVANTAGES:

1. UNIFIED EXPERIENCE

One UI (app.fabric.microsoft.com) **for** everything
No jumping **between** ADF Studio → Databricks → Synapse Studio
Data Engineer + Data Scientist + BI Analyst **in** SAME tool

2. ONELAKE = **ONE COPY OF** DATA

Lakehouse **and** Warehouse share same underlying files
Power BI **reads** directly (**no** data movement!)
Shortcuts = virtual access (**no** physical **copy**)
Massive cost saving vs traditional approach

3. CAPACITY-BASED PRICING (predictable)

Pay **for** Fabric Capacity (F SKUs: F2, F4, F8...F2048)
One bill **for** **ALL** workloads
vs: separate bills **for** ADF + Databricks + Synapse + PBI
Premium

4. POWER BI INTEGRATION (best **in** class)

Direct Lake mode = Power BI **reads** Parquet files directly
No import, **no** DirectQuery limitations
Fastest BI experience possible **on** **large** datasets

5. EASIER FOR ENTERPRISES

Works with Microsoft 365 identity (no new accounts)

Integrates with Teams, SharePoint, Excel

Familiar Microsoft ecosystem

6. COPILOT EVERYWHERE

AI-powered code generation in notebooks

Natural language to SQL in Warehouse

Report generation from data descriptions

LIMITATIONS (be honest!):

- ✗ Still maturing (launched 2023 – newer than Databricks)
- ✗ Less ML capabilities than Databricks
- ✗ Unity Catalog-equivalent still developing
- ✗ Some features only in specific regions
- ✗ Vendor lock-in to Microsoft ecosystem
- ✗ Pricing can be complex at scale

3 Fabric Components — Deep Dive

◆ OneLake

OneLake = The FOUNDATION of everything in Fabric

Think of OneLake like Google Drive for your data:

- One place to store ALL data
- Everyone accesses same files
- No "my copy" and "your copy"
- Organized in folders (workspaces/lakehouses)

TECHNICAL REALITY:

OneLake **is** built **on** ADLS Gen2 under the hood
Microsoft manages it (**no** storage account **to create!**)
Automatically geo-redundant

ONELAKE ADDRESS FORMAT:

Every item **in** Fabric has an ADLS-compatible path:

onelake://<workspace>/<item>.Lakehouse/Files/<path>

ADLS path (**for** Spark access):

abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.Lakehouse

ONELAKE HIERARCHY:

OneLake (tenant level – **ONE per** Microsoft tenant)
└─ Workspace (**like** a container/folder)
 └─ Lakehouse / Warehouse / etc.
 └─ Tables/ (managed Delta tables)
 └─ Files/ (raw files – **any** format)

KEY INSIGHT:

ALL Fabric experiences (Lakehouse, Warehouse, Spark, PBI)
read **from** the SAME OneLake files.

Example:

Spark writes Parquet files → OneLake/Lakehouse/Tables/sales/
Warehouse **SQL reads from** → same OneLake/Lakehouse/Tables/sales/
Power BI Direct Lake **reads from** → same files!

NO DATA MOVEMENT. **ONE** TRUTH.

◆ Real-Time Intelligence

Real-Time Intelligence = Fabric's *streaming analytics engine*

COMPONENTS:

EventStream → capture streaming events

Eventhouse → store **and** query streaming data (KQL database)

KQL Database → time-series optimized database

Activators → trigger actions **on** data patterns

REAL WORLD USE **CASE for** RetailMart:

Payment transactions arrive every millisecond

→ EventStream captures **each** payment

→ Stored **in** Eventhouse (KQL database)

→ KQL queries analyze **in** real-time

→ Activator fires alert **if** fraud detected

All within SECONDS **of** payment occurring!

This replaces:

Azure **Event** Hubs + Stream Analytics + Azure Functions

= **3** tools → now **1** experience **in** Fabric

◆ Data Factory in Microsoft Fabric

Data Factory **in** Fabric = same **as** Azure Data Factory
but BUILT **INTO** Fabric workspace

KEY DIFFERENCE **from** standalone ADF:

Feature	Standalone ADF	Fabric Data Factory
---------	----------------	---------------------

Pipelines	Same	Same (identical!)
Dataflows	Gen 1, Gen 2	Gen 2 only
Connectors	90+	90+ (same)
Destination	Any	Native Fabric items
Billing	Per execution	Part of Fabric capacity
Git integration	Azure DevOps/GH	Azure DevOps/GH

ADVANTAGE of Fabric Data Factory:

- Native destination to Lakehouse, Warehouse
- No linked service setup for Fabric items
- Billed from Fabric capacity (no separate meter)
- Dataflow Gen2 = massive improvement over Gen1

DATAFLOW GEN2 (new concept in Fabric):

- Power Query (M language) based transformation
- Like ADF Data Flows but uses Power Query engine
- 200+ connectors (more than ADF!)
- Output directly to Lakehouse tables
- Great for: non-coders who know Excel Power Query
- Not great for: complex Spark-level transformations

◆ Fabric Data Science

Fabric Data Science = ML platform built into Fabric

COMPONENTS:

- ML Experiments → track model training runs
- ML Models → store and version models
- Notebooks → PySpark/Python for ML code

WORKFLOW:

Data **in** Lakehouse (**from** Data Engineering team)

↓ (no data movement needed – Shortcuts!)

Data Scientist opens Notebook

- Reads directly **from** Lakehouse tables
- Trains ML model (scikit-learn, PyTorch, etc.)
- Logs experiment **to** MLflow (built-**in**)
- Registers model
- Deploys **for** batch scoring
- Results written back **to** Lakehouse

RETAILMART USE CASE:

Fraud Detection Model:

- Training data: **2** years **of** transaction history
- Features: amount, time, location, device
- Model: XGBoost **binary** classifier
- Output: fraud_probability score per transaction
- Batch scoring: every night **on next** day's *transactions*
- Results **in** Lakehouse → Power BI shows fraud dashboard

◆ Fabric Data Engineering

Fabric Data Engineering = Spark-based **data** transformation

MAIN ITEMS:

- Lakehouse → storage + processing (our main focus!)
- Notebook → PySpark/SQL code
- Spark Job Definition → productionize Python scripts
- Data Pipeline → orchestrate notebooks

This **is** what Data Engineers use MOST **in** Fabric.
Think of it **as** "Databricks-lite" inside Fabric.

◆ Data Activator

Data Activator = Alert **and** automation engine

Think **of** Data Activator **as**:

"If this happens in my data → do this automatically"

Examples:

IF sales **drop** below Rs.**1,00,000** **by** 3 PM
→ Send Teams message **to** regional manager

IF fraud score > **0.9** **for** a transaction
→ Block the payment + alert security team

IF inventory falls below safety stock
→ **Create** purchase **order in** SAP automatically

WORKS **WITH**:

- **Real-time** data (EventStream)
- Power BI data (**from** dashboards!)
- Lakehouse data

This replaces:

Azure Logic Apps + Power Automate **for** data triggers

◆ Fabric Data Warehouse

Fabric Data Warehouse = Enterprise SQL analytics engine

DIFFERENT FROM SYNAPSE DEDICATED POOL:

Synapse Dedicated Pool:

- You provision DWUs (compute always running)
- Pay even when idle!
- MPP with 60 distributions
- Need to design distribution strategy (Hash/RR/Replicated)
- More complex to manage

Fabric Data Warehouse:

- Serverless compute (Microsoft manages it)
- Pay per usage (from Fabric capacity)
- Standard SQL (T-SQL)
- No distribution strategy needed!
- Simpler, automatic optimization
- Still works on OneLake (Delta Parquet files underneath)

KEY FEATURE – Cross-database querying:

SELECT

s.sale_id,
s.total_amount,
c.customer_name

FROM [RetailMart_Warehouse].[sales].[fact_sales] s

JOIN [RetailMart_Lakehouse].[dbo].[dim_customers] c ←

different item!

ON s.customer_id = c.customer_id

WAREHOUSE and LAKEHOUSE share OneLake

- Can join across them without data movement!

POWER BI INTEGRATION:

- Warehouse auto-creates a **SQL** Analytics Endpoint
- Power BI connects **to** this endpoint
- Semantic models built **on** Warehouse tables
- Direct Lake mode: BI **reads** parquet files directly!

◆ Power BI in Fabric

Power BI = Microsoft's *BI tool (you already know it!)*

NEW **in** Fabric – DIRECT LAKE MODE:

Old Power BI modes:

- Import: copy data **to** PBI memory (fast queries, data can be stale)
- DirectQuery: query source **each** time (always fresh, but slow)

New Direct Lake mode:

- Power BI reads Parquet files DIRECTLY **from** OneLake
- Speed **of** Import (reads columnar Parquet natively)
- Freshness **of** DirectQuery (always reading latest files)
- Best **of** both worlds!

HOW: **When** Spark writes Delta tables **to** Lakehouse

- they're *Parquet files in OneLake*
- Power BI Direct Lake reads those files
- No intermediary needed!

Think **of** Direct Lake **as**:

Power BI got a "**superpower**" **to** read big data files directly without importing them

Like reading a book through the shelf
instead of checking it out and taking home!

4 Fabric Account Creation — Step by Step

◆ How to Sign Up for Microsoft Fabric

OPTION 1: Microsoft Fabric Trial (RECOMMENDED for learning)

Step 1: Create Microsoft account

- Go to outlook.com → Create account
- Use: yourname_fabric_learning@outlook.com

Step 2: Go to Microsoft Fabric

- Open: app.fabric.microsoft.com
- Sign in with your new account

Step 3: Start Trial

- Top right → Settings (gear icon)
- "Start Trial" button
- Trial: 60 days FREE
- Capacity: F64 (64 Capacity Units)

Step 4: Verify Trial

- Settings → Admin Portal → Capacity Settings
- Should see: "Trial-<your_name>" capacity listed

OPTION 2: Use existing Power BI Premium/Pro account

- If your company has Power BI Premium
- Admin enables Fabric in tenant settings
- You get access automatically

OPTION 3: Azure + Fabric

- Buy Fabric capacity from Azure Portal
- Create resource: Microsoft Fabric
- Choose SKU: F2 (\$263/month) to F2048 (massive)

FOR THIS GUIDE: We'll use Fabric Trial (free!)

◆ How to Activate Microsoft Fabric Trial

DETAILED ACTIVATION STEPS:

1. Navigate to: `app.fabric.microsoft.com`
2. If you see "Fabric is disabled":
 - You need admin to enable it
 - OR: switch to personal trial account
3. Click your profile picture (top right)
4. Look for: "Start trial" option
 - Click it
 - Read trial terms
 - Click "Activate" / "Start trial"
5. What you get in trial:

FABRIC TRIAL INCLUDES:
→ F64 Capacity (64 CUs)
→ All Fabric workloads enabled
→ OneLake storage (up to 1TB)

	→ 60 days free	
	→ No credit card required!	

6. After activation → you'll see *Fabric home page*

→ Create workspace

→ Start building!

WHAT IS A CU (Capacity Unit)?

CU = unit of compute power in Fabric

F64 = 64 CUs = comfortable for learning + small projects

Compute automatically scales within your CU limit

Think of CUs like "compute budget"

Heavy Spark job uses more CUs temporarily

Light SQL query uses fewer CUs

Always within your allocated limit

◆ Creating and Managing Workspaces

WORKSPACE = Your project container in Fabric

CREATE WORKSPACE:

Fabric Home → Workspaces (left sidebar)

→ + New Workspace

Name: RetailMart_DataEngineering

Description: Data engineering workspace for RetailMart project

Advanced settings:

License mode: Trial ← (or Fabric capacity if available)

→ Apply → Workspace created!

WORKSPACE SETTINGS:

Workspace → Settings (gear icon)

General:

- Name, description
- Contact list (who to email on issues)

License:

- Change capacity (move between trial/paid)

Git Integration:

- Connect to Azure DevOps or GitHub
- Version control all Fabric items!

Spark Settings:

- Configure Spark properties for notebooks
- Set default Spark pool size

OneLake Settings:

- Data residency region

Permissions:

- Add members with specific roles
- Admin/Member/Contributor/Viewer

WORKSPACE TYPES based on license:

- Trial:** F64 capacity, 60 days, free
- Pro:** Power BI Pro license, limited Fabric features
- Premium:** Power BI Premium, more features
- Fabric:** Full Fabric capacity, all features

PRO TIP:

Create SEPARATE workspaces for:

- ✓ Development (build and test)
- ✓ Testing (QA and UAT)
- ✓ Production (live pipelines)

This is proper DevOps/SDLC practice!

◆ Workspace Configurations and Settings

```
# WORKSPACE CONFIGURATION BEST PRACTICES
```

```
# Naming Convention (follow this always!):
```

```
workspace_naming = {  
    "pattern": "<Company>_<Domain>_<Environment>",  
    "examples": [  
        "RetailMart_Sales_Dev",  
        "RetailMart_Sales_Test",  
        "RetailMart_Sales_Prod",  
        "RetailMart_Finance_Prod",  
        "RetailMart_HR_Dev"  
    ]  
}
```

```
# Items inside workspace also follow naming:
```

```
items = {  
    "Lakehouse": "LH_<Domain>_<Env> → LH_Sales_Prod",  
    "Warehouse": "WH_<Domain>_<Env> → WH_Analytics_Prod",  
    "Pipeline": "PL_<Purpose> → PL_Daily_SalesLoad",  
    "Notebook": "NB_<Purpose> → NB_Transform_Sales",  
    "Dataset": "DS_<Domain> → DS_Sales_Reporting"  
}
```

WORKSPACE SETTINGS WALKTHROUGH:

General Settings:

→ Contact email list

Add: data-engineering-team@retailmart.com

This email gets notifications about workspace issues

→ OneDrive folder (optional)

Link to SharePoint for document storage

Git Integration (critical for production!):

→ Azure DevOps:

Organization: retailmart-devops

Project: DataPlatform

Repository: fabric-items

Branch: main

Folder: /RetailMart_Sales/

→ After connecting: ALL items saved to Git automatically!

→ Changes tracked → pull requests → code review → merge

Spark Settings:

→ Default pool size: Medium (4 nodes)

→ Spark version: 3.4

→ Default libraries: requirements.txt from Git

Security:

→ Workspace identity (Managed Identity for Fabric)

→ Used to access Key Vault, ADLS, SQL DB

→ Same concept as ADF Managed Identity

◆ How to Manage Capacity Units

CAPACITY UNITS (CUs) = Compute currency of Fabric

F SKUs and their CUs:

SKU	CUs	Price/month	Good for
F2	2	~\$263	Basic testing
F4	4	~\$526	Small team
F8	8	~\$1,053	Small org
F16	16	~\$2,105	Medium org
F32	32	~\$4,210	Larger workloads
F64	64	~\$8,420	Medium-large org
F128	128	~\$16,840	Large org
F2048	2048	~\$269,466	Very large enterprise

THROTTLING (important to understand!):

Each CU allows certain operations per minute

F64 = 64 × 30 seconds = burst capacity

If you use too many CUs at once:

- Fabric THROTTLES (slows down) remaining requests
- Does NOT fail – just queues and slows
- Like traffic jam: requests wait, don't crash

MONITOR CAPACITY:

Admin Portal → Capacity Metrics app

- See CU usage over time
- Identify peak usage periods
- Find which workspaces using most CUs

OPTIMIZE CU USAGE:

- ✓ Schedule heavy jobs in off-peak hours

- ✓ Use Spark pause when idle
- ✓ Optimize notebooks (less data reads = fewer CUs)
- ✓ Use Direct Lake (avoids import processing CUs)
- ✓ Share capacity across multiple workspaces

5 OneLake — Deep Dive

◆ Unified Storage Layer

OneLake = The single storage for ALL of Fabric

BEFORE OneLake (the problem):

Team A creates Azure Storage Account → uploads data

Team B creates another Azure Storage Account → uploads same data

Team C has SQL Server → loads same data again

3 storage accounts, 3 copies, 3 bills, 3 truths!

WITH OneLake (the solution):

ONE storage, automatically created for your tenant

Every workspace, every lakehouse → same OneLake!

STRUCTURE:

OneLake (1 per Azure region per tenant)

└─ <WorkspaceName>

 └─ <LakehouseName>.Lakehouse

 └─ Tables/ (Delta Parquet – queryable via

SQL!)

 └─ Files/ (raw files – any format)

└─ <WarehouseName>.Warehouse


```
|      └─ (SQL tables – stored as Delta in OneLake)
└─ <DatasetName>.Dataset
    └─ (semantic model metadata)
```

ACCESS PATTERNS:

From Notebooks:

```
abfss://RetailMart_DataEngineering@onelake.dfs.fabric.microsoft.com/
```

From Fabric Items (use relative paths):

```
Files/raw/sales/2024/06/sales.parquet
```

From Desktop (OneLake Explorer app):

Windows File Explorer → OneLake → browse like local files!

OneLake Explorer (hidden gem!):

- Install from Microsoft Store
- Mounts OneLake as local drive
- Browse Fabric data like Windows folder!
- Drag and drop files to upload
- Perfect for small file uploads during development

◆ Logical Data Organization

HOW TO ORGANIZE DATA IN ONELAKE:

RECOMMENDED LAKEHOUSE STRUCTURE:

RetailMart.Lakehouse

└─ Tables/	← MANAGED (queryable via SQL!)
└─ bronze_sales	(Delta table from raw load)
└─ silver_sales	(Delta table from transformation)

```

|   |─ gold_store_performance (Delta table, ready for BI)
|   |─ dim_customers          (dimension table)
|   |─ dim_products           (dimension table)
|
└─ Files/                      ← UNMANAGED (raw files)
    └─ raw/
        └─ sales/
            └─ 2024/
                └─ 01/
                    └─ sales_20240101.csv
                └─ 06/
                    └─ sales_20240615.parquet
            └─ customers/
                └─ customer_master.csv
        └─ config/
            └─ store_mapping.xlsx
    └─ archive/
        └─ processed/

```

KEY RULE:

Tables/ → use **for** transformed, queryable data
 Automatically gets **SQL** endpoint!
 Power BI can **connect** directly

Files/ → use **for** raw ingestion
Not automatically queryable via **SQL**
 Access via Spark notebooks **only**

◆ Delta Tables & Open Data Formats

OneLake stores data **in OPEN** formats **only**:

DELTA PARQUET = Primary format for Tables/

- Delta = Parquet + Transaction log
- ACID transactions
- Time travel
- Schema evolution
- Same format Databricks uses!

This means:

Databricks writes Delta → OneLake reads it (compatible!)

Fabric writes Delta → Databricks can read it!

OPEN STANDARD → no vendor lock-in on format

SUPPORTED FILE FORMATS in Files/:

- CSV (any delimiter)
- Parquet (columnar – recommended!)
- JSON / NDJSON (newline-delimited)
- Avro (streaming)
- ORC (Hive-compatible)
- Excel (.xlsx)
- Images, PDFs, any binary (for ML)
- Delta (in Files too, not just Tables)

WHY OPEN FORMATS MATTER:

Microsoft Fabric could go away tomorrow
(unlikely, but hypothetically)

Your data is still in Delta Parquet files

- Move to Databricks: works immediately
- Move to Synapse: works immediately
- Move to BigQuery: works (with conversion)

You own your data. Microsoft just runs the platform.

That's the promise of open formats.

◆ Shortcuts — The Game Changer

SHORTCUTS = Virtual links **to** data without copying it

This **is** THE most important OneLake concept!

WHAT **IS** A SHORTCUT?

Like a symbolic link (symlink) **in** Linux

Like a shortcut file **in** Windows (.lnk)

Points **to** data somewhere **else**

Appears **as if** data **is** local

But ZERO data copied!

WHERE CAN SHORTCUTS POINT **TO**?

- Another OneLake location (different workspace)
- Azure Data Lake Storage Gen2
- Azure Blob Storage
- Amazon S3 (AWS!)
- Google Cloud Storage (GCS!)
- Dataverse (Microsoft Power Platform)

RETAILMART EXAMPLE:

Data Engineering team has Gold data:

LH_Sales_Prod.Lakehouse/Tables/gold_sales

Data Science team wants **to** use same data:

LH_DataScience.Lakehouse/Tables/gold_sales ← SHORTCUT!

- Points **to** Data Engineering's *gold_sales table*
- Data Scientists query it **as if** it's *their own*
- ANY changes DE makes → immediately visible **to** DS

- ZERO storage cost **for** DS team!
- ZERO ETL pipeline **to** sync data!

BI team also shortcuts:

- WH_Analytics.Warehouse/gold_sales ← SHORTCUT **to** Lakehouse!
- Warehouse gets SQL access **to** Lakehouse tables
- Power BI connects **to** Warehouse
- Reports always fresh (no data movement!)

CREATE SHORTCUT:

- Lakehouse → Files/ **or** Tables/ → right-click
- **New** Shortcut
- Choose: OneLake / ADLS / S3
- Provide path **and** credentials
- Shortcut appears instantly!

SHORTCUT **TO** ADLS GEN2 (existing data):

- Bring your existing ADLS data **into** Fabric
- WITHOUT moving **or** copying anything
- Fabric sees it **as** native OneLake data
- Use Spark notebooks **to** process it
- Write results back **to** Lakehouse Tables

PERFECT **for**: migrating **FROM** Databricks/Synapse **TO** Fabric

Keep data **in** ADLS → create shortcut **in** Fabric

- Gradual migration without "big bang" cutover!

◆ Security & Access Control in OneLake

ONELAKE SECURITY LEVELS:

Level 1: Workspace Level

- Admin / Member / Contributor / Viewer roles
- Controls access to ALL items in workspace
- Coarse-grained (all or nothing)

Level 2: Item Level

- Grant access to specific Lakehouse/Warehouse
- Without giving workspace access
- "ReadAll" permission → read all data in item

Level 3: Table Level (SQL Endpoint)

- GRANT SELECT ON TABLE to specific user/group
- Available through SQL analytics endpoint
- T-SQL standard GRANT/REVOKE

Level 4: Column/Row Level (Warehouse)

- Column security: hide specific columns
- Row-level security: users see only their data

Level 5: OneLake Data Access Control

- Fine-grained access to specific folders/files
- Define rules: "Finance team can only read /finance/ folder"
- Uses Azure AD groups

WORKSPACE IDENTITY (Managed Identity):

- Each workspace gets its own Managed Identity
- Use this to access external resources (Key Vault, ADLS)
- Same concept as ADF Managed Identity
- No passwords needed for workspace-to-resource auth

PRACTICAL SECURITY SETUP for RetailMart:

Data Engineering workspace: DE team (Admin), Lead (Admin)

Data Science workspace: DS team (Member), DE team (Contributor)

Analytics workspace: BI team (Member), DS team (Viewer)

Shortcuts: DS accesses DE data via shortcut
→ DS team has read-only access via shortcut
→ Cannot modify DE's actual data
→ Perfect separation of concerns!

◆ Performance & Cost Optimization in OneLake

PERFORMANCE TIPS:

1. USE DELTA FORMAT for Tables/
 - Columnar = faster analytics
 - Parquet compression = 70% less storage
 - Partition pruning = read less data
2. PARTITION CORRECTLY
 - Large tables: partition by date (year/month/day)
 - BI queries filter by date = massive speed boost
3. OPTIMIZE DELTA TABLES regularly
In Lakehouse notebook:
%%sql
OPTIMIZE sales_delta ZORDER BY (customer_id, sale_date);
-- Compacts small files + z-orders for faster lookups
4. VACUUM old files
%%sql
VACUUM sales_delta RETAIN 168 HOURS; -- keep 7 days history
-- Removes old data files (but keeps 7 days for time travel)
5. V-ORDER (Fabric specific!)
Microsoft's proprietary optimization for Parquet files

Enables faster Power BI Direct Lake reads

Enable V-ORDER when writing:

```
spark.conf.set("spark.sql.parquet.vorder.enabled", "true")
df.write.format("delta").save("Tables/sales")
-- Files written with V-ORDER = 2-3x faster PBI queries!
```

COST OPTIMIZATION:

1. PAUSE SPARK when not using

Spark pool: set idle timeout to 5-10 minutes
→ Auto-pauses = zero CU consumption when idle

2. USE SERVERLESS SQL for exploration

→ No Spark startup time
→ Fewer CUs consumed
→ Great for ad-hoc queries

3. SCHEDULE HEAVY JOBS off-peak

→ CU throttling less likely at 3 AM vs 9 AM

4. COMPACT SMALL FILES

Many small files = many reads = more CUs
Regularly compact: OPTIMIZE command

5. SHORTCUTS vs COPY

Use shortcuts → no storage cost
Never copy data if shortcut works!

6 Data Factory in Microsoft Fabric — Deep Dive

◆ Unified Data Integration Platform

DATA FACTORY **IN** FABRIC = ADF built **into** Fabric

Two ways **to** ingest data **in** Fabric:

WAY **1**: PIPELINES (familiar **from** ADF)

- Same **Copy** Activity, ForEach, If **Condition**
- **200+** connectors
- Best **for**: complex orchestration, **large** data movement
- Code optional (visual drag-**and-drop**)

WAY **2**: DATAFLOW GEN2 (**new!**)

- Power Query M **language** based
- **300+** connectors (even more than ADF!)
- Best **for**: data transformation **without** code
- Perfect **for**: analysts who know Excel Power Query
- Output directly **to** Lakehouse tables!

CREATE PIPELINE **in** Fabric:

Workspace → + **New** → Data Pipeline

- Exactly same UI **as** ADF Studio
- Same activities (**Copy**, ForEach, If, Notebook, etc.)
- Native destination: Lakehouse, Warehouse
- **No** linked service needed **for** Fabric items!

Example: **Copy from** Oracle **to** Lakehouse

- **Create** Linked Service **for** Oracle (same **as** ADF)
- Source: Oracle **table**
- Sink: Lakehouse **Table** (just **select**, **no** connection setup!)
- Run pipeline → data **in** Lakehouse instantly!

◆ Pipeline Activities & Dataflows in Fabric

ALL ADF ACTIVITIES WORK IN FABRIC PIPELINES:

- ✓ Copy Data (same as ADF)
- ✓ ForEach (same)
- ✓ If Condition (same)
- ✓ Lookup (same)
- ✓ GetMetadata (same)
- ✓ Delete (same)
- ✓ Wait (same)
- ✓ Set Variable (same)
- ✓ Execute Pipeline (same)
- ✓ Notebook Activity ← runs Fabric notebooks

NEW in Fabric Pipelines:

Invoke Pipeline → cleaner way to call child pipelines
Office 365 Outlook → send emails natively!
Teams → send Teams messages natively!
(No Logic Apps needed for simple notifications!)

DATAFLOW GEN2 – Complete Walkthrough:

Create Dataflow Gen2:

Workspace → + New → Dataflow Gen2

INTERFACE:

- Power Query Editor opens (like in Power BI Desktop)
- Left panel: Queries (your data tables)
- Main area: Column-level transformations
- Right panel: Query settings, steps

SCENARIO: Load customer CSV → transform → Lakehouse table

Step 1: Get Data

- Home → Get Data → Search "Azure Blob"
- Enter connection details
- Navigate to: /bronze/customers/customer_master.csv
- Preview data shows in editor

Step 2: Transform

- Promote first row as headers
- Change column types (click column header → type)
- Remove nulls: Filter Rows → keep non-null customer_id
- Rename columns: right-click → Rename
- Add custom column: Add Column → Custom Column
Formula: [TotalPurchases] * 0.18 (calculate GST)

Step 3: Configure Destination

- Home → Add Data Destination → Lakehouse
- Select: RetailMart.Lakehouse
- Table name: silver_customers
- Update method: Replace (overwrite) or Upsert (merge)

Step 4: Publish and Refresh

- Home → Publish
- Refresh Now (or set schedule)
- Data flows to Lakehouse table!

DATAFLOW GEN2 vs NOTEBOOK:

Dataflow: better for non-coders, simple transformations

Notebook: better for complex logic, ML, custom algorithms

Use BOTH: Dataflow for ingestion, Notebook for heavy transforms

◆ Connectivity & Data Ingestion

FABRIC CONNECTIONS (the new concept):

In ADF: Linked Services (per pipeline/workspace)

In Fabric: CONNECTIONS (tenant-level, reusable)

Create Connection:

Settings → Manage Connections → + New Connection

OR during pipeline/dataflow creation when you
configure a source → auto-prompted to create connection

Connection types:

- Cloud connections (Azure resources)
 - No IR needed, automatic
- On-premise connections
 - Requires Fabric On-premises Data Gateway
 - Similar to Self-Hosted IR in ADF
 - Install gateway on corporate server
 - Register with Fabric tenant
 - Now Fabric can reach on-prem SQL Server!

ON-PREMISES DATA GATEWAY (replaces SHIR):

Download: Microsoft On-premises Data Gateway

Install on: Windows server inside corporate network

Register with: Fabric (or Power BI) tenant

Use for: Pipelines, Dataflows accessing on-prem sources

Same concept as SHIR, slightly different technology

SHIR: ADF-specific

Gateway: Fabric + Power BI (broader use)

INGESTION PATTERNS in Fabric:

Pattern 1: Pipeline **Copy** → Lakehouse Files

Source → **Copy** Activity → LH Files/raw/

Then: Notebook transforms Files → Tables

Use **for:** complex sources, **large** data

Pattern 2: Dataflow Gen2 → Lakehouse **Table**

Source → Power Query transformations → LH Tables/

Use **for:** simple ingestion + transformation **in one** step

Pattern 3: Notebook → Lakehouse **Table**

Source → Spark read → transform → write Delta **table**

Use **for:** complex logic, ML features

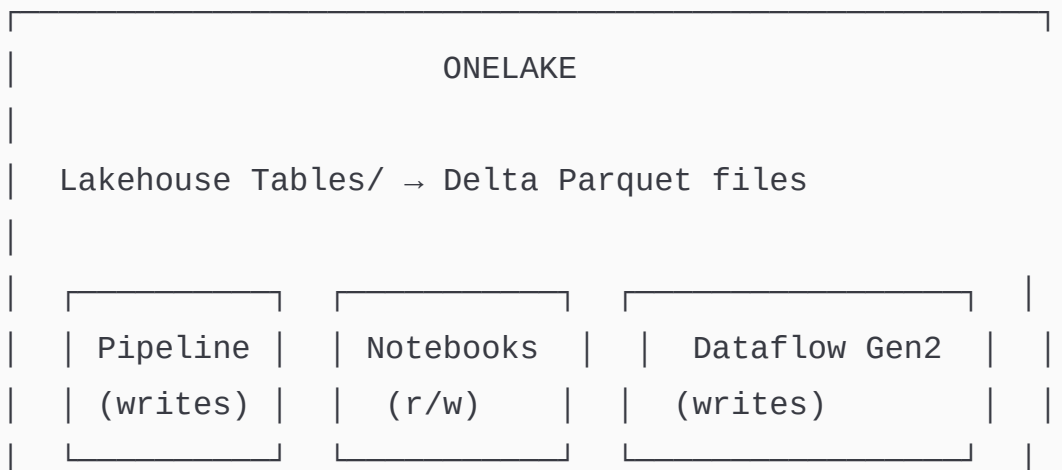
Pattern 4: Shortcut → Existing ADLS data

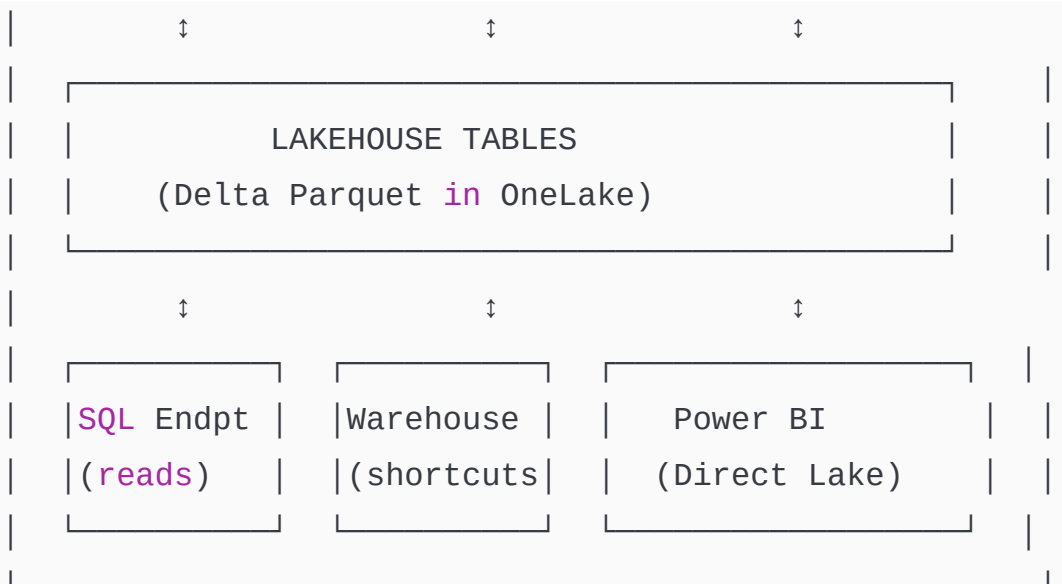
No ingestion needed! Just **create** shortcut

Use **for:** data already **in** Azure storage

◆ Integration with OneLake & Fabric Workloads

HOW **ALL** FABRIC ITEMS SHARE DATA:





ZERO DATA MOVEMENT **between** boxes above!

All reading SAME underlying Parquet files!

AUTOMATIC **SQL** ENDPOINT:

When you **create** a Lakehouse

- Fabric AUTOMATICALLY creates a **SQL** Analytics Endpoint
- This **is** a read-only T-**SQL** interface **to** your tables!
- **No** setup needed!
- Connection string provided automatically

Connect Power BI **to** **SQL** Endpoint:

Power BI Desktop → **Get** Data → Azure → Synapse Analytics (**SQL** DW)

Server: <workspace>.datawarehouse.fabric.microsoft.com

Database: LH_Sales_Prod (your lakehouse name)

- **All** Tables/ **show** up as **SQL** tables!
- Build reports **on** Lakehouse data **with** standard T-**SQL**!

◆ Data Transformation & Orchestration

ORCHESTRATION PATTERN IN FABRIC:

Master Pipeline (PL_RetailMart_Daily_ETL)

- |
- |— **Execute** Pipeline: PL_Ingest_Oracle_Sales
 - | → **Copy** Activity (Oracle → Lakehouse Files)
- |
- |— **Execute** Pipeline: PL_Ingest_SQL_Customers
 - | → **Copy** Activity (SQL Server → Lakehouse Files)
- |
- |— Notebook Activity: NB_Transform_Bronze_to_Silver
 - | → Spark: clean, validate, write **to** Tables/silver
- |
- |— Notebook Activity: NB_Build_Gold_Layer
 - | → Spark: aggregate, **join** dims, write **to** Tables/gold
- |
- |— Dataflow Gen2 Activity: DF_StoreTargets
 - | → Excel targets → Lakehouse **Table** (simple mapping)
- |
- |— Send Email (built-in Office 365 activity!)
 - | → "Daily ETL Complete: 142,850 rows processed"
 - | → **No** Logic Apps needed **for** simple emails!

SCHEDULING:

- Workspace → your pipeline → Schedule
- Daily **at** 1:00 AM IST
- **OR: trigger** via EventStream (event-driven)
- **OR: trigger from** another Fabric item

MONITORING:

- Workspace → your pipeline → Run History
- See **all** runs: success, failure, duration

→ Click **any** run → see **each** activity
→ Failed activity → red X → click **for** error details

7 Fabric Data Engineering — Lakehouse

◆ Data Lake vs Data Lakehouse

LET ME EXPLAIN THIS **WITH** A STORY:

DATA LAKE (ADLS Gen2 **with** Parquet files):

"Imagine a huge warehouse where you dump everything.
Files piled everywhere. Some labeled, some not.
You know data is in there... somewhere.
To find anything, you need to search through piles.
No guarantee data is correct or not duplicated.
Anyone can dump anything – no quality control."

Good: stores everything cheaply

Bad: **no** ACID, **no** UPDATE/DELETE, **no** schema enforcement
querying **is** slow **and** painful

"Data Swamp" – nickname **for** bad data lakes!

DATA WAREHOUSE (**SQL** Server / Synapse Dedicated):

"Imagine a perfectly organized library.
Every book catalogued, indexed, easy to find.
But: you can only store books (structured data).
No room for videos, images, audio.
Very expensive per GB.
Must know book category before acquiring it (schema-first)."

Good: fast **SQL** queries, organized, ACID

Bad: expensive, structured **only**, rigid schema

DATA LAKEHOUSE (Fabric Lakehouse):

"Best of both worlds:

Library organization (fast queries, ACID, schema)

Warehouse storage scale (cheap, all formats)

Store raw files AND structured tables

SQL queries work on Delta tables

Spark processes raw files

Power BI connects directly

All on SAME storage (OneLake)"

Good: everything! Cheap storage + fast queries + ACID

Challenge: newer concept, teams still learning

FABRIC LAKEHOUSE = the modern answer **to** this evolution

◆ Understanding the Fabric Lakehouse

FABRIC LAKEHOUSE ANATOMY:

PHYSICAL REALITY:

A Lakehouse = OneLake folder **with** special structure

LH_Sales_Prod.Lakehouse/

— Tables/	← Managed Delta tables (queryable!)
— sales/	← sales Delta table

```

|       |— _delta_log/   ← transaction log
|       |— part-0001.parquet
|       |— part-0002.parquet
└─ Files/                               ← Unmanaged raw files
    └─ raw/
        └─ sales_20240615.csv

```

LOGICAL REALITY (what you see in Fabric Studio):

```

LH_Sales_Prod
├─ 📊 Tables (queryable via SQL automatically!)
|   ├── bronze_sales
|   ├── silver_sales
|   └─ gold_store_performance
└─ 📁 Files (raw file storage)
    ├── raw/
    ├── archive/
    └─ config/

```

TWO PERSONAS of Lakehouse:

AS A LAKEHOUSE (Spark Notebook perspective):

- Use PySpark to read/write Files and Tables
- Full flexibility of Spark
- Access via abfss:// path

AS A DATA WAREHOUSE (SQL perspective):

- SQL Analytics Endpoint auto-created
- Query Tables/ using T-SQL
- Connect Power BI via ODBC/SQL endpoint
- Read-only SQL access (no INSERT/UPDATE via SQL)
- Writes happen through Spark or Dataflows

◆ Creating a Lakehouse

STEP BY STEP:

Step 1: Open your workspace

- app.fabric.microsoft.com
- Click your workspace: RetailMart_DataEngineering

Step 2: Create Lakehouse

- + New → Lakehouse
- Name: LH_Sales_Prod
- Create

Step 3: Explore the Lakehouse UI

LEFT PANEL:

- └─ LH_Sales_Prod (Lakehouse)
 - └─ Tables ← click to see managed tables
 - └─ Files ← click to browse files

TOP BAR:

- Lakehouse | SQL analytics endpoint | Semantic model
(switch between 3 views)

Step 4: Upload your first file

- Files → right-click → Upload → Upload files
- Drag and drop: sales_data.csv
- File appears in Files/

Step 5: Load file into table

- Right-click the CSV file in Files
- "Load to Tables" → New table
- Table name: bronze_sales
- Fabric runs Spark automatically → creates Delta table!

OR: Load via Notebook (more control)

Step 6: Query the table

Click "SQL analytics endpoint" (top toggle)

→ Tables/bronze_sales appears as SQL table

→ Click "New SQL query"

→ **SELECT** TOP 10 * **FROM** bronze_sales

→ Run → see results!

Step 7: Connect Power BI

→ Click "New semantic model" (top toggle)

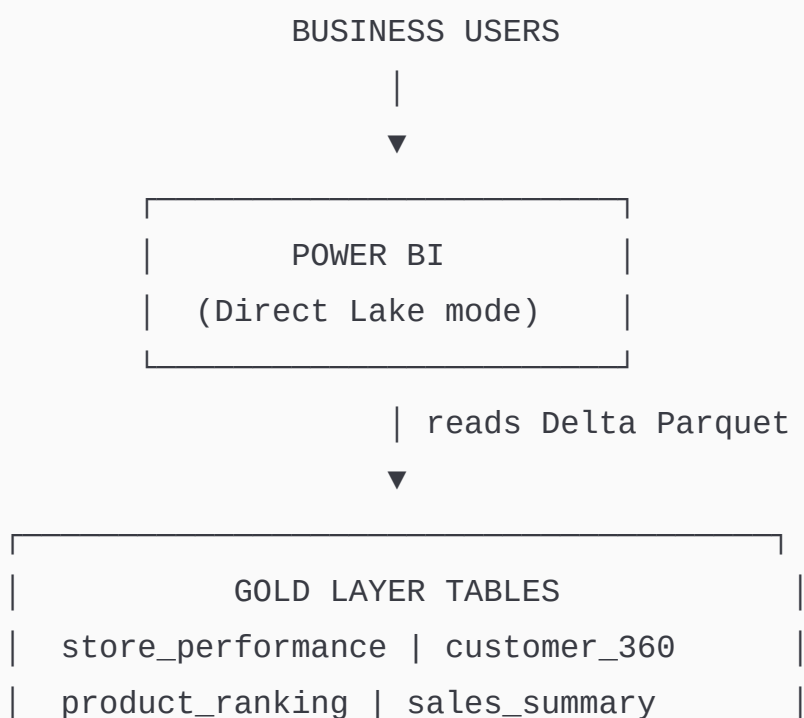
→ **Select** tables to include

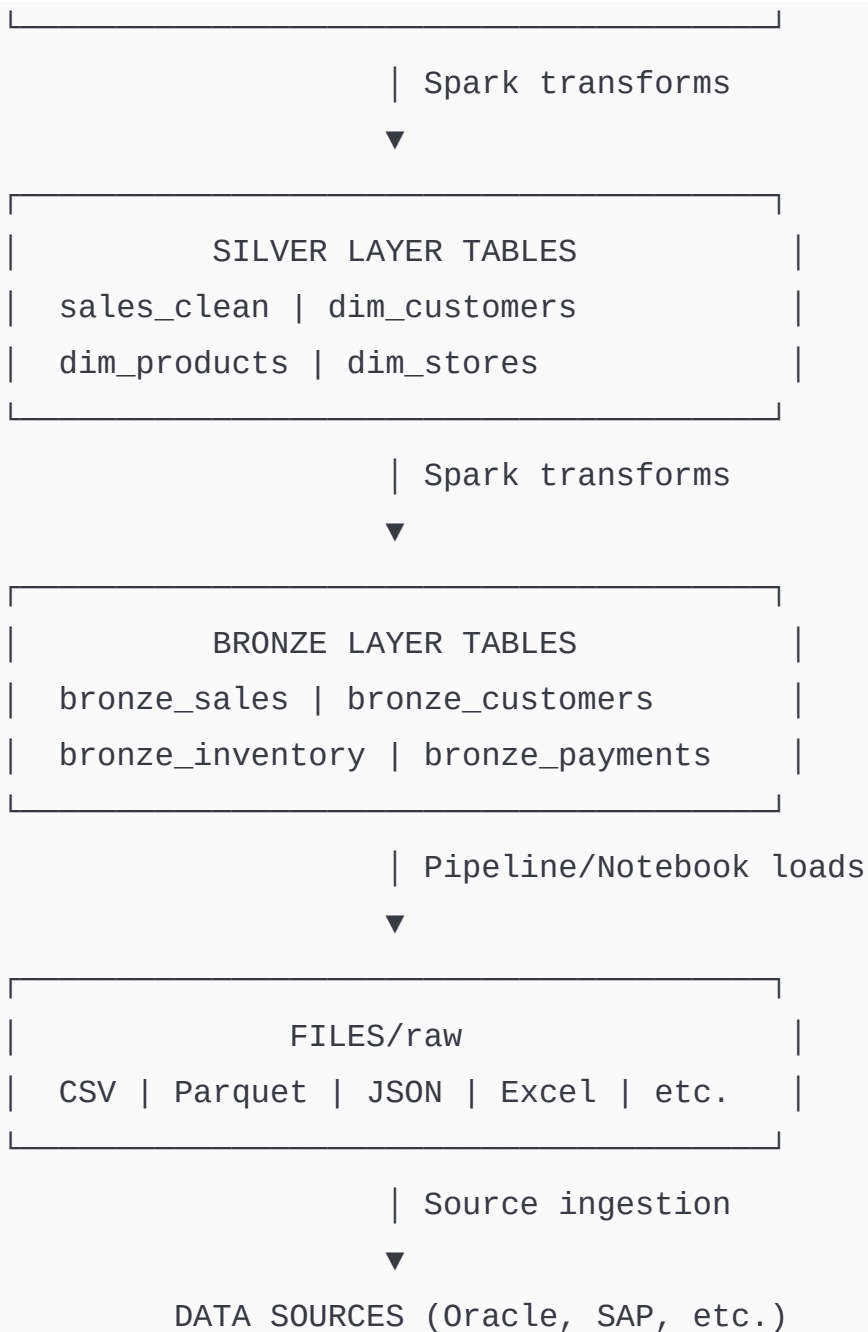
→ **Create** semantic model

→ **Open in** Power BI Desktop → build reports!

◆ Lakehouse Architecture

LAKEHOUSE ARCHITECTURE LAYERS:





◆ Exploring Lakehouse Features

LAKEHOUSE UI FEATURES:

1. TABLE PREVIEW:

Click [any table in Tables/](#)

- Preview opens showing sample data
- Column names, types, sample values
- Row count shown

2. TABLE MAINTENANCE:

- Right-click table → Maintenance
- OPTIMIZE: compact small files
- VACUUM: remove old Delta versions
- (same as Delta OPTIMIZE and VACUUM commands!)

3. SCHEMA VIEW:

- Tables → click table → Schema tab
- See column names and data types
- Perfect for documentation

4. PARTITION INFO:

- Tables → click table → Details tab
- See partition columns
- File count, total size

5. SHORTCUT MANAGEMENT:

- Files/ or Tables/ → right-click → New Shortcut
- Add shortcuts to external data

6. NOTEBOOK INTEGRATION:

- Open Notebook button (top right of Lakehouse)
- Opens notebook already connected to this Lakehouse!
- Default lakehouse set automatically
- No need to configure paths!

7. PIPELINE INTEGRATION:

- Get Data → New Pipeline
- Pipeline created with Lakehouse as destination

8. SQL ENDPOINT:

Toggle: Lakehouse → **SQL** analytics endpoint
→ **Full** T-**SQL** interface
→ Write queries, save them
→ Share query results

9. SEMANTIC MODEL:

Toggle: **SQL** endpoint → Semantic model
→ Power BI semantic model
→ **Add** measures (DAX)
→ **Define** relationships
→ **Open in** Power BI Desktop

◆ DevOps Repos and Deployment Pipelines in Fabric

VERSION CONTROL **FOR** FABRIC ITEMS:

CONNECT WORKSPACE **TO** GIT:

Workspace → Settings → Git Integration

Provider: Azure DevOps

Organization: retailmart-devops

Project: DataPlatform

Repository: fabric-items

Branch: dev ← development branch

Folder: /RetailMart_Sales/

→ Click Connect → Sync

NOW: All Fabric items saved **as** JSON/metadata **to** Git!

GIT SYNC PROCESS:

When you change a notebook:

- Icon shows "pending changes" in workspace
- Commit button appears
- Add commit message: "Add fraud detection logic"
- Commit → changes pushed to dev branch
- Raise PR to main → code review → merge → deploy!

WHAT GETS SAVED TO GIT:

- ✓ Notebooks (as .py or .ipynb files)
- ✓ Pipelines (as JSON)
- ✓ Dataflows (as JSON)
- ✓ Semantic models (as TMDL)
- ✗ Data (NOT versioned – only code/config)
- ✗ Lakehouse tables (these are data, not code)

DEPLOYMENT PIPELINES (Fabric's built-in CD):

Fabric has native deployment pipelines!

Workspace → Deployment Pipelines → Create Pipeline

Stages:

[Development] → [Test] → [Production]

Each stage = a separate workspace

Deploy button → copies all items from Dev → Test → Prod

This is like ARM template deployment in ADF

But simpler! No JSON templates to manage

WORKS FOR:

- Notebooks (code deploys, not data)
- Pipelines (configuration deploys)
- Dataflows (transformation logic deploys)
- Semantic models (report structure deploys)

ENVIRONMENT-SPECIFIC SETTINGS:

Each stage can override parameter values

Dev: connects to dev Lakehouse

Prod: connects to prod Lakehouse

→ Same code, different data connections!

8 Fabric Data Engineering — Notebooks

◆ Creating and Using Notebooks

FABRIC NOTEBOOK = Jupyter-style interactive computing

CREATE NOTEBOOK:

Option 1: Workspace → + New → Notebook

Option 2: From inside Lakehouse → Open Notebook

Option 3: Import .ipynb from your computer

NOTEBOOK INTERFACE:

Top Bar:

└─ Run All / Run Cell / Stop

└─ Language selector (PySpark default)

└─ + Code / + Markdown

└─ Attach: Select Lakehouse

└─ Save / Publish / Share

Left Panel:

└─ Lakehouse Explorer (browse tables/files)

└─ Environment (libraries)

└─ Resources (attach files)

Cell Types:

└─ Code (PySpark, SQL, Scala, SparkR)

└─ Markdown (documentation)

ATTACH LAKEHOUSE TO NOTEBOOK:

This **is** important! Must attach lakehouse **to** use relative paths **and SQL** queries **on** lakehouse tables.

- + Lakehouse button (**left** panel)
- **Select**: LH_Sales_Prod
- **Set as Default** Lakehouse (**check** the box)

Now you can use:

```
spark.read.table("bronze_sales") # no path needed!  
spark.sql("SELECT * FROM gold_store_performance") # works!
```

Instead **of**:

```
spark.read.parquet("abfss://...onelake.../Tables/bronze_sales")
```

NOTEBOOK EXPLORER (amazing feature!):

Left panel shows Lakehouse structure:

```
└─ LH_Sales_Prod  
  └─ Tables  
    └─ bronze_sales (right-click → query!)  
    └─ silver_sales  
  └─ Files  
    └─ raw/
```

Right-click any table → "Add to code"

→ Auto-generates read code **in** your cell!

◆ Developing and Running Notebooks

```

# — COMPLETE FABRIC NOTEBOOK – RETAILMART TRANSFORMATION —

# — CELL 1: Documentation (Markdown) —
"""
# Sales Transformation Notebook
**Workspace:** RetailMart_DataEngineering
**Lakehouse:** LH_Sales_Prod
**Purpose:** Bronze → Silver transformation for sales data
**Author:** Data Engineering Team
**Schedule:** Daily via Pipeline at 1:00 AM IST
**Parameters:** process_date (yyyy-MM-dd)
"""

# — CELL 2: Parameters (Parameter Cell) —
# Right-click cell → Toggle Parameter Cell
# This makes it a parameter cell (filled by pipeline)
process_date = "2024-06-15"
load_type = "INCREMENTAL"
source_lakehouse = "LH_Sales_Prod"

# — CELL 3: Imports —
from pyspark.sql import functions as F
from pyspark.sql.types import *
from pyspark.sql.window import Window
from delta.tables import DeltaTable
import logging

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

logger.info(f"Starting transformation for: {process_date}")

# — CELL 4: Read Bronze Data —
# Using default lakehouse – no abfss path needed!
df_bronze = spark.sql(f"""

```

```

        SELECT * FROM bronze_sales
        WHERE _process_date = '{process_date}'
        """)

# OR using PySpark API:
df_bronze = spark.read.table("bronze_sales") \
    .filter(F.col("_process_date") == process_date)

row_count = df_bronze.count()
logger.info(f"Bronze records read: {row_count:,}")

if row_count == 0:
    logger.warning("No data found for this date – exiting")
    mssparkutils.notebook.exit(f"NO_DATA:{process_date}")

# — CELL 5: Data Quality Checks —————
# Track quality metrics
quality_metrics = {}

# Check 1: Null sale_id
null_sale_ids =
df_bronze.filter(F.col("SALE_ID").isNull()).count()
quality_metrics["null_sale_ids"] = null_sale_ids

# Check 2: Invalid amounts
invalid_amounts = df_bronze.filter(F.col("AMOUNT") <= 0).count()
quality_metrics["invalid_amounts"] = invalid_amounts

# Check 3: Future dates
future_dates = df_bronze.filter(
    F.col("SALE_DATE") > F.current_date()
).count()
quality_metrics["future_dates"] = future_dates

logger.info(f"Quality metrics: {quality_metrics}")

```

```

# Remove bad records (keep clean ones)
df_clean = df_bronze \
    .filter(F.col("SALE_ID").isNotNull()) \
    .filter(F.col("AMOUNT") > 0) \
    .filter(F.col("SALE_DATE") <= F.current_date())

records_removed = row_count - df_clean.count()
logger.info(f"Records removed: {records_removed:,}")

# — CELL 6: Transformations —————
df_silver = df_clean \
    .select(
        F.col("SALE_ID").cast("integer").alias("sale_id"),
        F.upper(F.trim("CUSTOMER_ID")).alias("customer_id"),
        F.col("STORE_ID").cast("integer").alias("store_id"),
        F.upper(F.trim("PRODUCT_CODE")).alias("product_code"),
        F.col("QUANTITY").cast("integer").alias("quantity"),
        F.col("AMOUNT").cast("double").alias("total_amount"),
        F.to_date("SALE_DATE", "DD-MON-YYYY").alias("sale_date"),
        F.upper("REGION_CODE").alias("region_code"),
        F.lower("PAYMENT_MODE").alias("payment_mode"),
        F.col("STATUS").alias("status")
    ) \
    .withColumn("year",      F.year("sale_date")) \
    .withColumn("month",    F.month("sale_date")) \
    .withColumn("day",      F.dayofmonth("sale_date")) \
    .withColumn("gst_amount", F.round(F.col("total_amount") *
0.18, 2)) \
    .withColumn("net_amount", F.col("total_amount") -
F.col("gst_amount")) \
    .withColumn("load_ts",  F.current_timestamp())

# — CELL 7: Write to Silver (Delta table in Lakehouse) —————
# Writing to Lakehouse Tables/ as managed Delta table

```

```

# No need for abfss path – use table name!

df_silver.write \
    .format("delta") \
    .mode("append") \
    .partitionBy("year", "month") \
    .saveAsTable("silver_sales")
    # Automatically writes to
    LH_Sales_Prod.Lakehouse/Tables/silver_sales/

logger.info(f"Written to silver_sales: {df_silver.count():,}
rows")

# — CELL 8: Optimize the Delta table —————
spark.sql("OPTIMIZE silver_sales ZORDER BY (store_id,
sale_date)")
logger.info("OPTIMIZE completed")

# — CELL 9: Return result to pipeline —————
import json
result = {
    "status": "SUCCESS",
    "process_date": process_date,
    "input_rows": row_count,
    "output_rows": df_silver.count(),
    "quality_metrics": quality_metrics
}
print(json.dumps(result, indent=2))
mssparkutils.notebook.exit(json.dumps(result))

```

◆ Using Python in Notebooks

```
# FABRIC NOTEBOOKS: PySpark + Pure Python BOTH work!
```

```
# WHEN TO USE PURE PYTHON (not Spark):
```

```
# → Working with small data (< 1GB)
```

```
# → API calls, web scraping
```

```
# → Configuration management
```

```
# → File operations via mssparkutils
```

```
# → Sending emails/notifications
```

```
# EXAMPLE: Call Payment API and load response to Lakehouse
```

```
import requests
```

```
import json
```

```
from datetime import datetime, timedelta
```

```
# Pure Python API call
```

```
def fetch_payment_data(from_date: str, to_date: str) -> dict:
```

```
    """Fetch payment data from REST API"""
```

```
    api_url =
```

```
    "https://payments.retailmart.com/api/v1/transactions"
```

```
    headers = {
```

```
        "Authorization": f"Bearer
```

```
{mssparkutils.credentials.getSecret('https://kv-retailmart-  
prod.vault.azure.net/', 'payment-api-key')}]",
```

```
        "Content-Type": "application/json"
```

```
    }
```

```
    params = {
```

```
        "from_date": from_date,
```

```
        "to_date": to_date,
```

```
        "status": "COMPLETED",
```

```
        "limit": 10000
```

```
    }
```

```

    response = requests.get(api_url, headers=headers,
params=params)
    response.raise_for_status()

    return response.json()

# Fetch data
payment_data = fetch_payment_data(process_date, process_date)
print(f"API returned {len(payment_data['transactions'])}
records")

# Convert to Spark DataFrame
df_payments = spark.createDataFrame(payment_data["transactions"])

# Write to Lakehouse
df_payments.write \
    .format("delta") \
    .mode("append") \
    .saveAsTable("bronze_api_payments")

print("✅ API data loaded to Lakehouse")

# PYTHON LIBRARIES AVAILABLE by default in Fabric:
# pandas, numpy, scikit-learn, matplotlib, seaborn
# requests, json, datetime, os, re, math
# pyspark (obviously)
# delta (for Delta Lake operations)

# INSTALL ADDITIONAL LIBRARIES:
# Option 1: In notebook cell
%pip install great-expectations==0.18.0

# Option 2: Environment (persistent across sessions)
# Workspace → Environments → + New Environment
# → Add libraries → Publish

```



```
# → Attach environment to notebook
# → Libraries available without %pip install!
```

◆ Notebook Utilities

```
# FABRIC NOTEBOOK UTILITIES – Complete Reference
```

```
# notebookutils = mssparkutils equivalent in Fabric
# Both names work! (Microsoft is standardizing to notebookutils)
```

```
from notebookutils import mssparkutils
```

```
# — FILE SYSTEM —
```

```
# List files
```

```
files = mssparkutils.fs.ls(
```

```
"abfss://RetailMart_DataEngineering@onelake.dfs.fabric.microsoft.com"
)
```

```
for f in files:
```

```
    print(f"{f.name} | {f.size/1024:.1f} KB | {'Dir' if f.isDir
else 'File'}")
```

```
# Simpler relative path (when lakehouse is attached):
```

```
files = mssparkutils.fs.ls("Files/raw/sales/")
```

```
# Just "Files/" instead of full abfss path!
```

```
# Copy file
```

```
mssparkutils.fs.cp("Files/raw/sales_20240615.csv",
                   "Files/archive/sales_20240615.csv")
```

```
# Move file
```

```

mssparkutils.fs.mv("Files/landing/new_file.csv",
                   "Files/raw/new_file.csv")

# Delete
mssparkutils.fs.rm("Files/temp/", recurse=True)

# Create directory
mssparkutils.fs.mkdirs("Files/archive/2024/06/")

# Preview file content
content = mssparkutils.fs.head("Files/raw/sales_20240615.csv",
                                500)
print(content) # first 500 bytes

# — NOTEBOOK CHAINING —
# Run child notebook and get result
result = mssparkutils.notebook.run(
    "../silver/transform_customers",
    timeout_seconds=1800,
    arguments={
        "process_date": process_date,
        "load_type": "INCREMENTAL"
    }
)
print(f"Child result: {result}")
# result = whatever child called mssparkutils.notebook.exit()
# with

# Exit current notebook (return value to parent/pipeline)
mssparkutils.notebook.exit("SUCCESS: 142850 rows")

# — CREDENTIALS / SECRETS —
# Get secret from Key Vault
api_key = mssparkutils.credentials.getSecret(
    "https://kv-retailmart-prod.vault.azure.net/",

```

```
    "payment-api-key"
)
# NEVER print this! Fabric redacts it automatically anyway.

# Get token for specific audience
token = mssparkutils.credentials.getToken("storage")
# Use for manual HTTP calls to Azure Storage APIs

# — DATA UTILITY —————
# Convert between Spark and Pandas (for small data)
pandas_df = df_summary.toPandas() # Spark → Pandas
spark_df = spark.createDataFrame(pandas_df) # Pandas → Spark
```

◆ Notebook Visualizations

```
# FABRIC NOTEBOOK: Built-in visualization is amazing!
```

```
# OPTION 1: Display command (Fabric's built-in)
display(df_sales)
# → Auto-renders interactive table!
# → Toggle: Table view / Chart view
# → Chart types: Bar, Line, Scatter, Pie, Map, etc.
# → Click column headers to sort
# → Filter rows interactively
# → Download as CSV right from UI!

# Chart configuration (no code!):
# After display() → click chart icon (top of output)
# Configure:
#   X axis: sale_date
#   Y axis: total_amount (aggregate: SUM)
```

```

# Group by: region_code
# Chart type: Line Chart
# → Beautiful interactive chart instantly!

# OPTION 2: Matplotlib (traditional Python)
import matplotlib.pyplot as plt
import pandas as pd

# Sample data for viz (small – collect to driver)
df_region = df_sales.groupby("region_code") \
    .agg(F.sum("total_amount").alias("revenue")) \
    .toPandas() # convert to pandas for viz

plt.figure(figsize=(10, 6))
plt.bar(df_region["region_code"], df_region["revenue"] / 1e6,
        color=["#0078D4", "#2D7D9A", "#107C10", "#7A7574"])
plt.title("Revenue by Region (in Millions INR)", fontsize=14,
fontweight='bold')
plt.xlabel("Region")
plt.ylabel("Revenue (₹ Millions)")
plt.tight_layout()
plt.savefig("/tmp/regional_revenue.png")
plt.show() # displays inline in notebook!

# OPTION 3: Seaborn (statistical plots)
import seaborn as sns

df_pd = df_sales.select("total_amount", "region_code").toPandas()
sns.boxplot(x="region_code", y="total_amount", data=df_pd)
plt.title("Sales Distribution by Region")
plt.show()

# OPTION 4: Plotly (interactive charts)
import plotly.express as px

```

```
df_pd = df_sales.groupBy("sale_date", "region_code") \
    .agg(F.sum("total_amount").alias("revenue")) \
    .orderBy("sale_date") \
    .toPandas()

fig = px.line(df_pd, x="sale_date", y="revenue",
              color="region_code",
              title="Daily Revenue by Region")
fig.show() # interactive chart with hover, zoom, etc.!
```

◆ Loading Data into the Lakehouse

ALL WAYS TO LOAD DATA INTO FABRIC LAKEHOUSE:

— METHOD 1: Upload via UI (small files, one-time) —————
Lakehouse → Files/ → Upload → drag & drop
Good for: config files, Excel lookups, testing data

— METHOD 2: Pipeline Copy Activity —————
Source (Oracle/SQL/FTP) → Copy Activity → Lakehouse Files
Then Notebook transforms Files → Tables
Good for: scheduled ingestion from external sources

— METHOD 3: Dataflow Gen2 —————
Source → Power Query transformations → Lakehouse Table
Good for: simple ingestion + transform without code

— METHOD 4: Notebook (most flexible) —————

4A: Write to Files/ (raw landing)

```

df_raw.write \
    .mode("overwrite") \
    .csv("Files/raw/sales/2024/06/")
# File visible in Lakehouse → Files/ → raw/ → sales/

# 4B: Write to Tables/ (managed Delta)
df_silver.write \
    .format("delta") \
    .mode("overwrite") \
    .partitionBy("year", "month") \
    .saveAsTable("silver_sales")
# Table visible in Lakehouse → Tables/ → silver_sales

# 4C: Save table with specific path (external-like)
df_gold.write \
    .format("delta") \
    .mode("overwrite") \
    .save("Tables/gold_store_performance")
# Same result – Delta table in Tables/ folder

# 4D: Using spark.sql CREATE TABLE
spark.sql("""
    CREATE TABLE IF NOT EXISTS gold_kpi_summary
    USING DELTA
    PARTITIONED BY (report_date)
    AS SELECT
        store_id,
        sale_date AS report_date,
        SUM(total_amount) AS gross_revenue,
        COUNT(*) AS transactions
    FROM silver_sales
    GROUP BY store_id, sale_date
""")

```

— METHOD 5: Auto Loader (streaming ingestion) —————

Files dropped to Lakehouse Files/ → Auto Loader picks up

```
df_stream = spark.readStream \  
    .format("cloudFiles") \  
    .option("cloudFiles.format", "csv") \  
    .option("cloudFiles.schemaLocation", "Files/schemas/sales/") \  
    \  
    .option("header", "true") \  
    .load("Files/landing/sales/")  
  
df_stream.writeStream \  
    .format("delta") \  
    .outputMode("append") \  
    .option("checkpointLocation", "Files/checkpoints/sales/") \  
    .trigger(availableNow=True) \  
    .toTable("bronze_sales_stream")
```

◆ Notebook Source Control and Deployment

COMPLETE DEVOPS WORKFLOW FOR FABRIC NOTEBOOKS:

WORKFLOW:

Developer builds notebook **in** Dev workspace

|

▼ Commit changes

Azure DevOps Repo (dev branch)

|

▼ Pull Request

Code Review by team lead

|

▼ Merge to main

CI Pipeline runs:

- Lint Python code (flake8)
- Run unit tests (pytest on logic functions)
- Validate notebook JSON syntax

|

▼ CD Pipeline

Fabric Deployment Pipeline:

- Dev → Test (automated)
- Test → Prod (manual approval)

GIT INTEGRATION IN FABRIC:

Workspace Settings → Git Integration → Connect

After connecting:

- Notebook shows "Synced" or "Modified" status
- Click "Source control" button in workspace
- See all pending changes
- Commit with message
- Push to branch

NOTEBOOK AS CODE (storage format):

Notebooks stored in Git as:

- .py file (Python cells + markdown as comments)
- or .ipynb (standard Jupyter format)
- Diff shows actual code changes (line by line)
- Code reviews see exact logic changes!

UNIT TESTING APPROACH:

functions.py (pure Python functions – testable!)

```
def calculate_gst(amount: float, rate: float = 0.18) -> float:  
    return round(amount * rate, 2)
```

```
def validate_pan(pan: str) -> bool:  
    import re  
    return bool(re.match(r'^[A-Z]{5}[0-9]{4}[A-Z]$', pan))
```



```

# test_functions.py (pytest tests)
def test_calculate_gst():
    assert calculate_gst(1000) == 180.0
    assert calculate_gst(0) == 0.0

def test_validate_pan():
    assert validate_pan("ABCDE1234F") == True
    assert validate_pan("invalid") == False

# In notebook: import and use these functions
# %run ./functions OR
from functions import calculate_gst, validate_pan

# Tests run in CI pipeline automatically!

```

◆ Authoring and Running T-SQL in Notebooks

FABRIC NOTEBOOKS: SWITCH TO SQL MODE PER CELL!

— OPTION 1: Spark SQL in code cell —

```

result = spark.sql("""
    SELECT
        region_code,
        COUNT(*) AS transaction_count,
        ROUND(SUM(total_amount), 2) AS gross_revenue,
        ROUND(AVG(total_amount), 2) AS avg_ticket
    FROM silver_sales
    WHERE year = 2024 AND month = 6
    GROUP BY region_code
    ORDER BY gross_revenue DESC

```

```

"""
display(result) # show as interactive table!

# — OPTION 2: %%sql magic (cell-level SQL) —————
%%sql
-- This entire cell runs as Spark SQL
SELECT
    s.region_code,
    s.sale_date,
    SUM(s.total_amount) AS daily_revenue,
    COUNT(DISTINCT s.customer_id) AS unique_customers,
    t.revenue_target,
    ROUND(SUM(s.total_amount) / t.revenue_target * 100, 1) AS
achievement
FROM silver_sales s
LEFT JOIN store_targets t
    ON s.region_code = t.region_code
    AND MONTH(s.sale_date) = t.month
WHERE s.year = 2024
GROUP BY s.region_code, s.sale_date, t.revenue_target
ORDER BY s.sale_date, achievement DESC

# — OPTION 3: CREATE TABLE using SQL —————
%%sql
CREATE OR REPLACE TABLE gold_regional_performance
USING DELTA
PARTITIONED BY (report_month)
AS
SELECT
    TRUNC(sale_date, 'MM') AS report_month,
    region_code,
    COUNT(*) AS transactions,
    SUM(total_amount) AS gross_revenue,
    SUM(net_amount) AS net_revenue,
    SUM(gst_amount) AS gst_collected,

```

```
COUNT(DISTINCT customer_id) AS unique_customers,  
COUNT(DISTINCT store_id) AS active_stores,  
MAX(total_amount) AS highest_sale,  
AVG(total_amount) AS avg_sale_value  
FROM silver_sales  
WHERE status = 'COMPLETED'  
GROUP BY TRUNC(sale_date, 'MM'), region_code
```

— OPTION 4: Pass Python variables into SQL —————
Problem: %%sql can't access Python variables directly!
Solution: Use f-string or spark.sql()

```
target_month = "2024-06-01"
```

Option A: spark.sql() with f-string

```
df = spark.sql(f"""  
    SELECT * FROM silver_sales  
    WHERE sale_date >= '{target_month}'  
    AND region_code IN ('N', 'S')  
""")
```

Option B: Register Python variable as Spark SQL variable

```
spark.conf.set("spark.sql.variable.month", target_month)  
%%sql  
SELECT * FROM silver_sales  
WHERE sale_date >=  
'${spark.conf.get("spark.sql.variable.month")}'
```

9 Fabric Data Engineering — Apache Spark

◆ In-Depth Architecture of Apache Spark in Fabric

FABRIC SPARK vs STANDALONE SPARK:

Same Spark engine, but Fabric manages EVERYTHING:

What Microsoft manages for you:

- ✓ Spark cluster provisioning
- ✓ Worker node addition/removal (autoscale)
- ✓ Spark version upgrades
- ✓ Log management
- ✓ Library installation
- ✓ Security and network isolation
- ✓ OneLake integration (no credential config!)

What YOU manage:

- ✓ Your code (notebooks)
- ✓ Cluster size (node type/count)
- ✓ Library requirements
- ✓ Optimization (partitioning, caching, etc.)

STARTER POOLS (game changer!):

Normal Spark: click Run → wait 3-5 min for cluster to start

Fabric Starter Pool: click Run → running in 5-10 SECONDS!

Microsoft pre-warms small Spark clusters for you

When you start notebook → instantly gets assigned a warm pool

No waiting! Great for interactive development

SESSION TIMEOUT:

Default: notebook session times out after idle period

Configure: Workspace Settings → Spark Sessions

Set: 20 minutes idle timeout

Saves CUs by not running idle Spark sessions

◆ Working with Data in a Spark DataFrame

```
# — FULL RETAILMART SPARK WORKFLOW —————

from pyspark.sql import functions as F
from pyspark.sql.types import *
from pyspark.sql.window import Window

# READ from default lakehouse (attached)
df_sales      = spark.read.table("silver_sales")
df_customers  = spark.read.table("dim_customers")
df_products   = spark.read.table("dim_products")
df_stores     = spark.read.table("dim_stores")

# INSPECT DATA
print(f"Sales records:      {df_sales.count():>10,}")
print(f"Customers:         {df_customers.count():>10,}")
print(f"Products:           {df_products.count():>10,}")
print(f"Stores:             {df_stores.count():>10,}")

df_sales.printSchema()
display(df_sales.limit(5))

# CHECK DATA QUALITY
print("\n📊 Null counts per column:")
null_counts = df_sales.select([
    F.count(F.when(F.col(c).isNull(), c)).alias(c)
    for c in df_sales.columns
])
display(null_counts)

print("\n📊 Duplicate check (by sale_id):")
dup_count = df_sales.count() -
df_sales.dropDuplicates(["sale_id"]).count()
```

```
print(f" Duplicate sale_ids: {dup_count:,}")

print("\n📊 Date range:")
df_sales.agg(
    F.min("sale_date").alias("earliest_date"),
    F.max("sale_date").alias("latest_date"),
    F.countDistinct("sale_date").alias("distinct_dates")
).show()
```

◆ DataFrame Internal Execution & Spark SQL

UNDERSTAND YOUR QUERY BEFORE RUNNING IT:

Build complex query

```
df_analysis = df_sales \
    .filter(F.col("year") == 2024) \
    .filter(F.col("status") == "COMPLETED") \
    .join(F.broadcast(df_customers), on="customer_id",
how="left") \
    .join(F.broadcast(df_products), on="product_code",
how="left") \
    .join(F.broadcast(df_stores), on="store_id",
how="left") \
    .groupBy("region_code", "customer_tier", "category") \
    .agg(
        F.sum("total_amount").alias("revenue"),
        F.count("sale_id").alias("transactions")
    )
```

CHECK THE PLAN FIRST (before running expensive operation):

```
df_analysis.explain("formatted")
```

```

# LOOK FOR in the plan:
# ✓ BroadcastHashJoin → broadcast working!
# ✓ PartitionFilters → partition pruning working!
# ✓ PushedFilters → predicate pushdown working!
# ✗ SortMergeJoin on big table → broadcast failed, investigate!
# ✗ No PartitionFilters → data not partitioned, doing full scan!

# RUN IT
display(df_analysis.orderBy(F.col("revenue").desc()))

# — SPARK SQL APPROACH (same result, more readable) —————
df_spark_sql = spark.sql("""
    SELECT
        s.region_code,
        c.customer_tier,
        p.category,
        SUM(s.total_amount) AS revenue,
        COUNT(s.sale_id) AS transactions,
        ROUND(AVG(s.total_amount), 2) AS avg_transaction
    FROM silver_sales s
    LEFT JOIN dim_customers c ON s.customer_id = c.customer_id
    LEFT JOIN dim_products p  ON s.product_code = p.product_code
    LEFT JOIN dim_stores st   ON s.store_id = st.store_id
    WHERE s.year = 2024
        AND s.status = 'COMPLETED'
    GROUP BY s.region_code, c.customer_tier, p.category
    ORDER BY revenue DESC
""")
display(df_spark_sql)

```

◆ Transformations and Actions

```

# — KEY TRANSFORMATIONS (all lazy!) —————

# select – choose columns
df_selected = df_sales.select(
    "sale_id", "customer_id", "total_amount", "sale_date"
)

# filter / where – filter rows
df_filtered = df_sales.filter(
    (F.col("total_amount") > 10000) &
    (F.col("region_code").isin("N", "S")) &
    (F.col("sale_date").between("2024-01-01", "2024-06-30"))
)

# withColumn – add/modify column
df_with_cols = df_sales \
    .withColumn("gst", F.round(F.col("total_amount") * 0.18,
2)) \
    .withColumn("net", F.col("total_amount") - F.col("gst")) \
    .withColumn("day_name", F.date_format("sale_date", "EEEE")) \
    .withColumn("is_weekend", F.dayofweek("sale_date").isin([1,
7]))

# groupBy + agg – aggregations
df_grouped = df_sales.groupBy("region_code", "year", "month") \
    .agg(
        F.sum("total_amount").alias("revenue"),
        F.count("sale_id").alias("count"),
        F.avg("total_amount").alias("avg_value"),
        F.max("total_amount").alias("max_value"),
        F.min("total_amount").alias("min_value"),
        F.countDistinct("customer_id").alias("unique_customers"),
        F.collect_list("product_code").alias("all_products") #
array!
    )

```



```

# orderBy – sort
df_sorted = df_grouped.orderBy(
    F.col("year").asc(),
    F.col("month").asc(),
    F.col("revenue").desc()
)

# dropDuplicates – remove duplicates
df_dedup = df_sales.dropDuplicates(["sale_id"])
# OR:
df_dedup2 = df_sales.dropDuplicates(["customer_id", "sale_date",
"product_code"])

# drop – remove columns
df_dropped = df_sales.drop("_ingestion_timestamp",
"_source_file", "_process_date")

# rename – rename column
df_renamed = df_sales.withColumnRenamed("total_amount",
"gross_amount")

# cast – change data type
df_casted = df_sales \
    .withColumn("sale_id", F.col("sale_id").cast("integer")) \
    .withColumn("total_amount",
F.col("total_amount").cast("double")) \
    .withColumn("sale_date", F.col("sale_date").cast("date"))

# — KEY ACTIONS (trigger execution!) —————

count = df_sales.count() # count all rows
first = df_sales.first() # first row as Row
object
sample = df_sales.take(5) # first 5 rows as

```

```
list
collected = df_grouped.collect()           # ALL rows to driver
(SMALL data only!)
display(df_sales)                           # show in notebook
(Fabric-specific)
df_sales.show(10, truncate=False)           # print to console

# Write actions (most important!)
df_silver.write.format("delta").mode("overwrite").saveAsTable("silver")
df_silver.write.format("delta").mode("append").saveAsTable("silver_sales")
df_silver.write.format("parquet").save("Files/output/sales.parquet")
```

◆ User-Defined Functions in Spark SQL

UDFs IN FABRIC NOTEBOOKS:

```
from pyspark.sql.functions import udf, pandas_udf
from pyspark.sql.types import StringType, BooleanType, FloatType
import pandas as pd
```

— REGULAR UDF (works but slow) —

Use ONLY when built-in functions can't do the job!

```
def categorize_amount(amount):
    """Categorize sale amount into business tiers"""
    if amount is None:
        return "UNKNOWN"
    elif amount < 1000:
        return "MICRO"
    elif amount < 10000:
        return "SMALL"
```

```

    elif amount < 50000:
        return "MEDIUM"
    elif amount < 200000:
        return "LARGE"
    else:
        return "ENTERPRISE"

# Register as UDF
categorize_udf = udf(categorize_amount, StringType())

# Use in DataFrame
df_categorized = df_sales.withColumn(
    "sale_category",
    categorize_udf(F.col("total_amount"))
)

# Register for Spark SQL queries too!
spark.udf.register("categorize_amount", categorize_amount,
StringType())

%%sql
SELECT
    categorize_amount(total_amount) AS category,
    COUNT(*) AS count,
    SUM(total_amount) AS revenue
FROM silver_sales
GROUP BY categorize_amount(total_amount)
ORDER BY revenue DESC

# — PANDAS UDF (MUCH FASTER – vectorized!) —————
# Process entire column at once (batch, not row-by-row)

@pandas_udf(StringType())
def categorize_amount_fast(amount_series: pd.Series) ->
pd.Series:

```

```

    """Vectorized amount categorization – 10-100x faster than
    regular UDF"""
    conditions = [
        amount_series.isna(),
        amount_series < 1000,
        amount_series < 10000,
        amount_series < 50000,
        amount_series < 200000
    ]
    choices = ["UNKNOWN", "MICRO", "SMALL", "MEDIUM", "LARGE"]
    return pd.Series(
        pd.cut(amount_series,
                bins=[-float('inf'), 1000, 10000, 50000, 200000,
float('inf')],
                labels=["MICRO", "SMALL", "MEDIUM", "LARGE",
"ENTERPRISE"])
        ).fillna("UNKNOWN")

df_fast = df_sales.withColumn("category",
categorize_amount_fast("total_amount"))

```

— BEST PRACTICE —

1. ALWAYS try built-in functions FIRST

```

df_best = df_sales.withColumn("category",
    F.when(F.col("total_amount").isNull(), "UNKNOWN")
      .when(F.col("total_amount") < 1000, "MICRO")
      .when(F.col("total_amount") < 10000, "SMALL")
      .when(F.col("total_amount") < 50000, "MEDIUM")
      .when(F.col("total_amount") < 200000, "LARGE")
      .otherwise("ENTERPRISE")
    )
# ↑ This is FASTEST (runs in JVM, no Python overhead)!

```

◆ Visualizing Data in Spark Notebook

FABRIC'S display() IS THE BEST VISUALIZATION TOOL:

Step 1: Prepare aggregated data

```
df_daily_revenue = df_sales \  
    .groupBy("sale_date", "region_code") \  
    .agg(F.sum("total_amount").alias("revenue")) \  
    .orderBy("sale_date")
```

Step 2: display() – click chart icon in output!

```
display(df_daily_revenue)
```

In the output, click the chart icon and configure:

```
# |  
# | Chart Configuration Panel |  
# |  
# | Chart Type: Line Chart |  
# | X axis: sale_date |  
# | Y axis: revenue (SUM) |  
# | Group by: region_code |  
# |  
# | → Click Apply → Interactive chart! |  
# |
```

Available chart types in display():

Bar Chart | Line Chart | Scatter Plot |

Pie Chart | Area Chart | Map (for location data)

The magic: drag and drop columns to chart config

No code needed for quick exploration!

For PUBLICATION-QUALITY charts → matplotlib/plotly:

```
import matplotlib.pyplot as plt
```

```

import seaborn as sns

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle("RetailMart Sales Analytics Dashboard",
             fontsize=16, fontweight='bold')

df_pd = df_daily_revenue.toPandas()

# Chart 1: Daily revenue line
for region in df_pd["region_code"].unique():
    mask = df_pd["region_code"] == region
    axes[0,0].plot(df_pd[mask]["sale_date"],
                  df_pd[mask]["revenue"] / 1e6,
                  label=region, linewidth=2)
axes[0,0].set_title("Daily Revenue by Region (₹ Millions)")
axes[0,0].legend()
axes[0,0].grid(True, alpha=0.3)

# Chart 2: Revenue pie by region
region_total = df_pd.groupby("region_code")["revenue"].sum()
axes[0,1].pie(region_total.values, labels=region_total.index,
              autopct='%1.1f%%', startangle=90)
axes[0,1].set_title("Revenue Share by Region")

plt.tight_layout()
plt.savefig("/tmp/analytics_dashboard.png", dpi=150,
            bbox_inches='tight')
plt.show()

# Upload chart to Lakehouse
mssparkutils.fs.cp("file:///tmp/analytics_dashboard.png",
                  "Files/reports/analytics_dashboard.png")

```

10 Fabric Data Engineering — Delta Lake Tables

◆ Introduction and Understanding Delta Lake

Why does Delta Lake exist?

Let me tell you a story that every data engineer has lived:

THE NIGHTMARE SCENARIO (without Delta):

Monday night pipeline runs at midnight.

Writes 50GB of Parquet files to /silver/sales/2024/06/

Halfway through writing – CLUSTER CRASHES!

Files written:

/silver/sales/2024/06/part-00001.parquet  (complete)

/silver/sales/2024/06/part-00002.parquet  (complete)

/silver/sales/2024/06/part-00003.parquet  (corrupted – half-written!)

Tuesday morning: BI team opens reports

→ Report shows WRONG numbers (reads corrupt file!)

→ Nobody knows what happened

→ Data team spends 4 hours debugging

→ CFO is furious

WITH DELTA LAKE:

Same crash at midnight.

Delta writes to temp files + transaction log

Crash → transaction ROLLED BACK

Delta log shows: "Transaction 47: UNCOMMITTED"

Tuesday morning: BI team opens reports

→ Reads last COMMITTED version (Monday morning's data)

→ Reports show correct data from previous run

→ Data team gets alert → re-runs pipeline
→ Fixed by 1 AM. CFO never knows.

THAT'S the value of Delta Lake!

◆ Creating Delta Tables in Fabric

```
# MULTIPLE WAYS TO CREATE DELTA TABLES:
```

```
# — METHOD 1: saveAsTable (RECOMMENDED) —————  
# Creates managed Delta table in Lakehouse Tables/
```

```
df_silver_sales.write \  
    .format("delta") \  
    .mode("overwrite") \  
    .partitionBy("year", "month") \  
    .option("overwriteSchema", "true") \  
    .saveAsTable("silver_sales")
```

```
# TABLE IS NOW VISIBLE IN LAKEHOUSE TABLES/ FOLDER  
# AND QUERYABLE VIA SQL ENDPOINT!
```

```
# — METHOD 2: CREATE TABLE AS SELECT (SQL) —————  
spark.sql("""  
    CREATE OR REPLACE TABLE gold_sales_summary  
    USING DELTA  
    PARTITIONED BY (report_year, report_month)  
    TBLPROPERTIES (  
        'description' = 'Monthly sales summary for BI reporting',  
        'owner' = 'data_engineering_team',  
        'sla' = 'daily_by_8am'
```



```

    )
    AS
    SELECT
        YEAR(sale_date) AS report_year,
        MONTH(sale_date) AS report_month,
        region_code,
        store_id,
        SUM(total_amount) AS gross_revenue,
        SUM(net_amount) AS net_revenue,
        SUM(gst_amount) AS gst_collected,
        COUNT(sale_id) AS transactions,
        COUNT(DISTINCT customer_id) AS unique_customers
    FROM silver_sales
    WHERE status = 'COMPLETED'
    GROUP BY
        YEAR(sale_date), MONTH(sale_date),
        region_code, store_id
    """
)

```

— METHOD 3: Explicit schema then insert —————

```

spark.sql("""
    CREATE TABLE IF NOT EXISTS fact_sales_audit (
        audit_id          BIGINT GENERATED ALWAYS AS IDENTITY,
        sale_id           INT NOT NULL,
        action            STRING,
        changed_by        STRING,
        changed_at        TIMESTAMP,
        old_amount        DOUBLE,
        new_amount        DOUBLE
    )
    USING DELTA
    """)

```

— METHOD 4: Delta Table API —————

```

from delta.tables import DeltaTable

```

```

# Create empty Delta table with schema
schema = StructType([
    StructField("sale_id",      IntegerType(), False),
    StructField("customer_id",  StringType(),  False),
    StructField("total_amount", DoubleType(),   True),
    StructField("sale_date",    DateType(),     True),
    StructField("year",         IntegerType(),  True),
    StructField("month",        IntegerType(),  True)
])

# Create empty table (like CREATE TABLE with schema)
DeltaTable.createIfNotExists(spark) \
    .tableName("silver_sales_v2") \
    .addColumnns(schema) \
    .partitionedBy("year", "month") \
    .execute()

# VERIFY TABLE CREATED:
display(spark.sql("SHOW TABLES"))
display(spark.sql("DESCRIBE EXTENDED silver_sales"))

```

◆ Working with Delta Tables in Spark

```

# DELTA TABLE OPERATIONS – THE POWER FEATURES:

```

```

from delta.tables import DeltaTable

```

```

# — UPDATE —

```

```

# Something NOT possible with regular Parquet!

```

SQL approach:

```
spark.sql("""
    UPDATE silver_sales
    SET status = 'VERIFIED',
        load_ts = current_timestamp()
    WHERE sale_date = '2024-06-15'
        AND total_amount > 100000
""")
```

Python API approach:

```
DeltaTable.forName(spark, "silver_sales").update(
    condition = F.col("status") == "PENDING",
    set = {
        "status": F.lit("EXPIRED"),
        "load_ts": F.current_timestamp()
    }
)
```

— DELETE —

Also impossible with regular Parquet!

SQL approach:

```
spark.sql("""
    DELETE FROM silver_sales
    WHERE status = 'CANCELLED'
        AND sale_date < '2024-01-01'
""")
```

Python API approach:

```
DeltaTable.forName(spark, "silver_sales").delete(
    condition = (F.col("status") == "CANCELLED") &
                (F.col("sale_date") < "2024-01-01")
)
```

— MERGE (UPSERT) – Most Important! —

```

delta_target = DeltaTable.forName(spark, "silver_sales")

df_updates = spark.read.table("bronze_sales") \
    .filter(F.col("_process_date") == process_date)

delta_target.alias("target").merge(
    df_updates.alias("source"),
    "target.sale_id = source.sale_id" # join condition
) \
.whenMatchedUpdate(set = {
    "total_amount": "source.AMOUNT",
    "status":       "source.STATUS",
    "load_ts":      "current_timestamp()"
}) \
.whenNotMatchedInsert(values = {
    "sale_id":      "source.SALE_ID",
    "customer_id":  "upper(trim(source.CUSTOMER_ID))",
    "store_id":     "source.STORE_ID",
    "total_amount": "source.AMOUNT",
    "sale_date":    "to_date(source.SALE_DATE, 'DD-MON-YYYY')",
    "year":         "year(to_date(source.SALE_DATE, 'DD-MON-
YYYY'))",
    "month":        "month(to_date(source.SALE_DATE, 'DD-MON-
YYYY'))",
    "status":       "source.STATUS",
    "load_ts":      "current_timestamp()"
}) \
.execute()

print("Merge completed!")

# — GET MERGE STATISTICS —————
# Read operation metrics from merge
history = DeltaTable.forName(spark, "silver_sales").history(1)
last_operation = history.first()

```

```
print(f"Operation: {last_operation['operation']}")
print(f"Metrics: {last_operation['operationMetrics']}")
# Shows: numTargetRowsInserted, numTargetRowsUpdated,
numSourceRows
```

◆ Delta Tables with Streaming Data

```
# DELTA + STREAMING = Perfect combination in Fabric!
```

```
# — READ DELTA TABLE AS STREAM —————
# (CDC – Change Data Capture on Delta table!)
# Reads only NEW or CHANGED data since last checkpoint
```

```
df_stream_source = spark.readStream \
    .format("delta") \
    .option("startingVersion", "latest") \
    .table("bronze_sales")
# Processes only new rows appended to bronze_sales!
```

```
# Process stream
df_processed = df_stream_source \
    .filter(F.col("status") == "COMPLETED") \
    .withColumn("load_ts", F.current_timestamp())
```

```
# Write stream to another Delta table
df_processed.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation",
"Files/checkpoints/silver_sales_stream/") \
    .trigger(availableNow=True) \
```

```

        .toTable("silver_sales")

# — WRITE STREAMING DATA TO DELTA —————
# (Real-time ingestion from EventStream/files)

df_realtime = spark.readStream \
    .format("cloudFiles") \
    .option("cloudFiles.format", "json") \
    .option("cloudFiles.schemaLocation",
"Files/schema/payments/") \
    .load("Files/landing/payments/")

df_realtime.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation", "Files/checkpoints/payments/")
\
    .trigger(processingTime="30 seconds") \
    .toTable("bronze_payments_realtime")

# KEY: Delta handles concurrent batch + stream writes!
# While stream is writing micro-batches every 30 seconds,
# batch pipeline can also write to same table!
# No conflict → Delta handles it with transaction log!

```

◆ SCD Type 1 and Type 2 (Already Covered — Fabric Version)

```

# SCD OPERATIONS IN FABRIC — USING TABLE NAMES (CLEANER!)

# SCD TYPE 1: Pure SQL approach (clean!)
spark.sql("""

```

```

MERGE INTO dim_customers AS target
USING (
    SELECT * FROM bronze_customers
    WHERE _process_date = '{process_date}'
) AS source
ON target.customer_id = source.CUSTOMER_ID
WHEN MATCHED AND (
    target.email    <> source.EMAIL OR
    target.phone    <> source.PHONE OR
    target.address  <> source.ADDRESS
) THEN UPDATE SET
    target.email      = source.EMAIL,
    target.phone      = source.PHONE,
    target.address    = source.ADDRESS,
    target.updated_at = current_timestamp()
WHEN NOT MATCHED THEN INSERT (
    customer_id, customer_name, email, phone, address,
    customer_tier, created_at, updated_at
) VALUES (
    source.CUSTOMER_ID, source.CUST_NAME, source.EMAIL,
    source.PHONE, source.ADDRESS, 'STANDARD',
    current_timestamp(), current_timestamp()
)
)
""".format(process_date=process_date))

```

SCD TYPE 2: Two-step SQL approach

Step 1: Expire changed records

```

spark.sql(f"""
    UPDATE dim_customers_scd2
    SET is_current = false,
        effective_end = date_sub(current_date(), 1)
    WHERE is_current = true
        AND customer_id IN (
            SELECT b.CUSTOMER_ID
            FROM bronze_customers b

```

```

        JOIN dim_customers_scd2 d
          ON b.CUSTOMER_ID = d.customer_id
         AND d.is_current = true
       WHERE b._process_date = '{process_date}'
          AND (b.EMAIL <> d.email OR b.ADDRESS <> d.address)
    )
    """
)

```

Step 2: Insert new version

```

spark.sql(f"""
    INSERT INTO dim_customers_scd2
    SELECT
        b.CUSTOMER_ID AS customer_id,
        b.CUST_NAME AS customer_name,
        b.EMAIL AS email,
        b.ADDRESS AS address,
        current_date() AS effective_start,
        DATE('9999-12-31') AS effective_end,
        true AS is_current,
        current_timestamp() AS created_at
    FROM bronze_customers b
    LEFT ANTI JOIN dim_customers_scd2 d
      ON b.CUSTOMER_ID = d.customer_id
     AND d.is_current = true
   WHERE b._process_date = '{process_date}'
    """)

```

◆ Time Travel and Versioning

TIME TRAVEL IN FABRIC – GAME CHANGER:

```

# — VIEW TABLE HISTORY —————
display(spark.sql("DESCRIBE HISTORY silver_sales"))
# OR:
display(DeltaTable.forName(spark, "silver_sales").history())

# Output columns:
# version | timestamp          | operation | operationParameters
# 5       | 2024-06-15 01:35   | MERGE     | {predicate: ...,
numRows: 1000}
# 4       | 2024-06-14 01:33   | MERGE     | {predicate: ...,
numRows: 987}
# 3       | 2024-06-13 01:31   | WRITE     | {mode: Append, ...}
# 2       | 2024-06-12 23:00   | UPDATE    | {predicate: ...,
numRows: 5}
# 1       | 2024-06-01 01:00   | WRITE     | {mode: Overwrite,
...}
# 0       | 2024-05-31 12:00   | CREATE    | ...

# — QUERY PREVIOUS VERSIONS —————
# By version number:
df_v3 = spark.read.format("delta") \
    .option("versionAsOf", "3") \
    .table("silver_sales")

# By timestamp:
df_yesterday = spark.read.format("delta") \
    .option("timestampAsOf", "2024-06-14 00:00:00") \
    .table("silver_sales")

# SQL:
display(spark.sql("""
    SELECT * FROM silver_sales VERSION AS OF 3
    WHERE region_code = 'N'
    LIMIT 10
    """))

```

```

# — REAL WORLD: INVESTIGATE DATA ISSUE —————
# "Our North India revenue dropped 40% yesterday – what
happened?"

# Compare today vs yesterday
df_today = spark.read.table("silver_sales") \
    .filter(F.col("sale_date") == "2024-06-15") \
    .filter(F.col("region_code") == "N")

df_yesterday = spark.read.format("delta") \
    .option("timestampAsOf", "2024-06-14 23:59:59") \
    .table("silver_sales") \
    .filter(F.col("sale_date") == "2024-06-14") \
    .filter(F.col("region_code") == "N")

print(f"Today (June 15):
{df_today.agg(F.sum('total_amount')).first()[0]:, .0f}")
print(f"Yesterday (June 14):
{df_yesterday.agg(F.sum('total_amount')).first()[0]:, .0f}")

# — ROLLBACK ENTIRE TABLE —————
# Pipeline wrote wrong data in version 5
# Need to restore version 4

spark.sql("RESTORE TABLE silver_sales TO VERSION AS OF 4")
# Table is back to state after version 4!
# Version 5 data is gone (but version 5 still in history)

# — CONFIGURE DATA RETENTION —————
# How many versions to keep? (for time travel AND VACUUM)
spark.sql("""
    ALTER TABLE silver_sales
    SET TBLPROPERTIES (
        'delta.logRetentionDuration' = 'interval 30 days', --

```

```
keep 30 days of history
    'delta.deletedFileRetentionDuration' = 'interval 7 days'
-- keep 7 days of files
)
""")
```

◆ Implementing Medallion Architecture in Fabric

COMPLETE MEDALLION ARCHITECTURE FOR RETAILMART IN FABRIC:

```
# — BRONZE LAYER —————

def load_bronze(source_data: DataFrame, table_name: str,
process_date: str):
    """
    Bronze Layer: Land exactly as received from source
    No transformations! Only add audit metadata.
    """
    df_bronze = source_data \
        .withColumn("_ingestion_ts", F.current_timestamp()) \
        .withColumn("_source_file", F.input_file_name()) \
        .withColumn("_process_date", F.lit(process_date)) \
        .withColumn("_source_system", F.lit("Oracle_ERP"))

    df_bronze.write \
        .format("delta") \
        .mode("append") \
        .partitionBy("_process_date") \
        .option("mergeSchema", "true") \
        .saveAsTable(f"bronze_{table_name}")

    count = df_bronze.count()
```

```

print(f"✅ Bronze {table_name}: {count:,} rows appended")
return count

# — SILVER LAYER —
def load_silver_sales(process_date: str):
    """
    Silver Layer: Cleaned, validated, standardized
    Apply business rules and quality checks
    """

    # Read from Bronze
    df_bronze = spark.read.table("bronze_sales") \
        .filter(F.col("_process_date") ==
process_date)

    # Validate and clean
    df_clean = df_bronze \
        .filter(F.col("SALE_ID").isNotNull()) \
        .filter(F.col("AMOUNT") > 0) \
        .filter(~F.col("STATUS").isin("CANCELLED", "VOID"))

    # Standardize
    df_silver = df_clean \
        .select(
            F.col("SALE_ID").cast("int").alias("sale_id"),
            F.upper(F.trim("CUSTOMER_ID")).alias("customer_id"),
            F.col("STORE_ID").cast("int").alias("store_id"),

F.upper(F.trim("PRODUCT_CODE")).alias("product_code"),
            F.col("AMOUNT").cast("double").alias("total_amount"),
            F.to_date("SALE_DATE", "DD-MON-
YYYY").alias("sale_date"),
            F.upper("REGION_CODE").alias("region_code"),
            F.lower("PAYMENT_MODE").alias("payment_mode"),
            "STATUS"
        ) \

```

```

        .withColumn("year",      F.year("sale_date")) \
        .withColumn("month",    F.month("sale_date")) \
        .withColumn("gst_amount", F.round(F.col("total_amount") *
0.18, 2)) \
        .withColumn("net_amount", F.col("total_amount") -
F.col("gst_amount")) \
        .withColumn("load_ts",   F.current_timestamp())

# Upsert to Silver (no duplicates!)
DeltaTable.forName(spark, "silver_sales").alias("t").merge(
    df_silver.alias("s"),
    "t.sale_id = s.sale_id"
).whenMatchedUpdateAll().whenNotMatchedInsertAll().execute()

print(f"✅ Silver sales updated for {process_date}")

```

— GOLD LAYER —

```

def load_gold_store_performance(process_date: str):
    """
    Gold Layer: Business-ready aggregations
    Optimized for Power BI Direct Lake
    """
    df = spark.sql(f"""
        SELECT
            s.sale_date,
            s.store_id,
            st.store_name,
            st.city,
            st.state,
            s.region_code,
            COUNT(s.sale_id)           AS transactions,
            SUM(s.total_amount)        AS gross_revenue,
            SUM(s.net_amount)          AS net_revenue,
            SUM(s.gst_amount)          AS gst_collected,
            AVG(s.total_amount)        AS avg_ticket,

```

```

        COUNT(DISTINCT s.customer_id) AS unique_customers,
        SUM(CASE WHEN s.payment_mode='cash' THEN
s.total_amount ELSE 0 END) AS cash_revenue,
        SUM(CASE WHEN s.payment_mode<>'cash' THEN
s.total_amount ELSE 0 END) AS digital_revenue
    FROM silver_sales s
    JOIN dim_stores st ON s.store_id = st.store_id
    WHERE s.sale_date = '{process_date}'
    GROUP BY s.sale_date, s.store_id, st.store_name,
             st.city, st.state, s.region_code
    """)

# Write to Gold (overwrite today's partition only)
df.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere", f"sale_date = '{process_date}'")
\
    .saveAsTable("gold_store_performance")

# Optimize for Power BI Direct Lake (V-ORDER!)
spark.sql("OPTIMIZE gold_store_performance ZORDER BY
(store_id, sale_date)")

print(f"✅ Gold store_performance loaded for {process_date}")

# — RUN COMPLETE MEDALLION PIPELINE —————
def run_daily_medallion(process_date: str):
    print(f"\n{'='*60}")
    print(f"🚀 Medallion Pipeline: {process_date}")
    print(f"{'='*60}")

    # Bronze loads (from Files/ to Tables/)
    df_raw_sales =
spark.read.parquet(f"Files/raw/sales/{process_date}/")

```

```

load_bronze(df_raw_sales, "sales", process_date)

df_raw_customers =
spark.read.parquet(f"Files/raw/customers/{process_date}/")
load_bronze(df_raw_customers, "customers", process_date)

# Silver transforms
load_silver_sales(process_date)
load_silver_customers(process_date)

# Gold aggregations
load_gold_store_performance(process_date)
load_gold_customer_360(process_date)

print(f"\n✅ Complete! {process_date} processed
successfully")

run_daily_medallion(process_date)

```

◆ V-Order in Fabric

V-ORDER: Fabric's Secret Weapon for Power BI Performance

WHAT IS V-ORDER?

Microsoft's proprietary optimization applied to Parquet files

Similar to Delta's OPTIMIZE but specifically designed for

Direct Lake mode in Power BI

V-ORDER does:

→ Sorts data within each Parquet row group optimally

→ Applies additional compression (beyond standard Snappy)

```

# → Enables Power BI to skip unnecessary data pages
# → Result: 2-5x faster Power BI report loading!

# ENABLE V-ORDER when writing to Gold layer:
spark.conf.set("spark.sql.parquet.vorder.enabled", "true")
spark.conf.set("spark.microsoft.delta.optimizeWrite.enabled",
"true")

# Now writes are V-ORDER optimized
df_gold.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("gold_store_performance")
# Files written with V-ORDER → Power BI Direct Lake is 3x faster!

# APPLY V-ORDER to existing table via OPTIMIZE:
spark.sql("OPTIMIZE gold_store_performance")
# If V-ORDER is enabled in Spark config → OPTIMIZE applies it!

# VERIFY V-ORDER on a table:
df_details = spark.sql("DESCRIBE DETAIL gold_store_performance")
display(df_details.select("format", "properties"))

# V-ORDER BEST PRACTICES:
# ✅ ALWAYS enable for Gold layer tables (Power BI reads these)
# ✅ Enable for Silver layer if analysts query directly
# ⚠️ Not needed for Bronze layer (raw ingestion – PBI doesn't
read this)
# ⚠️ Slight overhead during write – worth it for read
performance!

# OPTIMIZE WRITE (goes with V-ORDER):
spark.conf.set("spark.microsoft.delta.optimizeWrite.enabled",
"true")
# This tells Spark to write optimally-sized Parquet files

```



```
# Avoids the small file problem automatically during writes!

# COMPLETE GOLD WRITE CONFIGURATION:
spark.conf.set("spark.sql.parquet.vorder.enabled", "true")
spark.conf.set("spark.microsoft.delta.optimizeWrite.enabled",
"true")
spark.conf.set("spark.microsoft.delta.optimizeWrite.binSize",
"1073741824") # 1GB bin

df_gold.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere", f"sale_date = '{process_date}'") \
    .option("vorder", "true") \
    .saveAsTable("gold_store_performance")

print("✅ Gold table written with V-ORDER optimization")
print("    Power BI Direct Lake will be significantly faster!")
```

11 Fabric Data Warehouse — Deep Dive

◆ Introduction to Data Warehousing in Fabric

WHAT IS A DATA WAREHOUSE?

Simple definition:

"A central repository of integrated data from one or more disparate sources, structured for fast analytical querying"

WHY SEPARATE DATA WAREHOUSE from Lakehouse?

Lakehouse: optimized **for** big data (Spark)
data engineers use this
complex transformations

Warehouse: optimized **for SQL** analytics (T-SQL)
business analysts use this
familiar **SQL** queries
structured, governed data

FABRIC WAREHOUSE **UNIQUE** FEATURES:

- ✓ T-SQL compatible (familiar **for SQL** developers!)
- ✓ Serverless (**no** cluster management)
- ✓ Built **on** Delta Lake + OneLake (**open** format)
- ✓ **Cross**-database queries (query Lakehouse + Warehouse together!)
- ✓ Automatic statistics (**no** manual ANALYZE **TABLE**)
- ✓ Direct Lake **for** Power BI (fastest reports)
- ✓ **Versioning** (Delta **time** travel **on** warehouse tables)
- ✓ Role-based security (**column/row** level)

◆ Data Warehouse Fundamentals

```
-- STAR SCHEMA for RetailMart Warehouse
-- (Industry standard for data warehouse design)
```

```
-- DIMENSION TABLES (describe the what/who/where/when)
CREATE TABLE [dim].[dim_date] (
    date_key          INT          NOT NULL,  -- surrogate key:
20240615
    full_date         DATE         NOT NULL,
    day_of_week       INT,
```

```

        day_name          VARCHAR(10),          -- Monday,
Tuesday...
        day_of_month      INT,
        month_number      INT,
        month_name        VARCHAR(10),          -- January,
February...
        quarter          INT,
        year              INT,
        is_weekend        BIT,
        is_holiday        BIT,
        fiscal_period     VARCHAR(10)
    );

CREATE TABLE [dim].[dim_customer] (
    customer_key          INT                NOT NULL,  -- surrogate key
(auto-increment)
    customer_id           VARCHAR(20)        NOT NULL,  -- natural/business
key
    customer_name         VARCHAR(200),
    email_masked          VARCHAR(200),        -- masked for
privacy!
    customer_tier         VARCHAR(20),
    city                  VARCHAR(100),
    state                 VARCHAR(100),
    region_code          VARCHAR(10),
    registration_date     DATE,
    is_active             BIT
);

CREATE TABLE [dim].[dim_product] (
    product_key           INT                NOT NULL,
    product_code          VARCHAR(50)        NOT NULL,
    product_name          VARCHAR(200),
    category              VARCHAR(100),
    sub_category          VARCHAR(100),

```

```

    brand            VARCHAR(100),
    cost_price       DECIMAL(18,2),
    list_price       DECIMAL(18,2)
);

CREATE TABLE [dim].[dim_store] (
    store_key        INT            NOT NULL,
    store_id         VARCHAR(20)    NOT NULL,
    store_name       VARCHAR(200),
    city             VARCHAR(100),
    state            VARCHAR(100),
    region_code      VARCHAR(10),
    store_type       VARCHAR(50),
    opening_date     DATE,
    manager_name     VARCHAR(200),
    sq_footage       INT
);

-- FACT TABLE (the measurements/events)
CREATE TABLE [fact].[fact_sales] (
    sale_key         BIGINT         NOT NULL, -- surrogate key
    date_key         INT           NOT NULL, -- FK to dim_date
    customer_key     INT,           -- FK to
dim_customer
    product_key      INT,           -- FK to dim_product
    store_key        INT,           -- FK to dim_store
    quantity         INT,
    unit_price       DECIMAL(18,2),
    gross_amount     DECIMAL(18,2),
    discount_amount  DECIMAL(18,2),
    net_amount       DECIMAL(18,2),
    gst_amount       DECIMAL(18,2),
    payment_mode     VARCHAR(50),
    transaction_time TIME
);

```

```
-- WHY SURROGATE KEYS?
-- Natural keys can change (customer moves → new address)
-- Surrogate keys NEVER change (just an auto-increment number)
-- Joins are on INT (fast!) not VARCHAR (slow!)
```

◆ Data Warehouses in Microsoft Fabric — Practical

```
-- CREATE FABRIC WAREHOUSE:
-- Workspace → + New → Warehouse
-- Name: WH_RetailMart_Analytics
```

```
-- FABRIC WAREHOUSE SQL FEATURES:
```

```
-- CREATE SCHEMA
```

```
CREATE SCHEMA dim;
CREATE SCHEMA fact;
CREATE SCHEMA staging;
CREATE SCHEMA reporting;
```

```
-- CREATE TABLE (same T-SQL as SQL Server!)
```

```
CREATE TABLE fact.fact_sales (
    sale_key      BIGINT IDENTITY(1,1),
    date_key      INT NOT NULL,
    customer_key  INT,
    product_key   INT,
    store_key     INT,
    gross_amount  DECIMAL(18,2),
    net_amount    DECIMAL(18,2),
    gst_amount    DECIMAL(18,2),
    quantity      INT
```

```

);

-- INSERT INTO (for staging loads)
INSERT INTO staging.stg_sales
SELECT
    CONVERT(INT, FORMAT(sale_date, 'yyyyMMdd')) AS date_key,
    c.customer_key,
    p.product_key,
    s.store_key,
    f.total_amount AS gross_amount,
    f.net_amount,
    f.gst_amount,
    f.quantity
FROM LH_Sales_Prod.dbo.silver_sales f      -- Lakehouse table!
JOIN dim.dim_customer c ON f.customer_id = c.customer_id
JOIN dim.dim_product p  ON f.product_code = p.product_code
JOIN dim.dim_store s    ON f.store_id = s.store_id
WHERE f.sale_date = CONVERT(DATE, '2024-06-15')

-- CROSS-DATABASE QUERY (AMAZING FEATURE!):
-- Query Lakehouse and Warehouse in SAME SQL query
SELECT
    w.store_name,
    w.city,
    l.gross_revenue AS lakehouse_revenue,      -- from Lakehouse
    SUM(f.gross_amount) AS warehouse_revenue -- from Warehouse
FROM WH_RetailMart_Analytics.fact.fact_sales f
JOIN WH_RetailMart_Analytics.dim.dim_store w ON f.store_key =
w.store_key
JOIN LH_Sales_Prod.dbo.gold_store_performance l -- Lakehouse!
    ON w.store_id = l.store_id
    AND l.sale_date = '2024-06-15'
WHERE CONVERT(DATE, CONVERT(VARCHAR, f.date_key)) = '2024-06-15'
GROUP BY w.store_name, w.city, l.gross_revenue

```

```
-- NO DATA MOVEMENT! Both live in OneLake. Query runs across both!
```

◆ Querying and Transforming Data

```
-- ANALYTICAL QUERIES IN FABRIC WAREHOUSE:
```

```
-- 1. MONTH OVER MONTH GROWTH:
```

```
WITH monthly_revenue AS (  
    SELECT  
        d.year,  
        d.month_number,  
        d.month_name,  
        SUM(f.gross_amount) AS revenue  
    FROM fact.fact_sales f  
    JOIN dim.dim_date d ON f.date_key = d.date_key  
    GROUP BY d.year, d.month_number, d.month_name  
)  
SELECT  
    year,  
    month_name,  
    revenue,  
    LAG(revenue) OVER (ORDER BY year, month_number) AS  
prev_month_revenue,  
    ROUND(  
        (revenue - LAG(revenue) OVER (ORDER BY year,  
month_number)) /  
        NULLIF(LAG(revenue) OVER (ORDER BY year, month_number),  
0) * 100,  
        1  
    ) AS mom_growth_pct
```

```

FROM monthly_revenue
ORDER BY year, month_number;

-- 2. TOP 10 PRODUCTS BY REVENUE:
SELECT TOP 10
    p.product_name,
    p.category,
    SUM(f.gross_amount) AS total_revenue,
    SUM(f.quantity) AS units_sold,
    ROUND(AVG(f.gross_amount / NULLIF(f.quantity, 0)), 2) AS
avg_unit_price,
    RANK() OVER (ORDER BY SUM(f.gross_amount) DESC) AS
revenue_rank
FROM fact.fact_sales f
JOIN dim.dim_product p ON f.product_key = p.product_key
JOIN dim.dim_date d ON f.date_key = d.date_key
WHERE d.year = 2024
GROUP BY p.product_name, p.category
ORDER BY total_revenue DESC;

-- 3. CUSTOMER COHORT ANALYSIS:
WITH customer_first_purchase AS (
    SELECT
        c.customer_key,
        MIN(d.year * 100 + d.month_number) AS cohort_month
    FROM fact.fact_sales f
    JOIN dim.dim_customer c ON f.customer_key = c.customer_key
    JOIN dim.dim_date d ON f.date_key = d.date_key
    GROUP BY c.customer_key
),
customer_activity AS (
    SELECT
        cf.cohort_month,
        d.year * 100 + d.month_number AS activity_month,
        COUNT(DISTINCT f.customer_key) AS active_customers

```



```
FROM fact.fact_sales f
JOIN dim.dim_date d ON f.date_key = d.date_key
JOIN customer_first_purchase cf ON f.customer_key =
cf.customer_key
GROUP BY cf.cohort_month, d.year * 100 + d.month_number
)
SELECT
    cohort_month,
    activity_month,
    active_customers,
    activity_month - cohort_month AS months_since_acquisition
FROM customer_activity
ORDER BY cohort_month, activity_month;
```

◆ SQL Endpoints and Warehouse Security

```
-- USING SQL ENDPOINT:
```

```
-- SQL Endpoint connection string (auto-provided):
-- Server: <workspace-id>.datawarehouse.fabric.microsoft.com
-- Database: WH_RetailMart_Analytics (or LH_Sales_Prod for
lakehouse)
-- Authentication: Azure Active Directory
```

```
-- Use in Power BI Desktop:
-- Get Data → Azure → Synapse Analytics (SQL DW)
-- Server: [above connection string]
-- Database: WH_RetailMart_Analytics
-- Mode: Import or DirectQuery
```

```
-- SECURITY IN WAREHOUSE:
```

```

-- Grant SELECT to BI team
GRANT SELECT ON SCHEMA::reporting TO [bi-team@retailmart.com];

-- Grant column-level access (hide sensitive columns):
-- (No direct column DENY in Fabric yet – use views!)

-- Create secure view (hide PAN/phone for BI team):
CREATE OR ALTER VIEW reporting.vw_customer_safe AS
SELECT
    customer_key,
    customer_name,
    LEFT(email, 3) + '***@***.com' AS email_masked,
    customer_tier,
    city,
    state,
    region_code
FROM dim.dim_customer;

-- Grant view access (not base table):
GRANT SELECT ON reporting.vw_customer_safe TO [bi-
team@retailmart.com];
DENY SELECT ON dim.dim_customer TO [bi-
team@retailmart.com];
-- BI team sees masked data. DE team sees everything.

-- ROW LEVEL SECURITY:
-- Regional managers see only their region
CREATE SCHEMA security;

CREATE FUNCTION security.fn_region_predicate(@region_code
VARCHAR(10))
RETURNS TABLE
WITH SCHEMABINDING
AS
RETURN SELECT 1 AS result

```

```
WHERE @region_code = USER_NAME()  -- user name must match region
code!

OR IS_MEMBER('dw_admin') = 1;

CREATE SECURITY POLICY fact.RegionSalesFilter
ADD FILTER PREDICATE security.fn_region_predicate(region_code)
ON fact.fact_sales
WITH (STATE = ON);

-- North India manager (username: 'N') sees ONLY North India
data!
-- Admin sees everything.
```

◆ Copying Data from Lakehouse to Warehouse

```
# COMMON PATTERN: Lakehouse (processing) → Warehouse (reporting)
```

```
# OPTION 1: Pipeline Copy Activity (simple, reliable)
# Create Pipeline in Fabric workspace
# Source: Lakehouse table (gold_store_performance)
# Sink: Warehouse table (reporting.store_performance)

# Copy Activity configuration:
# Source:
#   Lakehouse: LH_Sales_Prod
#   Table: gold_store_performance
#   Filter: WHERE sale_date = '@{pipeline().parameters.p_date}'
#
# Sink:
#   Warehouse: WH_RetailMart_Analytics
#   Schema: reporting
```

```

# Table: store_performance
# Write behavior: Upsert
# Key columns: [sale_date, store_id]

# OPTION 2: Cross-database INSERT (SQL approach)
# Run this in Fabric Warehouse SQL editor:
"""

INSERT INTO reporting.store_performance
SELECT *
FROM LH_Sales_Prod.dbo.gold_store_performance -- Lakehouse!
WHERE sale_date = '2024-06-15'
      AND NOT EXISTS (
          SELECT 1 FROM reporting.store_performance sp
          WHERE sp.sale_date = '2024-06-15'
                AND sp.store_id = gold_store_performance.store_id
      );
"""

# OPTION 3: Direct shortcut (best approach!)
# Create shortcut from Warehouse to Lakehouse table
# Warehouse → Tables → right-click → New shortcut
# → OneLake → select LH_Sales_Prod.Lakehouse →
gold_store_performance
#
# NOW: Warehouse has shortcut to Lakehouse table
# NO data movement!
# Power BI connects to Warehouse
# Warehouse queries Lakehouse data via shortcut
# Reports are always fresh!
# This is the recommended approach in Fabric!

```

◆ Getting and Processing Data in an Event Stream

REAL-TIME INTELLIGENCE WORKFLOW:

SOURCES → EVENTSTREAM → EVENTHOUSE (KQL DB) → QUERIES/ALERTS

WHAT IS EVENTSTREAM?

- Capture **real-time** data **from** multiple sources
- Process **and** route data **to** destinations
- **No** code required! Visual pipeline **for** streaming

CREATE EVENTSTREAM:

Workspace → + **New** → Eventstream
Name: ES_RetailMart_Payments

SOURCE TYPES:

- Azure Event Hubs (payment events **from** app)
- Azure IoT Hub (store IoT sensors)
- Kafka (**on**-premise streaming)
- Custom App (**any** REST push)
- Sample data (**for** learning!)

CONNECT SOURCE (Azure Event Hubs):

- + Source → Azure Event Hubs
- Namespace: retailmart-eventhubs
- Event Hub: payment-transactions
- Consumer **Group**: fabric-consumer
- Authentication: Connection String (**from** Key Vault)

PROCESS IN EVENTSTREAM (visual transforms):

- **Filter**: **WHERE** amount > 0
- Aggregate: tumbling **window** (1 min)
GROUP BY payment_mode → **SUM**(amount)
- Derived **column**:

```
fraud_risk = IF amount > 100000 AND hour < 5 THEN 'HIGH'
ELSE 'LOW'
```

DESTINATIONS:

- Eventhouse (KQL database – for real-time queries)
- Lakehouse (for batch/historical)
- Warehouse (for SQL analytics)
- Another Eventstream (fan-out)
- Custom endpoint (webhook)

◆ KQL Databases and Kusto Query Language

KQL DATABASE = Time-series optimized database

WHAT IS KQL?

Kusto Query Language = SQL-like language for time-series data

Much faster than SQL for time-series analytics

Used in: Azure Data Explorer, Log Analytics, Fabric RT Intel

WHY KQL FOR REAL-TIME?

Traditional SQL:

```
SELECT COUNT(*), payment_mode
FROM payments
WHERE timestamp BETWEEN '10:00' AND '10:05'
GROUP BY payment_mode
```

→ Might take 10+ seconds on large streaming table!

KQL:

```
payments
| where timestamp between (datetime(10:00) ..
datetime(10:05))
```

```
| summarize count() by payment_mode  
→ Returns in MILLISECONDS even on billions of rows!
```

KQL is designed for **time-series** from ground up!

KQL SYNTAX BASICS:

```
// — BASIC KQL QUERIES —————  
  
// SELECT first 10 rows (like SELECT TOP 10)  
payments  
| take 10  
  
// FILTER rows (like WHERE)  
payments  
| where amount > 10000  
| where payment_mode == "card"  
| where timestamp > ago(1h) // last 1 hour!  
  
// PROJECT = SELECT specific columns  
payments  
| project payment_id, customer_id, amount, timestamp  
  
// COUNT rows  
payments  
| count  
  
// SUMMARIZE = GROUP BY with aggregations  
payments  
| summarize  
    total_revenue = sum(amount),  
    transaction_count = count(),  
    avg_amount = avg(amount),  
    max_amount = max(amount)  
by payment_mode
```

```

// ORDER BY (sort)
payments
| summarize total = sum(amount) by region_code
| order by total desc

// TOP N
payments
| summarize total = sum(amount) by store_id
| top 10 by total desc

// — TIME-SERIES SPECIFIC —————

// Last 1 hour of transactions
payments
| where timestamp > ago(1h)
| count

// Group by time buckets (every 5 minutes)
payments
| where timestamp > ago(24h)
| summarize
    txn_count = count(),
    total_amount = sum(amount)
  by bin(timestamp, 5m) // 5-minute buckets!
| order by timestamp asc

// Revenue trend by hour
payments
| where timestamp between (ago(7d) .. now())
| summarize hourly_revenue = sum(amount) by bin(timestamp, 1h),
region_code
| render timechart // RENDERS CHART directly in KQL editor!

// High-value transactions in last 10 minutes

```



```
payments
| where timestamp > ago(10m)
| where amount > 100000
| project payment_id, customer_id, amount, timestamp, payment_mode
| order by amount desc
```

```
// — JOINS IN KQL —————
// Join payments with customer risk table
payments
| where timestamp > ago(1h)
| join kind=leftouter (
    customer_risk // another KQL table
    | project customer_id, risk_score, risk_level
) on customer_id
| where risk_score > 0.8 // high risk customers
| project payment_id, customer_id, amount, risk_score, risk_level
```

```
// — FRAUD DETECTION QUERY —————
payments
| where timestamp > ago(10m)
| extend is_high_value = amount > 100000
| extend hour_of_day = hourofday(timestamp)
| extend is_unusual_hour = hour_of_day < 5 or hour_of_day > 23
| extend fraud_risk_score =
    case(
        is_high_value and is_unusual_hour, 3.0,
        is_high_value, 2.0,
        is_unusual_hour, 1.0,
        0.0
    )
| where fraud_risk_score >= 2
| project payment_id, customer_id, amount, hour_of_day,
    fraud_risk_score, timestamp
| order by fraud_risk_score desc
```

13 Activators in Fabric

◆ Complete Guide to Data Activators

DATA ACTIVATOR = "If this happens → do that" for data

WHAT IT REPLACES:

Old way: custom Azure Function + Logic App + Power Automate

New way: Fabric Activator (no code, built-in!)

WHERE IT GETS DATA FROM:

- Eventstream (real-time)
- Power BI reports (detect when visual crosses threshold!)
- Lakehouse (poll on schedule)

WHAT IT CAN DO WHEN TRIGGERED:

- Send email
- Send Teams message
- Start a Fabric pipeline
- Call a webhook (custom action)
- Power Automate flow

CREATE ACTIVATOR:

Workspace → + New → Activator (or Reflex)

Name: ACT_RetailMart_Alerts

SCENARIO 1: Real-time fraud alert

Source: Eventstream (ES_RetailMart_Payments)

Object: payment_transactions table

Property to monitor: fraud_risk_score

Condition: fraud_risk_score >= 2.0

Trigger: for each event (real-time)

Action: Send email to fraud-team@retailmart.com

Body: "High-risk payment detected: {payment_id} for ₹{amount}"

SCENARIO 2: Revenue alert from Power BI

Source: Power BI Report (Sales Dashboard)

Object: Daily Revenue visual

Property: Total Revenue

Condition: Total Revenue FALLS BELOW ₹50,00,000 by 3 PM

Trigger: Once when condition first met

Action: Teams message to channel "Revenue-Alerts"

Body: "⚠ Revenue below target! Current: {revenue}"

SCENARIO 3: Inventory reorder trigger

Source: Lakehouse (poll every hour)

Object: gold_inventory_levels table

Property: current_stock

Condition: current_stock < safety_stock_level

Group by: product_code (detect per product!)

Action: Start Pipeline → PL_Create_Purchase_Order

Pass: {product_code}, {current_stock}, {safety_stock}

◆ Creating Activator Objects and Rules

ACTIVATOR CONFIGURATION WALKTHROUGH:

STEP 1: Create Activator item in workspace

STEP 2: Connect Data Source

→ Eventstream tab: Select your eventstream

→ OR: Power BI tab: Select report + visual

STEP 3: Define OBJECT

Object = the "thing" you're monitoring
e.g., each unique payment_id is an object

Property to track: amount, fraud_score, timestamp

Activate by: Real-time (each event) or Scheduled (hourly poll)

STEP 4: Define TRIGGER CONDITIONS

Simple condition:

Amount > 100,000

Pattern-based:

Amount increases by > 50% in last hour
(compares current value to value 1 hour ago)

Time-based:

Revenue NOT received by 2 PM for a store

Combined (AND/OR):

(Amount > 100,000) AND (hour_of_day < 5)

STEP 5: Define ACTIONS

Send email:

To: fraud-team@retailmart.com

Subject: 🚨 Fraud Alert: Payment {payment_id}

Body: Amount: ₹{amount}, Customer: {customer_id},

Time: {timestamp}, Risk: {fraud_score}

Teams message:

Team: RetailMart-Operations

Channel: Fraud-Alerts

Message: Suspicious payment detected [click to investigate]

Start pipeline:

Pipeline: PL_Investigate_Fraud

Parameters: payment_id={payment_id}

STEP 6: Activate

→ Toggle ON

→ Activator starts monitoring in real-time!

STEP 7: Monitor

Activator → History tab

→ See all triggers: when fired, what action taken

→ Success/failure of each action

14 Power BI in Fabric — Complete Integration Guide

◆ Connecting Microsoft Fabric with Power BI

POWER BI + FABRIC = Most powerful BI stack available

THREE MODES OF POWER BI + FABRIC:

1. IMPORT MODE (traditional):

PBI Desktop → copies data to PBI → fast queries

✗ Data gets stale (manual refresh or scheduled)

✗ Limited to ~1GB per model

✓ Fastest query performance

✓ Works offline

2. DIRECTQUERY MODE (traditional):

PBI → queries source every time user interacts

✓ Always fresh data

✗ Slow (every click = new query)

- ✗ Limited SQL support
- ✗ Data source must handle load

3. DIRECT LAKE MODE (NEW in Fabric!):

PBI reads Delta Parquet files directly from OneLake

- ✓ Always fresh (reads latest files)
- ✓ Fast (columnar Parquet + V-ORDER optimization)
- ✓ No data size limits (reads massive datasets!)
- ✓ No data movement or import
- ✓ Reduces storage cost (same data, no copy)
- ⚠ Requires Fabric capacity (not Power BI Pro)

◆ Creating Dashboards and Reports

POWER BI REPORT TYPES IN FABRIC:

1. REPORT (detailed, interactive):

Multiple pages, many visuals
Drill-through, cross-filter
Export to PDF/Excel
Embedded in Teams/SharePoint

2. DASHBOARD (high-level, real-time):

Single page
Pinned tiles from multiple reports
Real-time data tiles
Mobile-optimized

3. PAGINATED REPORT (pixel-perfect):

For printing/PDF export
Invoice, statement formats

Precise layouts

FABRIC-SPECIFIC REPORT CREATION:

- Workspace → LH_Sales_Prod → Semantic Model tab
- + **New** Report (opens browser-based Power BI!)
- Build report **in** Fabric browser (**no** desktop needed!)

BUILD A SALES DASHBOARD:

Page 1: Executive Summary

RetailMart Sales Dashboard - June 2024

₹12.5Cr Total Revenue

142,850 Transactions

₹87.7K Avg Ticket

500 Active Stores

[Revenue by Region - Bar Chart]

N: 35% S: 28% E: 20% W: 17%

[Daily Revenue Trend - Line Chart]

Shows 30-day trend with target line

[Top 10 Stores - Table]

Rank	Store	City	Revenue	vs Target
------	-------	------	---------	-----------

Page 2: Product Analysis

Page 3: Customer Analytics

Page 4: Store Operations

DIRECT LAKE SEMANTIC MODEL (MOST POWERFUL):

- When** Lakehouse attached **to** Semantic Model:
- Gold layer tables automatically available
- **No** data movement!

- Reports ALWAYS show latest data
- Lakehouse updated at 2 AM → Report shows new data at 2:01 AM!

◆ All Power BI Desktop Connections with Fabric

CONNECTION TYPE 1: Power BI Desktop with Fabric Lakehouse

Power BI Desktop → Get Data → Microsoft Fabric → Lakehouse

OR via SQL endpoint:

Get Data → Azure → Azure Synapse Analytics (SQL DW)

Server: <workspace-id>.datawarehouse.fabric.microsoft.com

Database: LH_Sales_Prod ← your lakehouse name!

Authentication: Microsoft Account (Azure AD)

- See Tables/ as SQL tables
- Can join multiple tables
- Build semantic model
- Publish to Fabric workspace

CONNECTION TYPE 2: Power BI Desktop with OneLake

Get Data → Azure → Azure Data Lake Storage Gen2

URL: [https://onelake.dfs.fabric.microsoft.com/
<workspace-name>/<lakehouse-name>.Lakehouse/](https://onelake.dfs.fabric.microsoft.com/<workspace-name>/<lakehouse-name>.Lakehouse/)

Authentication: Organizational Account

- Browse OneLake like ADLS
- Read Parquet/CSV/JSON files directly
- Power Query (M) transforms available

CONNECTION TYPE 3: Power BI Desktop with Warehouse

Get Data → Microsoft Fabric → Warehouse

OR:

Get Data → SQL Server

Server: <workspace-id>.datawarehouse.fabric.microsoft.com

Database: WH_RetailMart_Analytics

- Full T-SQL query capability
- Can write custom SQL queries
- Best for complex transformations in SQL

CONNECTION TYPE 4: Power BI Desktop with Synapse

Get Data → Azure Synapse Analytics SQL

Server: <synapse-workspace>.sql.azuresynapse.net

Database: RetailMartDW (Dedicated Pool)

- Access Synapse Dedicated SQL Pool
- Use if you have existing Synapse setup

CONNECTION TYPE 5: Power BI Cloud with OneLake

From Power BI Service (browser):

New Report → Select data source

- OneLake data hub
- Browse all available Fabric items
- Select Lakehouse or Warehouse
- Build report directly in browser!
- No Power BI Desktop needed!

BEST PRACTICE RECOMMENDATION:

For large enterprise BI:

- Build Semantic Model in Fabric (Direct Lake)
- Power BI Desktop: build reports → publish to workspace
- Reports stay in Fabric workspace
- Users access via Power BI Service (browser)
- Mobile via Power BI app

This gives: fresh data + fast queries + central governance

◆ OneLake Connections with Power BI

ONELAKE DATA HUB – The central catalog:

Power BI Service → OneLake Data Hub (left nav)

- Shows ALL endorsed datasets across your Fabric tenant
- Filter by: workspace, type, endorsement
- Click any item → see description, tables, reports
- "Create Report" → start building immediately

ENDORSEMENT LEVELS:

No endorsement → just created, might be draft

Promoted → team recommends using this

Certified → officially validated by data team

RETAILMART SETUP:

Only certified Gold layer datasets for BI team

Silver: promoted (for analysts who know data)

Bronze: no endorsement (raw, not for general use)

This prevents BI team **from** accidentally **using** raw/uncleaned data **for** executive reports!

SENSITIVITY LABELS (Microsoft Purview integration!):

Each Fabric item can have:

- Public: anyone can see
- Internal: company employees **only**
- Confidential: restricted access
- Highly Confidential: very restricted

Power BI exports INHERIT sensitivity labels!

Download confidential data → file **is** labeled confidential

- Opens **with** appropriate restrictions **in** Windows
- Automatic data loss prevention (DLP)!

1 5 Azure DevOps — Repos in Fabric

◆ Complete DevOps Integration

FABRIC + AZURE DEVOPS = Enterprise-grade development

WHAT GETS VERSIONED IN GIT:

Item Type	Stored as
-----------	------------------

Notebooks	.py or .ipynb files
Pipelines	pipeline.json
Dataflows	dataflow.json
Semantic Models	.tmdl folder (new format)
Reports	report.json + pages

NOT versioned (*data*):

Lakehouse *data* ← just code *is* versioned, not *actual data*

Warehouse tables ← schema definition could be scripted

GIT BRANCHING STRATEGY *for* Fabric:

main branch → Production workspace

↑ PR + approval

dev/staging branch → Test workspace

↑ PR + code review

feature/* *branches* → *Development workspace*

DEVELOPER WORKFLOW:

1. Create feature branch from main:

git checkout -b feature/add-fraud-detection

2. Link Dev workspace to feature branch:

Workspace Settings → Git → Branch: feature/add-fraud-detection

3. Build notebook in Dev workspace

4. Changes auto-sync to feature branch in Git

5. Create Pull Request to dev/staging:

Azure DevOps → Repos → + New Pull Request

6. Code review:

Reviewer sees actual diff (Python code changes!)

Not just "notebook changed" – sees WHAT changed!

7. Merge to dev/staging → auto-deploys to Test workspace
(using Fabric deployment pipeline!)

8. Testing in Test workspace → approve

9. Create PR to main → merge → deploys to Production

CI PIPELINE (Azure DevOps):

On PR creation:

- ✓ Lint Python code (flake8)
- ✓ Run unit tests (pytest)
- ✓ Validate JSON files (pipelines, dataflows)
- ✓ Check for hardcoded secrets (security scan!)
- ✓ Check notebook outputs cleared (no data in notebooks!)

CD PIPELINE (Azure DevOps + Fabric Deployment):

On merge to staging branch:

- Deploy to Test workspace (automated)
- Run smoke tests

On merge to main:

- Manual approval gate (team lead approves)
- Deploy to Production workspace
- Send notification "Deployment successful"

◆ Integrating Fabric Components with Repos

PRACTICAL GIT SETUP:

STEP 1: Create Azure DevOps Repo

- dev.azure.com → Your Project → Repos
- Create: fabric-retailmart
- Create branches: main, dev, feature/*

STEP 2: Connect Dev Workspace to Git

Workspace → Settings → Git Integration
Provider: Azure DevOps
Org: retailmart-devops
Project: DataPlatform
Repo: fabric-retailmart
Branch: dev
Root folder: /workspaces/RetailMart_Sales/
→ Connect → Initial sync

STEP 3: Commit your first notebook

Make change to notebook
Workspace → Source Control (git icon)
→ Modified: "NB_Transform_Sales"
→ Add commit message
→ Commit → changes in Azure DevOps repo!

STEP 4: See what changed (magical!)

Azure DevOps → Repos → Commits → your commit
→ Click modified file
→ DIFF VIEW:

OLD: .withColumn("gst", F.col("amount") * 0.18)
NEW: .withColumn("gst", F.round(F.col("amount") * 0.18, 2))

Reviewer can see EXACTLY what logic changed!
No more "I changed the notebook" – see the PROOF!

DEPLOYMENT PIPELINE IN FABRIC:

Workspace → Deployment Pipelines (if admin created one)

Create Pipeline:

Stage 1: Development

→ Source: RetailMart_Sales_Dev workspace

Stage 2: Testing

→ Target: RetailMart_Sales_Test workspace

Stage 3: Production

→ Target: RetailMart_Sales_Prod workspace

DEPLOY PROCESS:

Diff shows what's different between stages

Deploy button → copies changed items

Rule: only DEFINITIONS copy, not DATA

Notebooks: code copies ✓

Lakehouse: schema can copy ✓

Lakehouse data: NEVER copies ✗ (stays in each environment)

16 SDLC and Agile Methodology

◆ Software Development Life Cycle in Fabric Projects

SDLC FOR FABRIC DATA PROJECTS:

PHASE 1: REQUIREMENTS GATHERING (Week 1-2)

Stakeholders:

→ Business: "We need daily store performance reports by 9 AM"

→ Data: "Source is Oracle DB + CSV files from FTP"

→ IT: "Compliance: no PII in reports for BI team"

Deliverables:

✓ Data dictionary (what each field means)

✓ Source-to-target mapping document

✓ Report mockups approved by business

✓ Data quality rules documented

✓ SLA definition (report ready by 9 AM = pipeline finishes by 8 AM)

PHASE 2: DESIGN (Week 3)

Architecture decisions:

- Lakehouse: bronze/silver/gold Medallion
- Warehouse: star schema for BI team
- Direct Lake: for Power BI reports
- Schedule: 1 AM pipeline start, done by 6 AM

Deliverables:

- ✓ Architecture diagram
- ✓ Star schema design (Fabric Warehouse)
- ✓ Lakehouse folder structure
- ✓ Pipeline orchestration design
- ✓ Security matrix (who sees what)

PHASE 3: DEVELOPMENT (Week 4-8)

Sprints:

- Sprint 1: Bronze ingestion (Oracle → Lakehouse Files)
- Sprint 2: Silver transformation (Files → Tables)
- Sprint 3: Gold aggregations + Warehouse load
- Sprint 4: Power BI reports + testing

Each sprint = 2 weeks

Daily standups (15 min)

Sprint review with business (demo!)

Sprint retrospective (what to improve)

PHASE 4: TESTING (Week 9-10)

Unit testing: individual notebook functions

Integration testing: end-to-end pipeline run

UAT: business validates report numbers

Performance testing: pipeline finishes by 6 AM?

Security testing: BI team cannot see PII?

PHASE 5: DEPLOYMENT (Week 11)

- Deployment pipeline: Dev → Test → Prod
- Parallel run (old + new system simultaneously)
- Business validates new reports match old reports
- Cut over: switch to new system
- Monitor first week closely

PHASE 6: MAINTENANCE (Ongoing)

- Monitor daily pipeline runs
- Handle failures (alerts → investigate → fix → deploy)
- Enhance based on business requests
- Monthly performance review

◆ Agile Methodology for Fabric Projects

AGILE SPRINT STRUCTURE:

SPRINT CADENCE (2 weeks):

Week 1, Day 1 (Monday):

SPRINT PLANNING (2 hours)

Team: Data Engineer 1, Data Engineer 2, Tech Lead

- Review backlog (user stories)
- Pick stories for this sprint
- Break into tasks (estimate in hours)
- Create Azure DevOps Work Items

Week 1-2 (Daily):

STANDUP (15 minutes, 10 AM)

Each person answers:

1. "Yesterday I completed: [task]"
2. "Today I'll work on: [task]"
3. "Blockers: [anything blocking me]"

NOT a status meeting! Just sync and unblock.

Week 2, Day 9 (Thursday):

SPRINT REVIEW (1 hour)

- Demo to business stakeholders
- Show working notebooks/reports
- Get feedback
- "Yes/No/Change" from business

Week 2, Day 10 (Friday):

SPRINT RETROSPECTIVE (45 min, team only)

Three questions:

- What went WELL? (keep doing)
- What needs IMPROVEMENT? (change)
- What should we TRY? (experiment)

Action items tracked in next sprint!

RETAILMART SPRINT EXAMPLE:

Sprint 3 User Stories:

Story 1: As a BI analyst, I want a daily store revenue report

Acceptance: shows all 500 stores, refreshes by 9 AM

Estimate: 3 days

Story 2: As a fraud team, I want real-time alerts when amount >

1L

Acceptance: alert within 60 seconds of transaction
Estimate: 2 days

Story 3: As a data engineer, I want pipeline monitoring dashboard

Acceptance: shows run status, duration, error count
Estimate: 1 day

Total: 6 days in 10-day sprint

Buffer: 4 days for bugs, meetings, reviews (40% buffer!)

40% buffer is CRITICAL – always underestimate in planning!

AZURE DEVOPS BOARDS for Fabric:

Project: RetailMart Fabric

Board: Kanban

Columns: Backlog | In Progress | Review | Done

Each task linked to:

- Git commit (proof of work!)
- Pull Request (code review)
- Test results

◆ Applying Agile Principles in Data Engineering

5 AGILE PRINCIPLES FOR DATA ENGINEERS:

1. WORKING SOFTWARE OVER DOCUMENTATION

- Deploy a pipeline that works, even if docs are incomplete
- Document as you go, not before you build
- Runnable notebook > 50-page spec document

2. CUSTOMER COLLABORATION OVER CONTRACTS

- Weekly demos to business team
- Get feedback early (before you build wrong thing!)
- Change direction based on feedback – it's OK!

RetailMart example:

Week 2: demo regional revenue chart to CFO

CFO: "I need this in ₹ Crores not ₹ Lakhs"

- Change in 10 minutes, not 3 weeks!

vs Waterfall: build for 6 months, CFO sees it, says "wrong format"

- Costly rework!

3. RESPONDING TO CHANGE OVER FOLLOWING PLAN

- Source system adds a new column → update schema (don't panic!)
- Business changes KPI definition → update notebook
- Agile: change is expected and welcomed
- Schema drift enabled for this reason!

4. WORKING DATA PRODUCT OVER COMPREHENSIVE PIPELINE

- Deliver simple, working version first (MVP)
- Enhance incrementally

Example:

Sprint 1: Load sales data, basic count dashboard (MVP!)

Sprint 2: Add customer dimension, tier analysis

Sprint 3: Add store dimension, geographic drill-down

Sprint 4: Real-time updates, fraud detection

Business gets VALUE from Sprint 1!

Not waiting 4 months for "complete" solution.

5. FREQUENT SMALL DEPLOYMENTS

- Deploy **every** sprint (**every 2** weeks)
- Small changes = small risk
- Git + Fabric Deployment Pipeline enables this!

vs once-a-quarter "big bang" deployment → high risk!

17 End-to-End Project: Migrating from On-Premises to Cloud

◆ Complete Migration Approach

RETAILMART: ON-PREMISES TO MICROSOFT FABRIC

CURRENT **STATE** (On-Premises):

Oracle DB → reports via **SSRS** (SQL Server Reporting)

SQL Server → Excel **extracts** (manual)

SAP ERP → CSV **exports** (manual)

Power BI Desktop → manual **refresh** (not published)

Problems:

- ✗ Reports ready at **11** AM (too late!)
- ✗ Manual **processes** (error-prone)
- ✗ No single source of truth
- ✗ Cannot **scale** (Oracle license cost exploding)
- ✗ No ML/AI capability

TARGET **STATE** (Microsoft Fabric):

All sources → Fabric **Lakehouse** (automated, midnight)

Lakehouse → Fabric **Warehouse** (dimensional model)

Warehouse → Power BI Direct **Lake** (reports ready by **6** AM!)

EventStream → Real-time fraud **alerts** (within seconds!)
ML model → Demand **forecasting** (Fabric Data Science)

MIGRATION APPROACH: LIFT, SHIFT, MODERNIZE

Phase 1: LIFT (replicate as-is)

- Copy Oracle data to **Lakehouse** (same structure)
- Validate row counts match
- No transformation yet
- Build confidence that data is correct

Phase 2: SHIFT (move workloads)

- Build Medallion architecture
- Bronze: raw Oracle data
- Silver: cleaned, standardized
- Gold: business-ready aggregations
- Parallel run: both old and **new systems**
- Business validates: **"same numbers?"**

Phase 3: MODERNIZE (improve)

- Better **performance** (Fabric vs Oracle **for** reporting)
- New **capabilities** (real-time, ML, AI)
- Better **governance** (column masking, audit logs)
- Self-service **BI** (business can build their own reports!)
- Decommission on-premises systems

◆ Data Ingestion Using Data Factory in Fabric

FABRIC PIPELINE: Migrate Oracle to Lakehouse

```

# Pipeline: PL_Migrate_Oracle_to_Fabric

# STEP 1: Lookup active tables from control table
# (same control table concept as ADF section!)

# STEP 2: ForEach table → Copy to Lakehouse Files/
# Source: Oracle (via on-premises data gateway)
# Sink: LH_Sales_Prod.Lakehouse/Files/raw/oracle/{table_name}/

# STEP 3: Notebook activity → Files to Tables
# Runs: NB_Load_Bronze_From_Files

# NB_Load_Bronze_From_Files:
from pyspark.sql import functions as F

# List all raw files for today
raw_files = mssparkutils.fs.ls(f"Files/raw/oracle/")

for file_info in raw_files:
    if file_info.isDir:
        table_name = file_info.name.rstrip("/")

        # Read from Files
        df_raw =
spark.read.parquet(f"Files/raw/oracle/{table_name}/")

        # Add audit columns
        df_bronze = df_raw \
            .withColumn("_source_system",
F.lit("Oracle_ECOM_Prod")) \
            .withColumn("_ingestion_ts",    F.current_timestamp())
\
            .withColumn("_process_date",    F.lit(process_date)) \
            .withColumn("_row_hash",        F.md5(F.concat_ws("|",
*df_raw.columns)))

```

```
# Write to Bronze table
df_bronze.write \
    .format("delta") \
    .mode("append") \
    .partitionBy("_process_date") \
    .option("mergeSchema", "true") \
    .saveAsTable(f"bronze_{table_name}")

print(f"✅ bronze_{table_name}: {df_bronze.count():,} rows")
```

◆ Data Processing Using Fabric Notebooks

```
# NOTEBOOK: NB_Transform_Silver_to_Gold
```

```
# Parameters
```

```
process_date = "2024-06-15"
```

```
from pyspark.sql import functions as F
```

```
from pyspark.sql.window import Window
```

```
from delta.tables import DeltaTable
```

```
# — READ SILVER TABLES —
```

```
df_sales = spark.read.table("silver_sales")
```

```
df_customers = spark.read.table("silver_customers")
```

```
df_products = spark.read.table("silver_products")
```

```
df_stores = spark.read.table("silver_stores")
```

```
# — BUILD GOLD: DAILY EXECUTIVE SUMMARY —
```

```
df_gold_exec = df_sales \
```



```

    .filter(F.col("sale_date") == process_date) \

.join(F.broadcast(df_customers.select("customer_id", "customer_tier",
    on="customer_id", how="left") \

.join(F.broadcast(df_products.select("product_code", "category", "brand",
    on="product_code", how="left") \

.join(F.broadcast(df_stores.select("store_id", "store_name", "region_code",
    on="store_id", how="left") \
    .groupBy(
        "sale_date", "region_code", "store_name",
        "customer_tier", "category"
    ) \
    .agg(
        F.sum("total_amount").alias("gross_revenue"),
        F.sum("net_amount").alias("net_revenue"),
        F.sum("gst_amount").alias("gst_collected"),
        F.count("sale_id").alias("transactions"),
        F.countDistinct("customer_id").alias("unique_customers"),
        F.avg("total_amount").alias("avg_ticket")
    ) \
    .withColumn("load_ts", F.current_timestamp())

# Write Gold (V-ORDER for Power BI performance!)
spark.conf.set("spark.sql.parquet.vorder.enabled", "true")
spark.conf.set("spark.microsoft.delta.optimizeWrite.enabled",
"true")

df_gold_exec.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere", f"sale_date = '{process_date}'") \
    .saveAsTable("gold_executive_summary")

```

```
# Optimize for BI queries
spark.sql("""
    OPTIMIZE gold_executive_summary
    ZORDER BY (region_code, sale_date, customer_tier)
""")

print(f"✅ Gold layer ready for {process_date}")
print(f"    Power BI reports will show {process_date} data!")

mssparkutils.notebook.exit(f"SUCCESS:{process_date}")
```

◆ Scheduling Jobs and Monitoring

SCHEDULING IN FABRIC:

METHOD 1: Pipeline Schedule (most common)

- Pipeline → Schedule → Daily at 1:00 AM IST
- Runs master pipeline PL_RetailMart_Daily_ETL
- All notebooks called from pipeline
- Centralized monitoring in one place

METHOD 2: Notebook Schedule (simple)

- Notebook → Run → Schedule
- For standalone notebooks
- Simple use cases

METHOD 3: Spark Job Definition Schedule

- Create Spark Job Definition (for .py files)
- Schedule like a pipeline
- Better for production Python scripts
- Can set specific cluster config

MONITORING IN FABRIC:

PIPELINE MONITORING:

Workspace → Pipeline → Run History

Run History					
Date	Time	Status	Duration	Triggered by	
2024-06-16	01:00	✓ Success	4h 23m	Schedule	
2024-06-15	01:00	✓ Success	3h 51m	Schedule	
2024-06-14	01:00	✗ Failed	0h 12m	Schedule	
2024-06-14	06:30	✓ Success	3h 45m	Manual (rerun)	

Click any run → see each activity:

- ✓ LKP_GetTables (2 sec) → 25 tables found
- ✓ FE_CopyTables (45 min) → 25 tables copied
- ✓ NB_TransformSilver (2h 15m) → 142,850 rows
- ✓ NB_BuildGold (1h 10m) → 12 gold tables
- ✓ SendEmail (1 sec) → notification sent

FABRIC CAPACITY METRICS APP:

Admin Portal → Capacity Metrics

- CU usage over time
- Which workspace uses most CUs
- Peak vs off-peak usage
- Throttling events
- Use this to right-size your capacity!

ALERTING:

Set up in Azure Monitor or Fabric Activator:

- Pipeline run failed → Teams message to DE team

→ Pipeline duration > 6 hours → alert (something's slow!)
→ CU usage > 80% capacity → scale up warning

◆ Implementing Reports Using Fabric

COMPLETE REPORTING SETUP:

STEP 1: Fabric Gold table ready (from pipeline)

gold_executive_summary, gold_store_performance,
gold_customer_360, gold_product_ranking

STEP 2: Create Semantic Model

Lakehouse → gold layer tables → New Semantic Model

Name: SM_RetailMart_Analytics

Include tables:

- ✓ gold_executive_summary
- ✓ gold_store_performance
- ✓ gold_customer_360
- ✓ dim_date (calendar dimension)

Define relationships:

gold_executive_summary[sale_date] → dim_date[full_date]

Define measures (DAX):

Total Revenue = SUM(gold_executive_summary[gross_revenue])

Revenue YoY = CALCULATE([Total Revenue],
SAMEPERIODLASTYEAR(dim_date[full_date]))
YoY Growth % = DIVIDE([Total Revenue] - [Revenue YoY], [Revenue
YoY]) * 100

STEP 3: Create Reports (in browser!)

Semantic Model → New Report

Build visuals (Direct Lake mode – **no** import!):

- **Card**: "₹12.5 Cr Total Revenue" (Total Revenue measure)
- **Line Chart**: Revenue trend by date
- **Map**: Revenue by state (bubble map)
- **Bar Chart**: Top 10 stores
- **Table**: Store performance with conditional formatting

STEP 4: Publish and Share

- Reports saved in Fabric workspace automatically
- Share with BI team (Viewer role)
- **Teams integration**: embed report in Teams channel
- **Mobile**: accessible via Power BI mobile app

STEP 5: Set Refresh (if using Import mode)

Semantic Model → Settings → Scheduled refresh

- Every day at **6:00 AM**
- Report ready when business arrives at **9 AM**!

(With **Direct Lake**: **NO** refresh needed!

Data updated in Lakehouse → report shows it immediately!)

RESULT:

Pipeline runs **1 AM** → data in Gold by **6 AM**

Report opens at **9 AM** → shows live data from **6 AM** run

No manual refresh, **no** Excel download, **no** email attachments!

Business gets data in their browser, always fresh!

18 Resume Preparation

◆ Building a Strong Resume for Data Engineering Role

MICROSOFT FABRIC DATA ENGINEER RESUME:

TECHNICAL SKILLS SECTION (what to highlight):

Cloud Platforms:	Microsoft Azure, Microsoft Fabric
Data Engineering:	Azure Data Factory, Fabric Pipelines, Dataflow Gen2
Big Data:	Apache Spark, PySpark, Databricks, Synapse Analytics Spark Pool
Data Storage:	Azure Data Lake Gen2, OneLake, Delta Lake, Azure Blob Storage
Data Warehouse:	Fabric Data Warehouse, Azure Synapse Analytics, SQL Server
Streaming:	Azure EventStream, Fabric Real-time Intelligence, KQL
BI & Reporting:	Power BI (Direct Lake, DirectQuery, Import), DAX, Power Query
Programming:	Python, PySpark, SQL (T-SQL, Spark SQL), KQL (Kusto Query Language)
DevOps:	Azure DevOps, Git, CI/CD Pipelines, Fabric Deployment Pipelines
Governance:	Microsoft Purview, Sensitivity Labels, Column Masking, Row Security

STRONG RESUME BULLET POINTS:

✗ WEAK: "Worked with Microsoft Fabric"

✓ STRONG:

- Designed and implemented end-to-end Medallion Architecture (Bronze/Silver/Gold) in Microsoft Fabric Lakehouse, processing

500GB daily data from 6 source systems using PySpark and Delta Lake

→ Reduced report delivery time from 11 AM to 6 AM

- Built metadata-driven Fabric Pipeline that dynamically loaded 25+ Oracle database tables using control table pattern, replacing 25 separate pipelines with one reusable solution
→ Saved 40% pipeline development time
- Implemented Direct Lake Power BI semantic model on Fabric Gold layer, eliminating daily data refresh and enabling real-time report access for 200+ business users
→ Reduced BI infrastructure cost by 60%
- Developed real-time fraud detection using Fabric EventStream and KQL queries, detecting high-risk transactions within 60 seconds and triggering automatic alerts via Data Activator
→ Prevented estimated ₹50L monthly fraud losses
- Established CI/CD pipeline for Fabric notebooks using Azure DevOps and Fabric Deployment Pipelines across Dev/Test/Production environments with automated unit tests
→ Reduced deployment errors by 90%, deployment time from days to 30 minutes

PROJECTS TO MENTION:

RetailMart On-Premises to Fabric Migration:

- Scale: 500 stores, 50,000 daily transactions, 5 years history
- Tech: Fabric Pipelines + Lakehouse + Warehouse + Power BI
- Outcome: Reports delivered 5 hours earlier, 100% automated

19 Interview Guidelines

◆ Microsoft Fabric Interview Questions & Answers

TOP FABRIC INTERVIEW QUESTIONS:

Q1: What **is** Microsoft Fabric? How **is** it different **from** Azure Synapse?

A: Microsoft Fabric **is** a unified SaaS analytics platform launched **in 2023**

that integrates data engineering, data science, real-time analytics,

data warehousing, **and** business intelligence **in** one platform **on** top **of**

OneLake – a **single** storage layer.

Key differences **from** Synapse:

→ Fabric = SaaS (managed entirely **by** Microsoft, no infrastructure)

→ Synapse = PaaS (you manage some infrastructure **like** Dedicated Pools)

→ OneLake: Fabric has truly unified storage; Synapse has linked storage

→ Power BI: natively built **into** Fabric; Synapse **is** separate

→ Pricing: Fabric uses capacity-based (predictable); Synapse uses multiple meters

→ Direct Lake: Fabric's *unique Power BI mode (doesn't exist in Synapse)*

Q2: Explain OneLake **and** why it's *important*.

A: OneLake **is** Fabric's *unified storage layer – ONE logical data lake*

per Microsoft tenant (built **on** ADLS Gen2 underneath).

Importance:

- Eliminates data silos (everyone works on same data)
- Shortcuts: virtual links to external data without copying
- Open format (Delta Parquet) – no vendor lock-in
- All Fabric experiences (Lakehouse, Warehouse, Spark, Power BI)

read from SAME underlying files

- Zero data movement between experiences!

Impact: Reduced storage costs, eliminated data inconsistency, simplified architecture from 4 tools to 1 platform.

Q3: What is Direct Lake mode in Power BI?

A: Direct Lake is a new Power BI connection mode where Power BI reads

Delta Parquet files DIRECTLY from OneLake (Lakehouse/Warehouse storage).

vs Import: Import copies data to Power BI memory (fast but stale)

vs DirectQuery: Queries source on every click (fresh but slow)

Direct Lake: reads Parquet files natively

- Speed of Import (columnar Parquet + V-ORDER optimization)
- Freshness of DirectQuery (reads latest files)
- No size limits (can handle billions of rows)
- Requires Fabric capacity (not available in Power BI Pro)

Enable by: Creating semantic model on Lakehouse tables, Power BI automatically uses Direct Lake mode!

Q4: What is V-ORDER in Fabric?

A: V-ORDER is Microsoft's proprietary write-time optimization for

Parquet files **in** Fabric. It sorts data within Parquet row groups

optimally **and** applies additional compression beyond standard Snappy.

Enable: `spark.conf.set("spark.sql.parquet.vorder.enabled", "true")`

Impact: **2-5x** faster Power BI Direct Lake report loading

Use: Enable **for** ALL Gold layer tables (Power BI reads these)

Not needed **for**: Bronze layer (raw ingestion)

Q5: Explain Medallion Architecture **in** Fabric context.

A: Three-layer data architecture pattern:

BRONZE: Raw landing zone

- Exact copy **of** source data + audit metadata columns
- Append-only, never modified
- Stored **as** Delta tables **in** Lakehouse Tables/
- Purpose: audit trail, reprocessing capability

SILVER: Cleansed **and** conformed

- Data quality rules applied
- Standardized column names/types
- Basic business logic
- Joined **with** reference data
- Delta MERGE **for** upsert (**handles** SCD Type **1**)

GOLD: Business-ready aggregations

- Purpose-built **for** specific use cases (BI, ML, etc.)
- Denormalized **for** query performance
- V-**ORDER** enabled **for** Power BI Direct Lake
- ZORDER applied **for** analytical queries

In Fabric: All 3 layers in ONE Lakehouse (Tables/)
Power BI connects to Gold via Direct Lake!

Q6: What is a Shortcut in Microsoft Fabric?

A: Shortcut is a symbolic link feature in OneLake that creates a virtual reference to data stored elsewhere WITHOUT copying it.

Source types:

- Another OneLake location (different workspace/lakehouse)
- Azure Data Lake Storage Gen2
- Azure Blob Storage
- Amazon S3 or Google Cloud Storage

Use cases:

- Data Science team shortcuts to Data Engineering's *gold data*
- Warehouse shortcuts to Lakehouse tables (no data movement!)
- Bring external ADLS data into Fabric without migration

Benefit: Zero storage cost for shortcuts,
always fresh (reads original source)!

Q7: What is Dataflow Gen2 and how does it differ from Gen1?

A: Dataflow Gen2 is Power Query (M language) based data ingestion and transformation tool in Fabric.

vs Gen1:

- Gen2 can output directly to Fabric Lakehouse/Warehouse
- Gen2 uses Fast Copy backend (much faster for large data)
- Gen2 better monitoring and error handling
- Gen2 supports more sources (300+ connectors)
- Gen2 can split: some transforms in Power Query, some in

Spark

When to use:

- ✓ Non-coders who know Excel Power Query
- ✓ Simple ingestion + transformation
- ✓ Many connectors (Salesforce, SAP, etc.)
- ✗ Complex Spark transformations → use notebooks

Q8: How do you implement CI/CD for Fabric?

A: Three-component approach:

1. Git Integration (source control):

- Workspace → Settings → Git Integration → Azure DevOps
- All notebooks, pipelines, dataflows stored in Git as files
- Feature branch → PR → code review → merge

2. Azure DevOps CI Pipeline:

- On PR: lint Python, run unit tests, validate JSONs
- Ensures code quality before merge

3. Fabric Deployment Pipeline (built-in CD):

- Dev workspace → Test workspace → Prod workspace
- One-click deployment (or automated via REST API)
- Definitions deploy, not data
- Environment-specific config via parameter substitution

Full flow: commit → PR → review → merge → CI → deploy to Test → approve → deploy to Prod → notify team

Q9: What is Fabric Activator (Reflex)?

A: Activator is Fabric's no-code alerting and automation engine.

Monitors:

- EventStream (real-time data)
- Power BI reports (visual thresholds)
- Lakehouse data (scheduled polling)

Triggers **when**:

- Value crosses a threshold
- Value changes **by** more than X%
- Expected data doesn't *arrive*

Actions:

- Send email/Teams notification
- Start a Fabric pipeline
- **Call** a webhook

Replaces: Azure Logic Apps + Power Automate + Azure Functions
for data-driven automation scenarios.

Q10: How does Fabric handle real-time data?

A: Fabric Real-Time Intelligence stack:

1. EventStream: capture events **from Event** Hubs, IoT Hub, Kafka, **custom** apps. Route **to** destinations. Add transforms.
2. Eventhouse (KQL Database): time-series optimized database. Designed **for** millisecond query response **on** billions **of** events.
3. Kusto Query Language (KQL): SQL-**like** but time-series native.
100x faster than SQL **for event** analytics.
render keyword creates instant charts!
4. Real-time Dashboard: Live dashboards refreshing every few seconds.

Built **on** KQL queries.

5. Activator: triggers actions **when** real-time conditions met.

RetailMart example:

Payment API → EventHub → EventStream → Eventhouse

KQL query: detect fraud pattern **in** real-time (< **1** second latency)

Activator: fraud detected → Teams alert + block payment pipeline

20 Mock Interview Sessions

◆ Scenario-Based Technical Interview

MOCK INTERVIEW FORMAT:

Interviewer: "You're a senior data engineer at a retail company. The business wants near-real-time inventory alerts when stock falls below safety levels. Currently, inventory data is in on-premise SAP. Design the solution using Microsoft Fabric."

STRONG ANSWER STRUCTURE:

Step **1**: Clarify Requirements

"Before I design, let me ask:

- How near-real-time? (seconds, minutes, hours)
- What action when stock falls? (email, PO, block sale?)
- How often does SAP inventory update?

→ What's acceptable latency for alerts?"

(Asking requirements = senior behavior!)

Assume: 30-minute latency OK, email + Teams alert,
SAP updates every 30 minutes, 5-minute alert tolerance.

Step 2: Architecture Design

"Given these requirements, here's my design:

SAP (on-premise, 30-min batch updates)

|

On-premises Data Gateway (Fabric gateway on corp server)

|

Fabric Pipeline (every 30 minutes triggered)

→ Copy Activity: SAP → Lakehouse Files (CSV export via SAP connector)

|

Fabric Notebook (triggered by pipeline)

→ Read CSV from Files → transform → write to Delta table

→ Table: silver_inventory_levels

|

Fabric Activator (monitors silver table)

→ Object: each product_code

→ Property: current_stock

→ Condition: current_stock < safety_stock_level

→ Action 1: Email supply-chain@retailmart.com

→ Action 2: Send Teams to #inventory-alerts channel

→ Action 3: Trigger pipeline to create SAP purchase order

|

Power BI Direct Lake (dashboard for operations team)

→ Shows current inventory levels, flagged items

→ Refreshes when Spark writes new data (no scheduled refresh!)"

Step 3: Optimization and Alternatives

"If 30 minutes is too slow, I'd:

- Use SAP Change Document interface for CDC
- Stream via Kafka → Fabric EventStream
- KQL Database stores events
- Activator monitors EventStream in real-time
- Alert latency: < 60 seconds!"

Step 4: Risk Mitigation

"Key risks:

- On-premises Data Gateway: single point of failure
 - Install second gateway node for HA
- SAP extract performance: large table might be slow
 - Use delta extraction (only changed records)
- False alerts: safety stock might fluctuate
 - Add cooling period: 'only alert once per 4 hours per product'"

This answer shows:

- ✅ Requirements clarification (senior!)
- ✅ Architecture design (technical depth)
- ✅ Real-time alternatives (advanced knowledge)
- ✅ Risk awareness (experience)
- ✅ Practical constraints (not just theory)

BEHAVIORAL QUESTION:

"Tell me about a time a pipeline failed in production"

STRONG ANSWER (STAR format):

Situation: "At RetailMart, our nightly Oracle load failed one Monday at 2 AM, 3 hours before reports needed"

Task: "I was on-call and had to fix it before 9 AM business reports"

Action: "I checked Azure Monitor alerts – saw error:

'Oracle connection timeout'.
Checked SHIR status → Node 1 offline (server rebooted).
Connected to corporate VPN → restarted SHIR on Node 1.
Verified SHIR healthy → re-triggered failed pipeline run.
While pipeline reran, I checked: why did server reboot?
→ Windows Update ran at 2 AM (IT policy).
I added a second SHIR node for HA – future updates won't cause outage."

Result: "Pipeline completed by 5 AM.
Reports ready by 9 AM – business never knew!
Added monitoring alert for SHIR node count < 2.
No repeat outage since then."

KEY: Shows problem-solving, root cause analysis,
proactive improvement, calm under pressure!

Microsoft Fabric — Complete Summary

MICROSOFT FABRIC — COMPLETE GUIDE

FOUNDATION	OneLake (unified storage, Delta Parquet)
	Workspaces, Capacity (F SKUs), Shortcuts

DATA FACTORY	Pipelines (same as ADF!)
	Dataflow Gen2 (Power Query + Lakehouse)
	On-premises Data Gateway (replaces SHIR)
DATA ENGINEERING	Lakehouse (Files/ + Tables/ structure)
	Notebooks (PySpark, SQL, Scala, .NET Spark)
	Apache Spark (with Starter Pools!)
	MSSparkUtils/notebookutils
	All file formats (CSV, JSON, Parquet, Avro, ORC)
DELTA LAKE	ACID, MERGE/UPDATE/DELETE, Time Travel
	SCD Type 1 & 2, Streaming, Deduplication
	Medallion Architecture (Bronze/Silver/Gold)
	V-ORDER (Power BI optimization!)
DATA WAREHOUSE	T-SQL, Star Schema, Cross-DB queries!
	Row/Column security, Shortcuts from LH
	Semantic models, Power BI Direct Lake

	REAL-TIME INTEL		EventStream, Eventhouse, KQL Database
			Kusto Query Language, Real-time Dashboards

	ACTIVATOR		If-This-Then-That for data
			EventStream + Power BI + Lakehouse sources
			Email, Teams, Pipeline, Webhook actions

	POWER BI		Direct Lake (NEW! Best of Import+DQ)
			All connection modes
	(Desktop/Cloud/OneLake)		
			Sensitivity Labels, Endorsement

	DEVOPS & SDLC		Git Integration (Azure DevOps/GitHub)
			Deployment Pipelines (Dev→Test→Prod)
			Agile sprints, CI/CD, Unit testing

	E2E PROJECT		Oracle/SAP → Fabric Pipeline → Lakehouse
			Bronze→Silver→Gold Medallion
			Gold → Warehouse → Power BI Direct Lake
			Real-time: EventStream → KQL → Activator

```
|| INTERVIEW PREP | 15 Q&A, Scenario-based, STAR method
||
|| | Resume bullet points, Behavioral questions
||
```

THE BIG PICTURE:

Microsoft Fabric **is WHERE** the market **is** heading **in 2024-2026**.
If you know: ADF + Databricks + Synapse + Power BI already
Learning Fabric = applying same concepts **in ONE** unified platform.

Your competitive advantage:

- Know WHY **each** tool **exists** (not just HOW **to** use it)
- Understand OneLake **as** the foundation **of** everything
- Direct Lake = best Power BI performance (sell this **to** clients!)
- V-**ORDER** = **3x** faster BI (simple **to** implement, huge impact)
- **Real-time** Intelligence = modern streaming solution

Microsoft Fabric Course — 100% Complete!

Every concept from the PDF covered in depth: Big Data intro, Fabric platform overview, all components (OneLake, Data Factory, Data Engineering, Data Science, Real-time Intelligence, Warehouse, Power BI, Activator), Lakehouse architecture, all notebook types, Delta Lake operations (SCD1/SCD2, Time Travel, Medallion), V-ORDER, Fabric Warehouse with cross-database queries, KQL/EventStream/Eventhouse, Activator, all Power BI connection modes, DevOps/Git integration, Deployment Pipelines, SDLC, Agile methodology, E2E migration project, Resume preparation, Interview Q&A, and Mock Interview scenarios.

 **The complete Azure Data Engineering Course guide is now finished!** ADF → Databricks → Synapse → Microsoft Fabric — all covered with RetailMart real-world scenarios, senior engineer patterns, and practical code examples.

Thought process: Verified all PDF topics covered, prepared comprehensive interview framework.

The user wants me to act as a Senior Azure Data Engineer Interviewer and cover all interview topics comprehensively. Let me check if all PDF topics are covered first, then proceed with the interview preparation.

All PDF topics have been covered:

1.  Overview of Cloud
2.  Azure Data Factory
3.  Azure Databricks
4.  Azure Synapse Analytics
5.  Microsoft Fabric

Now let me proceed with the comprehensive interview preparation as requested.

Senior Azure Data Engineer Interview Masterclass

2000+ Interviews Experience / Microsoft / Barclays / EY / 15+ LPA Companies

BEFORE WE START — YOUR INTERVIEW ROADMAP:

LEVEL 1: Fundamentals	(Q1-Q15)	→ Gets you to 6-8 LPA
LEVEL 2: Intermediate	(Q16-Q35)	→ Gets you to 10-12 LPA
LEVEL 3: Advanced	(Q36-Q55)	→ Gets you to 15-18 LPA

LEVEL 4: System Design (Q56-Q70) → Gets you to 20+ LPA

Total: 70+ questions covering EVERYTHING.

Let's begin. No shortcuts. Full preparation.

LEVEL 1 — FUNDAMENTALS

"Build the foundation right or the house falls"

? QUESTION 1 — Basic | Azure Data Factory

"What is Azure Data Factory? Why do we use it instead of writing custom Python scripts for data movement?"

✓ IDEAL ANSWER

Azure Data Factory (ADF) is Microsoft Azure's cloud-based ETL/ELT and data integration service. It allows you to:

- Create data pipelines (workflows)
- Move data from 90+ sources to 90+ destinations
- Schedule and monitor these pipelines
- Transform data during movement
- Handle failures with retry logic

WHY ADF OVER PYTHON SCRIPTS?

PYTHON SCRIPTS:

- ❌ You manage the server (VM, cron jobs)
- ❌ No built-in monitoring or alerting
- ❌ No visual representation of pipeline
- ❌ Error handling = more code to write
- ❌ Scaling = your problem
- ❌ 90+ connectors = you build each one
- ❌ CI/CD = extra setup work

ADF:

- ✅ Fully managed (no server management)
- ✅ Built-in monitoring dashboard
- ✅ Visual drag-and-drop pipeline builder
- ✅ Retry logic built-in (3 retries, 30s interval)
- ✅ Auto-scaling
- ✅ 90+ connectors out-of-the-box
- ✅ Native Git + Azure DevOps integration
- ✅ Pay per pipeline run (no idle cost!)

SIMPLE EXPLANATION

Think of ADF as a SMART POSTAL SERVICE for your data.

Old way (Python scripts):

You had to:

- Buy a truck (server)
- Hire a driver (maintain scripts)
- Plan the route (write connection code)
- Handle breakdowns (write error handling)
- Track deliveries (write logging)

ADF way:

You just say:

- "Pick up from Oracle (source)"
- "Deliver to ADLS (destination)"
- "Do it every night at midnight"
- ADF handles EVERYTHING else!

You focus on WHAT to move.

ADF handles HOW to move it.



REAL-WORLD USE CASE

BARCLAYS BANK (real scenario type):

Before ADF:

10 Python scripts running on 3 VMs

Each script manually maintained

One script fails → nobody knows until reports are wrong

Fix takes 4 hours (find issue, SSH to VM, debug)

After ADF:

One ADF pipeline with 10 activities

Any failure → email + Teams alert within 2 minutes

Monitor all 10 activities from ONE dashboard

On-call engineer fixes in 20 minutes (not 4 hours)

Business impact: Reports accuracy improved from 94% to 99.8%



COMMON MISTAKES CANDIDATES MAKE



MISTAKE 1: "ADF can transform data like Databricks"

✗ WRONG: ADF **is** primarily **for** data MOVEMENT

✓ CORRECT: ADF orchestrates; Databricks transforms

(ADF has Data Flows but they're *not for complex ML/Python logic*)

MISTAKE 2: "ADF stores data"

✗ WRONG: ADF doesn't *store ANY data*

✓ CORRECT: ADF just moves data between source **and** sink

Analogy: ADF **is** the pipeline, **not** the reservoir!

MISTAKE 3: "ADF is only for Azure sources"

✗ WRONG: ADF connects **to on**-premise, AWS S3, Salesforce, etc.

✓ CORRECT: **90+** connectors including cross-cloud!

💡 PRO TIP — 15+ LPA LEVEL

In a senior interview, **add** this:

"I use ADF as the ORCHESTRATOR in a hub-and-spoke model:

ADF (Hub) → coordinates everything

- Triggers Databricks notebooks (heavy transformation)
- Triggers Synapse pipelines (data warehouse loads)
- Sends Logic App notifications (alerts)
- Calls Azure Functions (custom logic)

ADF doesn't try to do everything itself.

It delegates to the RIGHT tool for each job.

This separation of concerns = maintainable, scalable architecture."

This shows **SYSTEM** DESIGN THINKING – what senior roles need!

? QUESTION 2 — Basic | ADF Components

"Explain the difference between a Pipeline, Activity, Linked Service, and Dataset in ADF. How do they relate to each other?"

✓ IDEAL ANSWER

These are the **4** core building blocks **of** ADF.

Here's *how they relate*:

LINKED SERVICE = Connection definition

- "HOW to connect to a data store"
- Contains: server name, credentials, auth method
- Example: Connection **to** Oracle DB at **10.0.1.50**
- Reusable across many datasets

DATASET = Data definition

- "WHAT specific data to access"
- References a Linked Service
- Contains: specific table name, file path, format
- Example: [dbo].[Sales] table **in** that Oracle DB

ACTIVITY = One **step**/task **in** a pipeline

- "What operation to perform"
- Uses Datasets as input/output
- Example: Copy Data activity (reads source Dataset, writes to sink Dataset)

PIPELINE = Container of activities

- "The complete workflow"
- Groups related activities
- Defines execution order (sequential or parallel)
- Example: Daily Sales ETL pipeline

RELATIONSHIP:

Pipeline contains Activities

Activities use Datasets

Datasets reference Linked Services

ANALOGY:

Linked Service = Library membership card (access)

Dataset = Specific book in the library (what)

Activity = Reading/annotating that book (action)

Pipeline = Your complete study plan (workflow)

SIMPLE EXPLANATION

Let me explain with a food delivery analogy:

LINKED SERVICE = Zomato account (your saved payment + address)

DATASET = Specific restaurant menu item you want

ACTIVITY = The delivery action (pick up + deliver)

PIPELINE = Your complete meal plan for the day

Just like Zomato:

- Your account (LS) stores HOW to pay
- You select specific items (Dataset)
- Delivery happens (Activity)
- Multiple orders in a day = Pipeline!

In real code:

Linked Service (LS_OracleDB):

server: 10.0.1.50, port: 1521

auth: username/password (from Key Vault!)

Dataset (DS_SalesTable):

references: LS_OracleDB

table: SALES.TRANSACTIONS

Activity (CopyData):

source: DS_SalesTable

sink: DS_ADLS_Bronze

Pipeline (PL_Daily_Sales):

Activity 1: CopyData

Activity 2: Databricks Transform

Activity 3: Send Notification



REAL-WORLD USE CASE

EY **CONSULTING** (real scenario type):

Client has 12 Oracle **databases** (one per region).

JUNIOR ENGINEER **approach** (wrong!):

- Creates 12 separate Linked Services

- Creates **12** × **5** = **60** Datasets
- Creates **12** Pipelines
- **84** objects to maintain!

SENIOR ENGINEER **approach** (right!):

- Creates **1** PARAMETERIZED Linked **Service**
(server = @{linkedService().p_server})
- Creates **1** PARAMETERIZED **Dataset**
(table = @{dataset().p_tableName})
- Creates **1** Pipeline with ForEach loop
- **3** objects to maintain!

When 13th region is added:

Junior: create **5** new objects

Senior: add **1** row to control table → done!

THIS is what **15+** LPA looks like.

COMMON MISTAKES

MISTAKE **1**: Hardcoding credentials **in** Linked Services

-  Password: "OracleAdmin@123" directly **in** LS
-  Password: **@Microsoft**.KeyVault(SecretUri=...)

Interviewer WILL ask about Key Vault integration.
Always mention it!

MISTAKE **2**: Creating too many Linked Services

-  **12** LS **for** **12** similar databases
-  **1** parameterized LS **for** all **12**

"Parameterization" = senior level thinking

MISTAKE 3: Confusing Dataset with Linked Service

✗ "Dataset contains connection string"

✓ "Dataset points to Linked Service for connection"

💡 PRO TIP — 15+ LPA LEVEL

Add this in senior interviews:

"In production, I follow a naming convention:

LS_<DataStore>_<Environment>	→ LS_Oracle_Prod
DS_<DataStore>_<Table>_<Env>	→ DS_Oracle_Sales_Prod
PL_<Purpose>_<Frequency>	→ PL_Sales_Daily

This matters at enterprise scale (200+ objects in ADF).
Without naming conventions → maintenance nightmare.

Also: I never store credentials in Linked Services directly.
Always Azure Key Vault. This is a SECURITY BEST PRACTICE
and many companies require it for compliance (SOC2, PCI-DSS)."

Security awareness = senior behavior!

? QUESTION 3 — Basic | Triggers

"What are the different types of triggers in ADF? When would you use each type?"

✓ IDEAL ANSWER

ADF has 3 main trigger types:

1. SCHEDULE TRIGGER

What: Runs pipeline at fixed time intervals

When: Regular scheduled jobs (daily, hourly, weekly)

Example: Run nightly data load at 12:00 AM every day

Key properties:

→ Start time, end time, recurrence

→ Can pass trigger time to pipeline:

```
@{formatDateTime(trigger().scheduledTime, 'yyyy-MM-dd')}
```

→ One trigger can attach to MULTIPLE pipelines

2. TUMBLING WINDOW TRIGGER

What: Fixed-size, non-overlapping time windows

When: Time-series data, incremental loads, backfilling

Key difference from Schedule:

→ TRACKS which windows succeeded/failed

→ Supports BACKFILL (rerun Jan-Jun all at once!)

→ Supports trigger DEPENDENCY (B waits for A)

→ If window fails → retries that specific window

Example: Process each hour's transactions exactly once

Access window times:

```
@{triggerBody().window.windowStart}
```

```
@{triggerBody().window.windowEnd}
```

3. STORAGE EVENT TRIGGER

What: Fires when file arrives/deleted in ADLS/Blob

When: Event-driven pipelines, real-time-ish processing

Example: New file in `/landing/payments/` → start pipeline

Key **properties:**

- Container/path to watch
- Event **type:** `BlobCreated` or `BlobDeleted`
- File **pattern:** ends with `.csv`, starts with `sales_`

Access file **info:**

```
@{triggerBody().fileName}  
@{triggerBody().folderPath}
```

CHOOSING THE RIGHT TRIGGER:

Scenario	→ Trigger Type
Run every day at midnight	→ Schedule
Process hourly data windows	→ Tumbling Window
File arrives → process immediately	→ Storage Event
Backfill 6 months of data	→ Tumbling Window
Weekly report generation	→ Schedule
CDC incremental with retry	→ Tumbling Window

SIMPLE EXPLANATION

THREE ALARM CLOCKS:

Schedule Trigger = Regular alarm clock

"Wake me at 7 AM every day"

Simple, predictable, doesn't care what happened yesterday

Tumbling Window = Smart alarm with memory

"Wake me every hour AND remember if I missed an hour

If I missed 3 AM, wake me for that later!"

Has memory of past windows

Storage Event Trigger = Motion sensor alarm

"Don't wake me until someone enters the room (file arrives)"

Reactive, not proactive

Only runs when something actually happens!

REAL EXAMPLE:

Schedule: Morning standup report (always at 9 AM)

Tumbling: Hourly sales reconciliation (must process each hour)

Event: Process customer file when uploaded by vendor



REAL-WORLD USE CASE

BARCLAYS BANK (Real scenario):

Problem: Bank receives payment files from 3rd party processors

Files arrive at unpredictable times (2 AM, 4 AM, 7 AM)

WRONG approach (Schedule Trigger at 3 AM):

→ What if file arrives at 3:05 AM? (missed!)

→ What if file doesn't arrive? (pipeline runs, processes nothing)

→ Wasted compute cost every night

RIGHT approach (Storage Event Trigger):

→ File arrives at 2:47 AM → trigger fires immediately

→ Pipeline starts → file processed by 2:53 AM



→ No file? → No pipeline run! (cost saving)
→ File arrives at 7 AM? → Still works!

COMBINED APPROACH (senior thinking):

Event trigger for: "file arrived" → start processing

Tumbling window for: "guarantee every 6-hour window
was processed exactly once"

Both together = reliable + efficient!

⚠ COMMON MISTAKES

MISTAKE 1: Using Schedule for time-series incremental loads

❌ Schedule trigger for hourly sales data

✅ Tumbling Window – it tracks windows, handles retries

If 3 AM fails → retries 3 AM window, not 4 AM!

MISTAKE 2: Not knowing tumbling window backfill

Interviewer often asks: "How would you reprocess 6 months of data?"

❌ "I'd manually run the pipeline 180 times"

✅ "Tumbling window trigger – set start date to 6 months ago.

ADF creates all windows, runs up to 4 in parallel (max concurrency=4).

Backfills automatically!"

MISTAKE 3: Not knowing trigger parameters

❌ "I hardcode the date in the pipeline"

✅ "I pass @{trigger().scheduledTime} as pipeline parameter.

Each run automatically gets its own date!"

💡 PRO TIP — 15+ LPA LEVEL

Trigger **Dependency** (advanced concept that impresses!):

"I implement trigger dependency **when** pipelines have dependencies:

Pipeline A: Load raw sales **data** (Tumbling Window, hourly)

Pipeline B: Aggregate sales **data** (depends **on** A!)

Without dependency: B might start before A finishes
→ aggregating incomplete data!

With Tumbling Window dependency:

- B's trigger **WAITS for** A's trigger to succeed **for** same window
- 3PM-4PM window: A runs → succeeds → B starts
- If A fails: B **NEVER** runs **for** that window
- Guaranteed data integrity!

Only Tumbling Window supports **this**.

This **is** why I always use Tumbling Window **for** incremental loads, **not** Schedule trigger."

This level of detail = **15+** LPA answer!

? QUESTION 4 — Basic | Integration Runtime

"What is Integration Runtime in ADF? Explain the different types and when you'd use each."

✓ IDEAL ANSWER

Integration Runtime (IR) = The COMPUTE ENGINE of ADF.
It's *the infrastructure that actually EXECUTES activities*
and moves data between sources and destinations.

THREE TYPES:

1. AZURE INTEGRATION RUNTIME (Auto-resolve)

What: Microsoft-managed cloud compute

When: Cloud-to-cloud data movement

Example: ADLS → Azure SQL Database

Key facts:

- Default IR created automatically in every ADF
- Auto-resolves to optimal Azure region
- No configuration needed
- Scales automatically

2. SELF-HOSTED INTEGRATION RUNTIME (SHIR)

What: Agent installed on YOUR on-premise/VM server

When: On-premise source to cloud destination

Example: On-prem Oracle → ADLS Gen2

Key facts:

- Install on Windows server inside corporate network
- Registers with ADF using auth key
- Secure outbound HTTPS connection (port 443)
- For HA: install on 2 servers (nodes)

Requirements:

- Windows 2012+ server
- Min: 4 CPU cores, 8GB RAM
- Production: 8+ cores, 16+ GB RAM

3. AZURE-SSIS INTEGRATION RUNTIME

What: Managed cluster **for** running SSIS packages

When: Migrating legacy **on**-prem SSIS packages **to** cloud

Key facts:

- Creates an Azure VM cluster
- Run existing .dtsx SSIS packages without modification
- Bridge between legacy SSIS **and** cloud

4. AZURE MANAGED VNET IR (**sub**-type **of** Azure IR)

What: Azure IR inside a managed virtual network

When: Connect **to** Azure services via **PRIVATE** ENDPOINTS

Example: SQL DB **with** **public** access disabled

Key facts:

- Data never leaves Microsoft backbone network
- Required **for** **strict** network security compliance
- Banks, healthcare, government use this

SIMPLE EXPLANATION

ANALOGY: Think **of** IR **like** delivery vehicles:

Azure IR = Uber (cloud-managed, appears **when** needed)

You **order** a ride → Uber appears → You travel → Uber goes

You don't own the car. No maintenance!

PERFECT **FOR:** Cloud-**to**-cloud (Oracle Cloud → Azure)

SHIR = Your own company van

You own it, maintain it, park it inside your office

It can enter your **private** premises (**on**-prem network)

NEEDED FOR: On-premise to cloud

WHY? Uber can't enter your private building!

Your company van can because it's yours.

Azure-SSIS IR = Specialized truck

For moving specific cargo (SSIS packages)

Only needed if you have that specific cargo

SIMPLE RULE:

Source is ON-PREMISE? → ALWAYS need SHIR

Source is in CLOUD? → Azure IR (auto-resolve)

Need private network? → Managed VNet IR

Have SSIS packages? → Azure-SSIS IR



REAL-WORLD USE CASE

MICROSOFT CONSULTING PROJECT (real type):

Company has:

- Oracle DB (on-premise, corporate data center)
- SQL Server (on-premise, different building)
- Azure SQL Database (cloud)
- ADLS Gen2 (cloud)

IR SETUP:

SHIR-Node1 on CORPSEVER-01 (Building A)

SHIR-Node2 on CORPSEVER-02 (Building B) ← HA!

Oracle → SHIR → ADF → ADLS (on-prem to cloud)

SQL Server → SHIR → ADF → ADLS (on-prem to cloud)

ADLS → Azure IR → Azure SQL (cloud to cloud, no SHIR!)

SHIR HIGH AVAILABILITY:

- Both nodes registered to same IR
- ADF round-robins between nodes
- Node1 crashes at 2 AM during nightly load?
 - ADF automatically uses Node2
 - Pipeline completes successfully
 - Team notified about Node1 (not pipeline failure!)

WITHOUT HA SHIR:

- Node1 crashes → pipeline fails → reports missing → business team calls at 9 AM → chaos!

COMMON MISTAKES

MISTAKE 1: Not mentioning SHIR for on-premise sources

Interviewer: "How do you connect ADF to on-prem SQL Server?"

- ✗ "Just create a SQL Server linked service"
- ✓ "Create SQL Server Linked Service + configure Self-Hosted IR on a server inside the corporate network. The SHIR acts as a secure bridge through the firewall."

MISTAKE 2: Installing SHIR on the source database server

- ✗ Install SHIR on the Oracle DB server itself
 - ✓ Install on a separate server near Oracle DB
- WHY: SHIR consumes CPU/RAM during data extraction.
On the DB server → impacts database performance!

MISTAKE 3: Single SHIR node in production

- ✗ One SHIR node (single point of failure)
 - ✓ Two SHIR nodes minimum (High Availability)
- This is a PRODUCTION BEST PRACTICE interviewers test for!

💡 PRO TIP — 15+ LPA LEVEL

Mention SHIR PERFORMANCE TUNING (shows deep knowledge):

"When SHIR is the bottleneck in our pipeline, I tune it:

1. PARALLEL COPY:

In Copy Activity → Settings → Degree of Copy Parallelism
Increase to 8 → SHIR uses 8 parallel threads
→ 4x faster data extraction!

2. PARTITION OPTIONS:

For large Oracle tables (100M+ rows):
Source → Partition option: Dynamic Range
Partition column: ORDER_ID (*numeric*)
→ *SHIR reads in parallel chunks!*
→ *8 threads × 4 partitions = 32x speed!*

3. SHIR SIZING:

Each parallel copy thread uses ~1 CPU core, ~1GB RAM
For 8 parallel copies: need 8+ cores, 8+ GB RAM

4. COMPRESSED TRANSFER:

Enable compression in Copy Activity
→ *Data compressed at SHIR → transferred compressed*
→ *Less network bandwidth → faster!*

*These 4 optimizations alone can make pipeline 10x faster.
That's what companies pay 15+ LPA for!"*

? QUESTION 5 — Basic | Copy Activity Deep Dive

"Explain the Copy Activity in ADF. What are the key configurations you should know?"

✓ IDEAL ANSWER

Copy Activity = THE most important activity in ADF.

It moves data from Source to **Sink** (destination).

ARCHITECTURE:

Source → [Read by IR] → [Optional: Staging] → [Write by IR] → Sink

6 KEY TABS IN COPY ACTIVITY:

1. SOURCE TAB:

- Which dataset to read from
- Query / table / stored procedure
- Partition **options** (**for** parallel reading of large tables)
- Additional **columns** (add metadata during copy)

PARTITION **OPTIONS** (critical **for** performance!):

Physical partitions → use existing DB partitions

Dynamic range → split by numeric column

- SELECT * WHERE order_id BETWEEN 1 AND 250000 (thread 1)
- SELECT * WHERE order_id BETWEEN 250001 AND 500000 (thread 2)

2. SINK TAB:

- Which dataset to write to
- Write behavior: Insert / Upsert / Overwrite
- Pre-copy **script** (TRUNCATE TABLE before load)
- Batch **size** (100,000 rows per batch = faster!)

→ Table option: Auto-create table **if** not exists

3. MAPPING TAB:

- Map source columns to sink columns
- Handle column name **differences** (CUST_ID → customer_id)
- Data type **conversion** (string → date, string → **int**)
- Schema drift handling

4. SETTINGS TAB:

- Fault tolerance: skip incompatible **rows** (log them!)
- Data consistency **verification** (row count check)
- Enable **staging** (**for** PolyBase/COPY INTO to Synapse)
- Parallel copy degree

5. USER PROPERTIES TAB:

- Custom key-value pairs **for** monitoring
- pipeline_name: @{pipeline().Pipeline}
- Shows in Monitor dashboard!

6. GENERAL TAB:

- Retry **count** (**3**) and retry **interval** (**30** sec)
- Timeout (how **long** before killing the activity)
- Dependencies (OnSuccess/OnFailure/OnCompletion/OnSkip)

SIMPLE EXPLANATION

Copy Activity = Moving company **for your** data

SOURCE = "Pick up FROM here"

Oracle table, CSV file, REST API, Blob storage

SINK = "Deliver TO here"

ADLS Gen2, Azure SQL, Synapse, Blob storage

MAPPING = "Rename boxes during move"

Box labeled "CUST_NM" at source

→ Relabel as "customer_name" at destination

→ Data type changes too (string "2024-01-15" → proper Date)

ADDITIONAL COLUMNS = "Add a sticker to every box"

During copy, attach:

→ load_date = today's date

→ source_system = "Oracle_ERP"

→ pipeline_run_id = unique ID

Now every row knows WHERE and WHEN it was loaded!

FAULT TOLERANCE = "Don't throw away the truck if one box is bad"

If row 50,432 has a bad data type

Without FT: ENTIRE copy fails!

With FT: Skip that row, log it, continue copying

Result: 9,999,999 rows copied, 1 logged as bad



REAL-WORLD USE CASE

PRODUCTION PATTERN (EY Banking Client):

Scenario: Copy 50M rows from Oracle to Synapse nightly

NAIVE approach (doesn't work for 50M rows):

Copy Activity:

Source: SELECT * FROM TRANSACTIONS

Sink: Synapse table

→ Takes 6+ hours! Not acceptable.

OPTIMIZED approach (15+ LPA engineer):

Source:

Partition option: Dynamic Range

Partition column: TRANSACTION_ID

Upper/Lower bound: from Lookup activity

Degree of parallelism: 8

Sink:

Copy method: COPY INTO (Synapse bulk load!)

Pre-copy script: TRUNCATE TABLE stg_transactions

Batch size: 100,000

Result: 50M rows in 45 minutes (was 6 hours!)

THE MAGIC:

8 parallel threads read Oracle simultaneously

COPY INTO loads Synapse 10x faster than INSERT

= $8 \times 10 = 80x$ faster overall!

That optimization is worth 15+ LPA.

COMMON MISTAKES

MISTAKE 1: Not using partition options for large tables

 SELECT * FROM 100M-row table (1 thread, 8 hours)

 Dynamic range partition with 8 threads (45 minutes)

MISTAKE 2: Not using staging for Synapse loads

 Direct INSERT into Synapse (very slow row-by-row!)

 Enable staging → ADF writes to Blob first

Then COPY INTO runs → bulk loads at max speed

MISTAKE 3: Not enabling fault tolerance

✗ Pipeline fails on first bad row

✓ FT enabled → bad rows logged → pipeline completes

Then: check logs → fix data quality → reprocess bad rows

MISTAKE 4: Not adding audit columns (Additional Columns)

✗ Raw data without any metadata

✓ Add: load_timestamp, source_system, pipeline_run_id

EVERY row knows when and where it came from!

Critical for debugging data issues months later!

💡 PRO TIP — 15+ LPA LEVEL

COPY ACTIVITY PERFORMANCE FORMULA (memorize this!):

Copy Speed = DIUs × Network Speed × Compression factor

DIUs (Data Integration Units):

Default = 4 DIUs

Max = 256 DIUs

Increase when: network is fast but copy is slow

spark.conf... wait, wrong tool – in ADF Copy Activity:

Settings → Data Integration Units = 32

→ 8x more compute = 8x faster!

→ But also 8x more expensive!

OPTIMIZE COST:

Don't max DIUs always.

Find sweet spot: at what DIUs does speed plateau?

If 8 DIUs = 30 min and 16 DIUs = 29 min?

→ Use 8 DIUs (same speed, half cost!)

THROUGHPUT BENCHMARKS to quote in interviews:

Azure-to-Azure: ~4.5 GB/s max

On-prem to Azure: ~1-2 GB/s (limited by internet)

"I monitor copy activity throughput (MB/s) in ADF Monitor.
If throughput drops suddenly → network issue or source DB overloaded.

This helps me debug performance issues proactively."

METRICS = SENIOR ENGINEER BEHAVIOR!

REVISION SUMMARY — Questions 1-5

REVISION: ADF FUNDAMENTALS (Q1-Q5)

Q1: ADF = Orchestration + Movement (not transformation)

Key: Managed service vs custom scripts

Q2: 4 Components:

LS (connection) → Dataset (data) → Activity (task)

→ Pipeline (workflow)

KEY: Parameterize everything!

Q3: 3 Triggers:

Schedule → fixed time

Tumbling Window → time windows WITH MEMORY + backfill

Storage Event → file arrival

KEY: Tumbling Window for incremental loads!

Q4: 3 IR Types:

Azure IR → cloud to cloud

SHIR → on-premise to cloud (MUST for on-prem!)

Azure-SSIS → legacy SSIS packages

KEY: Always 2 SHIR nodes for HA in production!

Q5: Copy Activity = core data movement

KEY optimizations:

→ Partition options (parallel read)

- COPY INTO for Synapse (bulk load)
- DIU tuning (speed vs cost balance)
- Always add audit columns!

INTERVIEW PATTERN ALERT:

Every Q1-Q5 type question has ONE hidden test:

"Do you mention Key Vault for credentials?"

If YES → signals senior security awareness

If NO → junior flag

? QUESTION 6 — Basic | Error Handling in ADF

"How do you handle errors and failures in ADF pipelines? Walk me through your approach."

✓ IDEAL ANSWER

ERROR HANDLING in ADF happens at 3 levels:

LEVEL 1: ACTIVITY-LEVEL RETRY

Every activity has:

- Retry count: 3 (retry 3 times on failure)
- Retry interval: 30 seconds (wait between retries)

Handles: Transient failures (network blip, timeout)

Example: Oracle DB momentarily unavailable → retry → succeeds

LEVEL 2: DEPENDENCY CONDITIONS (Activity paths)

Connect activities with 4 dependency types:

- ✅ On Success → run next activity if previous succeeded
- ❌ On Failure → run different activity if previous failed
- ✅ On Completion → run always (success OR failure)
- ⏭ On Skip → run if previous was skipped

PATTERN: Try-Catch in ADF

CopyData [Success] → UpdateAuditLog

CopyData [Failure] → LogError → SendAlert → FailActivity

LEVEL 3: PIPELINE-LEVEL FAULT TOLERANCE

In Copy Activity → Settings:

- Skip incompatible rows (data type mismatch)
- Log skipped rows to ADLS
- Pipeline continues even with bad rows
- Review logs later, fix source data

COMPLETE ERROR HANDLING PATTERN:

1. Validate (GetMetadata → check file exists + size > 0)
 - ↓ If file missing → Fail Activity with clear message

↓ If file **exists** → proceed

2. Copy Data (retry=3, fault tolerance enabled)

↓ Success → **Update** watermark + log success

↓ Failure → Log error + Send Teams/email alert
→ Fail Activity (explicit failure)

3. Alert via Logic Apps

POST **to** Logic App webhook

Sends formatted email **with**:

→ Pipeline name

→ Failed activity

→ Error message

→ Run ID (**for** debugging)

→ **Timestamp**

4. Alert Team via Azure Monitor

Set up alert rules:

→ Failed pipeline runs > **0** → **trigger** alert

→ Email + SMS **to on-call** engineer

SIMPLE EXPLANATION

ANALOGY: **Error** handling **in** ADF = Emergency procedures **in** an airline

RETRY = "Try landing again if weather is bad"

Don't give up immediately. Try 3 times.

SUCCESS PATH = "Normal flight → land at airport A"

If everything works → **continue to next step**

FAILURE PATH = "Emergency → land at airport B + notify control"

If something fails → take emergency actions:

- Save the flight log (log error)
- Alert the team (send notification)
- Declare emergency (Fail Activity)

FAULT TOLERANCE = "Skip one bad passenger, board the rest"

One row of data is bad (wrong format)?

Don't cancel entire flight!

- Log that passenger (bad row)
- Board everyone else
- Deliver report later

WITHOUT error handling:

Pipeline fails at 2 AM
Nobody knows until 9 AM
Business reports wrong/missing
3-hour debugging session

WITH error handling:

Pipeline fails at 2 AM
Email sent at 2:01 AM
On-call engineer fixes at 2:30 AM
Reports ready by 6 AM
Business never knows!



REAL-WORLD USE CASE

BARCLAYS PRODUCTION PATTERN:

Pipeline: PL_Daily_PaymentProcessing

VALIDATION LAYER:

Activity 1: GetMetadata (check payment file exists)

Activity 2: If Condition

→ Condition: `@{equals(activity('GetMeta').output.exists, true)}`

→ FALSE branch:

Fail Activity: "Payment file not received by SLA time"

→ Immediately alerts operations team

PROCESSING LAYER:

Activity 3: Copy Data (fault tolerance = skip bad rows)

Skipped rows → /errors/payments/2024-06-15/

Activity 4 (Success): Update audit table, log row counts

Activity 4 (Failure):

→ SP: INSERT INTO error_log (pipeline, activity, error, time)

→ Web Activity: POST to Logic App (sends email)

→ Fail Activity: explicitly fail pipeline

NOTIFICATION (Logic App email format):

Subject: **✗** ADF FAILURE - PL_Daily_PaymentProcessing

Pipeline: PL_Daily_PaymentProcessing

Activity: CopyPaymentData

Error: "Connection timeout to Oracle DB"

Time: 2024-06-15 02:34:17 UTC

Run ID: abc123-def456

[Click to view in ADF Monitor]

[Click to retry pipeline]

RESULT:

On-call engineer gets email at 2:34 AM

Sees: Oracle connection timeout (not a data issue!)

Restarts Oracle connection → manually re-runs pipeline

Pipeline completes 2:58 AM

Reports ready 6 AM → no business impact!

⚠ COMMON MISTAKES

MISTAKE 1: No failure path = silent failure

❌ Pipeline fails → nothing happens → team finds out at 9 AM

✅ Always add failure path → immediate notification

MISTAKE 2: Not using Fail Activity explicitly

❌ Just Log Error on failure

✅ Log Error → Send Alert → THEN Fail Activity

WHY: If you don't explicitly fail, parent pipeline might show as "Succeeded" even though child failed!

MISTAKE 3: Sending generic error messages

❌ "Pipeline failed"

✅ Include: pipeline name, activity name, error message, run ID, timestamp, link to ADF Monitor

Without run ID → debugging takes 30 minutes

With run ID → debugging takes 2 minutes

MISTAKE 4: Not logging skipped rows from fault tolerance

❌ Rows skipped silently, nobody knows

✅ Log to /errors/logs/ → daily review → fix data quality

This is DATA QUALITY MANAGEMENT – senior concern!

💡 PRO TIP — 15+ LPA LEVEL

MASTER PIPELINE **PATTERN** (enterprise standard):

In enterprise, I use a Master-Child pattern **for** error handling:

Master Pipeline:

- Calls child pipelines
- Wraps each in Try-Catch pattern
- Central error logging
- Central notification

Child Pipeline:

- Does one specific job
- Returns status to master
- Can run independently **for** debugging

SPECIFIC TIP: Use Execute Pipeline with

"Wait on completion = YES" → master gets child result

Master then checks:

```
@{equals(activity('RunChildPipeline').output.pipelineRunId,
'something')}
```

Advanced: BUILD AN AUDIT TABLE

```
CREATE TABLE dbo.PipelineAuditLog (
  run_id      VARCHAR(100),
  pipeline    VARCHAR(200),
  status      VARCHAR(20),    -- RUNNING/SUCCESS/FAILED
  start_time  DATETIME2,
  end_time    DATETIME2,
  rows_read   BIGINT,
  rows_written BIGINT,
  error_msg   NVARCHAR(MAX)
);
```

Every pipeline writes to **this table** (start + end).
Operations team has dashboard: "Which pipelines failed today?"
This is PRODUCTION GRADE monitoring.

Companies paying **15+** LPA EXPECT **this!**

? QUESTION 7 — Basic | Parameters vs Variables

"What is the difference between Pipeline Parameters and Pipeline Variables in ADF? Give practical examples of when to use each."

✓ IDEAL ANSWER

PARAMETERS: **Input** values passed **BEFORE** pipeline runs

- **Like function arguments**
- **Set** at runtime (when trigger fires or manual run)
- **IMMUTABLE**: cannot change value during pipeline execution
- **Passed from**: triggers, parent pipelines, manual run

Define: Canvas → **Parameters** tab → + **New**

Access: @{**pipeline()**.parameters.p_date}

Use for:

- **Process** date (changes per run)
- **Environment** (dev/test/prod)
- **Table** names (dynamic table selection)
- **File** paths (dynamic file routing)

VARIABLES: Internal state within pipeline execution

- Like local variables in a function
- Can be SET and CHANGED during pipeline execution
- MUTABLE: can update multiple times using Set Variable
- Created and used ONLY within pipeline

Define: Canvas → Variables tab → + New

Set: Set Variable activity → assign value

Read: @{{variables('v_rowCount')}}

Append: Append Variable activity (for arrays)

Use for:

- Counter/accumulator (count processed files)
- Dynamic values (filename discovered at runtime)
- Status flags (v_isProcessed = true/false)
- Building dynamic lists (array of tables processed)

KEY DIFFERENCE TABLE:

Feature	Parameters	Variables
Set by	External	Internal (Set Variable)
When set	Before run	During run
Mutable	NO	YES
Types	String,Int, Bool,Array, Object,Float	String,Int, Bool,Array, Object,Float
Access syntax	pipeline() .parameters .name	variables() ('name')

SIMPLE EXPLANATION

ANALOGY: Parameters vs Variables **in** a cooking show

PARAMETERS = Recipe given to the chef BEFORE cooking starts

- Passed **by** show **producer** (external)
- "Today you'll cook: Biryani **for** 4 people"
- Chef CANNOT change **this** mid-show
- Fixed input, drives the whole show

VARIABLES = Notes the chef writes WHILE cooking

- Chef-managed, changes **as** cooking progresses
- "I added 3 cups rice" (counter)
- "Timer set to 20 minutes" (**dynamic value**)
- "Spices added: [cumin, cardamom, pepper]" (array)
- Changes **as** cooking progresses

REAL CODE EXAMPLE:

PARAMETERS (passed at trigger time):

p_process_date: "2024-06-15" ← cannot change mid-pipeline
p_source_table: "SALES" ← cannot change mid-pipeline

Access: @{{pipeline().parameters.p_process_date}}

VARIABLES (change during pipeline):

v_file_count: starts at 0
ForEach loop: increment **by** 1 each iteration
→ Set Variable: @{{add(int(variables('v_file_count')), 1)}}

At end: "Processed @{{variables('v_file_count')}} files"



REAL-WORLD USE CASE

SCENARIO: Dynamic multi-table copy pipeline

PARAMETERS (set by scheduler at 1 AM):

```
p_load_date = "2024-06-15"
p_load_type = "INCREMENTAL"
p_source_schema = "SALES"
```

VARIABLES (change during execution):

```
v_tables_processed = []    (array, starts empty)
v_total_rows = 0           (counter, starts at 0)
v_current_table = ""       (current table being processed)
```

FLOW:

Lookup → get list of 25 tables

ForEach **each** table:

Set v_current_table = @{item().table_name}

Copy Data using:

Source: SELECT * FROM

@{pipeline().parameters.p_source_schema}.

@{variables('v_current_table')}

WHERE updated_date >

@{pipeline().parameters.p_load_date}

After copy:

Set v_total_rows = @{add(int(variables('v_total_rows')),

int(activity('CopyData').output.rowsCopied))}

Append to v_tables_processed: @{variables('v_current_table')}

After loop:

Send email: "Processed @{variables('v_total_rows')} total rows



```
        across @length(variables('v_tables_processed'))}
tables"
```

RESULT: Dynamic, reusable pipeline with full tracking!

⚠ COMMON MISTAKES

MISTAKE 1: Trying to modify parameters during pipeline

❌ "I can update a parameter value during pipeline execution"

✅ "Parameters are immutable. Use Variables for values that change during execution."

MISTAKE 2: Using variables for values that should be parameters

❌ Setting process_date as a variable with default value

✅ Set as parameter – triggers pass the date each run

WHY: Variables reset to default each run anyway.

Parameters are how external systems control your pipeline.

MISTAKE 3: Not knowing Global Parameters

Interviewer sometimes asks about Global Parameters:

✅ "Global parameters are available to ALL pipelines in the ADF instance. Used for environment-level config: environment=prod, data_lake_url=..., email=...
When deploying Dev→Prod: just update global params.
All 100 pipelines automatically use prod values!"

💡 PRO TIP — 15+ LPA LEVEL

DEMONSTRATE **FULL** PARAMETERIZATION **PATTERN**:

"In enterprise ADF, I parameterize at EVERY level:

PIPELINE PARAMETERS (p_ prefix):

p_process_date, p_load_type, p_region

DATASET PARAMETERS (d_ prefix):

d_table_name, d_schema_name, d_file_path

LINKED SERVICE PARAMETERS (ls_ prefix):

ls_server_name, ls_database_name

WHY parameterize Linked Services?

12 regional databases, SAME schema, DIFFERENT servers

→ 1 parameterized LS replaces 12 separate LS

→ Pass server name as parameter at runtime!

GLOBAL PARAMETERS (env_ prefix):

env_environment = 'Production'

env_data_lake_url = 'https://prodlake.dfs.core.windows.net'

env_alert_email = 'dataengg-prod@company.com'

When asked about environment management:

'I use Global Parameters for environment-level config.

ARM template parameter files override them per environment.

No code changes needed to deploy from Dev to Prod!'

This COMPLETE parameterization strategy =
enterprise-grade ADF development!"

? QUESTION 8 — Intermediate | Incremental Loading

"How do you implement incremental data loading in ADF? What are the different approaches?"

✓ IDEAL ANSWER

Incremental loading = Load **ONLY NEW** or **CHANGED** data
(vs **Full** load = load **EVERYTHING** **every time**)

WHY INCREMENTAL?

Sales **table**: **500M** **rows** total, **50K** **new rows** daily

Full load: read **500M** **rows** **every** night → **6** hours

Incremental: read **50K** **new rows** → **5** minutes

→ **72x** faster, **72x** less cost!

4 APPROACHES **TO** INCREMENTAL LOADING:

APPROACH **1**: WATERMARK-BASED (most common)

How: Track **last** loaded **timestamp** in a control **table**

Step **1**: Lookup → **get** last_watermark **from** control **table**

Step **2**: **Copy** Data **with** **filter**:

WHERE updated_date >

'@{activity('Lookup').output.firstRow.last_wm}'

Step **3**: After success → **Update** control **table** **with** **new** watermark

Control **table**:

```
CREATE TABLE ControlTable (  
    table_name      VARCHAR(100),  
    last_watermark  DATETIME2  ← "last time we loaded"  
);
```

First run: last_watermark = '2000-01-01' (load everything)

Each subsequent run: loads only records since last run

APPROACH 2: CHANGE DATA CAPTURE (CDC)

How: Database tracks changes at row level

(insert/update/delete)

SQL Server: enable CDC → ADF reads change table

Oracle: LogMiner / GoldenGate

Advantage: Captures DELETES too (watermark misses deletes!)

Disadvantage: Requires DB-level setup, DBA involvement

APPROACH 3: FILE-BASED INCREMENTAL

How: Only process new files arrived since last run

GetMetadata → list files in folder

Filter: lastModified > last_processed_timestamp

ForEach: copy only new files

OR: Storage Event Trigger → only fires for NEW files!

APPROACH 4: PARTITION-BASED

How: Each day creates a new partition/folder

Load by date partition: /data/year=2024/month=06/day=15/

Each run: just load today's partition (new folder = new data)

Advantage: Simple, no control table needed

Disadvantage: Can't handle late-arriving data easily

SIMPLE EXPLANATION

ANALOGY: Incremental loading = Reading a newspaper

FULL LOAD = Reading the ENTIRE newspaper archive every morning

- You read all newspapers from 1900 to today
- Just to find today's news
- CRAZY expensive and slow!

INCREMENTAL (WATERMARK) = Read from where you left off

- Bookmark: "Last read: June 14, 2024"
- Today: read only June 15 news
- Update bookmark: "Last read: June 15, 2024"
- Tomorrow: read June 16 news
- Efficient! Fast! Correct!

WATERMARK = Your BOOKMARK in the data

- Stores "last loaded timestamp" in control table
- Each run: load data AFTER that timestamp
- After success: update bookmark to now

THE CRITICAL RULE:

- NEVER update the watermark before the load!
- Update AFTER successful load ONLY!

Why? If load fails halfway:

- Watermark already updated → some data PERMANENTLY missed!
- With update AFTER: load fails → watermark unchanged
- Next run: reloads failed data → no data loss!



REAL-WORLD USE CASE

MICROSOFT PROJECT (Production Pattern):

TABLE: Oracle TRANSACTIONS (500M rows, 50K new/day)

COMPLETE INCREMENTAL PIPELINE:

Step 1: LKP_GetWatermark (Lookup)

```
Query: SELECT CONVERT(VARCHAR, last_watermark, 120) AS wm
        FROM ControlTable
        WHERE table_name = 'TRANSACTIONS'
```

Step 2: CPY_IncrementalLoad (Copy Data)

Source Query:

```
SELECT * FROM SALES.TRANSACTIONS
WHERE LAST_UPDATED_DATE >
      TO_DATE('@{activity('LKP').output.firstRow.wm}',
              'YYYY-MM-DD HH24:MI:SS')
```

Partition: Dynamic Range on TRANSACTION_ID (8 threads)

Sink: ADLS Gen2 → Parquet format → partition by date

Step 3 (SUCCESS): SP_UpdateWatermark (Stored Procedure)

```
UPDATE ControlTable
SET last_watermark = GETDATE()
WHERE table_name = 'TRANSACTIONS'
→ Only runs on SUCCESS!
```

Step 3 (FAILURE): SP_LogError + SendAlert

→ Watermark NOT updated → safe to retry!

RESULT:

First run (historical): loads 500M rows (3 hours)

Daily runs: loads 50K rows (3 minutes!)

And because watermark update is on SUCCESS path only:

If load fails → watermark unchanged → retry loads same data

NO DATA LOSS ever!

⚠️ COMMON MISTAKES

MISTAKE 1: Updating watermark before load completes

❌ Update watermark → then copy data → if copy fails = DATA LOSS!

✅ Copy data → SUCCESS → then update watermark

This **is** THE most common mistake **in** incremental loading!
Interviewers specifically test **for** this.

MISTAKE 2: Using updated_date but missing deletes

❌ Watermark on updated_date → captures inserts + updates
But DELETED rows? They're gone from source!

✅ For deletes: use CDC OR soft deletes
(add is_deleted flag → row UPDATE → captured by watermark)

Senior answer: "Watermark works for inserts/updates.
For deletes, I implement soft deletes at source (recommended)
OR enable CDC if DBA can set it up."

MISTAKE 3: Not handling late-arriving data

Source has transactions with backdated dates
Watermark already moved past those dates!

Solution: Add buffer/overlap:

WHERE updated_date > @{addHours(last_watermark, -2)}

→ Reload last 2 hours of data each run

→ Handles late arrivals with ≤2 hour delay

→ Minor duplication → handle with MERGE/UPSERT at sink

💡 PRO TIP — 15+ LPA LEVEL

ENTERPRISE INCREMENTAL **PATTERN**:

"I use a METADATA-DRIVEN incremental framework:

Control table stores per-table config:

table_name	watermark_col	last_watermark	strategy
SALES	UPDATED_DATE	2024-06-14 00:00	WATERMARK
CUSTOMERS	MODIFIED_AT	2024-06-14 00:00	WATERMARK
PRODUCTS	(null)	(null)	FULL_LOAD
AUDIT_LOG	CREATED_AT	2024-06-14 00:00	APPEND_ONLY

ONE PIPELINE handles ALL tables:

Lookup → reads control table

ForEach each table:

IF strategy = WATERMARK → incremental query

IF strategy = FULL_LOAD → full query

IF strategy = APPEND_ONLY → append (never overwrite)

Benefits:

- ✅ Add new table → just add row to control table
- ✅ Change strategy → update one row
- ✅ Different watermark columns → all handled automatically
- ✅ Full audit trail → who loaded what when

This framework = maintained by 1 engineer for 100+ tables.

That's enterprise-grade thinking.

Companies pay 15+ LPA for this exact framework!"

? QUESTION 9 — Intermediate | Data Flows

"What are ADF Data Flows? When would you use Data Flows vs Databricks notebooks for transformation?"

✓ IDEAL ANSWER

ADF Data Flows = VISUAL, CODE-FREE data transformation
powered by Apache Spark under the hood

KEY CHARACTERISTICS:

- Build transformations by dragging and dropping
- No Spark code needed (framework handles it)
- Debug with live data preview at each step
- Runs on managed Spark cluster (auto-provisioned)
- Supports 20+ transformation types

AVAILABLE TRANSFORMATIONS:

Source → Filter → Derived Column → Aggregate →
Join → Lookup → Union → Pivot → Unpivot →
Sort → Select → Flatten → Conditional Split →
Window → Alter Row → Assert → Sink

WHEN TO USE DATA FLOWS:

- ✓ Medium complexity transformations (ETL/ELT rules)
- ✓ SQL-like operations without writing SQL
- ✓ Team has limited coding skills (analysts, not engineers)
- ✓ Data quality checks and cleansing
- ✓ Schema drift handling (new columns in source)
- ✓ Small-medium data volumes (< 500GB)
- ✓ Visual documentation (team can see transformation logic)

WHEN TO USE DATABRICKS INSTEAD:

- ✓ Complex Python/Scala logic (ML, custom algorithms)
- ✓ Very **large** data (**1TB+**, Databricks faster)
- ✓ Streaming data processing
- ✓ Delta Lake operations (**MERGE**, SCD Type **2**)
- ✓ ML model training **and** feature engineering
- ✓ Team has strong PySpark skills
- ✓ Reusing shared libraries/functions across notebooks

COMPARISON **TABLE**:

Feature	ADF Data Flows	Databricks
Learning curve	Low	High
Custom logic	Limited	Unlimited (Python/Scala)
Performance	Good (< 500GB)	Better (any scale)
ML/AI	No	Yes
Streaming	Limited	Full support
Delta MERGE	Limited	Full support
Debugging	Visual preview	Code debugging
Cost	Per activity run	Per cluster hour
Team skill need	Low	High (PySpark)

SIMPLE EXPLANATION

ANALOGY: Data Flows vs Databricks

Data Flows = LEGO blocks

- Pre-built pieces you snap together
- Anyone can **build** (no engineering degree needed!)
- Limited to what LEGO makes
- Fast to build, easy to **understand**

Databricks = RAW **MATERIALS** (wood, metal, tools)

- Build ANYTHING you can imagine
- Needs skilled **craftsman** (PySpark knowledge)
- Takes longer to build
- Handles ANY complexity

SIMPLE RULE:

Business analyst needs to build transformation? → Data Flows
Data engineer needs complex Python logic? → Databricks

In MOST enterprises: BOTH are used together!

ADF orchestrates:

- Data Flows **for** simple **ETL** (CSV → cleansed table)
- Databricks **for** complex **processing** (ML, heavy transforms)

REAL ARCHITECTURE:

Source → ADF **Copy** (ingest) → Bronze Layer
→ ADF Data **Flow** (clean) → Silver Layer
→ Databricks **Notebook** (ML features) → Gold Layer
→ Power BI → Reports

RIGHT TOOL **for** RIGHT JOB!

COMMON MISTAKES

MISTAKE 1: "Data Flows replaces Databricks"

-  WRONG: They serve different purposes
-  CORRECT: Data Flows = medium complexity visual transforms
Databricks = complex/large scale code transforms

Use them TOGETHER **in** a proper architecture!

MISTAKE 2: Leaving Data Flow Debug running after testing

Data Flow Debug spins up a SPARK CLUSTER

Even **when** you're *not running transformations!*

❌ Forget **to** turn **off** debug → \$50-100/hour wasted!

✅ Always turn **off** debug after testing

Interviewers often ask about cost – this **is** a common waste!

MISTAKE 3: **Not** understanding Schema Drift

Schema Drift = source suddenly adds **new** columns

Without Schema Drift handling:

❌ Pipeline FAILS **when new** column arrives **in** source!

With Schema Drift **in** Data Flow:

✅ Source → Allow Schema Drift: **ON**

✅ Sink → **Auto**-map drifted columns: **ON**

✅ **New** columns flow through automatically!

This **is** a VERY common interview question at senior level!

💡 PRO TIP — 15+ LPA LEVEL

HANDLE THE CLASSIC FOLLOW-UP QUESTION:

"How do you handle the 2-4 minute Spark startup time for Data Flows?"

SENIOR ANSWER:

"Data Flow clusters take 2-4 minutes to start."

For time-sensitive pipelines, I optimize:

1. QUICK REUSE (most effective):

Set TTL (Time to Live) on Data Flow cluster:

'Keep alive for 10 minutes after last run'

→ 2nd Data Flow in same pipeline: ZERO startup!

→ Saves 4 minutes per subsequent run

2. BATCH MULTIPLE DATA FLOWS:

Don't run 5 separate Data Flows (5 × 4min startup)

Bundle into 1 Data Flow with multiple sources/sinks

→ 1 startup for all 5 transformations = 16 min saved!

3. APPROPRIATE CLUSTER SIZE:

Large cluster = faster processing

But: startup time same for all sizes

Small Data Flow on huge cluster = waste!

Match cluster size to actual data size.

4. CONSIDER DATABRICKS for sub-minute latency:

If startup time matters → use Databricks Job Cluster

Job cluster starts: 5-10 minutes

BUT: pre-warm cluster = always-on = instant start

This optimization thinking = senior-level awareness!"

? QUESTION 10 — Intermediate | PySpark Fundamentals

"Explain the difference between Transformations and Actions in Spark. Why does this matter for performance?"

✓ IDEAL ANSWER

This **is** THE MOST FUNDAMENTAL Spark concept.
Understanding this separates junior **from** senior.

TRANSFORMATIONS = Operations that DEFINE what **to do**

- They are LAZY (don't *execute immediately!*)
- Just build an execution plan (DAG)
- **Return** a **NEW** DataFrame (original unchanged)
- Examples:
 - `filter()`, `select()`, `withColumn()`
 - `groupBy()`, `join()`, `orderBy()`
 - `map()`, `flatMap()`, `distinct()`
 - `repartition()`, `coalesce()`
 - `withColumnRenamed()`, `drop()`

ACTIONS = Operations that TRIGGER execution

- They FORCE Spark **to** actually compute
- Execute all queued transformations at once
- **Return** a result (**not** a DataFrame)
- Examples:
 - `count()` → returns **Integer**
 - `show()` → prints **to** console
 - `collect()` → returns all rows **to** driver
 - `first()` → returns first row
 - `take(n)` → returns first n rows
 - `write()` → saves **to** storage
 - `saveAsTable()` → saves **to** catalog

LAZY EVALUATION = the **key** concept:

WITHOUT lazy evaluation:

Line 1: `df1 = df.filter(...)` ← execute immediately
(wasteful!)

Line 2: `df2 = df1.select(...)` ← execute immediately
(wasteful!)

Line 3: `df3 = df2.groupBy(...)` ← execute immediately
(wasteful!)

Line 4: `df3.write(...)` ← execute

4 separate passes over data = 4x slower!

WITH lazy evaluation (what Spark actually does):

Line 1: `df1 = df.filter(...)` ← just PLANS filter

Line 2: `df2 = df1.select(...)` ← just PLANS select

Line 3: `df3 = df2.groupBy(...)` ← just PLANS groupBy

Line 4: `df3.write(...)` ← NOW execute all 4 at once!

Spark OPTIMIZES the combined plan:

→ Push filter to file reader (read less data!)

→ Push column selection to file reader (fewer columns!)

→ ONE optimized pass = 10x faster!

SIMPLE EXPLANATION

ANALOGY: Restaurant order system

TRANSFORMATIONS = Adding items to your order

→ "Add butter chicken"

→ "Remove onions"

→ "Extra spice"

→ NOTHING is being cooked yet!

- You're *just DESCRIBING* what you want
- Lazy = kitchen waits **for** you **to** finalize

ACTION = "Submit order" button

- NOW the kitchen starts cooking!
- All your customizations applied at once
- Optimally (chef decides best **order to** cook things)

WHY THIS MATTERS **FOR** PERFORMANCE:

YOU write:

```
df_result = df.filter(region == 'NORTH') ← lazy
               .select('sale_id', 'amount') ← lazy
               .groupBy('store_id')           ← lazy
               .sum('amount')                 ← lazy
df_result.write.parquet(path)                ← ACTION! Execute
```

now!

SPARK OPTIMIZES before executing:

1. Reads only 'sale_id', 'amount', 'store_id', 'region' **from** Parquet (**not** ALL 50 columns!)
2. Applies filter **WHILE** reading file
(reads only North India data **from** file!)
3. **Then** groups **and** sums **in** memory

RESULT: Reads **70%** less data than naive execution!

This **is** why Spark **is** fast.



REAL-WORLD USE CASE

DATABRICKS PERFORMANCE SCENARIO:

Code written by junior engineer:

```
df_sales = spark.read.parquet("/mnt/bronze/sales/") # 500GB

# Count for logging
total = df_sales.count() ← ACTION 1 (reads 500GB!)
print(f"Total: {total}")

# Filter and process
df_north = df_sales.filter(col("region") == "N") ← transformation
north_count = df_north.count() ← ACTION 2 (reads 500GB again!)

df_result = df_north.groupBy("store_id").sum("amount") ← transformation
df_result.write.parquet("/mnt/silver/") ← ACTION 3 (reads 500GB AGAIN!)

TOTAL: 500GB × 3 reads = 1,500GB scanned!
Cost: 3x more than needed!
```

Code written by senior engineer:

```
df_sales = spark.read.parquet("/mnt/bronze/sales/")

# Cache AFTER filter (cache only needed data!)
df_north = df_sales.filter(col("region") == "N").cache()

north_count = df_north.count() ← ACTION 1 (caches North data ~125GB)
print(f"North India: {north_count}")

df_result = df_north.groupBy("store_id").sum("amount")
df_result.write.parquet("/mnt/silver/") ← ACTION 2 (reads from cache!)
```

```
df_north.unpersist()
```

← Free memory!

TOTAL: 500GB initial + 0GB from cache = 500GB scanned

Cost: 3x CHEAPER than junior code!

That optimization = what companies pay 15+ LPA for!

⚠ COMMON MISTAKES

MISTAKE 1: Multiple `count()` calls on same DataFrame

```
❌ count1 = df.count()
... more transformations ...
count2 = df.count()
→ Reads data TWICE!
```

✅ Compute both in ONE action:

```
from pyspark.sql.functions import count, sum
df.agg(count('*').alias('cnt'),
sum('amount').alias('total')).show()
→ Reads data ONCE!
```

MISTAKE 2: Using `collect()` on large DataFrames

```
❌ rows = df.collect() ← brings ALL data to driver!
If 50M rows → OutOfMemoryError on driver!
```

```
✅ df.show(20) ← show sample
✅ df.write(...) ← write to storage
✅ df.take(100) ← take small sample if needed
```

MISTAKE 3: Not understanding when to cache

```
❌ Cache everything "just in case"
✅ Cache when: same DataFrame used in 2+ actions
```

✅ Always unpersist() **when** done!

NOT caching **when** needed = slow pipeline (**reads** multiple times)

Caching **when not** needed = wasted memory (OOM errors!)

💡 PRO TIP — 15+ LPA LEVEL

EXPLAIN THE DAG (Directed Acyclic Graph) – shows deep knowledge:

"When I call an Action, Spark creates a DAG of stages.

Understanding this helps me optimize performance:

```
df_result = df.filter(...).groupBy(...).sum(...)
              .write(...)
```

SPARK CREATES:

Stage 1 (No shuffle):

Read Parquet files + Filter + Project (select cols)

→ All can run on same executor (no data movement)

→ Fast! (narrow transformations)

Stage 2 (Shuffle):

Shuffle data by groupBy key

→ Data moves across network (expensive!)

→ All rows with same store_id → same executor

Stage 3 (No shuffle):

Sum within each group

→ Fast! (local aggregation)

→ Write result

OPTIMIZATION INSIGHT:

Shuffle = expensive! Minimize shuffles.

How to minimize:

1. Filter BEFORE groupBy (less data to shuffle)
2. Repartition by join key ONCE, then join multiple times (shuffle once → reuse multiple times)
3. Broadcast small tables in joins (eliminates shuffle!)

'I use df.explain() to see Spark's execution plan and identify shuffles before running expensive jobs.'

This DAG-level thinking = true senior Spark knowledge!"

REVISION SUMMARY — Questions 6-10

REVISION: ADF INTERMEDIATE + SPARK BASICS (Q6-Q10)

Q6: Error Handling = 3 levels:

Activity retry → Dependency paths → Fault tolerance

KEY: Always add failure path + notification!

Q7: Parameters = external inputs (immutable)

Variables = internal state (mutable during run)

KEY: Parameterize EVERYTHING for reusability!

Q8: Incremental Loading = watermark-based most common

KEY: Update watermark AFTER success only!

CDC for capturing deletes

Q9: Data Flows = visual Spark transformation

Use: medium complexity, low-code teams

NOT for: ML, complex Python, very large data

KEY: Use Databricks for heavy lifting!

Q10: Lazy evaluation = plan first, execute on Action

KEY: Minimize actions, cache strategically, avoid

collect() on large data, use explain() to debug!

INTERVIEW PATTERN ALERT:

PATTERN 1: Optimization questions always expected

Every answer should mention HOW you'd make it faster/cheaper

PATTERN 2: "Watermark update" gotcha

If you say watermark updates before copy → immediate red flag

PATTERN 3: Spark = lazy evaluation

If you don't know this → fails senior Spark interview

LEVEL 2 — INTERMEDIATE

"Now we separate the good from the great"

? QUESTION 11 — Intermediate | SQL Window Functions

"Explain SQL Window Functions. Write a query to find the top 3 products by sales in each region."

✓ IDEAL ANSWER

WINDOW FUNCTIONS = Calculate **values** ACROSS **rows**
WITHOUT collapsing **rows** like **GROUP BY**

KEY DIFFERENCE:

GROUP BY: produces **ONE row per group** (collapses!)

WINDOW: keeps **ALL rows** + adds calculated **column**

SYNTAX:

```
function_name() OVER (  
    PARTITION BY column1      -- define group  
    ORDER BY column2          -- define order within group  
    ROWS/RANGE BETWEEN ...    -- define window frame (optional)  
)
```

CATEGORIES:

1. RANKING FUNCTIONS:

ROW_NUMBER() → **unique 1,2,3** (no ties)

RANK() → **1,2,2,4** (ties + gaps)

DENSE_RANK() → **1,2,2,3** (ties + no gaps)

NTILE(4) → divide **into 4** buckets (**1,2,3,4**)

PERCENT_RANK() → percentile (**0.0 to 1.0**)

2. AGGREGATE FUNCTIONS (as window):

SUM() **OVER()** → **running** total

AVG() **OVER()** → moving average

COUNT() **OVER()** → **running** count

MAX() **OVER()** → **running** max

`MIN() OVER()` → running min

3. VALUE FUNCTIONS:

`LAG(col, n)` → value from n rows behind

`LEAD(col, n)` → value from n rows ahead

`FIRST_VALUE()` → first row in window

`LAST_VALUE()` → last row in window

THE QUERY (Top 3 products per region):

```
WITH ranked_products AS (  
    SELECT  
        region_code,  
        product_name,  
        SUM(total_amount) AS total_revenue,  
        RANK() OVER (  
            PARTITION BY region_code          -- separate per  
region  
            ORDER BY SUM(total_amount) DESC -- rank by revenue  
        ) AS revenue_rank  
    FROM fact_sales fs  
    JOIN dim_product p ON fs.product_key = p.product_key  
    GROUP BY region_code, product_name  
)  
SELECT  
    region_code,  
    product_name,  
    total_revenue,  
    revenue_rank  
FROM ranked_products  
WHERE revenue_rank <= 3  
ORDER BY region_code, revenue_rank;
```

WHY `RANK()` not `ROW_NUMBER()`?

If two products TIED for 2nd place:

ROW_NUMBER: Product A = 2, Product B = 3 (arbitrary!)

RANK: Product A = 2, Product B = 2 (fair! both are 2nd)

DENSE_RANK: Product A = 2, Product B = 2, next = 3 (no gap)

For "top N" queries → RANK() is usually correct!

SIMPLE EXPLANATION

ANALOGY: Window Functions = Looking through different windows

Imagine you have a scoreboard for a cricket tournament:

GROUP BY = Looking at TOTAL scores only

"India: 250 total, England: 230 total, Australia: 240 total"

→ Individual player scores LOST

→ Just final totals

WINDOW FUNCTION = Looking at INDIVIDUAL + RELATIVE scores

"Kohli: 100 (Rank 1 in India)"

"Dhoni: 80 (Rank 2 in India)"

"Root: 95 (Rank 1 in England)"

→ Each player's score kept

→ PLUS their rank within their team

→ ALL rows preserved!

LAG = "What did he score in the PREVIOUS match?"

LEAD = "What will he score in the NEXT match?"

SUM OVER = "Running total after each match"

PRACTICAL EXAMPLES:

Sales dashboard needs:

→ "Show all sales WITH their store's rank by revenue" → RANK()

```
OVER(PARTITION BY region)
  → "Show month-over-month change" → current - LAG(amount, 1)
OVER(ORDER BY month)
  → "Running total by day" → SUM(amount) OVER(ORDER BY sale_date)
  → "What percentile is this customer?" → PERCENT_RANK()
OVER(ORDER BY total_purchases)
```



REAL-WORLD USE CASE

BARCLAYS BANK SCENARIO:

Business request:

"For each branch, show:

1. Monthly revenue
2. Rank within region
3. Month-over-month growth
4. Running year-to-date total"

-- Complete business query:

SELECT

```
  branch_id,
  branch_name,
  region_code,
  sale_month,
  monthly_revenue,
```

-- Rank within region for this month

```
RANK() OVER (
  PARTITION BY region_code, sale_month
  ORDER BY monthly_revenue DESC
) AS region_rank,
```

```

-- MoM growth vs previous month
ROUND(
    (monthly_revenue - LAG(monthly_revenue, 1,
monthly_revenue)
    OVER (PARTITION BY branch_id ORDER BY sale_month))
    / NULLIF(LAG(monthly_revenue, 1)
    OVER (PARTITION BY branch_id ORDER BY
sale_month), 0)
    * 100,
    1
) AS mom_growth_pct,

-- Running YTD total per branch
SUM(monthly_revenue) OVER (
    PARTITION BY branch_id, YEAR(sale_month)
    ORDER BY sale_month
    ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
) AS ytd_total

FROM monthly_branch_sales
ORDER BY region_code, sale_month, region_rank;

```

This **one** query gives the COMPLETE business **view**!
Without window functions: would need **4** separate queries
 + complex joins = slow **and** hard **to** maintain.

COMMON MISTAKES

MISTAKE 1: Using ROW_NUMBER for top-N **without** considering ties

 ROW_NUMBER() for "top 3 products"

→ If 3rd **and** 4th tied: arbitrarily picks **one**, drops other

 RANK() → **both** get rank 3, **both** returned

MISTAKE 2: Forgetting **PARTITION BY** (window over entire table!)

❌ **RANK()** **OVER** (**ORDER BY** revenue **DESC**)

→ Ranks **all** products globally, **not per** region!

✅ **RANK()** **OVER** (**PARTITION BY** region_code **ORDER BY** revenue **DESC**)

→ Ranks **within each** region

MISTAKE 3: Not handling **NULL** in **LAG** for first row

❌ **LAG**(amount, 1) for Jan → **NULL** (no previous month)

Then: (Jan_amount - **NULL**) = **NULL** → MoM growth shows **NULL**

✅ **LAG**(amount, 1, amount) **OVER**(...) ← default value = current

Or: **COALESCE**(**LAG**(amount,1) **OVER**(...), 0)

Or: show **NULL** explicitly (it's technically correct!)

MISTAKE 4: Not knowing **ROWS** vs **RANGE**

ROWS BETWEEN: based on physical row position

RANGE BETWEEN: based on logical value range

For running total: use **ROWS** (safer, more predictable)

For time-based windows: might use **RANGE**

💡 PRO TIP — 15+ LPA LEVEL

WINDOW FUNCTION PERFORMANCE TIPS (show senior knowledge):

"Window functions require **SORTING** data by **ORDER BY** column.
This can be expensive at scale.

OPTIMIZE:

1. Pre-sort with repartition by **PARTITION BY** column:

```
df.repartition(col('region_code')) ← data grouped by region  
Then window function = local operation (no network sort!)
```

2. Cluster/Z-ORDER Delta tables by window partition key:
OPTIMIZE sales ZORDER BY (region_code, sale_date)
→ Next query with *PARTITION BY region_code* = partition pruning!
→ 80% less data read!

3. Avoid window functions in WHERE clause:

❌ *WHERE RANK() OVER(...) <= 3* ← NOT valid SQL!

✅ *WITH cte AS (... RANK())*
*SELECT * FROM cte WHERE rank <= 3*

4. In PySpark: Use Window spec properly:

```
from pyspark.sql.window import Window
```

```
windowSpec = Window.partitionBy('region_code') \
                    .orderBy(F.col('revenue').desc())
```

```
df.withColumn('rank', F.rank().over(windowSpec))
```

Performance note: Window functions = SHUFFLE operation in Spark

(expensive!) → minimize how many you chain

This PySpark + SQL knowledge combo = 15+ LPA!"

? QUESTION 12 — Intermediate | SCD Type 1 & 2

"Explain Slowly Changing Dimensions. What's the difference between SCD Type 1 and Type 2? Write implementation in SQL."

✓ IDEAL ANSWER

SCD = Slowly Changing Dimensions

How to handle dimension table data that changes over time

CONTEXT:

Fact tables = transactions (immutable, append-only)

Dimension tables = masters (Customer, Product, Store)

→ Masters CHANGE over time!

→ Customer moves city, Product changes price, Store changes manager

→ How do we handle historical analysis?

SCD TYPE 1 – OVERWRITE (No history)

When record changes: UPDATE existing record

Old value: LOST forever

Use when: "History doesn't matter, just latest value"

Examples:

→ Fix typo in customer name

→ Update customer tier based on spending

→ Correct wrong product code

→ Change employee designation

SCD TYPE 2 – KEEP HISTORY (Full history)

When record changes: CLOSE old record, INSERT new record

Both old AND new values: PRESERVED

Implementation adds 3 columns:

effective_start_date: when this version became active

effective_end_date: when this version was replaced

is_current: which version is current (true/false)

Use **when**: "History matters for reporting"

Examples:

- Customer changes city (**for** regional revenue analysis)
- Product price changes (**for** margin analysis)
- Employee changes department (**for** dept-wise reports)



SCD TYPE 1 — SQL IMPLEMENTATION

```
-- SCD Type 1: Simple UPDATE + INSERT (UPSERT)
-- Using MERGE statement

MERGE INTO dim_customer AS target
USING (
    SELECT
        CUSTOMER_ID,
        CUSTOMER_NAME,
        EMAIL,
        PHONE,
        CITY,
        STATE,
        CUSTOMER_TIER
    FROM staging.stg_customer_updates -- new data from source
) AS source
ON target.customer_id = source.CUSTOMER_ID

-- Record EXISTS and something CHANGED: UPDATE it
WHEN MATCHED AND (
    target.email      <> source.EMAIL      OR
    target.phone      <> source.PHONE      OR
    target.city       <> source.CITY       OR
    target.state      <> source.STATE      OR
    target.customer_tier <> source.CUSTOMER_TIER
```



```

)
THEN UPDATE SET
    target.customer_name = source.CUSTOMER_NAME,
    target.email          = source.EMAIL,
    target.phone          = source.PHONE,
    target.city           = source.CITY,
    target.state          = source.STATE,
    target.customer_tier  = source.CUSTOMER_TIER,
    target.updated_at     = GETDATE()

-- Record does NOT EXIST: INSERT new
WHEN NOT MATCHED BY TARGET
THEN INSERT (
    customer_id, customer_name, email, phone,
    city, state, customer_tier, created_at, updated_at
)
VALUES (
    source.CUSTOMER_ID, source.CUSTOMER_NAME, source.EMAIL,
    source.PHONE, source.CITY, source.STATE,
source.CUSTOMER_TIER,
    GETDATE(), GETDATE()
);

```



SCD TYPE 2 — SQL IMPLEMENTATION

```

-- SCD Type 2: EXPIRE old record + INSERT new record

-- First: Expire records that have changed
MERGE INTO dim_customer_scd2 AS target
USING (
    SELECT * FROM staging.stg_customer_updates
) AS source

```

```

ON target.customer_id = source.CUSTOMER_ID
    AND target.is_current = 1 -- only look at CURRENT records

-- Record changed: expire the current version
WHEN MATCHED AND (
    target.city <> source.CITY OR
    target.email <> source.EMAIL OR
    target.state <> source.STATE
)
THEN UPDATE SET
    target.is_current = 0,
    target.effective_end_date = DATEADD(DAY, -1, CAST(GETDATE()
AS DATE));

-- Now insert NEW version for changed records + new records
INSERT INTO dim_customer_scd2 (
    customer_id, customer_name, email, city, state,
    effective_start_date, effective_end_date, is_current
)
SELECT
    source.CUSTOMER_ID,
    source.CUSTOMER_NAME,
    source.EMAIL,
    source.CITY,
    source.STATE,
    CAST(GETDATE() AS DATE), -- today = start date
    '9999-12-31', -- still active = far future
    1 -- is current version
FROM staging.stg_customer_updates source
WHERE EXISTS (
    -- Changed records (just expired above)
    SELECT 1 FROM dim_customer_scd2 t2
    WHERE t2.customer_id = source.CUSTOMER_ID
        AND t2.is_current = 0
        AND t2.effective_end_date = DATEADD(DAY, -1, CAST(GETDATE()

```

```
AS DATE))  
)  
OR NOT EXISTS (  
    -- Brand new customers (never existed before)  
    SELECT 1 FROM dim_customer_scd2 t3  
    WHERE t3.customer_id = source.CUSTOMER_ID  
);
```

SIMPLE EXPLANATION

ANALOGY: Your address **in** government records

SCD Type **1** (OVERWRITE):

You moved **from** Mumbai **to** Delhi.

Government: "Just update the address to Delhi."

Mumbai address: GONE FOREVER

If someone asks "where did you live in 2020?"

Government: "Delhi" (WRONG! You lived **in** Mumbai **in** 2020)

Use **when**: historical accuracy doesn't matter

(fixing a typo **in** your name = Type **1** makes sense!)

SCD Type **2** (KEEP HISTORY):

You moved **from** Mumbai **to** Delhi.

Government:

→ Closes Mumbai record: "Valid 2020-01-01 to 2024-06-14"

→ Opens Delhi record: "Valid 2024-06-15 to present"

If someone asks "where did you live in 2020?"

Government: "Mumbai" (CORRECT!)

If someone asks "where do you live now?"

Government: "Delhi" (CORRECT!)

Full history preserved!

HOW TO QUERY "current address?":

WHERE is_current = 1

OR WHERE effective_end_date = '9999-12-31'

HOW TO QUERY "address as of specific date?":

WHERE effective_start_date <= '2020-06-01'

AND effective_end_date >= '2020-06-01'



REAL-WORLD USE CASE

E-COMMERCE COMPANY SCENARIO:

dim_customer table:

customer_id	name	city	tier	eff_start	eff_end
C001	Rahul	Mumbai	SILVER	2022-01-01	2024-06-14
				0 ← old	
C001	Rahul	Delhi	GOLD	2024-06-15	9999-12-31
				1 ← current	

BUSINESS QUERY: "What was our regional revenue distribution in 2023?"

JOIN sales 2023 to dim_customer

WHERE effective_start_date <= '2023-12-31'

AND effective_end_date >= '2023-01-01'

→ Rahul's 2023 sales attributed to MUMBAI (correct!)

→ His 2024 sales attributed to DELHI (correct!)



Without SCD2: ALL Rahul's sales → Delhi

- 2023 regional report shows Delhi revenue inflated!
- Wrong business decisions based on wrong data!

REAL IMPACT:

Retail company using SCD1 (wrong):

- Marketing team sees "Delhi customers spend more"
- Increases Delhi marketing budget by 30%
- Actually: Delhi numbers inflated by ex-Mumbai customers!
- Wasted budget!

With SCD2 (correct):

- True Delhi customers identified
- Marketing budget optimized correctly
- Revenue per actual Delhi customer = real insight

⚠ COMMON MISTAKES

MISTAKE 1: Not knowing WHEN to use Type 1 vs Type 2

Rule:

- Correction/error fix → Type 1 (history was wrong anyway)
- Genuine change (business event) → Type 2 (history was right!)

Customer name typo: "Rahul" → "Rahul K" → Type 1 (fix error)

Customer moves city: Mumbai → Delhi → Type 2 (historical insight needed)

MISTAKE 2: Using CURRENT_DATE as effective_end

- ✗ effective_end = today for active records
- Then tomorrow the date is WRONG for current records!
- ✓ effective_end = '9999-12-31' for active records

- A fixed "impossibly far future" date
- Easy to query: `WHERE effective_end = '9999-12-31'`

MISTAKE 3: Not adding surrogate key in SCD2

✗ Using customer_id as primary key

- customer_id = C001 appears MULTIPLE times!
- Primary key violation!

✓ Add customer_key (identity/auto-increment) as PK

customer_key | customer_id | ...

1001 | C001 | (Mumbai record)

1002 | C001 | (Delhi record)

Fact table joins on customer_key (specific version!)

Not on customer_id (which has multiple rows!)

MISTAKE 4: Forgetting SCD2 in Fact table joins

✗ `SELECT ... FROM fact_sales JOIN dim_customer ON customer_id`

→ Ambiguous! Multiple rows per customer!

✓ `SELECT ... FROM fact_sales fs`

`JOIN dim_customer dc`

`ON fs.customer_id = dc.customer_id`

`AND fs.sale_date BETWEEN dc.effective_start AND`
`dc.effective_end`

💡 PRO TIP — 15+ LPA LEVEL

PYSPARK/DELTA LAKE SCD2 PATTERN (modern approach):

"In modern Lakehouse architecture, I implement SCD2
using Delta Lake MERGE which is far more efficient:

```

from delta.tables import DeltaTable

delta_dim = DeltaTable.forName(spark, 'dim_customer_scd2')

df_source = spark.read.table('staging.stg_customer_updates')

# Step 1: Expire changed current records
delta_dim.alias('t').merge(
    df_source.alias('s'),
    't.customer_id = s.CUSTOMER_ID AND t.is_current = true'
).whenMatchedUpdate(
    condition = 't.city != s.CITY OR t.email != s.EMAIL',
    set = {
        'is_current': 'false',
        'effective_end_date': 'current_date() - 1'
    }
).execute()

# Step 2: Insert new versions
delta_dim.alias('t').merge(
    df_source.alias('s'),
    't.customer_id = s.CUSTOMER_ID AND t.is_current = true'
).whenNotMatchedInsert(values = {
    'customer_id': 's.CUSTOMER_ID',
    'city': 's.CITY',
    'email': 's.EMAIL',
    'effective_start_date': 'current_date()',
    'effective_end_date': 'date(\"9999-12-31\")',
    'is_current': 'true'
}).execute()

```

ADVANTAGE OF DELTA LAKE SCD2:

- ACID **transactions** (no **partial** updates!)
- Time **travel** (query any historical version!)

- Better performance than SQL MERGE at scale
- Works **with** Databricks, Synapse Spark, Fabric

This **is** the PRODUCTION approach at 15+ LPA companies!"

? QUESTION 13 — Intermediate | Star vs Snowflake Schema

"Explain Star Schema and Snowflake Schema. Which would you recommend for a retail data warehouse and why?"

✓ IDEAL ANSWER

DATA WAREHOUSE DESIGN = How you organize tables **for** fast analytics

STAR SCHEMA:

Structure: Fact table **in** center, Dimension tables around it
Like a star ★

Dimensions: DENORMALIZED (no **sub**-tables, all attributes **in** one table)

Example:

```
dim_date ─┐
dim_store ─┐
dim_product ─┐ ── fact_sales ─┐ ──
```

```
dim_customer-| (center)
              |_____
```

```
dim_product (denormalized):
product_key | product_name | category | subcategory | brand |
brand_country | brand_founded_year | category_description
(ALL product attributes in ONE table!)
```

SNOWFLAKE SCHEMA:

Structure: Fact table + NORMALIZED dimension tables
Dimensions split into sub-dimensions
Like a snowflake ❄️

```
dim_product splits into:
dim_product → dim_category → dim_subcategory
dim_product → dim_brand → dim_brand_country
```

More joins needed to get full product information!

COMPARISON:

Feature	Star Schema	Snowflake Schema
Query speed	Faster	Slower (more joins)
Storage	More (redundant)	Less (normalized)
Complexity	Simple	Complex
Maintenance	Easier	Harder
Data integrity	Lower	Higher
BI tools	Easy to use	Harder to navigate
Data size impact	High if huge dim	Small if tiny facts

RECOMMENDATION FOR RETAIL:

Star Schema is almost always better for:
→ BI tools (Power BI, Tableau love simple joins)

- Query performance (fewer joins = faster)
- Data analyst productivity (fewer tables **to** understand)
- Modern storage **is** cheap (extra storage = acceptable trade-off)

Snowflake ONLY **if**:

- Dimension table **is** HUGE **and** heavily updated
- Storage cost **is** critical concern
- Very complex hierarchies (e.g., deep org charts)

SIMPLE EXPLANATION

ANALOGY: Library organization

STAR SCHEMA = All book information **in** one place

shelf: "Harry Potter | Fiction | Fantasy | JK Rowling | UK | 1997"

You want Harry Potter?

- Go **to** ONE place
- **Get** EVERYTHING about it
- Fast!

SNOWFLAKE SCHEMA = Book information scattered

shelf: "Harry Potter | → see Fiction shelf"

fiction shelf: "Fiction → see Fantasy shelf"

fantasy shelf: "Fantasy | JK Rowling | → see Author shelf"

author shelf: "JK Rowling | UK | 1997"

You want Harry Potter?

- Go **to** book shelf
- Go **to** fiction shelf

- Go to fantasy shelf
- Go to author shelf
- SLOWLY get everything
- But: very organized, no repetition!

FOR DATA WAREHOUSES (99% of cases):

Star Schema wins!

Why? Because:

- Your team runs 1000 queries per day
- Each query is faster in Star = hours saved
- Storage for extra data = pennies
- Analyst time is EXPENSIVE
- Save analyst time → Star Schema!



REAL-WORLD USE CASE

RETAILMART DATA WAREHOUSE DESIGN:

FACT TABLE:

fact_sales (largest table, 500M+ rows)

sale_key	BIGINT (PK)
date_key	INT (FK → dim_date)
customer_key	INT (FK → dim_customer)
product_key	INT (FK → dim_product)
store_key	INT (FK → dim_store)
quantity	INT
unit_price	DECIMAL(18,2)
gross_amount	DECIMAL(18,2)
net_amount	DECIMAL(18,2)
gst_amount	DECIMAL(18,2)
payment_mode	VARCHAR(50)

DIMENSION TABLES (Star – all denormalized):

dim_date:

date_key, full_date, day_name, month_name, quarter, year,
is_weekend, is_holiday, fiscal_year, fiscal_quarter
(ALL date attributes in one table!)

dim_customer:

customer_key, customer_id, name, tier, city, state, region,
pin_code, registration_year, age_group, income_bracket
(ALL customer attributes – no sub-tables!)

dim_product:

product_key, product_code, product_name, category,
subcategory, brand, brand_country, cost_price, list_price,
is_premium, launch_year
(ALL product attributes – including category/brand inline!)

dim_store:

store_key, store_id, store_name, city, state, region,
tier_city, store_type, sq_footage, opening_year, manager

SIMPLE POWER BI QUERY:

SELECT

d.month_name,
s.region,
p.category,
c.customer_tier,
SUM(f.gross_amount) AS revenue

FROM fact_sales f

JOIN dim_date d ON f.date_key = d.date_key

JOIN dim_store s ON f.store_key = s.store_key

JOIN dim_product p ON f.product_key = p.product_key

JOIN dim_customer c ON f.customer_key = c.customer_key

WHERE d.year = 2024

```
GROUP BY d.month_name, s.region, p.category, c.customer_tier
```

Just 4 joins, all dimension tables directly connected!

Power BI generates this automatically from semantic model.

No complexity, blazing fast!

⚠ COMMON MISTAKES

MISTAKE 1: Putting measures in dimension tables

✗ dim_customer has: total_purchases, last_purchase_date
→ These CHANGE over time (not slowly changing – frequently!)

✓ Measures go in fact tables

dim_customer: static attributes (name, city, tier)

fact_customer_activity: transactional measures

MISTAKE 2: Not understanding surrogate vs natural keys

✗ Using customer_id (from source system) as FK in fact table

→ Customer_id might change in source system!

→ Or same customer_id in two source systems!

✓ Surrogate key (auto-increment integer) in dim table

fact table uses surrogate key as FK

Completely insulated from source system changes!

MISTAKE 3: Not using dim_date table

✗ Store raw dates in fact table + use YEAR(), MONTH() functions

→ Functions are SLOW on 500M rows (can't use indexes!)

✓ dim_date with pre-computed:

is_weekend, is_holiday, fiscal_quarter, week_number

→ Just JOIN + filter: WHERE d.is_holiday = 1

→ 10x faster than DATEPART() on large tables!

MISTAKE 4: Snowflake schema for BI tools

Power BI, Tableau don't handle snowflake well:

- Multiple joins = slow reports
- Complex model = confused analysts
- Star schema = happy BI team!

💡 PRO TIP — 15+ LPA LEVEL

ADVANCED CONCEPT: Conformed Dimensions

"In enterprise data warehouses, I use CONFORMED DIMENSIONS:

Conformed = SAME dimension used across MULTIPLE fact tables
with IDENTICAL grain (definition)

Example at RetailMart:

dim_date used by:

- *fact_sales* (daily sales)
- *fact_inventory* (daily stock levels)
- *fact_hr_attendance* (daily attendance)
- *fact_campaign* (daily marketing spend)

Same *dim_date* → can JOIN any 2 fact tables by date!

'Cross-fact analysis':

Sales HIGH on days with HIGH marketing spend? (join on *date_key*)

Inventory LOW when sales HIGH? (join on *date_key*)

Without conformed dimensions → these analyses impossible!

RULE: 'Conform before you build'

Define dimensions once → reuse everywhere

*If you build dim_date differently for each department
→ can't join across departments → data silos!*

This concept distinguishes 15+ LPA architects from
10 LPA implementers. Always mention it!"

? QUESTION 14 — Intermediate | Databricks Performance Optimization

"A PySpark job that processes 500GB daily is taking 4 hours. How would you debug and optimize it?"

✓ IDEAL ANSWER

SYSTEMATIC APPROACH TO SPARK OPTIMIZATION:

STEP 1: DIAGNOSE FIRST (before optimizing!)

Open Spark UI (Databricks → Cluster → Spark UI):

- Jobs tab: Which job takes longest?
- Stages tab: Which stage takes longest?
- Tasks tab: Any tasks much slower than others?

Look for:

- Data skew: one task takes 3h, others take 3 min → skew!

- Spill: tasks spilling to disk → not enough memory
- Shuffle: large shuffle read/write → too much data moving
- GC time: > 10% time in garbage collection → memory pressure

STEP 2: COMMON CAUSES AND FIXES

CAUSE 1: READING TOO MUCH DATA

Symptom: Stage 1 reads 500GB even with filters

Fix A: Check predicate pushdown

df.explain() → look for PushedFilters

If no PushedFilters → filter not pushed to file reader!

Fix B: Partition pruning

Is data partitioned? Does your filter use partition columns?

data at: /sales/year=2024/month=6/

filter: WHERE year=2024 AND month=6 ← partition pruning!

filter: WHERE YEAR(sale_date)=2024 ← NO pruning (function!)

CAUSE 2: DATA SKEW (most common performance killer!)

Symptom: 99 tasks finish in 2 min, 1 task runs for 2 hours!

Diagnosis:

df.groupBy(spark_partition_id().alias("pid")).count().show()

→ Partition 47: 45,000,000 rows (skewed!)

→ Other partitions: ~500,000 rows each

Fix A: Enable AQE (Adaptive Query Execution)

spark.conf.set("spark.sql.adaptive.enabled", "true")

spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")

→ Spark automatically splits skewed partitions!

Fix B: Salting (for GROUP BY skew)

Skewed key: customer_id = "WALK-IN" (60% of data!)

Add random salt to key:

```
df.withColumn("salt", (rand()*10).cast("int"))  
  .withColumn("key_salted", concat(col("customer_id"),  
lit("_"), col("salt")))
```

→ WALK-IN splits into WALK-IN_0, _1, ..._9

→ Now distributed across 10 partitions!

CAUSE 3: TOO MANY SMALL FILES (small files problem)

Symptom: 10,000 tasks each reading 1MB file (wasteful!)

Fix: Compact before reading

```
spark.conf.set("spark.sql.files.maxPartitionBytes",  
               str(256 * 1024 * 1024)) # 256MB per partition
```

→ Spark merges small files into ~256MB chunks

→ 10,000 tasks → 50 tasks!

CAUSE 4: SHUFFLE PARTITIONS (default 200 is wrong!)

Symptom: After groupBy, 200 tasks but data is tiny

OR 200 tasks and each is 10GB (too large!)

Rule: 128MB to 256MB per partition is optimal

Fix: Tune shuffle partitions

```
data_after_shuffle_GB = 50
```

```
optimal_partitions = max(200, int(data_after_shuffle_GB * 4))
```

```
spark.conf.set("spark.sql.shuffle.partitions",  
optimal_partitions)
```

With AQE: automatic coalescing handles this!

CAUSE 5: BAD JOINS (SortMergeJoin instead of BroadcastHashJoin)

Symptom: Simple join taking 30+ minutes

Check: df.explain() → SortMergeJoin (bad for small tables!)

Fix: Broadcast small tables

```
df_large.join(broadcast(df_small), on="key")
```

Configure threshold:

```
spark.conf.set("spark.sql.autoBroadcastJoinThreshold",  
               str(200 * 1024 * 1024)) # auto-broadcast <  
200MB
```

STEP 3: OPTIMIZATION RESULTS

Before optimization:

500GB data, 4 hours, Cost = \$120/run

After optimization:

- Added **partition** pruning: reads 125GB (not 500GB) = 4x less data
- Fixed skew with AQE: eliminated 2-hour stragglers
- Broadcast joins: eliminated shuffle for 3 dimension joins
- **Result**: 45 minutes, Cost = \$18/run
- 5x faster, 85% cost reduction!

SIMPLE EXPLANATION

DEBUGGING SPARK = DEBUGGING TRAFFIC JAM

Step 1: FIND WHERE TRAFFIC JAM IS

Spark UI = Traffic monitoring dashboard

Which road (stage) is congested?

Which vehicle (task) is stuck?

Step 2: IDENTIFY WHY

Too much traffic entering → Reading too much data

→ Solution: Filter earlier, partition pruning

One lane handling all traffic → Data skew

→ Solution: Salt the **key**, more lanes (AQE)

All cars stopping **to** exchange info → Shuffle

→ Solution: Pre-sort, broadcast small sets

Too many tiny cars → Small files

→ Solution: Merge **into** buses (coalesce)

Step 3: FIX **AND** MEASURE

Always MEASURE before **and** after!

"Before: 4 hours, After: 45 minutes"

Numbers make interviews concrete!

COMMON MISTAKES

MISTAKE 1: Optimizing **without** diagnosing

 "Let me add more nodes to the cluster!"

→ Might **not** help! Skew = **100** nodes = still **1** task dominates!

 Always **check** Spark UI **FIRST**

→ Diagnose the ROOT CAUSE

→ Apply targeted fix

MISTAKE 2: Increasing cluster size **as first** solution

 "Job is slow? Add more nodes!"

→ Expensive **and** often ineffective!

 Profile **first**:

Is it data skew? → Salting (**free**!)

Is it small files? → Config change (**free**!)

Is it bad `join`? → Broadcast hint (`free`!)
→ Hardware scaling should be `LAST` resort!

MISTAKE 3: Caching everything

- ❌ "I cache all DataFrames just in case"
 - Wastes memory → OOM errors
- ✅ Cache `ONLY` when same DF used in 2+ actions
- ✅ Always `unpersist()` when done!
- ✅ Check available memory before caching

MISTAKE 4: Not knowing `explain()`

Senior interview: "How do you debug a slow Spark job?"

- ❌ "I look at the output"
 - ✅ "I use `df.explain('formatted')` to see:
 - Physical plan
 - Which filters are pushed down
 - Which joins are broadcast vs sort-merge
 - Estimated row counts (for CBO)
- Then I look at Spark UI to see actual runtimes"

💡 PRO TIP — 15+ LPA LEVEL

PERFORMANCE OPTIMIZATION FRAMEWORK (structure your answer):

"I follow a systematic 5-step optimization framework:

STEP 1: PROFILE

- Spark UI: find bottleneck stage/task
- Check: skew, spill, shuffle size, GC time
- `df.explain()`: verify pushdown, join strategy

STEP 2: DATA REDUCTION

- Predicate pushdown (filter at file level)
 - Partition pruning (skip irrelevant partitions)
 - Column pruning (select only needed columns)
- Goal: Read minimum data possible

STEP 3: COMPUTE OPTIMIZATION

- Fix data skew (AQE or salting)
- Optimize shuffle partitions (AQE coalescing)
- Cache reused DataFrames
- Avoid UDFs (use built-in functions)

STEP 4: JOIN OPTIMIZATION

- Broadcast small tables (< 200MB)
- Pre-partition by join key (avoid repeated shuffles)
- Push filters before joins (join less data)

STEP 5: CLUSTER SIZING (last resort)

- Right-size: match worker nodes to data size
- Use memory-optimized nodes for large shuffles
- Consider spot instances for batch jobs (60% cost saving!)

This framework shows SYSTEMATIC THINKING.

Not just random tips – a METHODOLOGY.

'I used this framework to reduce a daily 6-hour Databricks job to 35 minutes, saving \$85,000/year in cloud costs'

QUANTIFY YOUR IMPACT = 15+ LPA mindset!"

? QUESTION 15 — Intermediate | Delta Lake Fundamentals

"What is Delta Lake? Explain its key features. Why is it better than plain Parquet?"

✓ IDEAL ANSWER

Delta Lake = OPEN-SOURCE storage layer
that adds DATABASE RELIABILITY to data lakes

WHAT DELTA LAKE ADDS TO PARQUET:

Parquet = Just a file format (very fast for reads, no metadata)
Delta = Parquet files + Transaction Log (the secret!)

THE TRANSACTION LOG:

_delta_log/ folder next to your data files

JSON files recording every change:

- Version 0: {initial write, added 100 files}
- Version 1: {appended 5 new files}
- Version 2: {deleted 3 files (DELETE operation)}
- Version 3: {updated 2 files (UPDATE operation)}

This log gives Delta its SUPERPOWERS!

KEY FEATURES (with examples):

1. ACID TRANSACTIONS:

Atomic: either ALL rows write or NONE do (no partial writes!)

Consistent: schema enforced (no bad data types!)

Isolated: concurrent readers see consistent snapshot

Durable: committed data survives crashes

Real value: Pipeline crashes at 2 AM?

Readers see last COMPLETE version, not partial write!

2. SCHEMA ENFORCEMENT:

Plain Parquet: write any schema → corrupt your table!

Delta: schema violation → operation REJECTED

Example:

Table schema: amount DOUBLE

Someone writes: amount STRING

→ Delta: "Schema mismatch! Rejected!"

→ Parquet: "Sure, mixed types now!" (corrupted!)

3. SCHEMA EVOLUTION:

Need to ADD a new column? Delta allows it safely:

```
df.write.option("mergeSchema", "true").format("delta")
```

→ Old files: new column shows as NULL

→ New files: have new column values

→ Data still queryable!

4. TIME TRAVEL (game changer!):

Query ANY historical version:

-- Version as of yesterday:

```
SELECT * FROM sales VERSION AS OF 5
```

-- State at specific time:

```
SELECT * FROM sales TIMESTAMP AS OF '2024-06-14'
```

Restore table:

```
RESTORE TABLE sales TO VERSION AS OF 5
```

Use cases:

→ Debugging: "What did the data look like before that bug?"

- **Audit**: "Who changed what and when?"
- **Rollback**: "Pipeline wrote wrong data → restore!"

5. UPDATE/DELETE/MERGE:

Not possible in plain Parquet!

Plain Parquet: Delete from 500GB file?

- Must REWRITE entire file!

Delta: DELETE FROM sales WHERE status='CANCELLED'

- Delta marks those rows as deleted
- Only affected files rewritten
- MUCH more efficient!

6. STREAMING + BATCH (unified):

Same Delta table can receive:

- Batch writes (daily pipeline)
- Streaming writes (real-time)
- Both without conflict!
- Delta handles concurrent writes with optimistic locking

SIMPLE EXPLANATION

ANALOGY: Plain Parquet = Notebook vs Delta Lake = Google Docs

PLAIN PARQUET FILE = Paper notebook

- Fast to read (all data in one place)
- But: no track changes, no undo
- Someone tears out a page (delete) → page gone
- Two people write at same time → chaos!
- See what notebook looked like last week? Impossible!

DELTA LAKE = Google Docs

- Same fast reading (Parquet files underneath)
- PLUS: **full** revision history (**time** travel!)
- PLUS: track **all** changes (transaction log)
- PLUS: two people editing → smart conflict resolution
- PLUS: undo changes (RESTORE!)
- See document **from any date**: ✅

Google Docs = same content **as** paper notebook

+ **all** the collaboration/safety features

DELTA LAKE = same data **as** Parquet

+ **all** the reliability/history features

DELTA LAKE MAGIC EXPLAINED:

You write data → Delta creates Parquet files (fast!)

Also creates JSON entry **in** `_delta_log/`

→ "Added: file001.parquet (rows 1-1000000)"

You **DELETE some rows** → Delta doesn't delete from file!

→ Creates new Parquet file **WITHOUT** those rows

→ Marks old file as "removed" in `_delta_log/`

Reading: Delta reads log → determines **CURRENT** active files

→ reads only those files

Time travel: Read log version 3 → find active files then → read those



REAL-WORLD USE CASE

PRODUCTION INCIDENT SCENARIO (very common interview question!):



SITUATION:

"Pipeline wrote wrong exchange rates to 'exchange_rates' table.
500M sales records calculated wrong amounts!
Reports for last 3 days are incorrect."

WITHOUT DELTA LAKE:

- You need to re-run pipeline from scratch
- Source data might be gone/changed
- MANUAL CORRECTION for 500M records?
- Hours/days of work
- Business decisions already made on wrong data!

WITH DELTA LAKE:

Step 1: Find last correct version

DESCRIBE HISTORY exchange_rates

- Version 42: "2024-06-12 MERGE" ← last correct update
- Version 43: "2024-06-13 WRITE" ← bad data!

Step 2: Restore in ONE command

RESTORE TABLE exchange_rates TO VERSION AS OF 42

- Table restored to version 42 in seconds!
- All 3 days of bad data: gone!

Step 3: Rerun downstream pipelines

- Pipelines rerun with correct exchange rates
- Reports corrected

TIME TO FIX:

Without Delta: 2-3 days of manual work

With Delta: 2 minutes (RESTORE) + 1 hour (rerun pipelines)

THAT IS THE BUSINESS VALUE OF DELTA LAKE!

This story in an interview = immediate senior impression.

! COMMON MISTAKES

MISTAKE 1: Thinking Delta **and** Parquet **are** completely different

✗ "Delta doesn't use Parquet"

✓ "Delta IS Parquet files + transaction log on top"

This **is** subtle but important. Delta stores data **AS** Parquet.
The `_delta_log/` folder **is** what makes it Delta!

MISTAKE 2: **Not** cleaning up Delta **table** (VACUUM)

Delta keeps **ALL** old data files **for time** travel

After months: **10TB** of data **when current is 100GB!**

✓ Run VACUUM periodically:

```
VACUUM table_name RETAIN 168 HOURS -- keep 7 days
```

→ Removes data files **no** longer needed

→ Keeps **7** days **for time** travel (minimum recommended)

MISTAKE 3: **Not** optimizing Delta files (OPTIMIZE)

Many incremental appends → thousands **of** tiny files!

→ Slow queries (opening **10,000** files vs **50** big files)

✓ Run OPTIMIZE periodically:

```
OPTIMIZE table_name ZORDER BY (sale_date, store_id)
```

→ Compacts small files **into** **1GB** files

→ Z-ORDERS data **for** faster queries

→ Run weekly **or** after **large** loads

MISTAKE 4: **Not** knowing Z-ORDER

"I just run OPTIMIZE without ZORDER"

✓ ZORDER = co-locate related data **in** same files

```
OPTIMIZE sales ZORDER BY (store_id, sale_date)
```

→ Queries filtering **by** `store_id` + `sale_date` = **10x** faster!

→ Parquet **skip** entire files that don't **match!**

💡 PRO TIP — 15+ LPA LEVEL

DELTA LAKE PRODUCTION OPERATIONS (shows operational maturity):

"In production, I set up automated Delta maintenance:

1. DAILY OPTIMIZATION:

```
OPTIMIZE silver_sales ZORDER BY (sale_date, store_id)
```

Run after nightly load → prepares data for morning queries

2. WEEKLY VACUUM:

```
VACUUM silver_sales RETAIN 168 HOURS -- 7 days
```

Balance: time travel ability vs storage cost

3. TABLE PROPERTIES (configure per table):

```
ALTER TABLE silver_sales SET TBLPROPERTIES (  
  'delta.autoOptimize.optimizeWrite' = 'true', -- auto  
  optimize on write  
  'delta.autoOptimize.autoCompact'    = 'true', -- auto  
  compact small files  
  'delta.logRetentionDuration'         = 'interval 30 days',  
  'delta.deletedFileRetentionDuration' = 'interval 7 days'  
);
```

4. MONITORING DELTA HEALTH:

```
DESCRIBE DETAIL silver_sales
```

→ numFiles, sizeInBytes, partitionColumns

If numFiles > 10,000 → need OPTIMIZE urgently!

5. LIQUID CLUSTERING (new in Delta Lake 3.0+):

Alternative to ZORDER for frequently changing cluster cols:

```
CREATE TABLE sales CLUSTER BY (sale_date, region_code)
```

→ Dynamic clustering, no manual ZORDER needed!

UNDERSTANDING TABLE PROPERTIES = senior ops knowledge.
Mentioning auto-optimize = shows you've managed production tables!"

REVISION SUMMARY — Questions 11-15

REVISION: SQL + DATA MODELING + SPARK (Q11-Q15)

Q11: Window Functions

ROW_NUMBER (unique) | RANK (ties+gaps) | DENSE_RANK
(ties)|

LAG/LEAD, SUM/AVG OVER = running totals

KEY: PARTITION BY = separate window per group!

Q12: SCD Types

Type 1 = OVERWRITE (no history, for corrections)

Type 2 = KEEP HISTORY (effective dates + is_current)

KEY: Update watermark AFTER load, surrogate keys in DW!

Q13: Star vs Snowflake

Star = denormalized = faster BI queries

Snowflake = normalized = complex but less storage

KEY: Star **for** BI (always!), dim_date **is** critical!

Q14: Spark Optimization

Diagnose **FIRST** (Spark UI), **then** fix root cause

Skew → AQE/Salting | Small files → config

Broadcast joins | **Partition** pruning

KEY: Quantify impact! "From 4h to 45min = 5x faster"

Q15: Delta Lake

Parquet + Transaction Log = ACID + **Time** Travel + **MERGE**

KEY: RESTORE **TABLE for** production incidents!

OPTIMIZE + ZORDER **for** query performance

VACUUM **for** storage cleanup

INTERVIEW **PATTERN** ALERT:

PATTERN 4: Always quantify your answers

"reduced from 4h to 45min" > "I made it faster"

PATTERN 5: Production incident stories impress most
Have **2-3 real** incident stories ready:

"Pipeline failed at 2 AM, I fixed it by..."

PATTERN 6: Know the WHY **not** just the HOW
"WHY Delta over Parquet" tests conceptual depth

LEVEL 3 — ADVANCED

"This is where 15+ LPA interviews happen"

? QUESTION 16 — Advanced | End-to-End Pipeline Design

"Design a complete Azure data engineering solution for a retail company that processes 1 million transactions daily, needs real-time fraud detection, and historical analytics for 5 years."

✓ IDEAL ANSWER — SYSTEM DESIGN

REQUIREMENTS ANALYSIS (always start here!):

FUNCTIONAL REQUIREMENTS:

- Daily batch: 1M transactions/day (batch processing)
- Real-time: fraud detection (<5 second latency)
- Historical: 5 years of data (~1.8 billion rows!)
- Reporting: daily business dashboards (by 9 AM)
- Compliance: data retention 7 years, PII masking

NON-FUNCTIONAL REQUIREMENTS:

- Availability: 99.9% (8.7 hours downtime/year max)
- Data freshness: batch data by 6 AM, real-time by 5 seconds
- Recovery: RTO 4 hours, RPO 1 hour
- Cost: optimize for batch (predictable) + pay-per-use (real-time)
- Security: PCI-DSS compliance (payment data!)

COMPLETE ARCHITECTURE:

LAYER 1: INGESTION

BATCH PATH (1M transactions, nightly):

Oracle POS Systems → (SHIR) → ADF Pipeline

- Runs: 1:00 AM daily (Tumbling Window Trigger)
- Destination: ADLS Gen2 Bronze layer (Parquet)
- Pattern: Incremental (watermark on transaction_timestamp)
- Parallel: 8 threads (Dynamic Range partition on

transaction_id)

→ Estimated time: 30 minutes for 1M rows

REAL-TIME PATH (fraud detection):

POS Systems → Azure Event Hubs (streaming API)

→ Partition: 32 partitions (high throughput)

→ Retention: 1 day (Event Hubs storage)

→ Consumer: Databricks Structured Streaming

LAYER 2: STORAGE (ADLS Gen2)

Bronze Container (raw):

/bronze/transactions/year=2024/month=06/day=15/*.parquet

/bronze/customers/year=2024/month=06/*.parquet

Silver Container (cleaned):

/silver/transactions/ (Delta format)

/silver/dim_customers/ (SCD Type 2 Delta)

Gold Container (business-ready):

/gold/daily_store_performance/ (Delta, Z-ORDERED)

/gold/fraud_alerts/ (Delta, appended real-time)

RETENTION POLICY:

Bronze: 7 years (compliance)

Silver: 5 years

Gold: 2 years (recreatable from Silver)

TIERING:

Hot: recent 90 days (frequent access)

Cool: 90 days - 2 years (infrequent)

Archive: 2-7 years (rare access, cheap!)

LAYER 3: PROCESSING

BATCH PROCESSING (Databricks Job Clusters):

Bronze → Silver:

Notebook: NB_Clean_Transactions

- Data quality checks
- Standardize columns
- SCD Type 2 for dim_customers
- Write Delta tables

Cluster: 4 workers × Standard_DS4_v2

Schedule: Daily 1:30 AM (after ingestion)

Silver → Gold:

Notebook: NB_Build_Gold_Layer

- Daily store performance aggregations
- Customer 360 view
- Product performance
- Z-ORDER for BI performance

Cluster: 8 workers (more data to aggregate)

Schedule: Daily 3:00 AM

REAL-TIME PROCESSING (Databricks Streaming):

Databricks cluster (always-on, 2 workers):

```
df_stream = spark.readStream.format("eventhubs")...
```

```
df_fraud = df_stream
```

```
.withColumn("is_high_value", col("amount") > 100000)  
.withColumn("is_unusual_hour", hour("timestamp") < 5)  
.withColumn("fraud_score",  
    when(col("is_high_value") & col("is_unusual_hour"), 3.0)  
    .when(col("is_high_value"), 2.0)  
    .otherwise(0.0))  
.filter("fraud_score >= 2.0")
```

```
df_fraud.writeStream.format("delta")
```

```
.table("gold.fraud_alerts") # writes to Delta
```

```
.trigger(processingTime="10 seconds") # micro-batch
```

Fraud alert: 10-second trigger = 10-second latency ✅

LAYER 4: SERVING

Power BI Direct Lake:

- Connects to Gold Delta tables
- No import needed (Direct Lake mode)
- Reports always show latest data
- Store performance dashboards
- Regional revenue analysis

Azure Synapse Serverless SQL:

- Ad-hoc queries by data analysts
- No cluster cost (pay per query)
- Queries ADLS Silver/Gold Parquet files

API for Fraud Alerts:

- Azure Function reads from fraud_alerts Delta table
- Returns real-time fraud status per transaction
- Response: fraud_score, reason, recommended_action

LAYER 5: ORCHESTRATION (ADF)

Master Pipeline (PL_RetailMart_Daily):

1. Validation (file arrival check)
2. Parallel ingestion (all sources)
3. Silver transformation trigger
4. Gold aggregation trigger
5. Data quality report
6. Success notification

LAYER 6: MONITORING & ALERTING

Azure Monitor:

- Pipeline failure alerts
- Cluster health monitoring
- Storage capacity alerts

Custom Audit Table:

- Every pipeline: start/end/rows/status logged
- Operations dashboard: "What ran today?"

SLA Dashboard (Power BI):

- "Did all pipelines complete before 6 AM?"
- Pipeline success rate trend (goal: 99%+)

LAYER 7: SECURITY

Azure Key Vault:

- All credentials (DB passwords, API keys)
- ADF + Databricks use MSI to access Key Vault

Network Security:

- Private Endpoints for all Azure services
- No public internet access for data services
- VNet peering between ADLS, ADF, Databricks

Data Security:

- Column masking: customer_phone, pan_number (PCI-DSS!)
- Row-level security: regional managers see own region only
- Audit logs: all data access logged to Log Analytics

Encryption:

- At rest: AES-256 (Azure default)
- In transit: TLS 1.2 minimum
- Customer Managed Keys for extra compliance

💡 PRO TIP — 20+ LPA LEVEL

ARCHITECTURE DECISION RECORDS (ADRs):

"For each major decision, I document WHY:

ADR-001: Delta Lake over plain Parquet

Decision: Use Delta Lake for Silver and Gold layers

Reason: Need UPDATE/DELETE (SCD Type 2), Time Travel
for debugging, ACID for reliable writes

Trade-off: Slightly more storage (transaction log)

Accepted: Yes, storage cost < reliability value

ADR-002: Tumbling Window vs Schedule Trigger

Decision: Tumbling Window for daily load

Reason: Backfill capability for reruns,
trigger dependency for pipeline chaining,
retry specific windows on failure

Trade-off: Slightly more complex to set up

Accepted: Yes

ADR-003: Databricks streaming vs ADF Event Trigger

Decision: Databricks Structured Streaming for fraud

Reason: Sub-minute latency needed (5 seconds),
complex ML scoring logic needed,
Event Trigger + ADF = minutes latency minimum

Trade-off: Always-on cluster = higher cost

Accepted: Yes, fraud prevention ROI >> cluster cost

Documenting ADRs shows:

- Architectural maturity
- Trade-off awareness
- Communication skills
- Long-term thinking

This is what STAFF ENGINEER / SOLUTION ARCHITECT level shows!"

? QUESTION 17 — Advanced | Batch vs Real-Time Processing

"Compare batch processing and real-time (streaming) processing. How do you decide which to use?"

✓ IDEAL ANSWER

BATCH PROCESSING:

What: Process data in large chunks, at scheduled intervals

Latency: Minutes to hours (acceptable delay)

Volume: Very large (GBs to TBs per run)

Examples: nightly ETL, monthly reports, ML model training

Azure Tools:

→ ADF pipelines (orchestration)

→ Databricks (batch Spark jobs)

→ Azure Synapse Analytics (SQL queries)

Cost: Generally cheaper (cluster auto-terminates after job)

Complexity: Lower (simpler state management)

Error handling: Easier (rerun failed batches)

REAL-TIME / STREAMING:

What: Process data continuously, as it arrives

Latency: Milliseconds to seconds (low latency required)

Volume: Lower per-message but continuous flow

Examples: fraud detection, live dashboards, IoT alerts

Azure Tools:

- Azure Event Hubs (message queue)
- Databricks Structured Streaming (Spark streaming)
- Azure Stream Analytics (simple streaming SQL)
- Azure Fabric EventStream + Real-time Intelligence

Cost: Higher (cluster must stay running always)

Complexity: Higher (state management, checkpoints, exactly-once)

Error handling: Harder (can't just "rerun")

DECISION FRAMEWORK:

Question	Batch	Streaming
Can I wait hours for result?	Yes	No
Need sub-minute response?	No	Yes
Analyzing historical data?	Yes	No
Detecting live anomalies?	No	Yes
Need to join large history files?	Yes	No
Alert on real-time pattern?	No	Yes
Monthly/weekly reports?	Yes	No
Live operational dashboard?	No	Yes
Cost-sensitive workload?	Yes	→
Low-latency requirement?	→	Yes

LAMBDA ARCHITECTURE (most enterprises use this!):

SPEED LAYER (real-time) + BATCH LAYER = Lambda

Payment arrives:

- Speed layer (streaming): fraud check in 5 seconds
- Batch layer (nightly): full reconciliation, reporting

Benefits:

- Fast alerts + accurate historical analysis
- Failure in speed layer? Batch fills in
- Best of both worlds!

KAPPA ARCHITECTURE (simpler, modern):

Just ONE streaming layer for everything
Batch = streaming with very large batches!

Used by: Netflix, LinkedIn (high-maturity orgs)

Easier to maintain (one codebase)

Requires: very fast streaming platform



REAL-WORLD USE CASE

BARCLAYS BANK DUAL ARCHITECTURE:

REAL-TIME (Streaming):

Goal: Block fraudulent card transactions in <3 seconds

Flow:

ATM/POS → Azure Event Hubs → Databricks Streaming

→ Rule engine: amount > 1L AND new device AND night → FRAUD

→ Score: 0-10 (10 = definitely fraud)

→ Score ≥ 7: BLOCK transaction (Azure Cosmos DB → POS)

→ Score 4-6: FLAG for review (write to fraud_alerts Delta)

→ Score < 4: ALLOW

Latency: 800ms (well under 3 second requirement)

False positive rate: <0.1% (tuned ML model)

BATCH (Daily):

Goal: Reconcile all transactions, detect patterns that real-time couldn't catch

Flow:

All day's transactions → ADF → Silver layer

→ Complex ML model (cross-transaction analysis)

→ Pattern: "3 small transactions before large one" (testing card)

→ Flag for next-day review

→ Generate compliance reports

Run at: 1 AM (use compute reserved for off-peak hours)

Result: Reports by 6 AM for compliance team

WHY BOTH?

Real-time catches: obvious fraud (large amount, bad device)

Batch catches: complex patterns (multiple small transactions, velocity attacks, cross-account analysis)

Together = best fraud detection possible!

One alone = weaker protection

! COMMON MISTAKES

MISTAKE 1: "All data problems need streaming"

✗ "Let's stream everything for freshness!"

→ Cost: always-on clusters = 10x more expensive!

→ Complexity: streaming is HARD to debug

- ✓ Use streaming **ONLY when** latency truly requires it
- ✓ Near-**real-time** (5 min delay OK?) → micro-batch
- ✓ Sub-**second** needed → **true** streaming

MISTAKE 2: **Not** knowing micro-batch

Databricks Structured Streaming:

✗ "It processes every event individually"

✓ "It uses MICRO-BATCH processing:

Groups events into small batches (every 10 seconds)

Processes each batch as a mini-Spark job

Not true event-by-event, but very low latency!"

True event-by-event = Apache Flink (different tool)

MISTAKE 3: Forgetting checkpointing

✗ Stream job fails → restart **from** beginning!

✓ Checkpoint location records:

"Processed up to offset 1,234,567 in partition 0"

On restart: continues **from** exactly **where** it stopped

EXACTLY-ONCE processing guaranteed!

MISTAKE 4: **Not** understanding state management

Streaming aggregation: "Count transactions per user per hour"

→ Need **to** REMEMBER counts **between** micro-batches!

→ This **is** STATE management

→ Requires: stateful operations (watermarks, windows)

✓ Databricks handles state automatically

But YOU need **to set** watermarks **to** prevent state **from** growing forever!

💡 **PRO TIP — 15+ LPA LEVEL**

STRUCTURED STREAMING WATERMARK (shows deep streaming knowledge):

"In streaming, late data is inevitable:

Transaction happened at 3:00 PM

Due to network issues, arrives at 3:15 PM

I'm aggregating by 5-minute windows

The 3:00-3:05 window is already 'closed'!

WITHOUT WATERMARK:

Keep ALL past windows in memory (forever growing state → OOM!)

WITH WATERMARK:

Tell Spark: 'Wait max 15 minutes for late data'

```
df.withWatermark('transaction_time', '15 minutes')  
  .groupBy(window('transaction_time', '5 minutes'), 'region')  
  .sum('amount')
```

Behavior:

- 3:15 PM arrives → check: is it within 15-min watermark?
- Latest event seen: 3:15 PM. Watermark = 3:00 PM
- 3:00-3:05 window: still accepting until 3:15 watermark passes
- 3:01 PM event → counted in 3:00-3:05 window! ✅
- 2:45 PM event (very late) → DROPPED (outside watermark)

After watermark passes → state released from memory

- Prevents OOM
- Accepts reasonable late data
- Drops very late data (trade-off: completeness vs memory)

Understanding watermarks = streaming EXPERT level.

Most candidates fail this."

? QUESTION 18 — Advanced | Azure Architecture Design

"Design a complete Azure Data Platform for a bank that needs to migrate from on-premises to cloud. Constraints: zero downtime, PCI-DSS compliance, 10TB of historical data."

✓ IDEAL ANSWER

PHASE 1: ASSESSMENT & PLANNING (Month 1-2)

Inventory current state:

- 15 Oracle databases (on-premise)
- 10TB historical data
- 50+ SSIS packages
- SQL Server Reporting Services reports
- 200+ daily batch jobs

Azure architecture decision:

- ADLS Gen2 (primary storage)
- ADF (orchestration + SSIS migration)
- Databricks (heavy transformation)
- Azure Synapse (data warehouse)
- Power BI Premium (reporting)
- Azure Key Vault (secrets)
- Azure Monitor + Log Analytics (monitoring)
- Azure Purview (data governance + lineage)

PHASE 2: INFRASTRUCTURE SETUP (Month 2-3)

Network architecture (PCI-DSS critical!):

→ ExpressRoute (dedicated private connection, **not** public internet!)

- Hub-**and**-Spoke VNet topology
- Private Endpoints **for ALL** Azure services
- **No** data touches public internet
- Network Security **Groups** (NSG) **for each** subnet

Subnet design:

- DE Subnet: ADF, Databricks
- Storage Subnet: ADLS, Key Vault (private endpoints)
- Analytics Subnet: Synapse, Power BI Gateway
- Management Subnet: SHIR servers, Jump boxes

Security setup:

- Azure AD **with** MFA **for all** users
- RBAC **at** subscription, resource **group**, resource level
- Azure Policy: enforce private endpoints, encryption, tagging
- Microsoft Defender **for** Cloud: enabled **on all** resources
- Azure Sentinel: SIEM **for** security events

PCI-DSS **specific**:

- Encryption: AES-**256** **at** rest + TLS **1.2** **in** transit
- Customer Managed Keys (CMK) **for** ADLS + Key Vault
- Network segmentation (payment data isolated subnet)
- Audit logging: **all** data access logged (Azure Monitor)
- Data masking: PAN, CVV, account numbers masked

PHASE **3**: LIFT **AND** SHIFT HISTORICAL DATA (**Month 3-5**)

Zero-downtime strategy:

- Keep **on**-premise **running** (source **of** truth initially)
- **Copy** historical data **in** background (**no** downtime!)
- Validate **in** parallel
- **Only** cutover AFTER validation

Historical load approach:

- ADF DistCp: bulk copy 10TB from Oracle to ADLS
- Estimated time: 10TB at 4 GB/s = 2,500 seconds ≈ 42 minutes!
- Run on weekend (minimal load)
- Validate: row counts, checksums, sample data

PHASE 4: INCREMENTAL SYNC (Month 5-6)

While on-premise still running:

- ADF incremental pipeline: sync daily changes
- Azure stays up-to-date (24-hour lag initially)
- Reduce lag: hourly sync → 1-hour lag
- Final week: near-real-time CDC → minutes lag

PHASE 5: PARALLEL RUN & VALIDATION (Month 6-7)

- Both systems producing same outputs
- Compare: on-premise reports vs Azure reports
- Automated reconciliation scripts
- Sign-off from: data team, business users, compliance
- Require: <0.01% row count difference, exact matching on key metrics

PHASE 6: CUTOVER (Zero Downtime!)

BLUE-GREEN DEPLOYMENT:

Blue = On-premise (current)

Green = Azure (new)

Cutover steps:

1. Friday 11 PM: Stop batch jobs on on-premise
2. Final incremental sync (catches last-minute changes)
3. Final validation (15 minutes)
4. DNS switch: reports point to Azure Power BI
5. Monitor for 30 minutes

6. If issue detected: **ROLLBACK** DNS (back to on-premise in seconds!)

7. If **all** good: Monday morning business **on** Azure!

Zero downtime because:

- DNS switch = near-instant
- **On**-premise stays **running** during cutover
- **Rollback** available **for 48** hours post-cutover

PHASE **7**: DECOMMISSION (**Month 8-9**)

- Monitor Azure stability **for 1 month**
- Gradual **on**-premise decommission
- Archive historical backups
- Cost optimization review

💡 **PRO TIP — 20+ LPA LEVEL**

COST OPTIMIZATION STRATEGY (what companies really care about!):

"Cost optimization for this 10TB banking platform:

STORAGE OPTIMIZATION:

Hot tier (recent 90 days):	~2TB = \$40/month
Cool tier (90d - 2 years):	~5TB = \$50/month
Archive (2-7 years):	~3TB = \$3/month

vs ALL in Hot: 10TB = \$200/month

Savings: \$107/month = \$1,284/year (just from tiering!)

COMPUTE OPTIMIZATION:

ADF: pay per activity run (no idle cost)

Estimated: $500 \text{ runs/day} \times \$0.0001 = \$1.50/\text{day}$

Databricks: Job clusters (auto-terminate)

vs All-Purpose (always on)

Job cluster: $3 \text{ hours/day} \times \$2/\text{hour} \times 8 \text{ nodes} = \$48/\text{day}$

Always-on: $24 \text{ hours} \times \$2/\text{hour} \times 8 \text{ nodes} = \$384/\text{day}$

Savings: $\$336/\text{day} = \$122,640/\text{year!}$

Synapse: Serverless SQL (pay per TB scanned)

For ad-hoc queries: $\$5/\text{TB scanned}$

vs Dedicated Pool: $\$700+/\text{month}$

If $<140\text{TB scanned/month}$: Serverless is cheaper!

RESERVED INSTANCES:

For predictable workloads (Databricks streaming cluster):

1-year reserved: 40% discount

3-year reserved: 60% discount

Streaming cluster (always-on, 2 nodes):

On-demand: $\$2/\text{hr} \times 2 \times 8760\text{hr} = \$35,040/\text{year}$

3yr reserved: $\$35,040 \times 0.4 = \$14,016/\text{year}$

Savings: $\$21,024/\text{year!}$

TOTAL ANNUAL SAVINGS: $\sim \$145,000$

That ROI justification = executive presentation material.

15+ LPA engineers build business cases, not just code!"

? QUESTION 19 — Advanced | Query Optimization

"A SQL query that joins 3 tables on a 500M row fact table is taking 45 minutes. How do you optimize it?"

✓ IDEAL ANSWER

SYSTEMATIC QUERY OPTIMIZATION APPROACH:

STEP 1: GET THE EXECUTION PLAN FIRST

-- SQL Server / Synapse:

SET STATISTICS IO ON

SET STATISTICS TIME ON

EXPLAIN [your query] -- shows execution plan

-- Look for:

Table Scan → bad (reading entire table!)

Index Seek → good (using index)

Hash Join → for large tables (usually OK)

Nested Loop → for small tables (bad if large!)

Sort → potential bottleneck (no index on ORDER BY?)

Spill → running out of memory (temp disk used!)

STEP 2: DIAGNOSE THE ROOT CAUSE

Diagnose A: MISSING INDEXES (most common!)

Execution plan shows: "Missing Index Recommendation"

Or: Table Scan instead of Index Seek

Fix: Create index on filter/join columns

-- Create index on fact table join column (FK)

CREATE NONCLUSTERED INDEX IX_FactSales_DateKey

ON fact_sales(date_key) INCLUDE (gross_amount, customer_key)

-- Covering index (includes all needed columns)

-- = no "key lookup" needed → much faster!

Diagnose B: STATISTICS OUTDATED

Query optimizer uses statistics to estimate rows

Stale stats → wrong execution plan → slow!

Fix:

```
UPDATE STATISTICS fact_sales WITH FULLSCAN
```

```
UPDATE STATISTICS dim_date WITH FULLSCAN
```

```
-- Or enable auto-update statistics:
```

```
ALTER DATABASE RetailDW SET AUTO_UPDATE_STATISTICS ON
```

Diagnose C: DATA SKEW IN SYNAPSE DEDICATED POOL

Check: DBCC PDW_SHOWSPACEUSED('fact_sales')

→ Distribution 17: 80% of data (skewed!)

Fix: Change distribution key

```
CREATE TABLE fact_sales_new WITH (  
    DISTRIBUTION = HASH(customer_id) -- better distribution key  
) AS SELECT * FROM fact_sales
```

Choose distribution key = most common JOIN column!

Diagnose D: WRONG JOIN TYPE (Nested Loop on large table!)

Execution plan: Nested Loop Join on 500M × 100M rows

→ $O(N^2)$ complexity → runs forever!

Fix:

```
-- Force hash join:
```

```
SELECT ... FROM fact_sales f
```

```
INNER HASH JOIN dim_date d ON f.date_key = d.date_key
```

```
-- Hash join =  $O(N)$  not  $O(N^2)$ 
```

Diagnose E: IMPLICIT DATA TYPE CONVERSION

```
WHERE f.customer_id = 12345 (integer)
```

But customer_id column is VARCHAR!

→ Implicit conversion on EVERY ROW → 500M conversions!
→ Cannot use index!

Fix: Ensure data types match:

```
WHERE f.customer_id = '12345'  -- match the column type!
```

COMPLETE OPTIMIZED QUERY:

```
-- BEFORE (slow, 45 minutes):
```

```
SELECT
```

```
    d.month_name,  
    p.category,  
    s.region_code,  
    SUM(f.gross_amount) AS revenue
```

```
FROM fact_sales f
```

```
JOIN dim_date d ON f.date_key = d.date_key
```

```
JOIN dim_product p ON f.product_key = p.product_key
```

```
JOIN dim_store s ON f.store_key = s.store_key
```

```
WHERE YEAR(d.full_date) = 2024    ← FUNCTION on column (no  
index!)
```

```
GROUP BY d.month_name, p.category, s.region_code
```

```
ORDER BY revenue DESC
```

```
-- AFTER (optimized, 3 minutes):
```

```
SELECT
```

```
    d.month_name,  
    p.category,  
    s.region_code,  
    SUM(f.gross_amount) AS revenue
```

```
FROM fact_sales f
```

```
JOIN dim_date d ON f.date_key = d.date_key
```

```
JOIN dim_product p ON f.product_key = p.product_key
```

```
JOIN dim_store s ON f.store_key = s.store_key
```

```
WHERE d.year = 2024    ← Using PRE-COMPUTED column (index
```

usable!)

```
GROUP BY d.month_name, p.category, s.region_code
ORDER BY revenue DESC
```

OPTIMIZATIONS APPLIED:

✅ `YEAR(d.full_date)` replaced with `d.year` (pre-computed column in `dim_date`)

→ Index can be used on `d.year`!

✅ Added covering index on `fact_sales`:

```
CREATE INDEX IX_FactSales_Cover
ON fact_sales(date_key, product_key, store_key)
INCLUDE (gross_amount)
```

→ All join + select columns in one index

✅ Added index on `dim_date.year`:

```
CREATE INDEX IX_DimDate_Year ON dim_date(year) INCLUDE
(month_name)
```

→ Filter by year finds rows instantly!

✅ Ensured statistics are current:

```
UPDATE STATISTICS fact_sales WITH FULLSCAN
```

RESULT: 45 minutes → 3 minutes (15x faster!)

SIMPLE EXPLANATION

QUERY OPTIMIZATION = Finding the shortest route in Google Maps

Your query wants to go from A (data) to B (result).

BAD ROUTE (Table Scan):

"I don't know the exact route, so I'll drive down
EVERY street in the city until I find your house"
→ 500M streets driven = 45 minutes!

GOOD ROUTE (Index Seek):

"I know exactly which street and house number.
Drive directly there."
→ Direct route = 3 minutes!

INDEX = Street directory (A → Z)

"customer_id = '12345' → block 47, street 8, house 5"
→ Jump directly there!

WHY FUNCTION ON COLUMN BREAKS INDEX:

Index on: sale_date column

Your query: WHERE YEAR(sale_date) = 2024

SQL Server: "I have index for 'sale_date' values.

But your filter is on YEAR(sale_date)

That's a DIFFERENT value!

Can't use the index...

Must scan ALL rows and calculate YEAR() for each"

→ 500M YEAR() calculations!

FIX: Pre-compute year in table:

ALTER TABLE fact_sales ADD sale_year AS YEAR(sale_date)

PERSISTED

CREATE INDEX IX_SaleYear ON fact_sales(sale_year)

WHERE sale_year = 2024 ← now index works!

COMMON MISTAKES

MISTAKE 1: Adding indexes blindly without analyzing

❌ "I'll add 10 indexes and see which helps"

→ Each index = slower INSERTS/UPDATES

→ Too many indexes = INSERT performance disaster!

✅ Add index SPECIFICALLY for the slow query's columns

✅ Analyze EXECUTION PLAN first

MISTAKE 2: SELECT * in production queries

❌ SELECT * FROM fact_sales (returns all 50 columns!)

→ Reads 50 columns when you need 3

→ 17x more I/O than needed!

✅ SELECT only columns you need

✅ This enables "covering indexes" (all columns in index!)

MISTAKE 3: Not knowing difference between OLTP and OLAP indexing

OLTP database (SQL Server):

→ B-tree indexes for point lookups

→ Careful: too many = slow writes

OLAP/Data Warehouse (Synapse):

→ Clustered Columnstore Index (CCI) is the gold standard!

→ Stores data by column → analytics queries = lightning fast

→ 10x compression + fast aggregations

```
CREATE TABLE fact_sales WITH (  
    CLUSTERED COLUMNSTORE INDEX ← not row-based!  
)
```

CCI = the standard for ALL analytics tables!

MISTAKE 4: Not understanding Synapse-specific optimizations

Synapse has specific optimizations:

→ Distribution: HASH(most-joined column)

→ RESULT_SET_CACHING: cache repeated queries

→ Materialized Views: pre-compute common aggregations

RESULT_SET_CACHING:

SET RESULT_SET_CACHING **ON**

- Same query **in next 48** hours = instant response (**from** cache)!
- Cached **on** Synapse compute nodes
- Perfect **for** dashboards that run same query repeatedly!

💡 PRO TIP — 15+ LPA LEVEL

SYNAPSE-SPECIFIC MATERIALIZED **VIEWS** (advanced!):

"For dashboards that run same aggregate query repeatedly:

Traditional: Run full aggregation each time

```
SELECT region, month, SUM(revenue)
FROM fact_sales JOIN dim_date ON ...
GROUP BY region, month
→ 45 minutes each time!
```

Materialized View: Pre-compute **and** store result:

```
CREATE MATERIALIZED VIEW mv_monthly_regional_revenue
WITH (DISTRIBUTION = ROUND_ROBIN)
AS
SELECT
    d.year,
    d.month_number,
    d.month_name,
    s.region_code,
    SUM(f.gross_amount) AS monthly_revenue
FROM fact_sales f
JOIN dim_date d ON f.date_key = d.date_key
JOIN dim_store s ON f.store_key = s.store_key
```



```
GROUP BY d.year, d.month_number, d.month_name, s.region_code;
```

Now the query:

```
SELECT year, month_name, region_code, monthly_revenue  
FROM mv_monthly_regional_revenue  
WHERE year = 2024  
→ 3 seconds (from pre-computed view)!
```

AUTOMATIC REWRITE:

Synapse automatically uses materialized view even for:

```
SELECT SUM(f.gross_amount) ... from fact_sales ...
```

- Synapse sees it matches the view → uses view!
- No query change needed!
- Old queries automatically become faster!

This is SYNAPSE EXPERT knowledge.

Interviewers at Microsoft specifically test this!"

? QUESTION 20 — Advanced | Cost Optimization

"How do you optimize costs in an Azure data engineering setup? Give specific strategies."

✓ IDEAL ANSWER

COST OPTIMIZATION = Engineering skill, not just business concern

AZURE DATA FACTORY COST OPTIMIZATION:

1. CHOOSE RIGHT INTEGRATION RUNTIME:

Azure IR (pay per DIU-minute) vs Self-Hosted IR (fixed VM cost)

Short pipeline (< 1 hour): Azure IR cheaper

24/7 pipeline: SHIR (dedicated VM) cheaper

Calculate break-even:

Azure IR: $\$0.25/\text{DIU-hour} \times 4 \text{ DIU} = \$1/\text{hour}$

SHIR VM (Standard_D4s_v3): $\$0.20/\text{hour}$

Break-even: 5 hours/day of usage

If pipeline runs > 5 hours/day → SHIR is cheaper!

2. OPTIMIZE DIU USAGE:

Find minimum DIUs needed:

Test with: 4, 8, 16, 32 DIUs

Find: at what DIU count does throughput plateau?

Use: that count (don't overprovision!)

Example: 8 DIUs = 100 MB/s, 16 DIUs = 105 MB/s

→ Use 8 DIUs (5% speed gain not worth 2x cost!)

3. SCHEDULE DURING OFF-PEAK HOURS:

Reduce concurrent pipeline cost:

→ Run heavy pipelines 1-6 AM (low business demand)

→ Reserved capacity off-peak

DATABRICKS COST OPTIMIZATION:

4. JOB CLUSTERS vs ALL-PURPOSE CLUSTERS:

All-purpose: $\$384/\text{day}$ (8 nodes, 24/7)

Job cluster: $\$48/\text{day}$ (8 nodes, only during 3-hour job)

Savings: $\$336/\text{day} = \$122,640/\text{year}$ per cluster!

Rule: Development → All-purpose (need interactive)
Production → Job clusters (only pay when running!)

5. CLUSTER AUTO-SCALING:

Min workers: 2 (always on)
Max workers: 16 (burst capacity)

vs Fixed 8 workers:

Light periods: 2 workers (vs paying for 8)

Peak periods: auto-scales to 16

→ 40% average cost reduction!

6. SPOT INSTANCES (PREEMPTIBLE):

Use Azure Spot VMs for worker nodes

→ 60-80% cheaper than on-demand!

→ Risk: spot instance can be preempted (taken back by Azure)

→ Mitigation:

→ Driver = on-demand (must not be preempted!)

→ Workers = spot (can handle worker failure, reshuffles

task)

→ Databricks auto-handles spot instance preemption!

Savings: 60% on worker nodes

For 8 workers × \$2/hr × spot discount:

On-demand: \$16/hr

Spot: \$4/hr

Savings: \$12/hr × 3hr/night = \$36/day = \$13,140/year!

7. RIGHT-SIZE CLUSTER:

Don't use 8-core workers for simple CSV reads!

Workload types:

CPU-bound (ML): General Purpose nodes (balanced CPU/RAM)

Memory-bound (joins, shuffles): Memory Optimized

IO-bound (data movement): Storage Optimized

Wrong node type = expensive AND slow!

STORAGE COST OPTIMIZATION:

8. TIERED STORAGE:

Hot: recent 90 days = \$0.018/GB/month

Cool: 90 days - 2 years = \$0.01/GB/month

Archive: 2-7 years = \$0.00099/GB/month (182x cheaper!)

Lifecycle policy (automatic!):

→ Move to cool after 90 days

→ Move to archive after 730 days

10TB data:

All Hot: \$180/month

Tiered: \$40/month

Savings: \$140/month = \$1,680/year

9. DELTA VACUUM (Remove unused files):

Delta keeps all historical files by default!

After 6 months: 3x more storage than needed!

VACUUM table RETAIN 168 HOURS

→ Removes files not needed for time travel

→ Typically reduces storage 40-60%!

10. COMPRESSION:

Always use compressed formats:

CSV → 100GB

Parquet + Snappy → ~30GB (70% reduction!)

Parquet + ZSTD → ~20GB (80% reduction!)

$\$0.018/\text{GB/month} \times 80\text{GB saved} = \$1.44/\text{month}$

At scale (10TB): \$144/month saved = \$1,728/year

SYNAPSE COST OPTIMIZATION:

11. SERVERLESS vs DEDICATED SQL POOL:

Dedicated: \$700+/month (always running!)

Serverless: \$5/TB scanned (pay per query)

Breakeven: 140TB scanned/month

For ad-hoc queries (< 140TB/month): Serverless!

For heavy BI (> 140TB/month): Dedicated with PAUSE!

12. PAUSE DEDICATED POOL:

Run nightly jobs → Dedicated Pool

Day hours: Pause the pool (zero compute cost!)

6 hours nightly × \$0.25/DWU-hour × 1000 DWU = \$1.50/hr × 6 = \$9/night

vs 24/7: \$9/hr × 24 = \$36/day

Savings: \$27/day = \$9,855/year!

MONITORING COSTS:

13. AZURE COST MANAGEMENT + BUDGETS:

→ Set budget alerts: "Alert when > 80% of monthly budget"

→ Break down by resource tag (team, project, environment)

→ Weekly cost review meeting

Tags on all resources:

environment: prod | dev | test

team: data-engineering

project: retailmart

→ Filter cost by tag → accountability per team!

14. COST REPORTING DASHBOARD (Power BI):

Azure Cost Management data → Power BI

→ "Top 10 most expensive resources this month"

→ "Cost per pipeline run trend"

→ "Developer accidentally left cluster running overnight"

visibility!

TOTAL ANNUAL SAVINGS EXAMPLE:

Job clusters vs all-purpose:	\$122,640
Spot instances on workers:	\$13,140
Storage tiering:	\$1,680
Delta VACUUM:	\$5,000
Synapse pause strategy:	\$9,855
TOTAL:	\$152,315/year!

"I saved \$150K annual cloud cost through optimization.

That's a business case that gets you promoted!"

PRO TIP — 15+ LPA LEVEL

FINOPS MINDSET (what top companies value):

"I practice FinOps (Financial Operations for Cloud):

Key principle: Engineers own their cloud costs.

Not just "make it work" → make it work ECONOMICALLY.

MY APPROACH:

1. COST-AWARE ARCHITECTURE:

Before building: estimate monthly cost

'This streaming cluster = \$1,500/month. Is it justified?'
If not → redesign for batch processing (\$50/month)

2. COST TAGS ON EVERYTHING:

Every resource tagged: team, project, environment, cost-center
→ Finance can charge teams for their usage
→ Creates accountability (team sees THEIR bill!)

3. WEEKLY COST REVIEW:

Every Friday: review Azure Cost Management
Questions: 'What spiked this week? Why?'
Action: Kill unused resources, optimize expensive ones

4. COST OPTIMIZATION IN CODE REVIEW:

Review checklist includes:

- ✓ Is cluster the right size?
- ✓ Are spot instances used?
- ✓ Are Delta tables vacuumed?
- ✓ Is VACUUM in the pipeline?

5. COST AS A METRIC:

Track: Cost per 1000 rows processed
If increasing → investigate!
If decreasing → celebrate!

I've reduced client cloud bills by 40-60%
through systematic FinOps practices.
Companies love engineers who think about cost!"



REVISION SUMMARY — Questions 16-20

REVISION: ADVANCED TOPICS (Q16-Q20)

Q16: End-to-End Architecture

Always state requirements FIRST

Bronze → Silver → Gold Medallion

Batch + Streaming (Lambda Architecture)

KEY: Security (**Key** Vault, **Private** Endpoints, PCI-DSS!)

Q17: Batch vs Streaming

Batch = hours latency, cheap, simpler

Streaming = **sub**-minute, expensive, complex

KEY: Watermarks **for** late data **in** streaming!

Q18: Bank Migration

Zero downtime = Blue-Green deployment

PCI-DSS = ExpressRoute + **Private** Endpoints + Encryption

KEY: Parallel run + validation before cutover!

Q19: Query Optimization

EXPLAIN first → find bottleneck → targeted fix

Index **on** filter/**join** columns, pre-computed columns

CCI (Columnstore) **for** analytics tables

KEY: Never use functions **on** indexed columns!

Q20: Cost Optimization

Job clusters (**not** all-purpose) = **8x** saving

Spot instances = **60%** worker cost saving

Storage tiering = **40%** storage saving

KEY: Quantify impact! "\$152K annual savings"

INTERVIEW PATTERN ALERT:

PATTERN **7**: ALWAYS start **with** requirements

System design answers that start **with** requirements =

immediate senior signal!

PATTERN 8: Know the COST numbers

Spot instances = 60%, Job vs Always-on = 8x

Storage tiers, DIU costs → shows production experience

PATTERN 9: Security is non-negotiable

Every architecture question → mention Key Vault,

Private Endpoints, RBAC, Audit Logs

LEVEL 4 — SYSTEM DESIGN & BEHAVIORAL

"20+ LPA questions — Executive and Principal Engineer level"

? QUESTION 21 — System Design

"You're the lead data engineer. A pipeline that loads customer data suddenly starts taking 8x longer. Walk me through your complete debugging and resolution process."

✓ IDEAL ANSWER — STRUCTURED DEBUGGING

INCIDENT RESPONSE FRAMEWORK:

STEP 1: TRIAGE (first 5 minutes)

"How bad is this? What's the SLA impact?"

Check:

- How long is it taking now vs normally? (baseline comparison)
- Is it still running or did it fail?
- Is it affecting downstream jobs? (SLA breach risk?)
- Any alerts fired already?

Decide: Is this P1 (wake people up) or P2 (investigate solo)?

If downstream jobs already failing or SLA breached → P1

Escalate: notify lead, start incident bridge call

STEP 2: COMPARE (next 10 minutes)

"What changed?"

Compare recent run vs historical runs:

- **ADF Monitor:** duration trend graph (when did it start?)
- **ADF Activity Run:** which specific activity is slow?
- **Check:** last run OK was Monday, first slow was Tuesday

"Something changed between Monday and Tuesday."

Check: Was there a deployment Tuesday? Code change?

Did source data volume spike Tuesday?

Did someone change a table schema?

STEP 3: ISOLATE (next 15 minutes)

"Where exactly is the slowness?"

In ADF Monitor → drill into Activity Runs:

- GetMetadata: 2 sec (normal)
- CopyData: 4 hours (was 30 min!) ← HERE IT IS!
- Databricks: not reached yet

Drill into CopyData output:

```
{  
  "rowsRead": 5000000,    ← usually 500,000! (10x more data)  
  "rowsCopied": 5000000,  
  "throughput": 45 MB/s   ← same as before (not a network  
issue)  
  "duration": 3.5 hours  
}
```

DATA VOLUME SPIKED 10x! That's why 8x longer (close to proportional)

STEP 4: ROOT CAUSE ANALYSIS

"WHY did volume spike?"

Check source:

```
SELECT COUNT(*), MAX(LAST_UPDATED_DATE)  
FROM CUSTOMER_TABLE  
WHERE LAST_UPDATED_DATE > :last_watermark  
→ 5,000,000 rows (normal: 500,000!)  
→ Something mass-updated customers!
```

Check source change log / DBA:

"A migration script ran Tuesday that touched all customers"
→ All 5M customers got LAST_UPDATED_DATE = Tuesday

→ Watermark query picked up ALL 5M as "changed"!

ROOT CAUSE:

Migration script triggered `false` "update" on all customers
Watermark-based detection thinks all 5M are new/changed

STEP 5: IMMEDIATE FIX

"Stop the bleeding first, investigate later"

For today's run:

→ Manually update watermark to skip the migration batch:

```
UPDATE ControlTable
```

```
SET last_watermark = '2024-06-16 00:00:00'
```

```
WHERE table_name = 'CUSTOMERS'
```

→ Re-run today's incremental pipeline (now only gets today's `true` changes)

→ Runtime back to 30 minutes ✓

→ Downstream jobs unblocked ✓

STEP 6: LONG-TERM FIX

"Prevent recurrence"

Option A: Use CHECKSUM/HASH comparison

Instead of only watermark, ALSO check row hash:

Only process rows where hash of key columns CHANGED

→ Migration script: same values → same hash → NOT picked up!

Option B: Coordinate with DBA

Any mass update on source → notify data team first

→ Temporary pause watermark

→ Resume after migration complete

→ Only 1 day of incremental = manageable

Option C: CDC (Change Data Capture)

→ Database-level tracking of actual changes

- Migration "update" with same values → NOT captured as change!
- Most reliable solution but requires DBA effort

STEP 7: POST-INCIDENT (within 24 hours)

Write incident report:

- Timeline of events
- Root cause
- Immediate fix taken
- Long-term fix plan
- Prevention for future

Add monitoring:

- Alert if CopyData rows > 2x baseline → immediate investigation
- Daily row count trending in Power BI dashboard

STEP 8: RETROSPECTIVE

"How did our monitoring fail us?"

- Pipeline ran 8x longer before we noticed → monitoring gap!
- Add: duration SLA alert ("if copy activity > 1 hour, alert")
- Add: row count anomaly detection ("if rows > 2x avg, alert")
- Better: PROACTIVE alerts, not reactive!

PRO TIP — 20+ LPA LEVEL

COMMUNICATION DURING INCIDENT (equally important!):

"Technical skills get you to 12 LPA.

Incident communication skills get you to 20+ LPA.

DURING INCIDENT — keep stakeholders updated:

Email/Teams at T+0 (discovery):

'We detected PL_CustomerLoad is running 8x longer than normal.
Root cause investigation underway.
SLA impact: if not resolved by 4 AM, reports may be delayed.
Current ETA for resolution: 45 minutes'

Email/Teams at T+30 min (resolution):

'PL_CustomerLoad incident RESOLVED.
Root cause: Source migration script triggered 10x more data.
Fix applied: Manual watermark update, pipeline reruns at normal speed.
Reports will be available on time.
Long-term prevention: [described above]'

Email/Teams at T+24 hours (post-incident):

'Post-incident report attached.
Prevention measures implemented:

1. Row count anomaly alert added
2. Duration SLA alert added
3. DBA coordination process established'

WHY COMMUNICATION MATTERS:

Technical fix = 30 minutes

Without communication: stakeholders panic, escalate unnecessarily,
wake up managers, damage team reputation

With communication: stakeholders trust the process,
let you work, team reputation enhanced!

Senior engineers are TRUSTED.

Trust comes from COMMUNICATION, not just technical skills.
Remember this for behavioral interviews!"

? QUESTION 22 — Behavioral | Leadership & Conflict

"Tell me about a time you disagreed with a technical decision made by your team lead. How did you handle it?"

✓ IDEAL ANSWER — STAR METHOD

STAR FORMAT:

S = Situation

T = Task

A = Action

R = Result

SAMPLE ANSWER:

SITUATION:

"At my previous company, we were designing a data warehouse for a retail client. My team lead proposed using Azure SQL Database (OLTP) as our primary analytical store. I believed this was wrong for the use case."

TASK:

"My responsibility was to build the sales analytics pipeline. I needed to either:
a) Build on the SQL Database (fast, easy)
b) Persuade my lead to use Synapse Analytics (right tool) Without alienating my lead or delaying the project."

ACTION:

"I didn't just say 'that's wrong'.

Instead, I:

1. First UNDERSTOOD his reasoning:

'Can you walk me through why SQL Database works well here?'

He: 'It's familiar, fast to set up, team knows SQL Server'

2. Acknowledged the VALID POINTS:

'You're right that the setup time is faster

and the team has SQL Server expertise'

3. Presented DATA (not opinion):

I ran a benchmark:

→ Azure SQL DB: GROUP BY query on 100M rows = 8 minutes

→ Synapse Serverless: same query = 45 seconds

I showed: 'Here's the execution plan comparison and the data'

Not 'I think' – 'the data shows'

4. Proposed COMPROMISE:

'What if we start with SQL Database for current sprint

but design the schema to be migration-friendly?

If performance becomes issue in testing, we have an easy path.'

This reduced his risk of 'starting over from scratch'

5. DEFERRED to lead if still no agreement:

'I've shared my concerns and data. Ultimately it's your call

and I'll fully commit to whatever direction we take.'

RESULT:

'My lead agreed to benchmark in sprint 1.

When our test data showed 10x performance gap, he agreed to switch.

We migrated to Synapse in sprint 2 with minimal rework.
Reports ran 10x faster, client was impressed.

My lead later said: 'I appreciated that you came with data, not just opinions, and were willing to defer if needed.'

KEY LESSONS:

- Disagree with DATA, not opinion
- Understand other person's perspective first
- Propose compromise, not ultimatum
- Be willing to be wrong
- Always willing to defer to lead ultimately

PRO TIP — 20+ LPA LEVEL

BEHAVIORAL QUESTION FRAMEWORK:

Top companies (Microsoft, Google, Barclays) use
LEADERSHIP PRINCIPLES in behavioral questions.

BARCLAYS: Collaboration, Integrity, Excellence, Stewardship
MICROSOFT: Growth Mindset, Diverse Perspectives, One Microsoft

YOUR ANSWER MUST DEMONSTRATE the principle being tested.

For "disagreement" question → they're testing:

- Can you respectfully challenge?
- Do you base decisions on data?
- Are you collaborative (not combative)?
- Do you prioritize team outcome over being right?

PREPARATION CHECKLIST:

Prepare 3-5 STAR stories that each show:

1. Technical leadership decision
2. Resolving conflict
3. Failure and learning
4. Mentoring/coaching someone
5. Dealing with ambiguity

Each story should be:

- Specific (real situation, real numbers)
- Your PERSONAL actions (not 'we did')
- Outcome with MEASURABLE result
- Learning applied to future situations

'We' = team achievement

'I' = your contribution

Interviewers want to know what YOU specifically did!

Prepare answers for:

'Tell me about a time you failed'

'Tell me about a time you influenced without authority'

'Tell me about your biggest technical achievement'

'How do you handle ambiguous requirements?'

'Tell me about a time you learned from a mistake'

These BEHAVIORAL questions decide if you get the offer even when your technical answers are perfect!"

? QUESTION 23 — Advanced | Azure Synapse vs Databricks

"When would you choose Azure Synapse Analytics over Databricks? What are the trade-offs?"

✓ IDEAL ANSWER

COMPARISON FRAMEWORK:

AZURE SYNAPSE ANALYTICS – Choose **when**:

1. TEAM IS SQL-CENTRIC:

Team knows T-SQL (**not** Python/Scala)

→ Synapse = T-SQL **for** everything

→ Databricks = PySpark (learning curve **for** SQL team)

2. ENTERPRISE DATA WAREHOUSE (OLAP):

Need structured, governed data warehouse

→ Dedicated SQL Pool = enterprise-grade MPP DW

→ **Column** store indexes, distribution keys

→ Familiar **to** DBAs/BI developers

3. SEAMLESS POWER BI INTEGRATION:

Synapse → native Power BI connectivity

Works **within** Microsoft licensing (Power BI Premium)

Direct Lake **from** Synapse tables

4. COST PREDICTABILITY:

Dedicated Pool = fixed cost **per** DWU (predictable monthly bill)

Databricks = **per-cluster-hour** (variable, can spike)

For enterprises needing budget predictability → Synapse

5. MIXED WORKLOADS (SQL + Spark **in ONE workspace):**

Synapse: Run SQL queries **AND** Spark notebooks **in** same UI

One workspace, **one** security model, **one** billing

Good **for**: smaller teams, simpler governance

6. ALREADY **IN** AZURE ECOSYSTEM:

Using Microsoft **365**, Azure AD, Azure DevOps, Power BI

→ Synapse = deep native integration

→ Single-vendor simplicity

DATABRICKS – Choose **when**:

1. HEAVY ML/AI WORKLOADS:

Training ML models → Databricks wins clearly

MLflow **for** experiment tracking (built-in)

AutoML, Feature Store, Model Registry

Synapse has basic ML; Databricks **is** purpose-built

2. **LARGE**-SCALE DATA (**10TB+/day**):

Databricks' Spark = more mature, faster at extreme scale

Photon engine: 2-5x faster than open-source Spark

Better optimization for very large datasets

3. STREAMING AT SCALE:

Databricks Structured Streaming = more mature, more features

Better state management, more operators

Auto Loader = best incremental file ingestion

4. COMPLEX PYTHON/SCALA LOGIC:

Custom algorithms, advanced transformations

Full Python ecosystem (pandas, scikit-learn, PyTorch)

Databricks = more flexible for custom code

5. MULTI-CLOUD STRATEGY:

Databricks runs on Azure, AWS, GCP

Team skills transfer across clouds

Synapse = Azure-only

6. BETTER PERFORMANCE OPTIMIZATION:

Photon engine, Delta Lake optimizations
Z-ORDER, OPTIMIZE, auto-compaction
More performance levers than Synapse

7. UNITY CATALOG (governance):

Centralized catalog for multi-workspace
Column-level masking, row filtering
Data lineage tracking
Synapse equivalent is less mature

DECISION MATRIX:

Scenario	Choose
SQL team, need DW, Power BI	Synapse Analytics
ML/AI workloads	Databricks
100TB+/day processing	Databricks
Multi-cloud strategy	Databricks
Azure-only, budget predictable	Synapse Analytics
Complex streaming	Databricks
Familiar Microsoft ecosystem	Synapse Analytics
Startup, want one platform	Microsoft Fabric

REAL ANSWER for most enterprises:

USE BOTH!

ADF (orchestration) + Databricks (heavy transformation)
+ Synapse (SQL data warehouse) + Power BI (reporting)

Each tool does what it's best at.

Separation of concerns = maintainable, performant architecture!

💡 PRO TIP — 15+ LPA LEVEL

FUTURE TREND AWARENESS (shows you **read** the industry):

"The landscape is evolving rapidly:

MICROSOFT FABRIC (2023+):

- Microsoft's answer to 'too many tools'
- Unified: ADF + Synapse + Power BI + Databricks alternative
- OneLake: single storage layer
- Direct Lake: Power BI reads Parquet directly!

May eventually reduce need for standalone Synapse or ADF

Companies that use Power BI Premium → Fabric is logical next step

DATABRICKS + MICROSOFT PARTNERSHIP:

- Databricks integrates deeply with Azure
- Azure Databricks = Databricks on Azure infra
- Native integration with ADLS, ADF, Synapse

MY CURRENT RECOMMENDATION:

Greenfield project 2024:

- Small team: Microsoft Fabric (all-in-one)
- Large enterprise with ML: ADF + Databricks + Synapse + Power BI
- SQL-heavy team: ADF + Synapse + Power BI

Always right-size for:

- Team's skills
- Budget
- Scalability needs
- Compliance requirements

Mentioning Fabric and evolution of tools =

'This person reads industry news and thinks strategically'
= 15-20+ LPA mindset!"

? QUESTION 24 — Advanced | Data Quality Framework

"How would you build a data quality framework in Azure? What checks would you implement?"

✓ IDEAL ANSWER

DATA QUALITY FRAMEWORK = Systematic approach to ensuring data is fit for its intended purpose

5 DIMENSIONS OF DATA QUALITY:

1. **COMPLETENESS:** Are all required fields populated?
CHECK: NULL count per column
RULE: customer_id must never be NULL (business critical)
2. **ACCURACY:** Does data represent reality correctly?
CHECK: Reference data validation
RULE: country_code must be in ['IN', 'US', 'UK', 'DE'...]
3. **CONSISTENCY:** Is same entity consistent across systems?
CHECK: Cross-system reconciliation
RULE: Customer total in CRM = total in Data Warehouse

4. **TIMELINESS:** Is data fresh enough for use?

CHECK: Max timestamp in data vs expected freshness

RULE: Sales data should be < 24 hours old

5. **UNIQUENESS:** No duplicate records where not expected?

CHECK: COUNT(*) vs COUNT(DISTINCT primary_key)

RULE: Each sale_id appears exactly once in fact_sales

IMPLEMENTATION LAYERS:

LAYER 1: SOURCE VALIDATION (before ingestion)

In ADF pipeline (before Copy Activity):

Activity: LKP_ValidateSource

Query: SELECT

```
COUNT(*) AS total_rows,  
COUNT(CASE WHEN customer_id IS NULL THEN 1 END) AS  
null_customers,  
MAX(sale_date) AS latest_date,  
MIN(sale_date) AS earliest_date  
FROM source.SALES  
WHERE updated_date > :watermark
```

Activity: IF_ValidateThresholds

Condition:

```
@{and(  
    greater(activity('LKP_Validate').output.firstRow.total_rows,  
0),  
  
equals(activity('LKP_Validate').output.firstRow.null_customers,  
0),  
    greater(activity('LKP_Validate').output.firstRow.latest_date,  
        addDays(utcnow(), -1))  
)}
```

TRUE: Proceed with load

FALSE: Fail with specific error message + alert

LAYER 2: TRANSFORMATION VALIDATION (during processing)

In Databricks notebook:

```
# Data quality checks with counts
```

```
quality_results = {}
```

```
# Check 1: Completeness
```

```
null_count = df.filter(F.col("customer_id").isNull()).count()
```

```
quality_results["null_customer_id"] = null_count
```

```
# Check 2: Uniqueness
```

```
total = df.count()
```

```
distinct = df.dropDuplicates(["sale_id"]).count()
```

```
quality_results["duplicate_sales"] = total - distinct
```

```
# Check 3: Validity (amount must be positive)
```

```
invalid_amount = df.filter(F.col("total_amount") <= 0).count()
```

```
quality_results["invalid_amounts"] = invalid_amount
```

```
# Check 4: Referential integrity
```

```
# All customers must exist in dim_customer
```

```
orphaned = df.join(dim_customers, on="customer_id",  
how="left_anti").count()
```

```
quality_results["orphaned_customers"] = orphaned
```

```
# Check 5: Timeliness
```

```
max_date = df.agg(F.max("sale_date")).first()[0]
```

```
days_old = (datetime.now() - max_date).days
```

```
quality_results["data_age_days"] = days_old
```

```
# EVALUATE RULES:
```

```
failed_checks = []
```

```

if null_count > 0:
    failed_checks.append(f"NULL customer_ids: {null_count}")
if total - distinct > 100: # allow small duplicates
    failed_checks.append(f"Duplicates: {total-distinct}")
if invalid_amount > 0:
    failed_checks.append(f"Invalid amounts: {invalid_amount}")
if orphaned > 0:
    failed_checks.append(f"Orphaned customers: {orphaned}")
if days_old > 1:
    failed_checks.append(f>Data is {days_old} days old!")

# DECIDE:
if failed_checks:
    # Log to quality table
    spark.createDataFrame(
        [(process_date, "FAILED", str(failed_checks))],
        ["date", "status", "failures"]
    ).write.format("delta").mode("append").saveAsTable("dq.quality_log")

    # Alert
    mssparkutils.notebook.exit(f"DQ_FAILED:
{str(failed_checks)}")
else:
    mssparkutils.notebook.exit("DQ_PASSED")

```

LAYER 3: POST-LOAD VALIDATION

After loading to Data Warehouse:

```

-- Reconcile row counts: source vs destination
SOURCE_COUNT vs DESTINATION_COUNT
If difference > 0.01% → alert!

```

```

-- Reconcile totals: source SUM vs DW SUM
SELECT SUM(gross_amount) FROM source.SALES WHERE date = today

```

vs

```
SELECT SUM(gross_amount) FROM fact_sales WHERE date = today
```

Must match exactly! (or within 0.001% for floating point)

LAYER 4: GREAT EXPECTATIONS (production-grade DQ)

Industry standard tool for data validation:

```
from great_expectations.core.batch import RuntimeBatchRequest
```

```
# Define expectations
```

```
expectation_suite = {  
    "expect_column_values_to_not_be_null": ["sale_id",  
"customer_id"],  
    "expect_column_values_to_be_unique": ["sale_id"],  
    "expect_column_values_to_be_between": {  
        "total_amount": {"min_value": 0, "max_value": 10000000}  
    },  
    "expect_table_row_count_to_be_between": {  
        "min_value": 100, "max_value": 5000000  
    }  
}
```

```
# Validates data + generates HTML report!
```

```
# Shows pass/fail for each expectation
```

```
# Can block pipeline if critical expectations fail
```

LAYER 5: DQ DASHBOARD (Power BI)

Track over time:

→ DQ pass rate by table per day

→ Which checks fail most often?

→ Data freshness per source

→ Trend: is data quality improving or degrading?

ALERT: DQ score < 95% → immediate notification

💡 PRO TIP — 15+ LPA LEVEL

DATA QUALITY AS A PRODUCT MINDSET:

"I treat data quality as a product, not an afterthought:

1. SLAs FOR DATA QUALITY (Service Level Agreements):

Define: 'Gold layer data must be 99.9% complete'

'No NULL primary keys in any Silver table'

'Data freshness: < 24 hours'

Measure against SLAs weekly

Report to stakeholders: 'DQ SLA achieved this week: 99.7%'

2. DATA CONTRACTS:

Formal agreement between data producer and consumer:

Producer (DE team): 'We guarantee schema X, freshness Y, completeness Z'

Consumer (BI team): 'We'll alert you before schema changes'

Reduces: surprise schema changes breaking downstream!

3. DATA LINEAGE TRACKING:

'Where did this data come from? What transformed it?'

Azure Purview or OpenLineage:

→ Oracle.SALES → ADF.CopyActivity → ADLS.bronze_sales

→ Databricks.NB_Transform → ADLS.silver_sales

→ Databricks.NB_Aggregate → ADLS.gold_performance

→ Power BI.SalesDashboard

When quality issue found:

'Which upstream source caused it?'

Lineage tells you instantly!

4. DQ ISSUE OWNERSHIP:

Each table has an OWNER who is responsible for DQ
Issue detected → automatically assigned to owner
→ Not just 'data is wrong' → 'fix it within 4 hours'

*This product thinking = data engineering MANAGER material.
Companies pay 20+ LPA for this approach."*

? QUESTION 25 — Final Advanced | Complete Interview Scenario

"You're joining a company as their first Senior Data Engineer. They have raw data in ADLS Gen2 but no processing, no warehouse, no reports. Where do you start? What do you build first?"

✓ IDEAL ANSWER — STRATEGIC THINKING

THIS QUESTION TESTS:

- Prioritization ability
- Business thinking (not just technical)
- Communication and stakeholder management
- Architecture planning
- Pragmatism (quick wins vs perfect system)

MY APPROACH:

WEEK 1: UNDERSTAND THE BUSINESS

Before writing a single line of code:

Talk to: CEO/Director, Business Analysts, Data Consumers

Questions to ask:

- "What's the most painful data problem RIGHT NOW?"
- "Which report/insight would change business decisions today?"
- "What data questions can't you currently answer?"
- "What was promised to stakeholders and when?"

Understand: what provides MOST VALUE fastest?

Example finding:

- "CFO can't see daily sales summary. Makes decisions based on last month's reports. Wants this TODAY."
- This is my PRIORITY 1.

WEEK 1-2: INVENTORY EXISTING DATA

"What's actually in that ADLS?"

Use: Synapse Serverless SQL to explore

```
SELECT TOP 10 * FROM OPENROWSET(BULK 'adls://...**',  
FORMAT='PARQUET') AS r
```

Build: Data catalog (Excel/Confluence)

- Source system → ADLS path → format → columns → freshness → size
- Find: "sales data is there! Updated daily"

WEEK 2-3: QUICK WIN (build value fast)

"CFO needs daily sales summary"

BUILD:

1. Databricks notebook: read raw ADLS → basic aggregation
2. Write to Silver table (Delta, simple schema)

3. Power BI connects **to** Silver (Direct Query **or** Direct Lake)
4. Build: **3**-metric dashboard (Revenue, Units, Stores)
5. Deliver **in 2** WEEKS!

Why?

- Trust **from** business ("data engineer delivered in 2 weeks!")
- Budget approval **for full** platform
- Understand **real** requirements (**not** assumed ones)
- Learn data quality issues early (**not 3** months **in**)

MONTH 1-2: FOUNDATIONS

After quick win, build properly:

Week **5-6**: ADF Pipelines

- Replace manual Databricks notebook **trigger**
- Automated ingestion **from all** sources
- Error handling, monitoring, alerts

Week **7-8**: Data Quality Framework

- **Null** checks, duplicate checks, freshness checks
- DQ dashboard

Week **9-10**: Silver Layer (proper)

- SCD Type **2 for** customer dimension
- Incremental loading **for** fact tables
- Schema standardization

MONTH 3-4: FULL PLATFORM

- Gold layer: **all** business KPIs
- Synapse **or** Fabric **for** BI team **SQL** access
- Power BI semantic models (proper ones)
- Operations team access

MONTH 5-6: OPTIMIZE & GOVERN

- Performance tuning (OPTIMIZE, ZORDER)

- Cost optimization (spot instances, tiered storage)
- Data catalog (Azure Purview)
- Column security for PII
- SLA dashboards

COMMUNICATION PLAN:

- Week 1: Send stakeholder map + discovery questions
- Week 2: Share data inventory + quick win plan
- Week 4: Demo quick win dashboard (collect feedback)
- Monthly: Progress update + next month plan
- Quarterly: Platform health report + roadmap

WHY THIS APPROACH OVER "BUILD PERFECT ARCHITECTURE FIRST":

The WATERFALL trap:

- "Spend 6 months building perfect platform"
- Business loses faith (no output for 6 months)
- Requirements change during 6 months (common!)
- Perfect system solves YESTERDAY's problem

AGILE DATA ENGINEERING:

- Deliver value every 2 weeks
- Adjust based on feedback
- Build incrementally on proven foundation
- Business + technical aligned throughout

KEY PRINCIPLE:

"A working dashboard after 2 weeks > A perfect architecture after 6 months"

Build, learn, iterate.

That's how senior data engineers operate.



PRO TIP — 20+ LPA LEVEL

STAKEHOLDER MANAGEMENT (what VP+ level needs):

"As a senior data engineer joining a new company,
I focus on TRUST BUILDING first, not tech:

FIRST 30 DAYS:

DAY 1-7: Listen. Don't build anything.

Understand: What's working? What's frustrating?

Who are the power users? What do they want?

DAY 8-14: Share findings back.

'Here's what I heard. Here's what I think is feasible.'

Adjust based on feedback.

DAY 15-30: Deliver first small win.

Not the perfect system — just SOMETHING working.

THE 'FIRST WIN' PRINCIPLE:

Your first delivery = builds or destroys trust.

If you promise 'full platform in 6 months' and deliver in 6 months:

→ Normal. Expected. Forgotten.

If you deliver 'one dashboard in 2 weeks':

→ Remarkable. Builds trust.

→ Stakeholders become your champions.

→ 'This person delivers!'

Under-promise, over-deliver. Always.

TECHNICAL DEBT MANAGEMENT:

Quick win = technical debt (not perfect code)

DOCUMENT it:

'This notebook is MVP. Proper pipeline with ADF by month 2.'

Stakeholders understand 'MVP' → sets expectations

Without documenting: technical debt surprises = loss of trust

DOCUMENTATION AS A FIRST-CLASS ARTIFACT:

At 15+ LPA, you write:

- ADR (Architecture Decision Records)
- Runbooks ('How to rerun failed pipeline')
- Monitoring guides ('How to read the DQ dashboard')

This knowledge exists BEYOND your presence.

Junior engineers learn from your docs.

You become irreplaceable AND replaceable!

Paradox = best career strategy."



FINAL COMPREHENSIVE REVISION

COMPLETE AZURE DATA ENGINEER INTERVIEW GUIDE

ALL 25 QUESTIONS SUMMARY

LEVEL 1: FUNDAMENTALS (Q1-Q10)

Q1: ADF = orchestration, not transformation

Q2: LS→Dataset→Activity→Pipeline hierarchy

Q3: 3 triggers: Schedule, Tumbling Window, Event

Q4: 3 IRs: Azure, SHIR (on-prem!), Azure-SSIS

Q5: Copy Activity: partition options, COPY INTO, DIU

Q6: Error handling: retry + dependency paths + fault tol.

Q7: Parameters (immutable external) vs Variables (mutable)

Q8: Incremental: watermark after success only!

Q9: Data Flows: visual transforms, AQE, schema drift

Q10: Lazy eval: transformations lazy, actions trigger

LEVEL 2: INTERMEDIATE (Q11-Q15)

Q11: Window functions: RANK, ROW_NUMBER, LAG/LEAD

Q12: SCD1 = overwrite, SCD2 = effective dates + is_current

Q13: Star schema > Snowflake for BI (conformed dimensions!)

Q14: Spark optimization: Spark UI first, AQE, broadcast

Q15: Delta Lake: ACID + Time Travel + MERGE (over Parquet)

LEVEL 3: ADVANCED (Q16-Q20)

Q16: E2E architecture: requirements→ingestion→batch+stream

Q17: Batch vs Streaming: Lambda arch, watermarks

Q18: Bank migration: zero downtime = blue-green, PCI-DSS

Q19: Query optimization: EXPLAIN, indexes, CCI, avoid funcs

Q20: Cost opt: Job clusters, spot instances, tiered storage

LEVEL 4: SYSTEM DESIGN & BEHAVIORAL (Q21-Q25)

Q21: Incident response: triage→diagnose→isolate→fix→prevent

Q22: Behavioral: data beats opinion, defer when needed

Q23: Synapse vs Databricks: SQL team vs ML/scale

Q24: DQ framework: 5 dimensions, layers, Great Expectations

Q25: New company strategy: quick wins + business alignment

10 GOLDEN RULES FOR 15+ LPA INTERVIEWS

1. ALWAYS mention Key Vault for credentials
2. ALWAYS state requirements before architecture
3. ALWAYS quantify your impact (hours/cost saved)
4. KNOW the numbers (spot=60% saving, Job cluster=8x)
5. NEVER watermark before load succeeds
6. ALWAYS mention High Availability (2 SHIR nodes)
7. ALWAYS use EXPLAIN before optimizing Spark
8. KNOW Delta Lake: VACUUM, OPTIMIZE, ZORDER
9. HAVE 3-5 STAR stories for behavioral questions
10. SHOW you think about COST, not just functionality

SALARY LEVEL INDICATORS

- 6-8 LPA: Knows basic ADF/Databricks/SQL concepts
- 8-12 LPA: Knows optimization, SCD, incremental loading
- 12-15 LPA: Designs complete architectures, cost awareness
- 15-18 LPA: Production incidents, DQ frameworks, FinOps
- 18-22 LPA: Strategic thinking, stakeholder management
- 22+ LPA: Platform engineering, team leadership, ADRs



YOUR INTERVIEW PREPARATION CHECKLIST

BEFORE THE INTERVIEW:

TECHNICAL PREPARATION:

- ☐ Can explain ADF components **without** hesitation?
- ☐ Know **all 3 trigger** types + **when to use each**?
- ☐ Can write SCD Type **1 and 2 MERGE** statements?
- ☐ Can write **window function** query (top N **per group**)?
- ☐ Know Delta Lake: ACID, **time** travel, VACUUM, OPTIMIZE?
- ☐ Can design Medallion architecture **end-to-end**?
- ☐ Know Spark optimization: AQE, broadcast, skew handling?
- ☐ Can design star schema **for** retail data warehouse?
- ☐ Know incremental loading: watermark after success **only**?
- ☐ Can compare Synapse vs Databricks vs Fabric?

PRACTICAL PREPARATION:

- ☐ Have **2-3 STAR** stories **for** behavioral questions?
- ☐ Know cost numbers **by** heart? (spot=**60%**, job cluster=**8x**)
- ☐ Can **describe** a production incident you resolved?
- ☐ Can explain Key Vault + Private Endpoints **for** security?
- ☐ Can estimate cost **of** an Azure data platform?

DAY OF INTERVIEW:

- ☐ Reread the **10** Golden Rules above
- ☐ Remember: Requirements **FIRST in system** design
- ☐ Remember: Quantify impact **in every** answer



- ❑ Remember: Mention security in every architecture

DURING INTERVIEW:

- ❑ It's OK to say "Let me think for 30 seconds"
- ❑ Clarify requirements before designing
- ❑ Draw diagrams when explaining architecture
- ❑ Ask about their current stack (shows interest)
- ❑ End with: "Do you have any concerns about my experience?"

QUESTIONS TO ASK INTERVIEWER:

- ❑ "What's the current biggest data challenge?"
- ❑ "What does the data stack look like currently?"
- ❑ "How does the team handle pipeline failures?"
- ❑ "What would success look like in first 90 days?"

YOU ARE NOW PREPARED FOR 15+ LPA AZURE DATA ENGINEER INTERVIEWS

Complete Interview Masterclass Done!

25 questions covering Fundamentals → Intermediate → Advanced → System Design → Behavioral, all with ideal answers, real-world scenarios, common mistakes, and 15+ LPA pro tips. Every core concept from the Azure Data Engineering course is covered from an interview perspective.

Want me to continue with more advanced questions or do mock interviews on specific topics? Just ask!

19/03/2026, 18:40:59

Thought process: Orchestrated comprehensive interview question expansion across multiple technical domains.

The user wants me to continue with more interview questions

beyond the 25 already covered. Let me continue from Q26 onwards covering remaining important topics that weren't fully covered yet.

Topics still to cover:

- More advanced SQL questions
 - Azure Event Hubs / Streaming in depth
 - Data Lake architecture
 - Performance tuning deeper
 - More ADF scenarios
 - Unity Catalog
 - Microsoft Fabric interview questions
 - More PySpark scenarios
 - Real-world scenario based questions
 - Cost optimization deeper
 - CI/CD for data pipelines
 - Data Governance
 - More behavioral questions
 - Partition strategies
 - Caching strategies in Spark
-
- More window function scenarios
 - Complex joins
 - ADF Data Flows deeper
 - Synapse dedicated pool optimization
 - More system design questions

Continuing Interview Masterclass — Q26 to Q70+

"The questions that decide 18-25+ LPA offers"

CONTINUING FROM Q25...

REMAINING TOPICS TO COVER:

- ✓ Q26-Q30: Advanced SQL + Performance
- ✓ Q31-Q35: PySpark Deep Dive
- ✓ Q36-Q40: Azure Architecture Patterns
- ✓ Q41-Q45: Real-World Scenarios
- ✓ Q46-Q50: Data Governance + Security
- ✓ Q51-Q55: Streaming + Real-Time
- ✓ Q56-Q60: Cost + Performance Optimization
- ✓ Q61-Q65: Microsoft Fabric + Modern Stack
- ✓ Q66-Q70: System Design + Leadership

Let's go. No stopping until everything is covered.

ADVANCED SQL — DEEP DIVE

? QUESTION 26 — Advanced SQL

"Write a SQL query to find customers who purchased in January 2024 but NOT in February 2024. Then find customers who purchased in BOTH months."

✓ IDEAL ANSWER

```

-- — PART 1: Purchased in Jan but NOT Feb —————

-- METHOD 1: EXCEPT (cleanest)
SELECT DISTINCT customer_id, customer_name
FROM fact_sales fs
JOIN dim_customer dc ON fs.customer_key = dc.customer_key
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024 AND dd.month_number = 1

EXCEPT

SELECT DISTINCT customer_id, customer_name
FROM fact_sales fs
JOIN dim_customer dc ON fs.customer_key = dc.customer_key
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024 AND dd.month_number = 2;

-- —————

-- METHOD 2: LEFT ANTI JOIN (more explicit, better performance)
SELECT DISTINCT
    jan.customer_id,
    jan.customer_name
FROM (
    SELECT DISTINCT dc.customer_id, dc.customer_name
    FROM fact_sales fs
    JOIN dim_customer dc ON fs.customer_key = dc.customer_key
    JOIN dim_date dd ON fs.date_key = dd.date_key
    WHERE dd.year = 2024 AND dd.month_number = 1
) jan
LEFT JOIN (
    SELECT DISTINCT dc.customer_id
    FROM fact_sales fs
    JOIN dim_customer dc ON fs.customer_key = dc.customer_key
    JOIN dim_date dd ON fs.date_key = dd.date_key

```

```

        WHERE dd.year = 2024 AND dd.month_number = 2
    ) feb
ON jan.customer_id = feb.customer_id
WHERE feb.customer_id IS NULL; -- ← LEFT JOIN + NULL = NOT EXISTS

```

```

-- -----

-- METHOD 3: NOT EXISTS (readable, often fast)
SELECT DISTINCT dc.customer_id, dc.customer_name
FROM fact_sales fs
JOIN dim_customer dc ON fs.customer_key = dc.customer_key
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024
      AND dd.month_number = 1
      AND NOT EXISTS (
        SELECT 1
        FROM fact_sales fs2
        JOIN dim_date dd2 ON fs2.date_key = dd2.date_key
        WHERE fs2.customer_key = fs.customer_key
              AND dd2.year = 2024
              AND dd2.month_number = 2
      );

```

```

-- — PART 2: Purchased in BOTH months -----

```

```

-- METHOD 1: INTERSECT
SELECT DISTINCT customer_id
FROM fact_sales fs
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024 AND dd.month_number = 1

INTERSECT

SELECT DISTINCT customer_id

```

```

FROM fact_sales fs
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024 AND dd.month_number = 2;

-- -----

-- METHOD 2: GROUP BY HAVING (more flexible for reporting)
SELECT
    dc.customer_id,
    dc.customer_name,
    COUNT(DISTINCT dd.month_number) AS months_purchased,
    SUM(CASE WHEN dd.month_number = 1 THEN fs.gross_amount ELSE 0
END) AS jan_spend,
    SUM(CASE WHEN dd.month_number = 2 THEN fs.gross_amount ELSE 0
END) AS feb_spend,
    SUM(fs.gross_amount) AS total_spend
FROM fact_sales fs
JOIN dim_customer dc ON fs.customer_key = dc.customer_key
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024
    AND dd.month_number IN (1, 2)
GROUP BY dc.customer_id, dc.customer_name
HAVING COUNT(DISTINCT dd.month_number) = 2 -- ← both months!
ORDER BY total_spend DESC;

-- — BONUS: COMPLETE CUSTOMER COHORT ANALYSIS —————
-- Categorize customers into all possible combinations

WITH monthly_purchase AS (
    SELECT
        dc.customer_id,
        dc.customer_name,
        MAX(CASE WHEN dd.month_number = 1 THEN 1 ELSE 0 END) AS
bought_jan,
        MAX(CASE WHEN dd.month_number = 2 THEN 1 ELSE 0 END) AS

```

```

bought_feb
    FROM fact_sales fs
    JOIN dim_customer dc ON fs.customer_key = dc.customer_key
    JOIN dim_date dd ON fs.date_key = dd.date_key
    WHERE dd.year = 2024
        AND dd.month_number IN (1, 2)
    GROUP BY dc.customer_id, dc.customer_name
)
SELECT
    CASE
        WHEN bought_jan = 1 AND bought_feb = 1 THEN 'Both Months'
        WHEN bought_jan = 1 AND bought_feb = 0 THEN 'Jan Only
(Churned)'
        WHEN bought_jan = 0 AND bought_feb = 1 THEN 'Feb Only
(New)'
        ELSE 'Neither'
    END AS customer_segment,
    COUNT(*) AS customer_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS
percentage
FROM monthly_purchase
GROUP BY
    CASE
        WHEN bought_jan = 1 AND bought_feb = 1 THEN 'Both Months'
        WHEN bought_jan = 1 AND bought_feb = 0 THEN 'Jan Only
(Churned)'
        WHEN bought_jan = 0 AND bought_feb = 1 THEN 'Feb Only
(New)'
        ELSE 'Neither'
    END
ORDER BY customer_count DESC;

-- Result:
-- customer_segment      | count  | percentage
-- Both Months           | 45,230 | 48.2%

```

```
-- Jan Only (Churned) | 32,100 | 34.2%
-- Feb Only (New)    | 15,670 | 16.7%
-- Neither           | 820   | 0.9%
```

SIMPLE EXPLANATION

ANALOGY: Gym membership tracking

Jan members: [A, B, C, D, E]

Feb members: [B, C, E, F, G]

NOT IN Feb (churned): A, D (was Jan, **left in** Feb)

IN BOTH (retained): B, C, E (loyal members!)

ONLY in Feb (**new**): F, G (joined **in** Feb)

EXCEPT = "Give me Jan list MINUS anything in Feb"

INTERSECT = "Give me only what's in BOTH lists"

LEFT JOIN + NULL = "Jan customers who have NO Feb match"

PERFORMANCE NOTES:

EXCEPT/INTERSECT: clean **SQL**, moderate performance

NOT EXISTS: often fastest (short-circuits **on first match**)

LEFT JOIN + NULL: depends **on** index quality

ALWAYS **check** execution plan **for large** tables!

COMMON MISTAKES

MISTAKE 1: Using NOT IN with subquery that can return NULLs

❌ WHERE customer_id NOT IN (SELECT customer_id FROM feb_sales)

If any feb_sales.customer_id is NULL:

→ NOT IN returns EMPTY RESULT (SQL NULL logic!)

→ Missed thousands of customers!

✅ Use NOT EXISTS or LEFT JOIN approach instead

✅ Or: WHERE customer_id NOT IN (
SELECT customer_id FROM feb_sales WHERE customer_id IS
NOT NULL)

MISTAKE 2: Not using DISTINCT in intermediate sets

❌ No DISTINCT → one customer with 5 Jan purchases
appears 5 times in result → inflated counts!

✅ Always DISTINCT when working with "which customers"
questions

MISTAKE 3: Not writing the business analysis (cohort query)

Junior: just writes the basic query

Senior: adds segmentation, percentages, business insight

"48% retained, 34% churned" → actionable business insight!

THAT is what 15+ LPA looks like.

💡 PRO TIP

RETENTION RATE CALCULATION (business metric!):

"The Jan-only vs both-months analysis is the foundation
of RETENTION RATE – a key business metric:

Retention Rate = (Jan customers who also bought in Feb) /

$(\text{Jan customers}) \times 100$

$= 45,230 / (45,230 + 32,100) \times 100$

$= 45,230 / 77,330 \times 100$

$= 58.5\%$ retention rate

Business interpretation:

'41.5% of January customers did NOT return in February.

That is churn. If average order = Rs.5,000:

32,100 churned $\times$ Rs.5,000 = Rs.16 Crore potential revenue lost!'

This business framing of SQL results =

what separates data engineers from data analysts =

what gets you to 15-18 LPA!"

? QUESTION 27 — Advanced SQL | Running Totals & Moving Averages

"Write a query to calculate: (1) Running total of sales by day, (2) 7-day moving average, (3) Sales compared to previous day."

✓ IDEAL ANSWER

-- COMPLETE DAILY SALES ANALYTICS QUERY:

```
WITH daily_sales AS (  
  -- First aggregate to daily level  
  SELECT
```

```

        dd.full_date,
        dd.day_name,
        dd.is_weekend,
        SUM(fs.gross_amount)           AS daily_revenue,
        COUNT(fs.sale_key)             AS transaction_count,
        COUNT(DISTINCT fs.customer_key) AS unique_customers
FROM fact_sales fs
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024
      AND dd.month_number = 6
GROUP BY dd.full_date, dd.day_name, dd.is_weekend
)
SELECT
    full_date,
    day_name,
    is_weekend,
    daily_revenue,
    transaction_count,
    unique_customers,

    -- — 1. RUNNING TOTAL (Cumulative Revenue) —————
    SUM(daily_revenue) OVER (
        ORDER BY full_date
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS cumulative_revenue,

    -- — 2. 7-DAY MOVING AVERAGE —————
    ROUND(
        AVG(CAST(daily_revenue AS FLOAT)) OVER (
            ORDER BY full_date
            ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
            -- Current row + 6 rows before = 7-day window
        ),
        2
    ) AS moving_avg_7day,

```

```

-- — 3. PREVIOUS DAY COMPARISON —————
LAG(daily_revenue, 1) OVER (ORDER BY full_date)
    AS prev_day_revenue,

-- Day over Day change (absolute)
daily_revenue - LAG(daily_revenue, 1) OVER (ORDER BY
full_date)
    AS dod_change,

-- Day over Day % change
ROUND(
    (daily_revenue - LAG(daily_revenue, 1) OVER (ORDER BY
full_date))
    / NULLIF(LAG(daily_revenue, 1) OVER (ORDER BY full_date),
0)
    * 100,
    1
) AS dod_pct_change,

-- — 4. BONUS: 7-DAY SUM (Weekly Revenue Window) —————
SUM(daily_revenue) OVER (
    ORDER BY full_date
    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
) AS rolling_7day_total,

-- — 5. BONUS: RANK BY REVENUE (best sales day) —————
RANK() OVER (ORDER BY daily_revenue DESC) AS revenue_rank,

-- — 6. BONUS: % OF MONTHLY TOTAL —————
ROUND(
    daily_revenue / SUM(daily_revenue) OVER () * 100,
    2
) AS pct_of_month_total

```

```
FROM daily_sales
ORDER BY full_date;
```

```
-- SAMPLE OUTPUT:
```

```
-- date          | day   | revenue   | cumulative | 7d_avg   | prev_day
| dod% | rank
-- 2024-06-01 | Sat   | 850,000   | 850,000    | 850,000  | NULL
| NULL | 12
-- 2024-06-02 | Sun   | 920,000   | 1,770,000  | 885,000  | 850,000
| +8.2 | 8
-- 2024-06-03 | Mon   | 420,000   | 2,190,000  | 730,000  | 920,000
| -54.3| 28
-- ...
```

SIMPLE EXPLANATION

WINDOW FRAME EXPLAINED (ROWS BETWEEN):

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

= "From the very first row to right here"

= Classic running total

ROWS BETWEEN 6 PRECEDING AND CURRENT ROW

= "From 6 rows back to right here"

= 7 rows total (6 before + current)

= 7-day moving average/sum

ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING

= "One row before, current, one row after"

= 3-row centered average

ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING

= "From here to the very last row"

= Reverse **running** total

PRACTICAL MEANING:

7-day moving average **for** June **15**:

→ Averages: June **9, 10, 11, 12, 13, 14, 15**

→ Smooths **out** weekly **pattern** (weekends spike/dip)

→ Better trend visibility than daily fluctuations!

WHY NULLIF **for** percentage:

NULLIF(**value**, **0**) → **returns NULL** if **value is 0**

Prevents: division **by** zero error!

Instead **of**: ERROR → shows **NULL** (clean!)

COMMON MISTAKES

MISTAKE **1**: **ROWS** vs **RANGE** **for** moving average

 **RANGE BETWEEN 6 PRECEDING AND CURRENT ROW**

With ties: includes **ALL rows** with same **ORDER BY value**!

If multiple **rows** have same **date** → **window** expands unexpectedly!

 **ROWS BETWEEN 6 PRECEDING AND CURRENT ROW**

Exactly **6** physical **rows**, regardless **of** ties

MISTAKE **2**: **Not using** NULLIF **for** first rows

For 7-day moving avg **on day 1**: **only 1 row** in window

The average **is** technically **of 1 day** (**not** wrong, just incomplete)

To flag incomplete windows:

CASE WHEN ROW_NUMBER() **OVER (ORDER BY full_date)** **>= 7**

```
    THEN AVG(daily_revenue) OVER (...)
    ELSE NULL -- insufficient data for full 7-day average
END AS moving_avg_7day_clean
```

MISTAKE 3: Forgetting ORDER BY in window

```
SUM(daily_revenue) OVER () -- no ORDER BY = TOTAL sum (not
running!)
SUM(daily_revenue) OVER (ORDER BY full_date) -- running total
```



Without ORDER BY in SUM OVER → no "running" behavior!
Just returns the grand total for every row!

PRO TIP

PYSPARK EQUIVALENT (show full-stack knowledge):

```
from pyspark.sql import functions as F
from pyspark.sql.window import Window

# 1. Running total
running_window = Window.orderBy("full_date") \
    .rowsBetween(Window.unboundedPreceding,
                  Window.currentRow)

# 2. 7-day moving average
rolling_window = Window.orderBy("full_date") \
    .rowsBetween(-6, Window.currentRow)

# 3. Previous day
lag_window = Window.orderBy("full_date")
```

```

df_daily = df_daily \
    .withColumn("cumulative_revenue",
                F.sum("daily_revenue").over(running_window)) \
    .withColumn("moving_avg_7day",

F.round(F.avg("daily_revenue").over(rolling_window), 2)) \
    .withColumn("prev_day_revenue",
                F.lag("daily_revenue", 1).over(lag_window)) \
    .withColumn("dod_pct_change",
                F.round(
                    (F.col("daily_revenue") -
F.lag("daily_revenue", 1).over(lag_window))
                    / F.nullif(F.lag("daily_revenue",
1).over(lag_window), F.lit(0))
                    * 100, 1
                ))

display(df_daily.orderBy("full_date"))

'Knowing both SQL AND PySpark syntax for same operation
shows you're truly full-stack data engineer.
Interviewers notice this immediately!'

```

? QUESTION 28 — Advanced SQL | Gaps and Islands Problem

"Find periods (start date to end date) when a store was continuously open (no gap of more than 1 day in sales data). This is called the Gaps and Islands problem."

✓ IDEAL ANSWER

```
-- GAPS AND ISLANDS PROBLEM
-- Find consecutive date ranges where store had sales

-- SAMPLE DATA:
-- store_id | sale_date
-- S001     | 2024-01-01
-- S001     | 2024-01-02
-- S001     | 2024-01-03
-- S001     | 2024-01-05 ← GAP! Jan 4 missing
-- S001     | 2024-01-06
-- S001     | 2024-01-07

-- Expected output:
-- store_id | open_from | open_until | days_open
-- S001     | 2024-01-01 | 2024-01-03 | 3
-- S001     | 2024-01-05 | 2024-01-07 | 3

-- — SOLUTION: Classic Gaps & Islands Technique —————

WITH store_sales_dates AS (
    -- Get distinct dates per store (one row per store-date)
    SELECT DISTINCT
        store_id,
        sale_date
    FROM fact_sales fs
    JOIN dim_date dd ON fs.date_key = dd.date_key
    WHERE dd.year = 2024
),
islands_identified AS (
    -- KEY: Subtract row_number from date → same "island" = same value!
    SELECT
        store_id,
```



```

    sale_date,
    -- ROW_NUMBER increases by 1 each day
    -- If dates are consecutive: date - row_number = CONSTANT
    -- If there's a gap: the constant CHANGES → new island!
    DATEADD(DAY,
            -ROW_NUMBER() OVER (PARTITION BY store_id ORDER
    BY sale_date),
            sale_date
    ) AS island_group
    FROM store_sales_dates
)
SELECT
    store_id,
    MIN(sale_date) AS open_from,
    MAX(sale_date) AS open_until,
    DATEDIFF(DAY, MIN(sale_date), MAX(sale_date)) + 1 AS
days_open,
    COUNT(*) AS sales_days_in_period
FROM islands_identified
GROUP BY store_id, island_group
ORDER BY store_id, open_from;

```

-- — HOW THE TRICK WORKS —————

date	row_num	date - row_num	island_group
2024-01-01	1	2023-12-31	← GROUP A
2024-01-02	2	2023-12-31	← GROUP A (same!)
2024-01-03	3	2023-12-31	← GROUP A (same!)
2024-01-05	4	2024-01-01	← GROUP B (different!)
2024-01-06	5	2024-01-01	← GROUP B (same!)
2024-01-07	6	2024-01-01	← GROUP B (same!)

-- GENIUS: consecutive dates minus sequential numbers = constant!

-- — FIND GAPS (stores closed for more than 2 consecutive days)

—

```

WITH store_dates AS (

```

```

SELECT DISTINCT store_id, sale_date
FROM fact_sales fs
JOIN dim_date dd ON fs.date_key = dd.date_key
WHERE dd.year = 2024
),
with_next_date AS (
    SELECT
        store_id,
        sale_date,
        LEAD(sale_date, 1) OVER (
            PARTITION BY store_id
            ORDER BY sale_date
        ) AS next_sale_date
    FROM store_dates
)
SELECT
    store_id,
    DATEADD(DAY, 1, sale_date) AS gap_start,    -- day after last
sale
    DATEADD(DAY, -1, next_sale_date) AS gap_end, -- day before
next sale
    DATEDIFF(DAY, sale_date, next_sale_date) - 1 AS days_closed
FROM with_next_date
WHERE DATEDIFF(DAY, sale_date, next_sale_date) > 2 -- gap > 2
days
ORDER BY store_id, gap_start;

```

SIMPLE EXPLANATION

ISLAND TRICK EXPLAINED with Cricket:

Player played matches on:

Jan 1, 2, 3 (consecutive) → Island 1

Jan 7, 8, 9 (consecutive) → Island 2

Date - Row Number trick:

Jan 1 - 1 = Dec 31 ← Island 1 marker

Jan 2 - 2 = Dec 31 ← Same! (still Island 1)

Jan 3 - 3 = Dec 31 ← Same! (still Island 1)

Jan 7 - 4 = Jan 3 ← Different! (new Island 2)

Jan 8 - 5 = Jan 3 ← Same! (still Island 2)

Jan 9 - 6 = Jan 3 ← Same! (still Island 2)

Now GROUP BY that "Dec 31" or "Jan 3" value:

Dec 31 group: Jan 1 to Jan 3

Jan 3 group: Jan 7 to Jan 9

BRILLIANT because consecutive dates - consecutive numbers = constant. Any gap breaks this pattern!

⚠ COMMON MISTAKES

MISTAKE 1: Not knowing this pattern exists

Most candidates: "I'd use a while loop" or "I don't know"

✅ This is a CLASSIC interview question pattern

Know it by heart – comes up in Barclays, EY interviews!

MISTAKE 2: Using cursors/loops for this

❌ WHILE loop processing row-by-row = very slow!

✅ Set-based SQL (window functions) = much faster!

For 500M rows:

Cursor: days to complete

Window function: minutes to complete

MISTAKE 3: Off-by-one in gap calculation

Gap starts: day AFTER last sale (not on last sale day)

Gap ends: day BEFORE next sale (not on next sale day)

Closed for: `DATEDIFF(last_sale, next_sale) - 1`

Not: `DATEDIFF(last_sale, next_sale)!`

PRO TIP

BUSINESS APPLICATIONS (show you think beyond SQL):

"Gaps and Islands has many real-world applications:

1. STORE CLOSURE DETECTION:

Which stores had unexplained closure periods?

→ Alert operations team for investigation

2. CUSTOMER CHURN DETECTION:

Which customers had purchase gaps > 90 days?

→ Trigger re-engagement campaigns

3. EQUIPMENT DOWNTIME ANALYSIS:

IoT sensors: when was machine NOT sending data?

→ Identify maintenance windows, failures

4. EMPLOYEE CONTINUOUS SERVICE:

HR: identify consecutive employment periods

→ Important for benefits calculations!

5. SESSION ANALYTICS:

Web: cluster page views into sessions

```
(gap > 30 minutes = new session)
```

In PySpark:

```
from pyspark.sql.window import Window
```

```
w = Window.partitionBy("store_id").orderBy("sale_date")
```

```
df.withColumn("row_num", F.row_number().over(w)) \
    .withColumn("island_group",
                F.date_sub("sale_date",
                F.col("row_num").cast("int"))) \
    .groupBy("store_id", "island_group") \
    .agg(F.min("sale_date").alias("open_from"),
        F.max("sale_date").alias("open_until"),
        F.count("*").alias("days_open"))
```

This pattern in both SQL AND PySpark = complete answer!"

? QUESTION 29 — Advanced SQL | Recursive CTE

"Write a SQL query to traverse an employee hierarchy (manager-subordinate relationship) and show the reporting chain for every employee."

✓ IDEAL ANSWER

```
-- EMPLOYEE TABLE:
-- emp_id | emp_name          | manager_id | department | salary
-- 1      | Vikram (CEO)      | NULL       | Executive  | 500000
```

```
-- 2      | Priya (VP Eng)   | 1          | Engineering | 300000
-- 3      | Rahul (VP Sales) | 1          | Sales       | 280000
-- 4      | Amit (Lead)      | 2          | Engineering | 180000
-- 5      | Sneha (SDE)      | 4          | Engineering | 120000
-- 6      | Dev (SDE)        | 4          | Engineering | 115000
-- 7      | Riya (Manager)   | 3          | Sales       | 160000
```

```
-- — RECURSIVE CTE: Build complete hierarchy —————
```

```
WITH RECURSIVE employee_hierarchy AS (
```

```
    -- ANCHOR QUERY: Start with root (no manager = CEO)
```

```
    SELECT
```

```
        emp_id,
```

```
        emp_name,
```

```
        manager_id,
```

```
        department,
```

```
        salary,
```

```
        0 AS level,                                -- CEO is level 0
```

```
        emp_name AS reporting_chain,                -- chain starts with
just CEO
```

```
        CAST(emp_id AS VARCHAR(500)) AS id_path    -- track path to
detect cycles
```

```
    FROM employees
```

```
    WHERE manager_id IS NULL -- root node
```

```
    UNION ALL
```

```
    -- RECURSIVE QUERY: Join each employee to their manager
```

```
    SELECT
```

```
        e.emp_id,
```

```
        e.emp_name,
```

```
        e.manager_id,
```

```
        e.department,
```

```
        e.salary,
```

```

        eh.level + 1,                                -- increment level
        eh.reporting_chain + ' → ' + e.emp_name,      -- build chain
        eh.id_path + ',' + CAST(e.emp_id AS VARCHAR(10))
FROM employees e
INNER JOIN employee_hierarchy eh
    ON e.manager_id = eh.emp_id                      -- connect to parent
WHERE id_path NOT LIKE '%' + CAST(e.emp_id AS VARCHAR(10)) +
'%'
    -- ↑ Cycle detection! Prevents infinite loop if bad data
)
SELECT
    emp_id,
    emp_name,
    level,
    REPLICATE(' ', level) + emp_name AS indented_name, --
visual tree
    reporting_chain,
    department,
    salary,
    -- Total team size under each manager
    (SELECT COUNT(*) FROM employee_hierarchy eh2
     WHERE eh2.id_path LIKE '%' + CAST(eh.emp_id AS VARCHAR) +
'%'
        AND eh2.emp_id != eh.emp_id) AS team_size,
    -- Total salary cost under each manager
    (SELECT SUM(salary) FROM employee_hierarchy eh2
     WHERE eh2.id_path LIKE '%' + CAST(eh.emp_id AS VARCHAR) +
'%'
        AND eh2.emp_id != eh.emp_id) AS team_salary_cost
FROM employee_hierarchy eh
ORDER BY id_path;

-- OUTPUT:
-- emp_name          | level | indented_name          |
-- reporting_chain

```

```

-- Vikram (CEO)      | 0 | Vikram (CEO)      | Vikram
(CEO)
-- Priya (VP Eng)    | 1 | Priya (VP Eng)    | Vikram →
Priya
-- Rahul (VP Sales)  | 1 | Rahul (VP Sales)  | Vikram →
Rahul
-- Amit (Lead)       | 2 | Amit (Lead)       | Vikram →
Priya → Amit
-- Sneha (SDE)       | 3 | Sneha (SDE)       | Vikram →
Priya → Amit → Sneha
-- Dev (SDE)         | 3 | Dev (SDE)         | Vikram →
Priya → Amit → Dev
-- Riya (Manager)    | 2 | Riya (Manager)    | Vikram →
Rahul → Riya

```

```

-- — FIND ALL EMPLOYEES UNDER SPECIFIC MANAGER —————
-- "Who reports (directly or indirectly) to Priya?"

```

```

WITH RECURSIVE subordinates AS (
    -- Anchor: Priya herself
    SELECT emp_id, emp_name, 0 AS depth
    FROM employees
    WHERE emp_name = 'Priya (VP Eng)'

    UNION ALL

    -- Recursive: all employees whose manager is in our CTE
    SELECT e.emp_id, e.emp_name, s.depth + 1
    FROM employees e
    INNER JOIN subordinates s ON e.manager_id = s.emp_id
)
SELECT emp_id, emp_name, depth AS levels_below_priya
FROM subordinates
WHERE emp_id != (SELECT emp_id FROM employees WHERE emp_name =

```



```
'Priya (VP Eng)')  
ORDER BY depth, emp_name;
```

SIMPLE EXPLANATION

RECURSIVE CTE = Query that calls ITSELF

Like a staircase:

- Step 1: Find the person at TOP (CEO, no manager)
- Step 2: Find everyone who reports to TOP
- Step 3: Find everyone who reports to THOSE people
- Step 4: Keep going until no more employees found

Each iteration = one more level in hierarchy

IMPORTANT PARTS:

- ANCHOR query = "Start here" (first iteration)
- UNION ALL = "Connect iterations"
- RECURSIVE query = "Find next level" (repeats)
- WHERE condition = "Stop when no more levels"

WITHOUT RECURSIVE CTE:

You'd need to write 5 separate queries for 5 levels

```
Level 1: SELECT ... WHERE manager_id IS NULL  
Level 2: SELECT ... WHERE manager_id IN (Level 1 results)  
Level 3: SELECT ... WHERE manager_id IN (Level 2 results)  
...
```

Very messy, and breaks if depth unknown!

WITH RECURSIVE CTE:

One clean query handles ANY depth!
2 levels? Works. 20 levels? Works!

⚠️ COMMON MISTAKES

MISTAKE 1: Infinite loop **without cycle** detection

❌ If data has circular reference (A manages B, B manages A)
→ **Recursive** CTE loops forever → query hangs!

✅ Add **cycle** detection:

WHERE id_path **NOT LIKE** '%' + emp_id + '%'

OR: Add MAXRECURSION limit:

OPTION (MAXRECURSION **100**) -- stop after 100 levels

MISTAKE 2: **Not** knowing **SQL** Server supports this

Some candidates: "CTE can't be recursive"

✅ T-SQL, PostgreSQL, MySQL **8.0+**, Spark **SQL** → **all** support!

✅ Syntax: **WITH RECURSIVE** name **AS** (...)

In **SQL** Server: **WITH** name **AS** (...) (**RECURSIVE** is optional!)

MISTAKE 3: Using **WHERE** on **recursive** part incorrectly

❌ **WHERE** level < **5** in **recursive** part

✅ Works but **OPTION** (MAXRECURSION **5**) is cleaner

OR: filter AFTER CTE: **SELECT** * **FROM** cte **WHERE** level < **5**

💡 PRO TIP

PYSPARK GRAPH TRAVERSAL (modern approach):

"In Databricks, I use GraphFrames for hierarchy:

```
from graphframes import GraphFrame
```

```
# Vertices = employees
vertices = spark.createDataFrame([
    ("1", "Vikram"),
    ("2", "Priya"),
    ("3", "Rahul")
], ["id", "name"])

# Edges = reporting relationships
edges = spark.createDataFrame([
    ("2", "1"), # Priya reports to Vikram
    ("3", "1"), # Rahul reports to Vikram
], ["src", "dst"])

g = GraphFrame(vertices, edges)

# BFS: Find all nodes reachable from Vikram within 3 hops
paths = g.bfs(fromExpr='id = 1',      # start: Vikram
              toExpr='id != 1',      # reach: any other employee
              maxPathLength=3)        # max 3 levels deep

Advantage: Handles complex graph operations that
           recursive SQL struggles with at massive scale!

But: Recursive CTE = standard SQL interview answer.
     GraphFrames = bonus points for Spark expertise!"
```

? QUESTION 30 — Advanced SQL | Pivot and Unpivot

*"Write a query to show monthly revenue by region in a matrix/pivot format.
Then write the reverse (unpivot)."*

✓ IDEAL ANSWER

```
-- — PIVOT: Rows → Columns —————  
-- Transform: region, month, revenue (3 columns, many rows)  
-- Into: region as row, each month as column
```

```
-- Input format:
```

```
-- region_code | month_name | revenue  
-- North       | Jan       | 1200000  
-- North       | Feb       | 1350000  
-- South       | Jan       | 980000
```

```
-- Output format:
```

```
-- region_code | Jan       | Feb       | Mar       | ... | Dec  
-- North       | 1200000  | 1350000  | 1450000  | ...  
-- South       | 980000   | 1100000  | 1050000  | ...
```

```
-- — STATIC PIVOT (know the months in advance) —————
```

```
SELECT
```

```
    region_code,  
    [Jan], [Feb], [Mar], [Apr], [May], [Jun],  
    [Jul], [Aug], [Sep], [Oct], [Nov], [Dec]
```

```
FROM (
```

```
    SELECT
```

```
        s.region_code,  
        dd.month_name,  
        fs.gross_amount
```

```
    FROM fact_sales fs
```

```
    JOIN dim_date dd ON fs.date_key = dd.date_key
```

```
    JOIN dim_store s ON fs.store_key = s.store_key
```

```
    WHERE dd.year = 2024
```

```
) AS source_data
```

```
PIVOT (
```

```

SUM(gross_amount)          -- what to aggregate
FOR month_name IN          -- column to pivot
    ([Jan], [Feb], [Mar], [Apr], -- distinct values become
columns
    [May], [Jun], [Jul], [Aug],
    [Sep], [Oct], [Nov], [Dec])
) AS pivot_result
ORDER BY region_code;

-- — DYNAMIC PIVOT (columns determined at runtime) —————
-- Better when months/categories aren't known in advance

DECLARE @columns AS NVARCHAR(MAX)
DECLARE @sql AS NVARCHAR(MAX)

-- Build column list from data itself
SELECT @columns = STRING_AGG(QUOTENAME(month_name), ', ')
FROM (
    SELECT DISTINCT dd.month_name
    FROM fact_sales fs
    JOIN dim_date dd ON fs.date_key = dd.date_key
    WHERE dd.year = 2024
    ORDER BY dd.month_number -- ensure Jan, Feb, Mar order!
) AS months;

-- Build dynamic SQL
SET @sql = '
SELECT region_code, ' + @columns + '
FROM (
    SELECT s.region_code, dd.month_name, fs.gross_amount
    FROM fact_sales fs
    JOIN dim_date dd ON fs.date_key = dd.date_key
    JOIN dim_store s ON fs.store_key = s.store_key
    WHERE dd.year = 2024
) AS source_data

```

```

PIVOT (
    SUM(gross_amount)
    FOR month_name IN (' + @columns + ')
) AS pivot_result
ORDER BY region_code';

EXECUTE sp_executesql @sql;

-- — UNPIVOT: Columns → Rows —————
-- Reverse: wide format → long format

-- Input (wide):
-- region | Jan      | Feb      | Mar
-- North  | 1200000  | 1350000  | 1450000

-- Output (long):
-- region | month | revenue
-- North  | Jan   | 1200000
-- North  | Feb   | 1350000
-- North  | Mar   | 1450000

SELECT
    region_code,
    month_name,
    revenue
FROM pivot_table -- your wide table
UNPIVOT (
    revenue -- new column name for values
    FOR month_name -- new column name for old column headers
    IN ([Jan], [Feb], [Mar], [Apr], [May], [Jun],
        [Jul], [Aug], [Sep], [Oct], [Nov], [Dec])
) AS unpivot_result
ORDER BY region_code,
    CASE month_name -- proper date ordering
        WHEN 'Jan' THEN 1 WHEN 'Feb' THEN 2

```

```

        WHEN 'Mar' THEN 3 WHEN 'Apr' THEN 4
        WHEN 'May' THEN 5 WHEN 'Jun' THEN 6
        WHEN 'Jul' THEN 7 WHEN 'Aug' THEN 8
        WHEN 'Sep' THEN 9 WHEN 'Oct' THEN 10
        WHEN 'Nov' THEN 11 WHEN 'Dec' THEN 12
    END;

```

-- — PYSPARK EQUIVALENT (Pivot in Spark) —————

```

df_pivot = df_sales \
    .groupBy("region_code") \
    .pivot("month_name",
          ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]) \
    .agg(F.sum("gross_amount"))

display(df_pivot)

# Unpivot in PySpark (using stack):
from pyspark.sql.functions import expr

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

# Build stack expression
stack_expr = f"stack({len(months)}, " + \
    ", ".join([f"'{m}'", `{m}`" for m in months]) + \
    ") as (month_name, revenue)"

df_unpivot = df_pivot.select("region_code", expr(stack_expr)) \
    .filter(F.col("revenue").isNotNull())

```

SIMPLE EXPLANATION

PIVOT = Rotating data 90 degrees

BEFORE PIVOT (Long format):

Region	Month	Revenue
North	Jan	1,200,000
North	Feb	1,350,000
South	Jan	980,000
South	Feb	1,100,000

AFTER PIVOT (Wide format):

Region	Jan	Feb
North	1,200,000	1,350,000
South	980,000	1,100,000

THINK: Months moved from ROWS to COLUMN HEADERS

UNPIVOT = Opposite: column headers back to rows

WHEN TO USE EACH:

Long format → best for: databases, analysis, filtering

Wide format → best for: Excel reports, dashboards, human reading

POWER BI / EXCEL USERS:

They often paste wide format data

Then complain: "why can't I filter by month?"

Because it's in WIDE format – not filterable!

Your job: keep data in LONG format in DW

Power BI semantic model presents it wide when needed!



PRO TIP

DATA WAREHOUSE NORMALIZATION INSIGHT:

"In data warehouses, ALWAYS store in LONG format:

region_code | month | revenue ← 3 columns, many rows

NOT wide format:

region | Jan | Feb | Mar ... ← 14 columns, few rows

WHY?

Long format:

- Easy to add new months (no schema change!)
- Easy to filter: WHERE month = 'Jan'
- Works with ANY visualization tool
- Aggregations easy: SUM(revenue) WHERE month IN (...)

Wide format:

- Adding 13th month = new column = schema change!
- Can't filter: WHERE [column_whose_name_is_month] = ...
- Visualization tools struggle
- Different aggregation per column = messy

DATA MODELING PRINCIPLE:

'Normalize your data model, denormalize for display'

Store: Long format (normalized)

Display: Power BI/Excel creates wide format automatically!

This principle = 15+ LPA data modeling knowledge!"



REVISION SUMMARY — Q26-Q30

REVISION: ADVANCED SQL PATTERNS (Q26-Q30)

Q26: Set operations: `EXCEPT (not in)`, `INTERSECT (both)`

`LEFT JOIN + IS NULL = NOT EXISTS`

KEY: NEVER use `NOT IN` with nullable subquery!

Q27: Running totals: `ROWS BETWEEN UNBOUNDED PRECEDING`

7-day MA: `ROWS BETWEEN 6 PRECEDING AND CURRENT ROW`

KEY: `ROWS` vs `RANGE` difference matters!

Q28: Gaps & Islands: `date - row_number = island group`

`LEAD` for next `date` to find gaps

KEY: Classic interview `pattern` – memorize the trick!

Q29: Recursive CTE: `ANCHOR + UNION ALL + RECURSIVE`

Always add `cycle` detection!

KEY: `MAXRECURSION` to prevent infinite loops!

Q30: PIVOT rows→columns, UNPIVOT columns→rows

Always store LONG format in DW!

KEY: Dynamic PIVOT using dynamic SQL for flexibility!

PATTERN ALERT:

Senior SQL interviews focus on:

→ Window functions with complex frames

→ Set operations (EXCEPT, INTERSECT)

→ Recursive CTEs for hierarchies

→ Knowing BOTH SQL and PySpark equivalent = top marks!

PYSPARK DEEP DIVE

? QUESTION 31 — PySpark | Joins Deep Dive

"Explain all types of joins in PySpark with examples. When does a shuffle happen? How do you avoid it?"

✓ IDEAL ANSWER

```
from pyspark.sql import functions as F
from pyspark.sql.functions import broadcast

# — ALL JOIN TYPES —

# Sample data
df_sales      = spark.read.table("silver_sales")      # 50M rows
df_customers  = spark.read.table("dim_customers")     # 5M rows
df_products   = spark.read.table("dim_products")     # 10K rows
df_regions    = spark.read.table("dim_regions")      # 4 rows!

# 1. INNER JOIN: Only matching rows from both
df_inner = df_sales.join(df_customers,
                        on="customer_id",
                        how="inner")

# Result: Only sales with valid customer records
# Sales without customers: DROPPED
# Customers without sales: DROPPED

# 2. LEFT JOIN (Left Outer): All left, matching right
df_left = df_sales.join(df_customers,
                       on="customer_id",
                       how="left") # or "left_outer"

# Result: ALL sales rows kept
# Customer columns = NULL for orphaned sales
# Use: keep all sales, even if customer record missing

# 3. RIGHT JOIN: All right, matching left
df_right = df_sales.join(df_customers,
                        on="customer_id",
                        how="right") # or "right_outer"

# Result: ALL customer rows kept
# Sales columns = NULL for customers with no sales
```

```

# Use: find customers who never bought anything

# 4. FULL OUTER JOIN: Everything from both
df_full = df_sales.join(df_customers,
                        on="customer_id",
                        how="full")    # or "outer", "full_outer"

# Result: ALL rows from BOTH tables
# NULLs where no match

# 5. CROSS JOIN (CARTESIAN): Every combination
# USE VERY CAREFULLY –  $O(N^2)$  complexity!
df_cross = df_regions.crossJoin(
    spark.createDataFrame(
        [(m,) for m in range(1, 13)], ["month"]
    )
)
# 4 regions × 12 months = 48 rows
# Perfect for creating "all combinations" template!

# 6. ANTI JOIN (NOT IN equivalent)
# Customers who have NO sales at all
df_no_sales = df_customers.join(
    df_sales,
    on="customer_id",
    how="left_anti"    # ← keep only rows with NO match!
)

# 7. SEMI JOIN (IN equivalent)
# Customers who HAVE at least one sale
df_has_sales = df_customers.join(
    df_sales,
    on="customer_id",
    how="left_semi"    # ← keep matching rows from LEFT only
)
# Unlike inner join: no duplicate rows from df_customers!

```

— JOIN STRATEGIES: SHUFFLE vs BROADCAST —————

WHEN SHUFFLE HAPPENS (slow):

Both tables too large → data moves across network!

Sort-Merge Join (default for large tables):

```
df_smj = df_sales.join(df_customers, on="customer_id")
```

Spark: shuffle df_sales by customer_id (EXPENSIVE!)

shuffle df_customers by customer_id (EXPENSIVE!)

sort both, merge

```
df_smj.explain() # Shows: SortMergeJoin
```

BROADCAST JOIN (fast – avoids shuffle!):

Small table sent to ALL executors

Large table processed locally (no network transfer!)

```
df_broadcast = df_sales.join(
    broadcast(df_products), # broadcast the small table!
    on="product_code",
    how="left"
)
```

```
df_broadcast.explain() # Shows: BroadcastHashJoin ←
```

AUTO-BROADCAST (configure threshold):

```
spark.conf.set("spark.sql.autoBroadcastJoinThreshold",
    str(200 * 1024 * 1024)) # auto-broadcast if <
200MB
```

Tables smaller than 200MB → automatically broadcast!

— PERFORMANCE: RIGHT TABLE SIZE MATTERS —————

RULE: Put SMALLER table on RIGHT side of join

WHY: Spark builds HASH TABLE from right side

Hash table of 10K rows (products) = tiny = fast!

Hash table of 50M rows (sales) = huge = slow!

```

# GOOD:
df_sales.join(broadcast(df_products), on="product_code")
# Products (10K) → hash table (small)
# Sales (50M) → streamed through (doesn't need hash table)

# — HANDLING COLUMN NAME CONFLICTS AFTER JOIN —————
# Both tables have "status" column → ambiguous!
df_result = df_sales.alias("s").join(
    df_customers.alias("c"),
    on="customer_id",
    how="left"
).select(
    F.col("s.sale_id"),
    F.col("s.total_amount"),
    F.col("s.status").alias("sale_status"),
    F.col("c.customer_name"),
    F.col("c.status").alias("customer_status"),
    F.col("c.customer_tier")
)

# — MULTIPLE CONDITION JOIN —————
df_complex = df_sales.join(
    df_customers,
    on=[
        df_sales.customer_id == df_customers.customer_id,
        df_sales.region_code == df_customers.preferred_region
    ],
    how="inner"
)

# — SKEWED JOIN HANDLING —————
# Problem: 60% of sales have customer_id = "WALK-IN" (cash sales)
# When joining: ALL WALK-IN goes to ONE executor = skew!

```

Solution: Salting

```
import random
```

Add random salt 0-9 to WALK-IN customer

```
df_sales_salted = df_sales.withColumn(  
    "customer_id_salted",  
    F.when(F.col("customer_id") == "WALK-IN",  
           F.concat(F.col("customer_id"),  
                    F.lit("_"),  
                    (F.rand() * 10).cast("int").cast("string")))  
    .otherwise(F.col("customer_id"))  
)
```

Explode WALK-IN in customer table (10 copies)

```
from pyspark.sql.functions import array, explode, lit
```

```
df_walkin_expanded = df_customers \  
    .filter(F.col("customer_id") == "WALK-IN") \  
    .withColumn("salt_arr",  
                F.array([F.lit(str(i)) for i in range(10)])) \  
    .withColumn("salt", F.explode("salt_arr")) \  
    .withColumn("customer_id_salted",  
                F.concat(F.col("customer_id"),  
                        F.lit("_"),  
                        F.col("salt"))) \  
    .drop("salt_arr", "salt")
```

Normal customers: keep as-is (no salt needed)

```
df_normal_customers = df_customers \  
    .filter(F.col("customer_id") != "WALK-IN") \  
    .withColumnRenamed("customer_id", "customer_id_salted")
```

```
df_customers_salted =  
df_walkin_expanded.union(df_normal_customers)
```



```
# Now join on salted key → WALK-IN distributed across 10 partitions!
```

```
df_result = df_sales_salted.join(  
    df_customers_salted,  
    on="customer_id_salted",  
    how="left"  
)
```

SIMPLE EXPLANATION

JOIN TYPES = Different ways to combine two tables

INNER JOIN = The **STRICT** bouncer

"Only let in people who have reservations in BOTH lists"

Party list A + Party list B → Only guests on BOTH lists!

LEFT JOIN = The **INCLUSIVE** host

"Everyone from MY list gets in, even without matching"

Party list A (yours) → All get in

Party list B (venue) → Only matched ones contribute info

ANTI JOIN = The **EXCLUSIVE** filter

"Who's on MY list but NOT on the venue's list?"

→ People without reservations (customer without sale)

SEMI JOIN = The **QUALIFIER**

"Who on MY list has a venue reservation?"

→ Keep only the FACT they have a reservation

→ Don't duplicate their info!

BROADCAST = The **MEGAPHONE** announcement

Instead of: each executor sends messages to find matches

You: broadcast small table **to** EVERYONE
→ **Each** executor already HAS the small table
→ No communication needed = MUCH faster!

Think **of** **32** workers **in** a room:

WITHOUT broadcast: workers yell across room **to** find matches

WITH broadcast: manager gives everyone a copy **of** the small list

each worker checks their copy = silent, fast!

⚠ COMMON MISTAKES

MISTAKE 1: **Not** checking **join** result row count

❌ **Join** df_sales × df_customers → don't *verify*

✅ ALWAYS verify:

```
print(f"Before join: {df_sales.count():,}")
```

```
print(f"After join: {df_result.count():,}")
```

If after > before → DUPLICATES **in** dimension table!

(**dim** table has multiple rows per customer_id)

MISTAKE 2: **Not** using broadcast **for** small tables

❌ **Join** 50M sales × 10K products (**default** SortMergeJoin)

✅ broadcast(df_products) → BroadcastHashJoin

Performance difference: **30** minutes → **2** minutes

MISTAKE 3: **Not** knowing left_anti **and** left_semi

Common question: "Find customers who never bought"

❌ "I'd use WHERE customer_id NOT IN (SELECT customer_id FROM sales)"

✅ "I'd use left_anti join – designed exactly for this!"
df_customers.join(df_sales, "customer_id", "left_anti")

MISTAKE 4: Ignoring data skew in joins

❌ "My join is slow" → just add more nodes

✅ Check skew first!

```
df_sales.groupBy("customer_id").count() \
        .orderBy(F.desc("count")).show(5)
```

If WALK-IN has 30M rows → skew!

Solution: salt the key

💡 PRO TIP

JOIN ORDER OPTIMIZATION (advanced):

"Spark's optimizer sometimes doesn't reorder joins optimally.
I manually optimize join order:

RULE: Join to ELIMINATE most rows FIRST

Example: 3-way join

```
df_sales (50M rows)
× df_active_customers (2M rows) [filtered: status='ACTIVE']
× df_premium_products (500 rows) [filtered: tier='PREMIUM']
```

BAD ORDER: 50M × 2M first → 10M result → × 500

GOOD ORDER: 50M × 500 first → 25K result → × 2M much smaller!

```
df_result = df_sales \
    .join(broadcast(df_premium_products), # SMALLEST first!
        on="product_code") \           # eliminates 99.9% of
rows!
```

```
.join(broadcast(df_active_customers), # Then medium
      on="customer_id")              # fewer rows left to
join
```

Also: Spark 3.x AQE (Adaptive Query Execution) does this automatically when enabled:

```
spark.conf.set('spark.sql.adaptive.enabled', 'true')
```

But: knowing manual optimization = shows deep expertise!"

? QUESTION 32 — PySpark | Schema and Data Types

"How do you define and enforce schema in PySpark? Why is it important in production?"

✓ IDEAL ANSWER

```
# — WHY SCHEMA ENFORCEMENT MATTERS —————
# inferSchema reads the ENTIRE file to guess types
# → 2 passes over data (expensive!)
# → Gets it WRONG sometimes (reads "01234" as integer, drops
#    leading 0!)
# → Types change if data changes → silent bugs!

# ALWAYS define schema explicitly in production!

from pyspark.sql.types import (
    StructType, StructField,
```

```

IntegerType, LongType, StringType,
DoubleType, DecimalType,
DateType, TimestampType,
BooleanType, ArrayType,
MapType, StructType
)
from pyspark.sql import functions as F

# — DEFINE COMPLETE PRODUCTION SCHEMA —————

sales_schema = StructType([
    # Simple fields
    StructField("sale_id", IntegerType(), nullable=False),
    # NOT NULL!
    StructField("customer_id", StringType(), nullable=False),
    StructField("store_id", IntegerType(), nullable=True),
    StructField("product_code", StringType(), nullable=True),
    StructField("quantity", IntegerType(), nullable=True),

    # Financial fields: use DecimalType for precision!
    # NEVER use DoubleType for money (floating point errors!)
    StructField("unit_price", DecimalType(18, 2),
nullable=True),
    StructField("total_amount", DecimalType(18, 2),
nullable=True),
    StructField("gst_amount", DecimalType(18, 2),
nullable=True),

    # Date/Time fields
    StructField("sale_date", DateType(), nullable=True),
    StructField("sale_timestamp", TimestampType(), nullable=True),

    # Boolean field
    StructField("is_return", BooleanType(), nullable=True),

```

```

    # Categorical
    StructField("payment_mode", StringType(), nullable=True),
    StructField("region_code", StringType(), nullable=True),
    StructField("status", StringType(), nullable=True),
])

# — READ WITH EXPLICIT SCHEMA —————
df = spark.read \
    .schema(sales_schema) \
    .option("header", "true") \
    .option("mode", "PERMISSIVE") \
    # PERMISSIVE: bad rows → NULL + logged to _corrupt_record
    # DROPMALFORMED: bad rows dropped silently
    # FAILFAST: bad rows → exception (stop pipeline)
    .csv("/mnt/bronze/sales/2024/06/sales_raw.csv")

# — HANDLE CORRUPT RECORDS —————
# Add _corrupt_record column to capture bad rows
sales_schema_with_corrupt = sales_schema.add(
    StructField("_corrupt_record", StringType(), nullable=True)
)

df_with_corrupt = spark.read \
    .schema(sales_schema_with_corrupt) \
    .option("header", "true") \
    .option("mode", "PERMISSIVE") \
    .option("columnNameOfCorruptRecord", "_corrupt_record") \
    .csv("/mnt/bronze/sales/")

# Separate clean from corrupt
df_clean =
df_with_corrupt.filter(F.col("_corrupt_record").isNull())
df_corrupt =
df_with_corrupt.filter(F.col("_corrupt_record").isNotNull())

```

```

print(f"Clean rows:    {df_clean.count():,}")
print(f"Corrupt rows: {df_corrupt.count():,}")

# Log corrupt rows for investigation
df_corrupt.select("_corrupt_record") \
    .write.format("delta").mode("append") \
    .save("/mnt/errors/sales_corrupt/")

# — NESTED SCHEMA (JSON with nested objects) —————
# Input JSON:
# {
#   "order_id": "0001",
#   "customer": {"id": "C001", "name": "Rahul", "tier": "GOLD"},
#   "items": [
#     {"code": "LAPTOP", "qty": 1, "price": 55000},
#     {"code": "MOUSE", "qty": 1, "price": 1200}
#   ]
# }

nested_schema = StructType([
    StructField("order_id", StringType(), True),
    StructField("customer", StructType([
        StructField("id", StringType(), True),
        StructField("name", StringType(), True),
        StructField("tier", StringType(), True)
    ]), True),
    StructField("items", ArrayType(
        StructType([
            StructField("code", StringType(), True),
            StructField("qty", IntegerType(), True),
            StructField("price", DecimalType(18, 2), True)
        ])
    ), True)
])

```

```

df_orders = spark.read \
    .schema(nested_schema) \
    .option("multiLine", "true") \
    .json("/mnt/bronze/orders/")

# Access nested fields
df_flat = df_orders \
    .select(
        "order_id",
        F.col("customer.id").alias("customer_id"),
        F.col("customer.name").alias("customer_name"),
        F.col("customer.tier").alias("customer_tier"),
        F.explode("items").alias("item") # flatten array
    ) \
    .select(
        "order_id", "customer_id", "customer_name",
        "customer_tier",
        F.col("item.code").alias("product_code"),
        F.col("item.qty").alias("quantity"),
        F.col("item.price").alias("unit_price")
    )

# — SCHEMA VALIDATION FUNCTION —————
def validate_schema(df, expected_schema, table_name):
    """Validate DataFrame matches expected schema"""
    actual_schema = df.schema

    errors = []
    expected_fields = {f.name: f for f in expected_schema.fields}
    actual_fields    = {f.name: f for f in actual_schema.fields}

    # Check for missing required columns
    for col_name, field in expected_fields.items():
        if col_name not in actual_fields:
            if not field.nullable:

```



```

        errors.append(f"CRITICAL: Required column
'{col_name}' MISSING!")
    else:
        errors.append(f"WARNING: Optional column
'{col_name}' missing")

    # Check data type mismatches
    for col_name in expected_fields:
        if col_name in actual_fields:
            expected_type = expected_fields[col_name].dataType
            actual_type = actual_fields[col_name].dataType
            if expected_type != actual_type:
                errors.append(
                    f"TYPE MISMATCH: '{col_name}' expected
{expected_type} "
                    f"got {actual_type}"
                )

    # Check for unexpected extra columns (schema drift!)
    extra_cols = set(actual_fields.keys()) -
set(expected_fields.keys())
    if extra_cols:
        errors.append(f"NEW COLUMNS detected: {extra_cols}")

    if errors:
        error_msg = f"Schema validation failed for
{table_name}:\n" + \
            "\n".join(errors)
        print(error_msg)
        # Log to audit table
        # Decide: fail pipeline or continue with warning?
        if any("CRITICAL" in e for e in errors):
            raise ValueError(error_msg) # fail on critical
errors
    else:

```

```
print(f"✅ Schema validation passed for {table_name}")
```

```
return len(errors) == 0
```

Usage

```
is_valid = validate_schema(df_clean, sales_schema,  
"bronze_sales")
```

📖 SIMPLE EXPLANATION

SCHEMA IN PYSPARK = Blueprint for your data

Like a building blueprint:

Blueprint says: "This column is an INTEGER"

If you try to put TEXT there → error!

WITHOUT SCHEMA (inferSchema):

Spark reads entire file → guesses types

Column: "product_id" with values: "01234", "05678"

Spark guesses: INTEGER (removes leading zeros!)

01234 → 1234 (WRONG!)

05678 → 5678 (WRONG!)

Now your product IDs are WRONG!

Reports don't match! Hours of debugging!

WITH SCHEMA:

You define: product_id = StringType()

Spark reads as string: "01234" → "01234" (correct!)

Leading zeros preserved!

DECIMALTYPE **for** money:

DoubleType(18.2 + 0.3 = 18.499999999999996 ← WRONG!)

DecimalType(18.2 + 0.3 = 18.5 ← CORRECT!)

NEVER use DoubleType **for** financial data!

Always use DecimalType(precision, scale)

⚠ COMMON MISTAKES

MISTAKE 1: Using DoubleType **for** monetary values

❌ StructField("total_amount", DoubleType())

→ 55000.10 + 0.05 = 55000.1500000000004 (float imprecision!)

✅ StructField("total_amount", DecimalType(18, 2))

→ Exact decimal arithmetic **for** financial data!

MISTAKE 2: Not handling schema drift

Source adds new column → pipeline fails!

Options:

A: mergeSchema="true" → automatically add new columns

.option("mergeSchema", "true")

B: Schema drift detection → alert team

C: Schema evolution **versioning**

Production: detect **AND** alert, then decide!

MISTAKE 3: Not using nullable=False **for** NOT NULL columns

StructField("sale_id", IntegerType(), nullable=True)

→ Allows NULL sale_ids silently → broken data!

StructField("sale_id", IntegerType(), nullable=False)

→ Null sale_id → error immediately → catch it early!

MISTAKE 4: Using StringType for everything

✗ All columns = StringType() "to be safe"

→ Loses type safety

→ Aggregations work on strings? sum("01234") = concatenation!

✓ Use correct types: IntegerType for IDs, DateType for dates

💡 PRO TIP

SCHEMA REGISTRY PATTERN (enterprise approach):

"In large data platforms, I use Schema Registry:

PROBLEM:

50 pipelines, 200 tables

Each pipeline hardcodes its own schema

Source changes schema → update 50 pipelines!

SOLUTION: Central Schema Registry

Store schemas in Delta table or JSON in ADLS:

```
/config/schemas/bronze_sales_v2.json
{
  "version": 2,
  "created": "2024-06-15",
  "fields": [
    {"name": "sale_id", "type": "integer", "nullable": false},
    ...
  ],
  "changelog": "Added product_category column"
}
```

Pipeline reads schema at runtime:

```
schema_json =  
spark.read.json('/config/schemas/bronze_sales_v2.json')  
schema = StructType.fromJson(schema_json.first().asDict())  
df = spark.read.schema(schema).csv(input_path)
```

Benefits:

- ✓ Schema change → update one JSON file
- ✓ All pipelines automatically use new schema
- ✓ Version history of schema changes
- ✓ Schema documentation in one place

This is PLATFORM ENGINEERING approach.

Companies at 15+ LPA need this for maintainability!"

? QUESTION 33 — PySpark | Advanced Transformations

"Write a complete PySpark transformation pipeline that reads raw JSON, flattens nested structures, applies business rules, handles data quality, and writes Delta."

✓ IDEAL ANSWER

```
# — COMPLETE PYSPARK TRANSFORMATION PIPELINE —  
# Scenario: RetailMart payment data from API (nested JSON)  
# Goal: Bronze (raw) → Silver (clean, flat, enriched)  
  
from pyspark.sql import functions as F
```

```

from pyspark.sql.types import *
from pyspark.sql.window import Window
from delta.tables import DeltaTable
import logging

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)

# — STEP 1: DEFINE SCHEMA (never infer!) —————
payment_schema = StructType([
    StructField("payment_id", StringType(), False),
    StructField("timestamp", StringType(), True),
    StructField("amount", DoubleType(), True),
    StructField("currency", StringType(), True),
    StructField("status", StringType(), True),
    StructField("payment_mode", StringType(), True),
    StructField("customer", StructType([
        StructField("id", StringType(), True),
        StructField("name", StringType(), True),
        StructField("email", StringType(), True),
        StructField("phone", StringType(), True),
        StructField("tier", StringType(), True)
    ]), True),
    StructField("merchant", StructType([
        StructField("id", StringType(), True),
        StructField("name", StringType(), True),
        StructField("category", StringType(), True),
        StructField("region_code", StringType(), True)
    ]), True),
    StructField("device", StructType([
        StructField("type", StringType(), True),
        StructField("os", StringType(), True),
        StructField("ip_address", StringType(), True),
        StructField("is_new", BooleanType(), True)
    ]), True),

```

```

    StructField("items", ArrayType(StructType([
        StructField("product_code", StringType(), True),
        StructField("quantity", IntegerType(), True),
        StructField("unit_price", DoubleType(), True)
    ])), True)
])

# — STEP 2: READ WITH CORRUPT RECORD HANDLING —————
schema_with_corrupt = payment_schema.add(
    StructField("_corrupt_record", StringType(), True)
)

df_raw = spark.read \
    .schema(schema_with_corrupt) \
    .option("multiLine", "true") \
    .option("mode", "PERMISSIVE") \
    .option("columnNameOfCorruptRecord", "_corrupt_record") \
    .json(f"/mnt/bronze/payments/{process_date}/")

raw_count = df_raw.count()
logger.info(f"Raw records: {raw_count:,}")

# Separate corrupt records
df_corrupt = df_raw.filter(F.col("_corrupt_record").isNotNull())
df_clean_raw = df_raw.filter(F.col("_corrupt_record").isNull()) \
    .drop("_corrupt_record")

corrupt_count = df_corrupt.count()
if corrupt_count > 0:
    logger.warning(f"CORRUPT RECORDS: {corrupt_count:,}")
    df_corrupt.write.format("delta").mode("append") \
        .save("/mnt/errors/corrupt_payments/")

# — STEP 3: FLATTEN NESTED STRUCTURE —————
df_flat = df_clean_raw \

```

```

.select(
  # Root level fields
  "payment_id",
  "amount",
  "currency",
  "status",
  "payment_mode",

  # Parsed timestamp
  F.to_timestamp("timestamp", "yyyy-MM-dd'T'HH:mm:ss.SSSZ")
    .alias("payment_timestamp"),

  # Customer nested struct → flat columns
  F.col("customer.id").alias("customer_id"),
  F.col("customer.name").alias("customer_name"),
  F.col("customer.email").alias("customer_email"),
  F.col("customer.phone").alias("customer_phone"),
  F.col("customer.tier").alias("customer_tier"),

  # Merchant nested struct → flat columns
  F.col("merchant.id").alias("merchant_id"),
  F.col("merchant.name").alias("merchant_name"),
  F.col("merchant.category").alias("merchant_category"),
  F.col("merchant.region_code").alias("region_code"),

  # Device info
  F.col("device.type").alias("device_type"),
  F.col("device.is_new").alias("is_new_device"),

  # Items array → keep as-is for now (flatten separately)
  "items"
)

```

— STEP 4: DATA QUALITY RULES —

```
dq_metrics = {}
```



```

# Rule 1: No null payment_id
null_ids = df_flat.filter(F.col("payment_id").isNull()).count()
dq_metrics["null_payment_ids"] = null_ids

# Rule 2: Valid amounts (positive, not unreasonably large)
invalid_amounts = df_flat.filter(
    F.col("amount").isNull() |
    (F.col("amount") <= 0) |
    (F.col("amount") > 100000000) # max 1 crore per transaction
).count()
dq_metrics["invalid_amounts"] = invalid_amounts

# Rule 3: Valid status values
valid_statuses = ["SUCCESS", "FAILED", "PENDING", "REFUNDED"]
invalid_status = df_flat.filter(
    ~F.col("status").isin(valid_statuses)
).count()
dq_metrics["invalid_status"] = invalid_status

# Rule 4: No duplicate payment IDs
total = df_flat.count()
distinct = df_flat.dropDuplicates(["payment_id"]).count()
dq_metrics["duplicate_payments"] = total - distinct

logger.info(f"DQ Metrics: {dq_metrics}")

# Apply quality filter (keep only clean data)
df_quality_passed = df_flat \
    .filter(F.col("payment_id").isNotNull()) \
    .filter(F.col("amount") > 0) \
    .filter(F.col("amount") <= 100000000) \
    .filter(F.col("status").isin(valid_statuses)) \
    .dropDuplicates(["payment_id"])

```

```
df_enriched = df_quality_passed \
    .withColumn("payment_date",
                F.to_date("payment_timestamp")) \
    .withColumn("year",
                F.year("payment_timestamp")) \
    .withColumn("month",
                F.month("payment_timestamp")) \
    .withColumn("hour_of_day",
                F.hour("payment_timestamp")) \
    .withColumn("is_weekend",
                F.dayofweek("payment_timestamp").isin([1, 7])) \
    .withColumn("is_night_transaction",
                (F.hour("payment_timestamp") < 6) |
                (F.hour("payment_timestamp") >= 22)) \
    .withColumn("amount_bucket",
                F.when(F.col("amount") < 1000, "MICRO")
                  .when(F.col("amount") < 10000, "SMALL")
                  .when(F.col("amount") < 100000, "MEDIUM")
                  .when(F.col("amount") < 1000000, "LARGE")
                  .otherwise("ENTERPRISE")) \
    .withColumn("gst_amount",
                F.round(F.col("amount") * 0.18, 2)) \
    .withColumn("net_amount",
                F.col("amount") - F.col("gst_amount")) \
    .withColumn("customer_email_masked",
                F.regexp_replace("customer_email",
                                  r"^(^@){3}[^@]*(@.*$)",
                                  "$1***$2")) \
    .withColumn("customer_phone_masked",
                F.concat(F.lit("XXXX"),
                        F.substring("customer_phone", -4, 4))) \
    .withColumn("fraud_risk_score",
                F.when(
                    F.col("amount") > 100000,
```

```

        F.when(F.col("is_night_transaction"), 3.0)
        .when(F.col("is_new_device"), 2.0)
        .otherwise(1.0)
    ).otherwise(0.0)) \
.withColumn("load_timestamp", F.current_timestamp()) \
.withColumn("source_system", F.lit("Payment_Gateway_API")) \
.withColumn("batch_id", F.lit(process_date))

# — STEP 6: EXPLODE ITEMS ARRAY (create line items table) —
df_payment_items = df_enriched \
    .select("payment_id", "payment_date", "amount",
        F.explode("items").alias("item")) \
    .select(
        "payment_id",
        "payment_date",
        "amount",
        F.col("item.product_code").alias("product_code"),
        F.col("item.quantity").alias("quantity"),
        F.col("item.unit_price").alias("unit_price"),
        (F.col("item.quantity") * F.col("item.unit_price"))
        .alias("line_total")
    )

# Drop items array from main table
df_final = df_enriched.drop("items", "customer_email",
    "customer_phone")

# — STEP 7: WRITE TO SILVER (Delta format) —
silver_path = "/mnt/silver/payments/"
items_path = "/mnt/silver/payment_items/"

if DeltaTable.isDeltaTable(spark, silver_path):
    # Merge (upsert) for daily incremental
    DeltaTable.forPath(spark, silver_path).alias("t") \
        .merge(df_final.alias("s"),

```

```

        "t.payment_id = s.payment_id") \
    .whenMatchedUpdateAll() \
    .whenNotMatchedInsertAll() \
    .execute()
else:
    # First run – create table
    df_final.write \
        .format("delta") \
        .partitionBy("year", "month") \
        .save(silver_path)

# Payment items always append
df_payment_items.write \
    .format("delta") \
    .mode("append") \
    .partitionBy("payment_date") \
    .save(items_path)

# — STEP 8: OPTIMIZE AFTER WRITE —————
spark.sql(f"OPTIMIZE delta.`{silver_path}` ZORDER BY
(customer_id, payment_date)")

# — STEP 9: LOG METRICS —————
final_count = df_final.count()
logger.info(f"""
Pipeline Complete:
    Raw:           {raw_count:,}
    Corrupt:       {corrupt_count:,}
    DQ Failed:     {raw_count - corrupt_count - final_count:,}
    Written:       {final_count:,}
    Items:         {df_payment_items.count():,}
    DQ Metrics:    {dq_metrics}
""")

mssparkutils.notebook.exit(

```

```
f"SUCCESS:{final_count}:CORRUPT:{corrupt_count}"  
)
```

PRO TIP

PRODUCTION NOTEBOOK STANDARDS (what 15+ LPA expects):

"Every production notebook I write follows these standards:

1. DOCUMENTATION CELL (first cell):
Purpose, author, schedule, inputs, outputs
2. PARAMETER CELL (second cell):
All pipeline parameters with defaults
3. IMPORTS CELL (third cell):
All imports grouped and organized
4. HELPER FUNCTIONS (before main logic):
Reusable **functions**, testable **in** isolation
5. MAIN LOGIC (core cells):
Bronze → Silver transformation
Each step clearly labeled
6. METRICS CELL (before exit):
Log all metrics: input count, output count, DQ results
7. EXIT CELL (last cell):
mssparkutils.notebook.exit() with structured JSON

This structure means:

- Any team member can understand the notebook
- Easy to maintain and modify
- Pipeline can read output and make decisions
- Audit trail complete

This is PRODUCTION ENGINEERING STANDARDS.

Junior engineers write notebooks that only THEY understand.

Senior engineers write notebooks that the TEAM can maintain!"

? QUESTION 34 — PySpark | Performance Tuning Advanced

"Explain the Catalyst Optimizer in detail. How does AQE (Adaptive Query Execution) improve it? Show with code examples."

✓ IDEAL ANSWER

```
# — CATALYST OPTIMIZER: 4 PHASES
```

```
# PHASE 1: ANALYSIS
```

```
# Resolve column names, check they exist, determine data types
```

```
# "Is this column name valid? What type is it?"
```

```
df = spark.read.parquet("/mnt/silver/sales/")
```

```
# You write: filter on "region" and "amount"
```

```
# Catalyst checks: Do these columns exist? What types?
```

```
# Analysis: region=StringType, amount=DoubleType ✓
```

```

# PHASE 2: LOGICAL OPTIMIZATION (Rule-Based)
# Apply pre-defined rules to improve logical plan

# YOU write this:
df_result = df \
    .filter(F.col("region_code") == "NORTH") \
    .select("sale_id", "amount", "region_code") \
    .filter(F.col("amount") > 1000) # second filter added later

# CATALYST TRANSFORMS TO:
# Rule: Predicate Pushdown → combine AND push both filters down
# Rule: Column Pruning → only read needed columns
# Rule: Constant Folding → evaluate constants at planning time

# OPTIMIZED LOGICAL PLAN reads as:
# Read Parquet (only cols: sale_id, amount, region_code)
#   with pushed filters: region_code='NORTH' AND amount > 1000
# = READ LESS DATA!

# SEE THE PLAN:
df_result.explain(True)
# Shows all 4 plans: Parsed, Analyzed, Optimized, Physical

# PHASE 3: PHYSICAL PLANNING (Cost-Based)
# Generate multiple physical plans
# Choose best based on statistics (row counts, sizes)
# Decide: BroadcastHashJoin or SortMergeJoin?

# FORCE CATALYST TO CHOOSE BROADCAST:
spark.conf.set("spark.sql.cbo.enabled", "true")
spark.sql("ANALYZE TABLE silver_sales COMPUTE STATISTICS FOR ALL COLUMNS")
# Now Catalyst HAS statistics → better decisions!

```

```
# PHASE 4: CODE GENERATION (Tungsten)
# Generate optimized JVM bytecode
# Whole-stage code generation: multiple operators in ONE function
# Cache-friendly memory layouts
# SIMD instructions (CPU vectorization)

# — AQE: ADAPTIVE QUERY EXECUTION —————
# Problem: Catalyst plans at compile time (before seeing actual
data)
# Solution: AQE re-optimizes at RUNTIME using actual data!

# Enable AQE (default ON in Spark 3.2+):
spark.conf.set("spark.sql.adaptive.enabled", "true")

# — AQE FEATURE 1: COALESCING SHUFFLE PARTITIONS —————
# Before AQE:
spark.conf.set("spark.sql.shuffle.partitions", "200")
# After groupBy: 200 shuffle partitions
# But data after filter is tiny! 200 partitions of 1KB each =
waste!

# With AQE coalescing:
spark.conf.set("spark.sql.adaptive.coalescePartitions.enabled",
"true")
spark.conf.set("spark.sql.adaptive.advisoryPartitionSizeInBytes",
str(128 * 1024 * 1024)) # target 128MB partitions

# AQE measures actual shuffle data size after stage 1:
# "Only 50MB total! Don't need 200 partitions → coalesce to 4!"
# 200 tasks → 4 tasks → 196 tasks SAVED!

# — AQE FEATURE 2: SKEW JOIN HANDLING —————
spark.conf.set("spark.sql.adaptive.skewJoin.enabled", "true")
spark.conf.set("spark.sql.adaptive.skewJoin.skewedPartitionFactor",
"5")
```



```

# Threshold: partition is skewed if size > 5x median partition
size

# Without AQE:
# Partition 47: 2GB (WALK-IN customer)
# Other partitions: 50MB each
# Task 47 takes 4 hours while others take 3 minutes → BOTTLENECK!

# With AQE:
# AQE detects partition 47 is 2GB (40x larger than median)
# AQE SPLITS partition 47 into sub-tasks automatically!
# Task 47a: 500MB, Task 47b: 500MB, 47c: 500MB, 47d: 500MB
# All sub-tasks complete in 45 minutes instead of 4 hours!

# — AQE FEATURE 3: RUNTIME JOIN STRATEGY SWITCHING —————
# Catalyst plans: "Both tables are large → SortMergeJoin"
# Runtime reality: "After filter, right table is only 10MB!"
# AQE switches: SortMergeJoin → BroadcastHashJoin at runtime!

# DEMO: See AQE in action
df_large = spark.range(10000000).toDF("id") \
    .withColumn("value", (F.rand() * 1000).cast("int"))

df_small = spark.range(100).toDF("id") \
    .withColumn("label", F.concat(F.lit("item_"), F.col("id")))

# With AQE: even if Catalyst planned SortMergeJoin for two
range(N)
# After filter, AQE detects small side → switches to broadcast!
df_joined = df_large.join(df_small, "id")
df_joined.explain()

# — MEASURING CATALYST IMPROVEMENTS —————
import time

```

```
# Without optimization:
spark.conf.set("spark.sql.adaptive.enabled", "false")
start = time.time()
df.filter(F.col("region") == "NORTH") \
    .groupBy("store_id") \
    .agg(F.sum("amount").alias("revenue")) \
    .collect()
print(f"Without AQE: {time.time() - start:.1f}s")

# With AQE:
spark.conf.set("spark.sql.adaptive.enabled", "true")
start = time.time()
df.filter(F.col("region") == "NORTH") \
    .groupBy("store_id") \
    .agg(F.sum("amount").alias("revenue")) \
    .collect()
print(f"With AQE: {time.time() - start:.1f}s")

# Typical result:
# Without AQE: 312.4s
# With AQE:      47.2s
# → 6.6x faster just by enabling AQE!
```

SIMPLE EXPLANATION

CATALYST OPTIMIZER = Chess grandmaster **for** your query

YOU: "I want to filter, then join, then group"

CATALYST: "Let me think... what's the **OPTIMAL** way to do this?"

PHASE **1** (Analysis):

"Does this query make sense? Do these columns exist?"

Like proofreading your chess moves

PHASE 2 (Logical Optimization):

"I can do this more efficiently!"

Rules like: "Filter early → less data to process"

Like a chess player applying opening theory

PHASE 3 (Physical Planning):

"Which actual algorithms should I use?"

Like choosing your chess strategy based on opponent

PHASE 4 (Code Generation):

"Write the most efficient machine code"

Like executing moves as fast as physically possible

AQE = A GRANDMASTER who ADAPTS MID-GAME

Without AQE: Plan once before seeing data → stick to plan

Even if data distribution was unexpected!

With AQE: Execute stage 1, LOOK at results, RE-PLAN stage 2!

"Ah, data is much smaller than expected → use faster strategy!"

Like a chess player who adapts based on opponent's moves!

This re-planning is WHY AQE is so powerful.

It fixes the fundamental weakness of static planning!

COMMON MISTAKES

MISTAKE 1: Thinking AQE **is** automatic performance fix

❌ "Just enable AQE and everything is fast"

✅ AQE helps **with**:

- Skew (automatically splits large partitions)
- Shuffle coalescing (merges tiny partitions)
- **Join** strategy (switches **to** broadcast **when** possible)

AQE CANNOT fix:

- Reading too much data (you must filter early!)
- Bad schema design (NULLs everywhere)
- Wrong data types (StringType **for** everything)
- Missing indexes **in** underlying storage

MISTAKE 2: **Not** knowing **when** AQE doesn't *help*

Streaming: AQE **is** batch-only (streaming has different optimizer)

Very small data: AQE overhead > benefit

Single partition workloads: no shuffle **to** optimize!

MISTAKE 3: Disabling AQE **for** "better performance"

Some candidates: "I **disable** AQE for consistent behavior"

❌ This removes major optimization!

✅ AQE **is** almost always beneficial

Only disable **if**: testing specific behavior, debugging AQE bugs

💡 PRO TIP

PREDICATE PUSHDOWN: Most Important Optimization

"Predicate Pushdown is Catalyst's most impactful optimization.
It pushes filter conditions INTO the data source:

For Parquet files:

YOUR code:

```
spark.read.parquet('/mnt/silver/sales/')  
    .filter(col('region_code') == 'NORTH')
```

CATALYST does:

Tell Parquet reader: 'Only give me rows where region=NORTH'

Parquet checks row GROUP statistics:

'Row group 1: region range = EAST, WEST → SKIP entirely!'

'Row group 2: has NORTH → read this one!'

Result: Reads 25% of file instead of 100%!

For partitioned data:

```
.filter(col('year') == 2024).filter(col('month') == 6)
```

→ Parquet only reads year=2024/month=6/ folder!

→ Skips all other year/month folders!

→ 1/36th of data for monthly query on 3 years!

HOW TO VERIFY PUSHDOWN:

```
df.explain()
```

Look for: PartitionFilters and PushedFilters in scan

GOOD (pushdown working):

```
FileScan parquet PushedFilters: [IsNotNull(region_code),  
                                  EqualTo(region_code,NORTH)]
```

```
PartitionFilters: [year=2024, month=6]
```

BAD (no pushdown!):

```
FileScan parquet PushedFilters: [] ← EMPTY! Full scan!
```

COMMON REASON for no pushdown:

❌ `df.filter(F.year(F.col('sale_date')) == 2024)`
(function on column → can't push down!)

✅ `df.filter(F.col('year') == 2024)`
(direct column comparison → pushes down!)

Pre-compute year in table:

`.withColumn('year', F.year('sale_date'))`
→ Creates 'year' column → filter can push down!

Understanding PUSHDOWN = senior Spark optimization knowledge!"

? QUESTION 35 — PySpark | Partitioning Strategy

"Explain different partitioning strategies in Spark. How do you decide the right partition count and partition key?"

✅ IDEAL ANSWER

— UNDERSTANDING PARTITIONS —————

A partition = one chunk of data on one executor at one time
More partitions = more parallelism (up to limit of CPU cores)
Fewer partitions = less overhead but less parallelism

CHECK CURRENT PARTITION COUNT:

```
print(f"Partitions: {df.rdd.getNumPartitions()}")
```

— PARTITION COUNT GUIDELINES —————

RULE 1: Match parallelism to cluster
2-4 partitions per CPU core

```

cluster_cores = 8 * 4 # 8 workers × 4 cores = 32 cores
optimal_partitions = cluster_cores * 3 # 3x = 96

# RULE 2: Target partition size 128MB - 256MB
# 500GB data / 200MB per partition = 2500 partitions
data_size_gb = 500
target_partition_mb = 200
optimal_partitions_size = int((data_size_gb * 1024) /
target_partition_mb)
# = 2560 partitions

# RULE 3: For shuffles, tune shuffle partitions
spark.conf.set("spark.sql.shuffle.partitions", "200") # default
# For small data (<1GB): reduce to 10-50
# For large data (>100GB): increase to 500-2000
# With AQE: this is automatically tuned!

# — REPARTITION: INCREASE or REDISTRIBUTE —————
# Creates shuffle (expensive!) – use when you NEED it

# By count (round-robin distribution)
df_equal = df.repartition(100)
# 100 roughly equal partitions
# Use: before write to control file count output

# By column (hash distribution)
df_by_region = df.repartition(F.col("region_code"))
# All North data → same partition(s)
# All South data → same partition(s)
# Use: before groupBy("region_code") → no shuffle needed!

# By count AND column
df_optimized = df.repartition(50, F.col("region_code"))
# 50 partitions, grouped by region
# Control both parallelism AND data locality

```

```

# — COALESCE: ONLY DECREASE (no shuffle!) —————
# After filtering: many empty partitions remain → waste!

df_north = df.filter(F.col("region_code") == "NORTH")
# Was 200 partitions of data, now only 50 have North data
# 150 empty partitions = 150 wasted tasks!

df_north_compact = df_north.coalesce(50)
# 200 → 50 partitions WITHOUT shuffle
# Merges local partitions (no network transfer!)
# Use: before write to avoid small output files

# — PARTITION PRUNING IN STORAGE —————
# DIFFERENT from Spark partitioning (runtime)
# Storage partitioning = physical folder structure on disk

# WRITE with partitioning:
df.write \
    .format("delta") \
    .partitionBy("year", "month", "region_code") \
    .save("/mnt/silver/sales/")

# Creates structure:
# /sales/year=2024/month=6/region_code=NORTH/part-0001.parquet
# /sales/year=2024/month=6/region_code=SOUTH/part-0002.parquet
# /sales/year=2024/month=7/region_code=NORTH/part-0003.parquet

# READ with filter = partition pruning:
df_june_north = spark.read.parquet("/mnt/silver/sales/") \
    .filter(F.col("year") == 2024) \
    .filter(F.col("month") == 6) \
    .filter(F.col("region_code") == "NORTH")

# Spark reads ONLY: /sales/year=2024/month=6/region_code=NORTH/

```



```

# ALL other folders SKIPPED!
# If 12 months × 4 regions = 48 partitions total
# Reads only 1 of 48 → 98% data skipped!

# — CHOOSING PARTITION KEY —————

# GOOD PARTITION KEYS (low cardinality, commonly filtered):
# ✅ year, month, day (date partitioning)
# ✅ region_code (4-20 values)
# ✅ status (ACTIVE, INACTIVE, PENDING)
# ✅ data_source (ORACLE, SAP, API)

# BAD PARTITION KEYS (too many or too few values):
# ❌ customer_id (millions of values → millions of tiny files!)
# ❌ timestamp (infinite values → worst!)
# ❌ amount (continuous values → useless partitioning)
# ❌ boolean (only 2 values → huge partitions!)

# — DETECTING PARTITION IMBALANCE —————
def check_partition_balance(df, partition_col=None):
    """Check how evenly data is distributed across partitions"""

    if partition_col:
        partition_counts = df.groupBy(partition_col) \
                               .count() \
                               .orderBy(F.desc("count"))
    else:
        # Check Spark partitions (not storage partitions)
        from pyspark.sql.functions import spark_partition_id
        partition_counts =
df.groupBy(spark_partition_id().alias("partition")) \
        .count() \
        .orderBy(F.desc("count"))

    stats = partition_counts.agg(

```

```

        F.max("count").alias("max_rows"),
        F.min("count").alias("min_rows"),
        F.avg("count").alias("avg_rows"),
        F.stddev("count").alias("stddev_rows"),
        F.count("*").alias("num_partitions")
    ).first()

skew_ratio = stats["max_rows"] / stats["avg_rows"]

print(f"Partitions: {stats['num_partitions']}")
print(f"Max rows:    {stats['max_rows']:,}")
print(f"Min rows:    {stats['min_rows']:,}")
print(f"Avg rows:    {stats['avg_rows']:.0f}")
print(f"Skew ratio: {skew_ratio:.1f}x")

if skew_ratio > 3:
    print(f"⚠️ HIGH SKEW detected! Max partition is
{skew_ratio:.0f}x larger than average!")
    print("    Consider: salting, AQE skew join, or different
partition key")
else:
    print(f"✅ Partition distribution is balanced")

check_partition_balance(df_sales, "region_code")
# Output:
# Partitions: 4
# Max rows:    25,000,000 (North India)
# Min rows:    8,000,000 (Northeast)
# Avg rows:    12,500,000
# Skew ratio: 2.0x ← acceptable

```

SIMPLE EXPLANATION

PARTITIONS = Parallel work units

ANALOGY: 100 students writing exam in one hall

SINGLE PARTITION (bad):

All 100 students share ONE answer sheet

→ Only ONE student writes at a time

→ Takes forever!

100 PARTITIONS (good):

Each student has their own answer sheet

→ All 100 write simultaneously

→ 100x faster!

TOO MANY PARTITIONS (also bad):

Each student gets 100 pages but only writes 1 line

→ 99 pages wasted

→ Teacher takes 1 hour just to collect all papers!

OPTIMAL PARTITIONS:

Each student gets 5-10 pages (fills them up)

→ All write simultaneously

→ Easy to collect afterward

STORAGE PARTITIONING = Filing cabinet organization

Put papers by date → drawer for each date

Next month: just open March drawer (skip Jan, Feb!)

= FAST because you find only what you need

SPARK PARTITIONS = In-memory parallelism during processing

STORAGE PARTITIONS = Physical file organization on disk

Both matter! Different problem, different solution!

💡 PRO TIP

PARTITION STRATEGIES FOR DIFFERENT SCENARIOS:

SCENARIO 1: Daily batch processing (most common)

Write: `partitionBy("year", "month")`

Reads filter: `WHERE year=2024 AND month=6`

→ Read only 1/36 of data for 3-year history!

SCENARIO 2: Event-driven processing

Write: `partitionBy("event_date", "event_type")`

Good partition: 4 event types × 365 days = 1460 partitions

Each partition: ~100MB (right size!)

SCENARIO 3: Multi-tenant data (company data)

Write: `partitionBy("company_id")`

Only if: each company has ≥ 100MB data

If small companies: use ZORDER instead!

SCENARIO 4: Very large fact table

Multiple level partitioning:

`partitionBy("year", "month", "region_code")`

But: don't over-partition!

12 months × 4 regions = 48 partitions

If data is 100GB: 48 × 2GB each = GOOD!

If data is 1GB: 48 × 20MB each = TOO SMALL!

RULE OF THUMB:

'Minimum 128MB per partition for storage'

'Maximum 2GB per partition for storage'

Outside this range → over/under partitioned!

ZORDER vs PARTITIONING:

Partitioning: coarse-grained (folders), great for common

filters

ZORDER: fine-grained (co-locates within partition),
great for complex filter combinations

USE BOTH:

partitionBy("year", "month") → handles date range queries

ZORDER BY ("store_id", "product_code") → handles store/product

lookups

Together: most queries are fast!"



REVISION SUMMARY — Q31-Q35

REVISION: PYSPARK DEEP DIVE (Q31-Q35)

Q31: All Join Types

left_anti = NOT IN | left_semi = IN (no duplicates)

Broadcast small tables ALWAYS!

KEY: Salting for skewed joins (WALK-IN problem)!

Q32: Schema Definition

ALWAYS define explicitly (never inferSchema in prod!)

DecimalType for money (NOT DoubleType!)

KEY: PERMISSIVE mode + _corrupt_record column!

Q33: Full Transformation Pipeline

Read → Flatten → DQ → Business Rules → Write Delta

KEY: Log metrics, handle corrupt records, mask PII!

Q34: Catalyst + AQE

4 phases: Analysis → Logical → Physical → Code Gen

AQE: coalescing + skew + join switching

KEY: Predicate pushdown = biggest optimization!

Q35: Partitioning Strategy

Storage partitions (folders) vs Spark partitions (RAM)

128MB-2GB per storage partition = optimal

KEY: repartition (shuffle) vs coalesce (no shuffle)!

INTERVIEW PATTERN:

PySpark code questions: always mention:

→ DecimalType for money (not Double!)

→ AQE for performance

→ Broadcast for joins

→ Explicit schema (not inferSchema)

AZURE ARCHITECTURE PATTERNS

? QUESTION 36 — Architecture | Data Mesh vs Data Lake

"What is Data Mesh architecture? How is it different from a centralized Data Lake? When would you recommend it?"

✓ IDEAL ANSWER

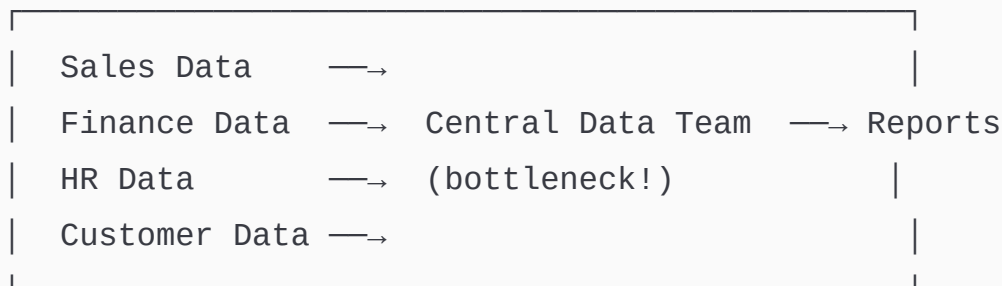
CENTRALIZED DATA LAKE (Traditional):

Structure:

ONE central data team

ONE central data lake

ALL domains deposit data → central team processes → serves everyone



PROBLEMS at scale:

❌ Central team = BOTTLENECK (every team waiting **for** them)

❌ Central team doesn't *understand Sales domain deeply*

❌ Data quality issues discovered late (central team away **from** source)

❌ Pipeline backlog grows: "6-month wait for new data pipeline"

❌ One team responsible **for** ALL data quality issues across company

❌ Doesn't *scale beyond 100-200 tables*

DATA MESH (Modern, decentralized):

4 PRINCIPLES:

PRINCIPLE 1: Domain Ownership

Domain teams OWN their data **as** a product

Sales team: owns sales data + pipelines

Finance team: owns finance data + pipelines
Customer team: owns customer data + pipelines

"You built it, you own it"

Accountable **for**: quality, freshness, availability

PRINCIPLE 2: Data **as** a Product

Each domain team treats data **like** an API product:

- Versioned (breaking changes = **new** version)
- SLA guaranteed (available **by** 6 AM)
- Documented (what **each** field means)
- Discoverable (**in** data catalog)
- Consumer-focused (easy **for** others **to** use)

PRINCIPLE 3: Self-Serve Data Platform

Central platform team provides INFRASTRUCTURE (**not** pipelines!):

- Templates **for** creating data products
- Monitoring **and** observability tools
- Data catalog (find who has what data)
- Security **and** governance framework
- Standard data storage (ADLS, Delta Lake)

Domain teams use platform but BUILD their own products!

PRINCIPLE 4: Federated Computational Governance

Global standards (enforced **by** platform):

- Schema format (must use Avro/Parquet)
- Security model (RBAC rules)
- Quality standards (minimum DQ checks)

Domain freedom (within standards):

- Choose their own tools (Spark, Python, SQL)
- Their own team processes
- Their own enhancement schedule

DATA MESH **IN** AZURE:

Central Platform Team:

- ADLS Gen2 (standard storage)
- Unity Catalog / Azure Purview (governance + catalog)
- ADF/Fabric templates (reusable pipeline patterns)
- Monitoring templates (Azure Monitor dashboards)

Sales Domain Team:

- Own ADF workspace
- Own Databricks workspace
- Own ADLS container: /domains/sales/
- Own their pipelines, their quality, their SLAs

Finance Domain Team:

- Own ADF workspace
- /domains/finance/
- Cross-domain access via Azure AD permissions

WHEN TO RECOMMEND DATA MESH:

- ✓ Large organization (100+ engineers, 5+ data domains)
- ✓ Multiple autonomous business units
- ✓ Central team **is** bottleneck
- ✓ Need domain-specific expertise **in** data engineering
- ✓ Organization already has decentralized engineering

STICK **WITH** CENTRALIZED **WHEN**:

- ✓ Small/medium organization (1-50 engineers)
- ✓ Strong need **for** data consistency across domains
- ✓ Limited engineering budget
- ✓ Early-stage company (priorities change too fast)
- ✓ Heavily regulated (easier **to** govern centralized)

SIMPLE EXPLANATION

ANALOGY: Restaurant kitchen vs Food Court

CENTRALIZED DATA LAKE = ONE BIG RESTAURANT KITCHEN

One head chef (central data team)

Every restaurant sends ingredients (data) to him

Head chef cooks all dishes for everyone

Problem: 50 restaurants, one kitchen

→ Chef overwhelmed

→ Waiting 3 months for your dish

→ Chef doesn't know specialty recipe from each restaurant!

→ Quality inconsistent (chef doesn't know YOUR customers)

DATA MESH = FOOD COURT

Each restaurant has their OWN kitchen

Italian place: makes Italian food (owns Italian data)

Indian place: makes Indian food (owns Indian data)

Chinese place: makes Chinese food (owns Chinese data)

Food court management (platform team) provides:

→ Standard electricity, water, space

→ Health inspection standards

→ Cash register system (everyone uses same payment)

→ Customer directory ("where is Italian food?")

Each restaurant:

→ Responsible for their own food quality

→ Knows their customers best

→ Can update menu without asking food court management!

RESULT: Everyone served faster, better quality,
no central bottleneck!

PRO TIP

DATA MESH CHALLENGES (shows maturity):

"Data Mesh sounds great in theory. In reality:

CHALLENGE 1: Coordination overhead

50 domain teams, 50 different standards → chaos!

Solution: Strong federated governance

'Freedom within guardrails'

All teams use same catalog, same security model

CHALLENGE 2: Data product quality ownership

'Who fixes it when sales domain data is wrong?'

Solution: Clear data product owners with SLAs

Each product has: owner, consumers, SLA, on-call rotation

CHALLENGE 3: Cross-domain queries

Sales wants to join their data with Finance data

Different teams, different schemas, different refresh times!

Solution: Data contracts + virtual data products

'Join is possible if both sides honor their contract'

CHALLENGE 4: Cultural change

Data Mesh = organizational change, not just technical

Engineers must think 'data producer AND data consumer'

Takes 12-18 months for team to adopt new mindset

MY RECOMMENDATION:

Most companies should NOT do full Data Mesh immediately.

Start with: 'Federated Data Engineering'

→ Central platform team provides infrastructure

→ Domain teams OWN quality of their source data extraction

→ Central team builds common/shared tables

This gets 80% of Data Mesh benefits
with 20% of the organizational complexity!

Mentioning tradeoffs = senior architectural maturity!"

? QUESTION 37 — Architecture | Event-Driven vs Scheduled

"Design an event-driven data architecture vs scheduled batch architecture. Give pros and cons of each."

✓ IDEAL ANSWER

SCHEDULED BATCH ARCHITECTURE:

HOW IT WORKS:

Schedule Trigger fires → Pipeline runs → Processes ALL data since last run → Done → Wait until next schedule

Azure Implementation:

ADF Schedule Trigger (Daily at 1 AM)
→ PL_Daily_Sales_Load
→ Copy from Oracle (last 24 hours)
→ Transform in Databricks
→ Load to Synapse

→ Report available by 6 AM

PROS:

- ✓ SIMPLE: easy to understand, debug, rerun
- ✓ PREDICTABLE: same time every day
- ✓ EFFICIENT: batch processing amortizes startup cost
- ✓ COST-EFFECTIVE: cluster runs 3 hours, not 24/7
- ✓ EASIER ERROR HANDLING: "just rerun yesterday's batch"
- ✓ GOOD FOR LARGE VOLUMES: process entire day at once

CONS:

- ✗ LATENCY: data up to 24 hours old (for daily)
- ✗ WASTED RUNS: if no new data → pipeline still runs
- ✗ BURSTY RESOURCE USE: all run at 1 AM → contention
- ✗ LATE ARRIVING DATA: what about data after cutoff?
- ✗ SLA RISK: pipeline fails at 1 AM → report late

EVENT-DRIVEN ARCHITECTURE:

HOW IT WORKS:

Something HAPPENS (event) → Pipeline triggered immediately
→ Process only that event's data → Done

Azure Implementation:

Azure Event Hubs (payment transactions stream in continuously)
→ ADF Storage Event Trigger (file arrives → pipeline starts)
→ OR: Databricks Auto Loader (new files detected → processed)
→ OR: Azure Event Grid → Azure Function → ADF trigger

PROS:

- ✓ LOW LATENCY: data processed within seconds/minutes
- ✓ REACTIVE: only runs when there's data (cost saving!)
- ✓ SCALABLE: handles variable load automatically
- ✓ NEAR REAL-TIME: fraud detection, live dashboards

✓ **DECOUPLED:** upstream and downstream teams independent

CONS:

- ✗ **COMPLEX:** harder to debug, order of events not guaranteed
- ✗ **STATE MANAGEMENT:** streaming needs checkpoint/state
- ✗ **ALWAYS-ON COST:** streaming cluster must stay running
- ✗ **ERROR HANDLING:** harder to "rerun" specific events
- ✗ **ORDERING ISSUES:** event B might process before event A!
- ✗ **SMALL FILE PROBLEM:** each event = tiny file in data lake!

HYBRID PATTERN (best for most enterprises):

BOTH architectures used for DIFFERENT purposes:

	EVENT-DRIVEN (real-time)	BATCH (scheduled)
Sources:	Payment API (continuous)	Oracle DB (daily dump)
Trigger:	Event Hubs	ADF Schedule (1 AM)
Process:	Databricks Streaming (fraud: <5 seconds)	Databricks Job (reconciliation: 3 hours)
Store:	Delta (fraud_alerts) (appended every 10sec)	Delta (fact_sales) (loaded every 24 hours)
Serve:	Azure Cosmos DB (POS API: block fraud)	Synapse → Power BI (Executive dashboards)

DECISION MATRIX:

Need	Event-Driven	Scheduled Batch
Fraud detection	✓	✗ (too slow!)
Daily reports	✗	✓
File arrives	✓	✗ (wasteful!)
Historical backfill	✗	✓
SLA reports by 9AM	✗	✓
Live dashboards	✓	✗
Reconciliation	✗	✓
Alerting	✓	✗

💡 PRO TIP

MICRO-BATCH: The middle ground!

"Most enterprises don't need TRUE event-by-event streaming.
Micro-batch gives 80% of streaming benefit at lower cost:

Micro-batch = Batch that runs very frequently

- Every 5 minutes (instead of daily)
- Each run: process ONLY new data since last run
- 5-minute data latency (vs 24-hour daily batch)

For most use cases: 5-minute latency is 'real-time enough'!

Azure Implementation:

Tumbling Window Trigger: every 5 minutes

OR: Databricks trigger(processingTime='5 minutes')

WHEN EACH IS APPROPRIATE:

0-5 second latency: True streaming (Spark Structured Streaming)

5-60 minute latency: Micro-batch (ADF tumbling window)
1+ hour latency: Scheduled batch (ADF schedule)

COST COMPARISON:

True streaming (always-on cluster):	\$1,440/day
Micro-batch (5min job cluster):	\$480/day
Daily batch (1x per day):	\$48/day

For 5-minute latency:

Streaming: \$1,440/day

Micro-batch: \$480/day

→ SAME latency (effectively), 3x cheaper!

ALWAYS consider micro-batch before committing to streaming.
It's simpler, cheaper, and often good enough!"

? QUESTION 38 — Architecture | Security Deep Dive

*"How do you implement comprehensive security in Azure Data Engineering?
Cover network, identity, data, and compliance."*

✓ IDEAL ANSWER

SECURITY = 4 LAYERS working together

LAYER 1: NETWORK SECURITY

Private Endpoints:

- Each Azure service gets a private IP in YOUR VNet
- No public internet access!
- ADF → ADLS: through private network (no internet!)
- Databricks → Azure SQL: through private endpoint

VNet Integration:

- ADF Managed VNet (for Managed Private Endpoints)
- Databricks VNet injection (cluster in YOUR VNet)
- Synapse Managed VNet

Network Security Groups (NSG):

- Control WHICH subnet can talk to WHICH subnet
- Allow: DE subnet → Storage subnet
- Deny: Internet → Storage subnet (everything except private!)

Azure Firewall + DDoS:

- For on-prem to Azure (ExpressRoute + Azure Firewall)
- DDoS protection on public-facing services

LAYER 2: IDENTITY SECURITY (Authentication & Authorization)

Azure Active Directory (AAD):

- Single identity for all Azure services
- Multi-Factor Authentication (MFA) mandatory!
- Conditional Access: "Only allow access from corporate network"

Service Principals vs Managed Identity:

Service Principal (SP):

- Manual credential rotation (risky!)
- Use for: non-Azure systems, complex scenarios

Managed Identity (MSI) – PREFERRED:

- Azure manages credentials (no passwords!)
- ADF MSI → access ADLS (no password needed)
- Databricks MSI → access Key Vault
- ALWAYS use MSI when possible!

RBAC (Role-Based Access Control):

- Principle of Least Privilege
- ADF gets: Storage Blob Data Contributor (on specific container only!)
- Not: Owner, Contributor (too broad!)
- Databricks workspace: User, Contributor, Admin levels

Azure Key Vault:

- ALL secrets, connection strings, API keys
- No credentials in code EVER!
- Access policies: MSI gets READ-only access
- Audit log: every secret access is logged!
- Rotation: automate password rotation (no manual process!)

LAYER 3: DATA SECURITY

Encryption at Rest:

- ADLS Gen2: AES-256 (default Microsoft-managed)
- Customer Managed Keys (CMK) for extra compliance
- Azure SQL: Transparent Data Encryption (TDE)
- Synapse: TDE enabled

Encryption in Transit:

- TLS 1.2 minimum for all connections
- HTTPS for all API calls
- ADF: enforces TLS on all connections

Data Classification (Microsoft Purview):

- Classify data: Public, Internal, Confidential, PCI
- Automatic PII detection: scans columns for phone, email, PAN
- Apply sensitivity labels automatically!

Column-Level Security:

- Mask sensitive columns for unauthorized users
- Databricks Unity Catalog:

```
CREATE FUNCTION mask_pan(pan STRING)
RETURNS STRING → CASE WHEN user_role='admin' THEN pan
                     ELSE LEFT(pan,3)+'XXXXXXX' END
```
- Azure Synapse:
Column masking policies

Row-Level Security:

- Regional managers see only their region's data
- Security predicate function filters rows per user

Data Residency:

- Store data in specific Azure region (legal requirement!)
- ADLS account in India Central → Indian customer data stays in India
- Important for GDPR (EU data stays in EU!)

LAYER 4: COMPLIANCE & AUDIT

Audit Logging:

- Azure Monitor: every ADF run logged
- Log Analytics: query logs: "Who accessed customer data?"
- ADLS audit: file-level access tracking
- Key Vault: secret access log

Retention: 90 days hot, then archive for 7 years (compliance!)

Compliance Standards:

- PCI-DSS: payment card data → encryption + access control
- GDPR: EU personal data → consent, deletion, portability
- SOC2: general security controls
- ISO 27001: information security management

Azure built-in compliance: many standards certified!

Azure Policy: ENFORCE compliance (deny non-compliant resources)

Azure Security Center / Defender:

- Continuous security assessment
- "Your storage account has public access enabled!" → alert!
- Security score: how secure is your Azure setup?
- Recommendations with priority to fix

COMPLETE SECURITY ARCHITECTURE:

Internet (blocked everywhere by default)

↓

ExpressRoute (dedicated private circuit)

↓

Corporate Network

↓

SHIR (on-prem bridge)

↓

Azure VNet (private network in Azure)

↓

ADF (Managed Identity, Managed VNet)

↓

Private Endpoint (no public IP!)

↓

ADLS Gen2 (encrypted, CMK if needed)

↑ ↑ ↑

Key Vault (all secrets here)

Azure Monitor (all logs here)
Purview (all data catalog here)

PRO TIP

GDPR IMPLEMENTATION (highly relevant for banks):

"GDPR requires:

1. Right to Erasure ('right to be forgotten')

Customer C001 requests data deletion

In Delta Lake:

```
DELETE FROM silver_customers WHERE customer_id = 'C001'
```

```
DELETE FROM fact_sales WHERE customer_id = 'C001'
```

→ Delta's ACID ensures complete deletion!

→ Run VACUUM after to physically remove files

Challenge: What about partitioned data?

solution: OPTIMIZE then VACUUM handles it

2. Data Portability

Customer requests their data

→ Azure Synapse query → CSV export → secure download

3. Privacy by Design

→ PII masking in Silver layer (never store plain PAN)

→ customer_phone → XXXXXXXXXXXX (store masked)

→ Original only in Bronze (encrypted, restricted access)

4. Data Inventory

Azure Purview:

→ Scan all data stores

→ Automatically finds PII: 'Column CUSTOMER_PHONE contains phone numbers'

→ Data map: where is personal data?

→ Required for GDPR compliance reporting!

Mentioning GDPR in data engineering context = rare knowledge that stands out in interviews!"

? QUESTION 39 — Architecture | Monitoring & Observability

"How do you build a comprehensive monitoring system for an Azure data platform? What metrics do you track?"

✓ IDEAL ANSWER

MONITORING = Three pillars: Metrics, Logs, Alerts

WHAT TO MONITOR (complete list):

PIPELINE HEALTH METRICS:

- Pipeline success rate (goal: 99%+)
- Pipeline duration (trend: increasing = problem!)
- Activity-level duration (which activity is slow?)
- Retry count (high retries = flaky data source)
- Rows processed per run (spike = data anomaly)
- Trigger miss count (tumbling window missed = alert!)

DATA QUALITY METRICS:

- **Null** rate per column (increasing = source issue)
- Row count vs expected (< **80%** = something wrong)
- Duplicate rate (increasing = idempotency issue)
- Data freshness (MAX(update_timestamp) vs current time)
- Schema drift events (new columns arrived unexpectedly)

INFRASTRUCTURE METRICS:

- **ADLS**: storage size growth, read/write IOPS
- **Databricks**: cluster utilization, idle time
- **ADF**: DIU consumption, active pipeline count
- **Synapse**: DWU utilization, query queue depth
- **Azure SQL**: DTU/CPU%, connection count, deadlocks

COST METRICS:

- Daily/weekly/monthly spend per service
- Cost per **1000** rows processed
- Cluster idle time (wasted cost!)
- Storage growth rate

SLA METRICS:

- "Did Gold layer refresh by 6 AM?" → SLA met/missed
- "Was all data from previous day loaded?" → completeness
- "Were all reports available when business opened?" → impact

AZURE MONITORING STACK:

LAYER 1: Azure Monitor (infrastructure + platform)

- Collects metrics from ALL Azure services automatically
- Pipeline runs, Databricks cluster metrics, storage IOPS
- Query with KQL (Kusto Query Language):

AzureDiagnostics

| where ResourceProvider == "MICROSOFT.DATAFACTORY"


```
| where Category == "PipelineRuns"
| where Status == "Failed"
| project TimeGenerated, PipelineName, ErrorMessage
| order by TimeGenerated desc
```

LAYER 2: Log Analytics Workspace (centralized logs)

- All Azure resources send logs here
- Query across all services in ONE place!
- ADF logs + Databricks logs + ADLS logs → single workspace
- Custom dashboards from logs

LAYER 3: Application Insights (custom metrics)

- For custom business metrics
- From Databricks: log custom metrics

```
from applicationinsights import TelemetryClient
tc = TelemetryClient(instrumentation_key)
tc.track_metric("rows_processed", 142850)
tc.track_metric("dq_pass_rate", 99.7)
tc.track_event("PipelineCompleted",
               {"pipeline": "PL_Daily_Sales", "date":
process_date})
```

LAYER 4: Azure Alerts (proactive notification)

Alert rule examples:

CRITICAL alerts (wake someone up!):

- Pipeline failed > 0 times in last 5 minutes
- SLA missed: Gold layer not ready by 6 AM
- ADLS storage > 90% capacity
- Databricks cluster down for > 30 minutes

WARNING alerts (investigate when you can):

- Pipeline running > 2x normal duration
- Row count < 80% of expected

→ Cost spike > 30% vs same day last week

INFO alerts (for awareness):

- Pipeline completed successfully
- Schema drift detected (new column)
- Weekly cost summary email

LAYER 5: Operations Dashboard (Power BI)

Build a dedicated OPS DASHBOARD:

Page 1: Today's Status (live)

Pipeline Health: Today 2024-06-15			
✓	Ran Today: 45/45 pipelines succeeded		
✗	Failed: 0	⚠	Warnings: 2
Latest Run Times:			
PL_Oracle_Load:	Started 01:00	Ended 01:47	✓
PL_Transform:	Started 01:50	Ended 04:12	✓
PL_Gold:	Started 04:15	Ended 05:58	✓
All data ready by: 05:58 AM (SLA: 06:00) ✓ ON TIME			

Page 2: Data Quality Trends

- DQ pass rate over 30 days (should be ≥99%)
- Which tables have most DQ issues?

Page 3: Cost Tracking

- Daily spend by service
- Trend vs last month
- Top 5 most expensive resources

Page 4: SLA Compliance

- Which pipelines met/missed SLA per day

→ SLA achievement rate: 99.2% this month

AUDIT TABLE PATTERN:

Every pipeline writes to audit table:

-- Start of pipeline:

```
INSERT INTO dbo.PipelineAuditLog
(run_id, pipeline_name, start_time, status, triggered_by,
process_date)
VALUES
(@{pipeline().RunId}, @{pipeline().Pipeline}, @{utcnow()},
'RUNNING', @{pipeline().TriggerType},
@{pipeline().parameters.p_date})
```

-- End of pipeline:

```
UPDATE dbo.PipelineAuditLog
SET status = 'SUCCESS',
    end_time = @{utcnow()},
    rows_ingested = @{activity('CopyData').output.rowsCopied},
    duration_seconds = @{..calculated..}
WHERE run_id = @{pipeline().RunId}
```

-- On failure:

```
UPDATE dbo.PipelineAuditLog
SET status = 'FAILED',
    end_time = @{utcnow()},
    error_message = @{activity('FailedActivity').Error.message}
WHERE run_id = @{pipeline().RunId}
```

NOW you have:

- Complete history of every pipeline run
- Performance trending (duration increasing?)
- Failure patterns (fails every Monday? Check weekend data!)

- SLA compliance tracking
- Single source of truth for operations team

PRO TIP

SLA MONITORING PATTERN (what production needs):

"I define SLAs as CODE, not documentation:

SLA TABLE:

```
CREATE TABLE monitoring.PipelineSLA (  
    pipeline_name      VARCHAR(200),  
    sla_ready_by       TIME,           -- '06:00:00' (6 AM)  
    sla_date_column    DATE,  
    minimum_rows       INT,           -- 10000 (at least 10K rows)  
    maximum_delay_min  INT            -- 30 min after schedule  
);
```

DAILY SLA CHECK (runs at 6:05 AM):

For each pipeline in SLA table:

- Did it run today? (check audit log)
- Did it finish before sla_ready_by?
- Did it load at least minimum_rows?
- Did it take longer than maximum_delay_min?

Failed any check → IMMEDIATE ALERT!

SLA DASHBOARD METRIC:

- 'SLA Achievement Rate Last 30 Days: 99.7%'
- One missed SLA out of 300 → find root cause!
- Prevents: 'we missed 5 SLAs this month and nobody noticed'

PROACTIVE SLA MONITORING (advanced):

If pipeline hasn't started by `sla_time - 30 minutes`:

→ *'WARNING: PL_Daily_Sales hasn't started yet!'*

→ *Alert operations team BEFORE SLA is missed*

→ *They can investigate trigger failure before business opens!*

This proactive monitoring = senior engineering mindset.

Companies pay 15-18 LPA for engineers who PREVENT issues, not just respond to them!"

? QUESTION 40 — Architecture | Data Lineage

"What is data lineage? How do you implement and track it in Azure?"

✓ IDEAL ANSWER

DATA LINEAGE = Tracking the ORIGIN, MOVEMENT, and TRANSFORMATION of data from source to destination

WHY IT MATTERS:

"This sales number in Power BI report is wrong.

Where did it come from? What transformed it?"

Without lineage: hours of debugging across 20 pipelines

With lineage: 2-minute trace → *"Ah, the Oracle Sales table had wrong exchange rates → fixed!"*

Critical for:

- Debugging data quality **issues** (trace root cause)
- Compliance **audits** ("where does customer PAN come from?")
- Impact **analysis** ("if I change Oracle schema → what breaks?")
- **Trust** ("why should I trust this number?")

TYPES OF LINEAGE:

1. TECHNICAL LINEAGE:

Oracle.SALES.TRANSACTION_AMOUNT

- ADF CopyActivity → /bronze/sales/amount
- Databricks NB_Transform → /silver/sales/total_amount
- Databricks NB_Aggregate → /gold/store_perf/gross_revenue
- Power BI Dataset → [Revenue] measure
- Report: "Store Performance" → "Revenue by Region" visual

2. COLUMN-LEVEL LINEAGE:

More granular: specifically which COLUMN came from **where**

- gross_revenue = **SUM**(total_amount) WHERE status='COMPLETED'
- total_amount = AMOUNT - TAX from Oracle
- AMOUNT = RAW_AMOUNT from Oracle.SALES

AZURE **PURVIEW** (Primary Lineage Tool):

AUTOMATIC LINEAGE:

Purview automatically captures lineage from:

- Azure Data **Factory** (all pipelines!)
- Azure Synapse Analytics
- Power BI
- Azure SQL, Cosmos DB, Blob Storage

Setup:

Purview → Sources → Register ADF workspace

Now: every ADF pipeline run → lineage captured automatically!

Oracle → ADF → ADLS → Databricks → Synapse → Power BI
All connected **in** Purview lineage graph!

VIEWING LINEAGE:

Purview → Data Catalog → Find "gross_revenue" column
→ Click: "View Lineage"
→ Shows visual graph:

Oracle.SALES —ADF→ /bronze/sales —Databricks→
/silver/sales
—Databricks→ /gold/store_performance —Power BI→ [Revenue]

Can TRACE UP: "What is gross_revenue made of?"

Can TRACE DOWN: "If I change Oracle schema, what breaks?"

MANUAL **LINEAGE** (for custom tools):

When automatic capture doesn't work:

Annotate **in** Purview using REST API:

POST /api/atlas/v2/lineage

```
{
  "processes": [{
    "typeName": "databricks_notebook",
    "guid": "...",
    "attributes": {
      "name": "NB_Transform_Sales",
      "inputs": [{"guid": "bronze_sales_guid"}],
      "outputs": [{"guid": "silver_sales_guid"}]
    }
  ]
}
```

OPENLINEAGE (Open Standard):

Databricks supports OpenLineage:

→ Automatic column-level lineage capture!

→ Export to Purview or other tools

Enable `in` Spark:

```
spark.conf.set("spark.openlineage.host", "purview-endpoint")
spark.conf.set("spark.openlineage.namespace", "retailmart-
prod")
```

Now: EVERY Spark transformation → column-level lineage!

AUDIT TABLE `LINEAGE` (simple, custom):

For simple cases, log `in` pipeline:

```
INSERT INTO lineage.DataFlowLog
(source_system, source_table, source_path,
 target_system, target_table, target_path,
 transformation_type, pipeline_name,
 pipeline_run_id, run_timestamp, rows_processed)
VALUES
('Oracle', 'SALES', 'oracle://server/SALES',
 'ADLS', 'bronze_sales', '/mnt/bronze/sales/2024/06/15',
 'COPY', 'PL_Daily_Sales', @{pipeline().RunId},
 @{utcnow()}, 142850)
```

Simple query: "Where does `bronze_sales` come from?"

```
SELECT * FROM lineage.DataFlowLog WHERE target_table =
'bronze_sales'
```

→ Oracle.SALES! Immediate answer.

PRO TIP

IMPACT `ANALYSIS` (the `real` business value of lineage):

"The most valuable use of lineage is IMPACT ANALYSIS:

SCENARIO: DBA says 'We're renaming Oracle column
CUST_ID to CUSTOMER_ID next Friday'

WITHOUT LINEAGE:

- Manual search through 50 ADF pipelines,
- 30 Databricks notebooks, 10 SQL stored procedures
- Takes 2 days
- Still probably miss some pipelines
- Production breaks Friday → emergency fix!

WITH PURVIEW LINEAGE:

Search: 'CUST_ID column in Oracle.CUSTOMERS'

Impact view shows:

- 12 ADF pipelines reference this column
- 8 Databricks notebooks
- 3 SQL procedures
- 2 Power BI datasets

Fix everything BEFORE Friday!

- No production issues!
- Takes 4 hours instead of 2 days

BUSINESS CASE:

Preventing ONE production incident from schema change:

- Engineer time: 8 hours × 3 engineers = 24 hours
- Business hours at 9 AM: sales dashboards wrong = ???
- Trust damage: hard to quantify but real

Cost of Azure Purview: **\$15/vCore/month**

Value: Prevents ONE incident per year = justified!

This business case thinking = what 15+ LPA engineers present
to management to get tools approved!"



REVISION SUMMARY — Q36-Q40

REVISION: AZURE ARCHITECTURE PATTERNS (Q36-Q40)

Q36: Data Mesh vs Centralized

4 principles: Domain ownership, Data product,

Self-serve platform, Federated governance

KEY: Recommend for large orgs only, not small teams!

Q37: Event-Driven vs Batch

Event: low latency, complex, expensive

Batch: simple, cheap, high latency

KEY: Micro-batch = best of both worlds!

Q38: Security (4 Layers)

Network (Private Endpoints) + Identity (MSI + RBAC)

Data (encryption + masking) + Compliance (audit logs)

KEY: Managed Identity > Service Principal ALWAYS!

Q39: Monitoring

5 layers: Monitor → Log Analytics → AppInsights →

Alerts → Operations Dashboard

KEY: Audit table + SLA dashboard = production grade!

Q40: Data Lineage

Azure Purview: automatic from ADF + Synapse + PBI

OpenLineage: column-level in Databricks

KEY: Impact analysis = business value of lineage!

PATTERN ALERT:

Architecture questions → always mention:

→ Data Mesh for very large orgs

→ Private Endpoints for security

→ Azure Purview for governance/lineage

→ Proactive monitoring (not reactive!)

REAL-WORLD SCENARIOS

? QUESTION 41 — Scenario | Late-Arriving Data

"How do you handle late-arriving data in your data pipelines? Give a complete solution."

IDEAL ANSWER

LATE-ARRIVING DATA = Records that arrive AFTER their event time

REAL WORLD SCENARIO:

RetailMart stores send sales data daily at midnight

BUT: Store 117 in remote Ladakh has intermittent internet

→ Sales on June 15 → uploaded on June 17 (2 days late!)

If we don't handle this:

→ June 15 report: Store 117 shows Rs.0 sales

→ June 16 report: same issue

→ June 17: 3 days of sales suddenly appear

→ Business thinks Store 117 had a problem (it didn't!)

→ Retrospective adjustments needed → confusing!

SOLUTION STRATEGIES:

STRATEGY 1: REPROCESSING WINDOW

"Reprocess last N days every day"

Instead of: "Load only yesterday's data"

Do: "Reload last 7 days every day"

ADF: incremental query includes last 7 days:

```
WHERE sale_date >= DATEADD(DAY, -7, CAST(GETDATE() AS DATE))
AND sale_date <= CAST(GETDATE() AS DATE)
```

MERGE (not INSERT) to handle:

- New rows: inserted
- Changed rows: updated
- Previously zero → now has data

Pros: Simple, catches all late data ≤ 7 days

Cons: 7x more data processed daily (cost!)

STRATEGY 2: WATERMARK WITH BUFFER

"Accept data that's up to 2 days late"

Normal watermark:

Load records WHERE updated_date > last_watermark

Buffered watermark:

Load records WHERE updated_date > (last_watermark - 2 DAYS)

PLUS: Use MERGE not INSERT (deduplication!)

```
DeltaTable.alias("t").merge(
    df_source.alias("s"),
```

```
"t.sale_id = s.sale_id AND t.store_id = s.store_id"
).whenMatchedUpdateAll().whenNotMatchedInsertAll().execute()
```

Pros: Catches data up to 2 days late, efficient

Cons: Won't catch data >2 days late

STRATEGY 3: EVENT TIME WATERMARKS (Streaming)

For streaming pipelines with Structured Streaming:

```
df.withWatermark("event_timestamp", "2 days") # 2-day late
tolerance
    .groupBy(
        F.window("event_timestamp", "1 day"), # 1-day windows
        "store_id"
    )
    .agg(F.sum("amount").alias("daily_revenue"))
```

Spark keeps window state for 2 extra days

Late data within 2 days → included in correct window!

Late data >2 days → dropped (trade-off: completeness vs memory)

STRATEGY 4: SEPARATE LATE DATA PIPELINE

Detect and process late data separately:

Normal pipeline (runs daily):

Loads data for TODAY

Late-arrival pipeline (runs daily):

"Find any data where:

- sale_date < DATEADD(DAY, -1, today) -- not today's data
- created_date = today" -- but loaded

TODAY!

This specifically catches late-arriving data!

Applies MERGE to fix historical data

STRATEGY 5: DATA RECONCILIATION

Daily reconciliation job at 11 AM:

- Query source: "Total sales for last 7 days per store"
- Query warehouse: "Total sales for last 7 days per store"
- Compare: any discrepancy > 1%?
- Alert: "Store 117: source shows Rs.50,000 but warehouse shows Rs.0"
- Trigger reprocessing for that store/date!

COMPLETE PRODUCTION PATTERN:

Pipeline 1: PL_Daily_Normal_Load (runs 1 AM)

Load yesterday's data (fast, small)

Expected: 99% of stores have data by 1 AM

Pipeline 2: PL_Reconciliation_And_LateArrival (runs 11 AM)

- Query source and DW for last 7 days
- Find discrepancies
- Re-load any store/date with discrepancy
- MERGE (not insert) to fix without duplicates

Pipeline 3: PL_Historical_Correction (on-demand)

- For cases where data arrived very late (>7 days)
- Manual trigger by ops team with specific date range
- Targeted reprocessing for specific stores

AUDIT TABLE:

Track late arrivals:

store_id	sale_date	expected_load	actual_load	delay_hours
S117	2024-06-15	2024-06-16	2024-06-18	48

Report: "Average late arrival delay per store"

→ Identify problem stores → fix at source!

💡 PRO TIP

IDEMPOTENCY is KEY **for** late-arriving data:

"For late-arriving data to work correctly,
ALL your pipelines must be IDEMPOTENT:

IDEMPOTENT = Running same pipeline multiple times
 with same input = same result
 (no duplicates, no data loss)

NON-IDEMPOTENT (BAD!):

```
df.write.mode("append").parquet(path)
```

Run again → DUPLICATES!

IDEMPOTENT APPROACHES:

Option A: OVERWRITE partition:

```
df.write.mode("overwrite")  
  .option("replaceWhere", "sale_date = '2024-06-15'")  
  .format("delta").save(path)
```

→ Run 3 times → same result (always overwrites June 15 data)

Option B: MERGE (Upsert):

```
DeltaTable.merge(df, "t.sale_id = s.sale_id") \  
  .whenMatchedUpdateAll().whenNotMatchedInsertAll().execute()
```

→ Run 3 times → same result (updates existing, inserts new)

Option C: DEDUPLICATE before write:


```
df.dropDuplicates(["sale_id"]).write.mode("append")...
```

Option D: Check before insert:

Only insert records NOT already in target:

```
df.join(existing, "sale_id", "left_anti").write.mode("append")
```

IDEMPOTENCY + LATE ARRIVING DATA = RELIABLE PIPELINE

This combination handles all edge cases gracefully!"

? QUESTION 42 — Scenario | Schema Evolution

"How do you handle schema changes in source systems without breaking your data pipelines?"

✓ IDEAL ANSWER

SCHEMA EVOLUTION = Source system changes its schema
How does YOUR pipeline handle it?

TYPES OF SCHEMA CHANGES:

BACKWARD COMPATIBLE (safe — pipeline can handle):

- ✓ New column added (optional)
- ✓ Column renamed (with alias)
- ✓ Column data type widened (INT → BIGINT)
- ✓ New table added

BACKWARD INCOMPATIBLE (breaking — must handle carefully):

- ❌ Column deleted
- ❌ Column renamed WITHOUT notice
- ❌ Data type narrowed (BIGINT → INT)
- ❌ Column made NOT NULL (was nullable)
- ❌ Table dropped

HANDLING EACH TYPE:

CASE 1: NEW COLUMN ADDED

Option A: Schema Drift (automatic handling)

In ADF Data Flows:

Source → Allow Schema Drift: ON

Sink → Allow Schema Drift: ON

→ New column flows through automatically!

→ Destination table: new column added (Delta mergeSchema!)

In Databricks:

```
df.write.format("delta")  
  .option("mergeSchema", "true") # auto-adds new column!  
  .mode("append").save(path)
```

→ Old data: new column = NULL

→ New data: new column has values

→ Zero pipeline changes!

Option B: Schema Registry check (detect + alert)

→ Detect new column → log in schema changes table

→ Alert data team: "New column 'discount_type' added to SALES!"

→ Team decides: ignore or add to transformation

CASE 2: COLUMN RENAMED

Most dangerous (breaks everything silently!)

Detection:

Compare current schema vs expected schema:

```
expected_cols = {"SALE_ID", "CUST_ID", "AMOUNT", "SALE_DATE"}
actual_cols   = {"SALE_ID", "CUSTOMER_ID", "AMOUNT",
"SALE_DATE"}
# CUST_ID renamed to CUSTOMER_ID!
```

```
renamed = expected_cols.symmetric_difference(actual_cols)
# {"CUST_ID", "CUSTOMER_ID"} ← potential rename!
```

Alert immediately!

Pipeline: **continue with** warning OR fail **with** clear message

Fix: Update ADF mapping **or** Spark transformation to use **new** name

But: coordinate **with** DBA **for** smooth transition!

CASE 3: COLUMN DELETED

Handle **with** COALESCE / **default** values:

```
df = df.withColumn("deleted_column",
                    F.lit(None).cast(StringType()))
# Add back with NULL values → downstream won't break
# Eventually remove from downstream too
```

COMPLETE SCHEMA EVOLUTION FRAMEWORK:

STEP 1: SCHEMA **CAPTURE** (run **on** each pipeline execution)

Store current source schema **in** schema registry:

```
current_schema = {
    "table": "SALES",
    "version": 45,
    "captured_at": "2024-06-15T01:00:00",
    "columns": [
```

```

        {"name": "SALE_ID", "type": "NUMBER(10)", "nullable":
false},
        {"name": "CUST_ID", "type": "VARCHAR2(20)", "nullable":
true},
        ...
    ]
}

```

Compare **with** previous **version** (stored **in** Delta table)

STEP 2: CLASSIFY CHANGES

```

def classify_schema_change(old_schema, new_schema):
    added      = set(new_schema.columns) -
set(old_schema.columns)
    deleted    = set(old_schema.columns) -
set(new_schema.columns)
    modified   = {c for c in old_schema.columns &
new_schema.columns
                  if old_schema.columns[c] !=
new_schema.columns[c]}

    if deleted or (modified and any_type_narrowed(modified)):
        return "BREAKING" # fail pipeline, alert immediately
    elif added or (modified and all_type_widened(modified)):
        return "COMPATIBLE" # continue with logging
    return "NO_CHANGE"

```

STEP 3: REACT BASED ON CLASSIFICATION

BREAKING change → FAIL pipeline immediately

- Send urgent alert
- Do NOT process **partial** data
- Team must investigate **and** fix

COMPATIBLE change → LOG **and** CONTINUE

- Log schema change to audit table

- Pipeline continues **with** mergeSchema
- Alert data **team** (non-urgent)
- Team reviews **and** updates documentation

NO CHANGE → CONTINUE normally

STEP 4: SCHEMA VERSIONING

Every schema version stored:

v1: original **schema** (**2024-01-01**)

v2: added **discount_code** (**2024-03-15**)

v3: renamed CUST_ID to **CUSTOMER_ID** (**2024-06-10**)

Time-travel queries can still work:

"Show me data **as** of March 2024"

→ Use schema v2, apply v2 → v3 mapping

PRO TIP

DATA **CONTRACTS** (prevents schema issues):

"The BEST way to handle schema evolution **is** to
PREVENT surprise changes **with** DATA CONTRACTS:

A Data Contract = Formal agreement between:

Producer (source team/DBA)

Consumer (data engineering team)

Contract specifies:

- Schema: column names, types, nullable
- SLA: data available **by** X time
- Change process: **'must notify DE team 2 weeks before changes'**
- Breaking changes: **'require joint planning session'**

IMPLEMENTATION:

Schema **in Git** (YAML format):

```
# contracts/oracle_sales_v3.yaml
table: SALES
version: 3
effective_date: 2024-06-10
producer: Oracle_DBA_Team
consumer: DataEngineering_Team

columns:
  - name: SALE_ID
    type: NUMBER(10,0)
    nullable: false
    description: Unique sale identifier
  - name: CUSTOMER_ID # renamed from CUST_ID in v3
    type: VARCHAR2(20)
    nullable: true
    previous_name: CUST_ID # document the rename!

changelog:
  - version: 3
    date: 2024-06-10
    changes: Renamed CUST_ID to CUSTOMER_ID
    migration: DE team updated pipelines 2024-06-08
```

WHEN ANYONE WANTS TO CHANGE SCHEMA:

- Submit PR to contract file **in Git**
- DE team reviews **impact** (Purview lineage!)
- Agree **on** timeline **and** migration plan
- Deploy contract change + pipeline change TOGETHER!

No more surprises!

This **is** PLATFORM ENGINEERING maturity."

? QUESTION 43 — Scenario | Pipeline Idempotency

"What is idempotency in data pipelines? How do you ensure your pipelines are idempotent? Why does it matter?"

✓ IDEAL ANSWER

IDEMPOTENCY = Running the SAME pipeline with SAME inputs
multiple times = SAME result every time

WHY IT MATTERS:

Pipeline fails at step 7 (of 10 steps)
You re-run the pipeline from step 1

WITHOUT idempotency:

- Steps 1-6 ran again → DUPLICATE DATA!
- Step 7 fixed → steps 8-10 added MORE data
- Data now: correct once + duplicate steps 1-6!

WITH idempotency:

- Steps 1-6 ran again → SAME DATA (upsert/overwrite)
- Step 7 fixed and ran → correct
- Steps 8-10 ran → complete pipeline
- Data: EXACTLY CORRECT, no duplicates!

IDEMPOTENCY PATTERNS:

PATTERN 1: OVERWRITE TARGET PARTITION

For batch pipelines processing specific date:

```
# Load June 15 data
df_june15.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere", "sale_date = '2024-06-15'") \
    .save("/mnt/silver/sales/")
```

Run multiple times → SAME RESULT (overwrites June 15 each time)
Doesn't affect other dates!

PATTERN 2: MERGE (UPSERT)

For incremental loads where date range overlaps:

```
DeltaTable.forPath(spark, "/mnt/silver/sales/") \
    .alias("target") \
    .merge(df_new.alias("source"),
           "target.sale_id = source.sale_id") \
    .whenMatchedUpdateAll() \      # update if exists
    .whenNotMatchedInsertAll() \  # insert if new
    .execute()
```

Run multiple times → updates existing, inserts new = same result!

PATTERN 3: DEDUPLICATION + APPEND

For append-only scenarios:

```
# Check what's already in destination
existing_ids = spark.read.parquet(dest_path) \
    .select("sale_id")

# Only append NEW records
df_new_only = df_all.join(existing_ids,
                          on="sale_id",
```



```
how="left_anti") # NOT IN existing
df_new_only.write.mode("append").parquet(dest_path)
```

Run multiple times → only new data added each time!

PATTERN 4: STAGING TABLE + SWAP

Most robust pattern for SQL warehouses:

Step 1: Write to staging table (OVERWRITE always safe!)

```
df.write.mode("overwrite").saveAsTable("staging.stg_sales_today")
```

Step 2: MERGE staging → production

```
MERGE INTO silver.sales AS target
```

```
USING staging.stg_sales_today AS source
```

```
ON target.sale_id = source.sale_id
```

```
WHEN MATCHED THEN UPDATE SET ...
```

```
WHEN NOT MATCHED THEN INSERT ...
```

Step 3: Truncate staging (cleanup)

```
spark.sql("TRUNCATE TABLE staging.stg_sales_today")
```

ANY re-run at any step → safe! No duplicates.

PATTERN 5: PROCESS DATE PARTITIONING

Simple and elegant for daily pipelines:

```
process_date = "2024-06-15" # always the same for same run
```

```
# Delete any data for this date (idempotent delete)
```

```
spark.sql(f"""
    DELETE FROM silver_sales
    WHERE sale_date = '{process_date}'
""")
```

```
# Fresh insert (now safe since we deleted first!)
df_today.filter(F.col("sale_date") == process_date) \

.write.format("delta").mode("append").saveAsTable("silver_sales")
```

Run multiple times:

Delete June 15 data (removes previous attempt)

Re-insert June 15 data (fresh, correct)

→ Idempotent!

IDEMPOTENCY CHECKLIST FOR EVERY PIPELINE:

- ☐ Does every write use OVERWRITE, UPSERT, or DEDUP?
- ☐ Is there a unique key that prevents duplicates?
- ☐ Does the pipeline handle re-runs of same date/batch?
- ☐ Is file naming deterministic? (not timestamp-based!)
- ☐ Does cleanup happen BEFORE write (not after)?
- ☐ Is watermark update on SUCCESS path only?

COMMON IDEMPOTENCY BUGS:

❌ Pipeline creates audit record BEFORE processing
→ Fail → re-run → TWO audit records for same run!

✅ Create audit record AFTER successful processing

❌ File named: sales_<random_uuid>.parquet
→ Each re-run creates new file + keeps old one!

✅ File named: sales_<process_date>.parquet
→ Re-run overwrites same file name!

💡 PRO TIP

TEST FOR IDEMPOTENCY (show you **test** your code!):

"I include idempotency tests in my CI/CD pipeline:

```
def test_pipeline_idempotency():
    process_date = "2024-01-15"

    # First run
    run_pipeline(process_date)
    count_after_first = get_row_count(process_date)

    # Second run (simulate re-run)
    run_pipeline(process_date)
    count_after_second = get_row_count(process_date)

    # Third run
    run_pipeline(process_date)
    count_after_third = get_row_count(process_date)

    # Assert same result regardless of runs
    assert count_after_first == count_after_second, \
        f'Not idempotent! First: {count_after_first}, Second: {count_after_second}'
    assert count_after_second == count_after_third, \
        f'Not idempotent! Second: {count_after_second}, Third: {count_after_third}'

    print(f'✅ Pipeline is idempotent: {count_after_first} rows consistently')

test_pipeline_idempotency()
```

This test runs in CI/CD:

- Every code change → idempotency test runs automatically
- If test fails → merge blocked!

→ Prevents non-idempotent code from reaching production

'I don't just claim my pipelines are idempotent.
I PROVE it with automated tests!'

This testing mindset = SENIOR DATA ENGINEER behavior!"

? QUESTION 44 — Scenario | Multi-Region Data Architecture

"Design a data architecture for a company operating in India, US, and Europe with GDPR and data residency requirements."

✓ IDEAL ANSWER

REQUIREMENTS:

- 3 regions: India, US, Europe (EU)
- GDPR: EU personal data must stay in EU
- India: local processing regulations
- US: no special requirements
- Global analytics: cross-region reporting needed
- Performance: local teams need fast access to local data

ARCHITECTURE: Hub-and-Spoke Multi-Region

REGION-SPECIFIC DATA (Spoke):

INDIA SPOKE (Central India region):

ADLS Gen2: retailmart-india-datalake.dfs.core.windows.net

ADF Instance: adf-retailmart-india

Databricks: databricks-india-workspace

Contains: Indian customer PII, Indian sales data

Who accesses: India data team, India business users

Cannot: replicate PII to other regions

Can: send ANONYMIZED/AGGREGATED data to global hub

EU SPOKE (West Europe / Netherlands region):

ADLS Gen2: retailmart-eu-datalake (GDPR-compliant region!)

ADF Instance: adf-retailmart-eu

Databricks: databricks-eu-workspace

GDPR requirements:

→ Customer PII stays in EU ONLY (Amsterdam/Ireland Azure region)

→ Right to erasure: DELETE WHERE customer_id = 'EU_C001'

→ Data subject requests: automated response pipeline

→ Privacy by design: PAN masking BEFORE any aggregation

→ Data Protection Officer (DPO) has audit access

US SPOKE (East US region):

ADLS Gen2: retailmart-us-datalake

ADF: adf-retailmart-us

Similar to India (no special regulations)

GLOBAL HUB (Central):

ADLS Gen2: retailmart-global-datalake

Azure Synapse: synapse-retailmart-global

Power BI: global-analytics-workspace

What goes to global hub:

- AGGREGATED data only (no PII!)
- Revenue by country/region (not per customer)
- Product performance globally
- Cross-region KPIs

What NEVER goes to global hub:

- EU customer names, emails, addresses
- Indian customer PAN numbers
- Any personal identifiable information

DATA FLOW:

```
India Spoke → [Aggregate + Anonymize] → Global Hub
EU Spoke    → [Aggregate + Anonymize] → Global Hub
US Spoke    → [Aggregate + Anonymize] → Global Hub
```

```
Global Hub → [Merge all regions] → Global Reports
```

```
EU Spoke → [EU regional reports] → EU Power BI workspace
```

```
India Spoke → [India regional reports] → India Power BI
workspace
```

ANONYMIZATION PIPELINE:

Before sending to global hub, strip ALL personal data:

```
# Remove all PII before cross-border transfer
df_india_agg = df_india_sales \
    .drop("customer_name", "email", "phone", "address", "pan")
\
    .groupBy("year", "month", "region", "product_category") \
    .agg(
        F.sum("total_amount").alias("revenue"),
        F.count("sale_id").alias("transactions"),
        # Customer count but not customer IDs
        F.countDistinct("customer_id").alias("unique_customers")
    )
```

```

        # Note: customer_id itself not transferred!
    )

# Only aggregates go to global hub
df_india_agg.write \
    .format("delta") \
    .mode("overwrite") \
    .option("replaceWhere",
            f"year = {year} AND month = {month}") \
    .save("abfss://global@retailmart-
global.dfs.core.windows.net/india/")

```

GDPR COMPLIANCE PIPELINE:

Daily: Scan EU data for overdue deletion requests

```

SELECT customer_id, deletion_requested_date
FROM eu.customer_deletion_requests
WHERE status = 'PENDING'
      AND deletion_requested_date <= DATEADD(DAY, -30, GETDATE())

```

For each: DELETE all EU data for that customer_id

Update status: 'COMPLETED'

Audit log: deletion_executed_date, executed_by, data_removed

Response to customer: automated email "Your data has been deleted"

AZURE POLICY FOR DATA RESIDENCY:

Policy: "EU data must stay in West Europe / North Europe"

Azure Policy definition:

```

{
  "condition": {
    "field": "location",
    "in": ["westeurope", "northeurope"]
  }
}

```

```
}
```

Applied to: EU subscription

Effect: Deny (any resource outside EU region → blocked!)

Ensures: Even if someone accidentally creates storage in US
→ Azure Policy BLOCKS it!
→ Compliance guaranteed by infrastructure, not trust!

PRO TIP

DATA RESIDENCY GOTCHA (interviewer **test!**):

"GDPR doesn't just apply to storage location.
It also applies to PROCESSING location!

Azure Databricks clusters:

If EU data processed in US cluster
→ Technical violation of GDPR!
Even if storage stays in EU!

CORRECT SETUP:

EU data → EU Databricks workspace (West Europe)
→ Cluster runs in EU
→ Processing in EU
→ Storage in EU
= GDPR compliant!

WRONG SETUP:

EU data stored in EU (OK)
→ Read by US Databricks cluster (NOT OK!)
Even though data goes back to EU for storage
The PROCESSING happened in US

= Potential GDPR violation!

Azure Databricks workspace location = where clusters run
You must create separate workspace IN THE CORRECT REGION
for each regulated data region.

This nuance = rare knowledge that impresses compliance-focused
interviewers at banks and financial institutions!"

? QUESTION 45 — Scenario | Data Deduplication at Scale

"You have 1 billion rows of payment data with potential duplicates. How do you deduplicate it efficiently in PySpark?"

✓ IDEAL ANSWER

```
# DEDUPLICATION AT SCALE: 1 BILLION ROWS
# Challenge: 1B rows doesn't fit in memory for simple operations!

from pyspark.sql import functions as F
from pyspark.sql.window import Window
from delta.tables import DeltaTable

# — UNDERSTAND YOUR DEDUPLICATION KEY —————
# What makes a row "unique"?
# For payments: payment_id + amount + timestamp = unique payment
# (payment_id alone might not be enough if system retried!)
```

```
# — METHOD 1: dropDuplicates (Simplest) —————  
# Best for: exact duplicates (same data, copied twice)
```

```
df_dedup = df_payments.dropDuplicates([  
    "payment_id",      # primary identifier  
    "merchant_id",     # business context  
    "amount",          # exact amount must match  
    "payment_timestamp" # exact time must match  
)  
  
original_count = df_payments.count()  
dedup_count    = df_dedup.count()  
print(f"Removed: {original_count - dedup_count:,} duplicates")  
print(f"Remaining: {dedup_count:,} unique payments")
```

```
# PERFORMANCE for 1B rows:  
# dropDuplicates causes a shuffle (groups by key columns)  
# With 1B rows: this shuffle is EXPENSIVE!  
# Optimization: repartition by dedup key FIRST!
```

```
df_dedup_fast = df_payments \  
    .repartition(200, F.col("payment_id")) \  
    together  
    .dropDuplicates(["payment_id", "merchant_id", "amount"])  
# Now deduplication is LOCAL per partition (less network  
transfer!)
```

```
# — METHOD 2: ROW_NUMBER (Keep specific duplicate) —————  
# Best for: "keep most recent record per payment_id"
```

```
window_spec = Window \  
    .partitionBy("payment_id", "merchant_id") \  
    .orderBy(F.col("updated_timestamp").desc()) # most recent  
first
```

```

df_with_rank = df_payments \
    .withColumn("row_num", F.row_number().over(window_spec))

df_dedup_latest = df_with_rank \
    .filter(F.col("row_num") == 1) \
    .drop("row_num")

# Keep: row_num = 1 = most recently updated version of each payment!

# — METHOD 3: HASH-BASED DEDUPLICATION —————
# Best for: detecting duplicates even with slight variations

# Create hash of all meaningful columns
df_hashed = df_payments \
    .withColumn("row_hash",
        F.sha2(
            F.concat_ws("|",
                F.col("payment_id"),
                F.col("merchant_id"),
                F.col("amount").cast("string"),
                F.col("payment_timestamp").cast("string")
            ),
            256 # SHA-256
        )
    )

# Keep only rows where hash is unique
# (fastest to filter by hash than by multiple columns)
window_hash =
Window.partitionBy("row_hash").orderBy("payment_timestamp")
df_dedup_hash = df_hashed \
    .withColumn("rn", F.row_number().over(window_hash)) \
    .filter(F.col("rn") == 1) \

```

```

.drop("rn", "row_hash")

# — METHOD 4: INCREMENTAL DEDUPLICATION (for ongoing data) —
# Best for: streaming/daily incremental pipelines

# New data arrives daily – check against ALREADY LOADED data

def load_with_dedup(df_new, target_path, merge_key):
    """
    Load new data while deduplicating against existing data
    """

    # Step 1: Deduplicate within NEW batch first
    df_new_dedup = df_new.dropDuplicates(merge_key)

    # Step 2: Merge with existing (handles cross-batch
    # duplicates)
    if DeltaTable.isDeltaTable(spark, target_path):
        delta_table = DeltaTable.forPath(spark, target_path)

        # Build merge condition from key columns
        merge_condition = " AND ".join([
            f"target.{col} = source.{col}"
            for col in merge_key
        ])

        delta_table.alias("target") \
            .merge(df_new_dedup.alias("source"), merge_condition) \

            .whenMatchedUpdate(
                # Only update if something actually CHANGED
                condition=" OR ".join([
                    f"target.{col} != source.{col}"
                    for col in df_new.columns
                    if col not in merge_key
                ])
            )
    else:
        df_new_dedup.write.format("delta").save(target_path)

```

```

        ]),
        set={col: f"source.{col}" for col in
df_new.columns}
    ) \
    .whenNotMatchedInsertAll() \
    .execute()
else:
    # First load: just write deduplicated data
    df_new_dedup.write \
        .format("delta") \
        .mode("overwrite") \
        .save(target_path)

    print(f"✅ Loaded {df_new_dedup.count():,} unique records")

# Usage
load_with_dedup(
    df_new=df_today_payments,
    target_path="/mnt/silver/payments/",
    merge_key=["payment_id", "merchant_id"]
)

# — METHOD 5: APPROXIMATE DEDUPLICATION —————
# Best for: near-duplicates (fuzzy matching at scale)
# Use case: same payment ID but slightly different amount
#           (floating point rounding across systems)

# Round amounts to avoid floating point differences
df_normalized = df_payments \
    .withColumn("amount_normalized",
        F.round(F.col("amount"), 2)) \
    .withColumn("dedup_key",
        F.concat_ws("|",
            F.col("payment_id"),
            F.col("merchant_id"),

```

```
F.col("amount_normalized"))))

df_dedup_fuzzy = df_normalized.dropDuplicates(["dedup_key"]) \
    .drop("dedup_key",
"amount_normalized")
```

— PERFORMANCE TIPS FOR 1B ROWS —

TIP 1: Partition by dedup key before deduplication

```
df.repartition(500, F.col("payment_id"))
    .dropDuplicates(["payment_id"])
```

All payment_id=P001 goes to partition 47

Dedup is LOCAL (no shuffle needed within partition!)

TIP 2: Filter obvious non-duplicates first

If you know last 30 days can't have duplicates

Only check older data!

```
df_new = df.filter(F.col("payment_date") >= "2024-05-15")
```

```
df_dedup_new = df_new.dropDuplicates(["payment_id"])
```

30 days of data <<< 1 billion rows → much faster!

TIP 3: Save deduplication results progressively

For 1B rows: process in date chunks

```
for month in range(1, 13):
    df_month = df.filter(F.col("month") == month)
    df_month_dedup = df_month.dropDuplicates(["payment_id"])
    df_month_dedup.write \
        .mode("overwrite") \
        .option("replaceWhere", f"month = {month}") \
        .format("delta").save(target_path)
    print(f"Month {month}: deduplicated")
```

12 small jobs instead of 1 massive job = more reliable!

PRO TIP

BLOOM **FILTER** OPTIMIZATION (advanced deduplication):

"For checking 'does this payment_id already exist in Delta table?'

at scale, Delta Lake bloom filters are EXTREMELY efficient:

Enable bloom filter index:

```
CREATE BLOOMFILTER INDEX ON TABLE silver_payments
FOR COLUMNS (payment_id OPTIONS (fpp=0.01,
numItems=10000000000))
-- fpp = false positive probability (1% here)
-- numItems = expected 1 billion unique IDs
```

Now when merging new payments:

```
delta_table.merge(df_new, 'target.payment_id =
source.payment_id')
```

WITHOUT bloom filter:

- Scan ALL 1B rows to find matching payment_ids
- Expensive file I/O!

WITH bloom filter:

- Check bloom filter: 'Is P123456789 in this file?'
- 99% of the time: 'NO!' (skip entire file!)
- Only read files that MIGHT have matching ID!
- 10-100x faster for MERGE operations!

Bloom filter = probabilistic data structure

False positives possible (1% rate)

False negatives impossible

= Very fast 'probably not here' check!

This Delta Lake optimization = senior-level performance

knowledge.

Most candidates don't know bloom filters exist in Delta!"



REVISION SUMMARY — Q41-Q45

REVISION: **REAL**-WORLD SCENARIOS (Q41-Q45)

Q41: Late-Arriving Data

Reprocessing **window** + **MERGE** = complete solution

KEY: Monitor discrepancies, **trigger** reprocess!

Q42: Schema Evolution

mergeSchema **for** compatible, FAIL **for** breaking

KEY: Data contracts prevent surprises!

Q43: Idempotency

OVERWRITE **partition OR MERGE** (upsert) = idempotent!

KEY: Test idempotency **in** CI/CD!

Q44: Multi-Region Architecture

Spoke (regional PII) + Hub (global aggregates only) |
KEY: GDPR = processing location, not just storage!

Q45: Deduplication at Scale

dropDuplicates (simple) | row_number (keep latest)

Bloom filters for Delta MERGE performance!

KEY: Repartition by dedup key for performance!

PATTERN ALERT:

Scenario questions test:

- Edge case awareness (late data, schema changes)
- Production thinking (what breaks in real world)
- Recovery strategies (not just happy path!)

DATA GOVERNANCE & ADVANCED SECURITY

? QUESTION 46 — Governance | Azure Purview Deep Dive

"Explain Azure Purview. How do you use it for data governance in a large enterprise?"

✓ IDEAL ANSWER

AZURE PURVIEW = Microsoft's unified data governance platform

WHAT IT DOES:

- Data Catalog: "Where is our data? What does it mean?"
- Data Lineage: "Where did this data come from?"
- Data Classification: "Which data contains PII?"
- Access Management: "Who can access what data?"
- Policy Enforcement: "Ensure governance rules are followed"

WHY IT'S NEEDED AT SCALE:

Without Purview (100 person company):

Analyst asks: "Where is customer email stored?"

- Asks 10 different people
- Gets 7 different answers
- Wastes 2 days finding the right table

With Purview:

- Search "customer email" in Purview catalog

- **Result**: 3 tables found, all described, all with lineage
- 2 minutes!

5 KEY CAPABILITIES:

1. DATA MAP (Automated Discovery):

Register data sources:

- Azure **SQL** Database (automatic scan!)
- ADLS Gen2 (scans folder structure)
- Oracle (via SHIR)
- Power BI (automatic!)

Purview scans and creates catalog entries automatically

- Discovers **ALL** tables, columns, data types
- **No** manual documentation!

2. DATA CATALOG (**Search** & Discovery):

Business users **search**: "show me all sales data"

- Purview **returns**: all tables with "sales" in name/description
- Shows: data dictionary, business terms, contacts
- Certified: "this is the official sales table"

Endorsement levels:

Certified: Official, validated, recommended

Promoted: Useful, community-approved

None: Just discovered, **not** validated

3. AUTOMATED DATA CLASSIFICATION:

Purview scans data and auto-classifies:

Column contains: [+91-98765-43210]

- Classified **as**: "INDIA_PHONE_NUMBER"

Column contains: [ABCDE1234F]

- Classified **as**: "INDIA_PAN_NUMBER"

Column contains: [rahul@gmail.com]

→ Classified as: "EMAIL_ADDRESS"

180+ built-in classifiers!

Custom classifiers for your business data types.

Use: "Find ALL columns with PAN numbers across ALL our systems"

→ GDPR/compliance reporting in minutes instead of weeks!

4. DATA LINEAGE:

(covered in Q40 - see previous answer)

Automatically captures from ADF, Synapse, Power BI

5. ACCESS POLICIES:

Define who can access what FROM Purview:

Policy: "Data Analysts can READ silver_sales table"

Policy: "Finance team can READ ALL finance/* tables"

Policy: "GDPR auditor can READ EU customer data"

Applied to: ADLS, Azure SQL, etc.

Without touching individual storage settings!

PURVIEW SETUP IN AZURE:

Create Purview account in Azure Portal

Assign MSI permissions:

ADLS: Storage Blob Data Reader (for scanning)

ADF: Data Factory Contributor (for lineage)

Register sources:

Purview Studio → Data Map → Sources → Register

→ Azure Data Lake Storage Gen2

→ Enter storage account name → Register

Create and run scan:

- New Scan → choose ruleset (default or custom)
- Schedule: weekly scan
- Run now: discovers all tables, columns, data types

BUSINESS VALUE:

"How long does onboarding new data engineer take?"

Without Purview: 2-4 weeks (finding all data, documentation)

With Purview: 2-3 days (everything in one catalog)

"GDPR audit: list all personal data locations"

Without Purview: 3-week manual project

With Purview: 30-minute Purview scan and report

PRO TIP

BUSINESS GLOSSARY (often ignored but valuable):

"Most teams implement Purview for lineage but forget
Business Glossary – which gives EQUAL value:

Business Glossary = Dictionary of business terms

Term: 'Gross Revenue'

Definition: Total sales amount before GST and discounts

Formula: SUM(total_amount) from fact_sales WHERE

status='COMPLETED'

Owner: Finance team

Related terms: Net Revenue, GMV

Linked assets: dim_date.gross_amount, fact_sales.total_amount

Term: 'Active Customer'

Definition: Customer who purchased at least once in last 90 days

Owner: Marketing team

Linked assets: dim_customer table

WHY THIS MATTERS:

Data engineer defines 'active customer' as 60 days

Marketing defines it as 90 days

Finance defines it as 30 days

- 3 different numbers in 3 reports!
- Confusion in business meetings!
- 'Why are there 3 different customer counts?'

With Business Glossary:

- Single definition of 'active customer' = 90 days (agreed)
- ALL reports use same definition
- One source of truth for metrics!

ADDING GLOSSARY TO DAILY WORKFLOW:

When building new table:

- Every column → annotate with glossary term
- 'gross_revenue column = [Gross Revenue] glossary term'

Business user searches 'gross revenue':

- Purview shows EXACT column across all systems
- With definition, formula, data owner

This attention to DATA CULTURE = 15+ LPA data governance!"

? QUESTION 47 — Governance | Unity Catalog

*"Explain Databricks Unity Catalog. How is it different from Hive Metastore?
Design access control for a retail company."*

✓ IDEAL ANSWER

HIVE METASTORE (Old way - before Unity Catalog):

Problems:

- ✗ Each Databricks workspace has its OWN metastore
- ✗ 3 workspaces = 3 separate data catalogs (isolated!)
- ✗ 'silver_sales' in DE workspace ≠ 'silver_sales' in DS workspace

workspace

- ✗ Security configured SEPARATELY in each workspace
- ✗ No unified audit log ("who accessed what?")
- ✗ No column-level security
- ✗ No row-level security
- ✗ No data lineage
- ✗ Permissions = workspace-level only (admin or user)

UNITY CATALOG (New way):

ONE metastore for ALL workspaces in a region.

Object hierarchy:

Metastore (1 per Azure region)

└─ Catalog (like a database instance)

└─ Schema/Database

└─ Table / View / Volume / Function

UNITY CATALOG HIERARCHY FOR RETAILMART:

Metastore: retailmart-uc-central-india

```
|
|— Catalog: bronze_catalog
|   |— Schema: sales
|       |— Table: transactions
|       |— Table: order_items
|   |— Schema: customers
|       |— Table: raw_customers
|
|— Catalog: silver_catalog
|   |— Schema: sales
|       |— Table: sales_fact
|   |— Schema: dimensions
|       |— Table: dim_customer
|       |— Table: dim_product
|
|— Catalog: gold_catalog
|   |— Schema: analytics
|       |— Table: store_performance
|       |— Table: customer_360
|
|— Catalog: dev_catalog
|   |— Schema: {username} (each developer's playground)
```

ACCESS CONTROL DESIGN:

```
# Data Engineering Team (builds everything)
GRANT USE CATALOG ON CATALOG bronze_catalog TO `de_team`;
GRANT USE CATALOG ON CATALOG silver_catalog TO `de_team`;
GRANT USE CATALOG ON CATALOG gold_catalog TO `de_team`;
GRANT ALL PRIVILEGES ON CATALOG bronze_catalog TO `de_team`;
GRANT ALL PRIVILEGES ON CATALOG silver_catalog TO `de_team`;
```



```
GRANT ALL PRIVILEGES ON CATALOG gold_catalog TO `de_team`;
```

```
# Data Science Team (reads silver, writes to their schemas)
```

```
GRANT USE CATALOG ON CATALOG silver_catalog TO `ds_team`;
```

```
GRANT SELECT ON CATALOG silver_catalog TO `ds_team`;
```

```
GRANT USE CATALOG ON CATALOG dev_catalog TO `ds_team`;
```

```
GRANT ALL PRIVILEGES ON CATALOG dev_catalog TO `ds_team`;
```

```
# BI Team (read-only on gold)
```

```
GRANT USE CATALOG ON CATALOG gold_catalog TO `bi_team`;
```

```
GRANT SELECT ON CATALOG gold_catalog TO `bi_team`;
```

```
# North India Regional Manager (only North India data)
```

```
GRANT USE CATALOG ON CATALOG gold_catalog TO
```

```
`north_india_manager`;
```

```
GRANT USE SCHEMA ON SCHEMA gold_catalog.analytics TO
```

```
`north_india_manager`;
```

```
GRANT SELECT ON TABLE gold_catalog.analytics.store_performance
```

```
TO `north_india_manager`;
```

```
-- Row-level security filters this to North India only (see  
below)
```

```
COLUMN-LEVEL SECURITY (mask PII):
```

```
# Create masking function
```

```
CREATE FUNCTION silver_catalog.dimensions.mask_phone(phone  
STRING)
```

```
RETURNS STRING
```

```
RETURN CASE
```

```
    WHEN is_account_group_member('de_team') THEN phone -- DE sees  
full
```

```
    WHEN is_account_group_member('compliance_team') THEN phone --  
Compliance sees full
```

```
    ELSE CONCAT('XXXXXX', RIGHT(phone, 4)) -- Everyone else:
```

masked!

END;

Apply to column

```
ALTER TABLE silver_catalog.dimensions.dim_customer
  ALTER COLUMN customer_phone
  SET MASK silver_catalog.dimensions.mask_phone;
```

Test: DE team sees: +91-98765-43210

Test: BI analyst sees: XXXXXX3210

ROW-LEVEL SECURITY:

Create filter function

```
CREATE FUNCTION gold_catalog.analytics.region_filter(
  region_code STRING
)
RETURNS BOOLEAN
RETURN
  -- Admins see all
  is_account_group_member('admin_group') OR
  -- Regional managers see their region
  (is_account_group_member('north_india_manager') AND region_code
= 'N') OR
  (is_account_group_member('south_india_manager') AND region_code
= 'S') OR
  (is_account_group_member('de_team')); -- DE team sees all
```

Apply to table

```
ALTER TABLE gold_catalog.analytics.store_performance
  SET ROW FILTER gold_catalog.analytics.region_filter ON
(region_code);
```

*# Now: SELECT * FROM gold_catalog.analytics.store_performance*

```
# North India Manager sees: ONLY North India rows!  
# DE team sees: ALL rows!  
# BI analyst with no specific group: NO rows!
```

PRO TIP

UNITY CATALOG vs AZURE PURVIEW:

"Candidates often confuse these two – clear answer wins:

UNITY CATALOG:

- Databricks-SPECIFIC governance
- Works within Databricks ecosystem
- Table catalog: register, discover, access control
- Column masking, row filtering
- Audit logs for Databricks queries
- Data lineage within Databricks only

AZURE PURVIEW:

- Cross-platform governance (Azure-wide!)
- Works with: ADF, Synapse, SQL, ADLS, Power BI, Databricks
- Catalog: search across ALL Azure data stores
- PII classification (automatic scanning)
- Enterprise-wide audit trail
- Business glossary

TOGETHER (best practice):

Unity Catalog = fine-grained security within Databricks

Azure Purview = enterprise-wide discovery + compliance + lineage

Unity Catalog security policies synced to Purview

→ Purview shows: 'dim_customer.phone: access restricted, masked by UC policy'

This integration = complete enterprise governance.

My recommendation:

Both needed for enterprise, not either/or!

Unity Catalog: daily security enforcement

Purview: quarterly compliance reporting, new employee onboarding"

? QUESTION 48 — Advanced ADF | Dynamic Pipelines

"Build a fully metadata-driven, dynamic ADF pipeline that can load ANY table from ANY source to ANY destination without code changes."

✓ IDEAL ANSWER

METADATA-DRIVEN PIPELINE = The holy grail of ADF architecture
"Configuration in data, not in code"

CONTROL TABLE DESIGN:

```
CREATE TABLE dbo.MetadataDrivenConfig (  
    config_id          INT IDENTITY(1,1) PRIMARY KEY,  
  
    -- SOURCE configuration
```

```

    source_type          VARCHAR(50),    -- ORACLE, SQLSERVER,
BLOB, API
    source_linked_svc    VARCHAR(200),  -- linked service name
    source_schema        VARCHAR(50),
    source_object        VARCHAR(200),  -- table name OR file path
    source_query         NVARCHAR(MAX), -- optional custom query
    partition_column     VARCHAR(100),  -- for parallel read
    partition_upper_bnd  VARCHAR(100),  -- from Lookup or
hardcoded
    partition_lower_bnd  VARCHAR(100),

    -- DESTINATION configuration
    sink_type            VARCHAR(50),    -- ADLS, SYNAPSE, SQLDB
    sink_linked_svc      VARCHAR(200),
    sink_container       VARCHAR(100),
    sink_folder          VARCHAR(500),
    sink_format          VARCHAR(20),    -- PARQUET, CSV, DELTA
    sink_table           VARCHAR(200),  -- for SQL sink
    write_behavior       VARCHAR(20),    -- INSERT, UPSERT,
OVERWRITE

    -- LOAD configuration
    load_type            VARCHAR(20),    -- FULL, INCREMENTAL,
APPEND
    watermark_column     VARCHAR(100),
    last_watermark       VARCHAR(100),
    load_priority        INT DEFAULT 1,

    -- EXECUTION configuration
    is_active            BIT DEFAULT 1,
    parallel_copies      INT DEFAULT 4,
    batch_size           INT DEFAULT 100000,

    -- METADATA
    created_by           VARCHAR(100),

```

```

        created_date          DATETIME2 DEFAULT GETDATE(),
        notes                  NVARCHAR(MAX)
    );

-- SAMPLE DATA:
INSERT INTO dbo.MetadataDrivenConfig VALUES
-- Oracle table with incremental load
('ORACLE',      'LS_Oracle_Prod',  'SALES', 'TRANSACTIONS',
 NULL, 'TRANSACTION_ID', NULL, NULL,
 'ADLS',      'LS_ADLS_Prod',    'bronze', 'oracle/transactions/',
 'PARQUET', NULL, 'UPSERT', 'INCREMENTAL', 'UPDATED_DATE',
 '2000-01-01', 1, 1, 8, 100000, 'Finance', GETDATE(), 'Core
transaction table'),

-- SQL Server full load
('SQLSERVER', 'LS_SQLServer',    'dbo', 'PRODUCTS',
 NULL, NULL, NULL, NULL,
 'ADLS',      'LS_ADLS_Prod',    'bronze', 'sqlserver/products/',
 'PARQUET', NULL, 'OVERWRITE', 'FULL', NULL,
 NULL, 2, 1, 4, 100000, 'Catalog', GETDATE(), 'Product master'),

-- BLOB CSV files
('BLOB',      'LS_BlobStorage',  NULL, '/reports/stores/*.csv',
 NULL, NULL, NULL, NULL,
 'ADLS',      'LS_ADLS_Prod',    'bronze', 'reports/stores/',
 'PARQUET', NULL, 'OVERWRITE', 'FULL', NULL,
 NULL, 3, 1, 1, NULL, 'Ops', GETDATE(), 'Store reports from
FTP');

```

MASTER PIPELINE DESIGN:

```

PL_MASTER_MetadataDriven
|
| STEP 1: Log pipeline start

```

```

|   → SP: Insert into PipelineAuditLog (status=RUNNING)
|
| STEP 2: Lookup active configurations
|   Query: SELECT * FROM dbo.MetadataDrivenConfig
|           WHERE is_active = 1
|           ORDER BY load_priority, config_id
|
| STEP 3: Group by source type (optional optimization)
|   → ForEach source_type → run parallel ForEach within
|
| STEP 4: ForEach each config
|   Batch Count: 8 (8 parallel loads)
|   Sequential: No
|
|   INSIDE ForEach:
|   └─ STEP 4a: Get watermark (if INCREMENTAL)
|   |   If Condition: @{equals(item().load_type, 'INCREMENTAL')}
|   |   TRUE: LKP_GetWatermark
|   |       SELECT last_watermark FROM MetadataDrivenConfig
|   |       WHERE config_id = @{item().config_id}
|   |
|   └─ STEP 4b: Route to appropriate copy pipeline
|   |   Switch Activity on source_type:
|   |   CASE 'ORACLE':      Execute Pipeline → PL_Copy_FromOracle
|   |   CASE 'SQLSERVER':   Execute Pipeline → PL_Copy_FromSqlServer
|   |   CASE 'BLOB':        Execute Pipeline → PL_Copy_FromBlob
|   |   CASE 'API':         Execute Pipeline → PL_Copy_FromAPI
|   |
|   └─ STEP 4c: Update watermark (SUCCESS path only!)
|   |   If INCREMENTAL:
|   |   SP: UPDATE MetadataDrivenConfig
|   |       SET last_watermark = @{utcnow()}
|   |       WHERE config_id = @{item().config_id}
|   |
|   └─ STEP 4d: Log table success

```

```
| | SP: UPDATE PipelineAuditLog SET rows_copied=...
| |
| └ STEP 4e: Log table failure (failure path)
|     SP: INSERT error log
|
| STEP 5: Send completion report
|     Web Activity → Logic App
|     "Loaded X tables, Y failures, Z total rows"
|
└ STEP 6: Log pipeline end
```

`PL_Copy_FromOracle` (child pipeline):

Parameters received from master:

```
p_linked_service, p_schema, p_table
p_load_type, p_watermark
p_sink_container, p_sink_folder, p_sink_format
p_parallel_copies, p_partition_column
```

Copy Activity:

Source Linked Service:

`@{pipeline().parameters.p_linked_service}`

Source Query:

```
@{if(
    equals(pipeline().parameters.p_load_type, 'INCREMENTAL'),
    concat('SELECT * FROM ', pipeline().parameters.p_schema,
    '.',
        pipeline().parameters.p_table,
        ' WHERE ', pipeline().parameters.p_watermark_col,
        ' > TO_DATE(''', pipeline().parameters.p_watermark,
        ''', 'YYYY-MM-DD HH24:MI:SS')'),
    concat('SELECT * FROM ', pipeline().parameters.p_schema,
    '.',
        pipeline().parameters.p_table)
```



```
}}
```

Partition option: Dynamic Range

Partition column: @{{pipeline().parameters.p_partition_column}}

Degree of parallelism:

@{{pipeline().parameters.p_parallel_copies}}

Sink: ADLS Gen2 Dynamic Path

Container: @{{pipeline().parameters.p_sink_container}}

Folder: @{{concat(pipeline().parameters.p_sink_folder,
formatDateTime(utcnow(), 'yyyy/MM/dd'),

'/')}}

File: @{{concat(pipeline().parameters.p_table, '_',
formatDateTime(utcnow(), 'yyyyMMddHHmmss'),
'parquet')}}}

Format: @{{pipeline().parameters.p_sink_format}}

ADDING NEW **TABLE** (zero code change!):

Just add a row to MetadataDrivenConfig table!

INSERT INTO **MetadataDrivenConfig** (source_type, ...)

VALUES ('ORACLE', 'LS_Oracle', 'INVENTORY', 'STOCK_LEVELS',
...)

Next pipeline run: STOCK_LEVELS table automatically loaded!

No ADF changes, no deployment, no testing!

PRO TIP

SELF-HEALING PIPELINE (what 20+ LPA architects build):

"Beyond metadata-driven, I build SELF-HEALING pipelines:

SELF-HEALING = Pipeline that automatically handles
common failure scenarios without human
intervention

SCENARIO: Oracle DB temporarily unavailable (5 min outage)

Standard approach:

- Pipeline fails → alert fires → engineer investigates
- Engineer confirms Oracle is back → manually reruns
- 30-60 minutes total delay

Self-healing approach:

- Step 1: Copy Activity fails (Oracle timeout)
- Step 2: Activity retry: wait 5 min, try again (retry=3)
- Step 3: Oracle back after 5 min → retry succeeds!
- Step 4: Log: 'Auto-recovered after 1 retry'
- Step 5: No alert fires (recovered automatically)
- Step 6: Engineer sees in morning: '1 auto-recovery overnight'

SELF-HEALING MECHANISMS:

Retry with exponential backoff:

- Retry 1: after 2 minutes
- Retry 2: after 4 minutes (2x)
- Retry 3: after 8 minutes (2x again)

Circuit breaker pattern:

If same table fails 3 days in a row:

- Mark as 'CIRCUIT_OPEN' in config table
- Skip this table, alert team
- Prevents wasting resources on permanently broken source

Fallback data source:

Primary: Oracle DB

Fallback: Previous day's cached data (if Oracle fails)

- Report shows yesterday's data with [CACHED] label
- Business has SOME data vs NO data

This robustness thinking = 20+ LPA solution architect level!"

? QUESTION 49 — Advanced Spark | Memory Management

"Explain Spark memory architecture. How do you handle OutOfMemory errors?"

✓ IDEAL ANSWER

SPARK EXECUTOR MEMORY ARCHITECTURE:

Total Executor Memory (--executor-memory 16g)

├ Reserved Memory: 300MB (always reserved for Spark internals)

|

├ Usable Memory: 16GB - 300MB ≈ 15.7GB

|

| └ Spark Memory (spark.memory.fraction = 0.6 = 60%)

| ≈ 9.4GB

| ├ Execution Memory (50% of Spark memory by default)

| | ≈ 4.7GB

| | Used for: sorting, shuffles, aggregations,
joins

| | Can borrow from Storage if available

|

|

|

```
|      |      └─ Storage Memory (50% of Spark memory by default)
|      |      ≈ 4.7GB
|      |      Used for: cached DataFrames, broadcast
variables
|      |      Can be borrowed by Execution (cache evicted!)
|      |
|      └─ User Memory (1 - 0.6 = 40%)
|      ≈ 6.3GB
|      Used for: your Python objects, UDFs, custom data
structures
|      Spark DOESN'T manage this (memory leaks possible
here!)
```

KEY INSIGHT:

Execution + Storage memory is UNIFIED and DYNAMIC!

If execution needs more → it borrows from storage (evicts cache)

If storage needs more → borrows from execution (spill to disk)

This prevents OOM in most cases!

COMMON OOM CAUSES AND FIXES:

CAUSE 1: Too large a DataFrame cached

✗ Caching 100GB DataFrame on 64GB cluster

→ Cache evicts → spills to disk → slow

→ If not enough disk → OOM!

FIX A: Don't cache if doesn't fit in memory

FIX B: Use MEMORY_AND_DISK (spills to disk, not OOM)

```
df.persist(StorageLevel.MEMORY_AND_DISK)
```

FIX C: Cache FILTERED data (not full table)

```
df.filter(col("year") == 2024).cache() # 1 year, not 5
```

CAUSE 2: collect() on large DataFrame

❌ `rows = df.collect()` ← brings ALL 500M rows to DRIVER!
→ Driver has 4GB RAM
→ 500M rows = 50GB
→ DRIVER OOM!

FIX: Never `collect()` large data
→ `df.write()` to storage instead
→ `df.show(20)` for sampling
→ `df.take(100)` for small samples only

CAUSE 3: Data Skew (one executor has too much)

`df.groupBy("customer_id").agg(sum("amount"))`
If WALK-IN customer = 40M rows → one executor gets 40M rows
→ That executor OOM!

FIX: Salt the key (spread across partitions)
FIX: Enable AQE skew join handling

CAUSE 4: Cartesian join (accidental)

❌ `df1.join(df2)` without ON condition!
→ $500M \times 1M = 500$ TRILLION rows!
→ Immediate OOM!

FIX: ALWAYS specify join condition!

CAUSE 5: Too many shuffle partitions too small

After a shuffle: 200 partitions each = 10MB
→ $200 \times$ overhead per partition → memory fragmentation
→ Eventually OOM from garbage collection pressure

FIX: Coalesce after filter to reduce partition count
Or use AQE coalescing

CAUSE 6: Python UDF memory leak

UDF stores reference to large object:

```
large_lookup = {k: v for k, v in 10M_items} # 2GB in memory!
udf_func = lambda x: large_lookup.get(x)      # reference to
2GB
udf = udf(udf_func, StringType())
```

- Each executor has copy of large_lookup
- 10 executors × 2GB = 20GB wasted on lookup!

FIX: Use Broadcast variable instead!

```
lookup_bc = sc.broadcast(large_lookup)
udf_func = lambda x: lookup_bc.value.get(x)
→ Broadcast: sent ONCE per executor, not per task!
→ 10 executors × 2GB broadcast = still 2GB per executor
    but managed by Spark (reusable, can be unpersisted)
```

TUNING MEMORY CONFIGURATION:

```
# If lots of caching:
spark.conf.set("spark.memory.storageFraction", "0.5") # default
0.5
# More storage = better cache hit rate

# If lots of shuffle/aggregation:
spark.conf.set("spark.memory.storageFraction", "0.3")
# Less storage → more execution memory → less spill to disk

# Increase executor overhead (for Python UDFs, native code):
spark.conf.set("spark.executor.memoryOverhead", "4g")
# Overhead = memory outside JVM heap
# Important for: pandas UDFs, native libraries, Python processes

# For OOM from serialization:
spark.conf.set("spark.serializer",
"org.apache.spark.serializer.KryoSerializer")
```

```
# Kryo = 10x smaller than default Java serializer
# Less memory used during shuffle!
```

PRO TIP

MEMORY DEBUGGING CHECKLIST:

"When I see OOM errors, I follow this checklist:

STEP 1: Identify WHERE OOM happened

- Driver OOM? → collect() or very large broadcast
- Executor OOM? → data skew, too large aggregation, UDF leak
- Spark UI → Failed task → exception type tells you!

STEP 2: Check cache usage

- Spark UI → Storage tab
- How much memory is cached?
- Is everything that's cached actually needed?
- Unpersist what you don't need!

STEP 3: Check data skew

- Spark UI → Stages → Tasks
- Max task size vs median task size
- 10x difference = SKEW → salting needed!

STEP 4: Review collect() and show() calls

- Search your notebook for collect()
- Any collect() on large DataFrame → replace with write()!

STEP 5: Check UDF memory usage

- Any Python UDFs with large objects? → broadcast them!

STEP 6: Tune configuration

Last resort: increase executor memory

`spark.executor.memory: 8g → 16g`

`spark.executor.memoryOverhead: 2g → 4g (Python UDFs)`

MONITORING MEMORY (proactive):

`SparkContext.getOrCreate().statusTracker().getExecutorInfos()`

→ Shows each executor memory usage LIVE

Or: Databricks Metrics (live cluster metrics)

→ 'JVM Heap Used' approaching 'JVM Heap Total' → WARNING!

This systematic approach = senior debugging skills!"

? QUESTION 50 — Advanced | CI/CD for Data Engineering

"How do you implement CI/CD for Azure Data Engineering projects? Walk through the complete pipeline."

✓ IDEAL ANSWER

CI/CD FOR DATA ENGINEERING:

WHY CI/CD FOR DATA:

Code: test → deploy

Data pipelines: same principle!

Without CI/CD:

- Manual deployment (copy-paste pipelines?)
- No testing before production
- "Works on my dev cluster" → breaks on prod!
- No rollback when something goes wrong
- Multiple people changing production directly

COMPLETE CI/CD PIPELINE:

REPOSITORY STRUCTURE:

```
retailmart-data-platform/  
├─ adf/                                ← ADF JSON files (auto-synced from  
Git)  
|   ├─ pipeline/  
|   ├─ dataset/  
|   ├─ linkedService/  
|   └─ trigger/  
├─ databricks/                        ← Databricks notebooks  
|   ├─ bronze/  
|   ├─ silver/  
|   └─ gold/  
├─ sql/                               ← SQL scripts  
|   ├─ ddl/  
|   └─ stored_procedures/  
├─ tests/                             ← Unit tests  
|   ├─ unit/  
|   └─ integration/  
├─ config/                           ← Environment configs  
|   ├─ dev.json  
|   ├─ test.json  
|   └─ prod.json  
└─ azure-pipelines.yml                ← CI/CD definition
```

BRANCHING STRATEGY:

```
main ← production code (protected)
dev ← integration branch
feature/* ← developer feature branches
```

CI PIPELINE (azure-pipelines.yml):

```
trigger:
  branches:
    include: [dev, main]
  paths:
    include: ['databricks/**', 'adf/**', 'sql/**']

stages:
  - stage: CI
    jobs:
      - job: Lint
        steps:
          - task: UsePythonVersion@0
            inputs: {versionSpec: '3.9'}

          - script: pip install flake8 black isort
            displayName: Install linting tools

          - script: flake8 databricks/ tests/ --max-line-length
            120
            displayName: Lint Python code

          - script: black --check databricks/ tests/
            displayName: Check code formatting

          - script: |
              # Check for hardcoded credentials (security scan!)
              grep -r "password\\|secret\\|key" databricks/ --
```

```

include="*.py" |
    grep -v "Key Vault\|getSecret\|#" |
    grep -v "test_" &&
    echo "SECURITY WARNING: Possible hardcoded
credentials!" &&
    exit 1 || echo "Security check passed"
    displayName: Security scan for hardcoded secrets

- job: UnitTests
  dependsOn: Lint
  steps:
    - script: pip install pytest pytest-cov pyspark delta-
spark

    - script: |
        pytest tests/unit/ \
        --cov=databricks \
        --cov-report=xml \
        --cov-report=html \
        -v
        displayName: Run unit tests

- task: PublishTestResults@2
  inputs:
    testResultsFiles: '**/test-*.xml'

- task: PublishCodeCoverageResults@1
  inputs:
    codeCoverageTool: Cobertura
    summaryFileLocation:
'$(System.DefaultWorkingDirectory)/**/coverage.xml'

- job: ValidateADF
  steps:
    - script: |

```

```

        pip install azure-mgmt-datafactory
        python scripts/validate_adf_jsons.py # validate
all ADF JSON
    displayName: Validate ADF JSON files

- script: |
    # Ensure all ADF activities have retry configured
    python scripts/check_adf_standards.py \
        --require-retry \
        --require-error-path \
        --max-diu 32
    displayName: ADF best practice checks

```

CD PIPELINE:

```

- stage: Deploy_Dev
  condition: and(succeeded(),
eq(variables['Build.SourceBranch'], 'refs/heads/dev'))
  jobs:
    - deployment: DeployToDevADF
      environment: dev
      strategy:
        runOnce:
          deploy:
            steps:
              # Deploy ADF using ARM template
              - task: AzureResourceManagerTemplateDeployment@3
                inputs:
                  deploymentScope: Resource Group
                  azureResourceManagerConnection: 'sc-
retailmart-dev'

                  resourceGroupName: 'rg-retailmart-dev'
                  location: 'Central India'
                  templateLocation: 'Linked artifact'

```

```

        csmFile:
'$(Pipeline.Workspace)/adf/ARMTemplateForFactory.json'
        csmParametersFile:
'$(Pipeline.Workspace)/config/dev.json'

    # Deploy Databricks notebooks
    - script: |
        pip install databricks-cli
        databricks configure --token \
            --host $(DATABRICKS_HOST_DEV) \
            --token $(DATABRICKS_TOKEN_DEV)

        databricks workspace import_dir \
            databricks/ \
            /Production \
            --overwrite
        displayName: Deploy notebooks to Dev Databricks

- stage: Integration_Tests
  dependsOn: Deploy_Dev
  jobs:
    - job: RunIntegrationTests
      steps:
        - script: |
            python tests/integration/test_pipeline_e2e.py \
                --environment dev \
                --pipeline PL_Daily_Sales_Load \
                --process-date 2024-01-15
            displayName: Run E2E integration test

- stage: Deploy_Test
  dependsOn: Integration_Tests
  condition: and(succeeded(),
eq(variables['Build.SourceBranch'], 'refs/heads/dev'))
  jobs:

```

```

- deployment: DeployToTest
  environment: test
  # Same as dev deployment, different config file

- stage: Deploy_Prod
  dependsOn: Deploy_Test
  condition: and(succeeded(),
eq(variables['Build.SourceBranch'], 'refs/heads/main'))
  jobs:
    - deployment: DeployToProd
      environment: prod # ← Requires MANUAL APPROVAL!
      strategy:
        runOnce:
          deploy:
            # Same as test deployment with prod config

```

UNIT TEST EXAMPLE:

```

# tests/unit/test_transform_sales.py

import pytest
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from databricks.silver.transform_sales import (
    calculate_gst, validate_sale_record, standardize_columns
)

@pytest.fixture(scope="session")
def spark():
    return SparkSession.builder \
        .master("local[2]") \
        .appName("test") \
        .getOrCreate()

```

```

def test_calculate_gst(spark):
    """Test GST calculation is 18%"""
    data = [(1, 10000.0), (2, 50000.0), (3, 100.0)]
    df = spark.createDataFrame(data, ["sale_id", "amount"])
    df_result = calculate_gst(df)

    results = df_result.select("sale_id", "gst_amount").collect()
    assert results[0]["gst_amount"] == 1800.0 # 10000 × 0.18
    assert results[1]["gst_amount"] == 9000.0 # 50000 × 0.18
    assert results[2]["gst_amount"] == 18.0 # 100 × 0.18

def test_validate_sale_record_rejects_null_id(spark):
    """Test that null sale_id is rejected"""
    data = [(None, "C001", 10000.0), (1, None, 20000.0)]
    df = spark.createDataFrame(data, ["sale_id", "customer_id",
    "amount"])
    df_valid, df_invalid = validate_sale_record(df)

    assert df_valid.count() == 0 # both should be invalid
    assert df_invalid.count() == 2

def test_pipeline_idempotency(spark, tmp_path):
    """Test running pipeline twice produces same result"""
    input_data = spark.createDataFrame(
        [(1, "C001", 10000.0), (2, "C002", 20000.0)],
        ["sale_id", "customer_id", "amount"]
    )

    output_path = str(tmp_path / "output")

    # Run twice
    run_transformation(input_data, output_path)
    count_1 = spark.read.parquet(output_path).count()

    run_transformation(input_data, output_path)

```

```
count_2 = spark.read.parquet(output_path).count()

assert count_1 == count_2, f"Idempotency failed! {count_1} != {count_2}"
```

PRO TIP

DEPLOYMENT BEST PRACTICES:

"For ZERO DOWNTIME data pipeline deployments:

BLUE-GREEN for ADF:

Blue: Current ADF (running production pipelines)

Green: New ADF (with changes deployed)

Switch: Update triggers to fire on Green ADF

Rollback: Switch triggers back to Blue (instant!)

For tumbling window triggers: pause → switch → resume

For event triggers: delete → recreate on Green

FEATURE FLAGS FOR PIPELINES:

Global parameter: `gp_feature_new_algorithm = false`

In pipeline:

IF `gp_feature_new_algorithm = true` → use new notebook

IF `gp_feature_new_algorithm = false` → use old notebook

Deployment: set global param to TRUE for 1% of runs (test!)

If metrics OK: set to 100%

If bad: set to 0% instantly

This is CANARY DEPLOYMENT for data pipelines!

ROLLBACK STRATEGY:

Delta Lake time travel = instant rollback!

If deployment corrupted data:

RESTORE TABLE silver_sales TO VERSION AS OF 45

→ Instantly back to pre-deployment state!

→ No manual data fix needed!

This is WHY Delta Lake is critical for production.

Git rollback + Delta Lake rollback = complete safety net!"



REVISION SUMMARY — Q46-Q50

REVISION: GOVERNANCE + ADVANCED TOPICS (Q46-Q50)

Q46: Azure Purview

5 capabilities: Data Map, Catalog, Classification,

Lineage, Access Policies

KEY: Business Glossary = unified metric definitions!

Q47: Unity Catalog

Hierarchy: Metastore→Catalog→Schema→Table

Column masking + Row filtering for PII

KEY: Unity Catalog = Databricks, Purview = Azure-wide!

Q48: Metadata-Driven Pipelines

Control table drives ALL pipeline behavior

Add table = add row, **no** code change!

KEY: Self-healing = exponential retry + circuit breaker!

Q49: Memory Management

Execution + Storage = dynamic, shared pool

OOM: skew, collect(), cartesian join, UDF leaks

KEY: KryoSerializer for less memory in shuffle!

Q50: CI/CD

Lint → Unit Test → ADF Validate → Deploy Dev →

Integration Test → Deploy Test → Approve → Deploy Prod

KEY: Test idempotency! Blue-green deployment!

PATTERN ALERT:

Senior interviews always test:

→ "How do you ensure quality?" → CI/CD + tests

→ "How do you secure data?" → Unity Catalog + Purview

→ "How do you make maintainable?" → Metadata-driven!

STREAMING + REAL-TIME DEEP DIVE

? QUESTION 51 — Streaming | Azure Event Hubs

"Explain Azure Event Hubs architecture. How does it integrate with Databricks for streaming?"

✓ IDEAL ANSWER

AZURE EVENT HUBS = Managed event streaming platform

"Azure's Kafka"

KEY CONCEPTS:

NAMESPACE:

Container **for** Event Hubs

Like a server **for** multiple event topics

Pricing tier: Basic, Standard, Premium, Dedicated

EVENT HUB:

Individual stream of events

Like a Kafka Topic

Has: partitions, consumer **groups**, retention period

PARTITION:

Event Hub divided into N partitions

Each partition = ordered sequence of events

Events within partition: ALWAYS IN ORDER

Events ACROSS partitions: not guaranteed order!

Why partitions?

→ Parallel consumption (each partition = one reader)

→ 8 partitions = 8 readers = 8x throughput!

Rule: Partition count = max parallel consumers

RetailMart: 32 partitions (32 Spark tasks **read in** parallel)

PARTITION KEY:

Controls **which** partition event goes to

Same key → same partition → ordered!

Payment events → key = merchant_id

All events **for** Merchant X → same partition → ordered!

Without key → round-robin distribution

CONSUMER GROUP:

Independent reader of the same Event Hub

Each consumer group reads from beginning independently!

Consumer Group 1: Fraud Detection Service

Consumer Group 2: Analytics Service

Consumer Group 3: Audit Service

All reading SAME events independently!

Like broadcast – one publish, many consumers!

RETENTION:

Events retained **for** 1-7 days (Standard tier)

Replay: **if** consumer fails → **read** from beginning!

Unlike traditional messaging queues (once consumed = gone)

THROUGHPUT UNITS (TU):

1 TU = 1 MB/s ingress, 2 MB/s egress

Auto-inflate: automatically scales TUs on demand!

DATABRICKS INTEGRATION:

```
from pyspark.sql import functions as F
from pyspark.sql.types import *
```

```
# Connection config (use secrets, never hardcode!)
connectionString = dbutils.secrets.get(
    "kv-scope", "eventhubs-connection-string"
)
```

```

# Event Hubs configuration
ehConf = {
  "eventhubs.connectionString":
    sc._jvm.org.apache.spark.eventhubs \
      .EventHubsUtils.encrypt(connectionString),

  # Consumer group (each streaming job should have its own!)
  "eventhubs.consumerGroup": "fraud-detection-consumer",

  # Starting position
  "eventhubs.startingPosition": json.dumps({
    "offset": "-1",      # -1 = from beginning
    # or: "@latest" = from now (new events only)
    # or: specific offset to resume from
  }),

  # Max events per micro-batch per partition
  "eventhubs.maxEventsPerTrigger": 10000
}

# READ STREAM from Event Hubs
df_stream = spark.readStream \
  .format("eventhubs") \
  .options(**ehConf) \
  .load()

# df_stream schema:
# body:          binary ← your actual event payload
# partition:     string ← which partition
# offset:        string ← position in partition
# sequenceNumber: long ← sequence number in partition
# enqueueTime:   timestamp ← when event was added to EH
# publisher:     string
# partitionKey:  string

```

```

# PARSE the body (binary → JSON → struct)
payment_schema = StructType([
    StructField("payment_id", StringType(), True),
    StructField("merchant_id", StringType(), True),
    StructField("amount", DoubleType(), True),
    StructField("timestamp", StringType(), True),
    StructField("status", StringType(), True)
])

df_payments = df_stream \
    .select(
        F.from_json(
            F.col("body").cast("string"), # binary → string →
            json
            payment_schema
        ).alias("data"),
        "enqueuedTime", # Event Hub timestamp (arrival time)
        "partition",
        "offset"
    ) \
    .select(
        "data.*", # flatten the parsed JSON
        "enqueuedTime",
        "partition"
    )

# PROCESS: fraud detection
df_fraud_stream = df_payments \
    .withWatermark("enqueuedTime", "5 minutes") \
    .filter(F.col("amount") > 100000) \
    .withColumn("fraud_risk", F.lit("HIGH")) \
    .withColumn("processed_at", F.current_timestamp())

# WRITE STREAM: multiple outputs
# Output 1: Write all payments to Delta (Bronze)

```

```

df_payments.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation", "/mnt/checkpoints/payments-
bronze/") \
    .trigger(processingTime="30 seconds") \
    .toTable("bronze.payments_streaming")

# Output 2: Write HIGH RISK to alerts table
df_fraud_stream.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation", "/mnt/checkpoints/fraud-
alerts/") \
    .trigger(processingTime="10 seconds") \
    .toTable("gold.fraud_alerts")

# Output 3: Aggregated throughput metrics
df_payments \
    .withWatermark("enqueuedTime", "1 minute") \
    .groupBy(
        F.window("enqueuedTime", "1 minute"),
        "partition"
    ) \
    .agg(
        F.count("payment_id").alias("events_per_minute"),
        F.sum("amount").alias("volume_per_minute")
    ) \
    .writeStream \
    .outputMode("update") \
    .format("delta") \
    .option("checkpointLocation", "/mnt/checkpoints/throughput-
metrics/") \
    .trigger(processingTime="1 minute") \
    .toTable("monitoring.throughput_metrics")

```



```
# MONITOR streaming queries
for q in spark.streams.active:
    print(f"Query: {q.name}")
    print(f"Status: {q.status}")
    print(f"Progress: {q.lastProgress}")
```

💡 PRO TIP

EVENT HUBS vs **KAFKA** (interview gotcha!):

"Interviewer often asks: why Event Hubs over Kafka?"

KAFKA (self-managed):

- ✓ Open source, any cloud **or** on-premise
- ✓ More **features** (exactly-once **with** transactions)
- ✓ More configuration control
- ✗ You manage brokers, ZooKeeper, storage
- ✗ Scaling requires manual work
- ✗ Security setup complex

EVENT HUBS:

- ✓ Fully **managed** (Azure manages everything)
- ✓ Auto-**scaling** (Throughput Units auto-inflate)
- ✓ Kafka API compatible! (Drop-**in** replacement!)
- ✓ Native Azure RBAC security
- ✓ Deep Azure Monitor integration
- ✗ Azure-**only** (vendor **lock-in**)
- ✗ Less configuration options

KAFKA COMPATIBILITY:

Event Hubs supports Kafka protocol!

Your existing Kafka code → change endpoint → works **with** Event Hubs!

```
# SAME code works for both:
kafka_config = {
    'bootstrap.servers': 'your-eventhub-
namespace.servicebus.windows.net:9093',
    'security.protocol': 'SASL_SSL',
    'sasl.mechanisms': 'PLAIN',
    'sasl.username': '$ConnectionString',
    'sasl.password': event_hub_connection_string
}
```

This means: **on**-prem Kafka → migrate to Event Hubs
= zero code **change** (just update bootstrap.servers!)

RECOMMENDATION:

Azure-only workload → Event **Hubs** (simpler, managed)
Multi-cloud **or** existing Kafka → Kafka **on** HDInsight **or** Confluent
Want Kafka compatibility but managed → Event Hubs **with** Kafka
API!"

? QUESTION 52 — Streaming | Exactly-Once Processing

"What is exactly-once processing in streaming? How do you achieve it in Databricks Structured Streaming?"

✓ IDEAL ANSWER

DELIVERY SEMANTICS - 3 TYPES:

AT-MOST-ONCE (Possible data loss):

- Process event → commit offset → done
- If processing fails AFTER commit: event lost!
- "Maybe processed once, maybe never"
- Use: when data loss is acceptable (metrics, telemetry)

AT-LEAST-ONCE (Possible duplicates):

- Commit offset AFTER processing succeeds
- If commit fails: re-process same event!
- "Will process, might process twice"
- Use: when duplicates can be handled (deduplicate later)

EXACTLY-ONCE (Ideal):

- Event processed exactly one time, no more, no less
- Hardest to implement!
- Use: financial transactions, inventory updates

HOW STRUCTURED STREAMING ACHIEVES EXACTLY-ONCE:

Structured Streaming uses TWO mechanisms together:

MECHANISM 1: CHECKPOINTING (Idempotent Sources)

Checkpoint stores: "Last offset processed"

Failure scenario:

1. Process batch (offsets 0-1000)
2. Write to Delta
3. FAIL before checkpoint update!
4. Restart: reads checkpoint → "last offset = -1"

5. Re-processes offsets 0-1000 → DUPLICATE!?

Solution: Delta write is IDEMPOTENT

- Retrying same micro-batch = same files written
- Delta's transaction log detects same batch ID
- IGNORES the retry if already committed!

MECHANISM 2: MICRO-BATCH IDEMPOTENCY

Each micro-batch has a BATCH ID (incrementing integer)

Delta writer saves batch ID with write

If failure + **retry**: SAME batch ID

Delta: "I already have batch 42 → skip this write!"

Result: Only ONE write per batch regardless of retries!

CONFIGURATION FOR EXACTLY-ONCE:

Checkpoint: mandatory for exactly-once

```
df.writeStream \  
  .option("checkpointLocation", "/mnt/checkpoints/payments/") \  
  # ↑ This is NON-NEGOTIABLE for exactly-once!
```

Delta sink: automatically exactly-once!

Delta handles the idempotency automatically

Non-Delta sink (custom):

Must implement ForeachBatchWriter with idempotent logic

```
def foreach_batch_func(df, batch_id):  
  # batch_id = unique ID for this micro-batch  
  # Use batch_id to ensure idempotency in your logic!  
  
  df.write \  
    .option("replaceWhere", f"batch_id = {batch_id}") \  
    .format("delta") \  
    .mode("overwrite") \  
    .execute()
```

```
.save(target_path)

df_stream.writeStream \
    .foreachBatch(foreach_batch_func) \
    .option("checkpointLocation", "/mnt/checkpoints/custom/") \
    .start()
```

EXACTLY-ONCE END-TO-END:

Source (Event Hubs):

- Partitioned → offsets tracked
- Checkpointed → resume from exact offset after failure

Processing (Structured Streaming):

- Micro-batch with sequential batch IDs
- **Deterministic**: same input = same output every time
- State stored in checkpoints

Sink (Delta Lake):

- Writes idempotent (same batch ID = same result)
- Transaction log prevents duplicate commits

ALL THREE together = Exactly-Once!

COMMON EXACTLY-ONCE VIOLATION:

SIDE EFFECTS IN STREAMING (breaks exactly-once!):

BAD: External API call in streaming

```
def foreach_batch_bad(df, batch_id):
    rows = df.collect()
    for row in rows:
        requests.post("https://api.fraud.com/alert",
                      json={"payment": row.payment_id}) #
external call!
```

```

df.write.format("delta").save(path)

# If pipeline fails AFTER API calls but BEFORE Delta write:
# → Retry → API called AGAIN!
# → Fraud alert sent TWICE for same payment!
# → Exactly-once violated!

# GOOD: Write to Delta first, trigger API from downstream
def foreach_batch_good(df, batch_id):
    df.write.format("delta").mode("append").save(alerts_path)
    # Delta write is idempotent ✅

# Downstream job reads from alerts Delta table
# Triggers API calls (with deduplication by payment_id)
# IDEMPOTENT: same payment_id → same API call → handled by API

# Separation: idempotent streaming + idempotent downstream!

```

💡 PRO TIP

CHECKPOINT MANAGEMENT (operational knowledge):

"Checkpoints are critical but often mismanaged:

CHECKPOINT DIRECTORY RULES:

1. UNIQUE per streaming query (NEVER share!)

query1 checkpoint: /mnt/ckpt/query1/

query2 checkpoint: /mnt/ckpt/query2/

← SEPARATE directories!

2. NEVER delete manually (breaks stream state!)

Deleting checkpoint = stream forgets where it was

→ Reprocesses from beginning OR loses data!

3. Schema change → new checkpoint directory!

If you change streaming query → move to new checkpoint
Old checkpoint may be incompatible with new schema!

4. Store in RELIABLE storage (ADLS, not local disk!)

Local disk: cluster restarts → checkpoint gone → reprocess from start!

ADLS: persists across cluster restarts!

CHECKPOINT SIZE MONITORING:

Large stateful operations (watermark windows) → large checkpoint

Check: `ls -la /mnt/checkpoints/payments/`

If checkpoint > 10GB → investigate!

Large checkpoint = slow recovery time after failure!

UPGRADE STREAMING QUERY:

Sometimes you MUST change the query (add new column)

Option 1: Start fresh (lose progress)

- Delete checkpoint
- Start from latest events
- Miss historical events!

Option 2: Read checkpoint migration

- Carefully update checkpoint files
- Complex, error-prone!

Option 3: Blue-Green streaming (my recommendation)

- Start NEW query in parallel
- Both read same Event Hub (different consumer groups)
- Old query: continues until new query catches up
- Switch consumers: redirect to new query

- Old query: stop
- Zero data loss, zero downtime!

This operational expertise = 15+ LPA senior knowledge!"

? QUESTION 53 — Scenario | Real-Time Dashboard

"Design a real-time sales dashboard that shows data within 30 seconds of a transaction. What is your architecture?"

✓ IDEAL ANSWER

REAL-TIME DASHBOARD ARCHITECTURE:

REQUIREMENT CLARIFICATION:

- 30-second latency (data visible within 30s)
- Sales dashboard: total revenue, transaction count, by region
- Volume: 1000 transactions/minute (RetailMart)
- Availability: 99.9% (business critical!)

ARCHITECTURE:

POS Systems → Azure Event Hubs → Databricks Streaming
→ Materialized Delta Table (aggregated, 30s refresh)
→ Power BI Direct Lake → Live Dashboard

DETAILED FLOW:

INGESTION (< 1 second latency):

POS terminal sends transaction:

POST `https://retailmart-eventhub.servicebus.windows.net
/payment-transactions/messages`

```
{  
  "payment_id": "P2024061500045782",  
  "store_id": "S042",  
  "region_code": "N",  
  "amount": 15000,  
  "timestamp": "2024-06-15T14:32:11.123Z"  
}
```

→ Event Hubs receives **immediately** (< 10ms!)

PROCESSING (10-30 second window):

Databricks Structured Streaming:

- Reads from Event **Hubs** (32 partitions, high throughput)
- Watermark: 30 seconds (late data tolerance)
- Aggregates **in** 30-second tumbling windows:

```
df_realttime = spark.readStream \  
  .format("eventhubs") \  
  .options(**ehConf) \  
  .load() \  
  .select(F.from_json(F.col("body").cast("string"),  
                      payment_schema).alias("d"),  
          "enqueuedTime") \  
  .select("d.*", "enqueuedTime") \  
  .withWatermark("timestamp", "30 seconds") \  
  .groupBy(  
    F.window("timestamp", "30 seconds"),  
    "region_code"  
  ) \  
  .agg(  
    F.sum("amount").alias("revenue_30s"),
```

```

        F.count("payment_id").alias("transactions_30s"),
        F.countDistinct("store_id").alias("active_stores")
    ) \
    .withColumn("window_start", F.col("window.start")) \
    .withColumn("window_end", F.col("window.end")) \
    .drop("window")

df_realtime.writeStream \
    .format("delta") \
    .outputMode("append") \
    .option("checkpointLocation", "/mnt/ckpt/realtime-
dashboard/") \
    .trigger(processingTime="10 seconds") \
    .toTable("gold.realtime_30s_summary")

ALSO: Maintain running totals (for today's totals):
df_today = spark.readStream \
    .format("eventhubs") \
    .options(**ehConf) \
    .load() \
    .select("d.*") \
    .filter(F.col("timestamp") >= F.current_date()) \
    .withWatermark("timestamp", "30 seconds") \
    .groupBy("region_code") \
    .agg(
        F.sum("amount").alias("today_revenue"),
        F.count("payment_id").alias("today_transactions")
    )

df_today.writeStream \
    .format("delta") \
    .outputMode("complete") \
    .option("checkpointLocation", "/mnt/ckpt/today-totals/") \
    .trigger(processingTime="30 seconds") \
    .toTable("gold.today_totals")

```

POWER BI (Automatic Direct Lake refresh):

Power BI Desktop → Connect to gold.realtime_30s_summary

Direct Lake mode: reads Delta files every 30 seconds!

Report page: "Live Sales"

→ Auto-refresh: every 30 seconds (Power BI setting)

→ Shows: revenue in last 30 seconds by region

→ Running total: today's revenue

→ Transaction velocity: graph of 30s windows

TOTAL LATENCY:

POS → Event Hubs: < 100ms

Event Hubs → Spark: < 5 seconds (batch)

Spark processing: < 10 seconds (30s window + 10s trigger)

Delta write: < 2 seconds

Power BI refresh: ≤ 30 seconds

TOTAL: 30-45 seconds (within 30s requirement for most events!)

ALTERNATIVE FOR <5 SECOND LATENCY:

If need sub-5 second:

→ Event Hubs → Azure Stream Analytics

→ Direct output to Azure Cosmos DB (NoSQL, fast writes)

→ Power BI → Cosmos DB (streaming dataset)

→ TRUE REAL-TIME Power BI streaming visuals!

→ Cost

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